

Complex Dynamics

From Local Normal Forms to Renormalization
and Modern Frontiers

A proof-oriented graduate textbook

2026 edition

Preface

Complex dynamics begins with a deceptively simple operation: choose a holomorphic or meromorphic map and iterate it. The resulting theory is not a catalogue of pictures of Julia sets. It is a network of mechanisms that repeatedly convert one kind of information into another: a local multiplier into a normal form, normality into a global stable/chaotic decomposition, potential theory into external coordinates, quasiconformal deformation into rigidity, real bounds into complex bounds, and renormalization into universality.

This book develops that network as a single proof-oriented narrative. It grows out of an earlier twenty-four-volume development, but the present edition is deliberately not a concatenation of twenty-four stand-alone texts. The material has been reorganized into five Parts with a common notation layer, a global result map, common proof conventions, and bridges that explain why one group of techniques is needed for the next.

The governing expository rule

For every major theorem we separate three questions:

1. What does the theorem say, with all hypotheses and normalizations visible?
2. Why should the conclusion be plausible before one reads the technical proof?
3. Where is the genuinely difficult mechanism in the proof, and what exactly does it contribute?

A citation is never used to conceal the answer to the third question. When a deep theorem is imported, its exact role is stated, the downstream deduction is written out, and the reader is told what would have to be proved in order to remove that import.

The intended reader has a solid course in one-variable complex analysis and the usual graduate mathematical maturity, but no prior course in complex dynamics is assumed. The book recalls dynamics-specific background when it first becomes necessary. The prerequisite floor includes the Riemann mapping theorem, normal families and Montel's theorem, basic covering-space theory, elementary measure theory, and standard facts about Banach spaces when they are needed. Quasiconformal mappings, polynomial-like maps, puzzles, renormalization, critical circle maps, parabolic implosion, and the several-variable objects used later are developed in the book rather than assumed.

The exposition is intentionally asymmetric. A two-line estimate that explains why a renormalization operator contracts may receive a page of interpretation; ten lines of algebraic bookkeeping may be compressed after the mechanism is transparent. This is not a lower standard of rigor. It is a choice about where explanation is most valuable. Hypotheses, domains, normalizations, constants, parameter dependence, and exceptional cases remain explicit throughout.

The five arcs

Part I begins with local holomorphic normal forms and then moves outward: from a single fixed point to the Fatou–Julia decomposition, from one polynomial to its basin of infinity and external rays, and finally from one dynamical plane to the quadratic parameter plane.

Part II introduces controlled flexibility. Quasiconformal maps and holomorphic motions let us deform dynamics; Sullivan’s theory shows how that flexibility leads back to rigidity and stability. Polynomial-like maps, straightening, puzzles, and Thurston theory then provide the two main languages for cutting a complicated system into controlled pieces.

Part III turns recurrence into scale. Real one-dimensional dynamics supplies the return combinatorics and distortion theory; real a priori bounds prevent geometric collapse; complex bounds place renormalization in a compact analytic setting; hyperbolicity then converts compactness and rigidity into universality. Critical circle maps show that the same architecture survives beyond the interval.

Part IV treats neutral dynamics. Small divisors lead to the Brjuno–Yoccoz theory; parabolic petals lead to Fatou coordinates and Ecalle–Voronin moduli; near-parabolic and pacman renormalization connect the two. The same Part also explains the measure-theoretic notion of “typical” real dynamics and the known local-connectivity regimes in the modern MLC programme.

Part V asks which mechanisms survive when compact polynomial geometry disappears. Transcendental maps have an essential singularity at infinity; Hénon maps have two escape directions and genuinely two-dimensional invariant currents. The final chapter is a source-sensitive frontier chapter: proved theorems, current preprints, and open problems are kept logically distinct.

A note on source status

The deepest parts of modern complex dynamics are too large to reproduce from first principles inside one book without turning this text into an entire library. We therefore use a proof-status convention described in the next section. The important point is that external input is never silent. Every formal result in the book has been checked against its declared source scope, and the final appendix records the principal provenance for each result.

The book also protects open problems from accidental promotion into theorem status. In particular, the general Mandelbrot local connectivity conjecture is discussed as an open problem; only the regimes whose local connectivity has actually been proved are stated as theorems.

How to use the book

A first reading should follow the chapter motivations, conceptual roadmaps, theorem statements, worked examples, and chapter summaries. On a second reading, use the

proof roadmaps to identify the mechanism of each argument before checking the estimates. The worked models and boundary cases serve as theorem stress tests: verify hypotheses on explicit models, locate the first failed hypothesis in a boundary case, and isolate the essential mechanism rather than repeating a whole proof mechanically.

The chapters are numbered continuously so that the book can be read linearly. Nevertheless, several branches are possible. A reader interested mainly in renormalization may move from Chapters 1–8 to Chapters 11–16; a reader interested in neutral dynamics can move from Chapters 1–6 to Chapters 17–19; and a reader interested in global parameter questions can follow Chapters 3–4, 8–9, 13–15, and 21. The global result map records these routes more precisely.

How to read the proofs

Long proofs in dynamics are difficult for two different reasons. Some contain a genuinely new idea; others are long because several familiar ideas must be composed without losing a normalization, a modulus bound, a critical orbit, or a parameter dependence. The reading strategy should be different in the two cases.

Proof-status convention

For exposition, each major result carries one of the following proof-status labels.

proved here

The principal argument is reconstructed in this book from the declared prerequisite floor.

derived here

The result is a transparent consequence of earlier results whose hypotheses are checked explicitly.

proved here modulo stated analytic input

The dynamical argument is given here, but one precisely stated analytic compactness, extension, or functional-analytic theorem is frozen as input.

proved here modulo stated foundation input

The proof is given modulo a theorem owned by the common dynamical-systems foundation.

frontier theorem from the literature

A current frontier theorem is recorded at exact source scope for orientation and comparison, but it is not used as a prerequisite for later proofs.

These labels do not rank mathematical importance. They tell the reader where the proof boundary lies. A five-page argument may be completely internal, while a one-paragraph corollary may depend on a deep external renormalization theorem.

Roadmap first, estimates second

Before entering a long proof, identify four pieces of data.

1. **The invariant object.** What is preserved: a basin, a modulus, a lamination, an itinerary, a renormalization class, a rotation number?
2. **The quantity that cannot degenerate.** Typical examples are a derivative, an annulus modulus, a cross ratio, a hyperbolic length, or the distance of a normalized map from the boundary of a function space.

3. **The compactness mechanism.** Is it Montel compactness, normal-family compactness, quasiconformal compactness, a real-bounds compactness statement, or compactness of a renormalization class?
4. **The rigidity or identification step.** Once a subsequence converges, why is the limit the desired object rather than merely some object?

This four-question test is especially useful in Chapters 5–16 and 19–21. Many proofs in those chapters have the schematic form

$$\begin{aligned} \text{a priori control} &\implies \text{compactness} \implies \text{limit object}, \\ &\implies \text{rigidity/uniqueness} \implies \text{quantitative conclusion}. \end{aligned}$$

The individual implementations differ, but the architecture is stable enough to be a learning tool.

Five recurring proof mechanisms

Normal families. In the earliest chapters, normality separates stable dynamics from chaos. Later, the same compactness idea appears after normalization in spaces of univalent or polynomial-like maps. When Montel’s theorem is invoked, ask what prevents the normal limit from being constant or degenerate.

Pullback. A local statement is often propagated backward through inverse branches. The danger is criticality. Every pullback argument should make visible where univalence is available and where a critical point forces a different construction.

Modulus and distortion. Annulus modulus is the currency that lets conformal geometry survive iteration. Real cross-ratio bounds, Koebe distortion, and complex modulus bounds are different manifestations of the same principle: a controlled collar prevents wild shape change.

Quasiconformal deformation. Quasiconformal flexibility is useful only when paired with a normalization and a rigidity statement. The typical route is

$$\text{Beltrami data} \rightarrow \text{integrating map} \rightarrow \text{deformed dynamics} \rightarrow \text{normalization} \rightarrow \text{rigidity}.$$

If the last arrow is missing, deformation alone does not prove uniqueness.

Renormalization. A return map is rescaled and regarded as a new point in a space of dynamical systems. The difficult questions are then: does renormalization stay in a compact class, is the combinatorial coding stable, and is the renormalization operator hyperbolic or contracting in the required directions?

What not to accept as a proof step

Phrases such as “standard”, “clearly”, or “by the theorem” are harmless only when they summarize steps that have already been displayed. They are not substitutes for an essential inference. In this book, a difficult imported theorem is named as an input;

a difficult internal inference is broken into the estimate, compactness argument, or geometric construction that makes it work.

When a proof uses a named theorem from earlier in the book, the reader should be able to answer: which hypothesis is being verified now, and which part of the earlier conclusion is being used? The worked examples repeatedly reinforce this habit.

Experiments, pictures, and proof

Julia sets, parameter planes, and renormalization pictures are powerful guides to conjecture, but a picture never certifies a theorem. Whenever a visual or numerical model is discussed, the logical role is explanatory: it illustrates geometry already justified by the mathematics or helps locate a hypothesis that should be tested. The proof itself is carried by the analytic, topological, combinatorial, or measure-theoretic argument.

Standing notation and conventions

Most notation is introduced where it first becomes meaningful. The following conventions are global and are worth fixing before the main text.

$\widehat{\mathbb{C}}$	The Riemann sphere. Rational maps are self-maps of $\widehat{\mathbb{C}}$ unless a different domain is stated.
$J(f)$ and $F(f)$	The Julia and Fatou sets of a one-variable map f .
$K(f)$	The filled Julia set when f is polynomial or polynomial-like and the context makes the definition meaningful.
g_K or G_f	A Green or escape-rate function. The precise normalization is always stated at first use.
$\text{mod}(A)$	The conformal modulus of an annulus A . When $U' \Subset U$ are topological disks, $\text{mod}(U \setminus \overline{U'})$ refers to the annular collar between them.
R_α	The rigid circle rotation of angle/rotation number α .
$\mathcal{R}$	A renormalization operator. The calligraphic letter is reserved for renormalization so that it is not confused with a rational map denoted by R .
W^s, W^u	Stable and unstable manifolds, when a hyperbolic invariant set or renormalization fixed point is under discussion.
μ	A probability or invariant measure unless a local theorem explicitly declares another meaning.
$P_T(\varphi)$	Topological pressure for a map T and potential φ when the dynamics is not otherwise unambiguous.

Normalization convention

Complex-dynamical theorems are often false as uniqueness statements until the natural Möbius, affine, or scalar freedom has been removed. We therefore state normalization data as part of the theorem, not as an afterthought. Typical examples include $\phi'(0) = 1$ for a local linearizer, three-point normalization for a quasiconformal map of the sphere, monic-centered normalization for polynomial families, and fixed external or Fatou-coordinate conventions for renormalization.

Parameter dependence

When constants are uniform over a class, the class and the allowed dependence are stated. In renormalization theory this distinction is crucial. A bound “depending only on the combinatorial type” is not the same as a universal bound independent of combinatorics. Likewise, a subsequential limit controlled by an Ecalle phase is not a phase-independent full-sequence limit.

Conformal versus topological language

The book frequently moves between a topological object and a conformally normalized representative. We keep those levels separate. An isotopy class of curves is not a specific geodesic; a polynomial-like map is not its straightened polynomial; a lamination is not a drawing of external rays; and a topological conjugacy is not automatically quasiconformal or conformal.

Map of the 105 principal results

This map is a navigation device, not a substitute for the theorem statements. It records the principal result of each proof unit and the proof-status boundary used in this edition. The exact hypotheses, normalizations, and exceptions appear in the chapters.

Reading the proof-status labels

proved here means that the principal proof is reconstructed in the book. derived here means that the result is deduced transparently from earlier results. proved here modulo stated analytic input and proved here modulo stated foundation input identify a single stated prerequisite boundary. frontier theorem from the literature marks a frontier theorem recorded at its current published or preprint scope; it is not used as a prerequisite for later proofs.

Chapter 1: Local Holomorphic Dynamics and Normal Forms

- **Koenigs linearization theorem.** proved here.
- **Böttcher local theorem.** proved here.
- **Poincaré linearization at repelling points.** proved here.
- **Multiplier classification of fixed points.** derived here.
- **Holomorphic persistence of simple periodic points.** proved here.
- **Small-divisor boundary for indifferent germs.** proved here.

Chapter 2: Fatou and Julia Theory for Rational Maps

- **Fatou-Julia dichotomy.** proved here.
- **Density of repelling periodic points in Julia set.** proved here.
- **Backward orbit characterization.** proved here.
- **Exceptional set theorem.** proved here.
- **Periodic Fatou component classification interface.** proved here.

Chapter 3: Polynomial Dynamics, Green Functions and External Rays

- **Global Böttcher coordinate for polynomials.** proved here.
- **Polynomial Green function.** proved here.
- **Brolin equidistribution theorem.** derived here.

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- **Rational equilibrium measure and measure of maximal entropy.** proved here modulo stated foundation input.
 - **External rays and landing under local connectivity.** proved here.
 - **Connectedness criterion for polynomials.** proved here.

Chapter 4: Parameter Spaces and the Mandelbrot Set: Foundations

- **Connectedness of the Mandelbrot set.** derived here.
- **Parameter Böttcher coordinate.** proved here.
- **Hyperbolic component multiplier map.** proved here.
- **Landing of rational parameter rays.** proved here.

Chapter 5: Quasiconformal Geometry and the Beltrami Equation

- **Measurable Riemann mapping theorem.** proved here modulo stated analytic input.
- **Modulus distortion under quasiconformal maps.** proved here modulo stated analytic input.
- **Quasisymmetric boundary values.** proved here.
- **Normal-family compactness for normalized QC maps.** proved here.

Chapter 6: Holomorphic Motions and Teichmüller Theory

- **Lambda lemma and Ślodkowski extension theorem.** proved here modulo stated analytic input.
- **Ahlfors-Bers holomorphic dependence.** proved here modulo stated analytic input.
- **Holomorphic motion of Julia sets on stable neighborhoods.** proved here.
- **Invariant-Beltrami quasiconformal deformation.** derived here.

Chapter 7: Sullivan Theory, No Wandering Domains and J-Stability

- **Sullivan no-wandering-domains theorem.** proved here modulo stated analytic input.
- **Global Fatou-component classification.** derived here.
- **Mañé-Sad-Sullivan J-stability theorem.** proved here.

Chapter 8: Polynomial-Like Maps, Straightening and Tuning

- **Straightening theorem.** proved here.
- **Consequences of hybrid equivalence on the filled Julia set.** derived here.
- **Normalized quadratic renormalization.** derived here.
- **Quadratic tuning and small Mandelbrot copies.** proved here.

Chapter 9: Yoccoz Puzzles, Principal Nests and Rigidity Machinery

- **Yoccoz puzzle construction.** proved here.
- **Puzzle Markov and first-return structure.** proved here.
- **Yoccoz shrinking theorem in the nonrenormalizable repelling case.** proved here.
- **Finite-renormalization local connectivity and combinatorial rigidity.** proved here.

Chapter 10: Thurston Theory of Branched Coverings

- **Thurston characterization theorem.** proved here modulo stated foundation input.
- **Levy cycle criterion for topological polynomials.** proved here.

Chapter 11: Interval and Unimodal Dynamics

- **Singer theorem.** proved here.
- **No wandering intervals.** proved here modulo stated foundation input.
- **Signed itinerary order and kneading admissibility.** proved here.
- **Milnor-Thurston kneading theorem.** proved here.
- **Entropy monotonicity in the real quadratic family.** proved here.

Chapter 12: Real A Priori Bounds

- **Real a priori bounds.** proved here.
- **Real Koebe principle.** proved here.
- **Cross-ratio inequality and deep C^3 replacement.** proved here.
- **Principal/restrictive-nest bounded geometry.** derived here.
- **Real rigidity precursor.** proved here.

Chapter 13: Complex Bounds and Polynomial-Like Renormalization

- **Complex a priori bounds for real quadratic renormalization.** proved here.
- **Compactness of normalized fixed-modulus germs.** proved here.
- **Pullback and hybrid-rigidity criterion from complex bounds.** proved here.
- **Kahn-Lyubich covering lemma.** proved here modulo stated analytic input.
- **Beau bounds for real-symmetric quadratic-like germs.** proved here.

Chapter 14: Renormalization Operator, Feigenbaum Fixed Point, and Horseshoe

- **Period-doubling Feigenbaum fixed point.** derived here.
- **Exact dynamical scaling at the period-doubling fixed point.** derived here.
- **Analyticity of fixed combinatorial renormalization branches.** proved here.
- **Full quadratic renormalization horseshoe - topological form.** proved here.

Chapter 15: Hyperbolicity, Rigidity, and Universality of Renormalization

- **Hyperbolicity of the bounded-type renormalization horseshoe.** proved here.
- **Exponential convergence along stable classes.** derived here.
- **Parameter universality by transversality.** proved here.
- **Rigidity for bounded combinatorics.** proved here.

Chapter 16: Critical Circle Maps and Renormalization

- **Real bounds for critical circle maps.** proved here.
- **Rigidity of critical circle maps.** proved here.
- **Hyperbolicity of critical-circle renormalization.** proved here.
- **Universal analytic convergence (arbitrary-type C^0 ; bounded-type C^r).** proved here.
- **Critical quasicircle renormalization extension.** proved here.

Chapter 17: Siegel, Brjuno, Yoccoz and Cremer Dynamics

- **Siegel linearization theorem.** proved here.

- **Brjuno sufficiency.** proved here.
- **Yoccoz sharpness for quadratic/germs.** proved here.
- **Critical point on Siegel boundary: scoped results.** proved here modulo stated analytic input.
- **Hedgehogs for nonlinearizable indifferent germs.** proved here.

Chapter 18: Parabolic Dynamics and Ecalle-Voronin Theory

- **Leau-Fatou flower theorem.** proved here.
- **Fatou coordinates.** proved here.
- **Ecalle-Voronin classification.** proved here.
- **Parabolic implosion.** proved here.

Chapter 19: Near-Parabolic, Inou-Shishikura and Pacman Renormalization

- **Inou-Shishikura near-parabolic invariant class.** proved here.
- **Pacman renormalization hyperbolicity.** proved here.
- **Self-similarity near neutral parameters.** derived here.
- **Uniform a priori bounds in neutral/satellite regimes.** proved here.
- **Approximation of Siegel by parabolic dynamics.** proved here.
- **Positive-area Feigenbaum Julia sets.** proved here.

Chapter 20: Typical Real Dynamics: Avila-Lyubich-de Melo-Moreira Programme

- **Jakobson positive-measure stochastic parameters.** proved here.
- **Regular-or-stochastic dichotomy.** proved here.
- **Typical nonregular maps are Collet-Eckmann.** proved here.
- **Phase-parameter relation.** proved here.
- **Collet-Eckmann invariant density and exponential mixing.** proved here.
- **Central limit theorem for Collet-Eckmann maps.** proved here.
- **Large deviations for Collet-Eckmann maps.** proved here.

Chapter 21: Mandelbrot Rigidity, Local Connectivity and the Modern MLC Programme

- **Classical Yoccoz MLC cases.** proved here.
- **Bounded-type satellite MLC theorem.** proved here.

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- **MLC at bounded-type Feigenbaum points.** proved here.
 - **Scaling universality near Feigenbaum parameters.** derived here.

Chapter 22: Transcendental Entire and Meromorphic Dynamics

- **Escaping set theorem.** proved here modulo stated analytic input.
- **Expansion in logarithmic coordinates.** proved here.
- **Dynamic rays for finite-order class B.** proved here.

Chapter 23: Hénon Maps and Several-Complex-Variable Dynamics

- **Polynomial automorphism/Hénon normal form.** proved here modulo stated foundation input.
- **Forward/backward Green functions.** proved here modulo stated analytic input.
- **Green currents and equilibrium measure.** proved here modulo stated analytic input.
- **Partial hyperbolicity and dominated splittings in Hénon dynamics.** proved here modulo stated analytic input.

Chapter 24: Contemporary Complex-Dynamics Frontiers and Open Problems

- **Unicritical disk-diffeomorphism renormalization.** frontier theorem from the literature.
- **Lim hyperbolicity theorem for critical quasicircle renormalization.** derived here.

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Part I

Local Dynamics, Julia Sets, and Parameter Geometry

Guide to Part I: from a fixed point to a parameter plane

The first four chapters are organized by a change of scale. The point is not merely that the objects become larger. Each change of scale forces a new mathematical language.

At the smallest scale, a holomorphic germ near a fixed point is governed to first order by its multiplier. When the multiplier is attracting or repelling, iteration amplifies one direction of time and suppresses the other; this makes a normalized limiting construction possible. At a superattracting point the derivative vanishes, so linearization is the wrong model and the local degree replaces the multiplier as the leading invariant. The first chapter therefore teaches a general lesson that will recur throughout the book: *before proving a normal form, identify the quantity that survives conjugacy and the normalization that removes the residual coordinate freedom.*

The second change of scale is from one orbit to all iterates of a rational map on the Riemann sphere. The central object is no longer a coordinate but a family, $\{R^n\}_{n \geq 0}$. Normal-family theory divides the sphere into a stable region and its chaotic complement. This is where Montel's theorem becomes dynamical: omitting a few values is converted into normality, and failure of normality is converted into density and recurrence phenomena.

The third scale change uses a special feature of polynomials: infinity is a superattracting fixed point. The local Böttcher coordinate from Chapter 1 can therefore be pushed outward until it becomes a global coordinate on the basin of infinity under the correct connectivity hypotheses. Potential theory packages the same information in an escape-rate function. External rays then translate complex-analytic geometry into angular combinatorics. This translation is the first prototype for the puzzle and lamination methods used much later.

Finally, Chapter 4 changes the variable that is being iterated. Instead of fixing $f_c(z) = z^2 + c$ and studying z , we vary c . The critical orbit gives a distinguished probe of the dynamics, and the dynamical Green function evaluated at that probe becomes a potential in parameter space. The Mandelbrot set is thus not introduced as a picture; it is introduced as the connectedness locus selected by the critical orbit.

The mathematical route through this Part

local normal forms
↓
normal families and $F(R), J(R)$
↓
Böttcher coordinate, Green function, external rays
↓
critical orbit and the Mandelbrot parameter plane.

The same objects will return later with greater force: external rays become puz-

zle boundaries, Green functions become parameter tools, and the multiplier-one boundary opens the parabolic theory of Part IV.

What to watch for

Three proof habits learned here should be carried through the rest of the book.

First, distinguish *local convergence* from *global continuation*. A Koenigs coordinate is local; a Poincaré function is produced only after a functional equation allows repeated continuation. A local Böttcher coordinate near infinity becomes global only after the topology of the basin is controlled.

Second, identify the exceptional set before applying a normal-family theorem. Montel's theorem is powerful because the omitted values are explicit. Later compactness arguments will have analogous exceptional sets: critical points, thin annuli, short curves, or degenerate normalized maps.

Third, never confuse a dynamical object with a parameter-space object simply because their pictures look similar. Dynamic rays live in a dynamical plane. Parameter rays live in the c -plane. Their relationship is deep and useful, but it must be proved.

Local Holomorphic Dynamics and Normal Forms

Why this chapter matters

Near a fixed or periodic point, a nonlinear holomorphic map can often be replaced by a much simpler model. The local multiplier tells us which model to expect: multiplication by λ in the attracting or repelling case, a monomial z^d at a superattracting point, and a small-divisor problem on the unit circle. This local classification is the first place where the rigidity of holomorphic maps becomes visible.

The chapter also explains two ways in which local information propagates. A Koenigs coordinate at a repelling fixed point extends to a global Poincaré function, while a simple periodic point persists holomorphically when the map varies. These mechanisms will later reappear in the global theory of Julia sets and parameter spaces.

Prerequisites. One-variable complex analysis, Taylor and Laurent series, the inverse and implicit function theorems, local uniform convergence, and basic rational maps on the Riemann sphere.

Learning goals

- Read the local geometry of a fixed point from its multiplier and local degree.
- Construct and normalize Koenigs and Böttcher coordinates rather than treating them as formal names.
- Understand how a repelling local coordinate produces a global Poincaré linearizer.
- Prove holomorphic persistence of a simple periodic point and identify multiplier 1 as the obstruction.
- Derive the formal small-divisor recursion for an irrationally indifferent germ and distinguish formal solvability from analytic convergence.

Conceptual roadmap

The hyperbolic and superattracting branches are local normal-form theorems,

$$\begin{aligned} 0 < |\lambda| \neq 1 &\longrightarrow \text{Koenigs coordinate,} \\ \lambda = 0 &\longrightarrow \text{Böttcher coordinate.} \end{aligned}$$

The repelling Koenigs coordinate then extends by the functional equation to a Poincaré function on all of $\mathbb{C}$. Independently, the implicit-function theorem controls simple periodic points in holomorphic families. On $|\lambda| = 1$, root-of-unity multipliers lead toward parabolic dynamics, while non-root-of-unity multipliers give a unique *formal* linearization whose denominators $\lambda^n - \lambda$ expose the analytic small-divisor problem studied later in Chapter 17.

How the main results fit together. Koenigs and Böttcher are the two local normal-form engines. The multiplier classification organizes all fixed points around those engines. The Poincaré theorem uses Koenigs but is global; periodic-point persistence is logically independent; the small-divisor theorem is a formal calculation that marks the boundary of the easy linearization theory.

1.1 Definitions

Definition 1.1 (Holomorphic germ). A *holomorphic germ* at z_0 is an equivalence class of holomorphic maps defined near z_0 , where two representatives are identified when they agree on a smaller neighborhood of z_0 .

Definition 1.2 (Multiplier). If $f(z_0) = z_0$, the *multiplier* is $\lambda = f'(z_0)$. If z_0 has exact period q , its multiplier is $(f^q)'(z_0)$.

Definition 1.3 (Analytic linearization). For a germ fixing 0 with multiplier λ , an *analytic linearization* is a local biholomorphism ϕ with $\phi(0) = 0$ such that

$$\phi \circ f \circ \phi^{-1}(w) = \lambda w.$$

When possible we normalize $\phi'(0) = 1$ to remove the scalar ambiguity.

Definition 1.4 (Fixed-point types). A fixed point is *attracting* if $|\lambda| < 1$, *repelling* if $|\lambda| > 1$, and *indifferent* if $|\lambda| = 1$. The case $\lambda = 0$ is called *superattracting*.

Definition 1.5 (Local degree and Böttcher coordinate). At a superattracting fixed point, choose a centered coordinate and write

$$f(z) = az^d + O(z^{d+1}), \quad a \neq 0,$$

with $d \geq 2$ minimal. This d is the *local degree*. A *Böttcher coordinate* is a local biholomorphism ϕ satisfying $\phi \circ f = \phi^d$.

Definition 1.6 (Poincaré function). For a rational map R with repelling fixed point p of multiplier λ , a *Poincaré function* is a nonconstant meromorphic map $\Psi : \mathbb{C} \rightarrow \hat{\mathbb{C}}$ satisfying

$$\Psi(0) = p, \quad \Psi(\lambda z) = R(\Psi(z)).$$

A derivative normalization in a local coordinate at p fixes its scalar ambiguity.

Definition 1.7 (Simple periodic point). A point p of exact period q is *simple* for the equation $f^q(z) = z$ if it is a simple zero of $f^q(z) - z$. Equivalently,

$$(f^q)'(p) \neq 1.$$

Definition 1.8 (Indifferent subtypes). An indifferent point is *rationally indifferent* when its multiplier is a root of unity and *irrationally indifferent* otherwise. An irrationally indifferent germ is *Siegel* if analytically linearizable and *Cremer* if not. If a root-of-unity multiplier has order q and f^q is not the identity germ, then f^q is a nontrivial multiplier-one germ, the parabolic regime treated later.

Definition 1.9 (Formal linearization). For $f(z) = \lambda z + O(z^2)$, a *formal linearization* is an invertible formal series $h(z) = z + O(z^2)$ satisfying $h \circ f = \lambda h$. Formal convergence is not assumed.

Definition 1.10 (Holomorphic family of self-maps). Let $\Lambda \subset \mathbb{C}^m$ and $U \subset \mathbb{C}$ be domains. A *holomorphic family of self-maps of U* is a holomorphic map $F : \Lambda \times U \rightarrow U$, written $f_\mu(z) = F(\mu, z)$. Its finite iterates $(\mu, z) \mapsto f_\mu^q(z)$ are jointly holomorphic.

1.2 Main results

Theorem 1.1 (Koenigs linearization theorem). *Let f be holomorphic near a fixed point z_0 with multiplier λ satisfying $0 < |\lambda| \neq 1$. There is a local biholomorphism ϕ with $\phi(z_0) = 0$, $\phi'(z_0) = 1$, and*

$$\phi(f(z)) = \lambda \phi(z).$$

With this derivative normalization the coordinate is unique; without it, the coordinate is unique up to multiplication by a nonzero constant.

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Attracting normalized iterates with a quadratic-error summability estimate; repelling case via the local inverse; power-series uniqueness.

The mathematical argument begins in the proof environment below.

Intuition (not part of the proof)

Normalize the iterates by the multiplier. The nonlinear error becomes quadratically small along an attracting orbit; the repelling case is reduced to the attracting inverse germ.

Proof. Translate the fixed point to the origin, so that

$$f(0) = 0, \quad f'(0) = \lambda, \quad 0 < |\lambda| \neq 1. \quad (1.1)$$

We first treat the attracting case $0 < |\lambda| < 1$.

Geometric convergence of normalized iterates.

Choose a number c such that

$$|\lambda| < c < 1, \quad c^2 < |\lambda|. \quad (1.2)$$

Such a choice is possible because $|\lambda| < \sqrt{|\lambda|}$. Taylor's formula gives

$$f(z) = \lambda z + R(z), \quad |R(z)| \leq M|z|^2 \quad (1.3)$$

on a sufficiently small disk. After shrinking that disk to D_r , continuity of $f(z)/z$ at the origin also gives

$$|f(z)| \leq c|z| \quad (z \in D_r). \quad (1.4)$$

Thus $f^n(D_r) \subset D_r$ and

$$|f^n(z)| \leq c^n|z|. \quad (1.5)$$

Define

$$\phi_n(z) := \lambda^{-n} f^n(z). \quad (1.6)$$

Then, using (1.3)–(1.5),

$$\begin{aligned} |\phi_{n+1}(z) - \phi_n(z)| &= |\lambda|^{-n-1} |f(f^n(z)) - \lambda f^n(z)| \\ &\leq \frac{M}{|\lambda|} \left(\frac{c^2}{|\lambda|} \right)^n |z|^2. \end{aligned} \quad (1.7)$$

The ratio $c^2/|\lambda|$ is strictly smaller than one. Hence $\sum_n (\phi_{n+1} - \phi_n)$ converges uniformly on D_r , so ϕ_n converges locally uniformly to a holomorphic function ϕ . Moreover $\phi'_n(0) = 1$ for every n , hence

$$\phi(0) = 0, \quad \phi'(0) = 1. \quad (1.8)$$

This establishes the geometric-convergence step.

Passing to the limit in

$$\phi_n(f(z)) = \lambda \phi_{n+1}(z) \quad (1.9)$$

gives Schroeder's equation

$$\phi(f(z)) = \lambda \phi(z). \quad (1.10)$$

By (1.8) and the holomorphic inverse-function theorem, ϕ is a biholomorphism between sufficiently small neighborhoods of the origin. Consequently

$$\phi \circ f \circ \phi^{-1}(w) = \lambda w, \quad (1.11)$$

which is the asserted analytic linearization in the attracting case.

Repelling reduction to the local inverse.

Suppose now that $|\lambda| > 1$. Since $f'(0) = \lambda \neq 0$, the inverse-function theorem gives a local inverse germ $g = f^{-1}$ fixing the origin, with

$$g'(0) = \lambda^{-1}, \quad |\lambda^{-1}| < 1. \quad (1.12)$$

Apply the attracting construction to g . After derivative normalization it gives a local

biholomorphism ϕ satisfying

$$\phi(g(w)) = \lambda^{-1}\phi(w). \quad (1.13)$$

Shrink the neighborhood so that both sides may be composed with f . Setting $w = f(z)$ in (1.13) yields

$$\phi(f(z)) = \lambda\phi(z), \quad (1.14)$$

so (1.11) holds in the repelling case as well. This proves the repelling case.

Uniqueness after normalization.

Let ϕ and ψ be two local linearizing coordinates with $\phi'(0) = \psi'(0) = 1$. Then

$$h := \psi \circ \phi^{-1} \quad (1.15)$$

is a germ fixing the origin with $h'(0) = 1$ and

$$h(\lambda w) = \lambda h(w). \quad (1.16)$$

Write $h(w) = w + \sum_{m \geq 2} b_m w^m$. Comparing coefficients in (1.16) gives

$$b_m(\lambda^m - \lambda) = 0 \quad (m \geq 2). \quad (1.17)$$

Because $0 < |\lambda| \neq 1$, one has $\lambda^{m-1} \neq 1$ for every $m \geq 2$; therefore every b_m vanishes. Hence $h(w) = w$ and $\phi = \psi$. Without the normalization of the derivative, the same argument shows that two linearizations differ by multiplication by a nonzero constant.

Returning from the translated coordinate to the original fixed point z_0 proves the theorem in full. $\square$

Theorem 1.2 (Böttcher local theorem). *Let f be holomorphic near 0 and suppose*

$$f(z) = az^d + O(z^{d+1}), \quad a \neq 0, \quad d \geq 2.$$

For every α satisfying $\alpha^{d-1} = a$, there is a unique local biholomorphism ϕ with $\phi(0) = 0$, $\phi'(0) = \alpha$, and

$$\phi(f(z)) = \phi(z)^d.$$

If the derivative is not prescribed, the Böttcher coordinate is unique up to multiplication by a $(d-1)$ st root of unity.

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Scale the leading coefficient, choose $\log u$ for $F(w) = w^d u(w)$, sum the geometrically weighted logarithmic orbit series, and prove normalized uniqueness.

The mathematical argument begins in the proof environment below.

Intuition (not part of the proof)

Factor the germ as $f(z) = z^d u(z)$ and solve the conjugacy equation additively after taking a local logarithm of u .

Proof. Write

$$f(z) = az^d + O(z^{d+1}), \quad a \neq 0, \quad d \geq 2. \quad (1.18)$$

Fix one number $\alpha \neq 0$ satisfying

$$\alpha^{d-1} = a. \quad (1.19)$$

Let $L(z) = \alpha z$ and set

$$F := L \circ f \circ L^{-1}. \quad (1.20)$$

Then

$$F(w) = w^d u(w), \quad u(0) = 1, \quad (1.21)$$

for a holomorphic function u near the origin.

Logarithmic Böttcher series.

Shrink to a disk D_r on which u has no zero and on which $F(D_r) \subset D_r$. Since D_r is simply connected and $u(0) = 1$, there is a holomorphic logarithm

$$\ell(w) := \log u(w), \quad \ell(0) = 0. \quad (1.22)$$

The function ℓ is bounded on a slightly smaller closed disk; fix such a disk and write $|\ell| \leq M$ there. Define

$$S(w) := \sum_{n=0}^{\infty} \frac{1}{d^{n+1}} \ell(F^n(w)). \quad (1.23)$$

Every orbit stays in the chosen disk and

$$\left| d^{-(n+1)} \ell(F^n(w)) \right| \leq M d^{-(n+1)}. \quad (1.24)$$

Hence the Weierstrass M -test gives locally uniform convergence of (1.23), so S is holomorphic. Put

$$\Phi(w) := w \exp S(w). \quad (1.25)$$

Because $S(0) = 0$, one has

$$\Phi(0) = 0, \quad \Phi'(0) = 1. \quad (1.26)$$

An index shift in (1.23) gives

$$dS(w) = \ell(w) + S(F(w)). \quad (1.27)$$

Therefore, using $F(w) = w^d e^{\ell(w)}$,

$$\Phi(w)^d = w^d e^{dS(w)} = F(w) e^{S(F(w))} = \Phi(F(w)). \quad (1.28)$$

This establishes the conjugacy identity (1.28).

By (1.26), Φ is a local biholomorphism. Returning to the original coordinate, define

$$\phi := \Phi \circ L. \quad (1.29)$$

Then

$$\phi(0) = 0, \quad \phi'(0) = \alpha, \quad (1.30)$$

and, by (1.20) and (1.28),

$$\phi(f(z)) = \phi(z)^d. \quad (1.31)$$

Thus existence is proved with the exact derivative normalization requested in the statement.

Normalized uniqueness.

Suppose Φ_1 and Φ_2 are two Böttcher coordinates for the normalized germ F and both have derivative one at the origin. The transition germ

$$h := \Phi_2 \circ \Phi_1^{-1} \quad (1.32)$$

satisfies

$$h(0) = 0, \quad h'(0) = 1, \quad h(w^d) = h(w)^d. \quad (1.33)$$

Write $h(w) = wv(w)$ with $v(0) = 1$. On a small disk choose the holomorphic logarithm $L_0 = \log v$ with $L_0(0) = 0$. Equation (1.33) becomes

$$L_0(w^d) = dL_0(w). \quad (1.34)$$

If L_0 were nonzero and $c_m w^m$ were its first nonzero term, the left side of (1.34) would have first nonzero degree dm , while the right side would have first nonzero degree m , impossible because $d \geq 2$. Hence $L_0 = 0$, so $h(w) = w$ and $\Phi_1 = \Phi_2$.

In the original coordinate, fixing $\phi'(0) = \alpha$ therefore gives a unique Böttcher coordinate. If the derivative is not fixed, the leading coefficient identity in (1.31) forces $\phi'(0)^{d-1} = a$; hence the possible derivatives are precisely the $(d-1)$ roots of a , and two choices differ by a $(d-1)$ st root of unity. This completes the proof. $\square$

Theorem 1.3 (Poincaré linearization at repelling points). *Let R be a rational map of degree at least two and let p be a repelling fixed point with multiplier λ , $|\lambda| > 1$. Fix a local coordinate ζ at p with $\zeta(p) = 0$. There exists a unique meromorphic map $\Psi : \mathbb{C} \rightarrow \hat{\mathbb{C}}$ satisfying*

$$\Psi(0) = p, \quad (\zeta \circ \Psi)'(0) = 1, \quad \Psi(\lambda z) = R(\Psi(z)).$$

If R is a polynomial, then Ψ is entire.

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Invert the local Koenigs coordinate and extend by $\Psi(z) = R^n(g(\lambda^{-n}z))$; prove compatibility, meromorphicity, the functional equation, and uniqueness.

The mathematical argument begins in the proof environment below.

Intuition (not part of the proof)

Invert the local Koenigs coordinate and repeatedly apply the rational map so that the functional equation extends the local solution over expanding disks.

Proof. Let R be a rational map of degree at least two, let p be a repelling fixed point, and let

$$\lambda = R'(p), \quad |\lambda| > 1, \quad (1.35)$$

where the multiplier is computed in any local coordinate. Fix once and for all a local coordinate ζ with $\zeta(p) = 0$; the derivative normalization in the statement is understood in this coordinate.

By the Koenigs linearization theorem, after scaling its linearizing coordinate, there is a local biholomorphism χ near p such that

$$\chi(p) = 0, \quad (\chi \circ \zeta^{-1})'(0) = 1, \quad \chi(R(w)) = \lambda \chi(w) \quad (1.36)$$

whenever both sides are defined. Let $g = \chi^{-1}$ near the origin. Then

$$g(0) = p, \quad (\zeta \circ g)'(0) = 1, \quad (1.37)$$

and, on a sufficiently small disk,

$$g(\lambda z) = R(g(z)). \quad (1.38)$$

Compatible global continuation.

Choose $\delta > 0$ so that g is defined on D_δ and (1.38) holds for $z \in D_{\delta/|\lambda|}$. For $n \geq 0$ and $|z| < |\lambda|^n \delta$, set

$$\Psi_n(z) := R^n(g(\lambda^{-n}z)). \quad (1.39)$$

If both Ψ_n and Ψ_{n+1} are defined, then

$$\begin{aligned} \Psi_{n+1}(z) &= R^n\left(R\left(g(\lambda^{-(n+1)}z)\right)\right) \\ &= R^n(g(\lambda^{-n}z)) = \Psi_n(z), \end{aligned} \quad (1.40)$$

by (1.38). Hence the maps glue on the increasing disks $D_{|\lambda|^n \delta}$ to a single map

$$\Psi : \mathbb{C} \longrightarrow \widehat{\mathbb{C}}. \quad (1.41)$$

Each Ψ_n is meromorphic because it is the composition of the holomorphic map g with the rational map R^n . Thus Ψ is meromorphic on all of $\mathbb{C}$. If R is a polynomial, every $R^n \circ g$ is holomorphic and there are no finite poles, so Ψ is entire. This establishes the compatible global continuation.

The local definition is $\Psi = g$, so (1.37) gives the required normalization. To verify the functional equation globally, fix z and take n large enough that the expressions below are defined. Using the $(n+1)$ st representation for $\Psi(\lambda z)$ and the n th representation

for $\Psi(z)$,

$$\begin{aligned}\Psi(\lambda z) &= R^{n+1}\left(g(\lambda^{-(n+1)}\lambda z)\right) \\ &= R\left(R^n(g(\lambda^{-n}z))\right) = R(\Psi(z)).\end{aligned}\tag{1.42}$$

Thus $\Psi(\lambda z) = R(\Psi(z))$ on $\mathbb{C}$.

Uniqueness of the normalized Poincare function.

Let $\tilde{\Psi}$ be another meromorphic solution with the same normalization. For z close to the origin its image is close to p , so

$$h(z) := \chi(\tilde{\Psi}(z))\tag{1.43}$$

is holomorphic and satisfies

$$h(0) = 0, \quad h'(0) = 1, \quad h(\lambda z) = \lambda h(z).\tag{1.44}$$

Write $h(z) = z + \sum_{m \geq 2} c_m z^m$. Comparing coefficients gives $c_m(\lambda^m - \lambda) = 0$. Since $|\lambda| > 1$, every coefficient c_m vanishes. Hence $h(z) = z$, so $\tilde{\Psi} = g = \Psi$ near the origin. For arbitrary z , choose n with $\lambda^{-n}z$ in that neighborhood and iterate the Poincare equation:

$$\tilde{\Psi}(z) = R^n(\tilde{\Psi}(\lambda^{-n}z)) = R^n(\Psi(\lambda^{-n}z)) = \Psi(z).\tag{1.45}$$

This proves uniqueness and completes the theorem. $\square$

Theorem 1.4 (Multiplier classification of fixed points). *For a nonconstant holomorphic germ at a fixed point, the multiplier gives the first local trichotomy. If $0 < |\lambda| < 1$ or $|\lambda| > 1$, Koenigs linearization applies; if $\lambda = 0$, Böttcher's theorem gives the superattracting monomial normal form. If $|\lambda| = 1$, then root-of-unity multipliers are rationally indifferent (with a parabolic iterate unless the germ has finite order), while non-root-of-unity multipliers are irrationally indifferent and divide into Siegel and Cremer cases according to analytic linearizability. No arithmetic linearization theorem for the irrationally indifferent case is claimed here; the sharp theory is owned by Chapter 17.*

Proof strategy (not part of the proof)

Proof status. derived here.

Strategic outline only. Multiplier case split; Koenigs/Böttcher normal forms; root-of-unity iterate; Siegel/Cremer classification only.

The mathematical argument begins in the proof environment below.

Intuition (not part of the proof)

Split only by the modulus and arithmetic type of the multiplier. The theorem is a taxonomy plus the Koenigs/Böttcher normal forms, not an arithmetic linearization criterion.

Proof. Translate the fixed point to the origin and write

$$f(z) = \lambda z + O(z^2).\tag{1.24}$$

The cases in the statement are exhaustive because every complex number λ satisfies exactly one of

$$\lambda = 0, \quad 0 < |\lambda| < 1, \quad |\lambda| > 1, \quad |\lambda| = 1. \quad (1.25)$$

If $0 < |\lambda| < 1$ or $|\lambda| > 1$, the hypotheses of *Koenigs linearization theorem* are satisfied. Therefore there is a local analytic coordinate in which f is exactly $w \mapsto \lambda w$. This gives the attracting and repelling hyperbolic cases.

Suppose next that $\lambda = 0$. Because the germ is assumed nonconstant, its Taylor series has a first nonzero term of some degree $d \geq 2$:

$$f(z) = az^d + O(z^{d+1}), \quad a \neq 0. \quad (1.26)$$

Thus the fixed point is superattracting of local degree d , and *Böttcher local theorem* gives a local coordinate conjugating f to $w \mapsto w^d$ (with the derivative normalization described there).

It remains to consider $|\lambda| = 1$. First suppose that λ is a root of unity, and let $q \geq 1$ be its order. By the chain rule,

$$(f^q)'(0) = \prod_{j=0}^{q-1} f'(f^j(0)) = \lambda^q = 1, \quad (1.27)$$

since every point in the orbit is the fixed point 0. There are now two possibilities. If f^q is the identity germ, then f has finite order. If f^q is not the identity germ, its Taylor series has the form

$$f^q(z) = z + cz^m + O(z^{m+1}) \quad (c \neq 0, m \geq 2), \quad (1.28)$$

so it is a nontrivial multiplier-one germ. This is precisely the parabolic regime whose petal and Ecalle–Voronin theory is developed later in the Part.

Finally suppose that $|\lambda| = 1$ and λ is not a root of unity. By definition the fixed point is irrationally indifferent. Exactly one of the following logical alternatives holds: the germ is analytically linearizable, or it is not. In the first case it is a Siegel germ and in the second it is a Cremer germ. This statement is only a classification: it does not assert an arithmetic criterion deciding which alternative occurs. The small-divisor mechanism is isolated in *Small-divisor boundary for indifferent germs*, and the sharp Brjuno–Yoccoz theory is postponed to Chapter 17.

All multiplier regimes listed in the statement have therefore been accounted for, with exactly the claimed normal-form conclusions and no stronger unit-circle theorem being used. $\square$

Theorem 1.5 (Holomorphic persistence of simple periodic points). *Let $\Lambda \subset \mathbb{C}^m$ and $U \subset \mathbb{C}$ be domains and let $F : \Lambda \times U \rightarrow U$ be a holomorphic family, with $f_\mu(z) = F(\mu, z)$. Suppose $p_0 \in U$ has exact period q for f_{μ_0} and multiplier $(f_{\mu_0}^q)'(p_0) \neq 1$. Then, after shrinking the parameter neighborhood of μ_0 , there is a unique holomorphic map $p(\mu)$ with $p(\mu_0) = p_0$ and $f_\mu^q(p(\mu)) = p(\mu)$; after a further shrink the exact period remains q .*

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Holomorphic IFT for $f_\mu^q(z) - z$; exclude proper divisors of q to preserve exact period.

The mathematical argument begins in the proof environment below.

Intuition (not part of the proof)

Write the periodic-point equation as $G(\mu, z) = f_\mu^q(z) - z$. Its z -derivative is the multiplier minus one, exactly the implicit-function nondegeneracy condition.

Proof. Let $\Lambda \subset \mathbb{C}^m$ and $U \subset \mathbb{C}$ be the parameter and phase domains from the statement, and write

$$F(\mu, z) = f_\mu(z).$$

Joint holomorphicity of finite iterates.

Because $F : \Lambda \times U \rightarrow U$ is holomorphic, each finite iterate

$$F_q(\mu, z) := f_\mu^q(z) \tag{1.29}$$

is jointly holomorphic on $\Lambda \times U$: this follows inductively from closure of holomorphic maps under composition.

Define

$$G(\mu, z) := F_q(\mu, z) - z. \tag{1.30}$$

The hypothesis that p_0 has period q for f_{μ_0} gives

$$G(\mu_0, p_0) = 0. \tag{1.31}$$

Differentiating with respect to the phase variable gives

$$\partial_z G(\mu_0, p_0) = (f_{\mu_0}^q)'(p_0) - 1. \tag{1.32}$$

The simple-periodic-point hypothesis says that the multiplier in (1.32) is not equal to 1, so

$$\partial_z G(\mu_0, p_0) \neq 0. \tag{1.33}$$

The holomorphic implicit-function theorem therefore yields neighborhoods $\Lambda_0 \ni \mu_0$ and $V \ni p_0$ and a unique holomorphic map

$$p : \Lambda_0 \longrightarrow V, \quad p(\mu_0) = p_0, \tag{1.34}$$

such that

$$G(\mu, p(\mu)) = 0, \quad \text{equivalently} \quad f_\mu^q(p(\mu)) = p(\mu) \tag{1.35}$$

for every $\mu \in \Lambda_0$. The same theorem also gives uniqueness among solutions lying in V after the neighborhoods are fixed sufficiently small.

Exact-period persistence.

It remains to verify that the period does not drop. If $q = 1$ there is nothing to prove. Suppose $q > 1$. For every proper positive divisor $r \mid q$, exactness of the original period gives

$$f_{\mu_0}^r(p_0) - p_0 \neq 0. \quad (1.36)$$

For such r , the function

$$H_r(\mu) := f_{\mu}^r(p(\mu)) - p(\mu) \quad (1.37)$$

is holomorphic, hence continuous, and $H_r(\mu_0) \neq 0$. Since q has only finitely many proper divisors, we may shrink Λ_0 once more so that

$$H_r(\mu) \neq 0 \quad \text{for every proper divisor } r \mid q \quad (1.38)$$

and every $\mu \in \Lambda_0$.

Equation (1.35) says that the period of $p(\mu)$ divides q . If it were strictly smaller, its exact period would be a proper divisor of q , in contradiction with (1.38). Hence $p(\mu)$ has exact period q throughout the shrunken parameter neighborhood. This proves the theorem. $\square$

Theorem 1.6 (Small-divisor boundary for indifferent germs). *Let $f(z) = \lambda z + \sum_{n \geq 2} a_n z^n$ with $|\lambda| = 1$ and λ not a root of unity. There is a unique formal series $h(z) = z + \sum_{n \geq 2} b_n z^n$ satisfying $h \circ f = \lambda h$. At degree n , the new unknown b_n is multiplied by the divisor $\lambda^n - \lambda$; hence formal solvability is automatic but analytic convergence is a genuine arithmetic small-divisor problem. The sharp Brjuno–Yoccoz criterion is deferred to Chapter 17.*

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Formal coefficient recursion with divisor $\lambda^n - \lambda$.

The mathematical argument begins in the proof environment below.

Intuition (not part of the proof)

Compare coefficients in $h \circ f = \lambda h$. At degree n the new coefficient is multiplied by $\lambda^n - \lambda$, so nonresonance gives formal uniqueness while small divisors threaten analytic convergence.

Proof. Seek a formal series

$$h(z) = z + \sum_{n \geq 2} b_n z^n \quad (1.39)$$

satisfying

$$h(f(z)) = \lambda h(z), \quad f(z) = \lambda z + \sum_{n \geq 2} a_n z^n. \quad (1.40)$$

We determine the coefficients b_n recursively.

Nonresonant formal coefficient recursion.

Fix $n \geq 2$ and suppose that $b_2, \dots, b_{n-1}$ have already been determined. Modulo terms of degree $> n$,

$$h(f(z)) = f(z) + \sum_{m=2}^n b_m f(z)^m. \quad (1.41)$$

The coefficient b_n occurs in degree n only through the term $b_n f(z)^n$, and its degree- n contribution is $b_n \lambda^n z^n$. Every other contribution of degree n depends only on

$$a_2, \dots, a_n, \quad b_2, \dots, b_{n-1}.$$

Consequently there is a polynomial expression Q_n , independent of b_n , such that

$$[z^n]h(f(z)) = b_n \lambda^n + a_n + Q_n(a_2, \dots, a_{n-1}, b_2, \dots, b_{n-1}). \quad (1.42)$$

On the other hand,

$$[z^n]\lambda h(z) = \lambda b_n. \quad (1.43)$$

Equating (1.42) and (1.43) yields the recursion

$$(\lambda^n - \lambda)b_n = -a_n - Q_n(a_2, \dots, a_{n-1}, b_2, \dots, b_{n-1}). \quad (1.44)$$

Because $|\lambda| = 1$, one has $\lambda \neq 0$. Moreover λ is not a root of unity, so for every $n \geq 2$

$$\lambda^n - \lambda = \lambda(\lambda^{n-1} - 1) \neq 0. \quad (1.45)$$

Thus (1.44) determines b_n uniquely from the previously determined coefficients. Starting with $n = 2$ and proceeding inductively proves both existence and uniqueness of the formal linearization.

The same formula also isolates the analytic obstruction. Write $\lambda = e^{2\pi i \theta}$ with $\theta \notin \mathbb{Q}$. Then

$$|\lambda^n - \lambda| = |e^{2\pi i(n-1)\theta} - 1|. \quad (1.46)$$

The irrational rotation $(n-1)\theta \pmod{1}$ returns arbitrarily close to 0, so the positive numbers in (1.46) have no uniform lower bound away from zero. Hence the formal recursion involves potentially very small divisors. Formal solvability by itself therefore supplies no geometric or exponential bound on the coefficients b_n and does not imply a positive radius of convergence.

This is exactly the small-divisor boundary asserted here. Determining which arithmetic conditions on θ force convergence, and which allow non-linearizable germs, requires the sharper Brjuno–Yoccoz theory owned by Chapter 17 and is not used in the present result. $\square$

1.3 Worked examples

Worked example: The quadratic germ $f_\lambda(z) = \lambda z + z^2$

The origin is fixed and has multiplier λ . The same family therefore crosses all of the local regimes in the chapter.

If $\lambda = 0$, then $f_0(z) = z^2$ already is its Böttcher normal form: the identity is a Böttcher coordinate with local degree 2.

If $0 < |\lambda| < 1$, Koenigs' theorem gives a unique normalized coordinate $\phi(z) = z + c_2 z^2 + O(z^3)$ with $\phi(f_\lambda(z)) = \lambda \phi(z)$. Comparing the coefficient of z^2 gives

$$1 + c_2 \lambda^2 = \lambda c_2, \quad c_2 = \frac{1}{\lambda(1 - \lambda)}.$$

The same coefficient formula remains formally valid for $|\lambda| > 1$, while the geometry changes from attraction to repulsion.

Now take $|\lambda| = 1$ with λ not a root of unity. A normalized formal linearization still starts with the same coefficient, but the higher recursion contains denominators $\lambda^n - \lambda$. They never vanish, yet they can become arbitrarily small. This is the exact distinction between *formal existence* and the analytic convergence problem.

Worked example: Persistence of a fixed point in a parameter family

Consider

$$f_\mu(z) = z^2 + \mu.$$

Fixed points solve $G(\mu, z) = z^2 - z + \mu = 0$. At $(\mu_0, p_0) = (0, 0)$,

$$\partial_z G(0, 0) = -1 \neq 0.$$

The holomorphic implicit-function theorem therefore produces a unique fixed-point branch near $\mu = 0$. Solving the quadratic equation identifies it explicitly as

$$p(\mu) = \frac{1 - \sqrt{1 - 4\mu}}{2},$$

where the square root is the branch equal to 1 at 0. Its multiplier is $2p(\mu)$, so the branch remains simple while $2p(\mu) \neq 1$. At the parameter where the multiplier reaches 1, the implicit-function derivative vanishes and the local continuation mechanism changes.

Boundary case: why the root-of-unity case must be separated

For $f(z) = z + z^2$, the multiplier is 1. The formal linearization recursion would require division by $\lambda^2 - \lambda = 0$ already at degree two, so the non-root-of-unity theorem does not apply. This is not a technical inconvenience: f is a nontrivial parabolic germ, whose local geometry is governed by attracting and repelling petals rather than a linear rotation.

By contrast, $f(z) = -z$ also has a root-of-unity multiplier, but f^2 is exactly the identity germ. Hence one must distinguish the finite-order case from the genuinely parabolic iterate.

There is a parallel multiplier-one boundary for periodic-point persistence. In the family $f_\mu(z) = z^2 + \mu$, a fixed point satisfies $p^2 - p + \mu = 0$ and has multiplier $2p$. At $p = 1/2$, hence $\mu = 1/4$, the multiplier is 1 and

$$\partial_z(f_\mu(z) - z) = 2z - 1 = 0.$$

The simple-root hypothesis in the persistence theorem is exactly what fails at this collision parameter.

Common pitfall

The labels “Siegel” and “Cremer” in this chapter are a dichotomy by analytic linearizability, not an arithmetic criterion. The theorem deciding linearizability from Brjuno arithmetic is deliberately postponed to Chapter 17.

1.4 Chapter summary

What to remember

- Hyperbolic nonzero multipliers are analytically linearized by Koenigs’ theorem; a derivative normalization makes the coordinate unique.
- Superattracting germs are conjugated by Böttcher coordinates to monomials, with the leading coefficient fixing the admissible derivative normalization.
- A repelling Koenigs coordinate can be continued through the functional equation to a global meromorphic Poincaré function.
- A periodic point persists holomorphically exactly when the fixed-point equation for the relevant iterate is simple; multiplier 1 is the obstruction.
- Irrationally indifferent germs are always formally linearizable, but the divisors $\lambda^n - \lambda$ make analytic convergence arithmetic and genuinely delicate.

Looking ahead

Chapter 2 moves from one local orbit to the global iteration of a rational map on the Riemann sphere. Normal families become the organizing principle: the Fatou set is the stable normality locus, the Julia set its chaotic complement, and Montel’s theorem turns omitted values into global information about backward orbits and repelling cycles.

Fatou and Julia Theory for Rational Maps

Why this chapter matters

A rational map is locally holomorphic everywhere on the Riemann sphere, but its iterates can behave in two radically different ways. On some regions the family $\{R^n\}$ is stable enough to possess locally convergent subsequences; on the complementary set arbitrarily small neighborhoods retain the full complexity of iteration. The Fatou and Julia sets encode this stable/unstable division.

The point of this chapter is not merely to define these sets. We build the global mechanisms that make them useful: exceptional points are completely classified; the Julia set is shown to be nonempty, perfect, and completely invariant; backward preimages of a nonexceptional point accumulate everywhere on the Julia set; periodic Fatou components admit only four analytic types; and repelling periodic points are dense in the Julia set. These are the basic topological and analytic engines for every later volume in complex dynamics.

Prerequisites. Chapter 1 local fixed-point theory; one-variable complex analysis on the Riemann sphere; Montel's theorem; the local mapping theorem and Riemann-Hurwitz; elementary hyperbolic-metric/Schwarz-Pick theory.

Learning goals

- Distinguish normality from pointwise orbit behavior and use it to define the Fatou and Julia sets.
- Prove why a rational map of degree at least two has at most two exceptional points and classify the exceptional configurations.
- Derive complete invariance, nonemptiness, and perfectness of the Julia set from normal-family arguments.
- Use omitted values to prove backward-orbit density without invoking repelling-cycle density circularly.
- Classify periodic Fatou components while keeping Sullivan's no-wandering theorem logically separate.
- Understand the two local mechanisms in the proof that repelling periodic points are dense: repelling Koenigs inverse branches and a minimal multiplier-one inverse sector.

Conceptual roadmap

The logical order matters. Exceptional-set counting comes first. It is then used in the topology of the Julia set. Once the Julia set is known to be perfect and completely invariant, Montel's theorem gives the backward-orbit characterization. Periodic Fatou components are classified independently of Sullivan non-wandering. Finally, backward-orbit density and the periodic-component classification feed the homoclinic inverse-branch argument proving density of repelling cycles.

exceptional set $\longrightarrow$ Fatou-Julia topology $\longrightarrow$ backward-orbit density,

Fatou-Julia topology $\longrightarrow$ periodic Fatou classification
 $\longrightarrow$ repelling-cycle density.

How the main results fit together. Do not reverse the proof order. Backward-orbit density is a Montel theorem consequence and is an input to repelling-cycle density, not a consequence of it. Likewise, the classification proved here concerns periodic Fatou components only; the statement that every Fatou component is eventually periodic belongs to Chapter 7.

2.1 Definitions

Definition 2.1 (Fatou set). For a rational map $R : \widehat{\mathbb{C}} \rightarrow \widehat{\mathbb{C}}$, the *Fatou set* $F(R)$ is the largest open subset on which the family of iterates $\{R^n : n \geq 0\}$ is normal in the spherical metric.

Definition 2.2 (Julia set). The *Julia set* of R is

$$J(R) := \widehat{\mathbb{C}} \setminus F(R).$$

Thus $J(R)$ is exactly the locus at which the iterates fail to form a normal family in every neighborhood.

Definition 2.3 (normal family). Let $U \subset \widehat{\mathbb{C}}$ be open. A family $\mathcal{F}$ of holomorphic maps $U \rightarrow \widehat{\mathbb{C}}$ is *normal* if every sequence in $\mathcal{F}$ has a subsequence converging locally uniformly on U in the spherical metric to a holomorphic map $U \rightarrow \widehat{\mathbb{C}}$. Constant limits, including the constant value ∞ , are allowed.

Definition 2.4 (exceptional point and exceptional set). For $a \in \widehat{\mathbb{C}}$ define its full backward orbit by

$$O^-(a) := \bigcup_{n \geq 0} R^{-n}(a).$$

The point a is *exceptional* if $O^-(a)$ is finite. The set of all exceptional points is denoted $E(R)$.

Definition 2.5 (completely invariant set). A subset $A \subset \widehat{\mathbb{C}}$ is *completely invariant* under R if

$$R(A) = A = R^{-1}(A).$$

Definition 2.6 (local degree and critical point). Let $p \in \widehat{\mathbb{C}}$ and choose local coordinates centered at p and $R(p)$. Then

$$R(z) = cz^{\nu_p} + O(z^{\nu_p+1}), \quad c \neq 0,$$

for a uniquely determined integer $\nu_p \geq 1$, the *local degree* of R at p . The point p is *critical* exactly when $\nu_p > 1$.

Definition 2.7 (Fatou component and periodic component). A *Fatou component* is a connected component U of $F(R)$. It is *periodic* if $R^q(U) = U$ for some $q \geq 1$; the least such q is its exact component period.

Definition 2.8 (periodic Fatou component types). Let U be a periodic Fatou component of exact period q and let $G = R^q|_U$ be its first-return map. It is an *attracting* (respectively *superattracting*) basin when G^n converges locally uniformly to a fixed point $p \in U$ with $0 < |G'(p)| < 1$ (respectively $G'(p) = 0$); a *parabolic basin* when G^n converges locally uniformly to a parabolic fixed point on ∂U ; a *Siegel disk* when G is analytically conjugate to an irrational rotation of a disk; and a *Herman ring* when G is analytically conjugate to an irrational rotation of an annulus.

Definition 2.9 (repelling periodic point). A point p of exact period q is *repelling* if its multiplier satisfies

$$|(R^q)'(p)| > 1.$$

2.2 Main results

Theorem 2.1 (Exceptional set theorem). **Statement scope.** Let $R : \widehat{\mathbb{C}} \rightarrow \widehat{\mathbb{C}}$ be a rational map of degree $d \geq 2$. Then $|E(R)| \leq 2$. If $E(R) = \{e\}$, a Möbius conjugacy sending e to ∞ turns R into a polynomial, and e is a superattracting fixed point. If $E(R) = \{e_0, e_\infty\}$, a Möbius conjugacy sending these points to 0 and ∞ turns R into $z \mapsto z^d$ or $z \mapsto z^{-d}$. In particular every exceptional point lies in the Fatou set.

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Finite completely invariant set; Riemann–Hurwitz cardinality bound; Möbius classification.

The mathematical argument begins in the proof environment below.

Proof setup. Write $d = \deg R \geq 2$.

Lemma 2.2 (Exceptional backward orbit is completely invariant). If $O^-(a)$ is finite, then $R^{-1}(O^-(a)) = O^-(a) = R(O^-(a))$.

Proof. Let a be exceptional and put $A = O^-(a)$. By definition A is finite and nonempty. If $x \in R^{-1}(A)$, then $R(x) \in R^{-n}(a)$ for some $n \geq 0$, hence $x \in R^{-(n+1)}(a) \subset A$. Thus

$$R^{-1}(A) \subset A. \quad (2.1)$$

Conversely, every $y \in A$ has at least one preimage because a nonconstant rational map is surjective on the Riemann sphere. Hence the restriction

$$R : R^{-1}(A) \longrightarrow A$$

is surjective, so finiteness gives $|R^{-1}(A)| \geq |A|$. Together with (2.1), this forces $R^{-1}(A) = A$. Applying R and using surjectivity of the displayed restriction then gives

$$R(A) = A = R^{-1}(A). \quad (2.2)$$

Thus $R|_A$ is a permutation. $\square$

Lemma 2.3 (Riemann-Hurwitz bound for finite complete invariance). *A finite completely invariant set for a rational map of degree at least two has cardinality at most two; consequently the exceptional set has cardinality at most two and is completely invariant.*

Proof. Let A be a finite completely invariant set, with $k = |A|$. Since $R|_A$ is a permutation, some iterate

$$G := R^q \quad (2.3)$$

fixes every point of A . Set $D = \deg G = d^q \geq 2$. Complete invariance gives

$$G^{-1}(p) = \{p\} \quad (p \in A). \quad (2.4)$$

The sum of the local degrees in a fiber of a degree- D rational map is D , so (2.4) implies that the local degree of G at each $p \in A$ equals D . Each such point therefore contributes $D - 1$ to the ramification sum. Riemann-Hurwitz for $G : \hat{\mathbb{C}} \rightarrow \hat{\mathbb{C}}$ gives

$$\sum_{x \in \hat{\mathbb{C}}} (\nu_x(G) - 1) = 2D - 2. \quad (2.5)$$

Consequently

$$k(D - 1) \leq 2D - 2 = 2(D - 1), \quad (2.6)$$

and $k \leq 2$.

If there were three distinct exceptional points a_1, a_2, a_3 , then

$$A := O^-(a_1) \cup O^-(a_2) \cup O^-(a_3)$$

would be a finite completely invariant set containing at least three points, contradicting Lemma 2.3. Hence

$$|E(R)| \leq 2. \quad (2.7)$$

Moreover, every point in the backward orbit of an exceptional point is itself exceptional: if $R^n(x) = a$ and $O^-(a)$ is finite, then $O^-(x) \subset O^-(a)$. Consequently, once (2.7) is known,

$$E(R) = \bigcup_{a \in E(R)} O^-(a)$$

is a finite union of finite completely invariant sets and is itself completely invariant. $\square$

Lemma 2.4 (Classification of exceptional configurations). *One exceptional point gives a polynomial after Möbius conjugacy; two exceptional points give a map conjugate to z^d or z^{-d} , and all exceptional points lie in the Fatou set.*

Proof. Assume first that $E(R) = \{e\}$. Conjugate by a Möbius map M sending e to ∞ and set $S = M \circ R \circ M^{-1}$. Total invariance of e says

$$S^{-1}(\infty) = \{\infty\}. \quad (2.8)$$

Thus S has no finite poles and is a polynomial. Its local degree at ∞ equals d , so in the coordinate $w = 1/z$ the fixed point $w = 0$ has multiplier 0; hence e is a superattracting fixed point.

Now suppose $E(R) = \{e_0, e_\infty\}$. After a Möbius conjugacy we may assume that this set is $\{0, \infty\}$. Complete invariance implies that all zeros and poles of the resulting rational function S lie in $\{0, \infty\}$. Therefore

$$S(z) = az^m \quad (2.9)$$

for some $a \neq 0$ and nonzero integer m . Since $\deg S = d$, necessarily $m = d$ or $m = -d$. A scaling $z \mapsto cz$ conjugates (2.9) to $z \mapsto z^d$ or $z \mapsto z^{-d}$ after choosing the appropriate nonzero root of a .

Finally, the exceptional points lie in the Fatou set. In the one-point case the exceptional point is a superattracting fixed point. For $z \mapsto z^d$, both 0 and ∞ are superattracting fixed points; for $z \mapsto z^{-d}$ they form a superattracting two-cycle, because the square is $z \mapsto z^{d^2}$. Normality is preserved by Möbius conjugacy, so the same conclusion holds for R . $\square$

Proof. The three lemmas give the cardinality, complete-invariance, classification, and Fatou-set assertions for the exceptional set. $\square$

Theorem 2.5 (Fatou-Julia dichotomy). **Statement scope.** *For a rational map R of degree at least 2, the Fatou set $F(R)$ is the maximal open set on which the iterates $\{R^n\}_{n \geq 0}$ form a normal family, and the Julia set $J(R) = \hat{\mathbb{C}} \setminus F(R)$ is a nonempty perfect set. Both sets are completely invariant:*

$$R(F(R)) = F(R) = R^{-1}(F(R)), \quad R(J(R)) = J(R) = R^{-1}(J(R)).$$

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Complete invariance of normality; degree-growth nonemptiness; Montel minimality; derived-set perfectness.

The mathematical argument begins in the proof environment below.

Proof setup. Let R have degree $d \geq 2$.

Lemma 2.6 (Complete invariance of Fatou and Julia sets). *The Fatou set and Julia set of a rational map are both completely invariant.*

Proof. By definition $F(R)$ is open. Suppose first that $R(z_0) \in F(R)$. Choose a neighborhood V of $R(z_0)$ on which $\{R^n\}$ is normal and a neighborhood U of z_0 with $R(U) \subset V$. Every sequence R^{n_j} with $n_j \geq 1$ can be written on U as

$$R^{n_j} = R^{n_j-1} \circ R, \quad (2.10)$$

and a locally uniformly convergent subsequence of R^{n_j-1} on V pulls back to one on U . The finitely many possible terms with exponent zero are irrelevant. Thus $z_0 \in F(R)$, proving

$$R^{-1}(F(R)) \subset F(R). \quad (2.11)$$

Conversely let $z_0 \in F(R)$ and put $w_0 = R(z_0)$. First choose a neighborhood W of z_0 with $W \Subset F(R)$. By the local mapping theorem, after choosing a sufficiently small disk V about w_0 , the component U of $R^{-1}(V)$ containing z_0 satisfies $U \Subset W$ and $R : U \rightarrow V$ is a surjective finite holomorphic map for which the inverse image in U of every compact subset of V is compact (that is, the restriction is proper). Take any sequence R^{n_j} with $n_j \rightarrow \infty$. Normality on U supplies a subsequence for which

$$R^{n_j+1} = R^{n_j} \circ R \quad (2.12)$$

converges locally uniformly on U . Because each function in (2.12) is constant on the fibers of $R : U \rightarrow V$, the limit is also constant on fibers and therefore descends to a holomorphic spherical limit on V . More concretely, for each compact $K \Subset V$, properness makes $R^{-1}(K) \cap U$ compact, and

$$\sup_{w \in K} d_{\text{sph}}(R^{n_j}(w), R^{n_k}(w)) \leq \sup_{z \in R^{-1}(K) \cap U} d_{\text{sph}}(R^{n_j+1}(z), R^{n_k+1}(z)). \quad (2.13)$$

Hence the descended subsequence converges uniformly on K . Therefore $w_0 \in F(R)$ and

$$R(F(R)) \subset F(R). \quad (2.14)$$

Together with (2.11) and surjectivity, this yields

$$R(F(R)) = F(R) = R^{-1}(F(R)). \quad (2.15)$$

Taking complements gives complete invariance of $J(R)$. □

Lemma 2.7 (Nonemptiness and infinitude of the Julia set). *For a rational map of degree at least two, the Julia set is nonempty and infinite.*

Proof. If $J(R) = \emptyset$, then $\{R^n\}$ is normal on the compact sphere. Hence some subsequence R^{n_j} converges uniformly in the spherical metric to a holomorphic map $H : \widehat{\mathbb{C}} \rightarrow \widehat{\mathbb{C}}$. Uniformly sufficiently close maps from the sphere to itself are homotopic (join corresponding non-antipodal points by the unique short spherical geodesic). Thus, for all large j ,

$$\deg(R^{n_j}) = \deg H. \quad (2.16)$$

But $\deg(R^{n_j}) = d^{n_j} \rightarrow \infty$, whereas $\deg H$ is a fixed nonnegative integer (zero if H is constant). This contradiction proves $J(R) \neq \emptyset$.

The Julia set is in fact infinite. If it were finite, complete invariance would imply that every $a \in J(R)$ has a finite full backward orbit, so $J(R) \subset E(R)$. The exceptional-set theorem gives $E(R) \subset F(R)$, contradicting $J(R) \cap F(R) = \emptyset$. $\square$

Lemma 2.8 (Minimal closed complete-invariance characterization). *The Julia set is contained in every closed completely invariant subset of the sphere containing at least three points.*

Proof. Let $K \subset \widehat{\mathbb{C}}$ be closed, completely invariant, and contain at least three points. Its complement $U = \widehat{\mathbb{C}} \setminus K$ is open and completely invariant. Every iterate $R^n|_U$ omits the same three chosen points of K . By Montel's theorem the family $\{R^n\}$ is normal on U , so $U \subset F(R)$. Hence

$$J(R) \subset K. \quad (2.17)$$

Thus $J(R)$ is contained in every closed completely invariant set with at least three points. $\square$

Lemma 2.9 (Derived-set perfectness mechanism). *The Julia set has no isolated points; equivalently, it is perfect.*

Proof. Let J' denote the set of accumulation points of $J(R)$. Since $J(R)$ is infinite and compact, J' is nonempty and closed, and $J' \subset J(R)$.

We claim that J' is completely invariant. If $x_n \in J(R)$ are distinct and $x_n \rightarrow x$, then only finitely many x_n can have the same image under the finite-to-one map R . Passing to a subsequence, the points $R(x_n)$ are distinct; by continuity they converge to $R(x)$, so $R(x) \in J'$. Thus $R(J') \subset J'$.

For backward invariance, suppose $R(x) = y \in J'$. Choose distinct $y_n \in J(R)$ with $y_n \rightarrow y$. The local mapping theorem, in coordinates at x and y , gives points $x_n \rightarrow x$ with $R(x_n) = y_n$ after passing to a tail. Complete backward invariance of $J(R)$ implies $x_n \in J(R)$. Since $y_n \neq y$, we can take $x_n \neq x$, and therefore $x \in J'$. Hence $R^{-1}(J') \subset J'$, and surjectivity gives complete invariance.

If J' were finite, then every point of J' would have finite backward orbit contained in J' , hence would be exceptional. But the exceptional-set theorem gives $E(R) \subset F(R)$ while $J' \subset J(R)$, a contradiction. Therefore J' contains at least three points. Applying (2.17) with $K = J'$ gives $J(R) \subset J'$. Since already $J' \subset J(R)$, we conclude $J' = J(R)$. Thus $J(R)$ is perfect.

All assertions in the Fatou-Julia dichotomy now follow from the definition, (2.15), nonemptiness, and perfectness. $\square$

Proof. Lemmas 2.6–2.9 give the complete-invariance, nonemptiness, infinitude, minimality, and perfectness assertions of the Fatou–Julia dichotomy. $\square$

Theorem 2.10 (Backward orbit characterization). **Statement scope.** *Let R be a rational map of degree $d \geq 2$ and let $a \in \widehat{\mathbb{C}}$ be nonexceptional. Then*

$$J(R) \subseteq \overline{O^-(a)}.$$

If $a \in J(R)$, then $\overline{O^-(a)} = J(R)$.

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Three omitted backward values plus Montel; complete invariance for equality.

The mathematical argument begins in the proof environment below.

Proof setup. Let a be nonexceptional. Then its full backward orbit $O^-(a)$ is infinite, so choose three distinct points

$$b_1, b_2, b_3 \in O^-(a). \quad (2.18)$$

For each i there is $m_i \geq 0$ with

$$R^{m_i}(b_i) = a. \quad (2.19)$$

Lemma 2.11 (Three omitted backward values force normality). *If a is nonexceptional, then every neighborhood of every Julia point meets the full backward orbit $O^-(a)$. If moreover $a \in J(R)$, then $\overline{O^-(a)} = J(R)$.*

Proof. Fix $z \in J(R)$ and let U be an arbitrary neighborhood of z . Suppose, toward a contradiction, that

$$U \cap O^-(a) = \emptyset. \quad (2.20)$$

We claim that every iterate $R^n|_U$ omits each of the three values b_i . Indeed, if $R^n(u) = b_i$ for some $u \in U$, then by (2.19)

$$R^{n+m_i}(u) = a, \quad (2.21)$$

so $u \in O^-(a)$, contradicting (2.20).

Thus the family $\{R^n|_U : n \geq 0\}$ omits the same three points of the Riemann sphere. Montel’s theorem makes this family normal on U . This contradicts $z \in J(R)$. Hence every neighborhood of every Julia point meets $O^-(a)$, and therefore

$$J(R) \subseteq \overline{O^-(a)}. \quad (2.22)$$

If in addition $a \in J(R)$, complete backward invariance of the Julia set from gives

$$O^-(a) \subset J(R). \quad (2.23)$$

Since $J(R)$ is closed,

$$\overline{O^-(a)} \subset J(R). \quad (2.24)$$

Combining (2.22) and (2.24) proves equality. $\square$

Proof. The lemma is exactly the backward-orbit density assertion of the theorem. $\square$

Theorem 2.12 (Periodic Fatou component classification interface). **Statement scope.** *Periodic Fatou components are attracting or superattracting basins, parabolic basins, Siegel disks, or Herman rings. This is a classification of periodic components only; Sullivan’s no-wandering theorem, proved in Chapter 7, is required before arbitrary Fatou components may be declared eventually periodic.*

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. First return on a hyperbolic Riemann surface; iteration trichotomy; Snail Lemma; rotation-domain classification; no Sullivan no-wandering. *The mathematical argument begins in the proof environment below.*

Lemma 2.13 (First return on a periodic Fatou component). *Let U be a periodic Fatou component of exact period q for a rational map R of degree at least two, and put $G = R^q$. Then $G(U) = U$, and U is a hyperbolic Riemann surface.*

Proof. The equality in (2.30), rather than mere inclusion, follows as follows. A rational map is open, so the image of a Fatou component is open and connected inside a Fatou component. If K is compact in that target component, then a sequence in U mapping into K cannot accumulate on $\partial U \subset J(R)$, because complete invariance gives $R(\partial U) \subset J(R)$ whereas $K \subset F(R)$. Thus the restriction from U to the target component is proper. Its image is therefore both open and closed in the connected target component, hence is the whole component. Iterating this statement gives (2.30).

Because $J(R)$ is infinite by the Fatou–Julia dichotomy, the complement of U contains at least three points. Hence U is a hyperbolic Riemann surface: its universal covering surface is the unit disk, and it carries the complete Poincare metric. $\square$

Lemma 2.14 (Iteration on a hyperbolic Riemann surface). *Let S be a hyperbolic Riemann surface and $g : S \rightarrow S$ holomorphic. Exactly one of the following occurs:*

- (a) *all iterates converge locally uniformly to a unique attracting fixed point in S ;*
- (b) *the iterates are compactly divergent in S (equivalently, every orbit eventually leaves every compact subset of S);*
- (c) *g is an automorphism of finite order;*
- (d) *S is conformally a disk, a punctured disk, or an annulus and g is conjugate to an irrational rotation.*

If, in case (b), S is a domain in the sphere and g extends holomorphically across a neighborhood of $\bar{S}$, then all iterates converge locally uniformly in the spherical metric to one boundary fixed point.

Proof. Schwarz–Pick makes every holomorphic self-map nonexpanding for the Poincare metric. Suppose first that some orbit has an accumulation point $a \in S$.

Choose $n_j \nearrow \infty$ with $g^{n_j}(z_0) \rightarrow a$. Montel compactness on the universal cover, followed by descent, gives a subsequential limit of difference iterates

$$g^{n_{j+1}-n_j} \longrightarrow h : S \longrightarrow S, \quad h(a) = a. \quad (2.31)$$

If g is not a Poincare local isometry, the strict form of Schwarz-Pick shows that h strictly decreases the Poincare distance between distinct points. Consequently h has at most one fixed point. Since g commutes with every limit of its iterates, $g(a)$ is also fixed by h , so $g(a) = a$. In a local disk coordinate at a , strict Schwarz-Pick gives $|g'(a)| < 1$. The Poincare distance from a decreases along every nonconstant orbit; if an orbit did not converge to a , a compactness/subsequence argument would produce a second fixed point of a limit map exactly as in (2.31). Hence all orbits converge to a , locally uniformly on compact subsets. This is case (a).

It remains to consider the equality case of Schwarz-Pick. Then g is a Poincare local isometry and therefore a covering map. The same recurrence argument gives a sequence of powers converging locally uniformly to the identity: lift a limit map fixing an accumulation point to the disk; it is an elliptic disk isometry, and a subsequence of its powers converges to the identity. Pulling this subsequence back to powers of g yields

$$g^{m_j} \longrightarrow \text{id}_S. \quad (2.32)$$

Equation (2.32) forces g to be one-to-one: if $g(x) = g(y)$ then $g^{m_j}(x) = g^{m_j}(y)$ for $m_j \geq 1$, and passage to the limit gives $x = y$. Thus g is an automorphism.

We next classify an automorphism satisfying (2.32) by an explicit uniformization argument. Let $\pi : \mathbb{D} \rightarrow S$ be the universal covering with deck group Γ , and fix a lift G of g . Choose a compact disk $K \Subset S$ and one lifted copy $K^* \Subset \mathbb{D}$. For every j there is a deck transformation γ_j such that

$$G_j := \gamma_j \circ G^{m_j}$$

converges to the identity uniformly on K^* and hence locally uniformly on $\mathbb{D}$. Since g^{m_j} is an automorphism, G_j normalizes Γ . For each fixed $\gamma \in \Gamma$, $G_j \gamma G_j^{-1} \in \Gamma$ and tends to γ . Discreteness of Γ therefore implies

$$G_j \gamma = \gamma G_j \quad (2.33)$$

for all sufficiently large j . If one such G_j is the identity, then $g^{m_j} = \text{id}_S$ and g has finite order.

Assume otherwise. Deck transformations of the disk are nonelliptic. Two commuting nonidentity disk automorphisms have the same fixed point set on $\partial\mathbb{D}$ when one of them is nonelliptic. By (2.33), any two nonidentity elements of Γ therefore have the same boundary fixed point set. Hence either Γ is trivial, or all its nonidentity elements lie in the one-dimensional centralizer of a fixed parabolic or hyperbolic element. A discrete subgroup of such a centralizer is infinite cyclic. Consequently

$$S \cong \mathbb{D}, \quad \mathbb{D}^*, \quad \text{or an annulus.} \quad (2.34)$$

On the disk an automorphism whose powers recur to the identity must be elliptic;

after moving its interior fixed point to 0 it is a rotation. Automorphisms of $\mathbb{D}^*$ are rotations, while an infinite-order recurrent automorphism of an annulus is a rotation (the other conformal automorphisms reverse the two ends and have finite order modulo rotations). Since we have excluded finite order, the rotation angle is irrational. This proves cases (c) and (d).

If no orbit has an accumulation point in S , then every orbit leaves every compact subset, which is case (b). Finally suppose $S \subset \widehat{\mathbb{C}}$ and g extends holomorphically across a neighborhood of $\overline{S}$. Join a base point z_0 to $g(z_0)$ by a path $p : [0, 1] \rightarrow S$ and extend it by $p(t+1) = g(p(t))$. Let δ be the Poincaré diameter of $p([0, 1])$. Schwarz-Pick gives Poincaré diameter at most δ for every segment $p([n, n+1])$. Compact divergence makes these segments leave every compact subset of S . A bounded-Poincaré-diameter set approaching ∂S has spherical diameter tending to zero: otherwise two subsequences at a fixed positive spherical separation would approach distinct boundary points while remaining a bounded hyperbolic distance apart, contradicting completeness and the boundary blow-up of the Poincaré metric. Thus

$$\text{diam}_{\text{sph}} p([n, n+1]) \rightarrow 0. \quad (2.35)$$

The cluster set of the tail of a continuous path is $\bigcap_{T>0} \overline{p([T, \infty))}$, hence is connected. If $p(t_j) \rightarrow \xi$ with $t_j \rightarrow \infty$, then (2.35) and $p(t_j+1) = g(p(t_j))$ give $g(\xi) = \xi$. The cluster set is therefore a connected subset of the finite fixed-point set of g , so it is one point $\{\xi\} \subset \partial S$. Hence $p(t) \rightarrow \xi$. For any $z \in S$,

$$d_S(g^n(z), g^n(z_0)) \leq d_S(z, z_0),$$

and the same boundary comparison forces $g^n(z) \rightarrow \xi$; the estimate is uniform when z ranges over a compact subset. Therefore $g^n \rightarrow \xi$ locally uniformly and continuity gives $g(\xi) = \xi$. This proves the addendum. $\square$

Lemma 2.15 (Snail lemma). *Let f be holomorphic near 0, with $f(0) = 0$ and multiplier λ . Suppose there is a continuous path $p : [0, \infty) \rightarrow \mathbb{C} \setminus \{0\}$ tending to 0 and satisfying*

$$f(p(t)) = p(t+1). \quad (2.36)$$

Then either $|\lambda| < 1$ or $\lambda = 1$.

Proof. The convergence in (2.36) excludes $|\lambda| > 1$ by the repelling case of the inverse function theorem (equivalently, by Koenigs linearization). Assume for a contradiction that $|\lambda| = 1$ and $\lambda \neq 1$. For large t choose a continuous logarithmic lift

$$p(t) = e^{P(t)}, \quad \Re P(t) \rightarrow -\infty. \quad (2.37)$$

Since $f(z) = \lambda z(1 + O(z))$, a lift of f on a sufficiently far left half-plane has the form

$$F(Z) = Z + ic + \eta(Z), \quad \eta(Z) \rightarrow 0 \quad (\Re Z \rightarrow -\infty), \quad (2.38)$$

where $c \in \mathbb{R} \setminus \{0\}$ and $e^{ic} = \lambda$. After changing the lift by an integral multiple of $2\pi i$, (2.36) becomes $P(t+1) = F(P(t))$. Replacing c by $-c$ if necessary, assume $c > 0$. Choose X_0 so

far to the left that F is univalent there and

$$\Im(F(Z) - Z) > c/2 \quad (\Re Z < X_0). \quad (2.39)$$

For t_0 large, the lifted tail $P([t_0, \infty))$ lies in this half-plane. It is a proper curve whose image under F is the same curve with its initial segment removed. In the region above its initial height, take the component V_0 lying on the left-hand side of this oriented curve. The univalence of F and (2.39) imply

$$F(V_0) \Subset_{\text{mloc}} V_0, \quad (2.40)$$

and every forward orbit in V_0 has real part tending to $-\infty$. Therefore

$$V := \exp(V_0) \cup \{0\}$$

is a hyperbolic neighborhood of 0 with $f(V) \subsetneq V$ and all orbits in V tending to 0. Conjugate V to the disk by a Riemann map sending 0 to 0. Schwarz's lemma is strict because the inclusion in (2.40) is proper, so $|f'(0)| < 1$, contradicting $|\lambda| = 1$. Hence the only unit-modulus possibility compatible with (2.36) is $\lambda = 1$. $\square$

Proof of the periodic Fatou component classification. Let U be a periodic Fatou component of exact period q , and set

$$G := R^q.$$

By Lemma 2.13, $G : U \rightarrow U$ is a holomorphic self-map of a hyperbolic Riemann surface. Apply Lemma 2.14. In case (a) of Lemma 2.14, U is the immediate basin of the attracting fixed point of G ; multiplier zero is the superattracting subcase. In case (b), the spherical extension is automatic because G is rational, so G^m converges locally uniformly on U to a boundary fixed point $\xi \in \partial U \subset J(R)$. Choose a path from z to $G(z)$ in U and extend it by the dynamics. Lemma 2.15 shows that the multiplier of G at ξ is either of modulus < 1 or is equal to 1. The first alternative would make ξ an attracting fixed point and hence put a whole neighborhood of ξ in the Fatou set, impossible because $\xi \in J(R)$. Thus $G'(\xi) = 1$ and U is a parabolic basin.

Case (c) of Lemma 2.14 cannot occur. If G^m were the identity on U , the identity theorem for rational maps would give $R^{qm} = \text{id}$ on the whole sphere, contradicting

$$\deg(R^{qm}) = d^{qm} > 1. \quad (2.41)$$

In case (d), a disk rotation is exactly a Siegel disk and an annulus rotation is a Herman ring. It remains only to exclude the punctured disk. Suppose $\phi : \mathbb{D}^* \rightarrow U$ conjugates $z \mapsto e^{2\pi i \theta} z$ to G , with $\theta \notin \mathbb{Q}$. A univalent map from a punctured disk to the sphere cannot have an essential singularity at the puncture: otherwise the great Picard theorem would force infinitely many repetitions of values, contradicting injectivity. Hence ϕ extends meromorphically, and therefore holomorphically as a sphere-valued map, across 0; write $a = \phi(0)$. The conjugacy identity extends across the puncture and gives $G(a) = a$. The extended disk $\phi(\mathbb{D})$ is a neighborhood on which the iterates are conjugate to a normal family of rotations, so $a \in F(R)$. This connected Fatou neighborhood strictly enlarges U , contradicting that U was a Fatou component. Thus the punctured-

disk alternative cannot occur.

Applying this fixed-component classification to the first return map $G = R^q$ gives exactly the four periodic types stated: attracting/superattracting, parabolic, Siegel, and Herman. Nothing in the proof says that an arbitrary Fatou component is eventually periodic; that separate assertion is Sullivan's no-wandering theorem and remains assigned to Chapter 7. $\square$

Theorem 2.16 (Density of repelling periodic points in Julia set). **Statement scope.** *For every rational map R of degree at least two, the repelling periodic points of R are dense in $J(R)$.*

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Holomorphic fixed-point formula; homoclinic inverse orbit; minimal multiplier-one inverse sector; compact inverse branch gives a repelling periodic point.

The mathematical argument begins in the proof environment below.

Lemma 2.17 (Holomorphic fixed-point alternative). *Let R be a rational map of degree $d \geq 2$. The map R has a fixed point $p \in J(R)$ whose multiplier λ satisfies*

$$|\lambda| > 1 \quad \text{or} \quad \lambda = 1. \quad (2.39)$$

Proof. First suppose that every fixed point is simple. Conjugate R by a Mobius map so that ∞ is not fixed; multipliers are unchanged by conjugacy. Then all fixed points are finite. For

$$\omega(z) = \frac{dz}{z - R(z)}, \quad (2.40)$$

the residue at a simple fixed point p is

$$\text{Res}_p \omega = \frac{1}{1 - R'(p)}. \quad (2.41)$$

Since $R(\infty) \neq \infty$, we have $R(z) = O(1)$ as $z \rightarrow \infty$, so $\omega = (z^{-1} + O(z^{-2}))dz$ and $\text{Res}_\infty \omega = -1$. The residue theorem gives the holomorphic fixed-point formula

$$\sum_{R(p)=p} \frac{1}{1 - R'(p)} = 1. \quad (2.42)$$

There are $d + 1$ fixed points counted with multiplicity, hence exactly $d + 1$ distinct ones in the present simple case. For $|\lambda| \leq 1$, $\lambda \neq 1$,

$$\Re \frac{1}{1 - \lambda} = \frac{1 - \Re \lambda}{|1 - \lambda|^2} \geq \frac{1}{2}, \quad (2.43)$$

with equality precisely when $|\lambda| = 1$. If every fixed point satisfied $|\lambda| \leq 1$, the real part of the left side of (2.42) would be at least $(d + 1)/2 > 1$, a contradiction. Hence a repelling fixed point exists.

If some fixed point is multiple as a zero of $R(z) - z$, then differentiating shows its multiplier is 1. Thus (2.39) always holds for at least one fixed point.

It remains to put such a point in the Julia set. If $|\lambda| > 1$ and the iterates were normal near p , some subsequence R^{n_j} would converge locally uniformly there. Since every term takes the finite value p at p , ordinary holomorphic convergence holds in a local coordinate, so derivatives at p would remain bounded. But

$$(R^{n_j})'(p) = \lambda^{n_j} \longrightarrow \infty,$$

a contradiction. Hence a repelling fixed point lies in $J(R)$.

Now suppose $\lambda = 1$. If $p \in F(R)$, the Fatou component containing p is fixed by R . The periodic-component classification just proved leaves four possibilities. It cannot be an attracting basin because the multiplier is not of modulus < 1 ; it cannot be a parabolic basin because the parabolic fixed point of that case lies on the boundary of the component; a Siegel disk has an irrational unit multiplier at its interior fixed center; and a Herman ring has no fixed point. Thus $p \notin F(R)$, so $p \in J(R)$. This proves the lemma. $\square$

Lemma 2.18 (Minimal inverse sector at a multiplier-one point). *Let f be a nonidentity holomorphic germ fixing 0 with $f'(0) = 1$. Then for some $k \geq 1$ and $a \neq 0$,*

$$f(z) = z + az^{k+1} + O(z^{k+2}). \quad (2.44)$$

There is a punctured neighborhood N of 0 with the following property: if $z \in J(R) \cap N$ in the rational-map application $f = R$, then the local inverse germ $g = f^{-1}$ can be iterated indefinitely at z , all inverse iterates remain in N , and

$$g^n(z) \longrightarrow 0. \quad (2.45)$$

Proof. The inverse function theorem gives

$$g(z) = z - az^{k+1} + O(z^{k+2}). \quad (2.46)$$

On the punctured neighborhood set

$$W(z) := -\frac{1}{kaz^k}. \quad (2.47)$$

A direct binomial expansion of (2.44)–(2.46) gives

$$W(f(z)) = W(z) + 1 + O(|W(z)|^{-1/k}), \quad W(g(z)) = W(z) - 1 + O(|W(z)|^{-1/k}). \quad (2.48)$$

Shrink N so that $|W(z)|$ is large and both error terms in (2.48) have modulus $< 1/4$. If $\Re W(z) > -1/2$, then after one forward step $\Re W(f(z)) > 1/4$, and thereafter

$$\Re W(f^n(z)) \geq \Re W(f(z)) + \frac{3}{4}(n-1) \quad (n \geq 1). \quad (2.49)$$

The same inequalities hold uniformly on a sufficiently small neighborhood of such a z , so the forward iterates converge locally uniformly to 0 there; hence z is a Fatou

point. Consequently a Julia point in N must satisfy $\Re W(z) \leq -1/2$. For such a point the inverse estimate gives

$$\Re W(g^n(z)) \leq \Re W(z) - \frac{3}{4}n, \quad (2.50)$$

so all inverse iterates remain in N and converge to 0. This is (2.45). $\square$

Proof of the density theorem. Write $d = \deg R \geq 2$. By Lemma 2.17, fix a Julia fixed point p with the stated alternative. Let U be an arbitrary open set meeting $J(R)$. Since $J(R)$ is perfect, after shrinking U if necessary we may assume

$$p \notin \overline{U}, \quad (2.51)$$

while $U \cap J(R) \neq \emptyset$. The point p is nonexceptional because the exceptional-set theorem places $E(R)$ in the Fatou set. The backward-orbit characterization therefore supplies an integer $r > 0$ and

$$z_r \in U \cap J(R), \quad R^r(z_r) = p. \quad (2.52)$$

Choose a small neighborhood N of p , disjoint from $\overline{U}$. The Julia point z_r is likewise nonexceptional, so applying backward-orbit density to z_r and the open set $N \cap J(R)$ gives $q > r$ and

$$z_q \in N \cap J(R), \quad R^{q-r}(z_q) = z_r. \quad (2.53)$$

We extend (2.52)–(2.53) to a backward orbit

$$\cdots \mapsto z_{q+2} \mapsto z_{q+1} \mapsto z_q \mapsto \cdots \mapsto z_r \mapsto \cdots \mapsto z_0 = p \quad (2.54)$$

which converges backward to p . If p is repelling, Koenigs linearization provides a local inverse branch fixing p and contracting to p ; apply it repeatedly to z_q after taking N small enough. If $R'(p) = 1$, translate p to 0 and apply Lemma 2.18. Because $z_q \in J(R)$, the inverse branch in (2.45) produces the required tail. Thus in either case

$$z_j \longrightarrow p \quad (j \rightarrow \infty). \quad (2.55)$$

$\square$

Lemma 2.19 (Homoclinic inverse orbit creates a repelling cycle). *For the orbit (2.54), every sufficiently small neighborhood of its finite segment can be chosen so that some inverse branch of a high iterate maps a hyperbolic domain compactly into itself. Its fixed point is repelling for the corresponding forward iterate and its orbit passes through U .*

Proof. Choose simply connected neighborhoods $V_j \ni z_j$ along a long finite segment of (2.54) so small that

$$R : V_j \longrightarrow V_{j-1} \quad (2.56)$$

is proper and has no critical point other than the central point z_j when that point itself is critical. Take $V_r \Subset U$. Along the tail near p the inverse branch is univalent and its iterates converge to p , so the neighborhoods can be pulled back so that, for arbitrarily large m ,

$$\overline{V_m} \subset V_0. \quad (2.57)$$

If none of the central points in the finite chain is critical, then $R^m : V_m \rightarrow V_0$ is biholomorphic. Its inverse $h : V_0 \rightarrow V_m \Subset V_0$ is a strict contraction for the Poincare metric.

If some central points are critical, shrink the neighborhoods in (2.56). Every critical value of the restricted composition $R^m : V_m \rightarrow V_0$ is then the central value p : a critical point at an intermediate stage is carried by the remaining part of the orbit to p . Choose a simple slit S from p to ∂V_0 , disjoint from $\overline{V_m}$. On

$$X := V_0 \setminus S \quad (2.58)$$

the branched covering has no branch values. The component Y of its inverse image that follows the chosen orbit is therefore mapped biholomorphically onto X , and by (2.57)

$$Y \subset V_m \Subset X. \quad (2.59)$$

Thus in both the branched and unbranched cases we have a holomorphic inverse branch

$$h : X \longrightarrow Y \Subset X, \quad (2.60)$$

where X is simply connected and hyperbolic.

For completeness, conjugate X to the unit disk. Since the image of h is relatively compact, it lies in a disk $|w| \leq \rho < 1$. Writing the conjugated map as $T = \rho S_0$ with $S_0 : \mathbb{D} \rightarrow \mathbb{D}$, Schwarz-Pick gives the hyperbolic derivative bound

$$\|DT\|_{\text{hyp}} \leq \rho \frac{1 - |S_0|^2}{1 - \rho^2 |S_0|^2} \leq \rho < 1. \quad (2.61)$$

The complete Poincare metric and the contraction theorem yield a unique fixed point y of h . Since h is an inverse branch of R^m ,

$$R^m(y) = y, \quad |(R^m)'(y)| = |h'(y)|^{-1} > 1, \quad (2.62)$$

where at a fixed point the Euclidean and Poincare multiplier moduli agree. Moreover the successive images of Y under the chosen branch itinerary pass through $V_r \Subset U$. Hence the periodic orbit of y meets U . $\square$

Completion of the density theorem. Applying Lemma 2.19 to the homoclinic orbit (2.54) produces a repelling periodic point whose orbit meets the originally prescribed open set U . Since periodicity implies that every point on this orbit is itself a repelling periodic point, U contains a repelling periodic point. The open set U was arbitrary among neighborhoods meeting $J(R)$, so the repelling periodic points are dense in $J(R)$. $\square$

2.3 Worked examples

Worked example: The complete picture for $R(z) = z^2$

The points 0 and ∞ are superattracting fixed points. If $|z| < 1$, then $R^n(z) = z^{2^n} \rightarrow 0$; if $|z| > 1$, then $R^n(z) \rightarrow \infty$. Hence

$$F(R) = \{|z| < 1\} \cup \{|z| > 1\} \cup \{\infty\}, \quad J(R) = \{|z| = 1\}.$$

The two Fatou components are superattracting basins. Both 0 and ∞ are exceptional, and the exceptional-set theorem predicts exactly this two-point configuration: after the present normalization the map already is $z \mapsto z^2$.

Periodic points on the unit circle solve $z^{2^q} = z$, hence $z^{2^q-1} = 1$. At such a point,

$$(R^q)'(z) = 2^q z^{2^q-1} = 2^q,$$

so every nonzero periodic point is repelling. Roots of unity of orders $2^q - 1$ become dense in the unit circle, giving a concrete model of the general density theorem.

Worked example: A parabolic boundary point for $R(z) = z^2 + \frac{1}{4}$

The fixed-point equation is

$$z^2 - z + \frac{1}{4} = 0,$$

so $p = 1/2$ is a double fixed point and $R'(p) = 1$. Thus the implicit-function simplicity condition from Chapter 1 fails exactly at this parameter. In the translated coordinate $w = z - 1/2$,

$$R(z) - \frac{1}{2} = w + w^2,$$

which is the simplest nontrivial multiplier-one germ. The fixed point lies on the Julia set, while a neighboring Fatou component is drawn toward it along a parabolic direction. This is the parabolic alternative in the periodic Fatou classification, not an attracting basin with $|\lambda| < 1$.

Boundary case: periodic classification is not no-wandering

The theorem in this chapter starts with a Fatou component U for which $R^q(U) = U$. It then classifies the possible first-return dynamics on U . Nothing in that proof shows that an arbitrary Fatou component must eventually become periodic. That additional global assertion is Sullivan's no-wandering theorem and is proved only in Chapter 7. Using the periodic classification as if it already implied eventual periodicity would therefore be a genuine logical gap.

Common pitfall

Backward-orbit density and repelling-periodic density are closely related but are not interchangeable proof tools. The clean dependency direction is: prove backward-orbit density directly from Montel, then use it to construct the inverse orbit needed for the repelling-cycle argument.

2.4 Chapter summary

What to remember

- A rational map of degree at least two has at most two exceptional points; one exceptional point gives a polynomial model and two give $z^{\pm d}$ after Möbius conjugacy.
- The Fatou and Julia sets are completely invariant, the Julia set is nonempty and perfect, and it is the minimal closed completely invariant set with at least three points.
- The full backward orbit of every nonexceptional target accumulates on the Julia set; for a target in the Julia set its closure is exactly the Julia set.
- Every periodic Fatou component is an attracting/superattracting basin, a parabolic basin, a Siegel disk, or a Herman ring. Sullivan no-wandering is not part of this theorem.
- Repelling periodic points are dense in the Julia set. The proof combines a fixed-point index alternative, local inverse dynamics near a repelling or multiplier-one fixed point, backward-orbit density, and a compact inverse-branch contraction.

Looking ahead

Chapter 3 specializes to polynomial dynamics. The basin of infinity admits a global Böttcher coordinate and Green function; these analytic objects lead to external rays, equilibrium measures, and equidistribution of preimages. The Fatou-Julia machinery developed here supplies the global dynamical background for those potential-theoretic constructions.

Polynomial Dynamics, Green Functions and External Rays

Why this chapter matters

Polynomial dynamics has a distinguished global direction: infinity is a superattracting fixed point. That single fact turns the local Böttcher normal form from Chapter 1 into a global geometric tool. When the filled Julia set is connected, the entire basin of infinity can be uniformized by the exterior disk; even without connectedness, the escape rate produces a continuous Green function on the plane.

These analytic coordinates do much more than describe escape. Their boundary geometry produces external rays, their Laplacian produces the equilibrium measure, and normalized inverse images of any nonexceptional target converge to that same measure. To make the last assertion logically correct, this chapter also proves the general rational-map natural-measure theorem: a rational map of degree $d \geq 2$ has a unique measure of maximal entropy, with entropy $\log d$. The polynomial Brolin theorem is then obtained by identifying the rational natural measure with the Riesz measure of the polynomial Green function.

Prerequisites. Chapter 1 local Böttcher theory; Chapter 2 Fatou-Julia and exceptional-set theory; the Riemann mapping theorem and elementary conformal geometry; Foundation F13 potential theory; Birkhoff's theorem and the F14 variational principle for entropy.

Learning goals

- Globalize the Böttcher coordinate precisely when the filled Julia set is connected.
- Construct the polynomial Green function from escape rates without assuming connectedness.
- Prove the critical-orbit criterion for connectedness by two different mechanisms.
- Distinguish the forward Böttcher map from its inverse at the boundary and use the correct Carathéodory extension theorem to define landing rays.
- Construct the balanced natural measure of a rational map from inverse images and prove preimage equidistribution.
- Understand why entropy $\log d$ comes from the d inverse branches and why branch imbalance contradicts maximal entropy.

- Identify the polynomial equilibrium/Brolin measure as $(2\pi)^{-1}\Delta G_P$ and derive Brolin equidistribution from the rational theorem.

Conceptual roadmap

There are three interacting chains. The geometric chain is

connected $K(P) \rightarrow$ global Böttcher coordinate
 $\rightarrow$ boundary landing under local connectivity.

The escape/potential chain is

local Böttcher germ at infinity $\rightarrow G_P \rightarrow \frac{1}{2\pi}\Delta G_P$.

The measure-theoretic chain is

contracting inverse branches $\rightarrow$ balanced rational measure
 $\rightarrow h = \log d$
 $\rightarrow$ unique maximal-entropy measure.

Only after the rational chain is complete do we identify its polynomial specialization with the Riesz measure of G_P . This order avoids the false implication “polynomial Brolin theorem $\Rightarrow$ general rational maximal-entropy theorem.”

How the main results fit together. Two scope warnings are essential. First, the external-ray theorem extends the *inverse* Böttcher coordinate continuously to the circle; it does not assert a continuous inverse boundary conjugacy when several angles can land at one point. Second, the Green function and Brolin measure exist even when $K(P)$ is disconnected, although the global exterior-disk Böttcher isomorphism need not exist.

3.1 Definitions

Definition 3.1 (filled Julia set). For a polynomial P of degree at least two, the *filled Julia set*

$$K(P) := \{z \in \mathbb{C} : \{P^n(z)\}_{n \geq 0} \text{ is bounded}\}$$

is the set of points with bounded forward orbit. Its boundary is the Julia set $J(P)$.

Definition 3.2 (Böttcher coordinate at infinity). If $P(z) = a_d z^d + O(z^{d-1})$, a *Böttcher coordinate at infinity* is a conformal coordinate ϕ on a neighborhood of infinity satisfying

$$\phi(P(z)) = \phi(z)^d, \quad \frac{\phi(z)}{z} \rightarrow c, \quad c^{d-1} = a_d.$$

Once the root c is fixed, this normalization removes the root-of-unity ambiguity.

Definition 3.3 (polynomial Green function). The *polynomial Green function*, or escape-rate function, is

$$G_P(z) := \lim_{n \rightarrow \infty} d^{-n} \log^+ |P^n(z)|, \quad \log^+ t := \max\{\log t, 0\}.$$

It is normalized to vanish on $K(P)$ and to have logarithmic growth at infinity.

Definition 3.4 (equilibrium measure). The *equilibrium* or *Brolin measure* of P is

$$\mu_P := \frac{1}{2\pi} \Delta G_P$$

in the distributional sense. Chapter 3 proves that this is a probability supported on $J(P)$, that it equals the logarithmic equilibrium measure of $K(P)$, and that it is the natural balanced/maximal-entropy measure of the polynomial.

Definition 3.5 (basin of infinity). The *basin of infinity* is

$$A_\infty(P) := \widehat{\mathbb{C}} \setminus K(P).$$

Equivalently, it consists of ∞ together with the finite points whose forward orbits tend to infinity.

Definition 3.6 (normalized preimage measure and normalized pullback). Let $R : \widehat{\mathbb{C}} \rightarrow \widehat{\mathbb{C}}$ be rational of degree $d \geq 2$. For $a \in \widehat{\mathbb{C}}$, the *normalized n -th preimage measure* is

$$\mu_{n,a} := \frac{1}{d^n} \sum_{R^n(z)=a} \deg_z(R^n) \delta_z,$$

where every preimage is counted with its local multiplicity. For a probability measure ν , its *normalized pullback* is

$$T_R^* \nu := \frac{1}{d} R^* \nu.$$

Equivalently, its dual operator on continuous functions is

$$(\mathcal{L}_R f)(a) := \frac{1}{d} \sum_{R(z)=a} \deg_z(R) f(z).$$

Definition 3.7 (balanced measure). A Borel probability measure μ for a degree- d rational map R is *balanced* if

$$R^* \mu = d\mu, \quad \text{equivalently} \quad T_R^* \mu = \mu.$$

For the non-atomic balanced measure constructed below, this is equivalent to the constant-Jacobian identity

$$\mu(R(A)) = d \mu(A)$$

on every Borel set A on which R is injective.

Definition 3.8 (measure of maximal entropy). For a continuous self-map T of a compact metric space, an invariant Borel probability measure μ is a *measure of maximal*

entropy if

$$h_\mu(T) = h_{\text{top}}(T),$$

where metric and topological entropy are the notions fixed in the Foundation thermodynamic-formalism interface.

Definition 3.9 (external ray and landing). Assume $K(P)$ is connected and let

$$\phi : A_\infty(P) \rightarrow \widehat{\mathbb{C}} \setminus \overline{\mathbb{D}}, \quad \psi := \phi^{-1}.$$

For $\theta \in \mathbb{R}/\mathbb{Z}$, the *external ray of angle θ* is

$$\mathcal{R}_\theta := \{\psi(re^{2\pi i\theta}) : r > 1\}.$$

It *lands* at $z \in J(P)$ if

$$\lim_{r \downarrow 1} \psi(re^{2\pi i\theta}) = z.$$

Definition 3.10 (Carathéodory loop). When the inverse Böttcher coordinate ψ extends continuously to $\partial\mathbb{D}$, its boundary restriction

$$\gamma(e^{2\pi i\theta}) := \lim_{r \downarrow 1} \psi(re^{2\pi i\theta})$$

is the *Carathéodory loop* (or boundary landing map). It need not be injective unless stronger boundary hypotheses, such as a Jordan boundary, are imposed.

3.2 Main results

Theorem 3.1 (Global Böttcher coordinate for polynomials). **Statement scope.** Let $P(z) = a_d z^d + O(z^{d-1})$ have degree $d \geq 2$ and connected filled Julia set. Then there is a conformal isomorphism

$$\phi : A_\infty(P) \longrightarrow \widehat{\mathbb{C}} \setminus \overline{\mathbb{D}}$$

satisfying $\phi(P(z)) = \phi(z)^d$ and $\phi(z)/z \rightarrow c$ with $c^{d-1} = a_d$. The coordinate is unique after fixing this leading normalization.

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Uniformize the simply connected basin by the exterior disk; conjugate P to a proper disk self-map and use the finite-Blaschke-product classification to force a monomial.

The mathematical argument begins in the proof environment below.

Proof setup. Let

$$P(z) = a_d z^d + a_{d-1} z^{d-1} + \cdots + a_0, \quad d \geq 2, \quad a_d \neq 0,$$

and let

$$A_\infty(P) := \widehat{\mathbb{C}} \setminus K(P)$$

be the basin of infinity, with the point ∞ included.

Lemma 3.2 (Basin of infinity is a simply connected pointed domain). *If $K(P)$ is connected, then the basin of infinity $A_\infty(P)$ is a simply connected proper domain of the sphere containing ∞ , and hence admits a conformal map to the exterior disk.*

Proof. The filled Julia set is full: $\mathbb{C} \setminus K(P)$ has no bounded component. Indeed, choose an escape radius R so that $K(P) \subset \overline{D(0, R)}$ and $P(K(P)) = K(P)$. If U were a bounded complementary component, then $\partial U \subset K(P)$. For each n , the holomorphic function P^n is bounded by R on ∂U , hence by the maximum-modulus principle by R throughout U . Thus every point of U would have bounded orbit, contradicting $U \cap K(P) = \emptyset$.

Since $K(P)$ is connected by hypothesis and is full, the standard planar simply-connectedness criterion shows that the domain $A_\infty(P)$ is simply connected. It is a proper domain of the sphere containing ∞ .

By the Riemann mapping theorem there is therefore a conformal isomorphism

$$\phi_0 : A_\infty(P) \longrightarrow \hat{\mathbb{C}} \setminus \overline{\mathbb{D}} \quad (3.1)$$

with $\phi_0(\infty) = \infty$. Its Laurent expansion at infinity has the form

$$\phi_0(z) = c_0 z + c_1 + O(z^{-1}), \quad c_0 \neq 0. \quad (3.2)$$

□

Lemma 3.3 (Exterior-disk conjugate is a monomial). *In an exterior-disk uniformization, the polynomial acts as a unit multiple of w^d ; after a rotation this yields a global Böttcher coordinate with the stated leading-coefficient normalization and uniqueness.*

Proof. Complete invariance of $K(P)$ gives $P(A_\infty) = A_\infty$. Hence

$$Q := \phi_0 \circ P \circ \phi_0^{-1}$$

is a proper holomorphic self-map of the exterior disk of degree d , fixes ∞ , and has ∞ as its only preimage of ∞ .

Invert the exterior disk and set

$$q(\zeta) := \frac{1}{Q(1/\zeta)}. \quad (3.3)$$

Then $q : \mathbb{D} \rightarrow \mathbb{D}$ is a proper holomorphic map of degree d and $q^{-1}(0) = \{0\}$. A proper holomorphic self-map of the unit disk is a finite Blaschke product. Since its only zero is 0, counted with total multiplicity d , it follows that

$$q(\zeta) = \eta \zeta^d, \quad |\eta| = 1. \quad (3.4)$$

Equivalently,

$$Q(w) = \eta^{-1} w^d. \quad (3.5)$$

Choose $\rho \in \partial\mathbb{D}$ with $\rho^{d-1} = \eta^{-1}$ and put $\phi = \rho\phi_0$. Then (3.5) becomes

$$\phi(P(z)) = \phi(z)^d \quad (z \in A_\infty(P)). \quad (3.6)$$

Thus ϕ is the desired global Böttcher coordinate.

Finally write

$$\phi(z) = cz + O(1) \quad (z \rightarrow \infty).$$

Substituting this and $P(z) = a_d z^d + O(z^{d-1})$ into (3.6) gives

$$ca_d = c^d, \quad \text{hence} \quad c^{d-1} = a_d. \quad (3.7)$$

Therefore $\phi(z)/z \rightarrow c$, with precisely the leading-coefficient normalization required in the statement. If two such coordinates satisfy the same functional equation, their quotient in the exterior coordinate extends to an automorphism commuting with $w \mapsto w^d$ and fixing infinity, hence is multiplication by a $(d-1)$ -st root of unity. Fixing the chosen root c removes this ambiguity. $\square$

Proof. Lemma 3.2 provides the global exterior uniformization, and Lemma 3.3 gives the Böttcher functional equation, normalization, and uniqueness. $\square$

Theorem 3.4 (Polynomial Green function). **Statement scope.** *For every polynomial P of degree $d \geq 2$, the limit*

$$G_P(z) = \lim_{n \rightarrow \infty} d^{-n} \log^+ |P^n(z)|$$

exists locally uniformly (indeed uniformly on $\mathbb{C}$), is continuous on $\mathbb{C}$, harmonic and strictly positive on $A_\infty(P) \setminus \{\infty\}$, vanishes exactly on $K(P)$, and satisfies $G_P(P(z)) = dG_P(z)$.

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Bound the one-step log-plus defect and telescope for global uniform convergence; use the local Böttcher germ at infinity for positivity, harmonicity, and asymptotics.

The mathematical argument begins in the proof environment below.

Proof. Write

$$P(z) = a_d z^d + O(z^{d-1}), \quad d \geq 2,$$

and put $L(z) = \log^+ |z|$. Since

$$L(P(z)) - dL(z)$$

is continuous on $\mathbb{C}$ and tends to $\log |a_d|$ as $|z| \rightarrow \infty$, there is a constant C such that

$$|L(P(z)) - dL(z)| \leq C \quad (z \in \mathbb{C}). \quad (3.8)$$

Define

$$G_n(z) := d^{-n} L(P^n(z)). \quad (3.9)$$

Then (3.8) gives

$$|G_{n+1}(z) - G_n(z)| = d^{-(n+1)} |L(P^{n+1}(z)) - dL(P^n(z))| \leq C d^{-(n+1)}. \quad (3.10)$$

The Weierstrass criterion therefore shows that G_n converges uniformly on all of $\mathbb{C}$ to a continuous nonnegative function G_P .

Shifting the sequence yields the functional equation

$$G_P(P(z)) = \lim_{n \rightarrow \infty} d^{-n} L(P^{n+1}(z)) = dG_P(z). \quad (3.11)$$

If $z \in K(P)$, its orbit is bounded, hence $G_n(z) \rightarrow 0$ and

$$G_P(z) = 0. \quad (3.12)$$

Now let $z \notin K(P)$. Its orbit tends to infinity. Apply the local Böttcher theorem at the superattracting fixed point ∞ : on some neighborhood U of infinity there is a holomorphic coordinate ϕ with

$$\phi(P(w)) = \phi(w)^d, \quad \phi(w) = cw + O(1), \quad c^{d-1} = a_d. \quad (3.13)$$

Shrink U so that $|\phi| > 1$ there. Choose N with $P^N(z) \in U$. For $w \in U$, iteration of (3.13) gives

$$d^{-m} \log |\phi(P^m(w))| = \log |\phi(w)|. \quad (3.14)$$

Since $\log |\phi(\xi)| - \log |\xi| \rightarrow \log |c|$ at infinity, dividing that bounded asymptotic error by d^m shows

$$\lim_{m \rightarrow \infty} d^{-m} \log |P^m(w)| = \log |\phi(w)|. \quad (3.15)$$

Consequently, by (3.11),

$$G_P(z) = d^{-N} \log |\phi(P^N(z))| > 0. \quad (3.16)$$

Thus the zero set of G_P is exactly $K(P)$.

Formula (3.16) also proves harmonicity on the basin of infinity: every escaping point has a neighborhood whose sufficiently high iterate lies in U , and there G_P is a positive constant multiple of the harmonic function $\log |\phi \circ P^N|$. Finally (3.15) with $w = z \rightarrow \infty$ gives

$$G_P(z) = \log |z| + \frac{\log |a_d|}{d-1} + o(1), \quad (3.17)$$

which records the usual normalization at infinity. □

Proposition 3.5 (Connectedness criterion for polynomials). **Statement scope.** *For a polynomial P of degree at least two, the filled Julia set $K(P)$ is connected if and only if every finite critical point of P has bounded forward orbit, equivalently belongs to $K(P)$.*

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Connected K gives a global Böttcher conjugacy and hence no finite basin critical point. Conversely, pull back a forward-invariant exterior disk; Riemann-Hurwitz makes every pullback a disk, and K is the nested intersection of their connected complements.

The mathematical argument begins in the proof environment below.

Proof setup. Suppose first that $K(P)$ is connected. By the global Böttcher theorem there is a conformal isomorphism

$$\phi : A_\infty(P) \longrightarrow \widehat{\mathbb{C}} \setminus \overline{\mathbb{D}}$$

with

$$\phi(P(z)) = \phi(z)^d. \quad (3.18)$$

For a finite point $z \in A_\infty(P)$, differentiation gives

$$\phi'(P(z))P'(z) = d\phi(z)^{d-1}\phi'(z). \quad (3.19)$$

The conformal map has nonzero derivative and $|\phi(z)| > 1$, so the right-hand side is nonzero. Hence $P'(z) \neq 0$. Thus every finite critical point lies in $K(P)$ and therefore has bounded orbit.

Conversely suppose every finite critical point of P has bounded orbit, hence belongs to $K(P)$. Choose $R > 0$ so large that

$$|z| > R \implies |P(z)| > R. \quad (3.20)$$

Let

$$U_0 := \{z : |z| > R\} \cup \{\infty\}, \quad U_n := P^{-n}(U_0), \quad V_n := \widehat{\mathbb{C}} \setminus U_n. \quad (3.21)$$

Then $U_0 \subset U_1 \subset \dots$ by (3.20), so

$$V_0 \supset V_1 \supset \dots, \quad K(P) = \bigcap_{n \geq 0} V_n. \quad (3.22)$$

Lemma 3.6 (Exterior pullbacks are disks). *If all finite critical points of P have bounded orbit, then every successive pullback $U_n = P^{-n}(U_0)$ of a sufficiently small disk-neighborhood of infinity is a topological disk.*

Proof. Every U_n is a topological disk containing ∞ .

We prove this by induction. It is true for U_0 . Suppose U_n is a disk. Let W be the component of $P^{-1}(U_n)$ containing ∞ . Since the local degree of a degree- d polynomial at ∞ is d , the proper map

$$P : W \longrightarrow U_n$$

has degree d . Thus it already accounts for all d preimages of a generic point of U_n , and consequently $P^{-1}(U_n) = W$ is connected. No finite critical point lies in W , because $W \subset A_\infty(P)$ whereas every finite critical point belongs to $K(P)$. The only ramification

in W is therefore ∞ , with ramification index $d - 1$. Riemann–Hurwitz gives

$$\chi(W) = d\chi(U_n) - (d - 1) = d - (d - 1) = 1. \quad (3.23)$$

Hence $W = U_{n+1}$ is a topological disk. $\square$

Proof. It follows that each V_n , the complement of a disk containing infinity, is a compact connected topological disk. The family is nested and nonempty. Therefore its intersection in (3.22) is connected. Thus $K(P)$ is connected.

The forward implication follows from the global Böttcher coordinate. For the converse, Lemma 3.6 makes the complements V_n nested compact connected disks; their intersection is $K(P)$, hence is connected. $\square$

Theorem 3.7 (External rays and landing under local connectivity). **Statement scope.** *Let P have degree $d \geq 2$. If $J(P)$ is connected and locally connected, then the inverse Böttcher coordinate*

$$\psi = \phi^{-1} : \widehat{\mathbb{C}} \setminus \overline{\mathbb{D}} \longrightarrow A_\infty(P)$$

extends continuously to $|w| = 1$. Consequently every external ray lands, the boundary map $\gamma : \mathbb{R}/\mathbb{Z} \rightarrow J(P)$ is continuous and onto, and

$$P(\gamma(\theta)) = \gamma(d\theta \pmod{1}).$$

No injectivity of γ is asserted without a stronger boundary hypothesis.

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Global Böttcher + locally-connected-boundary Carathéodory extension for the inverse map; external-ray landing and boundary semiconjugacy.

The mathematical argument begins in the proof environment below.

Proof setup. Let P have degree $d \geq 2$ and suppose that $J(P)$ is connected and locally connected. Since $K(P)$ is obtained from $J(P)$ by adjoining the bounded complementary components of $J(P)$, each of whose boundary lies in $J(P)$, the set $K(P)$ is connected. Hence by the global Böttcher theorem there is a conformal isomorphism

$$\phi : A_\infty(P) \longrightarrow \widehat{\mathbb{C}} \setminus \overline{\mathbb{D}}, \quad \phi(P(z)) = \phi(z)^d. \quad (3.24)$$

Write

$$\psi := \phi^{-1} : \widehat{\mathbb{C}} \setminus \overline{\mathbb{D}} \longrightarrow A_\infty(P). \quad (3.25)$$

Lemma 3.8 (Carathéodory extension in the locally connected case). *A conformal map from the disk onto a simply connected proper domain with locally connected boundary extends continuously to the closed disk and maps the unit circle onto the boundary.*

Proof. Let $U \subset \widehat{\mathbb{C}}$ be a simply connected proper domain and let $\Psi : \mathbb{D} \rightarrow U$ be conformal. If ∂U is locally connected, then Ψ has a unique continuous extension

$$\overline{\Psi} : \overline{\mathbb{D}} \longrightarrow \overline{U},$$

and $\overline{\Psi}(\partial\mathbb{D}) = \partial U$.

Here is the boundary argument needed in the present application. Fix $\zeta \in \partial\mathbb{D}$. For arbitrarily small $r > 0$, the component D_r of $\mathbb{D} \cap D(\zeta, r)$ whose closure contains ζ can be cut off by a crosscut $C_r \subset \mathbb{D}$ with endpoints on $\partial\mathbb{D}$ tending to ζ . Conformal area and Cauchy–Schwarz on a family of circular crosscuts allow the C_r to be chosen so that the spherical length of $\Psi(C_r)$ tends to zero. Consequently the two endpoints of $\Psi(C_r)$ on ∂U approach one another.

Local connectivity of the compact set ∂U gives the following uniform form: for every $\varepsilon > 0$ there is $\delta > 0$ such that any two boundary points at spherical distance below δ lie in a connected subset of ∂U of diameter below ε . For r small, join the two endpoints of $\Psi(C_r)$ by such a boundary continuum L_r . The crosscut $\Psi(C_r)$ together with L_r separates from a fixed interior base point the image $\Psi(D_r)$, and its spherical diameter tends to zero. Therefore

$$\text{diam}_{\text{sph}} \Psi(D_r) \longrightarrow 0. \quad (3.26)$$

Thus $\Psi(z)$ has a unique limit as $z \rightarrow \zeta$ inside $\mathbb{D}$. Define that limit to be $\overline{\Psi}(\zeta)$. Estimate (3.26), uniformly on sufficiently short boundary arcs by compactness of $\partial\mathbb{D}$, proves continuity of the extension. Every boundary value lies in ∂U , while every point of ∂U is a cluster value of $\Psi(z_n)$ for some $z_n \rightarrow \partial\mathbb{D}$; compactness then gives $\overline{\Psi}(\partial\mathbb{D}) = \partial U$. Uniqueness follows from density of $\mathbb{D}$ in its closure. $\square$

Proof. Apply the lemma after inversion to the exterior map (3.25). Since

$$\partial A_\infty(P) = J(P)$$

is locally connected, ψ extends continuously to

$$\overline{\psi} : \{w : |w| \geq 1\} \cup \{\infty\} \longrightarrow A_\infty(P) \cup J(P), \quad \overline{\psi}(\partial\mathbb{D}) = J(P). \quad (3.27)$$

This is the direction of boundary extension that is valid without assuming that $J(P)$ is a Jordan curve; the forward map ϕ need not extend as a single-valued continuous inverse on $J(P)$ when several angles land together.

For $\theta \in \mathbb{R}/\mathbb{Z}$, define the external ray

$$\mathcal{R}_\theta := \{\psi(re^{2\pi i\theta}) : r > 1\}. \quad (3.28)$$

By continuity in (3.27),

$$\lim_{r \downarrow 1} \psi(re^{2\pi i\theta}) = \overline{\psi}(e^{2\pi i\theta}) \in J(P). \quad (3.29)$$

Thus every external ray lands. The boundary map

$$\gamma : \mathbb{R}/\mathbb{Z} \longrightarrow J(P), \quad \gamma(\theta) := \overline{\psi}(e^{2\pi i\theta}), \quad (3.30)$$

is continuous and onto.

Finally (3.24) is equivalent on $|w| > 1$ to

$$P(\psi(w)) = \psi(w^d). \quad (3.31)$$

Putting $w = re^{2\pi i\theta}$ and letting $r \downarrow 1$ yields

$$P(\gamma(\theta)) = \gamma(d\theta \pmod{1}). \quad (3.32)$$

Hence the Carathéodory boundary map semiconjugates angle d -tupling to $P|_{J(P)}$, as claimed.

Applying Lemma 3.8 to the inverse Böttcher map gives the continuous landing map and, by passing the functional equation to the boundary, the angle-doubling semiconjugacy claimed in the theorem. $\square$

Theorem 3.9 (Rational equilibrium measure and measure of maximal entropy). **Statement scope.** *Let $R : \widehat{\mathbb{C}} \rightarrow \widehat{\mathbb{C}}$ be rational of degree $d \geq 2$. There is a unique invariant probability measure μ_R of maximal entropy, and*

$$h_{\mu_R}(R) = h_{\text{top}}(R) = \log d.$$

It is balanced, charges no exceptional point, and for every nonexceptional a ,

$$\frac{1}{d^n} \sum_{R^n(z)=a} \deg_z(R^n) \delta_z \xrightarrow{*} \mu_R.$$

Proof strategy (not part of the proof)

Proof status. proved here modulo stated foundation input.

Strategic outline only. Dimension-one inverse-branch contraction and target comparison; balanced natural measure; constant Jacobian and entropy $\log d$; orbit-graph entropy upper bound; Briend-Duval/Lyubich branch-imbalance uniqueness. *The mathematical argument begins in the proof environment below.*

Proof setup. Let

$$R : \widehat{\mathbb{C}} \longrightarrow \widehat{\mathbb{C}}, \quad \deg R = d \geq 2.$$

For $a \in \widehat{\mathbb{C}}$ define the normalized n -th preimage measure

$$\mu_{n,a} := \frac{1}{d^n} \sum_{R^n(z)=a} \deg_z(R^n) \delta_z. \quad (3.33)$$

Equivalently, if

$$(\mathcal{L}_R f)(a) := \frac{1}{d} \sum_{R(z)=a} \deg_z(R) f(z), \quad (3.34)$$

then $\int f d\mu_{n,a} = \mathcal{L}_R^n f(a)$. The symmetric sum in (3.34) is continuous when f is continuous, so the dual normalized pullback

$$T^* \nu := \frac{1}{d} R^* \nu \quad (3.35)$$

is weak-* continuous on probability measures and $(T^*)^n \delta_a = \mu_{n,a}$.

Lemma 3.10 (Many contracting inverse branches). *Let C be the finite critical set of R and put*

$$V_\ell := \bigcup_{j=1}^{\ell} R^j(C).$$

For every $\varepsilon > 0$ one can choose ℓ with the following property. If

$$\Delta \Subset \tilde{\Delta} \Subset \widehat{\mathbb{C}} \setminus V_\ell$$

are simply connected Jordan domains, then for all sufficiently large n there are at least $(1 - \varepsilon)d^n$ inverse branches g of R^n on $\tilde{\Delta}$ for which

$$\text{diam}_{\text{sph}} g(\Delta) \leq C_{\Delta, \tilde{\Delta}, \varepsilon} d^{-n/2}. \quad (3.36)$$

Proof. To prove this, first note that all d^ℓ inverse branches of R^ℓ exist on $\tilde{\Delta}$. If b_m branches of R^m have survived, their image domains are pairwise disjoint. At most $v := |R(C)| \leq 2d - 2$ of these image domains can contain a critical value of R ; every other image domain admits all d inverse branches of one further iterate. Thus

$$b_{m+1} \geq d(b_m - v), \quad m \geq \ell,$$

and hence

$$\frac{b_n}{d^n} \geq 1 - v \sum_{m=\ell}^{n-1} d^{-m}. \quad (3.37)$$

Choose ℓ so that the tail on the right is below $\varepsilon/2$.

For the surviving branches, the domains $g(\tilde{\Delta})$ are disjoint, so the sum of their spherical areas is at most the area of the sphere. All but at most $(\varepsilon/2)d^n$ therefore have area $O_\varepsilon(d^{-n})$. We use the elementary conformal area-diameter estimate

$$\text{diam}_{\text{sph}} h(\Delta)^2 \leq C(\Delta, \tilde{\Delta}) \text{Area}_{\text{sph}} h(\tilde{\Delta}) \quad (3.38)$$

for every univalent h on $\tilde{\Delta}$. Indeed, after conformally identifying $\tilde{\Delta}$ with the disk, Δ lies in a fixed smaller disk; Koebe distortion bounds the derivative on that smaller disk by a fixed multiple of its L^2 norm on the whole disk, while the latter is precisely Euclidean area with multiplicity. Passing between Euclidean and spherical charts on sets of small area changes only the fixed constant. Combining (3.37)–(3.38) proves (3.36). $\square$

Lemma 3.11 (Fixed postcritical truncations have vanishing preimage mass). *If a is*

nonexceptional and $F \subset \widehat{\mathbb{C}}$ is finite, then

$$\mu_{n,a}(F) \longrightarrow 0. \quad (3.39)$$

Proof. It is enough to fix $z \in F$. Unless $R^n(z) = a$, its contribution is zero. If this equality holds for infinitely many n , the forward orbit of z eventually enters a periodic orbit containing a . Along a period- q cycle the multiplicity grows by the product of the q local degrees. If that product were d^q , every point of the cycle would be totally ramified, so the cycle would be finite and completely invariant; its points would be exceptional by the exceptional-set theorem. Since a is nonexceptional, the product is strictly smaller than d^q . Hence

$$d^{-n} \deg_z(R^n) \longrightarrow 0,$$

and summing over the finite set F proves (3.39). $\square$

Lemma 3.12 (Preimage distributions forget the nonexceptional target). *For nonexceptional a, b ,*

$$\mu_{n,a} - \mu_{n,b} \xrightarrow{*} 0. \quad (3.40)$$

Proof. Fix $f \in C(\widehat{\mathbb{C}})$ and $\eta > 0$. Choose $\varepsilon > 0$ so small that the total contribution of 3ε bad mass to integrals of f is below $\eta/3$, and then choose ℓ from Lemma 3.10. If $x, y \notin V_\ell$, join them by an arc avoiding the finite set V_ℓ and choose

$$\Delta \Subset \widetilde{\Delta} \Subset \widehat{\mathbb{C}} \setminus V_\ell$$

containing the arc. Match the good inverse branches from Lemma 3.10. On every matched branch, (3.36) and uniform continuity of f give

$$|f(g(x)) - f(g(y))| < \eta/3$$

for n large. The unmatched mass is at most ε , so

$$\left| \int f d\mu_{n,x} - \int f d\mu_{n,y} \right| < \eta \quad (3.41)$$

uniformly for x, y in any fixed finite subset of $\widehat{\mathbb{C}} \setminus V_\ell$.

For arbitrary nonexceptional a, b , use the semigroup identity

$$\mu_{n+m,a} = \int \mu_{n,x} d\mu_{m,a}(x). \quad (3.42)$$

By Lemma 3.11, choose m so that both $\mu_{m,a}(V_\ell)$ and $\mu_{m,b}(V_\ell)$ are below ε . The remaining supports are finite and lie outside V_ℓ ; compare every term there with a fixed base point outside V_ℓ by (3.41). Then let $n \rightarrow \infty$ and finally $\varepsilon \rightarrow 0$. This proves (3.40). $\square$

Lemma 3.13 (Balanced equilibrium measure and preimage convergence). *There is a probability measure μ_R such that*

$$T^* \mu_R = \mu_R, \quad \mu_R(E(R)) = 0, \quad (3.43)$$

and it is the unique probability with these two properties. Moreover

$$\mu_{n,a} \xrightarrow{*} \mu_R \text{ for every } a \notin E(R). \quad (3.44)$$

Proof. Choose a small trapping neighborhood U of the exceptional set with $R(U) \subseteq U$; when $E(R) = \emptyset$ take $U = \emptyset$. Such a neighborhood exists from the explicit one- and two-point exceptional models in the exceptional-set theorem. Choose $a \notin U$ and form the Cesàro averages

$$\nu_N := \frac{1}{N} \sum_{n=0}^{N-1} \mu_{n,a}. \quad (3.45)$$

By compactness of probability measures on the sphere, a subsequence converges to some μ_R . Since

$$T^* \nu_N - \nu_N = \frac{1}{N} (\mu_{N,a} - \delta_a) \xrightarrow{*} 0,$$

weak-* continuity of T^* gives (3.43). Also $\mu_{n,a}(U) = 0$ for every n , because $R(U) \subset U$ while $a \notin U$; hence $\mu_R(E(R)) = 0$.

Suppose α and β are two balanced probabilities charging no exceptional point. Then for $f \in C(\hat{\mathbb{C}})$,

$$\int f d\alpha - \int f d\beta = \iint \left(\int f d\mu_{n,x} - \int f d\mu_{n,y} \right) d\alpha(x) d\beta(y). \quad (3.46)$$

The integrand tends to zero by Lemma 3.12 for $\alpha \times \beta$ -almost every pair and is uniformly bounded; dominated convergence gives $\alpha = \beta$. Finally, for nonexceptional a ,

$$\int f d\mu_R - \int f d\mu_{n,a} = \int \left(\int f d\mu_{n,y} - \int f d\mu_{n,a} \right) d\mu_R(y), \quad (3.47)$$

which tends to zero by Lemma 3.12 and dominated convergence. This proves (3.44).

The balance equation also implies invariance, because

$$R_* R^* \nu = d\nu$$

for every measure ν , counting local degree. Thus $R_* \mu_R = \mu_R$. $\square$

Lemma 3.14 (Non-atomicity and constant Jacobian). *The measure μ_R is non-atomic. Moreover, whenever A is Borel and R is injective on A ,*

$$\mu_R(R(A)) = d \mu_R(A). \quad (3.48)$$

Proof. If $\mu_R\{x\} > 0$, balance on the singleton $\{x\}$ gives

$$\deg_x R \mu_R\{R(x)\} = d \mu_R\{x\}. \quad (3.49)$$

Since $\deg_x R \leq d$, equation (3.49) shows that atomic mass is nondecreasing along the forward orbit:

$$\mu_R\{R(x)\} \geq \mu_R\{x\} > 0.$$

A probability measure has only finitely many atoms of mass at least $\mu_R\{x\}$, so the

forward orbit of x is eventually periodic. On a periodic cycle of period q , multiplying (3.49) around the cycle gives

$$\prod_{j=0}^{q-1} \deg_{R^j(x)} R = d^q.$$

Every factor is at most d , hence every factor equals d . Each point of the cycle is therefore the unique preimage (with multiplicity d) of the next point; the cycle is finite and completely invariant and hence exceptional. This contradicts $\mu_R(E(R)) = 0$. Thus μ_R has no atoms, and in particular the finite set of critical values has zero measure.

For injective A , the definition of pullback and the critical-value null set give

$$d\mu_R(A) = R^*\mu_R(A) = \int \sum_{\substack{z \in A \\ R(z)=y}} \deg_z R d\mu_R(y) = \mu_R(R(A)),$$

which is (3.48). $\square$

Lemma 3.15 (Orbit-graph area and topological entropy). *The topological entropy satisfies*

$$h_{\text{top}}(R) \leq \log d. \quad (3.50)$$

Proof. Let ω be the spherical area form normalized by $\int \omega = 1$, and let

$$\Gamma_n := \{(z, Rz, \dots, R^{n-1}z) : z \in \widehat{\mathbb{C}}\} \subset (\widehat{\mathbb{C}})^n.$$

For the product Kähler form $\Omega_n = \sum_{j=0}^{n-1} \pi_j^* \omega$,

$$\text{Area}_{\Omega_n}(\Gamma_n) = \sum_{j=0}^{n-1} \int (R^j)^* \omega = \sum_{j=0}^{n-1} d^j \leq \frac{d^n}{d-1}. \quad (3.51)$$

If E is (n, ε) -separated for the Bowen metric, the corresponding points of Γ_n are ε -separated in the product metric. The monotonicity estimate for a complex curve in a product of round spheres gives a constant $c_\varepsilon > 0$, independent of n , such that each disjoint product ball of radius $\varepsilon/3$ centered on Γ_n contains at least c_ε of Ω_n -area of the curve. This estimate follows in a normal coordinate ball by projecting the complex tangent line to a coordinate factor on which its squared projection has size at least $1/n$ and summing the n coordinate contributions in Ω_n ; the resulting lower bound is dimension independent. Thus

$$c_\varepsilon |E| \leq \text{Area}_{\Omega_n}(\Gamma_n) \leq \frac{d^n}{d-1}.$$

Taking $n^{-1} \log$, then the supremum over separated sets and finally $\varepsilon \downarrow 0$, proves (3.50). $\square$

Lemma 3.16 (The balanced measure has entropy $\log d$). *The balanced measure satisfies*

$$h_{\mu_R}(R) = \log d. \quad (3.52)$$

Proof. Triangulate the sphere so that every critical value lies in the one-skeleton S and so that $\mu_R(S) = 0$; the latter is achieved by a small perturbation of the finitely many edges because μ_R is non-atomic. Over every open triangle, R has exactly d conformal inverse branches. Label them measurably and let $U_1, \dots, U_d$ be the unions of the correspondingly labelled inverse components.

By (3.48), images of μ_R -null injective arcs are null. Consequently the full forward and backward orbit of S is null. Remove it and call the remaining completely invariant full-measure set X . On X , each $R : U_j \cap X \rightarrow X$ is a measurable bijection and (3.48) gives

$$\mu_R(U_j \cap X) = \frac{1}{d}. \quad (3.53)$$

For the partition $\mathcal{P} = \{U_1 \cap X, \dots, U_d \cap X\}$, every nonempty n -cylinder is mapped bijectively by R^n onto X . Iterating (3.48), each such cylinder has measure d^{-n} ; in fact all d^n words occur. Hence

$$H_{\mu_R} \left(\bigvee_{k=0}^{n-1} R^{-k} \mathcal{P} \right) = n \log d,$$

so $h_{\mu_R}(R) \geq \log d$. By the Foundation variational principle at zero potential and Lemma 3.15,

$$h_{\mu_R}(R) \leq h_{\text{top}}(R) \leq \log d,$$

proving (3.52) and at the same time

$$h_{\text{top}}(R) = \log d. \quad (3.54)$$

□

Lemma 3.17 (Ergodic reduction at maximal entropy). *Let ν be invariant. Under its ergodic decomposition*

$$\nu = \int \nu_\xi d\tau(\xi),$$

metric entropy is affine:

$$h_\nu(R) = \int h_{\nu_\xi}(R) d\tau(\xi). \quad (3.55)$$

Proof. For completeness, for every finite measurable partition $\mathcal{Q}$, conditioning its n -block entropy on the invariant sigma-algebra gives

$$H_\nu \left(\bigvee_{j=0}^{n-1} R^{-j} \mathcal{Q} \middle| \mathcal{I} \right) = \int H_{\nu_\xi} \left(\bigvee_{j=0}^{n-1} R^{-j} \mathcal{Q} \right) d\tau(\xi).$$

The chain rule shows that conditioning on the invariant sigma-algebra changes block entropy only by a sublinear term. Divide by n , pass to the subadditive limit, and then take the supremum over a countable family of finite partitions generated by a countable basis of the sphere. Monotone approximation gives (3.55).

Thus if $h_\nu(R) = \log d$, then (3.50) implies $h_{\nu_\xi}(R) = \log d$ for τ -almost every ergodic component. It is therefore enough to prove uniqueness among ergodic maximal-entropy measures. □

Lemma 3.18 (A positive-measure set carries the entropy of an ergodic measure). *Let ν be ergodic and let Y be Borel with $\nu(Y) > 0$. If $h_{\text{top}}(R, Y)$ denotes Bowen separated-set entropy restricted to Y , then*

$$h_{\text{top}}(R, Y) \geq h_\nu(R). \quad (3.56)$$

Proof. We give the localization argument explicitly, because an average block-entropy bound by itself does not imply (3.56).

Fix $\eta > 0$. Choose a finite measurable partition $\mathcal{Q} = \{Q_1, \dots, Q_m\}$ with ν -null boundaries and

$$h_\nu(R, \mathcal{Q}) > h_\nu(R) - \eta. \quad (3.56a)$$

Since ν is ergodic and $\nu(Y) > 0$, the invariant union $\bigcup_{j \geq 0} R^{-j}Y$ has full measure. Hence, for every small $\delta > 0$, there is a fixed integer r such that

$$Z := \bigcup_{j=0}^{r-1} R^{-j}Y, \quad \nu(Z) > 1 - \delta. \quad (3.56b)$$

Choose compact cores $K_i \subset Q_i$ such that, with $B := \widehat{\mathbb{C}} \setminus \bigcup_i K_i$, one has $\nu(B) < \delta^2$. Put

$$G_n := \left\{ x : \sum_{j=0}^{n-1} \mathbf{1}_B(R^j x) \leq \delta n \right\}.$$

By invariance and Markov's inequality, $\nu(G_n) \geq 1 - \delta$ for every n . Therefore $Z_n := Z \cap G_n$ has mass $a_n := \nu(Z_n) \geq 1 - 2\delta$. Let $\nu_n := a_n^{-1}\nu|_{Z_n}$. If $a_n < 1$, write $\nu = a_n\nu_n + (1 - a_n)\xi_n$. The entropy-of-a-mixture inequality gives

$$H_\nu(\mathcal{Q}^n) \leq \log 2 + a_n H_{\nu_n}(\mathcal{Q}^n) + (1 - a_n)n \log m.$$

Consequently

$$\liminf_{n \rightarrow \infty} \frac{1}{n} H_{\nu_n}(\mathcal{Q}^n) \geq h_\nu(R, \mathcal{Q}) - \varepsilon_m(\delta), \quad (3.56c)$$

where $\varepsilon_m(\delta) \rightarrow 0$ as $\delta \downarrow 0$.

If at least two compact cores are nonempty, let $\rho > 0$ be their minimum mutual distance; otherwise choose any $\rho > 0$. Fix $0 < \varepsilon < \rho$. Let $E_n \subset Z_n$ be a maximal (n, ε) -separated set. It spans Z_n in the Bowen metric. If $x \in Z_n$ is assigned to $y \in E_n$ with $d_n(x, y) < \varepsilon$, then the $\mathcal{Q}$ -names of x and y can differ only at times when at least one orbit visits B , hence in at most $2\delta n$ positions. Thus one Bowen ball around a point of E_n meets at most

$$B_{m,\delta}(n) := \sum_{0 \leq j \leq \lfloor 2\delta n \rfloor} \binom{n}{j} (m-1)^j$$

positive- ν_n atoms of $\mathcal{Q}^n$ (with $B_{1,\delta}(n) = 1$). Since the entropy of a probability on N atoms is at most $\log N$,

$$H_{\nu_n}(\mathcal{Q}^n) \leq \log |E_n| + \log B_{m,\delta}(n). \quad (3.56d)$$

The standard binomial estimate gives

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log B_{m,\delta}(n) \leq H_2(2\delta) + 2\delta \log(m-1) =: b_m(\delta),$$

with the evident value 0 for $m = 1$, and $b_m(\delta) \rightarrow 0$ as $\delta \downarrow 0$. From (3.56a), (3.56c), and (3.56d), choosing δ sufficiently small yields

$$h_{\text{top}}(R, Z) \geq h_\nu(R) - 2\eta. \quad (3.56e)$$

This is the compact-core/Hamming conversion behind the Foundation variational principle, now applied to the high-measure set Z ; the only dynamical input needed to make Z high-measure is ergodicity and Birkhoff's theorem.

It remains to return from Z to Y . For each fixed $0 \leq j < r$, compactness gives, for every $\varepsilon > 0$, a finite number $N_{j,\varepsilon}$ of d_j -balls of radius $\varepsilon/2$ covering the sphere. Partition any (n, ε) -separated subset of $R^{-j}Y$ according to this finite cover. Inside one class no separation occurs during the first j iterates, so after applying R^j its image is $(n-j, \varepsilon)$ -separated in Y . Hence fixed finite prefixes do not change exponential separated-set growth, and

$$h_{\text{top}}(R, R^{-j}Y) \leq h_{\text{top}}(R, Y).$$

A finite union has separated-set entropy equal to the maximum of the component rates, so (3.56b) gives

$$h_{\text{top}}(R, Z) \leq h_{\text{top}}(R, Y). \quad (3.56f)$$

Letting $\eta \downarrow 0$ in (3.56e)–(3.56f) proves (3.56). $\square$

Lemma 3.19 (Branch imbalance contradicts maximal entropy). *Let ν be an ergodic invariant probability with $h_\nu(R) = \log d$. Then $\nu = \mu_R$.*

Proof. First ν is non-atomic: an ergodic invariant measure with an atom is supported on a finite periodic orbit and has entropy zero. Hence $\nu(E(R)) = 0$. If ν were balanced, Lemma 3.13 would give $\nu = \mu_R$. Assume therefore that ν is not balanced.

Choose a triangulation with critical values in its one-skeleton and with skeleton null for both μ_R and ν . Over each open triangle label the d inverse components so that the component of largest ν -mass is labelled 1. Refining the triangulation and approximating a Borel injectivity set witnessing failure of the balance identity shows that the global first branch U_1 satisfies

$$\nu(U_1) > \frac{1}{d}. \quad (3.57)$$

(If every sufficiently fine inverse-branch partition gave exactly $1/d$ to each branch, monotone approximation would give $R^*\nu = d\nu$ on every injectivity set and hence balance.) Choose

$$\frac{1}{d} < \sigma < \nu(U_1)$$

and an open $O \in U_1$ with $\nu(O) > \sigma$.

By Birkhoff's theorem there is m such that the set

$$Y := \{z : \#\{0 \leq j < n : R^j z \in O\} \geq \sigma n \text{ for every } n \geq m\} \quad (3.58)$$

has positive ν -measure. Lemma 3.18 therefore gives

$$h_{\text{top}}(R, Y) \geq \log d. \quad (3.59)$$

We now derive the opposite strict inequality.

Remove from Y the null set

$$\bigcup_{j \geq 0} R^{-j}(S),$$

where S is the triangulation skeleton. Invariance of ν and $\nu(S) = 0$ show that the remaining set, still denoted Y , has positive measure and every one of its forward iterates has a well-defined inverse-branch label. Hence every length- n itinerary of a point of Y uses symbol 1 at least σn times. Let N_n be the number of words in $\{1, \dots, d\}^n$ with that property. For every $t > 1$,

$$N_n t^{\sigma n} \leq (t + d - 1)^n,$$

so

$$N_n \leq ((t + d - 1)t^{-\sigma})^n. \quad (3.60)$$

Because $\sigma > 1/d$, the derivative at $t = 1$ of $\log(t + d - 1) - \sigma \log t$ is $1/d - \sigma < 0$. Thus for some $t > 1$ there is $\rho < 1$ with

$$(t + d - 1)t^{-\sigma} = d\rho. \quad (3.61)$$

For a fixed itinerary α , the corresponding part $\Gamma_n(\alpha)$ of the orbit graph has area at most n : on each branch component R^j is injective and therefore

$$\int_{\Gamma_n(\alpha)} \Omega_n = \sum_{j=0}^{n-1} \int_{A_\alpha} (R^j)^* \omega \leq n. \quad (3.62)$$

The same local monotonicity argument used in Lemma 3.15 converts area into a bound for separated points of Y inside each itinerary piece. Equations (3.60)–(3.62) yield

$$h_{\text{top}}(R, Y) \leq \limsup_{n \rightarrow \infty} \frac{1}{n} \log(n(d\rho)^n) = \log(d\rho) < \log d, \quad (3.63)$$

contradicting (3.59). Hence ν is balanced and therefore equals μ_R . $\square$

Proof of Theorem 3.9. Lemma 3.13 constructs the unique balanced probability μ_R charging no exceptional point, proves its invariance, and gives the stated preimage convergence for every nonexceptional target. By Lemma 3.16,

$$h_{\mu_R}(R) = h_{\text{top}}(R) = \log d.$$

Hence μ_R is a measure of maximal entropy.

Let ν be any invariant probability with $h_\nu(R) = \log d$. By Lemma 3.17, almost every

ergodic component of ν has entropy $\log d$. Lemma 3.19 identifies each such component with μ_R . Therefore $\nu = \mu_R$, proving uniqueness and completing all assertions of the theorem. $\square$

Theorem 3.20 (Brolin equidistribution theorem). **Statement scope.** Let P be a polynomial of degree $d \geq 2$, let

$$\mu_P = \frac{1}{2\pi} \Delta G_P$$

be its equilibrium/Brolin measure, and let $a \notin E(P)$. Then

$$\frac{1}{d^n} \sum_{P^n(z)=a} \deg_z(P^n) \delta_z \xrightarrow{*} \mu_P.$$

Moreover μ_P is the logarithmic equilibrium measure of $K(P)$.

Proof strategy (not part of the proof)

Proof status. derived here.

Strategic outline only. Identify $(2\pi)^{-1} \Delta G_P$ with the logarithmic equilibrium measure, prove balance from $G_P \circ P = dG_P$, then specialize rational preimage convergence.

The mathematical argument begins in the proof environment below.

Proof setup. Let P be a polynomial of degree $d \geq 2$ and let G_P be the escape-rate function from the polynomial Green-function theorem.

Lemma 3.21 (Green Riesz measure is the balanced equilibrium probability). *The measure $\nu_P = (2\pi)^{-1} \Delta G_P$ is a probability supported on $J(P)$, equals the logarithmic equilibrium measure of $K(P)$, and satisfies $P^* \nu_P = d\nu_P$.*

Proof. The uniform limit construction in the polynomial Green-function theorem writes G_P as a locally uniform limit of the subharmonic functions

$$d^{-n} \log^+ |P^n(z)|.$$

Hence G_P is subharmonic. By the Riesz-measure theorem its book-normalized Riesz measure

$$\nu_P := \frac{1}{2\pi} \Delta G_P \tag{3.64}$$

is a positive Radon measure. Since G_P is harmonic on $A_\infty(P)$ and identically zero on the interior of $K(P)$, the measure is supported on

$$\partial K(P) = J(P). \tag{3.65}$$

The infinity asymptotic from the polynomial Green-function theorem is

$$G_P(z) = \log |z| + \frac{\log |a_d|}{d-1} + o(1). \tag{3.66}$$

For a subharmonic function with logarithmic growth $m \log |z| + O(1)$, the total mass of $(2\pi)^{-1} \Delta u$ is m : apply Green's identity on a large circle avoiding the negligible set of radii at which the radial derivative is inconvenient, or equivalently use the Riesz decomposition from the Riesz-decomposition theorem and compare the coefficient of $\log |z|$ at infinity. Thus

$$\nu_P(\mathbb{C}) = 1. \quad (3.67)$$

The compact set $K(P)$ is full, and G_P is positive harmonic on its complement, continuous on the whole plane with boundary value zero, and has the logarithmic asymptotic (3.66). Therefore $K(P)$ is regular for the Green function at infinity and, by the uniqueness clause of the Green-function theorem, G_P is the potential-theoretic Green function of $K(P)$. If $\mu_{K(P)}$ denotes the unique logarithmic equilibrium probability from the equilibrium-measure theorem, then the representation in the Green-function theorem gives distributionally

$$\Delta G_P = 2\pi \mu_{K(P)}. \quad (3.68)$$

Consequently

$$\nu_P = \mu_{K(P)}. \quad (3.69)$$

This proves the equivalence of the two descriptions in the definition of the polynomial equilibrium measure.

We next prove balance. For a holomorphic map F and a locally integrable subharmonic u , distributional Laplacians obey

$$\Delta(u \circ F) = F^*(\Delta u), \quad (3.70)$$

where the pullback counts local degree. To verify (3.70), first convolve u in local charts with smooth approximate identities, use the classical chain rule and change of variables on the complement of the finite critical set, and pass to the limit in distributions; local models $F(z) = z^m$ show that the limiting multiplicity is exactly the local degree.

Using $G_P \circ P = dG_P$ from the polynomial Green-function theorem and (3.70),

$$P^* \nu_P = \frac{1}{2\pi} P^*(\Delta G_P) = \frac{1}{2\pi} \Delta(G_P \circ P) = d\nu_P. \quad (3.71)$$

Thus ν_P is balanced. The equilibrium measure has finite logarithmic energy by the equilibrium-measure theorem, so it has no atoms; in particular it charges no exceptional point. This completes the lemma. $\square$

Corollary 3.22 (Brolin convergence by rational specialization). *For every nonexceptional target a , the normalized degree-counted preimage measures converge weak-* to the polynomial equilibrium measure.*

Proof. Apply the rational equilibrium/maximal-entropy theorem to the rational map P . That theorem constructs the unique balanced probability which charges no exceptional point and proves that for every $a \notin E(P)$,

$$\frac{1}{d^n} \sum_{P^n(z)=a} \deg_z(P^n) \delta_z \xrightarrow{*} \mu_P^{\text{dyn}}. \quad (3.72)$$

By the Green–Riesz identification lemma, $\nu_P = (2\pi)^{-1}\Delta G_P$ is itself a balanced probability charging no exceptional point. Uniqueness in the rational equilibrium/maximal-entropy theorem therefore gives

$$\mu_P^{\text{dyn}} = \nu_P = \mu_{K(P)}. \quad (3.73)$$

By the definition of the polynomial equilibrium measure this common measure is precisely the equilibrium/Brolin measure μ_P . Substituting (3.73) into (3.72) proves the Brolin equidistribution theorem. $\square$

Proof. The first result identifies the Green, equilibrium, and balanced measures; the corollary supplies Brolin equidistribution, proving the theorem. $\square$

3.3 Worked examples

Worked example: The monomial model $P(z) = z^2$

Here

$$K(P) = \overline{\mathbb{D}}, \quad J(P) = \partial\mathbb{D}, \quad A_\infty(P) = \{|z| > 1\} \cup \{\infty\}.$$

The normalized global Böttcher coordinate is simply $\phi(z) = z$, so

$$G_P(z) = \log^+ |z|.$$

Distributionally,

$$\mu_P = \frac{1}{2\pi} \Delta \log^+ |z|$$

is normalized angular measure on the unit circle. Thus for every continuous f ,

$$\int f d\mu_P = \frac{1}{2\pi} \int_0^{2\pi} f(e^{it}) dt.$$

Take the nonexceptional target $a = 1$. The n -th preimages are the 2^n -th roots of unity, all simple, hence

$$\mu_{n,1} = 2^{-n} \sum_{k=0}^{2^n-1} \delta_{e^{2\pi i k/2^n}}.$$

These are Riemann sums for angular measure, so the Brolin convergence is visible directly in this model. The rational maximal-entropy theorem says the same measure is the unique invariant probability with entropy $\log 2$.

Worked example: The Chebyshev model $P(z) = z^2 - 2$

The change of variables

$$z = w + w^{-1}$$

conjugates squaring outside the unit disk to P because

$$P(w + w^{-1}) = (w + w^{-1})^2 - 2 = w^2 + w^{-2}.$$

Solving for the exterior branch gives the global Böttcher coordinate

$$\phi(z) = \frac{z + \sqrt{z^2 - 4}}{2}, \quad \phi(z) \sim z \quad (z \rightarrow \infty),$$

with the square root chosen so that $\sqrt{z^2 - 4} \sim z$. Hence

$$K(P) = J(P) = [-2, 2], \quad G_P(z) = \log \left| \frac{z + \sqrt{z^2 - 4}}{2} \right|$$

outside the interval and $G_P = 0$ on it.

On the unit circle put $w = e^{2\pi i \theta}$. The inverse Böttcher boundary map is

$$\gamma(\theta) = w + w^{-1} = 2 \cos(2\pi \theta).$$

It is continuous and onto but not injective: angles θ and $1 - \theta$ land together. The semiconjugacy is the elementary identity

$$P(2 \cos t) = 2 \cos(2t).$$

The equilibrium measure is therefore the pushforward of angular measure,

$$d\mu_P(x) = \frac{dx}{\pi \sqrt{4 - x^2}}, \quad -2 < x < 2.$$

This example is the simplest concrete reason the external-ray theorem must speak about continuous extension of the inverse Böttcher map rather than a boundary homeomorphism.

Boundary case: What fails for $P(z) = z^2 - 3$

The unique finite critical point is 0, and

$$0 \mapsto -3 \mapsto 6 \mapsto 33 \mapsto \dots,$$

so the critical orbit escapes. By the connectedness criterion, $K(P)$ is disconnected. Consequently the basin of infinity is not conformally the exterior disk in the way required by the global Böttcher theorem, and the external-ray landing theorem for locally connected Julia sets cannot be invoked.

What survives is important. The local Böttcher germ near infinity still exists, the escape-rate Green function is still defined and continuous on the plane, the rational natural measure still exists, and the Brolin theorem still gives preimage equidistribution. Thus disconnectedness destroys the global uniformizing coordinate, not the potential-theoretic or measure-theoretic constructions.

Common pitfall

Do not omit local multiplicities in a general preimage measure. A critical preimage represents several sheets of the degree- d^n covering. The normalization $d^{-n} \sum \deg_z(P^n) \delta_z$ keeps total mass exactly one for every target and every n .

3.4 Chapter summary**What to remember**

- If $K(P)$ is connected, the basin of infinity is simply connected and the local Böttcher coordinate globalizes to a conformal isomorphism with the exterior disk.
- For every polynomial, connected or not, the normalized escape rates converge uniformly to a continuous Green function satisfying $G_P \circ P = dG_P$, with zero set exactly $K(P)$.
- The filled Julia set is connected exactly when all finite critical points have bounded forward orbit.
- Under local connectivity, the inverse Böttcher coordinate extends continuously to the unit circle; every external ray lands and the boundary map semi-conjugates angle multiplication by d to polynomial dynamics.
- Every rational map of degree $d \geq 2$ has a balanced natural probability obtained as the common asymptotic distribution of nonexceptional preimages. It has entropy $\log d$ and is the unique measure of maximal entropy.
- For polynomials, this natural measure equals $(2\pi)^{-1} \Delta G_P$, equivalently the logarithmic equilibrium measure of $K(P)$; this is Brodin's theorem.

Looking ahead

Chapter 4 moves from the dynamical plane to parameter space. For the quadratic family $P_c(z) = z^2 + c$, the connectedness criterion turns the Mandelbrot set into the set of parameters for which the critical orbit of 0 is bounded. The Böttcher and Green-function machinery from Chapter 3 then reappears in parameter space, where it supplies the exterior coordinate and potential of the Mandelbrot set.

Parameter Spaces and the Mandelbrot Set: Foundations

Why this chapter matters

Chapter 3 studied one polynomial at a time. Chapter 4 asks what happens when the polynomial itself moves through the quadratic family

$$f_c(z) = z^2 + c.$$

The decisive observation is that the dynamical Green function can be evaluated at the critical value. Outside the Mandelbrot set this produces a holomorphic exterior coordinate in the *parameter* plane. Once that coordinate is constructed correctly, connectedness of the Mandelbrot set becomes a consequence rather than an assumption.

The same bridge between the parameter plane and the dynamical plane sends a parameter ray of angle θ to the statement that the critical value lies on the dynamic ray of angle θ . Rational dynamic-ray rigidity can therefore be transported to parameter space. Periodic angles land at parabolic parameters; strictly preperiodic angles land at Misiurewicz parameters. These landing pairs organize the boundaries of hyperbolic components and ultimately force the attracting-cycle multiplier to uniformize each component by the unit disk.

Wakes and limbs are introduced at the end of the chapter because they are already visible from parameter-ray landing. The actual tuning theorem is deliberately postponed to Chapter 8, where polynomial-like maps and straightening have been defined. This separation prevents a parameter-space chapter from silently using later quasiconformal machinery.

Prerequisites. Chapter 1 multiplier classification; Chapter 2 periodic Fatou-component theory; Chapter 3 polynomial Green functions, the critical-orbit connectedness criterion, and external rays; elementary covering-space and planar topology.

Learning goals

- Construct the parameter Böttcher map $\Phi_M(c) = \phi_c(c)$ without assuming the Mandelbrot set is already connected.
- Deduce compactness and connectedness of M from the exterior uniformization.
- Use the critical-value/dynamic-ray bridge to prove landing of every rational parameter ray.

- Distinguish periodic angles, which land at parabolic parameters, from strictly preperiodic angles, which land at Misiurewicz parameters.
- Prove that the attracting-cycle multiplier gives a conformal coordinate on every hyperbolic component.
- Read characteristic parameter-ray pairs geometrically as wakes and limbs, while keeping the later tuning theorem outside the logical scope of Chapter 4.

Conceptual roadmap

The parameter-uniformization chain is

$$\text{escape rate } G_c \longrightarrow \Phi_M(c) = \phi_c(c) \longrightarrow \mathbb{C} \setminus M \cong \mathbb{C} \setminus \overline{\mathbb{D}} \longrightarrow M \text{ connected.}$$

The combinatorial chain is

$$\begin{aligned} \text{parameter ray} &\longrightarrow \text{critical value on the same dynamic ray} \\ &\longrightarrow \text{rational landing} \longrightarrow \text{characteristic ray pairs / wakes.} \end{aligned}$$

Finally,

$$\text{rational landing} + \text{fullness of } M \longrightarrow \text{unique hyperbolic root} \longrightarrow \lambda_H : H \cong \mathbb{D}.$$

Tuning is not part of this dependency chain: Chapter 4 supplies its parameter combinatorics, while Chapter 8 supplies the polynomial-like straightening mechanism.

How the main results fit together. The order of the theorems matters. The parameter Böttcher theorem is proved before Mandelbrot connectedness; using connectedness to globalize the parameter coordinate would be circular. Likewise, rational parameter-ray landing is established before the multiplier-map theorem and is one of its proof engines.

4.1 Definitions

Definition 4.1 (Mandelbrot set). For the quadratic family

$$f_c(z) = z^2 + c,$$

the *Mandelbrot set* is

$$M := \{c \in \mathbb{C} : \{f_c^n(0)\}_{n \geq 0} \text{ is bounded}\}.$$

Definition 4.2 (hyperbolic component). A parameter $c \in M$ is *hyperbolic* if the finite critical point 0 is attracted to an attracting periodic orbit. A *hyperbolic component* is a connected component of this open hyperbolic locus; the exact period of its attracting orbit is constant on the component and is called the period of the component.

Definition 4.3 (Misiurewicz parameter). A quadratic parameter c is a *Misiurewicz parameter* if the critical point 0 (equivalently the critical value $c = f_c(0)$) is strictly preperiodic and its eventual periodic orbit is repelling.

4.2 Uniformizing the escape locus

Theorem 4.1 (Parameter Böttcher coordinate). **Statement scope.** For $c \notin M$ let ϕ_c be the normalized Böttcher coordinate of $z^2 + c$ on the univalent escape domain containing the critical value, with $\phi_c(z)/z \rightarrow 1$ at infinity. Then

$$\Phi_M(c) := \phi_c(c)$$

is a conformal isomorphism

$$\mathbb{C} \setminus M \longrightarrow \mathbb{C} \setminus \overline{\mathbb{D}},$$

normalized by $\Phi_M(c)/c \rightarrow 1$ as $c \rightarrow \infty$.

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Quadratic escape-rate continuity; maximal univalent Böttcher domain above the critical Green level; local holomorphic parameter dependence; properness and simple parameter-infinity degree one.

The mathematical argument begins in the proof environment below.

Proof setup. Put

$$f_c(z) = z^2 + c, \quad G_c(z) := \lim_{n \rightarrow \infty} 2^{-n} \log^+ |f_c^n(z)|. \quad (4.1)$$

The existence and basic dynamical properties of G_c are the quadratic case of the polynomial Green-function theorem. We first record the parameter-uniform consequences that will be used below.

Lemma 4.2 (Uniform escape rate and compactness of the nonescaping locus). *The function*

$$(c, z) \longmapsto G_c(z)$$

is continuous on $\mathbb{C}^2$ and locally uniform limits in (4.1) may be taken on compact subsets of parameter-dynamical space. Moreover

$$M = \{c : G_c(0) = 0\} = \{c : G_c(c) = 0\} \quad (4.2)$$

is compact, and

$$G_c(c) = 2G_c(0). \quad (4.3)$$

Proof. Indeed, on every compact subset of $\mathbb{C}^2$ there is a constant A such that

$$|\log^+ |z^2 + c| - 2 \log^+ |z|| \leq A. \quad (4.4)$$

Therefore the sequence $2^{-n} \log^+ |f_c^n(z)|$ is locally uniformly Cauchy by the same telescoping estimate as in the polynomial Green-function theorem; this proves continuity. The functional equation

$$G_c(f_c(z)) = 2G_c(z) \quad (4.5)$$

gives (4.3). Since $G_c(z) = 0$ exactly on the filled Julia set, (4.2) follows. It also shows that M is closed. Finally, if $|c| > 2$, then

$$|f_c(c)| = |c^2 + c| \geq |c|(|c| - 1) > |c|,$$

and once an orbit has modulus greater than $|c| > 2$ its modulus grows strictly; thus $0 \mapsto c \mapsto f_c(c) \mapsto \dots$ escapes. Hence $M \subset \{|c| \leq 2\}$ and M is compact. $\square$

Lemma 4.3 (The critical value lies in the univalent Böttcher domain). *If $c \notin M$ and*

$$h(c) := G_c(0) > 0,$$

then the normalized Böttcher coordinate at infinity extends uniquely and conformally to the Green superlevel

$$V_c := \{z : G_c(z) > h(c)\}. \quad (4.6)$$

In particular, $c \in V_c$, because $G_c(c) = 2h(c)$.

Proof. We make the extension and the absence of monodromy explicit. For $t > h(c)$ write

$$U_t := \{z : G_c(z) > t\} \cup \{\infty\}.$$

Choose $T > h(c)$ so large that the normalized local Böttcher germ is already a conformal isomorphism

$$\phi_c : U_T \longrightarrow \{w \in \widehat{\mathbb{C}} : |w| > e^T\} \cup \{\infty\}.$$

Suppose the coordinate has been constructed on U_{2s} for some $s > h(c)$. The functional equation for the Green function gives

$$f_c^{-1}(U_{2s}) = U_s. \quad (4.6a)$$

Let W be the component of U_s containing infinity. The proper map $f_c : W \rightarrow U_{2s}$ has local degree two at infinity. Since the total degree of f_c is two, this component already exhausts the full mapping degree; hence there is no second component and $W = U_s$. Thus every superlevel above $h(c)$ is connected.

The finite critical point 0 is not in U_s , because $G_c(0) = h(c) < s$. After removing infinity,

$$f_c : U_s \setminus \{\infty\} \longrightarrow U_{2s} \setminus \{\infty\}$$

is therefore a connected unbranched two-sheeted cover. Via the already constructed coordinate on U_{2s} , its target is the annulus $|w| > e^{2s}$. The squaring map

$$q : \{|w| > e^s\} \longrightarrow \{|w| > e^{2s}\}, \quad q(w) = w^2,$$

is the connected two-sheeted cover of that annulus. The classification of connected covers of an annulus, together with the normalization at infinity, gives a unique cov-

ering isomorphism

$$\phi_c : U_s \setminus \{\infty\} \longrightarrow \{|w| > e^s\}$$

satisfying

$$\phi_c(f_c(z)) = \phi_c(z)^2, \quad \frac{\phi_c(z)}{z} \longrightarrow 1 \quad (z \rightarrow \infty).$$

It extends conformally across infinity. Starting from U_T and applying this pullback step to the levels $2^{-1}T, 2^{-2}T, \dots$ proves the result for those levels; for an arbitrary $t > h(c)$ first choose n with $2^n t > T$ and perform the same finite downward induction. The normalized coordinates agree on overlaps, so their union is a single conformal coordinate on

$$V_c = \bigcup_{t > h(c)} U_t = \{G_c > h(c)\} \cup \{\infty\}.$$

Its modulus is fixed by

$$\log |\phi_c(z)| = G_c(z). \quad (4.7)$$

This proves the asserted univalent extension without an implicit monodromy assumption. Notice that (4.3) places the critical value $c = f_c(0)$ strictly above the critical Green level, exactly where we need to evaluate it.

Define

$$\Phi_M(c) := \phi_c(c), \quad c \notin M. \quad (4.8)$$

By (4.7),

$$\log |\Phi_M(c)| = G_c(c) = 2G_c(0) > 0, \quad (4.9)$$

so Φ_M takes values in $\mathbb{C} \setminus \overline{\mathbb{D}}$. $\square$

Lemma 4.4 (Holomorphic parameter dependence). *The function Φ_M is holomorphic on $\mathbb{C} \setminus M$.*

Proof. Fix $c_0 \notin M$. By (4.9), choose N and a neighborhood U of c_0 so that $f_c^N(c)$ belongs, for every $c \in U$, to one fixed neighborhood of infinity on which the normalized local Böttcher coordinate is given by the uniformly convergent product

$$\phi_c(z) = z \prod_{j=0}^{\infty} \left(1 + \frac{c}{(f_c^j(z))^2} \right)^{2^{-j-1}}. \quad (4.10)$$

For $|z|$ large and c in a fixed compact set, the factors are uniformly close to 1; choosing the logarithm near 1 makes (4.10) jointly holomorphic in (c, z) . The functional equation gives

$$\Phi_M(c)^{2^N} = \phi_c(f_c^N(c)). \quad (4.11)$$

The right-hand side is holomorphic and nonzero on U . Shrinking U if necessary, there is a unique holomorphic 2^N -th root agreeing at c_0 with the normalized continuation from infinity. It is precisely Φ_M . Thus Φ_M is holomorphic near every $c_0 \notin M$. $\square$

Lemma 4.5 (Properness and degree one). *The map*

$$\Phi_M : \mathbb{C} \setminus M \longrightarrow \mathbb{C} \setminus \overline{\mathbb{D}}$$

is proper and has mapping degree one.

Proof. First consider the parameter infinity. Substituting $z = c$ in (4.10) gives, for $|c|$ sufficiently large,

$$\frac{\Phi_M(c)}{c} = \prod_{j=0}^{\infty} \left(1 + \frac{c}{(f_c^j(c))^2} \right)^{2^{-j-1}}. \quad (4.12)$$

Because $f_c^j(c)$ grows at least quadratically after the first step, the logarithm of the product tends uniformly to 0 as $c \rightarrow \infty$. Hence

$$\frac{\Phi_M(c)}{c} \rightarrow 1. \quad (4.13)$$

Next let $c_n \notin M$ tend to a finite boundary point $c_* \in \partial M$. Continuity from Lemma 4.2 and (4.9) give

$$\log |\Phi_M(c_n)| = 2G_{c_n}(0) \rightarrow 2G_{c_*}(0) = 0,$$

so $|\Phi_M(c_n)| \rightarrow 1$. Together with (4.13), this proves properness: the preimage of a compact annulus $1 + \varepsilon \leq |w| \leq R$ stays away from both ∂M and infinity and is therefore compact.

Write

$$F(c) := \frac{1}{\Phi_M(c)}.$$

The map extends across the parameter infinity by $F(\infty) = 0$, and (4.13) shows that this zero is simple in the local coordinate $1/c$. On every connected component of $\mathbb{C} \setminus M$, properness makes F a proper holomorphic map to $\mathbb{D}$. The component containing infinity is therefore onto and has mapping degree equal to the multiplicity of the only preimage of 0, namely one. Thus it is biholomorphic to $\mathbb{D}$.

There can be no other component. Indeed, on any finite component U the proper map $F : U \rightarrow \mathbb{D}$ would again have open-and-closed image and hence would be onto, so it would have a zero in U . But $F(c) = 1/\phi_c(c)$ is never zero at a finite parameter. Therefore $\mathbb{C} \setminus M$ has just the component containing infinity, and F has degree one globally. Equivalently, Φ_M is a conformal isomorphism onto $\mathbb{C} \setminus \overline{\mathbb{D}}$. $\square$

Proof of Theorem 4.1. Lemma 4.2 gives compactness of M and continuity of the escape rate in parameter-dynamical space. Lemma 4.3 shows that for every $c \notin M$ the normalized Böttcher coordinate can be evaluated at the critical value, defining $\Phi_M(c) = \phi_c(c)$. By Lemma 4.4, this map is holomorphic, and Lemma 4.5 proves that it is a proper degree-one map onto $\mathbb{C} \setminus \overline{\mathbb{D}}$, hence a conformal isomorphism. The normalization $\Phi_M(c)/c \rightarrow 1$ at parameter infinity is part of the same properness lemma. This proves the theorem. $\square$

Theorem 4.6 (Connectedness of the Mandelbrot set). **Statement scope.** *The Mandelbrot set*

$$M = \{c : \text{the orbit of } 0 \text{ under } z^2 + c \text{ is bounded}\}$$

is compact and connected.

Proof strategy (not part of the proof)

Proof status. derived here.

Strategic outline only. Compactness from the parameter escape-rate function; connectedness from the parameter Böttcher isomorphism and the planar simply-connected-complement criterion.

The mathematical argument begins in the proof environment below.

Lemma 4.7 (Simply connected complement forces connectedness). *Let $K \subset \widehat{\mathbb{C}}$ be compact and suppose $\widehat{\mathbb{C}} \setminus K$ is simply connected. Then K is connected.*

Proof. Suppose instead that $K = A \cup B$ with A, B nonempty disjoint compact sets. Because the sphere is normal, choose disjoint neighborhoods of A and B with compact closures. A standard polygonal approximation/separation argument on the sphere supplies a Jordan curve

$$\gamma \subset \widehat{\mathbb{C}} \setminus K$$

that separates A from B : one complementary component of γ meets A and the other meets B . Pick $a \in A$ and $b \in B$. After a Möbius change of coordinate sending b to infinity, γ has nonzero winding number about a . Winding number is invariant under homotopies in $\widehat{\mathbb{C}} \setminus \{a, b\}$, hence in the smaller set $\widehat{\mathbb{C}} \setminus K$. Consequently γ cannot be contracted there, contradicting simple connectivity. $\square$

Proof of Theorem 4.6. Compactness follows from Lemma 4.2: the escape-rate characterization

$$M = \{c : G_c(0) = 0\}$$

makes M closed, while $|c| > 2$ forces the critical orbit to escape, so $M \subset \{|c| \leq 2\}$.

By the parameter Böttcher theorem,

$$\Phi_M : \mathbb{C} \setminus M \xrightarrow{\cong} \mathbb{C} \setminus \overline{\mathbb{D}} \quad (4.14)$$

is conformal. After adjoining infinity, this identifies

$$\widehat{\mathbb{C}} \setminus M \quad \text{with} \quad \widehat{\mathbb{C}} \setminus \overline{\mathbb{D}}. \quad (4.15)$$

The latter is simply connected. Lemma 4.7 applied to $K = M$ therefore gives connectedness. Together with compactness, this proves the theorem. $\square$

4.3 External parameter rays

Definition 4.4 (parameter ray). Using the parameter Böttcher isomorphism Φ_M proved above, the *external parameter ray* of angle $\theta \in \mathbb{R}/\mathbb{Z}$ is

$$\mathcal{R}_M(\theta) := \Phi_M^{-1}\{re^{2\pi i\theta} : r > 1\}.$$

Its *impression* is the intersection of the closures of its tails as $r \downarrow 1$; the ray *lands* if this impression consists of one point.

Theorem 4.8 (Landing of rational parameter rays). **Statement scope.** Every external parameter ray of the quadratic Mandelbrot set whose angle is rational lands. If the angle is periodic under the doubling map $t \mapsto 2t \pmod{1}$, its landing point is parabolic; if the angle is strictly preperiodic under doubling, its landing point is a Misiurewicz parameter.

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Dynamic-ray landing/stability; parameter/dynamic critical-value bridge; algebraic count of parabolic ray-period parameters; characteristic periodic ray-pair and kneading-separator induction; preperiodic stability and finite Misiurewicz equation.

The mathematical argument begins in the proof environment below.

Proof setup. Let

$$\mathcal{R}_M(\theta) := \Phi_M^{-1}\{re^{2\pi i\theta} : r > 1\} \quad (4.16)$$

be the parameter ray of angle θ . If $c \in \mathcal{R}_M(\theta)$, then by definition of Φ_M ,

$$\phi_c(c) = e^{G_c(c) + 2\pi i\theta}. \quad (4.17)$$

Thus the critical value $c = f_c(0)$ lies on the dynamical external ray of the same angle at positive potential $G_c(c)$. This elementary identity is the bridge from dynamical rays to parameter rays.

We first isolate the two dynamical facts used throughout.

Lemma 4.9 (Rational dynamic rays and repelling stability). *Let $f_c(z) = z^2 + c$ with $c \in M$.*

1. *If θ is periodic under doubling, the dynamic ray $\mathcal{R}_c(\theta)$ lands at a periodic point which is either repelling or parabolic.*
2. *If θ is strictly preperiodic, $\mathcal{R}_c(\theta)$ lands at a preperiodic point; unless its eventual cycle is parabolic, that eventual cycle is repelling.*
3. *If a periodic or preperiodic ray lands at a repelling (pre)periodic point whose forward orbit does not meet the critical point, then both the landing point and the ray, parametrized by Green potential away from zero, vary continuously with c in a parameter neighborhood. If a finite pullback meets a precritical point, the same statement holds after the first such collision with the two local inverse continuations retained.*

Proof. For (1), $c \in M$ implies $K(f_c)$ is connected by the polynomial critical-orbit connectedness criterion, so the Böttcher inverse defines the ray for all positive potentials. Let the angle have period q . Choose a fundamental ray segment between potentials t and $2^q t$. In the hyperbolic metric of $\mathbb{C} \setminus K(f_c)$, its pullbacks under f_c^q all have the same length, because f_c^q is a covering away from the filled Julia set. Their Euclidean diameters tend to zero as the segments approach the boundary: otherwise a sequence of fixed positive Euclidean diameter at bounded hyperbolic length would meet a compact

subset of the domain, contradicting that the potentials tend to zero. The nested chain of ray segments therefore has a single endpoint z . The functional equation for rays gives $f_c^q(z) = z$. A periodic Fatou point cannot lie on the Julia boundary, so the multiplier is not attracting. The neutral irrational case is excluded by the Snail Lemma from the Chapter 2 periodic-Fatou engine; hence z is repelling or parabolic.

For (2), apply a sufficiently high forward iterate so that $2^m\theta$ is periodic, use (1), and pull the landing point back along the prescribed ray branches. The ray itself selects a unique pullback at every noncritical stage, and at a critical pullback the two one-sided continuations meet at the critical point; in either case the original ray has a landing point.

For (3), the implicit-function theorem continues a repelling periodic point holomorphically. A local inverse branch of the relevant return map is uniformly contracting on one fixed small disk. A ray segment of fixed positive potential varies holomorphically with the parameter through the Böttcher coordinate and, for the base parameter, eventually lies in that disk. Iterating the contracting inverse branch therefore proves continuous landing at the continued point. Pulling back finitely many times proves the preperiodic statement; if a pullback hits the critical point, the two square-root branches are retained and vary continuously as an unordered pair. This proves the lemma. $\square$

Lemma 4.10 (Parabolic counting at fixed ray period). *For $n > 1$, let a_n be the number of angles of exact period n under doubling on $\mathbb{R}/\mathbb{Z}$. The number p_n of quadratic parameters having a parabolic orbit of exact ray period n is finite and satisfies*

$$p_n \leq \frac{a_n}{2}. \quad (4.18)$$

For $n = 1$ there is exactly one such parameter, $c = 1/4$.

Proof. For $n > 1$, Möbius inversion applied to the $2^n - 1$ solutions of $2^n\theta = \theta \pmod{1}$ gives the exact-period count $a_n = 2s_n$ with $s_n \in \mathbb{N}$. Consider the exact-period- n factor $Q_n(c, z) = 0$ of $f_c^n(z) - z = 0$. A generic parameter fiber contains $2s_n$ exact-period points, grouped into $2s_n/n$ cycles. On the normalization of $Q_n = 0$, the multiplier

$$\Lambda(c, z) = (f_c^n)'(z) \quad (4.19)$$

is proper: for $|c| \rightarrow \infty$, exact-period points satisfy $|z| = |c|^{1/2}(1 + o(1))$, and hence

$$|\Lambda(c, z)| = |4c|^{n/2}(1 + o(1)). \quad (4.20)$$

Counting the sheets of the exact-period curve with this pole order gives mapping degree ns_n for Λ on Q_n ; this is the same elementary degree count as for a large multiplier value. Thus the multiplier value 1 occurs at ns_n points of Q_n , counting multiplicity. Projection to the parameter coordinate identifies the n points on each orbit, so there are at most $s_n = a_n/2$ parameter values.

A parabolic orbit of orbit period k and multiplier $e^{2\pi ip/q}$ has ray period $n = qk$, and it becomes multiplier 1 precisely when regarded as an n -periodic orbit. It is therefore counted in this multiplier-1 fiber, proving (4.18). For $n = 1$, solving the fixed-point and

multiplier-one equations $z^2 + c = z$ and $2z = 1$ gives $z = 1/2$, $c = 1/4$. $\square$

Lemma 4.11 (Periodic parameter rays land). *If θ has exact doubling period n , then $\mathcal{R}_M(\theta)$ lands at a parabolic parameter of ray period n .*

Proof. The impression

$$I(\theta) := \bigcap_{r>1} \overline{\Phi_M^{-1}\{se^{2\pi i\theta} : 1 < s < r\}} \quad (4.21)$$

is nonempty, compact, and connected: it is the nested intersection of closures of connected ray tails in the compact sphere. Properness of Φ_M puts it in ∂M . Fix $c_0 \in I(\theta)$. By Lemma 4.9, the dynamic ray of angle θ for f_{c_0} lands at a repelling or parabolic periodic point.

It cannot land repelling. If it did, Lemma 4.9(3) would make the complete low-potential ray land stably at the continued repelling point for all nearby parameters. But for c on the parameter ray, (4.17) puts the critical value on that same dynamic ray. Since the angle is periodic, repeatedly pulling the ray backward through the angle cycle must encounter the critical point (equivalently the ray continuation “bounces” at a precritical point); such a ray cannot at the same time be an unbranched stable ray landing at a repelling cycle. Hence every $c_0 \in I(\theta)$ is parabolic, and the dynamic ray lands on its parabolic orbit.

The ray period is n : the angle itself has exact period n , and the first return on the finite collection of rays landing at the parabolic orbit permutes those rays; the ray period is exactly orbit period times the order of this permutation. By Lemma 4.10 there are only finitely many such parameters. The connected set $I(\theta)$ is thus a singleton. This proves periodic parameter-ray landing.

For the preperiodic case we need the following strengthening of periodic landing; it will also be reused for hyperbolic components. $\square$

Lemma 4.12 (Periodic ray pairs and kneading separators). *For every parabolic quadratic parameter of ray period $n > 1$, exactly two periodic parameter rays of exact period n land there; they are the two characteristic angles of the parabolic orbit. At ray period 1, the root $c = 1/4$ is the landing point of the single parameter ray of angle 0 (represented by the two sides 0 and 1 in the lifted interval convention). Moreover periodic landing pairs, together with this period-one boundary ray, separate parameter space into regions on each of which any prescribed finite initial block of the critical kneading sequence is constant.*

Proof. Here is the proof in the amount needed later. For a periodic angle ϑ , cut the angle circle at the two inverse angles

$$\frac{\vartheta}{2}, \quad \frac{\vartheta + 1}{2}$$

and label the two complementary arcs 0 and 1; the itinerary of ϑ under doubling, with a boundary symbol $*$ when an iterate hits an endpoint, is its kneading sequence. In a dynamical plane, the corresponding two inverse rays land at preimages of the periodic landing point and give the same two-symbol partition of the Julia set away from the critical boundary. Under parameter perturbation, a finite itinerary can change only

when the critical value crosses one of the relevant dynamic rays. By (4.17), that crossing occurs precisely on the parameter ray of the same angle. Consequently, once all periodic ray pairs of periods below n are known, their union cuts the parameter plane into finitely many regions on which every kneading prefix of length $< n$ is constant.

Now let several period- n parameter rays land at one parabolic parameter c . Their dynamic rays land on the parabolic orbit by Lemma 4.11. Project the parabolic orbit to the periodic Fatou component containing the critical value along arcs in the full filled Julia set. The projections all meet at the unique parabolic boundary point adjacent to the critical value; hence any parameter ray landing at c has its dynamic ray at the *characteristic* point. Among the rays landing at this point, consider the sectors between consecutive rays. A sector not containing the critical point maps homeomorphically and its angular width doubles; the unique sector containing the critical point maps two-to-one, while the shortest sector in every cycle of sectors must contain the critical value. The shortest such sectors are nested, so there is a unique innermost one. Its two boundary rays are the characteristic rays. If two periodic rays at the characteristic point had the same kneading sequence without being these two boundary rays, follow the cyclic permutation of sectors: at the first iterate where one partition puts a ray in the critical side and the other does not, the two kneading symbols differ. Thus at most the two characteristic angles can have the same kneading data compatible with landing at c .

We now induct on n . At period one, the positive-real parameter ray of angle $0 = 1$ lands at $c = 1/4$; it supplies the two boundary sides needed by the kneading partition even though there is only one ray on the angle circle. Assume the pairing statement for smaller periods. Lemma 4.11 says every one of the a_n period- n rays lands at a ray-period- n parabolic parameter. The preceding kneading/sector argument says at most two such rays land at any one parameter. Hence at least $a_n/2$ distinct parabolic parameters are hit. Lemma 4.10 says there are at most $a_n/2$ such parameters altogether. For $n > 1$, equality follows: every such parabolic parameter is hit and exactly two rays land there, necessarily the characteristic pair. Together with the period-one base case, the finite-prefix constancy statement established in the first paragraph therefore has actual landing boundaries at every inductive stage. This proves Lemma 4.12. $\square$

Proof of Theorem 4.8. Now let θ be strictly preperiodic under doubling and let $c_0 \in I(\theta)$. We first show that c_0 is not parabolic. If it were, Lemma 4.12 would give its characteristic periodic landing boundary (a pair when the ray period is > 1 , and the two-sided period-one boundary at $c_0 = 1/4$). The one-sided periodic kneading sequences adjacent to that boundary are periodic. The kneading sequence of the strictly preperiodic angle θ is strictly preperiodic, so it differs from each of those periodic sequences at some finite position. By the finite-prefix separator part of Lemma 4.12, a periodic landing pair of sufficiently high but finite period separates $\mathcal{R}_M(\theta)$ from c_0 . A ray cannot cross that closed separator. This contradicts $c_0 \in I(\theta)$. Thus no point of the impression is parabolic.

By Lemma 4.9, the dynamic ray $\mathcal{R}_{c_0}(\theta)$ therefore lands at a repelling preperiodic point z_0 . We claim

$$z_0 = c_0, \tag{4.22}$$

where on the right c_0 is regarded as the critical value in the dynamical plane of f_{c_0} . Choose parameters $c_j \in \mathcal{R}_M(\theta)$ with $c_j \rightarrow c_0$. Their parameter potentials

$$t_j := \log |\Phi_M(c_j)| = G_{c_j}(c_j)$$

tend to zero, and (4.17) says precisely that the dynamic ray point of angle θ and potential t_j is c_j . If the forward orbit of z_0 avoids the critical point, Lemma 4.9(3) gives continuous dependence of the entire ray and landing point, and letting $t_j \downarrow 0$ yields (4.22). If a finite forward image of z_0 lies on the backward critical orbit, apply stability to the first forward image beyond the collision and pull back. The two possible inverse continuations meet at the critical point, and the branch containing the points c_j again converges to c_0 ; hence (4.22) still holds. Were the landing point a proper backward iterate of the critical value, the equality after this pullback would force the critical value to be periodic, contrary to strict preperiodicity of the angle. Thus the landing point is exactly the critical value.

Since dynamic rays respect angle doubling,

$$f_{c_0}^m(c_0)$$

is periodic once $2^m\theta$ becomes periodic. The point c_0 itself cannot be periodic: the first return near a repelling or parabolic periodic point permutes the finite set of landing rays, so every ray landing at a periodic point has periodic angle. Therefore the critical value, and equivalently the critical point, is strictly preperiodic to a repelling cycle. By definition, c_0 is a Misiurewicz parameter.

Finally, if θ has preperiod m and eventual period q , every $c_0 \in I(\theta)$ satisfies

$$f_{c_0}^{m+q}(c_0) = f_{c_0}^m(c_0). \quad (4.23)$$

The two sides are polynomials in c_0 of different degrees before cancellation, so their difference is not identically zero. Hence (4.23) has only finitely many solutions. The impression $I(\theta)$ is connected and contained in this finite set, so it is a singleton. Thus the preperiodic parameter ray lands, and its landing point is Misiurewicz.

The periodic and strictly preperiodic cases together prove the theorem. $\square$

4.4 Hyperbolic components

Definition 4.5 (center of a hyperbolic component). A *center* of a hyperbolic component is a parameter at which the attracting cycle is superattracting, equivalently its multiplier is 0.

Definition 4.6 (root of a hyperbolic component). For a hyperbolic component H of period n , a *root* is a boundary parameter obtained by letting the multiplier of the attracting n -cycle tend to 1; equivalently, the limiting n -return has a parabolic fixed point of multiplier 1. In a satellite component the underlying parabolic orbit may have smaller orbit period while its ray period equals n .

Theorem 4.13 (Hyperbolic component multiplier map). **Statement scope.** For every hyperbolic component H of the quadratic Mandelbrot set, the multiplier of the attracting cycle defines a conformal isomorphism

$$\lambda_H : H \longrightarrow \mathbb{D}.$$

The unique parameter at which the multiplier is 0 is the center of H .

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Holomorphic proper multiplier map; parabolic density and boundary-in-M; rational-ray landing-pattern uniqueness of the root; fullness/simple connectivity; slit-lift contradiction forcing degree one; unique zero=center.

The mathematical argument begins in the proof environment below.

Proof setup. Let H be a hyperbolic component of period n . For $c \in H$, write

$$\mathcal{O}_c = \{z_0(c), \dots, z_{n-1}(c)\}$$

for its attracting orbit and

$$\lambda_H(c) := (f_c^n)'(z_0(c)) \quad (4.24)$$

for its multiplier. The value is independent of which point on the cycle is chosen.

Lemma 4.14 (Holomorphicity, nonconstancy, and properness). The map

$$\lambda_H : H \longrightarrow \mathbb{D} \quad (4.25)$$

is nonconstant, holomorphic, proper, and hence a finite-degree surjection.

Proof. The cycle is simple throughout H : a multiple periodic point would have multiplier 1 and therefore would not be attracting. The implicit-function theorem consequently continues its points locally holomorphically, and the symmetric product (4.24) is single-valued and holomorphic on H . It is not constant. We spell out the algebraic-at-infinity check that will also be used below. The local periodic-point branch lies on an irreducible component of

$$f_c^n(z) - z = 0.$$

No such component is vertical in the c -line, so its projective closure has branches with $|c| \rightarrow \infty$. Write $z_j = f_c^j(z_0)$ along an n -periodic orbit and put $C = |c|$. For C sufficiently large, every periodic orbit satisfies

$$|z_j| > \frac{1}{2}\sqrt{C} \quad (0 \leq j < n). \quad (4.24a)$$

Indeed, if $|z_j| \leq \frac{1}{2}\sqrt{C}$, then $|z_{j+1}| \geq C - |z_j|^2 \geq 3C/4 > 2\sqrt{C}$; but once $|w| > 2\sqrt{C}$, the inequality $|w^2 + c| > |w|$ persists and the orbit escapes, contradicting periodicity. There-

fore on every branch over parameter infinity

$$|(f_c^n)'(z_0)| = 2^n \prod_{j=0}^{n-1} |z_j| \geq C^{n/2} \longrightarrow \infty. \quad (4.24b)$$

The algebraic multiplier is thus nonconstant on the irreducible component, and its restriction λ_H is nonconstant.

To prove properness, suppose $c_j \in H$ leaves every compact subset of H while $|\lambda_H(c_j)| \leq r < 1$. Since $H \subset M$ for a quadratic hyperbolic map (the unique finite critical point is attracted to the finite attracting cycle), compactness of M gives a subsequence $c_j \rightarrow c_* \in \partial H$. Passing to a subsequence of cycle points gives an n -periodic limiting cycle for f_{c_*} with multiplier of modulus at most r . It is attracting, so the implicit-function theorem and attraction persist near c_* ; hence c_* belongs to the same hyperbolic component, a contradiction. Therefore inverse images of compact subsets of $\mathbb{D}$ are compact in H . A nonconstant proper holomorphic map has open and closed image, so $\lambda_H(H) = \mathbb{D}$, and every fiber is finite with a constant total multiplicity. $\square$

Lemma 4.15 (Boundary, landing pattern, and unique root). *The boundary of H is contained in ∂M . There exists a unique parameter $r_H \in \partial H$ obtained as a limit of points with multiplier tending to 1; at r_H the limiting cycle is parabolic. This parameter is the root of H .*

Proof. First fix $b \in \partial H$ and choose $c_j \rightarrow b$ in H . After passing to a subsequence, the attracting cycles converge to an n -periodic limiting cycle and $\lambda_H(c_j) \rightarrow \zeta \in \partial \mathbb{D}$; if $|\zeta| < 1$, the attracting cycle would persist across b , contradicting $b \in \partial H$.

If ζ is a root of unity, then an iterate of the limiting cycle has multiplier 1, so b is parabolic. If ζ is not a root of unity, then $\zeta \neq 1$ and the limiting periodic point is a simple zero of $f_c^n(z) - z$. The implicit-function theorem therefore continues that cycle and its multiplier as a holomorphic function $\tilde{\lambda}(c)$ on a full parameter neighborhood of b . This local multiplier is nonconstant by the periodic-branch at-infinity check above: its germ lies on an irreducible periodic-point component, and (4.24a)–(4.24b) rule out constancy on that component. Since roots of unity are dense on the unit circle and a nonconstant holomorphic map is open, there are parameters arbitrarily close to b at which $\tilde{\lambda}$ is a root of unity. Those parameters are parabolic for a suitable iterate. Thus every boundary point of H is a limit of parabolic parameters.

By Theorem 4.8 and Lemma 4.12, every quadratic parabolic parameter is the landing point of a periodic parameter ray and therefore lies in ∂M . Since ∂M is closed, we obtain

$$\partial H \subset \partial M. \quad (4.26)$$

Properness and surjectivity give a sequence $c_j \in H$ with $\lambda_H(c_j) \rightarrow 1$. Every limit point $r \in \partial H$ carries a parabolic limiting cycle with multiplier 1 for the n -th return; call such a point a root candidate. We prove all root candidates coincide.

Inside H , every periodic dynamic ray lands on a repelling periodic orbit: external rays land on $J(f_c)$, whereas the only nonrepelling finite cycle is the attracting one. By Lemma 4.9, the landing relation among any fixed finite collection of periodic rays is locally constant, hence constant throughout connected H . Approach a root candidate

from inside H . All repelling landing relations persist; the only new collision is the parabolic one. The sector-width argument from Lemma 4.12 identifies, from the interior landing pattern alone, the unique pair of periodic rays bounding the smallest sector adjacent to the critical-value Fatou component. In the limit these become exactly the characteristic rays of the parabolic orbit. Consequently every root candidate of the same component has the same two characteristic angles.

Lemma 4.12 says that the corresponding two periodic *parameter* rays land together at exactly the parameter carrying those characteristic angles. Hence two root candidates with the same characteristic pair must be equal. We have therefore obtained a unique root r_H . $\square$

Lemma 4.16 (Fullness and simple connectivity of the component). *The set M is full, and H is simply connected.*

Proof. By the parameter Böttcher theorem, $\mathbb{C} \setminus M$ is conformally the exterior disk and is connected, so M is full. Equation (4.26) shows that H is a connected component of $\text{int } M$: if it were a proper open subset of a larger interior component, its relative boundary would contain a point of $\text{int } M$, contrary to (4.26).

Let $\gamma \subset H$ be a Jordan curve and let B be its bounded complementary component. If B contained a point of $\mathbb{C} \setminus M$, connectedness of $\mathbb{C} \setminus M$ would give a path from that point to infinity avoiding M ; such a path would have to cross $\gamma \subset M$, impossible. Thus $B \subset M$. Since B is open, $B \subset \text{int } M$; because B meets H along its boundary and lies on the same side of that boundary, it belongs to the same interior component H . Every Jordan curve in H therefore bounds in H , and the standard Jordan decomposition of loops shows that H is simply connected. $\square$

Lemma 4.17 (The multiplier degree is one). *The proper map $\lambda_H : H \rightarrow \mathbb{D}$ has degree one.*

Proof. Let its degree be d . Suppose $d > 1$. Since H is simply connected, a proper holomorphic map of degree d has $d-1$ critical points counted with multiplicity (equivalently, conjugate by a Riemann map to a finite Blaschke product and apply Riemann-Hurwitz). Choose a critical value $v \in \mathbb{D}$ and a simple arc

$$\alpha : [0, 1) \longrightarrow \mathbb{D}, \quad \alpha(0) = v, \quad \alpha(t) \longrightarrow 1, \quad (4.27)$$

that avoids all other critical values.

At a critical point over v , at least two lifts of the slit $\alpha((0, 1))$ emanate. Properness forces each lift to leave every compact subset of H as $t \rightarrow 1$; every cluster point is a root candidate, hence by Lemma 4.15 both lifts end at the unique root r_H . Two adjacent lifts together with their common endpoints therefore contain a Jordan curve $\Gamma \subset H \cup \{r_H\}$; cutting it a tiny distance before the root and joining the endpoints inside H gives a Jordan curve $\Gamma_\varepsilon \Subset H$ whose interior U_ε contains one full sheet of

$$\lambda_H^{-1}(\mathbb{D} \setminus \alpha). \quad (4.28)$$

In particular, for points $w_k \in \mathbb{D} \setminus \alpha$ tending to an arc of $\partial\mathbb{D}$ away from 1, there are points $x_k \in U_\varepsilon$ with $\lambda_H(x_k) = w_k$.

Properness says x_k must leave every compact subset of H as $|w_k| \rightarrow 1$; hence a subsequence tends to ∂H . But by Lemma 4.16 the interior of the compact Jordan curve $\Gamma_\varepsilon \in H$ is contained in H , so $\overline{U_\varepsilon} \in H$. This is a contradiction. Therefore $d = 1$. $\square$

Proof of Theorem 4.13. By Lemma 4.14, the multiplier map $\lambda_H : H \rightarrow \mathbb{D}$ is a nonconstant proper holomorphic surjection. Lemma 4.15 gives the unique root $r_H \in \partial H$, and Lemma 4.16 shows that H is simply connected. Lemma 4.17 then implies that λ_H has degree one and therefore is a conformal isomorphism.

Its unique preimage of 0 is the unique center. Indeed,

$$\lambda_H(c) = \prod_{j=0}^{n-1} 2z_j(c),$$

so the multiplier vanishes exactly when one point of the attracting orbit is the critical point 0, equivalently when the cycle is superattracting. This proves all assertions of the theorem. $\square$

4.5 Wakes, limbs, and tuning

Definition 4.7 (wake). When two periodic parameter rays land at the same parabolic boundary point, their union with that landing point separates the parameter plane. The component not containing parameter infinity is the *wake* cut out by that landing pair. A *limb* is the intersection of the closure of such a wake with M , with the combinatorial attachment convention specified by the characteristic ray pair.

Remark 4.1 (Wakes, limbs, and the tuning interface). **Scope.** The parameter-space work of Chapter 4 supplies the hyperbolic components, characteristic parameter-ray pairs, wakes, and limbs that form the combinatorial input to tuning. The actual Douady-Hubbard tuning theorem is proved later in Chapter 8, after polynomial-like maps, quasiconformal straightening, and renormalization language are available. No result of Chapter 8 is used as a black box in Chapter 4.

4.6 Worked examples

Worked example: The main cardioid from the multiplier coordinate

Inside the period-one hyperbolic component, the attracting fixed point z satisfies

$$z = z^2 + c.$$

Its multiplier is $\lambda = 2z$, hence $z = \lambda/2$ and therefore

$$c = z - z^2 = \frac{\lambda}{2} - \frac{\lambda^2}{4}.$$

For $|\lambda| < 1$ this parametrizes the main cardioid. Formula (4.W1) is the inverse of the multiplier map in this component.

At $\lambda = 0$ one obtains $c = 0$. The critical point is fixed,

$$0 \mapsto 0,$$

so $c = 0$ is the center. As $\lambda \rightarrow 1$, one obtains $c \rightarrow 1/4$ and the fixed point tends to $z = 1/2$. Indeed

$$f_{1/4}(1/2) = 1/2, \quad f'_{1/4}(1/2) = 1,$$

so $c = 1/4$ is the parabolic root of the main cardioid. Thus the abstract theorem that $\lambda_H : H \rightarrow \mathbb{D}$ is a conformal isomorphism is completely explicit for the main cardioid.

Worked example: Two rational-ray landmarks: $c = 1/4$ and $c = -2$

The angle 0 is fixed under doubling. Its parameter ray lands at the period-one parabolic parameter $c = 1/4$, matching the periodic-angle half of the rational landing theorem.

The angle $1/2$ is strictly preperiodic:

$$\frac{1}{2} \mapsto 0 \mapsto 0.$$

At $c = -2$ the critical orbit is

$$0 \mapsto -2 \mapsto 2 \mapsto 2,$$

and the eventual fixed point is repelling because

$$f'_{-2}(2) = 4.$$

Hence $c = -2$ is a Misiurewicz parameter. The parameter ray of angle $1/2$ lands there, illustrating the strictly preperiodic half of the theorem.

The contrast is structural: periodic external combinatorics produces a parabolic collision, whereas strictly preperiodic combinatorics produces a finite critical relation to a repelling orbit.

Boundary case: Why tuning is not yet a Chapter 4 theorem

A wake can already be defined from two periodic parameter rays landing at the same parabolic parameter. This uses only the parameter-ray theory proved in Chapter 4. By contrast, saying that a wake contains a small homeomorphic copy of M obtained by tuning requires a polynomial-like restriction, hybrid equivalence, and the straightening theorem.

Those notions are not part of the Chapter 4 prerequisite language. Treating tuning as a theorem here would therefore either use undefined terminology or import Chapter 8 as a forward black box. The book instead records the wake/limb

combinatorics here and proves the formal tuning theorem in Chapter 8 after the required quasiconformal and polynomial-like machinery has been developed.

Common pitfall

Do not infer that every boundary point of the Mandelbrot set is the landing point of a rational parameter ray. Chapter 4 proves landing for rational angles and identifies the corresponding parabolic or Misiurewicz parameters. General local connectivity of M is a much deeper frontier question treated later in Chapter 21.

4.7 Chapter summary

What to remember

- The critical escape rate gives a parameter exterior coordinate $\Phi_M(c) = \phi_c(c)$, defined on the univalent Böttcher domain containing the critical value and normalized by $\Phi_M(c)/c \rightarrow 1$.
- This coordinate is a conformal isomorphism from $\mathbb{C} \setminus M$ to the exterior unit disk. Consequently the Mandelbrot set is compact, full, and connected.
- A parameter ray is transported to the dynamical plane by the identity saying that the critical value lies on the dynamic ray of the same angle.
- Every rational parameter ray lands: periodic angles at parabolic parameters and strictly preperiodic angles at Misiurewicz parameters.
- Characteristic periodic ray pairs determine roots, wakes, and limbs. They also provide the separation mechanism used in the preperiodic landing proof.
- On every hyperbolic component, the multiplier of the attracting cycle is a conformal coordinate onto the unit disk; its unique zero is the center and the multiplier-one boundary point is the root.
- Chapter 4 stops at the wake/limb interface. The formal tuning theorem is owned by Chapter 8, after polynomial-like maps and straightening are available.

Looking ahead

Chapter 5 develops quasiconformal maps and the measurable Riemann mapping theorem. These tools will later feed the straightening theorem in Chapter 8. Before that, Chapter 6 and Chapter 7 develop holomorphic motions and Sullivan stability, providing the deformation language needed throughout modern parameter-space dynamics.

Part II

Quasiconformal Deformation, Rigidity, and Topological Models

Guide to Part II: controlled deformation and rigidity

The central tension of Part II is simple to state: complex analysis is rigid, while the problems we want to solve require us to change the geometry. Quasiconformal mappings resolve the tension by allowing a bounded amount of anisotropic distortion. The Beltrami coefficient records that distortion, and the measurable Riemann mapping theorem turns the coefficient back into a coordinate change.

That flexibility is useful only because it is controlled. Three-point normalization prevents a sequence of quasiconformal maps of the sphere from drifting through the Möbius group. Modulus bounds prevent annuli from collapsing. Holomorphic motions encode parameter dependence while keeping every slice injective. These are not technical accessories: they are the compactness and normalization mechanisms on which the later rigidity theory rests.

Sullivan’s no-wandering-domain theorem is a model of how deformation becomes rigidity. A hypothetical wandering component produces too much quasiconformal freedom. But rational maps of fixed degree form only a finite-dimensional family. The contradiction is not “quasiconformal maps are rigid”; it comes from constructing an infinite-dimensional deformation space and proving that the dynamical constraints would force it through a finite-dimensional bottleneck.

Polynomial-like maps then localize the polynomial world. A return map defined only on $U' \Subset U$ can carry the full dynamics of a smaller polynomial. Straightening says that, after quasiconformal surgery, this local polynomial has a genuine polynomial model. The modulus of $U \setminus \overline{U'}$ is the geometric collar that makes this localization stable.

Yoccoz puzzles refine localization further. External rays and equipotentials produce dynamically defined pieces whose pullbacks form a nested microscope around the critical orbit. The key question becomes whether the pieces shrink. Their shrinking is not a formal consequence of nesting: one needs modulus control, recurrence analysis, and rigidity.

Thurston theory attacks a complementary problem. Instead of starting with an analytic map and cutting it into pieces, it starts with a topological branched covering and asks whether there exists a rational map in the same combinatorial class. The pull-back map on Teichmüller space and the multicurve obstruction convert a topological realization problem into a contraction problem.

The mathematical route through this Part

Beltrami data $\longrightarrow$ quasiconformal map $\longrightarrow$ holomorphic motion

$\downarrow$
deformation/stability

$\swarrow$
straightening and puzzles

$\searrow$
Thurston pullback and obstructions.

The two branches meet again in rigidity: controlled geometry plus an absence of deformation directions forces a unique analytic model.

The recurring diagnostic

Whenever Part II introduces a deformation, ask two questions. What is being allowed to move? What normalization prevents the motion from becoming meaningless? Whenever it asserts rigidity, ask a third: which possible deformation direction has been ruled out?

This diagnostic prevents a common misunderstanding. Quasiconformal methods do not weaken the problem by replacing holomorphic equivalence with a softer notion. They create a larger space in which one can move, and then use analytic or dynamical rigidity to show that the motion was forced into a much smaller class.

Quasiconformal Geometry and the Beltrami Equation

Why this chapter matters

Complex dynamics repeatedly asks us to change coordinates without destroying the complex-analytic structure we ultimately want to study. Conformal changes of coordinates are too rigid for this purpose: they cannot usually absorb a prescribed distortion field. Quasiconformal maps provide the controlled flexibility that is missing. Their infinitesimal eccentricity is recorded by a Beltrami coefficient μ , and the measurable Riemann mapping theorem says that every measurable field with $\|\mu\|_\infty < 1$ can actually be integrated to a global change of coordinates.

The theorem is powerful only after two geometric facts are understood. First, quasiconformal maps distort extremal length by at most their dilatation. This turns a differential inequality into a conformally natural geometric estimate. Second, three-point normalization prevents families of quasiconformal sphere maps from collapsing, giving the compactness needed to pass from smooth Beltrami coefficients to merely measurable ones. The logical order in this chapter therefore runs from modulus to compactness and only then to the measurable Riemann mapping theorem.

A second branch reaches the boundary. Modulus control forces a quasiconformal disk map to have quasisymmetric boundary values, while the Beurling-Ahlfors construction extends every orientation-preserving quasisymmetric circle map back to a quasiconformal disk map. Together these results form the deformation toolkit used later for holomorphic motions, Sullivan stability, polynomial-like straightening, Thurston theory, and quasiconformal surgery.

Prerequisites. Graduate real and complex analysis; weak first derivatives and basic Sobolev convergence; Cauchy integral techniques; elementary conformal maps and normal-family compactness.

Learning goals

- Translate between the analytic Beltrami definition of quasiconformality and geometric extremal-length distortion.
- Explain why fixing $0, 1, \infty$ is the correct compactness normalization on the Riemann sphere.
- Prove the measurable Riemann mapping theorem through the Cauchy-

Beurling operator method and compactness, rather than using it as a black box.

- Track precisely where the strict condition $\|\mu\|_\infty < 1$ enters the proof.
- Derive quasisymmetric boundary values from quadrilateral modulus and construct the converse extension by the Beurling–Ahlfors averaging formula.
- Understand the basic surgery mechanism: integrate an invariant Beltrami coefficient so that a quasiregular model becomes holomorphic in the new coordinates.

Conceptual roadmap

The analytic-geometric spine is

Beltrami/differential distortion $\rightarrow$ extremal-length distortion
 $\rightarrow$ normalized QC compactness
 $\rightarrow$ measurable Riemann mapping theorem.

The boundary branch is

modulus distortion $\rightarrow$ boundary extension and quasisymmetry
 $\longleftrightarrow$ Beurling–Ahlfors extension.

Finally, the later dynamical use is

invariant Beltrami field $\xrightarrow{\text{MRMT}}$ QC coordinate change
 $\rightarrow$ holomorphic/rational model.

How the main results fit together. The theorem order matters logically. Compactness is proved before the measurable Riemann mapping theorem because the measurable-coefficient approximation uses it. The boundary theorem is a separate branch: it uses the modulus machinery but is not needed to construct the Beltrami solution.

5.1 Definitions

Definition 5.1 (quasiconformal map). Let $\Omega, \Omega' \subset \mathbb{C}$ be domains. An orientation-preserving homeomorphism $f : \Omega \rightarrow \Omega'$ is *K-quasiconformal* if it is absolutely continuous on almost every horizontal and vertical line (ACL), has weak first derivatives locally in L^2 , and

$$|Df(z)|_{\text{op}}^2 \leq K J_f(z)$$

for almost every z . Here J_f is the Jacobian and $|Df|_{\text{op}}$ is the operator norm. Equivalently, $|\mu_f| \leq k = (K - 1)/(K + 1)$ almost everywhere.

Definition 5.2 (Beltrami coefficient). For an ACL map with weak Wirtinger deriva-

tives $f_z, f_{\bar{z}}$ and $f_z \neq 0$ almost everywhere, its *Beltrami coefficient* is

$$\mu_f = \frac{f_{\bar{z}}}{f_z}.$$

Values on the null set where the quotient is undefined are immaterial and are set to zero. The Beltrami equation with coefficient μ is $f_{\bar{z}} = \mu f_z$.

Definition 5.3 (extremal length and modulus). For a family Γ of locally rectifiable curves in a planar domain, its extremal length is

$$\lambda(\Gamma) = \sup_{\rho} \frac{(\inf_{\gamma \in \Gamma} \int_{\gamma} \rho \, ds)^2}{\iint \rho^2 \, dA},$$

where the supremum is over nonnegative Borel densities with finite positive area. Our modulus convention is $\text{Mod}(\Gamma) = 1/\lambda(\Gamma)$, with the usual extended-value conventions.

Definition 5.4 (quasisymmetry). A homeomorphism $h : X \rightarrow Y$ of metric spaces is η -quasisymmetric if $\eta : [0, \infty) \rightarrow [0, \infty)$ is a homeomorphism and, for all distinct x, a, b ,

$$\frac{d_Y(h(x), h(a))}{d_Y(h(x), h(b))} \leq \eta\left(\frac{d_X(x, a)}{d_X(x, b)}\right).$$

On $\mathbb{R}$, for an increasing homeomorphism fixing infinity, this is equivalent to the adjacent-interval condition

$$M^{-1} \leq \frac{h(x+t) - h(x)}{h(x) - h(x-t)} \leq M \quad (x \in \mathbb{R}, t > 0)$$

for some $M \geq 1$.

Definition 5.5 (ACL and Wirtinger data). ACL means absolutely continuous on almost every line parallel to each coordinate axis. For an ACL map $f = u + iv$ we write

$$f_z = \frac{1}{2}(f_x - if_y), \quad f_{\bar{z}} = \frac{1}{2}(f_x + if_y), \quad J_f = |f_z|^2 - |f_{\bar{z}}|^2.$$

At points of differentiability the singular values of Df are $|f_z| + |f_{\bar{z}}|$ and $|f_z| - |f_{\bar{z}}|$.

Definition 5.6 (normalized quasiconformal map). A quasiconformal self-homeomorphism of the Riemann sphere is *normalized* when it fixes 0, 1, and ∞ . Uniform convergence of sphere maps is always with respect to the chordal (spherical) metric

$$\chi(z, w) = \frac{|z - w|}{\sqrt{1 + |z|^2} \sqrt{1 + |w|^2}}, \quad \chi(z, \infty) = \frac{1}{\sqrt{1 + |z|^2}}.$$

Definition 5.7 (Cauchy and Beurling transforms). For $g \in C_c^\infty(\mathbb{C})$, the principal Cauchy transform and the Beurling transform are

$$\mathcal{C}g(z) = \frac{1}{\pi} \int_{\mathbb{C}} \frac{g(\zeta)}{z - \zeta} \, dA(\zeta), \quad \mathcal{S}g(z) = -\frac{1}{\pi} \text{p.v.} \int_{\mathbb{C}} \frac{g(\zeta)}{(z - \zeta)^2} \, dA(\zeta).$$

Here p.v. means the symmetric Cauchy principal value, obtained by removing $D(z, \varepsilon)$ and letting $\varepsilon \downarrow 0$. Their L^p extensions and distributional identities $\partial_{\bar{z}} \mathcal{C}g = g$, $\partial_z \mathcal{C}g = \mathcal{S}g$ are

proved below before they are used for rough coefficients. When a value normalization at a finite base point is preferable, one subtracts the corresponding additive constant; the derivatives are unchanged.

Definition 5.8 (principal Beltrami solution). If μ has compact support, a *principal solution* of $f_{\bar{z}} = \mu f_z$ is a quasiconformal homeomorphism satisfying $f(z) = z + o(1)$ as $z \rightarrow \infty$. This normalization is unique when such a solution exists.

Definition 5.9 (quasiregular map and invariant Beltrami coefficient). A nonconstant $W_{\text{loc}}^{1,2}$ map is K -quasiregular if it satisfies the same differential distortion inequality as a K -quasiconformal map but is not assumed injective. A Beltrami coefficient is *invariant* for a quasiregular dynamical model F when the pullback complex structure agrees almost everywhere with the original one, equivalently when the Beltrami composition formula makes F holomorphic after integration of that coefficient.

Definition 5.10 (marked quadrilateral and crossing family). A *marked quadrilateral* is a Jordan domain together with four distinct boundary points in positive cyclic order. Its *crossing family* is the family of curves joining one chosen pair of opposite boundary sides. Its extremal length is the extremal length of that crossing family; swapping to the other pair of opposite sides reciprocates the value. For a Euclidean rectangle of width w and height h , the family joining the vertical sides has extremal length w/h .

Definition 5.11 (Lusin N , N^{-1} , and zero-modulus curve families). A map has *Lusin's N property* if it sends planar Lebesgue-null sets to null sets, and has N^{-1} if the preimage of every null set is null. A curve family is called a *2-modulus-zero family* when its analytic 2-modulus $\text{Mod} = 1/\lambda$ is zero; adding or removing such an exceptional family does not alter extremal length.

Definition 5.12 (proper map). A continuous map $F : X \rightarrow Y$ between locally compact Hausdorff spaces is *proper* if $F^{-1}(C)$ is compact for every compact $C \subset Y$. This is the meaning of “proper” in the covering arguments below.

5.2 Main results

Proposition 5.1 (Modulus distortion under quasiconformal maps). **Statement scope.** Let $f : \Omega \rightarrow \Omega'$ be a K -quasiconformal homeomorphism and Γ a family of locally rectifiable curves in Ω . Then

$$K^{-1}\lambda(\Gamma) \leq \lambda(f\Gamma) \leq K\lambda(\Gamma).$$

Equivalently the same two-sided factor- K bound holds for the reciprocal modulus convention fixed above.

Proof strategy (not part of the proof)

Proof status. proved here modulo stated analytic input.

Strategic outline only. Planar QC differential calculus; ACL area/curve formulas; quadrilateral modulus characterization; pull back admissible metrics and apply the inverse.

The mathematical argument begins in the proof environment below.

Proof setup. We first isolate the analytic facts that convert the differential definition of quasiconformality into a statement about curve families.

Lemma 5.2 (Planar quasiconformal differential calculus). *At almost every differentiability point of an orientation-preserving ACL map f , put $a = f_z$ and $b = f_{\bar{z}}$. The singular values of the real-linear map Df are*

$$L_f = |a| + |b|, \quad \ell_f = |a| - |b|, \quad (5.1)$$

and

$$J_f = L_f \ell_f = |a|^2 - |b|^2. \quad (5.2)$$

Consequently

$$L_f^2 \leq K J_f \iff \frac{L_f}{\ell_f} \leq K \iff |\mu_f| \leq \frac{K-1}{K+1}. \quad (5.3)$$

The pointwise inverse linear map has the same singular-value ratio. Lemma 5.3 supplies the Sobolev inverse theorem needed to promote this almost-everywhere linear calculation to the inverse homeomorphism; once that bridge is in place, f^{-1} is again K -quasiconformal. If f and g are quasiconformal and the Sobolev chain rule is applied at a point where both maps are differentiable, then

$$(g \circ f)_z = (g_w \circ f)f_z + (g_{\bar{w}} \circ f)\overline{f_z}, \quad (g \circ f)_{\bar{z}} = (g_w \circ f)f_{\bar{z}} + (g_{\bar{w}} \circ f)\overline{f_{\bar{z}}}. \quad (5.4)$$

Proof. Indeed, the real-linear map $z \mapsto az + b\bar{z}$ sends the unit circle to an ellipse with semiaxes $|a| + |b|$ and $|a| - |b|$, which proves (5.1)–(5.3). At a point where Df is invertible, the inverse linear map has singular values L_f^{-1} and ℓ_f^{-1} in reverse order, so their ratio is unchanged. Formula (5.4) is the real Sobolev chain rule rewritten in Wirtinger coordinates.

□

Lemma 5.3 (Area and curve formulas at QC regularity). *Let $f : \Omega \rightarrow \Omega'$ be a quasiconformal homeomorphism. Then f and f^{-1} have Lusin's N property, and for every nonnegative Borel function ψ ,*

$$\int_{f(E)} \psi(w) dA(w) = \int_E \psi(f(z)) J_f(z) dA(z) \quad (5.5)$$

for measurable $E \subset \Omega$. Moreover, outside a curve family of extremal length zero, every locally rectifiable γ satisfies

$$\int_{f \circ \gamma} \sigma ds \leq \int_{\gamma} (\sigma \circ f) L_f ds \quad (5.6)$$

for every nonnegative Borel density σ .

Proof. We prove the measure-theoretic bridge without deriving Lusin N from mere almost-everywhere differentiability. Fix a compact square $Q \Subset \Omega$ and a null set $E \subset Q$.

Given $\varepsilon > 0$, absolute continuity of the integral gives an open set $O \Subset \Omega$ with $E \subset O$ and

$$\int_O |Df|^2 dA < \varepsilon. \quad (5.5a)$$

Cover E by Whitney squares Q_j whose doubled squares lie in O and have bounded overlap. By ACL and Fubini, for each j one can choose a concentric square Q_j^* with $Q_j \subset Q_j^* \subset 2Q_j$ such that

$$\text{length } f(\partial Q_j^*)^2 \leq C \int_{2Q_j} |Df|^2 dA. \quad (5.5b)$$

Indeed, integrate the boundary L^2 energy over the one-parameter family of concentric squares and then use Cauchy-Schwarz on the selected boundary. Because f is a homeomorphism, $f(\partial Q_j^*)$ is a Jordan curve. The planar isoperimetric inequality therefore gives

$$A(f(Q_j^*)) \leq C \int_{2Q_j} |Df|^2 dA. \quad (5.5c)$$

Bounded overlap and (5.5a) imply $A(f(E)) \leq C\varepsilon$. Letting $\varepsilon \downarrow 0$ proves Lusin's N property. Notice that this energy-covering argument, rather than the invalid implication "a.e. differentiable $\Rightarrow N$ ", is the essential point.

We now use one explicitly frozen analytic input from planar Sobolev theory. Its exact form is the following: an orientation-preserving ACL homeomorphism of bounded distortion satisfying $L_f^2 \leq KJ_f$ has $J_f > 0$ almost everywhere, its inverse belongs to $W_{\text{loc}}^{1,2}$, and

$$Df^{-1}(f(z)) = (Df(z))^{-1} \quad \text{for a.e. } z. \quad (5.5d)$$

Thus the pointwise calculation of Lemma 5.2 makes f^{-1} analytically K -quasiconformal. Repeating the energy-covering argument for f^{-1} gives N^{-1} . The Sobolev area formula for an approximately differentiable injective map with property N then yields (5.5), first for characteristic functions, then for simple functions, and finally by monotone convergence.

Finally, ACL plus the Sobolev upper-gradient theorem gives $|(f \circ \gamma)'| \leq L_f(\gamma)|\gamma'|$ for every rectifiable curve outside a 2-modulus-zero family. Integrating yields (5.6); the same statement is available for f^{-1} . Removing or adding a zero-modulus exceptional family does not change extremal length, so these are exactly the curve formulas needed for the extremal problem. □

Lemma 5.4 (Quadrilateral modulus characterization). *Suppose a homeomorphism h between planar domains satisfies*

$$K^{-1}\lambda(\Gamma_Q) \leq \lambda(h\Gamma_Q) \leq K\lambda(\Gamma_Q) \quad (5.7)$$

for the crossing family of every Euclidean quadrilateral Q compactly contained in the domain. Then h is ACL, belongs locally to $W^{1,2}$, and is K -quasiconformal in the analytic sense of the definition of quasiconformal map.

Proof. We recall the standard short proof because this converse will be used in the compactness theorem. Apply (5.7) to finitely many disjoint thin rectangles based on subintervals of a fixed horizontal line. The rectangle formula $\lambda = w/h$ and Cauchy-Schwarz bound the sum of squared image increments by a constant times the total area of the image strips. Letting the strip height tend to zero and using Fubini proves absolute continuity on almost every horizontal line; the vertical argument is identical. The same estimate gives local L^2 -integrability of the weak derivatives. At an almost-everywhere point of differentiability, rotate the source and target so that the derivative has singular values $L \geq \ell > 0$. Apply (5.7) to squares shrinking to that point with sides in the singular directions. After rescaling, the source squares converge to a square and their images converge to the rectangle of aspect ratio L/ℓ . Continuity of rectangle extremal length yields $L/\ell \leq K$. Hence $|Dh|_{\text{op}}^2 \leq K J_h$ almost everywhere, as claimed. $\square$

Proof of Theorem 5.1. It is convenient to use reciprocal modulus for one line of the argument. Let σ be admissible for $f\Gamma$, meaning

$$\int_{f \circ \gamma} \sigma \, ds \geq 1$$

for every curve outside a modulus-zero exceptional family. Set

$$\rho(z) = \sigma(f(z)) L_f(z).$$

By Lemma 5.3, ρ is admissible for Γ , while Lemma 5.2 and the area formula give

$$\int_{\Omega} \rho^2 \, dA \leq K \int_{\Omega} \sigma(f(z))^2 J_f(z) \, dA(z) = K \int_{\Omega'} \sigma(w)^2 \, dA(w). \quad (5.8)$$

Taking the infimum over admissible σ yields

$$\text{Mod}(\Gamma) \leq K \text{Mod}(f\Gamma). \quad (5.9)$$

The inverse conclusion in Lemma 5.2, justified by Lemma 5.3, gives the reverse inequality after applying the same estimate to f^{-1} :

$$\text{Mod}(f\Gamma) \leq K \text{Mod}(\Gamma). \quad (5.10)$$

Reciprocating (with the standard conventions at 0 and ∞) gives

$$K^{-1} \lambda(\Gamma) \leq \lambda(f\Gamma) \leq K \lambda(\Gamma),$$

which proves the theorem. The quadrilateral converse used later in the chapter is recorded separately as Lemma 5.4. $\square$

Theorem 5.5 (Normal-family compactness for normalized QC maps). **Statement scope.** *For fixed $K < \infty$, the family of normalized K -quasiconformal self-homeomorphisms of $\hat{\mathbb{C}}$ is compact in the uniform spherical topology. More generally, a locally uniform homeomorphic limit of K -quasiconformal maps is K -quasiconformal; without a three-point normalization a sequence may degenerate after subselection.*

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Ring-modulus equicontinuity for normalized maps; Arzela–Ascoli for maps and inverses; exact- K closedness from continuity of quadrilateral modulus and the analytic characterization.

The mathematical argument begins in the proof environment below.

Proof setup. Write χ for spherical distance.

Lemma 5.6 (Uniform ring control for normalized maps). *For every $K < \infty$, normalized K -quasiconformal sphere maps and their inverses have a common spherical modulus of continuity.*

Proof. For every $K < \infty$ there is a modulus of continuity $\omega_K : [0, \infty) \rightarrow [0, \infty)$ with $\omega_K(t) \rightarrow 0$ as $t \rightarrow 0$ such that every normalized K -quasiconformal sphere homeomorphism satisfies

$$\chi(f(z), f(w)) \leq \omega_K(\chi(z, w)). \quad (5.11)$$

The same estimate holds for f^{-1} .

To prove this, first recall that the family of curves crossing a round annulus $A(r, R)$ has extremal length

$$\lambda(A(r, R)) = \frac{1}{2\pi} \log \frac{R}{r}. \quad (5.12)$$

If two continua E, F in the sphere have large relative separation, a round annulus separating them with modulus tending to infinity is obtained by centering at a point of the smaller continuum and comparing its diameter with the distance to the other continuum. Conversely, if both continua have spherical diameter bounded below and their mutual distance is bounded below, every annulus separating them has modulus bounded above. These two elementary comparisons are obtained by monotonicity from (5.12).

Assume (5.11) fails. Then there are normalized K -QC maps f_n and points z_n, w_n with $\chi(z_n, w_n) \rightarrow 0$ but $\chi(f_n(z_n), f_n(w_n)) \geq \varepsilon > 0$. After a subsequence $z_n, w_n \rightarrow z$. Among the three fixed points $0, 1, \infty$, choose two, say a, b , that stay a definite spherical distance from z . A round annulus separating the tiny continuum joining z_n to w_n from a fixed continuum joining a to b has modulus tending to infinity. By the modulus-distortion theorem its image has modulus at least K^{-1} times this quantity. But the image inner continuum has diameter at least ε , while the outer continuum still joins the fixed points a and b ; the reverse ring comparison gives a uniform upper bound, a contradiction. The inverse is normalized and K -QC by planar quasiconformal differential calculus, so the same argument applies to f^{-1} . $\square$

Lemma 5.7 (Closedness of the normalized class). *A uniform spherical limit of normalized K -quasiconformal sphere homeomorphisms is a normalized K -quasiconformal sphere homeomorphism.*

Proof. If normalized K -quasiconformal sphere maps f_n converge uniformly in spherical metric to f , then f is a normalized K -quasiconformal sphere homeomorphism.

By the uniform ring-control lemma the inverse family is equicontinuous as well. Arzela–Ascoli gives, after passing to a further subsequence, uniform convergence $f_n^{-1} \rightarrow g$. Uniform convergence and

$$f_n \circ f_n^{-1} = \text{id}, \quad f_n^{-1} \circ f_n = \text{id}$$

imply $f \circ g = g \circ f = \text{id}$; hence f is a homeomorphism fixing $0, 1, \infty$.

It remains to retain the *same* constant K , not merely some distortion constant. Let Q be a Jordan quadrilateral compactly contained in a planar chart. Uniform convergence of f_n and f_n^{-1} carries the four sides of Q to the corresponding sides of $f(Q)$. The modulus of a Jordan quadrilateral is continuous under such uniform convergence: choose normalized conformal maps of the disk onto the quadrilaterals, use the kernel theorem on compacta and the boundary homeomorphisms to identify the four limiting marked points. Therefore

$$\lambda(f_n \Gamma_Q) \longrightarrow \lambda(f \Gamma_Q). \quad (5.13)$$

The factor- K inequalities for f_n pass to the limit. The converse direction of quadrilateral modulus characterization now shows that f is analytically K -quasiconformal. $\square$

Proof. The uniform ring-control lemma makes the normalized class equicontinuous on the compact sphere, so Arzela–Ascoli gives a uniformly convergent subsequence from any sequence. The closedness lemma shows that its limit belongs to the same normalized K -QC class. Thus the class is compact.

For the stated local variant, the same proof is applied on compactly contained subdomains, with a diagonal subsequence; once the limit is known to be a homeomorphism, the quadrilateral argument gives the exact K bound. Without three-point normalization, postcomposition by Möbius transformations can send two normalizing points together, producing constant degeneration on compacta away from the exceptional pole. This explains precisely the degeneration clause and also why three fixed points exclude it.

Lemma 5.6 gives equicontinuity of the normalized class and its inverse class; Arzelà–Ascoli plus Lemma 5.7 gives compactness. The same argument on compactly contained subdomains yields the local variant, while loss of three-point normalization permits the stated Möbius degeneration. $\square$

Theorem 5.8 (Measurable Riemann mapping theorem). **Statement scope.** Let $\mu \in L^\infty(\mathbb{C})$ satisfy $\|\mu\|_\infty < 1$. There exists an orientation-preserving quasiconformal homeomorphism $f : \mathbb{C} \rightarrow \mathbb{C}$ solving $f_{\bar{z}} = \mu f_z$ almost everywhere. After imposing $f(0) = 0$, $f(1) = 1$, and $f(\infty) = \infty$, the solution is unique.

Proof strategy (not part of the proof)

Proof status. proved here modulo stated analytic input.

Strategic outline only. Near-L2 Cauchy/Beurling L_p package; Neumann-series nonhomogeneous Beltrami solver; smooth principal homeomorphisms via the Ahlfors–Bers integrating factor; measurable compact-support approximation; trun-

cation plus normalized QC compactness; weak $W^{1,2}$ passage; normalized uniqueness.

The mathematical argument begins in the proof environment below.

Proof setup. Let

$$k := \|\mu\|_\infty < 1, \quad K := \frac{1+k}{1-k}. \quad (5.14)$$

The proof is divided into five analytic steps and a final uniqueness argument.

Lemma 5.9 (Cauchy-Beurling transform package). *Let $1 < p < \infty$. The Beurling transform S , initially defined on $C_c^\infty(\mathbb{C})$, extends boundedly to $L^p(\mathbb{C})$. On L^2 it is an isometry. If $C_p = \|\mathcal{S}\|_{L^p \rightarrow L^p}$, then*

$$\lim_{p \rightarrow 2} C_p = 1. \quad (5.15)$$

For compactly supported $g \in L^p$ with $p > 2$, the Cauchy transform is continuous, vanishes at infinity, and satisfies

$$\partial_{\bar{z}} \mathcal{C}g = g, \quad \partial_z \mathcal{C}g = Sg \quad (5.16)$$

in distributions, together with the Morrey estimate

$$|\mathcal{C}g(z) - \mathcal{C}g(w)| \leq A_{p,R} \|g\|_p |z - w|^{1-2/p} \quad (5.17)$$

when $\text{supp } g \subset D(0, R)$.

Proof. We include the singular-integral argument because (5.15) is used later in the proof. The kernel

$$B(z) = -\frac{1}{\pi z^2}$$

has zero mean on every centered annulus and obeys $|\nabla B(z)| \leq C|z|^{-3}$. Fourier transformation gives

$$\widehat{\mathcal{S}g}(\xi) = m(\xi) \widehat{g}(\xi), \quad |m(\xi)| = 1 \quad (\xi \neq 0),$$

so Plancherel yields $\|\mathcal{S}g\|_2 = \|g\|_2$.

For completeness, the Calderón-Zygmund step is as follows. Given $g \in L^1 \cap L^2$ and height $\alpha > 0$, perform the dyadic CZ decomposition $g = u + \sum_Q b_Q$, where

$$|u| \leq C\alpha, \quad \|u\|_2^2 \leq C\alpha \|g\|_1, \quad \int_Q b_Q = 0, \quad \sum_Q |Q| \leq \alpha^{-1} \|g\|_1. \quad (5.18)$$

The L^2 estimate and Chebyshev give

$$|\{|Su| > \alpha/2\}| \leq C\alpha^{-1} \|g\|_1. \quad (5.19)$$

Outside the triples $3Q$, cancellation and the kernel derivative bound imply

$$\int_{\mathbb{C} \setminus 3Q} |Sb_Q(z)| dA(z) \leq C \|b_Q\|_1. \quad (5.20)$$

Summing (5.20), using (5.18), and adding the area of $\bigcup 3Q$ gives the weak $(1, 1)$ inequality

$$|\{|Sg| > \alpha\}| \leq C\alpha^{-1}\|g\|_1. \quad (5.21)$$

Marcinkiewicz interpolation with the L^2 isometry gives boundedness for $1 < p \leq 2$; duality gives $2 \leq p < \infty$. Finally fix $q > 2$. Interpolating the L^2 norm 1 with the finite L^q norm yields, for $2 < p < q$,

$$C_p \leq C_q^{\theta(p)}, \quad \theta(p) \rightarrow 0 \quad (p \downarrow 2),$$

so $\limsup_{p \downarrow 2} C_p \leq 1$. Applying the same statement to the inverse Fourier multiplier gives the reverse liminf, proving (5.15).

The distributional identities (5.16) follow from $\partial_{\bar{z}}(\pi^{-1}z^{-1}) = \delta_0$ and differentiation of the Cauchy kernel. To obtain (5.17), write the difference of kernels, split the plane into $|\zeta - z| \leq 2|z - w|$ and its complement, and apply Hölder's inequality. The two integrals are respectively bounded by constants times $|z - w|^{1-2/p}$; compact support controls the harmless far tail. The same kernel estimate shows $Cg(z) = o(1)$ at infinity.

Because $k < 1$, (5.15) lets us choose once and for all

$$p > 2 \quad \text{so close to 2 that} \quad kC_p < 1. \quad (5.22)$$

□

Lemma 5.10 (Nonhomogeneous Beltrami solver). *Assume for the moment that μ has compact support. If $\sigma \in L^p$ has compact support, then*

$$\omega_{\bar{z}} = \mu\omega_z + \sigma, \quad \omega(z) = o(1) \quad (z \rightarrow \infty), \quad (5.23)$$

has a unique solution with $\omega_z, \omega_{\bar{z}} \in L^p$. Moreover the solution depends continuously on μ (under a.e. convergence with a common L^∞ bound k) and on σ in L^p .

Proof. Set $q = \omega_z$. By (5.16), (5.23) is equivalent to

$$q = S(\mu q) + S\sigma. \quad (5.24)$$

The operator $q \mapsto S(\mu q)$ has norm at most $kC_p < 1$, so the Neumann series gives a unique $q \in L^p$ and

$$\|q\|_p \leq \frac{C_p}{1 - kC_p} \|\sigma\|_p. \quad (5.25)$$

Then

$$\omega = \mathcal{C}(\mu q + \sigma) \quad (5.26)$$

has the asserted derivatives and decay. If $\mu_n \rightarrow \mu$ a.e., $\|\mu_n\|_\infty \leq k$, and $\sigma_n \rightarrow \sigma$ in L^p , subtracting (5.24) and using the uniform inverse bound $(1 - kC_p)^{-1}$ plus dominated convergence shows $q_n \rightarrow q$ in L^p ; (5.17) then gives local uniform convergence of the solutions.

□

Lemma 5.11 (Smooth compact-support principal solution). *If $\mu \in C_c^\infty(\mathbb{C})$ and $\|\mu\|_\infty \leq k$, then the equation $f_{\bar{z}} = \mu f_z$ has a principal K -quasiconformal homeomorphic solution.*

Proof. Apply Lemma 5.10 with $\sigma = \mu$ and put

$$f_0(z) = z + \omega(z). \quad (5.27)$$

Then f_0 solves the homogeneous Beltrami equation and is asymptotic to z at infinity. It remains to prove injectivity; this is the point at which one must not mistake an elliptic weak solution for a homeomorphism.

Apply Lemma 5.10 again, now with $\sigma = \mu_z$, to obtain

$$\lambda_{\bar{z}} = \mu \lambda_z + \mu_z, \quad \lambda(z) \rightarrow 0 \quad (z \rightarrow \infty). \quad (5.28)$$

Smoothness of μ and the integral equation bootstrap λ to C^1 . Set $\rho = e^\lambda$. Equation (5.28) gives

$$\rho_{\bar{z}} = \mu \rho_z + \mu_z \rho = (\mu \rho)_z. \quad (5.29)$$

Hence the C^1 one-form determined by

$$F_z = \rho, \quad F_{\bar{z}} = \mu \rho \quad (5.30)$$

is closed; the plane is simply connected, so it integrates to a global C^1 map F . Its Jacobian is

$$J_F = |\rho|^2(1 - |\mu|^2) > 0. \quad (5.31)$$

Thus F is an orientation-preserving local homeomorphism. We still need a justified asymptotic at infinity: $\lambda(z) \rightarrow 0$ alone would only control the derivative and does not by itself imply $F(z) = z + b + o(1)$.

Because (5.26) represents λ as the Cauchy transform of a compactly supported L^p function, in fact

$$\lambda(z) = O(|z|^{-1}), \quad \rho(z) - 1 = e^{\lambda(z)} - 1 = O(|z|^{-1}). \quad (5.32a)$$

Since $p > 2$, $\rho - 1 \in L^p(\mathbb{C})$. The weak principal solution $f_0 = z + \omega$ from (5.27) has $(f_0 - z)_z = \omega_z \in L^p$. Hence $G := F - f_0$ satisfies $G_z \in L^p$ and, because F and f_0 solve the same homogeneous Beltrami equation,

$$G_{\bar{z}} = \mu G_z. \quad (5.32b)$$

Applying the contraction equation (5.24) with $\sigma = 0$ to G_z gives $G_z = 0$, and then (5.32b) gives $G_{\bar{z}} = 0$. Thus G is a constant, say b . Since $f_0(z) = z + o(1)$ by construction,

$$F(z) = z + b + o(1) \quad (z \rightarrow \infty). \quad (5.32)$$

This proves properness. A proper local homeomorphism of $\mathbb{C}$ onto its (open-and-closed) image is a covering of $\mathbb{C}$; the target is simply connected, so the covering has one sheet. Thus F is a homeomorphism of the plane, and $F - b = f_0$ is the required principal homeomorphism. Finally (5.3) gives the maximal dilatation $K = (1 + k)/(1 - k)$. $\square$

Lemma 5.12 (Measurable compact-support coefficient). *If μ is merely measurable, compactly supported, and $\|\mu\|_\infty \leq k < 1$, then it has a principal K -quasiconformal homeomorphic solution.*

Proof. Choose $\mu_n \in C_c^\infty$ supported in one fixed disk, with $\|\mu_n\|_\infty \leq k$, $\mu_n \rightarrow \mu$ almost everywhere and in L^p . Let f_n be the principal maps from Lemma 5.11. Lemma 5.10 applied to $\sigma = \mu_n$ shows that the corresponding derivatives $q_n = (f_n - z)_z$ converge in L^p , and (5.17) gives

$$f_n \longrightarrow f \quad \text{locally uniformly,} \quad (5.33)$$

where f solves $f_{\bar{z}} = \mu f_z$ and is principal at infinity.

We verify that the limiting solution remains a homeomorphism. Normalize

$$F_n(z) = \frac{f_n(z) - f_n(0)}{f_n(1) - f_n(0)}. \quad (5.34)$$

These are normalized K -QC sphere homeomorphisms, so the normal-family compactness for normalized qc maps gives a subsequence converging uniformly to a normalized K -QC homeomorphism F . The denominators in (5.34) cannot tend to zero: the uniform L^p bound in Lemma 5.10 and the fixed support give a uniform estimate $f_n(z) = z + O(1/|z|)$ for large $|z|$, while $f_n(0)$ stays uniformly bounded by the Cauchy-transform estimate. If the denominator tended to zero, (5.34) and the uniform convergence of F_n would force $f_n(R) - f_n(0) \rightarrow 0$ at a fixed sufficiently large R , contradicting $f_n(R) = R + O(R^{-1})$. Thus, after a subsequence, the denominator tends to a nonzero number, and (5.33) expresses f as an affine postcomposition of the homeomorphism F . Hence f is itself a K -QC homeomorphism. Uniqueness of the principal normalization makes the conclusion independent of the chosen approximation. $\square$

Lemma 5.13 (Arbitrary measurable coefficient). *For every $\mu \in L^\infty(\mathbb{C})$ with $\|\mu\|_\infty \leq k < 1$, there is a normalized K -quasiconformal sphere homeomorphism solving $f_{\bar{z}} = \mu f_z$ almost everywhere on $\mathbb{C}$.*

Proof. Let

$$\mu_R = \mu \mathbf{1}_{D(0,R)}.$$

By Lemma 5.12, μ_R has a principal solution p_R . Affinely normalize it by

$$f_R(z) = \frac{p_R(z) - p_R(0)}{p_R(1) - p_R(0)}. \quad (5.35)$$

Each f_R fixes $0, 1, \infty$, has Beltrami coefficient μ_R , and is K -quasiconformal. By the normal-family compactness for normalized qc maps choose $R_j \rightarrow \infty$ such that

$$f_{R_j} \longrightarrow f \quad (5.36)$$

uniformly in spherical metric, where f is a normalized K -QC homeomorphism.

It remains to identify its coefficient. Fix a disk $D(0, m)$. For $R_j > m$,

$$(f_{R_j})_{\bar{z}} = \mu(f_{R_j})_z \quad \text{a.e. on } D(0, m). \quad (5.37)$$

On a slightly larger disk, uniform spherical convergence and the fact that the limit sends finite points to finite points give a common Euclidean bound on the images. By the distortion inequality and the area formula,

$$\int_{D(0, m)} |Df_{R_j}|^2 dA \leq K \int_{D(0, m)} J_{f_{R_j}} dA \leq K A(f_{R_j}(D(0, m))) \leq C_m. \quad (5.38)$$

Thus, after a subsequence, the Wirtinger derivatives converge weakly in L^2 to the weak derivatives of the uniform limit f . Testing (5.37) against $\varphi \in C_c^\infty(D(0, m))$ and using that $\mu\varphi \in L^2$ gives

$$\int (f_{R_j})_{\bar{z}} \varphi dA = \int (f_{R_j})_z (\mu\varphi) dA \longrightarrow \int f_z (\mu\varphi) dA.$$

Hence $f_{\bar{z}} = \mu f_z$ on $D(0, m)$. Since m was arbitrary, the equation holds almost everywhere on the plane. $\square$

Lemma 5.14 (Normalized uniqueness). *Two quasiconformal homeomorphisms of the sphere with the same Beltrami coefficient differ by postcomposition with a Möbius transformation. Hence two solutions fixing $0, 1, \infty$ coincide.*

Proof. Let f and g have the same coefficient μ and put $h = g \circ f^{-1}$. The composition and inverse formulas from Lemma 5.2 give, almost everywhere,

$$h_{\bar{w}} = 0. \quad (5.39)$$

The map h is locally $W^{1,2}$. Weyl's lemma for the $\bar{\partial}$ operator follows by convolving with a smooth approximate identity: the convolutions are holomorphic on slightly smaller domains because their $\bar{\partial}$ derivative is zero; local L^1 convergence then identifies h almost everywhere with a holomorphic map. Continuity makes the identification pointwise. Thus h is a conformal sphere homeomorphism, hence Möbius. If f and g both fix $0, 1, \infty$, then h fixes those three points and is the identity. $\square$

Proof of Theorem 5.8. Lemma 5.13 supplies a normalized solution for every measurable coefficient with $\|\mu\|_\infty < 1$, and Lemma 5.14 supplies normalized uniqueness. This is the measurable Riemann mapping theorem. $\square$

Theorem 5.15 (Quasisymmetric boundary values). **Statement scope.** *Every orientation-preserving K -quasiconformal self-homeomorphism of the unit disk extends to an orientation-preserving homeomorphism of the closed disk whose boundary value on S^1 is η_K -quasisymmetric. Conversely, every orientation-preserving η -quasisymmetric homeomorphism of S^1 has a $K(\eta)$ -quasiconformal extension to the disk. Only dependence on the indicated control data is asserted.*

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Boundary extension by semiannulus/ring modulus; quantitative quasisymmetry by half-plane quadrilateral moduli; Beurling-Ahlfors averaging extension with explicit derivative/Jacobian estimates; Möbius transfer to the circle.

The mathematical argument begins in the proof environment below.

Lemma 5.16 (Boundary extension of a disk QC map). *Every K -quasiconformal homeomorphism $f : \mathbb{D} \rightarrow \mathbb{D}$ extends uniquely to a homeomorphism $\bar{f} : \bar{\mathbb{D}} \rightarrow \bar{\mathbb{D}}$.*

Proof. Fix $\zeta \in S^1$. A sequence tending to ζ cannot have an image subsequence converging to an interior point $w \in \mathbb{D}$, because continuity of f^{-1} would then force the original subsequence to converge to $f^{-1}(w) \in \mathbb{D}$. Hence every cluster point lies on S^1 .

We show that the cluster set is a singleton. Let $0 < r < R < 1/4$ and consider the semiannulus

$$A_{r,R} = \{z \in \mathbb{D} : r < |z - \zeta| < R\}.$$

Reflection across the tangent line after a local conformal chart, or direct comparison with a half-annulus, gives

$$\lambda(\Gamma_{r,R}) \asymp \log(R/r), \quad (5.40)$$

for the family $\Gamma_{r,R}$ separating the two boundary crosscuts; the constants are absolute. Thus the extremal length tends to infinity as $r \downarrow 0$. By the modulus-distortion theorem the same is true, up to the factor K , for the image family.

If the cluster set of f at ζ contained two distinct points of S^1 , choose disjoint boundary arcs around them. Arbitrarily small source caps at ζ would have images meeting both arcs, hence containing a continuum of spherical diameter bounded below. On the other hand, for r small those image caps stay a fixed distance from the interior point $f(0)$. The elementary round-annulus comparison used in the uniform ring-control lemma then gives a uniform upper bound for the extremal length of a family separating that image continuum from a fixed neighborhood of $f(0)$, contradicting (5.40). Therefore $f(z)$ has a unique limit as $z \rightarrow \zeta$. Compactness of S^1 and the same ring estimate make the resulting boundary map continuous.

Apply the same argument to f^{-1} . Its boundary extension is inverse to the boundary extension of f , because the identities $f^{-1}(f(z)) = z$ pass to radial (or arbitrary interior) limits. Thus the boundary extension is a homeomorphism. Orientation is preserved because cyclic order of three boundary points is the limit of cyclic order of nearby interior crosscuts. $\square$

Lemma 5.17 (QC boundary values are quantitatively quasisymmetric). *For each K there is $M(K) < \infty$ such that the boundary value of every K -quasiconformal self-homeomorphism of the upper half-plane fixing infinity satisfies*

$$M(K)^{-1} \leq \frac{h(x+t) - h(x)}{h(x) - h(x-t)} \leq M(K) \quad (5.41)$$

for all $x \in \mathbb{R}$ and $t > 0$.

Proof. By Lemma 5.16, the boundary value h exists. Normalize the source by $A(z) = (z - x)/t$ and the target by

$$B(w) = \frac{w - h(x)}{h(x) - h(x - t)}.$$

The conjugated map is still K -quasiconformal and its boundary sends $-1, 0, 1, \infty$ to $-1, 0, r, \infty$, where

$$r = \frac{h(x + t) - h(x)}{h(x) - h(x - t)}.$$

Let Q_1 denote the marked half-plane quadrilateral $(-1, 0, 1, \infty)$ and Q_r the corresponding quadrilateral $(-1, 0, r, \infty)$. the modulus-distortion theorem gives

$$K^{-1}\lambda(Q_1) \leq \lambda(Q_r) \leq K\lambda(Q_1). \quad (5.42)$$

The conformal modulus of a four-point half-plane quadrilateral is a continuous strictly monotone function of its cross-ratio; in this normalization it tends to one endpoint of the extended interval as $r \downarrow 0$ and to the other as $r \rightarrow \infty$. Hence the compact interval in (5.42) forces $M(K)^{-1} \leq r \leq M(K)$, proving (5.41). The adjacent-interval criterion is equivalent to an η_K -quasisymmetry function. Möbius charts transfer the result to S^1 ; compactness of the circle makes the finitely many chart controls into one function η_K . $\square$

Lemma 5.18 (Beurling-Ahlfors extension). *Let $h : \mathbb{R} \rightarrow \mathbb{R}$ be increasing and M -quasisymmetric. Then h has a $K(M)$ -quasiconformal extension $H : \mathbb{H} \rightarrow \mathbb{H}$.*

Proof. For $z = x + iy$, $y > 0$, define

$$U(x, y) = \frac{1}{2y} \int_{x-y}^{x+y} h(s) ds, \quad (5.43)$$

$$V(x, y) = \frac{1}{2y} \left(\int_x^{x+y} h(s) ds - \int_{x-y}^x h(s) ds \right) > 0, \quad H = U + iV. \quad (5.44)$$

Because h is continuous, these expressions are C^1 for $y > 0$ even when h is singular as a one-dimensional function. Put

$$a = h(x + y) - h(x), \quad b = h(x) - h(x - y), \quad s = a + b.$$

Quasisymmetry gives

$$M^{-1} \leq a/b \leq M. \quad (5.45)$$

We shall also use the elementary lower bound

$$V(x, y) \geq \frac{s}{4(M+1)}.$$

Indeed, on $t \in [y/2, y]$ the integrand $h(x+t) - h(x-t)$ is at least $h(x+y/2) - h(x-y/2)$, and the adjacent-interval estimate at the two half-scale midpoints shows that this latter increment is at least $s/(M+1)$.

We now give the distortion computation explicitly. Translation and positive affine

rescaling in source and target do not change the distortion, so for the calculation we may take $x = 0$, $y = 1$, and $h(0) = 0$. Write

$$A = \int_0^1 h(t) dt, \quad B = - \int_{-1}^0 h(t) dt. \quad (5.46)$$

Monotonicity gives $0 \leq A \leq a$ and $0 \leq B \leq b$. Direct differentiation of (5.43)-(5.44), in this normalization, gives

$$U_x = \frac{a+b}{2}, \quad V_x = \frac{a-b}{2}, \quad (5.47)$$

$$U_y = \frac{a-b-A+B}{2}, \quad V_y = \frac{a+b-A-B}{2}. \quad (5.48)$$

Hence every entry of DH is bounded in absolute value by s , so

$$|DH|_{\text{op}} \leq 2s. \quad (5.49)$$

More importantly, the Jacobian has the exact form

$$\begin{aligned} J_H &= U_x V_y - U_y V_x \\ &= ab - \frac{bA + aB}{2} = \frac{b(a-A) + a(b-B)}{2}. \end{aligned} \quad (5.50)$$

This identity makes positivity quantitative. Apply the adjacent-interval estimate at $x = 1/2$, $t = 1/2$. If $c = h(1/2)$, then

$$\frac{a-c}{c} \geq M^{-1}, \quad \text{hence} \quad c \leq \frac{M}{M+1}a.$$

Therefore

$$A \leq \frac{1}{2}c + \frac{1}{2}a \leq \left(1 - \frac{1}{2(M+1)}\right)a. \quad (5.51)$$

Applying the same argument to the reflected interval $[-1, 0]$ yields

$$B \leq \left(1 - \frac{1}{2(M+1)}\right)b. \quad (5.52)$$

Substitution into (5.50) gives

$$J_H \geq \frac{ab}{2(M+1)}. \quad (5.53)$$

Since (5.45) implies

$$ab \geq \frac{M}{(M+1)^2}(a+b)^2,$$

we obtain

$$J_H \geq c_M s^2 > 0. \quad (5.54)$$

Undoing the source scale y multiplies the derivative by y^{-1} , so (5.49) and (5.54) become

$$|DH(x, y)|_{\text{op}} \leq 2\frac{s}{y}, \quad J_H(x, y) \geq c_M \left(\frac{s}{y}\right)^2. \quad (5.55)$$

Consequently

$$|DH|_{\text{op}}^2 \leq K(M)J_H, \quad (5.56)$$

and H is locally quasiconformal. Notice that this computation uses only monotonicity and the adjacent-interval quasisymmetry bound; no differentiability of h on the boundary has been assumed.

The averages in (5.43)–(5.44) satisfy $U(x, y) \rightarrow h(x)$ and $V(x, y) \rightarrow 0$ as $y \downarrow 0$. Moreover (5.47) gives $U_x > 0$, and (5.54) makes H a local orientation-preserving homeomorphism. We next verify properness rather than appealing to it implicitly. If $H(x_n, y_n)$ stays in a compact subset of $\mathbb{H}$, then $U(x_n, y_n)$ and $V(x_n, y_n)$ are bounded while $V(x_n, y_n)$ is bounded away from zero. The lower bound above makes

$$s_n = h(x_n + y_n) - h(x_n - y_n)$$

uniformly bounded. Since $U(x_n, y_n)$ lies between the two endpoint values $h(x_n - y_n)$ and $h(x_n + y_n)$, both endpoint values stay in a fixed compact interval. The properness of the line homeomorphism h therefore bounds $x_n - y_n$ and $x_n + y_n$, hence bounds both x_n and y_n . Finally y_n cannot tend to zero: on a bounded x -interval, uniform continuity of h gives $s_n \rightarrow 0$, and then $0 < V(x_n, y_n) \leq s_n/2 \rightarrow 0$, contradicting compactness of the image set. Thus preimages of compact subsets of $\mathbb{H}$ are compact.

A proper local homeomorphism of $\mathbb{H}$ is a covering onto an open-and-closed subset of $\mathbb{H}$; the image is nonempty and $\mathbb{H}$ is connected, so it is onto. Since $\mathbb{H}$ is simply connected, the covering has one sheet. Hence H is a homeomorphism of $\mathbb{H}$ onto itself. This proves the extension lemma. $\square$

Proof of Theorem 5.15. Now let $h : S^1 \rightarrow S^1$ be orientation-preserving and η -quasisymmetric. Choose Möbius maps carrying one source point and its image to infinity. The conjugated boundary map of $\mathbb{R}$ satisfies an adjacent-interval bound $M = M(\eta)$, so the Lemma 5.18 gives a $K(M)$ -QC half-plane extension. Conjugating back gives a $K(\eta)$ -quasiconformal self-homeomorphism of the disk with boundary value h . Together with the Lemma 5.16 and the Lemma 5.17, this completes both directions. $\square$

5.3 Worked examples

Worked example: The affine stretch realizes the extremal distortion constant

Fix $K \geq 1$ and define

$$f_K(x + iy) = x + iKy.$$

In complex notation,

$$f_K(z) = \frac{1+K}{2}z + \frac{1-K}{2}\bar{z},$$

so

$$(f_K)_z = \frac{1+K}{2}, \quad (f_K)_{\bar{z}} = \frac{1-K}{2}, \quad \mu_{f_K} = \frac{1-K}{1+K}.$$

Hence

$$|\mu_{f_K}| = \frac{K-1}{K+1} \quad \text{and} \quad \frac{1+|\mu_{f_K}|}{1-|\mu_{f_K}|} = K.$$

The singular values of the derivative are 1 and K , while the Jacobian is K . Thus

$$|Df_K|_{\text{op}}^2 = K^2 = KJ_{f_K},$$

so the differential quasiconformal inequality is sharp.

The same map makes the modulus estimate sharp. Let $R = [0, w] \times [0, h]$ and let Γ_h be the curves joining the two horizontal sides. Then

$$\lambda(\Gamma_h) = \frac{h}{w}, \quad \lambda(f_K \Gamma_h) = \frac{Kh}{w} = K\lambda(\Gamma_h).$$

For the family Γ_v joining the two vertical sides,

$$\lambda(\Gamma_v) = \frac{w}{h}, \quad \lambda(f_K \Gamma_v) = \frac{w}{Kh} = K^{-1}\lambda(\Gamma_v).$$

Both endpoints of the two-sided modulus theorem therefore occur already for an affine map.

Finally, f_K fixes 0, 1, and ∞ . Thus it is also the normalized solution of the constant Beltrami equation with coefficient $(1-K)/(1+K)$. In this model the measurable Riemann mapping theorem is visible without approximation or compactness.

Worked example: Why three-point normalization matters

Every Möbius transformation is 1-quasiconformal. Nevertheless the sequence

$$m_n(z) = nz$$

does not have a homeomorphic locally uniform limit on the sphere: away from 0 it converges spherically to ∞ , while $m_n(0) = 0$. The family is therefore not compact inside the class of sphere homeomorphisms.

This is not a failure of quasiconformal distortion control; it is the residual three-real-dimensional conformal freedom. If each map fixes 0, 1, and ∞ , that freedom disappears. Ring-modulus estimates then give a common modulus of continuity for both maps and inverses, and Arzelà–Ascoli produces a homeomorphic quasiconformal limit.

Boundary case: A homeomorphism with no uniform quasiconformal constant

Consider

$$F(x + iy) = x + iy^3.$$

It is an orientation-preserving homeomorphism of the plane and belongs locally to $W^{1,2}$. Away from $y = 0$,

$$F_z = \frac{1 + 3y^2}{2}, \quad F_{\bar{z}} = \frac{1 - 3y^2}{2},$$

so

$$|\mu_F(x + iy)| = \frac{|1 - 3y^2|}{1 + 3y^2}.$$

As $y \rightarrow 0$, this ratio approaches 1. Consequently

$$\|\mu_F\|_\infty = 1$$

in the essential-supremum sense, and no finite K satisfies the quasiconformal distortion inequality globally. Thus the strict hypothesis $\|\mu\|_\infty < 1$ in the measurable Riemann mapping theorem is genuinely essential: the associated Neumann-series operator ceases to have a uniform contraction margin, and the factor- K modulus estimates cease to exist.

Common pitfall

Do not confuse “ $|\mu(z)| < 1$ almost everywhere” with the uniform condition $\|\mu\|_\infty < 1$. Pointwise strict inequality can hold while the essential supremum is one. The measurable Riemann mapping theorem in this chapter requires a uniform gap from one.

5.4 Chapter summary

What to remember

- A planar quasiconformal map can be described analytically by a Beltrami coefficient or geometrically by bounded distortion of extremal length; the two viewpoints are quantitatively equivalent at the level needed in this book.
- Three-point-normalized K -quasiconformal sphere homeomorphisms form a compact family. Controlling the maps and their inverses prevents a limit from collapsing.
- The measurable Riemann mapping theorem is proved by combining near- L^2 control of the Beurling transform, a Neumann-series solver, an integrating-factor homeomorphism argument for smooth coefficients, compactness for measurable coefficients, and truncation for noncompact support.
- Two solutions of the same Beltrami equation differ by a conformal postcomposition; fixing $0, 1, \infty$ removes this ambiguity.
- Quasiconformal disk maps have quasisymmetric boundary values. Conversely, the Beurling-Ahlfors averaging construction extends every orientation-preserving quasisymmetric boundary map quasiconformally.

- The strict bound $\|\mu\|_\infty < 1$ is essential for uniform finite distortion; merely having $|\mu| < 1$ pointwise is not enough.
- In later surgery arguments, an invariant Beltrami coefficient is integrated and the Beltrami composition formula makes the conjugated quasiregular model holomorphic.

Looking ahead

Chapter 6 uses this deformation theory to study holomorphic motions and Teichmüller space. Chapter 7 then applies those deformation tools to Sullivan stability and rigidity. In Chapter 8, the measurable Riemann mapping theorem becomes a central proof engine in straightening polynomial-like maps, after which the tuning theorem postponed from Chapter 4 can finally be proved at its natural logical location.

Holomorphic Motions and Teichmüller Theory

Why this chapter matters

A holomorphic family often begins with very little geometric information: we may know only how a finite set of periodic points moves. Holomorphic motions turn that sparse information into a global deformation tool. Their definition is strikingly weak—holomorphy is required only in the parameter and injectivity only slice by slice—yet the lambda lemma forces continuity and quasiconformal geometry in the spatial variable. Slodkowski’s theorem goes further: a motion of an arbitrary subset of the sphere extends, over the same parameter disk, to a motion of the entire sphere.

The second deformation engine comes from the Beltrami equation. Chapter 5 proved that every coefficient with norm strictly below one has a unique normalized quasiconformal integration. Here we ask what happens when the coefficient itself varies holomorphically. The Ahlfors–Bers theorem says that the normalized solution varies holomorphically as well. This supplies analytic coordinates on deformation spaces and explains why quasiconformal surgery can produce genuinely holomorphic families of rational maps.

These two engines meet in dynamics but are logically independent. Repelling periodic points can be continued by the implicit-function theorem; density of repelling cycles and the lambda lemma then move the whole Julia set. Separately, an invariant Beltrami coefficient can be integrated so that the conjugated map is rational. Keeping those two chains distinct is essential for avoiding the circular dependency that often hides inside informal accounts of stability.

Prerequisites. Chapter 2 density of repelling periodic points; Chapter 5 quasiconformal geometry, normalized compactness, and the measurable Riemann mapping theorem; graduate complex analysis including Schwarz–Pick, Montel, Hurwitz, one-variable Runge separation, Schwarz reflection, and the holomorphic implicit-function theorem; standard real-analytic complexification, boundedness of the Hilbert transform on Hölder spaces, and Banach-space ODE existence/uniqueness.

Learning goals

- State precisely what a holomorphic motion does and does not assume.
- Separate the classical lambda lemma from Slodkowski’s whole-sphere extension theorem.

- Understand the finite-motion/analytic-disc mechanism behind the Slodkowski extension theorem.
- Prove Ahlfors–Bers holomorphic dependence from the normalized Beltrami solution operator rather than from the lambda lemma.
- Use a dense holomorphically moving set of repelling cycles to obtain a holomorphic Julia-set conjugacy.
- Integrate an invariant Beltrami coefficient and verify directly that the conjugated map is rational.
- Interpret these constructions in the marking model of Teichmüller space without importing later extremal or rigidity theorems.

Conceptual roadmap

There are two independent spines:

finite/partial holomorphic motion



Slodkowski extension



lambda-lemma QC control



motion of a dense repelling set



Julia-set conjugacy

and

Chapter 5 normalized MRMT



Ahlfors–Bers holomorphic dependence



holomorphic Beltrami deformations



rational conjugate from an invariant coefficient

Teichmüller space records such quasiconformal changes together with their marking data. Chapter 7 will add the converse stability criteria and identify the bifurcation locus.

How the main results fit together. Do not use the lambda lemma to prove Ahlfors–Bers dependence, and do not use Julia-set stability to prove invariant-Beltrami straightening. Those are different mechanisms. The theorem on Julia sets in this chapter assumes repelling-periodic stability; Chapter 7 proves the deeper equivalence with the usual J-stability and critical-passivity criteria.

6.1 Definitions

Definition 6.1 (holomorphic motion). Let Λ be a complex manifold, let $\lambda_0 \in \Lambda$, and let $E \subset \widehat{\mathbb{C}}$. A *holomorphic motion* of E over (Λ, λ_0) is a map

$$h : \Lambda \times E \longrightarrow \widehat{\mathbb{C}}$$

such that $h(\lambda_0, z) = z$ for every $z \in E$, each slice $h_\lambda = h(\lambda, \cdot)$ is injective on E , and $\lambda \mapsto h(\lambda, z)$ is holomorphic for every fixed $z \in E$. No continuity in z is included in the definition; it is a theorem.

Definition 6.2 (Teichmüller space of a marked surface). Fix an oriented Riemann surface X of finite type, including its prescribed punctures or marked points. A *marking* is an orientation-preserving quasiconformal homeomorphism $f : X \rightarrow Y$ to another Riemann surface of the same topological type. Two markings $f_i : X \rightarrow Y_i$ are equivalent if there is a conformal isomorphism $c : Y_1 \rightarrow Y_2$ such that $c \circ f_1$ is isotopic to f_2 relative to the prescribed marking data. The set of equivalence classes is $T(X)$. Its Teichmüller distance is

$$d_T([f_1], [f_2]) = \frac{1}{2} \inf_h \log K(h),$$

where $h : Y_1 \rightarrow Y_2$ ranges over quasiconformal maps in the induced marking class. This volume uses this marking model; no extremal-map existence theorem is silently built into the definition.

Definition 6.3 (classical λ -lemma). In this book, the *classical λ -lemma* means the closure, continuity, and quasiconformal-slice consequences for a holomorphic motion over a disk: the motion extends uniquely to the closure of the moved set, the extended motion is jointly continuous, and every fixed-parameter slice is a quasiconformal embedding and admits a quasiconformal sphere extension. The stronger assertion that the entire motion extends holomorphically to the whole sphere is called the Slodkowski extension theorem.

Definition 6.4 (quasiconformal deformation). A *quasiconformal deformation* of a rational map f is a family obtained from Beltrami coefficients μ_t with $\|\mu_t\|_\infty < 1$ by taking their normalized integrations Φ_t and setting $g_t = \Phi_t \circ f \circ \Phi_t^{-1}$ whenever the invariance condition on μ_t makes g_t holomorphic. The coefficient family, its normalization, and its invariance equation are part of the data.

Definition 6.5 (normalized holomorphic motion). A holomorphic motion over $(\mathbb{D}, 0)$ is *normalized* when the moved set contains $0, 1, \infty$ and every slice fixes those three points. Any motion of three distinguished points can be normalized by postcomposition with the unique parameter-dependent Möbius transformation sending their images to $0, 1, \infty$.

Definition 6.6 (Slodkowski extension). A *Slodkowski extension* of a holomorphic motion $h : \mathbb{D} \times E \rightarrow \widehat{\mathbb{C}}$ is a holomorphic motion $H : \mathbb{D} \times \widehat{\mathbb{C}} \rightarrow \widehat{\mathbb{C}}$ over the *same* parameter disk such that $H|_{\mathbb{D} \times E} = h$.

Definition 6.7 (holomorphic L^∞ family). A map $t \mapsto \mu_t$ from a complex parameter domain to L^∞ is *holomorphic* when it is Fréchet-holomorphic as a map into the complex

Banach space L^∞ . In finite-dimensional parameter domains this is equivalent to local boundedness together with holomorphy after every continuous linear functional on L^∞ is applied.

Definition 6.8 (normalized Beltrami solution). For a sphere Beltrami coefficient μ with $\|\mu\|_\infty < 1$, its *normalized Beltrami solution* is the unique quasiconformal sphere homeomorphism w^μ solving $(w^\mu)_z = \mu(w^\mu)_z$ almost everywhere and fixing $0, 1, \infty$.

Definition 6.9 (periodic-type-stable parameter neighborhood). Let $f : U \times \widehat{\mathbb{C}} \rightarrow \widehat{\mathbb{C}}$ be a holomorphic family of fixed-degree rational maps. A simply connected neighborhood U of λ_0 is *periodic-type stable for the repelling spectrum* if every repelling periodic point of f_{λ_0} has a single-valued holomorphic continuation throughout U , distinct continuations never collide, remain repelling, and for every $\lambda \in U$ these continuations exhaust all repelling periodic points of f_λ . The Mañé-Sad-Sullivan stable locus has this property; the equivalence with the other standard J-stability criteria is proved later in Chapter 7.

Definition 6.10 (holomorphic motion conjugacy of Julia sets). A *holomorphic-motion conjugacy of Julia sets* for a family f_λ is a holomorphic motion $h : U \times J(f_{\lambda_0}) \rightarrow \widehat{\mathbb{C}}$ such that

$$h_\lambda(J(f_{\lambda_0})) = J(f_\lambda), \quad h_\lambda \circ f_{\lambda_0} = f_\lambda \circ h_\lambda \quad \text{on } J(f_{\lambda_0}).$$

Definition 6.11 (pullback and invariant Beltrami coefficient). If f is holomorphic and μ is a Beltrami coefficient, define off the critical set

$$(f^*\mu)(z) = \mu(f(z)) \frac{\overline{f'(z)}}{f'(z)}.$$

Values on the finite critical set are irrelevant. The coefficient is *f-invariant* when $f^*\mu = \mu$ almost everywhere.

6.2 Main results

Theorem 6.1 (λ -lemma and Ślodkowski extension theorem). **Statement scope.** Every holomorphic motion $h : \mathbb{D} \times E \rightarrow \widehat{\mathbb{C}}$ extends to a holomorphic motion of all of $\widehat{\mathbb{C}}$ over the same disk. Consequently it extends uniquely to $\overline{E}$, the extension is jointly continuous, and every slice is the restriction of a quasiconformal sphere homeomorphism.

Proof strategy (not part of the proof)

Proof status. proved here modulo stated analytic input.

Strategic outline only. Finite radial structures; polynomial-hull collar control and Schwarz reflection; Banach-ODE attached-disc exhaustion; finite extension; holomorphic axiom of choice; maximal extension; then punctured-sphere metric control for the classical lambda-lemma consequences.

The mathematical argument begins in the proof environment below.

Proof setup. It is enough to work over the unit parameter disk $\mathbb{D}$ with base point 0. A general simply connected parameter disk is reduced to this case by a Riemann map, and a different base point is moved to 0 by a disk automorphism. The proof has two logically different parts. First we prove the whole-sphere extension theorem. Only afterwards do we derive the classical continuity and quasiconformality conclusions usually called the λ -lemma.

The analytic tools used below are standard ones: real-analytic complexification and the inverse-function theorem, one-variable Runge separation, Schwarz reflection across a real-analytic arc, boundedness of the Hilbert transform on C^α for $0 < \alpha < 1$, and the local existence/uniqueness theorem for ordinary differential equations in a Banach space. The finite-dimensional radial interpolation needed below is written out explicitly. The research-level content is therefore reconstructed rather than imported.

Lemma 6.2 (Radial structures for a finite motion). *Let*

$$a_0 \equiv 0, a_1, \dots, a_N : \overline{\mathbb{D}} \longrightarrow D(0, R)$$

be pairwise disjoint holomorphic graphs which are real-analytic on a neighborhood of $\mathbb{D}$. After enlarging R , one can choose, for every $\zeta \in \partial\mathbb{D}$, a real-analytic radial structure

$$X_\zeta^a, \quad a \in \partial D(0, R), \quad (6.1)$$

consisting of pairwise disjoint simple arcs from 0 to a , depending real-analytically on (a, ζ) , so that the boundary trace $\zeta \mapsto a_j(\zeta)$ is carried by one distinguished leaf of the structure.

Proof. Start with the straight radial foliation at the base parameter. Write $\zeta = e^{i\theta}$ and let $r \in [0, 1]$. The finitely many traces $a_j(r\zeta)$ remain pairwise disjoint. Put

$$\rho(w) = R^2 - |w|^2$$

and, after enlarging R so that every marked trace stays strictly inside the circle, define the real-analytic interpolation functions

$$L_j(r, \zeta, w) = \frac{\rho(w)}{\rho(a_j(r\zeta))} \prod_{k \neq j} \frac{|w - a_k(r\zeta)|^2}{|a_j(r\zeta) - a_k(r\zeta)|^2}.$$

The denominators are uniformly nonzero by compactness and disjointness. Thus $L_j = 0$ on $|w| = R$ and $L_j(r, \zeta, a_i(r\zeta)) = \delta_{ij}$. Hence

$$Y(r, \zeta, w) = \sum_{j=0}^N \partial_r a_j(r\zeta) L_j(r, \zeta, w)$$

is a bounded real-analytic nonautonomous vector field on a neighborhood of the relevant compact set, vanishes on the outer circle, and satisfies

$$Y(r, \zeta, a_j(r\zeta)) = \partial_r a_j(r\zeta) \quad (0 \leq j \leq N).$$

Extend it boundedly outside the closed disk if necessary.

The associated nonautonomous ODE has a global flow $\Phi_{r,\zeta}$ on the closed disk. It fixes $\partial D(0, R)$, begins at the identity, and uniqueness gives

$$\Phi_{r,\zeta}(a_j(0)) = a_j(r\zeta).$$

Apply $\Phi_{1,\zeta}$ to the straight segments from 0 to the outer circle. The resulting arcs have all the asserted properties. Since every $\Phi_{1,\zeta}$ is a diffeomorphism, different leaves meet only at the common center 0. $\square$

For $a \in \partial D(0, R)$ let $\mathcal{F}^a$ be the set of functions $g \in A(\mathbb{D})$ such that

$$g(\zeta) \in X_\zeta^a \quad (\zeta \in \partial \mathbb{D}), \quad (6.2)$$

and put $\mathcal{F} = \bigcup_{|a|=R} \mathcal{F}^a$. The next lemma is the continuation estimate that was still imported in the preceding edition.

Lemma 6.3 (Uniform reflection and compactness of attached discs). *There exist $\varepsilon > 0$ and $K < \infty$, depending only on the radial structure, such that every $g \in \mathcal{F}$ extends holomorphically to $D(0, 1 + \varepsilon)$ and satisfies*

$$\sup_{|z| < 1 + \varepsilon} |g(z)| \leq K. \quad (6.3)$$

Consequently $\mathcal{F}$ and every $\mathcal{F}^a$ are compact in $C^{1/2}(\partial \mathbb{D})$. In addition, the only attached disc which meets the common center on the boundary is the zero disc, and the only disc in $\mathcal{F}^a$ which meets the outer endpoint a on the boundary is the constant disc $g \equiv a$.

Proof. Let

$$X^a = \bigcup_{\zeta \in \partial \mathbb{D}} \{\zeta\} \times X_\zeta^a \subset \partial \mathbb{D} \times \mathbb{C}.$$

The real-analytic parametrization of Lemma 6.2 shows that, uniformly in a , a neighborhood of X^a admits a holomorphic coordinate change

$$(z, w) \mapsto (z, B_z^a(w)) \quad (6.4)$$

which sends X^a to $\partial \mathbb{D} \times [0, 1]$. This follows by complexifying the real-analytic map $(\zeta, s) \mapsto (\zeta, b(a, \zeta, s))$ and using the holomorphic inverse function theorem. The only point that requires care is to know that all attached graphs enter one common coordinate neighborhood near the boundary.

We prove precisely that collar statement. Let $\hat{X}^a$ denote the polynomial hull of X^a in $\mathbb{C}^2$. If $g \in \mathcal{F}^a$, then its graph lies in $\hat{X}^a$: for every polynomial $P(z, w)$, the function $z \mapsto P(z, g(z))$ is holomorphic, so the maximum principle bounds it by its boundary values on X^a .

For $\zeta_0 \in \partial \mathbb{D}$ the fiber of this hull is exactly the radial arc:

$$\{w : (\zeta_0, w) \in \hat{X}^a\} = X_{\zeta_0}^a. \quad (6.5)$$

Indeed, if $w_0 \notin X_{\zeta_0}^a$, the simple arc $X_{\zeta_0}^a \subset \mathbb{C}$ is polynomially convex. Hence one can choose a polynomial $p(w)$ with

$$|p(w_0)| > 1, \quad \sup_{w \in X_{\zeta_0}^a} |p(w)| < 1.$$

By continuous dependence of the leaves the same strict upper bound holds for ζ near ζ_0 . On the remaining compact part of the boundary set, let M bound $|p|$. For n large the polynomial

$$P_n(z, w) = \left(\frac{1 + \bar{\zeta}_0 z}{2} \right)^n p(w) \quad (6.6)$$

has modulus < 1 on all of X^a but modulus > 1 at (ζ_0, w_0) . Thus $(\zeta_0, w_0) \notin \hat{X}^a$, proving (6.5).

Now allow a to vary. The set

$$\mathcal{Y} = \{(a, z, w) : |a| = R, (z, w) \in \hat{X}^a\} \quad (6.7)$$

is compact. Boundedness is immediate. For closedness, if $a_n \rightarrow a$ and $(z_n, w_n) \in \hat{X}^{a_n}$ converges to (z, w) , then for every polynomial P ,

$$|P(z, w)| \leq \lim_{n \rightarrow \infty} \sup_{X^{a_n}} |P| = \sup_{X^a} |P|,$$

so $(z, w) \in \hat{X}^a$. By (6.5), the part of $\mathcal{Y}$ over $|z| = 1$ is exactly the real-analytic boundary surface. Hence compactness gives $\rho < 1$ such that every attached graph over $\rho \leq |z| \leq 1$ lies in one common coordinate neighborhood on which (6.4) and its inverse are defined.

For such a graph set

$$h(z) = B_z^a(g(z)), \quad \rho < |z| < 1.$$

This is holomorphic in the annulus and has real boundary values in $[0, 1]$ on $|z| = 1$. Schwarz reflection across the unit circle gives

$$h(z) = \overline{h(1/\bar{z})}, \quad 1 < |z| < \rho^{-1}. \quad (6.8)$$

Applying the inverse of (6.4) reconstructs g holomorphically across the circle. All coordinate neighborhoods came from compact families, so after shrinking the common reflected collar we obtain one $\varepsilon > 0$ and one bound K for every $g \in \mathcal{F}$. This proves (6.3).

Cauchy estimates on a slightly smaller disk now give a uniform C^1 bound on $\mathbb{D}$. Thus every sequence in $\mathcal{F}$ has a subsequence converging in $C^{1/2}(\partial\mathbb{D})$, and the closed attachment condition (6.2) passes to the limit. Hence $\mathcal{F}$ is compact.

It remains to record the endpoint rigidity used below. We prove the center case; the outer endpoint is identical after replacing g by $a - g$ and using the terminal tangent of the leaf. Let $g \in \mathcal{F}^a$ and suppose $g(\zeta_0) = 0$. Along the boundary write

$$g(\zeta) = r(\zeta) \tau_0(\zeta), \quad r(\zeta) \geq 0, \quad (6.9)$$

where $\tau_0(\zeta)$ is the unit initial tangent of X_ζ^a at the center. The tangent field has winding

number zero because the radial structures were obtained by an isotopy from straight radial segments. Choose a real-analytic argument α with $\tau_0 = e^{i\alpha}$. The standard Hilbert-transform construction produces a nowhere-vanishing holomorphic function H on a neighborhood of $\mathbb{D}$ whose boundary direction is $H/|H| = \tau_0$. Therefore

$$k = g/H$$

is holomorphic on $\mathbb{D}$, extends across the boundary by (6.3), and has real nonnegative boundary values. Schwarz reflection extends k to the Riemann sphere; it is holomorphic also at infinity because the reflected function tends to $\overline{k(0)}$ there. Hence k is constant. Since $k(\zeta_0) = 0$, it is identically zero and $g \equiv 0$. The same argument at the terminal tangent shows that if $g(\zeta_0) = a$, then $g \equiv a$. $\square$

Lemma 6.4 (ODE exhaustion of an attached-disc class). *Fix $a \in \partial D(0, R)$. The nonconstant elements of $\mathcal{F}^a$ form one Banach-ODE orbit joining the zero disc to the constant disc a . For every $\zeta_0 \in \partial\mathbb{D}$, evaluation*

$$g \mapsto g(\zeta_0) \tag{6.10}$$

is a homeomorphism from $\mathcal{F}^a$ onto the leaf $X_{\zeta_0}^a$.

Proof. Let

$$E = C^{1/2}(\partial\mathbb{D}) \cap A(\mathbb{D})$$

and let T be the Hilbert transform on $C^{1/2}$, normalized to have mean zero. Away from the center write $A(\zeta, w)$ for the relative tangent angle of the leaf through w :

$$\tau(\zeta, w) = \frac{w}{|w|} e^{iA(\zeta, w)}. \tag{6.11}$$

The same isotopy argument as above gives a single real-analytic branch of A on $\partial\mathbb{D} \times (D(0, R + \delta) \setminus \{0\})$ for a small $\delta > 0$.

For g with no boundary zero define

$$\mathcal{V}(g) = g \exp(-T[A(\cdot, g(\cdot))] + iA(\cdot, g(\cdot))). \tag{6.12}$$

The Nemytskii map $g \mapsto A(\cdot, g(\cdot))$ is locally Lipschitz on every set whose boundary values stay a positive distance from 0; boundedness of T on $C^{1/2}$ and the Banach-algebra property of $C^{1/2}$ therefore make $\mathcal{V}$ locally Lipschitz as a map into E . The Banach-space ODE

$$\dot{g}_t = \mathcal{V}(g_t) \tag{6.13}$$

has unique local solutions.

If $g_t \in \mathcal{F}^a$, then on the boundary

$$\dot{g}_t(\zeta) = |g_t(\zeta)| e^{-T[A(\cdot, g_t)](\zeta)} \tau(\zeta, g_t(\zeta)). \tag{6.14}$$

Thus every boundary point moves in the positive tangent direction of its leaf. In particular the attachment condition is preserved as long as the point has not reached an

endpoint.

Consider a maximal orbit inside $\mathcal{F}^a \setminus \{0, a\}$. If, in one time direction, every boundary value remained in a compact subarc separated from both endpoints, then Lemma 6.3 would give compactness in E and local ODE existence at every limiting disc; the orbit could then be continued. Hence at a maximal end some boundary value approaches an endpoint. Compactness again gives a limiting attached disc, and the endpoint-rigidity part of Lemma 6.3 forces that limit to be either 0 or the constant disc a . Formula (6.14) fixes the orientation: backward time tends to 0, forward time tends to a .

For a fixed ζ_0 , (6.14) shows that evaluation along the orbit is strictly monotone in the oriented leaf coordinate. It therefore maps the orbit homeomorphically onto the open arc $X_{\zeta_0}^a \setminus \{0, a\}$, and the two limiting discs add the endpoints. Finally, two different ODE orbits in $\mathcal{F}^a$ cannot occur: both approach the constant disc a , and local uniqueness for (6.13), run backward from a point sufficiently near that disc, makes them coincide. This proves (6.10). $\square$

Lemma 6.5 (Slodkowski attached-disc foliation). *For the radial structures constructed in Lemma 6.2 there is a family of analytic discs whose graphs are pairwise disjoint and fill $\mathbb{D} \times D(0, R)$. More precisely, evaluation at one boundary parameter labels the discs uniquely, and for every $\lambda \in \mathbb{D}$ the evaluation map from the label disk to $D(0, R)$ is a homeomorphism.*

Proof. Fix $\zeta_0 \in \partial\mathbb{D}$. The radial structure at ζ_0 foliates $D(0, R)$ by the leaves $X_{\zeta_0}^a$, all sharing only the center. By Lemma 6.4, each $u \in D(0, R)$ therefore labels a unique attached disc g^u with

$$g^u(\zeta_0) = u. \quad (6.15)$$

The compactness lemma and the homeomorphism statement in Lemma 6.4 show that $u \mapsto g^u$ is continuous in $C^{1/2}$.

Take $u \neq v$. For every boundary point ζ , the points $g^u(\zeta)$ and $g^v(\zeta)$ lie on different locations in the radial foliation and therefore do not coincide. Hence $g^u - g^v$ has no boundary zero. The winding number of this boundary difference is locally constant in (u, v) on $D(0, R)^2 \setminus \{u = v\}$, which is connected. For two distinct labels on the outer circle the discs are constant, so the winding number is zero. The argument principle gives

$$g^u(\lambda) \neq g^v(\lambda) \quad (\lambda \in \mathbb{D}). \quad (6.16)$$

Thus, for every λ , the evaluation map

$$\phi_\lambda(u) = g^u(\lambda) \quad (6.17)$$

is continuous and injective on the closed disk and is the identity on its boundary. Invariance of domain makes the interior image open; the image is also compact and has boundary equal to the outer circle. Equivalently, its boundary degree is one. Hence ϕ_λ is a homeomorphism of $\overline{D(0, R)}$ onto itself. The graphs are therefore pairwise disjoint and fill the bidisk. The distinguished discs carrying the original marked boundary traces are, by uniqueness in (6.15), exactly the original holomorphic graphs a_j . $\square$

Lemma 6.6 (Finite holomorphic-motion extension). *Every holomorphic motion of a finite subset of $\widehat{\mathbb{C}}$ over $\mathbb{D}$ extends to a holomorphic motion of the whole sphere.*

Proof. First suppose the moving coordinate functions extend holomorphically past $\overline{\mathbb{D}}$. After a fixed Möbius change of the spatial coordinate, put one marked point at infinity and all remaining graphs in $D(0, R)$. Let g^u be the attached discs supplied by Lemma 6.5 and define

$$\phi_\lambda(u) = g^u(\lambda), \quad |u| \leq R.$$

Each ϕ_λ is a homeomorphism of the disk, and the outer boundary is fixed by the radial structure. Define

$$F_\lambda(w) = \begin{cases} \phi_\lambda \circ \phi_0^{-1}(w), & |w| \leq R, \\ w, & |w| \geq R. \end{cases} \quad (6.18)$$

The formulas agree on $|w| = R$. For fixed w , the first line is one of the attached analytic discs after a constant relabelling, hence is holomorphic in λ . For fixed λ , (6.18) is injective, fixes the exterior disk, and moves every original marked point along its prescribed graph. Thus it is the required whole-sphere motion in the boundary-regular case.

For a finite motion defined only on the open parameter disk, apply the regular case to the rescaled motions on $r\overline{\mathbb{D}}$, $r < 1$. If the finite set has at least three points, normalize three moving graphs to $0, 1, \infty$. For every compact parameter disk $|\lambda| \leq s < 1$, Schwarz-Pick applied to four-point cross-ratios of the resulting whole-sphere motions gives a common quasimetric modulus, independent of $r > s$. Hence the slices form an equicontinuous family in the spherical metric. Choose $r_n \uparrow 1$ and diagonalize first on a countable dense set of spatial points and then use this common equicontinuity to extend the limit to every point of the sphere. For each spatial point the convergence is locally uniform in λ , so the limit graph is holomorphic. Hurwitz applied after the three-point normalization preserves disjointness of distinct graphs, and the base slice is the identity. This gives the finite extension on all of $\mathbb{D}$. With one marked point a Möbius translation gives an extension directly; with two, a parameter-dependent affine map does so. $\square$

Lemma 6.7 (Holomorphic axiom of choice). *Let $h : \mathbb{D} \times E \rightarrow \widehat{\mathbb{C}}$ be a holomorphic motion and let $w_0 \notin E$. There exists a holomorphic map $g : \mathbb{D} \rightarrow \widehat{\mathbb{C}}$ such that*

$$g(0) = w_0, \quad g(\lambda) \notin h_\lambda(E) \quad (\lambda \in \mathbb{D}). \quad (6.19)$$

Proof. Direct the finite subsets $A \subset E$ by inclusion and restrict attention to those containing three fixed points of E ; if E has fewer than three points, adjoin auxiliary fixed points first. Lemma 6.6 applied to the finite motion on $A \cup \{w_0\}$ gives a holomorphic section g_A through w_0 disjoint from the moving graphs of A . Let M_λ be the Möbius map taking the three chosen moving points to $0, 1, \infty$. Then $M_\lambda(g_A(\lambda))$ omits $0, 1, \infty$, so the family is normal. Its compact-open closure is compact; consequently the net has a convergent subnet, with holomorphic limit G .

Fix $x \in E$. Eventually the directed finite sets contain x , and therefore

$$M_\lambda(g_A(\lambda)) - M_\lambda(h_\lambda(x)) \neq 0.$$

Hurwitz's theorem says that the limiting difference is either nowhere zero or identically zero. At $\lambda = 0$ it is nonzero because $w_0 \neq x$, so it is nowhere zero. This holds for every $x \in E$. Undoing M_λ gives the required section g . $\square$

Lemma 6.8 (Slodkowski maximal extension). *Every holomorphic motion $h : \mathbb{D} \times E \rightarrow \widehat{\mathbb{C}}$ extends to a holomorphic motion of all of $\widehat{\mathbb{C}}$ over the same parameter disk.*

Proof. Partially order extensions of h by inclusion of their base fibers. The union of a chain is again a holomorphic motion: injectivity and pointwise holomorphy are checked on one member of the chain containing the finitely many points in question. Zorn's lemma therefore gives a maximal extension $H : \mathbb{D} \times E_{\max} \rightarrow \widehat{\mathbb{C}}$. If $E_{\max} \neq \widehat{\mathbb{C}}$, choose w_0 outside it. Lemma 6.7 gives a holomorphic section through w_0 disjoint from every moving graph of $E_{\max}$; adjoining that graph contradicts maximality. Hence $E_{\max} = \widehat{\mathbb{C}}$. $\square$

Lemma 6.9 (Classical λ -lemma consequences). *For a whole-sphere holomorphic motion H , each slice H_λ is a quasiconformal sphere homeomorphism. Moreover the motion is jointly continuous; therefore the restriction of an arbitrary motion to E has a unique extension to $\overline{E}$.*

Proof. Normalize three moving points by a parameter-dependent Möbius map. Fix $r < 1$ and an ordered quadruple (z_0, z_1, z_2, z_3) of distinct points whose base cross-ratio is a fixed harmonic value, say $[z_0, z_1; z_2, z_3] = -1$. Then

$$q(\lambda) = [H_\lambda(z_0), H_\lambda(z_1); H_\lambda(z_2), H_\lambda(z_3)] \quad (6.20)$$

is holomorphic and omits $0, 1, \infty$. Schwarz-Pick for the complete Poincaré metric of the thrice-punctured sphere gives

$$d_{\mathbb{C} \setminus \{0,1\}}(q(\lambda), -1) \leq d_{\mathbb{D}}(\lambda, 0), \quad |\lambda| \leq r. \quad (6.21)$$

Thus the image cross-ratio of every harmonic quadruple lies in one compact subset of $\widehat{\mathbb{C}} \setminus \{0, 1, \infty\}$ depending only on r . The four-point criterion for quasimetricity gives a distortion function η_r such that all slices with $|\lambda| \leq r$ are η_r -quasimetric in the spherical metric. By the quadrilateral-modulus criterion of Chapter 5 this is equivalent to a uniform quasiconformal bound.

In particular every H_λ is continuous. It is already injective by the definition of holomorphic motion, so compactness of the sphere and invariance of domain make it onto. Hence every slice is a quasiconformal sphere homeomorphism. The same uniform quasimetric estimate supplies a common modulus of continuity on $|\lambda| \leq r$. Since pointwise parameter dependence is holomorphic, hence continuous, the two-term estimate

$$\chi(H_\lambda(z), H_{\lambda_0}(z_0)) \leq \chi(H_\lambda(z), H_\lambda(z_0)) + \chi(H_\lambda(z_0), H_{\lambda_0}(z_0))$$

proves joint continuity.

Now start with the original motion h on E . Lemma 6.8 supplies a whole-sphere extension H , hence restriction gives a continuous extension to $\bar{E}$. If two such extensions existed on $\bar{E}$, they would agree on the dense set E and therefore everywhere by continuity. This is the classical closure part of the λ -lemma, while the preceding maximal extension lemma is the stronger Slodkowski theorem. $\square$

Proof of Theorem 6.1. Lemma 6.2 supplies the finite radial geometry. Lemmas 6.3 and 6.4 give the reflection compactness and global Banach-ODE continuation that produce the attached-disc foliation in Lemma 6.5. This yields the finite extension lemma. Normal families then give the holomorphic axiom of choice, Zorn's lemma gives whole-sphere extension, and the punctured-sphere Schwarz–Pick argument gives the classical continuity and quasiconformal conclusions. Thus all research-level steps of the Slodkowski extension argument are internal to the chapter, modulo the standard analytic tools stated at the beginning of the proof. $\square$

Theorem 6.10 (Ahlfors–Bers holomorphic dependence). **Statement scope.** *Let $t \mapsto \mu_t$ be a holomorphic L^∞ family of sphere Beltrami coefficients on a complex parameter domain, locally uniformly satisfying $\|\mu_t\|_\infty < 1$. Then the normalized solutions w^{μ_t} depend holomorphically on t : for every z , $t \mapsto w^{\mu_t}(z)$ is holomorphic, and locally the dependence is holomorphic in a normalized $W^{1,p}$ -Hölder Banach topology for some $p > 2$.*

Proof strategy (not part of the proof)

Proof status. proved here modulo stated analytic input.

Strategic outline only. Normalized Cauchy/Beurling solver; near-zero Neumann inversion; identify via MRMT uniqueness; change the base coefficient by the Beltrami composition law.

The mathematical argument begins in the proof environment below.

Proof setup. Let $\Omega \subset \mathbb{C}^m$ be a parameter domain and let $t \mapsto \mu_t$ be holomorphic as a map into L^∞ Beltrami differentials on the sphere. Assume locally

$$\sup_{t \in U} \|\mu_t\|_\infty \leq k < 1. \quad (6.9)$$

Write w_t for the unique normalized solution of $(w_t)_{\bar{z}} = \mu_t(w_t)_z$ furnished by the measurable Riemann mapping theorem in Chapter 5. We prove that $t \mapsto w_t(z)$ is holomorphic for every z , and in fact obtain the local Banach-space form of the assertion.

Lemma 6.11 (Normalized spherical $\bar{\partial}$ solver). *For every $p > 2$ there is a bounded normalized right inverse $\mathcal{P}$ for $\bar{\partial}$ on spherical Beltrami differentials, with $\partial\mathcal{P}$ bounded on L^p and the corresponding $W^{1,p}$ -Hölder estimate.*

Proof. For every $p > 2$ there is a bounded linear operator $\mathcal{P}$ from L^p Beltrami differentials on the sphere to normalized $W^{1,p}$ vector fields such that

$$\bar{\partial}(\mathcal{P}\beta) = \beta, \quad (\mathcal{P}\beta)(0) = (\mathcal{P}\beta)(1) = 0, \quad \mathcal{P}\beta = o(z^2) \quad (z \rightarrow \infty). \quad (6.10)$$

Moreover $S_0 := \partial\mathcal{P}$ is bounded on the same L^p space, and $\mathcal{P}\beta$ is locally $C^{1-2/p}$ with a norm bounded by $C_p\|\beta\|_p$.

In one affine chart the normalized Cauchy kernel is

$$(\mathcal{P}\beta)(z) = \frac{1}{\pi} \int_{\mathbb{C}} \beta(\zeta) \frac{z(z-1)}{\zeta(\zeta-1)(z-\zeta)} dA(\zeta), \quad (6.11)$$

first for smooth differentials supported away from $0, 1, \infty$. The factor in front removes the three-dimensional holomorphic-vector-field ambiguity and makes the integral convergent at infinity. Distributional differentiation gives $\bar{\partial}\mathcal{P}\beta = \beta$. In a chart at infinity the kernel has the same Calderón–Zygmund singular part as the ordinary Cauchy kernel; the difference is smooth and finite rank. A two-chart partition of unity and The Cauchy–Beurling estimates proved in Volume 5 therefore give the L^p boundedness of S_0 , while Morrey’s inequality gives the stated Hölder estimate. Density extends (6.10)–(6.11) to arbitrary L^p Beltrami differentials. $\square$

Lemma 6.12 (Ahlfors–Bers dependence near the zero coefficient). *For coefficients in a sufficiently small L^∞ ball, the normalized Beltrami solution depends Banach-holomorphically on the coefficient in the local $W^{1,p}$ -Hölder topology.*

Proof. There is $\varepsilon_p > 0$ such that for $\|\nu\|_\infty < \varepsilon_p$ the normalized solution g^ν depends holomorphically on ν as a normalized local $W^{1,p}$ - $C^{1-2/p}$ map.

Put $g^\nu(z) = z + v(z)$ in the affine chart and set $q = v_{\bar{z}}$. The Beltrami equation and the normalized dbar-solver lemma imply

$$v = \mathcal{P}q, \quad q = \nu(1 + S_0q). \quad (6.12)$$

Choose $\varepsilon_p\|S_0\|_{L^p \rightarrow L^p} < 1$. Then

$$q(\nu) = (I - M_\nu S_0)^{-1}\nu = \sum_{n \geq 0} (M_\nu S_0)^n \nu \quad (6.13)$$

converges in L^p , locally uniformly for ν in the L^∞ ball. Every term is a continuous homogeneous polynomial in ν , so (6.13) is a Banach-holomorphic map. Hence $v(\nu) = \mathcal{P}q(\nu)$ is Banach-holomorphic.

There is no hidden existence issue in this calculation. Chapter 5 already gives the normalized quasiconformal solution g^ν . Its higher-integrability estimate, proved there by choosing an exponent close enough to 2, places its weak derivatives in the present L^p space after ε_p is decreased if necessary. Thus its q satisfies (6.12), and Neumann-series uniqueness forces it to be exactly (6.13). The analytic vector-field solution is therefore the MRMT homeomorphism, not merely a formal weak solution. $\square$

Lemma 6.13 (Change of base Beltrami coefficient). *Composition with the normalized solution at a base coefficient transforms nearby coefficients Banach-holomorphically to coefficients near zero.*

Proof. Fix a coefficient μ_0 with $\|\mu_0\|_\infty < 1$ and let w_0 be its normalized solution. If w^μ denotes the normalized solution for a nearby coefficient μ , then

$$g^\mu := w^\mu \circ w_0^{-1}$$

is normalized and, at $w = w_0(z)$, has Beltrami coefficient

$$\nu_\mu(w_0(z)) = \frac{(w_0)_z(z)}{(w_0)_z(z)} \frac{\mu(z) - \mu_0(z)}{1 - \overline{\mu_0(z)}\mu(z)} \quad \text{a.e.} \quad (6.14)$$

The formula is the composition/inverse identity proved in the Chapter 5 planar QC calculus. The first factor has modulus one and is fixed. Because $\|\mu_0\|_\infty < 1$, inversion of $1 - \overline{\mu_0}\mu$ is holomorphic in the commutative Banach algebra L^∞ on a neighborhood of μ_0 . Pullback by the fixed homeomorphism w_0 is bounded linear on L^∞ . Hence

$$\mu \longmapsto \nu_\mu \quad (6.15)$$

is Banach-holomorphic and $\nu_{\mu_0} = 0$. $\square$

Proof. Fix $t_0 \in \Omega$. By the change-of-base lemma, $t \mapsto \nu_{\mu_t}$ is an L^∞ -holomorphic map vanishing at t_0 . After shrinking the parameter neighborhood its norm is below ε_p , so the near-zero dependence lemma shows that

$$t \longmapsto g_t := w_t \circ w_{t_0}^{-1} \quad (6.16)$$

is holomorphic in the normalized local $W^{1,p}$ -Hölder topology. Composition on the right by the fixed map w_{t_0} gives $w_t = g_t \circ w_{t_0}$. Point evaluation is continuous in the Hölder topology, hence $t \mapsto w_t(z)$ is holomorphic for every z .

The same argument works along every complex line in a Banach parameter space; for the finite-dimensional parameter domains used in this volume, separate holomorphy plus the local bounds above gives ordinary joint holomorphy. This is precisely the normalized Ahlfors–Bers holomorphic-dependence conclusion.

Fixing a parameter t_0 , Lemma 6.13 reduces the family to one near the zero coefficient, and Lemma 6.12 gives local Banach-holomorphic dependence. Point evaluation is continuous in the Hölder topology, yielding the normalized Ahlfors–Bers holomorphic-dependence theorem. $\square$

Definition 6.12 (Teichmüller marking model). **Statement scope.** For the fixed reference surface X , Chapter 6 uses the marking quotient $T(X)$ and the distance d_T defined above. Beltrami coefficients with essential norm < 1 represent quasiconformal changes of conformal structure after integration; coefficients representing equivalent markings determine the same point. This row is a canonical definition/model, not a separate theorem and not a claim of existence or uniqueness of extremal Teichmüller maps.

Theorem 6.14 (holomorphic motion of Julia sets on stable neighborhoods). **Statement scope.** Let f_λ be a fixed-degree holomorphic family and let U be a simply connected periodic-type-stable neighborhood of λ_0 in the sense of the periodic-type-stability definition above. Then the repelling periodic points of f_{λ_0} move holo-

morphically on U ; this motion extends uniquely to a holomorphic-motion conjugacy $J(f_{\lambda_0}) \rightarrow J(f_\lambda)$, and every slice admits a quasiconformal sphere extension. The full equivalence with J -stability and critical passivity is deferred to Chapter 7.

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Continue repelling cycles by IFT; use the periodic-type-stable hypothesis for noncollision/exhaustion; extend the dense repelling motion using the lambda lemma; extend conjugacy by continuity.

The mathematical argument begins in the proof environment below.

Proof setup. Let $f : U \times \widehat{\mathbb{C}} \rightarrow \widehat{\mathbb{C}}$ be a holomorphic family of rational maps of fixed degree $d \geq 2$, let $\lambda_0 \in U$, and shrink U to the simply connected periodic-type-stable neighborhood specified in the definition of periodic-type stability above. Thus every repelling periodic point of f_{λ_0} has a single-valued holomorphic continuation through U , distinct continuations do not collide, and for every $\lambda \in U$ these continuations exhaust the repelling periodic points of f_λ .

Lemma 6.15 (Local continuation of a repelling cycle). *A repelling periodic point in a holomorphic rational family has a locally unique holomorphic continuation, with exact period and repelling character preserved after shrinking the parameter neighborhood.*

Proof. If p_0 has exact period n for f_{λ_0} and multiplier $\rho_0 = (f_{\lambda_0}^n)'(p_0)$ with $|\rho_0| > 1$, then p_0 has a unique holomorphic continuation as a period- n point near λ_0 .

In a local coordinate write

$$F(\lambda, z) = f_\lambda^n(z) - z.$$

Then

$$\partial_z F(\lambda_0, p_0) = \rho_0 - 1 \neq 0.$$

The holomorphic implicit-function theorem gives a unique holomorphic solution $p(\lambda)$ of $F(\lambda, p(\lambda)) = 0$. Exact period remains n after shrinking, because a drop of period would force one of the finitely many holomorphic functions $f_\lambda^m(p(\lambda)) - p(\lambda)$, $m < n$, to acquire zeros accumulating at λ_0 . Repelling character is open because the multiplier varies continuously. The periodic-type-stability hypothesis is precisely what permits these local germs to be continued through all of U without collision or monodromy. $\square$

Proof. Let P_0 be the set of repelling periodic points of f_{λ_0} . For $p \in P_0$ denote its global continuation by $p(\lambda)$ and define

$$h : U \times P_0 \longrightarrow \widehat{\mathbb{C}}, \quad h(\lambda, p) = p(\lambda). \quad (6.17)$$

At λ_0 this is the identity. Each parameter function is holomorphic, and the no-collision clause makes each h_λ injective. Thus (6.17) is a holomorphic motion of P_0 .

By the density theorem for repelling periodic points proved in Volume 2,

$$\overline{P_0} = J(f_{\lambda_0}). \quad (6.18)$$

Apply the lambda lemma and Slodkowski extension theorem proved above. The motion extends uniquely and jointly continuously to $J(f_{\lambda_0})$, and each slice admits a quasiconformal sphere extension. We next identify its image. Periodic-type stability gives

$$h_\lambda(P_0) = P_\lambda, \quad (6.19)$$

where P_λ is the full set of repelling periodic points of f_λ . Since h_λ is a homeomorphism onto its image on the compact set $J(f_{\lambda_0})$,

$$h_\lambda(J(f_{\lambda_0})) = h_\lambda(\overline{P_0}) = \overline{h_\lambda(P_0)} = \overline{P_\lambda} = J(f_\lambda), \quad (6.20)$$

again by density of repelling cycles.

Finally, the dynamical identity holds first on P_0 . Indeed, if $p(\lambda)$ is the continuation of p , then $f_\lambda(p(\lambda))$ is the continuation of $f_{\lambda_0}(p)$ on the same periodic orbit, so uniqueness in the local continuation lemma gives

$$h_\lambda(f_{\lambda_0}(p)) = f_\lambda(h_\lambda(p)). \quad (6.21)$$

Both sides are continuous in p , and P_0 is dense in the Julia set. Therefore (6.21) extends to every $z \in J(f_{\lambda_0})$. We have obtained a holomorphic motion of Julia sets with compatible dynamical conjugacy and quasiconformal slice extensions.

The Mañé-Sad-Sullivan periodic-type-stable locus $H(f)$ satisfies the hypothesis used here. The converse implications and the equivalence with critical passivity/J-stability belong to Chapter 7; they are not used in this proof.

Using Lemma 6.15 and periodic-type stability, the repelling cycles form a holomorphic motion. Density of repelling cycles and the λ -lemma extend it to the Julia set; continuity then extends the dynamical conjugacy from the dense repelling set. This proves the asserted Julia-set motion and quasiconformal slice extension. $\square$

Theorem 6.16 (invariant-Beltrami quasiconformal deformation). **Statement scope.** *Let f be rational of degree $d \geq 2$ and let μ satisfy $\|\mu\|_\infty < 1$ and $f^*\mu = \mu$ almost everywhere. If Φ is the normalized integration of μ , then $\Phi \circ f \circ \Phi^{-1}$ is rational of degree d . If $t \mapsto \mu_t$ is a holomorphic locally uniformly bounded family of such f -invariant coefficients, the conjugates form a holomorphic family in Rat_d ; multipliers of periodically continued cycles with multiplier different from 1 vary holomorphically.*

Proof strategy (not part of the proof)

Proof status. derived here.

Strategic outline only. MRMT plus pullback invariance and Beltrami composition to straighten; Ahlfors–Bers for parameter dependence; finite rational interpolation plus IFT for multipliers.

The mathematical argument begins in the proof environment below.

Proof setup. Let f be a rational map of degree $d \geq 2$ and let μ be a measurable Beltrami coefficient on the sphere with $\|\mu\|_\infty < 1$. Off the finite critical set define

$$(f^*\mu)(z) = \mu(f(z)) \frac{\overline{f'(z)}}{f'(z)}, \quad (6.22)$$

and assign arbitrary values on the critical set, which is measure zero. Assume $f^*\mu = \mu$ almost everywhere. Let Φ be the normalized MRMT solution with Beltrami coefficient μ .

Lemma 6.17 (Invariant-Beltrami straightening). *If $f^*\mu = \mu$ almost everywhere and Φ integrates μ , then $\Phi f \Phi^{-1}$ is a rational map of the same degree.*

Proof. The map

$$g = \Phi \circ f \circ \Phi^{-1} \quad (6.23)$$

is rational of degree d .

Because f is holomorphic, the chain rule gives, away from its critical set,

$$\mu_{\Phi \circ f}(z) = \mu_\Phi(f(z)) \frac{\overline{f'(z)}}{f'(z)} = (f^*\mu)(z) = \mu(z) = \mu_\Phi(z). \quad (6.24)$$

Thus $\Phi \circ f$ and Φ have the same Beltrami coefficient. The composition/inverse formula from the Chapter 5 planar QC calculus therefore gives

$$\mu_{(\Phi \circ f) \circ \Phi^{-1}} = 0 \quad \text{a.e.} \quad (6.25)$$

The conjugate is locally $W^{1,2}$, so Weyl's lemma for $\bar{\partial}$ makes it holomorphic. A holomorphic self-map of the Riemann sphere is rational. Since (6.23) is a topological conjugacy, topological degree is preserved; hence $\deg g = d$. $\square$

Lemma 6.18 (Holomorphic rational interpolation). *A pointwise-holomorphic fixed-degree rational family obtained by conjugacy has locally holomorphic homogeneous coefficients, recoverable from $2d + 1$ generic evaluations.*

Proof. Suppose $t \mapsto \Phi_t(z)$ is holomorphic for every z and $g_t = \Phi_t \circ f \circ \Phi_t^{-1}$ has degree d for all t near t_0 . Then $t \mapsto g_t$ is a holomorphic map into the coefficient space Rat_d .

Choose $2d + 1$ points $z_1, \dots, z_{2d+1}$ so that $x_j(t_0) = \Phi_{t_0}(z_j)$ are distinct and avoid the poles needed for an affine coefficient chart. Put

$$x_j(t) = \Phi_t(z_j), \quad y_j(t) = \Phi_t(f(z_j)). \quad (6.26)$$

Both are holomorphic and satisfy $g_t(x_j(t)) = y_j(t)$. Write

$$g_t(z) = \frac{A_t(z)}{B_t(z)}, \quad A_t(z) = \sum_{k=0}^d a_k(t) z^k, \quad B_t(z) = \sum_{k=0}^d b_k(t) z^k. \quad (6.27)$$

The interpolation equations

$$A_t(x_j(t)) - y_j(t)B_t(x_j(t)) = 0, \quad 1 \leq j \leq 2d+1, \quad (6.28)$$

form a holomorphic $(2d+1) \times (2d+2)$ matrix. At t_0 its kernel is one-dimensional: two rational maps of degree at most d agreeing at $2d+1$ distinct non-pole points have cross-multiplied difference of degree at most $2d$ with too many zeros, so they coincide. A nonzero maximal minor therefore remains nonzero near t_0 . Normalize one coefficient to be 1 in that projective chart and solve (6.28) by Cramer's rule. All normalized coefficients are holomorphic. The degree cannot drop because every g_t is topologically conjugate to f . This proves the lemma. $\square$

Proof. L^∞ family with a local uniform norm bound below one, and assume $f^*\mu_t = \mu_t$ almost everywhere for every t . Let Φ_t be the normalized solutions. The Ahlfors-Bers theorem proved above says that $t \mapsto \Phi_t(z)$ is holomorphic for every z . the invariant-straightening lemma gives rational maps

$$g_t = \Phi_t \circ f \circ \Phi_t^{-1}$$

of the same degree, and the rational-interpolation lemma makes their rational coefficients holomorphic in local projective coordinates.

Finally let p_{t_0} be a periodic point of exact period n of g_{t_0} with multiplier not equal to 1. The equation $g_t^n(z) - z = 0$ has nonzero z -derivative at (t_0, p_{t_0}) , so the holomorphic implicit-function theorem gives a local holomorphic continuation p_t . Since the coefficients of g_t are holomorphic,

$$t \mapsto (g_t^n)'(p_t) \quad (6.29)$$

is holomorphic. This is the precise controlled-multiplier conclusion asserted for quasiconformal deformation families.

For an invariant holomorphic Beltrami family, Ahlfors-Bers dependence gives pointwise-holomorphic conjugacies, Lemma 6.17 gives rational conjugates of fixed degree, and Lemma 6.18 gives holomorphic rational coefficients. The implicit-function theorem then gives holomorphic continuation of periodic points with multiplier different from one and holomorphic dependence of their multipliers. $\square$

6.3 Worked examples

Worked example: A four-point motion that cannot be Möbius

Let

$$E = \{0, 1, \infty, 2\}$$

and, for $|\lambda| < 1$, set

$$h_\lambda(0) = 0, \quad h_\lambda(1) = 1, \quad h_\lambda(\infty) = \infty, \quad h_\lambda(2) = 2 + \frac{\lambda}{4}.$$

At the base parameter 0 this is the identity. Every point moves holomorphically. The fourth point remains in the disk $D(2, 1/4)$, which is disjoint from $\{0, 1\}$, so every slice is injective. Hence this is a normalized holomorphic motion. It cannot be produced by a Möbius family. A Möbius transformation fixing 0, 1, and ∞ is the identity, but the fourth point moves when $\lambda \neq 0$. Ślodkowski's theorem therefore supplies genuinely nonconformal sphere homeomorphisms H_λ extending these four prescribed values. The classical lambda lemma then says each H_λ is quasiconformal and the full motion is jointly continuous. This tiny model contains the key logical surprise: four holomorphic graphs and slice injectivity already force a global quasiconformal deformation of the sphere.

Worked example: An explicit Ahlfors-Bers family

Fix a real number $0 < k < 1$ and define, for $|t| < 1$,

$$w_t(z) = \frac{z + tk\bar{z}}{1 + tk}.$$

The denominator is nonzero because $|tk| < 1$. The map fixes 0 and 1, and its affine growth fixes ∞ on the sphere. Its Wirtinger derivatives are

$$(w_t)_z = \frac{1}{1 + tk}, \quad (w_t)_{\bar{z}} = \frac{tk}{1 + tk},$$

so

$$\mu_{w_t} = tk, \quad K(w_t) = \frac{1 + |tk|}{1 - |tk|}.$$

Thus w_t is exactly the normalized solution for the constant coefficient $\mu_t = tk$. For each fixed z ,

$$t \mapsto \frac{z + tk\bar{z}}{1 + tk}$$

is holomorphic. This is the Ahlfors-Bers conclusion in a model where the solution can be written down explicitly.

Worked example: An invariant Beltrami coefficient for z^d

Let $f(z) = z^d$ and set, away from 0 and ∞ ,

$$\mu(z) = k \frac{z}{\bar{z}}, \quad |k| < 1.$$

Then

$$\mu(z^d) = k \frac{z^d}{\bar{z}^d}, \quad \frac{\overline{f'(z)}}{f'(z)} = \frac{\bar{z}^{d-1}}{z^{d-1}},$$

and therefore

$$(f^*\mu)(z) = k \frac{z^d}{\bar{z}^d} \frac{\bar{z}^{d-1}}{z^{d-1}} = k \frac{z}{\bar{z}} = \mu(z).$$

So μ is invariant.

For real $k \in (-1, 1)$ write

$$\beta = \frac{2k}{1-k}.$$

Then the radial stretch

$$\Phi(z) = z|z|^\beta$$

has

$$\mu_\Phi(z) = \frac{\beta}{\beta+2} \frac{z}{\bar{z}} = k \frac{z}{\bar{z}}.$$

Moreover

$$\Phi(z^d) = z^d |z|^{d\beta} = (\Phi(z))^d.$$

Hence the straightened rational map is again z^d . The example shows that an invariant Beltrami structure can be nonzero even when its deformation is a self-conjugacy of the same rational map.

Boundary case: Collision at the edge of parameter space

On $E = \{0, 1\}$ define

$$h_\lambda(0) = 0, \quad h_\lambda(1) = 1 - \lambda, \quad |\lambda| < 1.$$

This is a holomorphic motion on the open unit disk because $1 - \lambda \neq 0$ there. But as $\lambda \rightarrow 1$, the two moved points collide. Thus one cannot extend this same motion as an injective holomorphic motion through the boundary parameter $\lambda = 1$.

The example does not contradict Ślodkowski's theorem: that theorem preserves the given *open* parameter disk. It does not promise extension across a parameter where injectivity itself fails. Correspondingly, the quasiconformal constants of extensions need not stay bounded as the parameter approaches such a collision.

Common pitfall

"Holomorphic motion" does not mean that $z \mapsto h_\lambda(z)$ is holomorphic. Except at the base parameter, a slice is generally only quasiconformal after the lambda lemma is proved. The adjective "holomorphic" refers to the parameter variable.

6.4 Chapter summary

What to remember

- A holomorphic motion assumes pointwise holomorphy in the parameter and slice injectivity, but no spatial continuity. The lambda lemma supplies that missing regularity.
- Ślodkowski's theorem extends any holomorphic motion over the unit disk to the whole Riemann sphere over the same disk. Its proof can be organized as finite extension, holomorphic choice, and a maximality argument.

- Cross-ratios of four moving points take values in the triply punctured sphere. Schwarz–Pick control of those cross-ratios gives bounded eccentricity and hence quasiconformal slices.
- The Ahlfors–Bers theorem is independent of the lambda lemma in the dependency graph. Near the zero Beltrami coefficient it is a Neumann-series statement for the normalized $\bar{\partial}$ solver; a composition formula reduces an arbitrary base coefficient to zero.
- The Teichmüller space used here remembers a quasiconformal deformation together with its marking, modulo conformal equivalence and marking-preserving isotopy.
- On a periodic-type-stable parameter neighborhood, repelling cycles move holomorphically. Their density in the Julia set and the lambda lemma extend that motion to a dynamical conjugacy of Julia sets.
- If $f^*\mu = \mu$, integration of μ makes $\Phi \circ f \circ \Phi^{-1}$ holomorphic and therefore rational. Holomorphic families of invariant coefficients give holomorphic rational families.

Looking ahead

Chapter 7 uses these tools to prove the full Mañé–Sad–Sullivan J-stability criterion, relate stability to critical passivity and bifurcation, and develop Sullivan’s rigidity/deformation arguments. Chapter 8 will use the same invariant Beltrami mechanism in the straightening theorem for polynomial-like maps.

Sullivan Theory, No Wandering Domains and J-Stability

Why this chapter matters

Two of the most important global facts in one-dimensional complex dynamics are negative statements: rational maps have no wandering Fatou components, and bifurcations cannot occupy an open region of parameter space. Neither fact is visible from local fixed-point theory. Both become accessible only after we combine the quasiconformal deformation machinery of the previous chapters with finite-dimensionality and normal-family arguments.

Sullivan's no-wandering theorem begins with a deliberately impossible object: a Fatou component whose forward images never repeat. A late image of such a component would be simply connected and would move univalently forever. That univalence lets us place arbitrary Beltrami data on one component and spread it over its whole grand orbit. Integrating the resulting invariant structures would produce infinitely many independent infinitesimal deformations of a rational map. But the space of degree- d rational maps has complex dimension only $2d + 1$. The contradiction is a model example of rigidity by dimension.

The stability theorem of Mañé–Sad–Sullivan has a different flavor. A single repelling cycle always continues locally by the implicit-function theorem, so its continuation says almost nothing about stability. True J-stability is a simultaneous statement: all periodic types fit together into a holomorphic motion of the Julia set, equivalently no neutral cycle bifurcates, and equivalently every critical orbit is passive. This theorem identifies the critical set as the finite collection of dynamical sensors that detects every bifurcation.

Prerequisites. Volume 2: Fatou–Julia theory, density of repelling periodic points, and periodic Fatou-component classification; Volume 5: the measurable Riemann mapping theorem; Volume 6: Ahlfors–Bers holomorphic dependence, holomorphic motions, and invariant-Beltrami straightening; graduate complex analysis including Montel, Hurwitz, covering spaces, the Riemann mapping theorem, and the holomorphic implicit-function theorem.

Learning goals

- Define wandering and preperiodic Fatou components without confusing images of sets with the Fatou components containing them.
- Follow Sullivan's finite-versus-infinite dimensional contradiction from a hypothetical wandering component to invariant Beltrami deformations.

- Use no wandering together with the earlier periodic classification to classify every Fatou component globally.
- Distinguish automatic continuation of one repelling cycle from genuine J-stability.
- Prove the equivalence between Julia-set holomorphic motion, absence of unpersistent neutral cycles, and passivity of marked critical points.
- Explain why the J-stable locus is open and dense while density of hyperbolicity remains a separate open conjecture.
- Recognize invariant line fields as deformation directions rather than as a theorem of nonexistence.

Conceptual roadmap

The chapter has two independent spines. The no-wandering spine is

hypothetical wandering component $\implies$ eventual disk and univalence
 $\Downarrow$
 invariant Beltrami directions $\implies$ infinitesimal rational deformations
 $\Downarrow$
 infinite-dimensional test space $\implies \dim_{\mathbb{C}} \text{Rat}_d = 2d + 1 \implies$ contradiction.

The stability spine is

periodic-type stability $\iff$ Julia motion
 $\Updownarrow$
 critical passivity $\implies$ open dense stable locus.

The second spine ends at J-stability. It does *not* continue with an arrow to density of hyperbolicity.

How the main results fit together. The periodic Fatou classification was already proved in Volume 2. This chapter uses Sullivan’s theorem only to show that every Fatou component eventually reaches one of those periodic types. Likewise, the stability theorem is logically independent of no wandering: its engine is periodic analytic continuation, holomorphic motion, and critical-orbit normality.

7.1 Definitions

Definition 7.1 (wandering Fatou component). Let U be a Fatou component and let U_n denote the Fatou component containing $f^n(U)$. The component U is *wandering* if the components $U_0, U_1, U_2, \dots$ are pairwise distinct. It is *preperiodic* if $U_m = U_n$ for some $m < n$.

Definition 7.2 (J-stability). A parameter λ_0 in a holomorphic family (f_λ) is *J-stable* if

there is a neighborhood $V \ni \lambda_0$ and a holomorphic motion

$$h : V \times J(f_{\lambda_0}) \longrightarrow \widehat{\mathbb{C}}$$

centered at λ_0 such that $h_\lambda(J(f_{\lambda_0})) = J(f_\lambda)$ and $h_\lambda \circ f_{\lambda_0} = f_\lambda \circ h_\lambda$ on $J(f_{\lambda_0})$ for every $\lambda \in V$.

Definition 7.3 (stability and bifurcation loci). For a fixed holomorphic family, the *stability locus* is the set of J-stable parameters. Its complement in parameter space is the *bifurcation locus*. Thus “bifurcation” in this volume means failure of local dynamical holomorphic motion of the Julia set.

Definition 7.4 (invariant line field). A measurable line field on a measurable set $E \subset \widehat{\mathbb{C}}$ may be encoded by a measurable Beltrami differential μ with $|\mu| = 1$ almost everywhere on E , where multiplying μ by a positive real scalar does not change the unoriented line. It is *f*-invariant if $f^*\mu = \mu$ almost everywhere on the part of E where the pullback is defined.

Definition 7.5 (critically marked family). A degree- d holomorphic family is *critically marked* on a parameter domain V if there are holomorphic maps $c_1, \dots, c_{2d-2} : V \rightarrow \widehat{\mathbb{C}}$ which enumerate the critical points of f_λ , counted with multiplicity, for every $\lambda \in V$. Locally, such a marking is always available after a finite branched cover of parameter space.

Definition 7.6 (passive and active critical points). A marked critical point $c(\lambda)$ is *passive* at λ_0 if the family of meromorphic parameter maps

$$\{\lambda \mapsto f_\lambda^n(c(\lambda)) : n \geq 0\}$$

is normal on some neighborhood of λ_0 . Otherwise it is *active*. Passivity of the critical value is equivalent, since deleting or adding the first iterate does not affect normality.

Definition 7.7 (persistent and unpersistent neutral cycle). Let $p(\lambda)$ be a holomorphic periodic branch with multiplier $\rho(\lambda)$. It is *neutral* at λ_* if $|\rho(\lambda_*)| = 1$. A neutral branch is *persistent* on a neighborhood if ρ is locally constant with values on the unit circle; it is *unpersistent neutral* if it is neutral at a parameter but this local persistence fails.

Definition 7.8 (hyperbolic rational map). For a rational map f , let

$$P(f) = \overline{\bigcup_{n \geq 1} f^n(\text{Crit}(f))}$$

be its postcritical set. The map is *hyperbolic* if $P(f) \cap J(f) = \emptyset$. Hyperbolicity implies J-stability, but the converse is not built into the definition and is false for general holomorphic families.

7.2 Main results

Theorem 7.1 (Sullivan no-wandering-domains theorem). **Statement scope.** *Every Fatou component of a rational map of degree at least two is eventually periodic. Equivalently, rational maps have no wandering Fatou components.*

Proof strategy (not part of the proof)

Proof status. proved here modulo stated analytic input.

Strategic outline only. Baker eventual simple connectivity and univalence; invariant Beltrami spreading; Ahlfors-Bers infinitesimal deformation; repelling-density rigidity; infinite-dimensional boundary-rigid Beltrami test space versus $\dim_{\mathbb{C}} \text{Rat}_d = 2d+1$.

The mathematical argument begins in the proof environment below.

Proof setup. Let f be a rational map of degree $d \geq 2$. Suppose, for contradiction, that U is a wandering Fatou component, and write

$$U_n = f^n(U), \quad n \geq 0.$$

After replacing U by a sufficiently late forward image, we will show that every U_n is simply connected and that

$$f : U_n \longrightarrow U_{n+1} \tag{7.1}$$

is a conformal isomorphism.

Lemma 7.2 (Baker eventual-disk lemma). *If U is a wandering Fatou component of a rational map, then after replacing U by a sufficiently late forward image, every component $U_n = f^n(U)$ is simply connected and every restriction $f : U_n \rightarrow U_{n+1}$ is a conformal isomorphism.*

Proof. Only finitely many Fatou components can contain a critical point of f . Discarding finitely many U_n , we may assume that none contains a critical point. Then every map $f : U_n \rightarrow U_{n+1}$, and hence every iterate $f^n : U_0 \rightarrow U_n$, is an unbranched proper map and therefore a covering map.

Conjugate by a Möbius map so that $\infty \in U_0$. The pairwise disjoint domains U_n , $n \geq 1$, lie in $\mathbb{C} \setminus U_0$, so their spherical areas tend to zero. The iterates form a normal family on U_0 . Any locally uniform subsequential limit must be constant: a nonconstant holomorphic limit would map a compact set onto a set with positive area, while the corresponding images lie in domains whose areas tend to zero. Consequently

$$\text{diam } f^n(K) \longrightarrow 0 \tag{7.2}$$

for every compact $K \Subset U_0$.

Take a loop $\gamma \Subset U_0$, put $\gamma_n = f^n(\gamma)$, and let B_n be the union of the bounded components of $\mathbb{C} \setminus \gamma_n$. The shrinking in (7.2) gives $\text{diam } B_n \rightarrow 0$. For all sufficiently large n , openness of f and $\partial f(B_n) \subset \gamma_{n+1}$ force

$$f(B_n) \subset B_{n+1}. \tag{7.3}$$

Hence all forward images of such a B_n avoid the fixed domain U_0 , which contains three distinct points. Montel's theorem makes the iterates normal on B_n , so $B_n \subset F(f)$. Since B_n meets U_n and its boundary lies in U_n , it belongs to the same Fatou component: $B_n \subset U_n$. Therefore γ_n is null-homotopic in U_n . Lifting that null-homotopy through the covering $f^n : U_0 \rightarrow U_n$ shows that γ is null-homotopic in U_0 . Thus a late U_0 is simply

connected, and the same argument applies to every tail. A covering between simply connected source and target has degree one, so (7.1) is a conformal isomorphism.

Shift indices once and for all so this conclusion holds for every $n \geq 0$. Define the *grand orbit* of U by

$$\text{GO}(U) = \{z \in \widehat{\mathbb{C}} : f^r(z) \in U_s \text{ for some } r, s \geq 0\}. \quad (7.4)$$

□

Lemma 7.3 (Invariant Beltrami spreading). *Every L^∞ Beltrami differential μ on the eventual wandering disk U extends linearly over its grand orbit to an f -invariant sphere coefficient $\widehat{\mu}$ with $\|\widehat{\mu}\|_\infty = \|\mu\|_\infty$.*

Proof. Take any L^∞ Beltrami differential μ on U . Transport it from $U = U_0$ to every U_n through the conformal isomorphisms above, then pull it back through all inverse levels of the grand orbit, and put it equal to zero off $\text{GO}(U)$. Univalence along the forward chain makes the definition consistent almost everywhere. Holomorphic pullback preserves the essential norm, so the resulting coefficient $\widehat{\mu}$ satisfies

$$f^*\widehat{\mu} = \widehat{\mu}, \quad \|\widehat{\mu}\|_\infty = \|\mu\|_\infty. \quad (7.5)$$

□

Lemma 7.4 (Infinitesimal deformation operator). *The normalized Ahlfors–Bers deformation associated with $t\widehat{\mu}$ defines a complex-linear map*

$$D : L^\infty(U) \longrightarrow T_f \text{Rat}_d, \quad D\mu = v \circ f - f'v,$$

where $\bar{\partial}v = \widehat{\mu}$ and $v(0) = v(1) = v(\infty) = 0$.

Proof. For small t , set $\mu_t = t\widehat{\mu}$ and let Φ_t be the normalized Beltrami solution fixing $0, 1, \infty$. Ahlfors–Bers holomorphic dependence from the preceding volume says that $t \mapsto \Phi_t(z)$ is holomorphic for each z . Invariance (7.5) and the invariant-Beltrami straightening theorem give

$$f_t = \Phi_t \circ f \circ \Phi_t^{-1} \in \text{Rat}_d, \quad (7.6)$$

with holomorphic dependence on t .

Let

$$v(z) = \left. \frac{d}{dt} \right|_{t=0} \Phi_t(z).$$

The first variation of the Beltrami equation gives $\bar{\partial}v = \widehat{\mu}$, with $v(0) = v(1) = v(\infty) = 0$. Expanding $\Phi_t = \text{id} + tv + O(t^2)$ and $\Phi_t^{-1} = \text{id} - tv + O(t^2)$ yields the infinitesimal rational deformation

$$D\mu = \left. \frac{d}{dt} \right|_{t=0} f_t = v \circ f - f'v. \quad (7.7)$$

Thus $D : L^\infty(U) \rightarrow T_f \text{Rat}_d$ is complex linear. □

Lemma 7.5 (Trivial infinitesimal deformation vanishes on the Julia set). *If $D\mu = 0$, then the normalized infinitesimal vector field v vanishes on $J(f)$.*

Proof. If $D\mu = 0$, then

$$v(f(z)) = f'(z)v(z). \quad (7.8)$$

For a repelling periodic point p of exact period q , with multiplier $\rho = (f^q)'(p)$, iterating (7.8) around the cycle gives $v(p) = \rho v(p)$. Since $|\rho| > 1$, $v(p) = 0$. Repelling periodic points are dense in the Julia set by Volume 2, and v is continuous, so

$$v = 0 \quad \text{on } J(f). \quad (7.9)$$

□

Lemma 7.6 (Boundary-rigid Beltrami test space). *Every simply connected wandering disk U carries an infinite-dimensional subspace $N(U) \subset L^\infty(U)$ of compactly supported Beltrami differentials such that*

$$\bar{\partial}v = \mu \in N(U), \quad v|_{\partial U} = 0 \implies \mu = 0.$$

Proof. We now construct infinitely many Beltrami directions on U which cannot give a trivial infinitesimal deformation. Begin with $U = \mathbb{D}$. For $k \geq 0$ put

$$\nu_k(z) = \bar{z}^k \frac{d\bar{z}}{dz}.$$

A continuous global vector-field solution of $\bar{\partial}V_k = \nu_k$ is

$$V_k(z) = \frac{1}{k+1} \begin{cases} \bar{z}^{k+1}, & |z| < 1, \\ z^{-(k+1)}, & |z| \geq 1. \end{cases} \quad (7.10)$$

If a finite linear combination $\sum_{k=0}^m a_k \nu_k$ were $\bar{\partial}v$ in the disk with $v = 0$ on the unit circle, then $V - v$ would be holomorphic in the disk and would have boundary values

$$\sum_{k=0}^m \frac{a_k}{k+1} z^{-(k+1)}.$$

A holomorphic disk function has no negative Fourier modes, so every a_k must vanish.

To make the coefficients compactly supported, rescale these modes to $|z| < 1/2$ and set them equal to zero on the outer annulus. If the associated vector field vanishes on $|z| = 1$, it is holomorphic on that outer annulus; its Laurent coefficients vanish as one approaches the boundary, so the vector field vanishes throughout the annulus. The preceding negative-mode argument on the inner disk then forces the coefficient to be zero. We obtain an infinite-dimensional space $N(\mathbb{D})$ with this boundary-rigidity property.

For general simply connected U , transport $N(\mathbb{D})$ by a Riemann map $\psi : \mathbb{D} \rightarrow U$. The supports remain compactly contained in U . Because the coefficients vanish near the boundary, the pulled-back vector field is holomorphic in an outer annulus, and the same Laurent argument applies without requiring a Carathéodory extension of ψ . Thus we get an infinite-dimensional space $N(U)$ such that

$$\bar{\partial}v = \mu \in N(U), \quad v|_{\partial U} = 0 \implies \mu = 0. \quad (7.11)$$

□

Proof. By Lemma 7.2, we may replace the wandering component by a late image on which the forward chain is conformally isomorphic. Lemmas 7.3–7.5 then provide the infinitesimal deformation operator and its Julia-set vanishing property.

Restrict the operator of Lemma 7.4 to the space $N(U)$ of Lemma 7.6. If $D\mu = 0$, Lemma 7.5 gives $v = 0$ on $J(f)$, hence on ∂U ; boundary rigidity then forces $\mu = 0$. Therefore

$$D|_{N(U)} : N(U) \hookrightarrow T_f \text{Rat}_d \quad (7.12)$$

is injective.

This is impossible. A degree- d rational map is represented by a pair of coprime polynomials of degree at most d , up to common scale. Hence Rat_d is a Zariski-open subset of $\mathbb{P}^{2d+1}$, so

$$\dim_{\mathbb{C}} T_f \text{Rat}_d = 2d + 1 < \infty,$$

whereas $N(U)$ is infinite-dimensional. The contradiction eliminates the wandering component. Every Fatou component is therefore eventually periodic. □

Theorem 7.7 (Global Fatou-component classification). **Statement scope.** *Every Fatou component is preperiodic to a periodic component, and the eventual periodic component is an attracting or superattracting basin, a parabolic basin, a Siegel disk, or a Herman ring. The periodic classification itself is the Chapter 2 theorem; the new Chapter 7 input is Sullivan no wandering.*

Proof strategy (not part of the proof)

Proof status. derived here.

Strategic outline only. Immediate two-parent corollary: Sullivan no wandering plus the Chapter 2 classification of periodic Fatou components.

The mathematical argument begins in the proof environment below.

Proof. Let U be an arbitrary Fatou component of a rational map f of degree at least two. Sullivan’s theorem proved above gives integers $m < n$ for which the Fatou components containing $f^m(U)$ and $f^n(U)$ coincide. Thus the forward component orbit of U is eventually periodic. Let V be a periodic component in the eventual cycle.

Volume 2 already classified periodic Fatou components. Applied to the first return map on V , that theorem says that V is an attracting or superattracting basin, a parabolic basin, a Siegel disk, or a Herman ring. Consequently the original component U eventually maps to a component of exactly one of these four periodic types.

The logical division is important: the local periodic classification belongs to Volume 2; Sullivan’s theorem supplies precisely the missing global fact that an arbitrary component eventually enters the periodic regime. □

Theorem 7.8 (Mañé–Sad–Sullivan J-stability theorem). **Statement scope.** *Let (f_λ) be a holomorphic family of rational maps of fixed degree $d \geq 2$. After a finite local critical-marking cover, the following are equivalent near a parameter λ_0 : the Julia set moves by a dynamical holomorphic motion; there are no unpersistent neutral cycles; and every marked critical point is passive. On a sufficiently small stable neighborhood the*

Julia motion has the standard MSS extension to a locally uniformly quasiconformal conjugacy on a neighborhood of the Julia set. These conditions define the J-stable locus. The J-stable locus is open and dense; the bifurcation locus is closed with empty interior.

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Periodic analytic continuation and neutral crossings; Chapter 6 Julia motion; attracting-basin critical capture; Montel-Hurwitz activity/passivity criterion; active-critical capture; finite critical marking; density of the J-stable locus.

The mathematical argument begins in the proof environment below.

Proof. Let $(f_\lambda)_{\lambda \in \Lambda}$ be a holomorphic family of rational maps of fixed degree $d \geq 2$, and fix λ_0 . Work on a simply connected coordinate neighborhood U of λ_0 . After a finite local branched cover, the critical points can be labeled by holomorphic maps

$$c_1, \dots, c_{2d-2} : U \longrightarrow \widehat{\mathbb{C}}, \quad (7.13)$$

counted with multiplicity. Local stability is unchanged by this finite base change, so it is enough to prove the theorem for the critically marked family.

Periodic types and Julia motion.

For each period n , periodic points form the finite analytic set

$$\mathcal{P}_n = \{(\lambda, z) : f_\lambda^n(z) = z\}. \quad (7.14)$$

A simple repelling point continues uniquely by the holomorphic implicit-function theorem. A repelling branch can cease to be simple and repelling only by meeting the unit circle in multiplier space: a collision has multiplier 1, while a change from repelling to attracting crosses $|\rho| = 1$. Therefore, if no neutral cycle is unpersistent, every repelling branch remains simple and repelling. The fixed total number of roots of (7.14), counted with multiplicity, prevents a new repelling branch from appearing without another such crossing. Hence the entire repelling spectrum is stable in the periodic-type sense of Volume 6. The Julia-motion theorem of that volume now supplies a dynamical holomorphic motion of the Julia sets.

We shall also use the standard neighborhood form of the MSS construction. After shrinking the stable parameter disk, there is a neighborhood N_* of the base Julia set and embeddings

$$H_\lambda : N_* \longrightarrow \widehat{\mathbb{C}}, \quad H_{\lambda_*} = \text{id}, \quad (7.14a)$$

holomorphic in λ , locally uniformly quasiconformal in the space variable, and satisfying $H_\lambda \circ f_{\lambda_*} = f_\lambda \circ H_\lambda$ wherever both sides are defined. This is obtained in the same lifting construction: cover the compact Julia set by finitely many disks on which the marked local branches have fixed degree, lift the repelling-point motion through those branches (using the marked critical local normal form at critical disks), and use density of repelling points for uniqueness on overlaps. A finite number of lifts gives a collar

motion; Slodkowski extension then gives the quasiconformal slices, with a common local dilatation bound after shrinking the parameter disk.

Conversely, suppose the Julia sets move dynamically and a periodic branch $p(\lambda)$ is neutral but unpersistent at some parameter. Its multiplier is a nonconstant holomorphic function. By the open mapping theorem, arbitrarily nearby values make the same branch repelling and attracting. Start at a repelling parameter. Dynamical Julia motion carries that repelling point to a holomorphic periodic branch throughout the stable neighborhood, and uniqueness of analytic continuation identifies it with $p(\lambda)$. The moved point always lies in the Julia set. At an attracting parameter, however, $p(\lambda)$ lies in the Fatou set. This contradiction proves

$$\text{dynamical Julia motion} \iff \text{no unpersistent neutral cycle.} \quad (7.15)$$

Attracting cycles detect critical points.

Let A be an immediate basin of period q and put $g = f^q|_A$. If g had no critical point in A , it would be an unbranched proper self-cover of the hyperbolic Riemann surface A , hence a local isometry for the Poincaré metric. At the attracting fixed point the Poincaré derivative norm would be one, contradicting $|g'(p)| < 1$. Thus g has a critical point. By the chain rule, the full attracting basin contains a critical point of f .

The critical-activity criterion.

Suppose first that λ_0 is unstable. By (7.15), arbitrarily close to it a periodic branch crosses from repelling through a nonpersistent neutral parameter to attracting. On an attracting subneighborhood, Step 2 gives a marked critical point captured by that cycle. A subsequence of the critical orbit therefore converges locally uniformly to the attracting periodic branch.

If the critical-orbit family were normal across the whole neighborhood, a subsequence would converge there. The identity theorem would force its limit to be the same periodic branch everywhere. At a nearby parameter where that branch is repelling this is impossible: in a linearizing coordinate, a forward orbit can converge to a repelling fixed point only by eventually landing on it, and an isolated landing relation breaks normality under parameter perturbation. Thus some marked critical point is active. Instability is therefore detected by critical activity.

Conversely, suppose the Julia set moves dynamically on a neighborhood. Fix a marked critical point c . Since the base Julia set is perfect, choose three distinct points a_1, a_2, a_3 in it which the base critical orbit avoids. The classical Mañé–Sad–Sullivan moving-target lemma says that, after shrinking the stable neighborhood, the critical orbit avoids the three disjoint moving Julia graphs $h_\lambda(a_j)$. An isolated first collision would be precisely the critical-relation instability excluded by the dynamical motion. A parameterwise Möbius map sends the three moving graphs to $0, 1, \infty$. Picard–Montel therefore makes

$$\{\lambda \mapsto f_\lambda^n(c(\lambda)) : n \geq 0\} \quad (7.16)$$

normal. Every marked critical point is passive.

We have proved

$$\text{J-stability} \iff \text{all marked critical points are passive.} \quad (7.17)$$

Together with (7.15), this gives all three local criteria.

Density of the stable locus.

We need one more Montel observation. If a marked critical point $c(\lambda)$ is active at λ_* , follow a short backward orbit and choose holomorphic branches

$$c_{-2}(\lambda), \quad c_{-1}(\lambda), \quad c(\lambda)$$

which are distinct near λ_* and satisfy

$$f_\lambda(c_{-2}(\lambda)) = c_{-1}(\lambda), \quad f_\lambda(c_{-1}(\lambda)) = c(\lambda). \quad (7.18)$$

If the forward orbit of $c(\lambda)$ avoided all three moving targets on a neighborhood, moving-target Montel would make it normal, contradicting activity. Hence arbitrarily near λ_* an iterate of c hits one of the three targets. Applying at most two more iterates makes a critical point periodic. Its cycle has multiplier zero and is superattracting. The same argument with three points on a continued repelling cycle also gives nearby parameters at which an active critical point is strictly preperiodic to a repelling cycle.

Now take any nonempty parameter neighborhood W . If all critical points are passive somewhere in W , (7.17) gives a stable parameter. Otherwise choose an active critical point and perturb inside W until it becomes superattracting. Its attracting basin persists under a small further perturbation, so that marked critical point is passive on a neighborhood of the new parameter. If another critical point is active, repeat inside the intersection of the passivity neighborhoods already obtained. There are only $2d - 2$ marked critical points counted with multiplicity, so after finitely many steps all of them are passive. The resulting parameter is J-stable.

Every nonempty open set therefore contains a J-stable parameter. Since J-stability is local by definition, the stable locus is open and dense, and the bifurcation locus is closed with empty interior.

This last conclusion must not be confused with density of hyperbolicity. A stable family may have persistent neutral dynamics; hyperbolicity asks for a strictly stronger condition. $\square$

Conjecture 7.9 (Density of hyperbolicity: status). **Statement scope.** *Hyperbolic parameters form an open subset of the J-stable locus. As of September 2026, the conjecture that hyperbolic rational maps are dense is open in full generality, already for the quadratic polynomial family. No later result in this book uses density of hyperbolicity as an established theorem unless a separately sourced subclass is specified.*

7.3 Worked examples

Worked example: The fixed-point hyperbolic component of $z^2 + c$

Let

$$f_c(z) = z^2 + c$$

and suppose p is a fixed point. Then

$$p = p^2 + c, \quad \rho = f'_c(p) = 2p.$$

Hence $p = \rho/2$ and

$$c = p - p^2 = \frac{\rho}{2} - \frac{\rho^2}{4}. \tag{7.E1}$$

Thus the multiplier ρ itself parametrizes the fixed-point hyperbolic component by the unit disk $|\rho| < 1$.

Indeed, $|\rho| < 1$ means that p is attracting. The immediate attracting basin contains the unique finite critical point 0, so the critical orbit is passive throughout this component. The stability theorem therefore gives a dynamical holomorphic motion of the Julia set.

At $\rho = 1$, formula (7.E1) gives

$$p = \frac{1}{2}, \quad c = \frac{1}{4}.$$

The fixed point has multiplier 1 and is parabolic. Because the multiplier varies nontrivially with the parameter, this neutral cycle is not persistent: it is the simplest local model of a bifurcation boundary.

Worked example: A J-stable family with persistent neutral dynamics

Fix an irrational number θ and set

$$\alpha = e^{2\pi i \theta}.$$

For $\lambda \neq 0$ consider

$$f_\lambda(z) = \alpha z + \lambda z^2.$$

Define $L_\lambda(z) = \lambda z$. Then

$$\begin{aligned} L_\lambda \circ f_\lambda \circ L_\lambda^{-1}(w) &= \lambda \left(\alpha \frac{w}{\lambda} + \lambda \frac{w^2}{\lambda^2} \right) \\ &= \alpha w + w^2. \end{aligned} \tag{7.E2}$$

Thus every map in any parameter region contained in $\mathbb{C}^*$ is conformally conjugate to the same quadratic polynomial. The family is therefore J-stable: a holomorphic conjugacy is already available explicitly.

Nevertheless 0 is a fixed point of multiplier α , so it is irrationally neutral for every parameter. The neutral multiplier is persistent rather than bifurcating. Such a family is not hyperbolic, because hyperbolic rational maps have no neutral periodic cycles.

This example separates two notions that are easy to conflate:

$$\text{J-stable} \not\Rightarrow \text{hyperbolic}.$$

The density of J-stability proved in this chapter therefore does not settle the density-of-hyperbolicity conjecture.

Worked example: The negative Fourier modes in Sullivan's deformation argument

On the unit disk, consider the Beltrami coefficient

$$\nu_k(z) = \bar{z}^k \frac{d\bar{z}}{dz}.$$

A global solution of $\bar{\partial}V_k = \nu_k$ is

$$V_k(z) = \frac{1}{k+1} \begin{cases} \bar{z}^{k+1}, & |z| < 1, \\ z^{-(k+1)}, & |z| \geq 1. \end{cases}$$

On the unit circle, where $\bar{z} = z^{-1}$, the boundary trace is exactly the negative Fourier mode

$$\frac{1}{k+1} z^{-(k+1)}.$$

Suppose

$$\nu = \sum_{k=0}^m a_k \nu_k$$

were $\bar{\partial}v$ for a vector field v vanishing on the unit circle. Then $V - v$ would be holomorphic in the disk but would have boundary values containing only the negative modes

$$\sum_{k=0}^m \frac{a_k}{k+1} z^{-(k+1)}.$$

A holomorphic disk function has no negative Fourier coefficients. Therefore all $a_k = 0$.

This calculation is the linear-algebraic heart of Sullivan's proof. It gives arbitrarily many independent Beltrami directions which cannot disappear as trivial infinitesimal conjugacies. A wandering component would therefore force an infinite-dimensional subspace into the finite-dimensional tangent space of Rat_d .

Boundary case: A neutral cycle can be stable or bifurcating

The phrase “neutral cycle” by itself does not decide stability. In the quadratic fixed-point example, the multiplier passes nontrivially through 1, so the neutral parameter is a bifurcation point. In the isotrivial family $f_\lambda(z) = \alpha z + \lambda z^2$, the neutral multiplier α is constant and every map is conformally conjugate to every other one, so the neutral cycle is persistent and the family is stable.

The theorem therefore uses *unpersistent neutral cycle*, not merely “neutral cycle.”

Common pitfall

Do not read “repelling periodic points move holomorphically” as a complete stability theorem. Each simple repelling point moves locally by the implicit-function theorem even near many bifurcation parameters. Stability requires a single neighborhood on which the entire periodic structure is compatible and the resulting dense motion extends to the whole Julia set.

7.4 Chapter summary

What to remember

- A wandering Fatou component would have pairwise distinct forward component images. Baker’s lemma shows that sufficiently late images would be simply connected and connected by conformal isomorphisms.
- Univalence along that hypothetical wandering orbit lets arbitrary Beltrami data on one component spread to an invariant coefficient on its grand orbit.
- Integrating those invariant coefficients produces infinitesimal deformations in $T_f \text{Rat}_d$. A boundary-rigid infinite-dimensional test space injects into this tangent space if a wandering component exists, contradicting $\dim_{\mathbb{C}} \text{Rat}_d = 2d + 1$.
- Sullivan’s theorem therefore makes every Fatou component preperiodic. Combining this with the periodic classification from Volume 2 gives the global Fatou-component classification.
- J-stability is equivalent to absence of unpersistent neutral cycles and, after local critical marking, to passivity of every critical point.
- Critical points are the bifurcation sensors because every attracting cycle captures critical dynamics, while Montel normality of all critical orbits prevents periodic-type bifurcation.
- The stable locus is open and dense. Its complement, the bifurcation locus, is closed and has empty interior.
- Density of J-stability is not density of hyperbolicity. Persistent neutral stable families show that the notions differ, and density of hyperbolicity remains open in general.

- Invariant line fields encode measurable invariant deformation directions. This chapter prepares that interface but does not assume a general no-line-field theorem.

Looking ahead

The next rigidity chapters use the same two engines in more structured settings. Polynomial-like maps and straightening turn local return dynamics into genuine polynomials; invariant Beltrami structures and line fields become tests for whether a quasiconformal deformation is dynamically meaningful or rigid. The stability/bifurcation distinction established here will remain the parameter-space backdrop for those arguments.

Polynomial-Like Maps, Straightening and Tuning

Why this chapter matters

A polynomial-like map is a local polynomial: it has the proper finite-degree mapping behavior of a polynomial, but only on a disk compactly contained in a larger disk. This apparently modest change of viewpoint is one of the decisive ideas in complex dynamics. A high iterate of a polynomial can reproduce a smaller quadratic system inside one part of its own dynamical plane. Once that return map is recognized as polynomial-like, the straightening theorem turns it back into an actual polynomial.

The proof is a model application of quasiconformal surgery. We first sew the local map to a power map near infinity, then alter the complex structure so the sewn map becomes holomorphic. The measurable Riemann mapping theorem integrates that structure. The output is not merely similar to a polynomial: it is hybrid equivalent, meaning conformal almost everywhere on the filled Julia set. The same mechanism becomes a parameter-space microscope. A quadratic-like first return has a unique normalized straightening $z^2 + c$. As the original parameter varies through a characteristic renormalization family, the parameter c ranges through the entire Mandelbrot set. Inverting this straightening map produces a small copy of M inside M : Douady-Hubbard tuning.

Learning goals

By the end of this chapter you should be able to:

- define polynomial-like maps, filled Julia sets, hybrid equivalence, and fundamental annuli;
- explain why straightening existence does not require connectedness and why uniqueness does;
- reconstruct the quasiregular-sewing and invariant-Beltrami proof of straightening;
- distinguish a.e. conformality on a filled set from pointwise conformality at its boundary;
- normalize a connected quadratic-like renormalization to a unique parameter $c_R \in M$;
- explain why a winding-one quadratic-like family has a connectedness locus homeomorphic to M ;
- describe tuning as inverse straightening and track its characteristic binary

block substitution.

Conceptual roadmap

The first spine is the one-map surgery:

polynomial-like map $\implies$ quasiregular global extension
 $\Downarrow$
 invariant complex structure $\implies$ MRMT integration
 $\Downarrow$
 polynomial straightening $\implies$ hybrid equivalence.

The second spine is the parameter construction:

quadratic-like first return $\implies$ normalized parameter $c_R \in M$
 $\Downarrow$
 winding-one family $\implies$ degree-one straightening
 $\Downarrow$
 tuning embedding $\implies$ small copy of M .

Connectedness enters the first spine only at uniqueness. It enters the second spine because a quadratic parameter belongs to M exactly when its critical orbit is bounded.

How the main results fit together. The straightening theorem is stronger than the common shorthand “connected polynomial-like maps straighten”: every polynomial-like map straightens. The connectedness hypothesis is what makes the polynomial representative rigid. Likewise, “renormalizable” names the existence of a polynomial-like first return; the nontrivial conclusion used later is that a connected quadratic-like return has a unique normalized parameter in the Mandelbrot set.

8.1 Definitions

Definition 8.1 (polynomial-like map). A *polynomial-like map of degree d* is a proper holomorphic map

$$f : U' \longrightarrow U$$

of degree $d \geq 2$ between topological disks with $\overline{U'} \Subset U$. Its filled Julia set is

$$K(f) = \{z \in U' : f^n(z) \in U' \text{ for every } n \geq 0\},$$

and its Julia set is $J(f) = \partial K(f)$.

Definition 8.2 (hybrid equivalence). Two polynomial-like maps f and g are *hybrid equivalent* if there is a quasiconformal conjugacy h between neighborhoods of $K(f)$ and $K(g)$ whose Beltrami coefficient satisfies $\mu_h = 0$ almost everywhere on $K(f)$.

Definition 8.3 (straightening). A *straightening* of a polynomial-like map f is a polynomial P of the same degree which is hybrid equivalent to f . For a connected quadratic-like map, its *normalized straightening parameter* is the unique c for which the straightening is $P_c(z) = z^2 + c$.

Definition 8.4 (external class). For a polynomial-like map with connected filled Julia set, the dynamics in a fundamental annulus can be uniformized to an expanding degree- d circle covering. Its conformal conjugacy class, modulo rotation of the circle, is the *external class* of the polynomial-like map.

Definition 8.5 (fundamental annulus). For a polynomial-like restriction $f : V' \rightarrow V$ with Jordan boundaries, the annulus

$$A = V \setminus \overline{V'}$$

is a *fundamental annulus*. Replacing the original domains by smaller ones without changing $K(f)$ is called taking an equivalent polynomial-like restriction.

Definition 8.6 (quadratic-like renormalization). A map F is *quadratic-like renormalizable of period p* on disks $U' \Subset U$ if

$$R = F^p : U' \longrightarrow U$$

is a degree-two polynomial-like map with connected filled Julia set. The map R is the *renormalization*; its unique normalized straightening parameter is the *renormalization parameter*.

Definition 8.7 (Mandelbrot-like and semi-tunable families). A *Mandelbrot-like family* is an analytic family of quadratic-like maps over a parameter disk, with marked critical point and a holomorphically varying tubing of the fundamental annuli, whose connectedness locus is compactly contained in the parameter disk and whose parametric degree is one. By the Douady-Hubbard boundary-degree formula, the last condition can be checked by showing that the marked critical-value section has winding number one around a parameter boundary enclosing the connectedness locus. A *semi-tunable family* is the corresponding one-root-degenerate situation: the family is Mandelbrot-like away from one parabolic pinching parameter, the straightening maps the punctured connectedness locus to $M \setminus \{1/4\}$, and its inverse extends continuously over the omitted cusp. In both cases the *connectedness locus* is the set of parameters for which the marked critical point belongs to the filled Julia set.

Definition 8.8 (tuning embedding and small Mandelbrot copy). For a hyperbolic component $H \subset M$, the *tuning embedding* $\tau_H : M \rightarrow M$ is the inverse of the normalized straightening map of the characteristic quadratic-like renormalization family attached to H , with normalization $\tau_H(0) = c_H$ at the center and $\tau_H(1/4) = r_H$ at the root. Its image M_H is the corresponding *small Mandelbrot copy*.

8.2 Main results

Definition 8.9 (Polynomial-like localization). **Statement scope.** Polynomial-like maps retain the polynomial notions of filled Julia set and Julia set while allowing the

dynamics to be studied on a compactly contained local domain. Equivalent restrictions which contain $K(f)$ define the same local dynamical germ and the same filled Julia set.

Theorem 8.1 (Straightening theorem). **Statement scope.** *Every polynomial-like map $f : U' \rightarrow U$ of degree $d \geq 2$ is hybrid equivalent to a polynomial P of degree d . If $K(f)$ is connected, then P is unique up to affine conjugacy.*

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Smooth equivalent restriction; quasiregular sewing; invariant Beltrami pullback; Chapter 5 MRMT; polynomial identification; connected hybrid rigidity.

The mathematical argument begins in the proof environment below.

Proof setup. Let

$$f : U' \longrightarrow U$$

be a polynomial-like map of degree $d \geq 2$. We first construct a polynomial hybrid equivalent to f , without assuming that $K(f)$ is connected.

Lemma 8.2 (Smooth fundamental-annulus restriction). *There are Jordan disks*

$$V' \Subset V, \quad K(f) \subset V', \quad (8.1)$$

with real-analytic boundaries such that $f : V' \rightarrow V$ is proper of degree d and has the same filled Julia set as the original polynomial-like map.

Proof. Choose Riemann maps $\alpha : \mathbb{D} \rightarrow U'$ and $\beta : \mathbb{D} \rightarrow U$. The map

$$B := \beta^{-1} \circ f \circ \alpha : \mathbb{D} \longrightarrow \mathbb{D}$$

is proper holomorphic of degree d , hence a finite Blaschke product. Its critical set is finite. Choose $r < 1$ sufficiently close to 1 so that all critical values of B lie in D_r and $\alpha^{-1}(K(f)) \subset B^{-1}(D_r)$. This is possible because both sets in question are compact in the unit disk. Above the collar $\{r < |w| < 1\}$ the Blaschke product is then unbranched. Its restriction to $\partial\mathbb{D}$ is the connected degree- d covering of the circle, so the preimage of this outer collar is one annulus. Consequently $B^{-1}(D_r)$ is a single Jordan disk D'_r bounded by one real-analytic level curve $|B| = r$, and the proper map $B : D'_r \rightarrow D_r$ has the full degree d . In particular the whole compact set $\alpha^{-1}(K(f))$, not merely one of its components when $K(f)$ is disconnected, lies in D'_r . Put

$$V = \beta(D_r), \quad V' = \alpha(D'_r).$$

The boundaries are real analytic because the Riemann maps are holomorphic on neighborhoods of the compact circles used above. The original filled set is contained in V' . Conversely, a point whose orbit under the restricted map never leaves V' certainly never leaves U' . Hence the restricted and original filled Julia sets coincide. $\square$

Replacing (U', U) by (V', V) does not alter the dynamical germ near $K(f)$. Set

$$A := V \setminus \overline{V'}.$$

This is the *fundamental annulus* of the chosen restriction.

Lemma 8.3 (Quasiregular polynomial extension). *There is an orientation-preserving quasiregular branched covering*

$$F : \widehat{\mathbb{C}} \longrightarrow \widehat{\mathbb{C}} \quad (8.2)$$

of topological degree d such that $F = f$ on V' , F is holomorphic on V' and on $\widehat{\mathbb{C}} \setminus \overline{V}$, and ∞ is a fixed point of local degree d .

Proof. The boundary map $f : \partial V' \rightarrow \partial V$ is an analytic orientation-preserving covering of degree d . Choose analytic parametrizations of the two boundaries. Lifting their angular coordinates to $\mathbb{R}$ shows that, after an orientation-preserving analytic change of coordinate on the two circles, the boundary covering is $\zeta \mapsto \zeta^d$.

Fix $R > 1$. Map $\partial V'$ to $|w| = R$ and ∂V to $|w| = R^d$ so that

$$H(f(z)) = H(z)^d \quad (z \in \partial V'). \quad (8.3)$$

The two analytic boundary homeomorphisms are quasimetric. Interpolating their lifted angular coordinates linearly in logarithmic radius gives a quasiconformal homeomorphism of A onto $\{R < |w| < R^d\}$. On the unbounded component outside V , continue H conformally to $\{|w| > R^d\}$, with $H(\infty) = \infty$. Thus

$$H : \widehat{\mathbb{C}} \setminus V' \longrightarrow \widehat{\mathbb{C}} \setminus \overline{D_R}$$

is quasiconformal, conformal outside V , and satisfies (8.3). Define

$$F(z) = \begin{cases} f(z), & z \in V', \\ H^{-1}(H(z)^d), & z \notin V'. \end{cases} \quad (8.4)$$

Equation (8.3) makes the two definitions agree continuously on $\partial V'$. On A , the inner H is quasiconformal while $H(z)^d$ lies outside $|w| = R^d$, where H^{-1} is conformal; hence this piece is quasiregular. Outside V , both H and H^{-1} are conformal, so F is holomorphic there. The quasiregular sewing lemma across analytic curves gives a global quasiregular map. Formula (8.4) has degree d , and near infinity it is conformally conjugate to $w \mapsto w^d$. $\square$

Lemma 8.4 (Invariant Beltrami straightening produces a polynomial). *For the quasiregular extension F of Lemma 8.3, there is an F -invariant measurable Beltrami coefficient which vanishes almost everywhere on $K(f)$. Integrating it conjugates F to a degree- d polynomial hybrid equivalent to f .*

Proof. We now change the complex structure so that F becomes holomorphic. Give the exterior of V the standard structure. On A put the pullback by F of the standard

structure. If $z \in V' \setminus K(f)$, let $n(z) \geq 1$ be the first integer for which

$$f^{n(z)}(z) \in A.$$

Put on z the pullback by $f^{n(z)}$ of the structure already defined on A . Finally, put the standard structure on $K(f)$. In Beltrami notation this gives a measurable coefficient μ satisfying

$$\|\mu\|_\infty < 1, \quad F^*\mu = \mu \quad \text{a.e.}, \quad \mu = 0 \quad \text{a.e. on } K(f). \quad (8.5)$$

Indeed every orbit outside $K(f)$ meets the fundamental annulus at most once before entering the exterior model, and pullback by the holomorphic pieces preserves the absolute value of a Beltrami coefficient. Thus the essential bound on A propagates without growth.

Apply the measurable Riemann mapping theorem from Volume 5 to obtain a quasiconformal homeomorphism Φ integrating μ , normalized by $\Phi(\infty) = \infty$ and two further points. Set

$$P := \Phi \circ F \circ \Phi^{-1}. \quad (8.6)$$

The invariance equation (8.5) is exactly the Beltrami composition condition which says that P has zero complex dilatation almost everywhere. Hence P is holomorphic on the sphere away from its pole at infinity. It has degree d , and because ∞ is fixed with local degree d , it has no finite pole. Therefore P is a polynomial of degree d . On a neighborhood of $K(f)$ we have $F = f$, so (8.6) gives

$$\Phi \circ f = P \circ \Phi,$$

and (8.5) gives $\mu_\Phi = 0$ almost everywhere on $K(f)$. Thus f and P are hybrid equivalent. This proves existence for arbitrary $K(f)$. $\square$

Lemma 8.5 (Hybrid rigidity for connected polynomial filled sets). *If two degree- d polynomials with connected filled Julia sets are hybrid equivalent, then they are affinely conjugate.*

Proof. Let h be a hybrid conjugacy near $K(P)$. Since $K(P)$ and $K(Q)$ are connected, their basins of infinity have normalized Böttcher coordinates ϕ_P, ϕ_Q satisfying

$$\phi_P(Pz) = \phi_P(z)^d, \quad \phi_Q(Qz) = \phi_Q(z)^d. \quad (8.7)$$

The external dynamics therefore admits a conformal conjugacy

$$e = \phi_Q^{-1} \circ R_\omega \circ \phi_P, \quad (8.8)$$

where the rotation $R_\omega(w) = \omega w$ is chosen with $\omega^{d-1} = 1$ to match the orientation/prime-end marking induced by h .

Choose a small equipotential annulus contained in the neighborhood on which h is defined. Interpolate once, quasiconformally, between h on its inner boundary and e on its outer boundary. Pull this interpolation back under P and Q . Because those maps are holomorphic coverings of the basin of infinity, every pullback has the same

dilatation bound. At the n th stage the transition annulus is the n th pullback of the original one and hence has Green potential of order d^{-n} ; it approaches $J(P)$. We obtain normalized quasiconformal homeomorphisms H_n with one uniform dilatation bound such that $H_n = h$ on $K(P)$ and $H_n = e$ at every fixed point of the basin of infinity once n is large enough.

The normalized QC compactness theorem from Volume 5 gives a locally uniform subsequence converging to a quasiconformal homeomorphism H . Pointwise,

$$H = h \quad \text{on } K(P), \quad H = e \quad \text{on } \widehat{\mathbb{C}} \setminus K(P). \quad (8.9)$$

The two formulas are compatible because they are obtained as a single uniform limit. The map H conjugates P to Q : this holds on $K(P)$ by h , on the basin by e , and hence everywhere by continuity. Its Beltrami coefficient vanishes almost everywhere on $K(P)$ because h is hybrid, and vanishes on the complement because e is conformal. Thus $\mu_H = 0$ almost everywhere on the sphere. Weyl's lemma, or the normalized uniqueness part of the Volume 5 Beltrami theorem, makes H Möbius. Since $H(\infty) = \infty$, it is affine. This proves the lemma. $\square$

Proof. Apply Lemma 8.2 to replace the original polynomial-like representative by one with an analytic fundamental annulus. Lemma 8.3 gives a global quasiregular degree- d extension, and Lemma 8.4 straightens that extension to a polynomial hybrid equivalent to f . This proves existence for arbitrary $K(f)$.

If $K(f)$ is connected and P, Q are two straightenings, composing one hybrid conjugacy with the inverse of the other gives a hybrid equivalence between P and Q near their connected filled Julia sets. Lemma 8.5 then makes them affinely conjugate. This proves uniqueness. $\square$

Theorem 8.6 (Consequences of hybrid equivalence on the filled Julia set). **Statement scope.** *A hybrid conjugacy sends $K(f)$ onto $K(g)$ and $J(f)$ onto $J(g)$. It is conformal on every component of $\text{int } K(f)$. Hence holomorphic dynamical data, including multipliers, are preserved wherever the conjugacy is conformal on a neighborhood of the periodic orbit. No blanket pointwise conformality or boundary-multiplier claim is inferred merely from $\mu = 0$ almost everywhere on $K(f)$.*

Proof strategy (not part of the proof)

Proof status. derived here.

Strategic outline only. Conjugacy invariance of K and J ; Weyl lemma on $\text{int } K$; chain rule only under genuine neighborhood conformality.

The mathematical argument begins in the proof environment below.

Proof. Let

$$f : U'_f \rightarrow U_f, \quad g : U'_g \rightarrow U_g$$

be polynomial-like maps and let h be a hybrid conjugacy between them on a neighborhood N_f of $K(f)$, with image N_g . Shrink N_f if necessary so that whenever $z, f(z) \in N_f$ we have

$$h(f(z)) = g(h(z)). \quad (8.10)$$

Filled and Julia sets are conjugacy invariants.

The hybrid conjugacy satisfies

$$h(K(f)) = K(g), \quad h(J(f)) = J(g). \quad (8.11)$$

Indeed, if $z \in K(f)$, every forward iterate of z remains in $K(f)$ and hence in the conjugacy neighborhood. Iterating (8.10) gives

$$h(f^n(z)) = g^n(h(z)) \quad (n \geq 0),$$

so the orbit of $h(z)$ remains in the target polynomial-like domain and $h(z) \in K(g)$. Applying the same argument to h^{-1} gives equality of the filled sets. Since a homeomorphism preserves boundaries,

$$h(\partial K(f)) = \partial K(g),$$

which is the Julia-set assertion. $\square$

The definition of hybrid equivalence also gives

$$\mu_h = 0 \quad \text{a.e. on } K(f). \quad (8.12)$$

This statement is measure-theoretic; it should not be silently strengthened to pointwise conformality at every boundary point.

Conformality on interior components.

The restriction of h to every component W of $\text{int } K(f)$ is holomorphic and injective, hence conformal.

On W , equation (8.12) says $\bar{\partial}h = 0$ almost everywhere. A quasiconformal map lies in $W_{\text{loc}}^{1,2}$, so Weyl's lemma gives a holomorphic representative on W . Continuity identifies that representative with h , and injectivity gives nonvanishing derivative. $\square$

Consequently the conjugacy preserves all holomorphic dynamical data that can be read inside such an interior component. In particular, suppose p is a periodic point of exact period q and h is conformal on a neighborhood of the orbit of p . Differentiating

$$h \circ f^q = g^q \circ h$$

at p gives

$$h'(p)(f^q)'(p) = (g^q)'(h(p))h'(p).$$

Because $h'(p) \neq 0$,

$$(f^q)'(p) = (g^q)'(h(p)). \quad (8.13)$$

Thus multipliers are preserved whenever the needed conformality has actually been established (for example for attracting cycles lying in the interior of the filled set). Equation (8.12) alone does not justify differentiating at an arbitrary boundary periodic point. This is exactly the scope asserted in the theorem. $\square$

Definition 8.10 (External class of a connected polynomial-like map). **Statement scope.** For connected $K(f)$, uniformizing the fundamental annulus and pulling back gives an expanding degree- d circle covering defined up to conformal circle conjugacy.

This external class records how the local polynomial-like dynamics is attached to its outside.

Theorem 8.7 (Normalized quadratic renormalization). **Statement scope.** *If a first return $R = F^p : U' \rightarrow U$ is quadratic-like with connected filled Julia set, then R has a unique normalized quadratic straightening*

$$P_{c_R}(z) = z^2 + c_R, \quad c_R \in M.$$

The parameter c_R depends only on the hybrid class of the renormalization and is called its renormalization parameter.

Proof strategy (not part of the proof)

Proof status. derived here.

Strategic outline only. Straighten the connected quadratic-like return; uniquely normalize to $z^2 + c$; use connectedness criterion to show c in M .

The mathematical argument begins in the proof environment below.

Proof. Let F be a holomorphic dynamical system for which there are topological disks

$$U' \Subset U$$

and an integer $p \geq 1$ such that

$$R := F^p : U' \longrightarrow U \tag{8.14}$$

is a polynomial-like map of degree two with connected filled Julia set $K(R)$. This restriction is the quadratic-like renormalization under consideration.

By the straightening theorem, R is hybrid equivalent to a quadratic polynomial Q , unique up to affine conjugacy. We now remove that affine ambiguity.

Unique monic-centered quadratic representative.

Every quadratic polynomial is affinely conjugate to a unique polynomial

$$P_c(z) = z^2 + c. \tag{8.15}$$

Write $Q(z) = az^2 + bz + e$ with $a \neq 0$. Translation by the critical point $-b/(2a)$ removes the linear term, and a subsequent nonzero scaling makes the leading coefficient one. Thus Q is affinely conjugate to some P_c . If an affine map $L(z) = \alpha z + \beta$ conjugates P_c to $P_{c'}$, comparison of the quadratic and linear coefficients in

$$L \circ P_c = P_{c'} \circ L$$

gives $\alpha = 1$ and $\beta = 0$, hence $c = c'$. $\square$

Let c_R denote the unique parameter obtained from the monic-centered normalization lemma. Hybrid equivalence maps the unique critical point of the quadratic-like map to the unique critical point of the straightened quadratic, with the same bounded/nonescaping dynamical status. Because $K(R)$ is connected, the critical point belongs to $K(R)$; equivalently the critical orbit of P_{c_R} is bounded. The quadratic connect-

edness criterion from Volume 3 therefore gives

$$c_R \in M. \quad (8.16)$$

We call P_{c_R} the normalized straightening of the renormalization and c_R its *renormalization parameter*. Uniqueness in the straightening theorem together with the monic-centered normalization lemma shows that c_R depends only on the hybrid class of the polynomial-like first return, not on auxiliary affine coordinates. $\square$

Theorem 8.8 (Quadratic tuning and small Mandelbrot copies). **Statement scope.** *For every hyperbolic component H of the quadratic Mandelbrot set there is a canonically normalized Douady-Hubbard tuning embedding*

$$\tau_H : M \longrightarrow M.$$

It is a homeomorphism onto a small Mandelbrot copy M_H , satisfies $\tau_H(0) = c_H$ and $\tau_H(1/4) = r_H$, and the period- p polynomial-like renormalization of $P_{\tau_H(c)}$ is hybrid equivalent to P_c . On the rational characteristic-ray skeleton, tuning carries parameter ray pairs, wakes, and limbs by the standard characteristic block substitution.

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Characteristic quadratic-like family; the degree-two tubing-straightening regularity proved in this chapter; boundary quadratic QC rigidity and MSS positive-local-degree assembly; compact ramified covering and winding-one degree; punctured satellite root extension; inverse straightening; rational block-substitution combinatorics.

The mathematical argument begins in the proof environment below.

Proof setup. Write

$$P_c(z) = z^2 + c, \quad M = \{c : K(P_c) \text{ is connected}\}.$$

Let H be a hyperbolic component of M . If H is the main cardioid, define τ_H to be the identity and every assertion is immediate. We therefore assume that the center c_H has exact period $p \geq 2$.

Let r_H be the root of H . By the parameter-ray theory of Volume 4 there is a characteristic rational ray pair

$$\mathcal{R}_M(\theta_H^-), \quad \mathcal{R}_M(\theta_H^+) \quad (8.17)$$

landing at r_H and cutting out the characteristic wake W_H . The two angles have periodic binary itineraries of period p ; denote by A_H^0, A_H^1 the two length- p binary blocks determined by the two sides of the critical sector. For a binary angle $\theta = .\epsilon_1\epsilon_2\epsilon_3\dots$ define its *H-tuned angle* by block substitution

$$T_H(\theta) := .A_H^{\epsilon_1}A_H^{\epsilon_2}A_H^{\epsilon_3}\dots \quad (8.18)$$

At dyadic angles use the one-sided convention determined by the corresponding pa-

parameter ray; the two conventions meet at the root exactly as in the usual binary description of M .

Lemma 8.9 (Characteristic quadratic-like family). *Associated with H is a parameter puzzle domain $\Lambda_H \subset W_H$ and, for $c \in \Lambda_H$ away from the root in the satellite case, Jordan disks*

$$U'_c \Subset U_c$$

depending holomorphically through their ray/equipotential boundaries such that

$$R_c := P_c^p : U'_c \longrightarrow U_c \quad (8.19)$$

is a degree-two polynomial-like map with marked critical point 0. Its connectedness locus

$$\mathcal{M}_H^\circ := \{c \in \Lambda_H : 0 \in K(R_c)\} \quad (8.20)$$

contains c_H , and the critical-value/tubing coordinate winds once around the fundamental annulus as c goes once around $\partial\Lambda_H$. In the primitive case the family is polynomial-like through its root; in the satellite case the only missing connectedness-locus point is r_H , where the annulus pinches parabolically.

Proof. Inside the wake W_H , the characteristic periodic orbit born from the center continues holomorphically until the two characteristic rays collide at the root. Volume 4 gives both ingredients needed here: rational dynamic rays at repelling periodic points move holomorphically while their combinatorics stays fixed, and the corresponding rational parameter rays land at the parameter where the characteristic collision occurs.

Choose an equipotential level small enough that, at the center, the two characteristic dynamic rays together with this equipotential bound a central puzzle disk U_{c_H} containing 0. Pull the boundary back along the period- p critical itinerary and let U'_{c_H} be the pullback component containing 0. Exactly one critical point of P_c lies in the central piece. Riemann-Hurwitz therefore gives

$$\deg(P_c^p : U'_c \rightarrow U_c) = 2. \quad (8.21)$$

Compact containment at the center is strict. Holomorphic motion of the finitely many bounding rays and the equipotential therefore preserves $U'_c \Subset U_c$ on a parameter puzzle neighborhood. Continue the parameter puzzle until one of its boundary relations becomes characteristic; by the landing theorem from Volume 4 the only such terminal relation in the characteristic wake is the root relation. This produces Λ_H and the family (8.19). Choose the exterior parameter equipotential as part of $\partial\Lambda_H$. On that boundary the marked critical value lies outside the polynomial-like domain, so in the primitive case the connectedness locus is compactly contained in Λ_H . In the satellite case the same is true after adjoining the unique pinching parameter r_H to the punctured locus.

For a degree-two polynomial-like map, connectedness of the filled set is equivalent to nonescape of its unique critical point. Hence (8.20) is exactly the connectedness locus of the family. The parameter-puzzle boundary is defined by the same characteristic ray pair and an exterior parameter equipotential. Under the parameter Böttcher coordinate from Volume 4, the critical value crosses this boundary once. Equivalently,

the critical-value position in a holomorphic tubing of the fundamental annuli has winding number one. This is the parameter-degree computation in the Douady–Hubbard Mandelbrot-like-family construction.

If H is primitive, the characteristic pieces remain compactly nested at the root and the family extends polynomial-like there. If H is satellite, the period- p first-return annulus pinches at the parabolic root, so the polynomial-like domains are defined on the punctured parameter puzzle; no other point pinches because a second pinching relation would give a second landing parameter for the same characteristic rational ray pair, contrary to Volume 4. Thus the root is the unique omitted point. $\square$

We now prove the parameter form of straightening needed for the tuning argument. The analytic-family regularity and the later topological-degree argument are logically different, so we keep them separate.

Lemma 8.10 (Boundary quadratic quasiconformal rigidity). *If $c_0 \in \partial M$ and P_{c_0} is quasiconformally equivalent to P_{c_1} , then $c_1 = c_0$.*

Proof. Restrict the Beltrami coefficient μ of a quasiconformal conjugacy to $K(P_{c_0})$ and extend it by zero outside. It is invariant under P_{c_0} . Integrating $t\mu$ by Ahlfors–Bers gives

$$\Phi_t \circ P_{c_0} \circ \Phi_t^{-1} = P_{u(t)}, \quad (8.22a)$$

with $u(t)$ holomorphic. Since the coefficient vanishes in the basin of infinity, $u(t) \in M$. The open mapping theorem and $u(0) = c_0 \in \partial M$ force u to be constant. At the parameter corresponding to the original coefficient the remaining conjugacy is hybrid, so connected hybrid rigidity gives $c_1 = c_0$. Thus

$$c_0 \in \partial M, \quad P_{c_0} \sim_{qc} P_{c_1} \implies c_1 = c_0. \quad (8.22b)$$

$\square$

Proposition 8.11 (Degree-two family regularity for tubing-normalized straightening). *Let $(R_\lambda)_{\lambda \in \Lambda}$ be an analytic family of degree-two polynomial-like maps over a Riemann surface, equipped locally with a horizontally analytic tubing, and let*

$$\tilde{\chi} : \Lambda \longrightarrow \mathbb{C}$$

be the tubing-normalized straightening coordinate. Then:

1. $\tilde{\chi}$ is continuous;
2. for every $c \in M$, the fibre $\tilde{\chi}^{-1}(c)$ is a complex-analytic subset of the parameter surface;
3. on every MSS-stable component contained in the connectedness locus, $\tilde{\chi}$ is holomorphic;
4. if $\tilde{\chi}$ is nonconstant, a parameter of the connectedness locus is bifurcating exactly when its straightening parameter belongs to ∂M .

Proof. We prove the four statements from the quasiconformal and stability tools of Chapters 5–7.

MRMT continuity under L^1 control. Suppose sphere Beltrami coefficients μ_n, μ satisfy

$$\|\mu_n\|_\infty, \|\mu\|_\infty \leq k < 1, \quad \|\mu_n - \mu\|_{L^1} \rightarrow 0. \quad (8.22f)$$

Their normalized solutions converge uniformly in the spherical metric. Indeed, the common bound gives normalized quasiconformal compactness. Every subsequence has a further coefficient subsequence converging almost everywhere; the weak Beltrami equation then passes to the uniform limit by the Chapter 5 L^p solver. Normalized uniqueness identifies every cluster point with the solution for μ .

Continuity on a stable component. Fix a relatively compact parameter disk W in the MSS-stable locus. The neighborhood form of the MSS trivialization gives a neighborhood V of the base Julia set and embeddings

$$T_\lambda : V \rightarrow \mathbb{C}, \quad T_{\lambda_0} = \text{id}, \quad (8.22g)$$

holomorphic in λ , uniformly quasiconformal in z , and conjugating the dynamics wherever both sides are defined. Put

$$B_\lambda = T_\lambda(V) \cup K(R_\lambda), \quad B_{\lambda,n} = R_\lambda^{-n}(B_\lambda). \quad (8.22h)$$

For large n , the collar $B_{\lambda,n} \setminus K(R_\lambda)$ is the image under T_λ of the corresponding base collar. The base collars decrease to the empty set outside the filled Julia set and hence have area tending to zero.

The common quasiconformal bound and the Chapter 5 $W^{1,p}$ estimate give, for some $p > 2$,

$$\sup_{\lambda \in W} \|DT_\lambda\|_{L^p} \leq C_W. \quad (8.22i)$$

Thus

$$\text{area}(T_\lambda(E)) \leq C_W^2 \text{area}(E)^{1-2/p}, \quad (8.22j)$$

and therefore

$$\sup_{\lambda \in W} \text{area}(B_{\lambda,n} \setminus K(R_\lambda)) \rightarrow 0. \quad (8.22k)$$

Truncate the invariant straightening coefficient after n pullback levels. For fixed n the truncated coefficient depends continuously in L^1 on λ , while the tail satisfies

$$\|\mu_\lambda - \mu_{\lambda,n}\|_{L^1} \leq k \text{area}(B_{\lambda,n} \setminus K(R_\lambda)). \quad (8.22l)$$

Hence the full coefficient is L^1 -continuous, and (8.22f) gives continuity of the normalized straightening map and of $\tilde{\chi}$ on W .

Analyticity of the fibres. Fix $c \in M$. Compare the given tubing with a fixed external tubing of P_c . After transporting the resulting annular complex structure to a fixed disk and integrating it by Ahlfors–Bers, the external identification becomes a jointly holomorphic function $H(\lambda, z)$ on a fixed annulus $r < |z| < 1$. The map R_λ is hybrid equivalent to P_c exactly when H_λ extends holomorphically through the inner disk. The

boundary winding is one, so such an extension has degree one and is univalent; the dynamical identity then extends by the identity theorem.

For $r < \rho < 1$ write

$$H(\lambda, z) = \sum_{j \in \mathbb{Z}} a_j(\lambda) z^j, \quad a_j(\lambda) = \frac{1}{2\pi i} \int_{|z|=\rho} H(\lambda, z) z^{-j-1} dz. \quad (8.22o)$$

The coefficients are holomorphic in λ , and extension is equivalent to $a_{-1} = a_{-2} = \dots = 0$. Near a point of the fibre, either all negative coefficients vanish identically, or one of them is nonzero as a holomorphic function and, after shrinking, its only zero under consideration is the base point. Near a point outside the fibre one coefficient stays nonzero. Hence the fibre is locally either all, empty, or discrete, and is therefore an analytic subset of the Riemann surface.

Allowing the target quadratic parameter c to vary gives in the same way an analytic graph

$$\Gamma = \{(\lambda, c) : R_\lambda \sim_{\text{hyb}} P_c\}. \quad (8.22q)$$

On a stable connectedness component, connected hybrid rigidity makes this graph single-valued and the continuity just proved makes its section continuous. A continuous function with analytic graph is holomorphic. This proves the third assertion.

Global continuity in degree two. If the critical point escapes, escape persists and the tubing external critical-value coordinate $v(\lambda)$ satisfies

$$\tilde{\chi}(\lambda) = \Phi_M^{-1}(v(\lambda)), \quad (8.22r)$$

with Φ_M the quadratic parameter Böttcher coordinate. Hence $\tilde{\chi}$ is holomorphic there. If the parameter is interior to the connectedness locus, the critical orbit is locally bounded, hence passive; MSS stability and the stable-continuity argument apply.

At a boundary parameter λ_0 , the critical orbit is active, so nonpersistent neutral parameters accumulate at λ_0 . At such a neutral parameter the straightened quadratic also has a neutral periodic orbit: quasiconformal conjugacy preserves the attracting/repelling versus neutral local topological type. Every neutral parameter of the standard quadratic family belongs to ∂M . To see this without invoking density of hyperbolicity, note first that a persistent neutral branch would have constant unit-modulus multiplier by the open mapping theorem; for fixed period and fixed multiplier the equations $P_c^q(z) = z$ and $(P_c^q)'(z) = \rho$ have only finitely many parameter solutions, since their resultant is not identically zero (outside M the quadratic map is hyperbolic). Thus a neutral branch is nonpersistent and hence bifurcating. The quadratic critical orbit is passive on $\text{int } M$ because its iterates are locally uniformly bounded by 2, and passive on $\mathbb{C} \setminus M$ by persistent escape; it is active on ∂M . The Montel-Hurwitz criterion therefore gives the standard identity $\text{Bif}(P) = \partial M$.

Normalized quasiconformal compactness now gives subsequential limits of the straightenings of neutral parameters approaching λ_0 . Their marked critical values stay in a fixed compact set, so the normalized quadratic parameters also have subsequential limits, every one quasiconformally equivalent to the straightening of R_{λ_0} . Since the approximating parameters lie on ∂M , boundary quadratic rigidity (Lemma 8.10) forces the straightening of λ_0 itself to be that same boundary limit. Repeating the

compactness argument for an arbitrary sequence approaching λ_0 shows that every subsequential quadratic limit equals it. Thus $\tilde{\chi}$ is continuous at λ_0 .

Connectedness versus bifurcation. For a degree-two polynomial-like family, the unique critical point is passive in the interior and exterior of the connectedness locus and active on its boundary; this is exactly the Montel-Hurwitz activity criterion of Chapter 7. Hence the bifurcation locus is the boundary of the connectedness locus. The same argument for $z^2 + c$ gives the quadratic bifurcation locus ∂M . Continuity shows that a parameter mapping to $\text{int } M$ is interior to the connectedness locus. Conversely an interior connectedness parameter is stable and $\tilde{\chi}$ is holomorphic there. If its value were on ∂M , stable quasiconformal conjugacy and boundary rigidity would force $\tilde{\chi}$ to be locally constant. Its analytic boundary fibre would then contain an open set and hence, on the connected parameter surface, would be the whole surface, contrary to nonconstancy. This proves the fourth assertion and completes the proposition. $\square$

Lemma 8.12 (Topological holomorphy from family regularity). *Assume Proposition 8.11. If $\tilde{\chi}$ is not constant, then, for every $c \in M$, the fibre $\tilde{\chi}^{-1}(c)$ is discrete and every point of that fibre has positive local topological degree. Equivalently, $\tilde{\chi}$ is topologically holomorphic over M .*

Proof. Boundary quadratic quasiconformal rigidity is Lemma 8.10.

By Proposition 8.11, every fibre over M is an analytic subset of a one-dimensional parameter surface. A non-discrete fibre therefore contains a nonempty open set; an analytic subset of a connected Riemann surface with nonempty interior is the whole surface. Hence $\tilde{\chi}$ would be constant. All fibres are therefore discrete.

Fix $\lambda_0 \in \tilde{\chi}^{-1}(M)$, put $c_0 = \tilde{\chi}(\lambda_0)$, and choose a closed disk $\Delta \Subset \Lambda$ containing no other point of the fibre over c_0 . Continuity makes

$$i_{\lambda_0}(\tilde{\chi}) = \text{wind}(\tilde{\chi}(\partial\Delta), c_0) \quad (8.22c)$$

well defined. If $c_0 \in \text{int } M$, the family-regularity proposition puts λ_0 in the stable locus and makes $\tilde{\chi}$ holomorphic near it. Nonconstancy of the fibre then says that (8.22c) is the positive zero multiplicity of $\tilde{\chi} - c_0$.

If $c_0 \in \partial M$ but λ_0 were stable, the Julia motion would give quasiconformal equivalences between R_{λ_0} and all nearby R_λ . After straightening we would obtain $P_{c_0} \sim_{qc} P_{\tilde{\chi}(\lambda)}$, and (8.22b) would make $\tilde{\chi}$ locally constant, contradicting discreteness of the fibre. Thus this case cannot occur.

Finally suppose $c_0 \in \partial M$ and λ_0 is bifurcating. The MSS criterion supplies arbitrarily nearby parameters on a nonpersistent neutral multiplier branch and, across it, an attracting parameter $\lambda_a \in \Delta$. For λ_a sufficiently close, continuity gives

$$\text{wind}(\tilde{\chi}(\partial\Delta), c_0) = \text{wind}(\tilde{\chi}(\partial\Delta), c_a), \quad c_a = \tilde{\chi}(\lambda_a). \quad (8.22d)$$

The attracting cycle lies in the interior of the filled set, where the hybrid straightening is conformal, so P_{c_a} is attracting and $c_a \in \text{int } M$. Choose a nearby regular value $w \in \text{int } M$ in the local open image of λ_a . Every preimage of w is stable and $\tilde{\chi}$ is holomorphic

there. Additivity of Brouwer degree over the finitely many preimages in Δ gives

$$\text{wind}(\tilde{\chi}(\partial\Delta), w) = \sum_{\lambda \in \Delta \cap \tilde{\chi}^{-1}(w)} \deg_{\lambda}(\tilde{\chi}, w) = \sum_{\lambda \in \Delta \cap \tilde{\chi}^{-1}(w)} \text{ord}_{\lambda}(\tilde{\chi} - w) > 0. \quad (8.22e)$$

Winding number is locally constant away from the boundary image, and (8.22d)–(8.22e) give $i_{\lambda_0}(\tilde{\chi}) > 0$. Thus every fibre point over M has positive local degree. $\square$

Lemma 8.13 (Degree-one mandelbrot-like family lemma). *Let (R_{λ}) be a nonconstant analytic quadratic-like family over a parameter disk Λ , equipped with a holomorphic tubing of its fundamental annuli. Assume that*

$$\mathcal{M}_R = \{\lambda : K(R_{\lambda}) \text{ is connected}\} \Subset \Lambda$$

and that the critical-value section has winding number one around the tubing coordinate on the boundary of a disk containing $\mathcal{M}_R$. Then the normalized straightening parameter

$$\chi(\lambda) \in M, \quad R_{\lambda} \sim_{\text{hyb}} P_{\chi(\lambda)}, \quad (8.22)$$

is a homeomorphism $\mathcal{M}_R \rightarrow M$.

Proof. Use the tubing to form the full straightening coordinate $\tilde{\chi} : \Lambda \rightarrow \mathbb{C}$. On $\mathcal{M}_R$ connected hybrid rigidity identifies its value with the unique parameter $\chi(\lambda) \in M$, so $\mathcal{M}_R = \tilde{\chi}^{-1}(M)$. The boundary critical-value section has winding one; in the tubing coordinate this is impossible for a constant straightening coordinate (a constant coordinate has boundary degree zero). Hence $\tilde{\chi}$ is not constant, which is the hypothesis required in Lemma 8.12. Therefore it is topologically holomorphic over M .

Compactness of $\mathcal{M}_R$ makes $\chi : \mathcal{M}_R \rightarrow M$ a proper ramified covering in the topological sense. Its parametric degree δ is the sum of the positive local degrees in a fiber and is constant because M is connected. The boundary-degree formula identifies δ with the winding number of the marked critical-value section. Hence

$$\delta = 1. \quad (8.23)$$

In particular the covering is nonempty over every point of M , and each fiber contains exactly one point, of local degree one. The degree-one local form of topological holomorphy is a homeomorphism on the connectedness locus. Therefore χ is a continuous bijection $\mathcal{M}_R \rightarrow M$; compactness of the source and Hausdorffness of the target make it a homeomorphism. No density of hyperbolic parameters or centers is used. $\square$

Lemma 8.14 (Primitive and punctured-satellite straightening). *For a primitive hyperbolic component H , normalized straightening gives a homeomorphism $\chi_H : \mathcal{M}_H \rightarrow M$. For a satellite component it gives a homeomorphism on the punctured loci*

$$\chi_H : \mathcal{M}_H^{\circ} \longrightarrow M \setminus \{1/4\}.$$

Proof. Apply Lemma 8.13 to the primitive characteristic quadratic-like family. We obtain a homeomorphism

$$\chi_H : \mathcal{M}_H \longrightarrow M. \quad (8.24)$$

Its inverse is the desired tuning map in the primitive case.

For a satellite component the polynomial-like annulus degenerates at the root, so the compact version of Lemma 8.13 must *not* be applied verbatim. Instead work on the punctured puzzle $\Lambda_H \setminus \{r_H\}$. The same straightening coordinate is topologically holomorphic over M , and on every compact subdisk avoiding the root the local-degree computation is the one in Lemma 8.13. The only loss of properness occurs at the pinching parameter r_H . Along the central hyperbolic component the multiplier of the straightened fixed point tends to 1 exactly when the characteristic period- p cycle of the original polynomial becomes parabolic. The hyperbolic-component multiplier theorem and characteristic-ray landing theorem of Volume 4 identify that unique source limit as r_H and the corresponding target limit as the main-cardioid cusp $1/4$.

Hence, for every compact set $L \subseteq M \setminus \{1/4\}$, inverse images of L stay a positive distance from r_H : otherwise a sequence approaching the root would, by the multiplier limit just identified, have straightening parameters tending to $1/4$, contrary to compactness of L away from the cusp. Excise a sufficiently small circle around r_H . Its straightening image lies in a small neighborhood of $1/4$, so it has winding number zero about every $c \in L$, while the outer parameter boundary still has the critical-value winding one. The topological degree over L is therefore one. The degree-one ramified-covering argument of Lemma 8.13 now gives a unique local homeomorphic preimage for every $c \in L$. Exhausting $M \setminus \{1/4\}$ by such compact sets proves that the punctured straightening is a homeomorphism

$$\chi_H : \mathcal{M}_H^\circ \longrightarrow M \setminus \{1/4\}. \quad (8.24s)$$

This is precisely the semi-tunable case: the single omitted source point and the single omitted target point are both parabolic pinching points. $\square$

Lemma 8.15 (Satellite root extension). *In the satellite case the inverse straightening*

$$\chi_H^{-1} : M \setminus \{1/4\} \longrightarrow \mathcal{M}_H^\circ$$

extends continuously by

$$\chi_H^{-1}(1/4) = r_H, \quad (8.25)$$

and the extension is injective.

Proof. Let $c_n \rightarrow 1/4$ in $M \setminus \{1/4\}$ and set $\lambda_n = \chi_H^{-1}(c_n)$. Compactness of the closed characteristic wake gives cluster points of (λ_n) . A cluster point in the interior of the parameter puzzle would have a nondegenerate polynomial-like limit, whose straightening would equal $P_{1/4}$ by the continuity argument in Lemma 8.13; its period- p characteristic cycle would therefore be parabolic. Volume 4 says the characteristic ray pair has exactly one such landing parameter, namely r_H . Hence every cluster point is r_H , proving convergence and therefore continuity of (8.25). Since r_H was the only omitted source point and the punctured inverse was injective, the extension remains injective. $\square$

Define in both primitive and satellite cases

$$\tau_H := \chi_H^{-1} : M \longrightarrow M, \quad M_H := \tau_H(M). \quad (8.26)$$

The image lies in the global Mandelbrot set because connectedness of the renormalized critical orbit implies boundedness of the original critical orbit. The map is continuous and injective. Since M is compact and the parameter plane is Hausdorff, τ_H is a homeomorphism onto M_H . By its definition through straightening,

$$P_c^p : U'_{\tau_H(c)} \rightarrow U_{\tau_H(c)} \quad \text{is hybrid equivalent to} \quad P_c \quad (8.27)$$

for every $c \in M$. Also

$$\tau_H(0) = c_H, \quad \tau_H(1/4) = r_H, \quad (8.28)$$

which fixes the canonical normalization.

It remains to identify the combinatorics.

Lemma 8.16 (Characteristic block substitution). *For every rational parameter angle θ for which the source parameter ray lands at $c \in M$, the two parameter rays at the one-sided tuned angles $T_H(\theta)$ land at the corresponding tuned parameter $\tau_H(c)$, with the usual identification when θ is dyadic. Consequently characteristic ray pairs, wakes, and rational limbs are carried to their H -tuned counterparts.*

Proof. The two characteristic rays in (8.17) divide the dynamical plane into the two sectors labeled 0 and 1. One iterate of the renormalized return corresponds to p iterates of the original polynomial. Therefore an itinerary symbol ϵ_j for the straightened quadratic is replaced by the length- p characteristic block $A_H^{\epsilon_j}$ in the original dynamics. This is exactly the substitution (8.18).

For a rational angle the relevant dynamic rays are preperiodic or periodic and land by the ray theorem already proved in Volumes 3–4. Their landing pattern is stable throughout a parameter puzzle until a characteristic relation is met. The parameter-ray landing theorem then identifies the unique parameter at which that relation occurs. Hence the source ray at θ and the tuned ray(s) at $T_H(\theta)$ determine parameters related by τ_H . Applying the same statement to both angles of a characteristic pair gives the wake correspondence, and taking the component cut off by such a pair gives the limb correspondence. $\square$

Proof. Lemma 8.9 supplies the characteristic quadratic-like family. Lemma 8.14 identifies its straightening with M in the primitive case and with the punctured Mandelbrot set in the satellite case; Lemma 8.15 fills the unique satellite puncture. Thus the inverse straightening defines the continuous injective map τ_H in (8.26), with the normalization (8.28), and compactness makes it a homeomorphism onto the small copy M_H . Equation (8.27) is the defining hybrid equivalence. Finally Lemma 8.16 identifies rational ray pairs, wakes, and limbs under the tuning substitution. These statements are exactly the assertions of the tuning theorem. $\square$

8.3 Worked examples

Worked example: A quadratic polynomial viewed as a quadratic-like map

Let $P_c(z) = z^2 + c$ and choose $R > 0$ so large that

$$R^2 - |c| > R. \tag{8.E1}$$

Put $V = D(0, R)$ and let

$$V' = P_c^{-1}(V) = \{z : |z^2 + c| < R\}.$$

For $|z| = R$, (8.E1) gives

$$|P_c(z)| \geq R^2 - |c| > R,$$

so $\overline{V'} \subset D(0, R) = V$. The map $P_c : V' \rightarrow V$ is proper of degree two. If $K(P_c)$ is connected, the unique critical point 0 has bounded orbit and therefore belongs to V' ; the component of $P_c^{-1}(V)$ containing it contains the whole filled Julia set. Thus this restriction is quadratic-like.

Its straightening is visibly P_c itself: the identity map is already a hybrid conjugacy. Uniqueness says that every other polynomial straightening is affinely conjugate to P_c , and monic-centered normalization recovers exactly the same parameter c .

Worked example: Normalizing an arbitrary quadratic

Take

$$Q(z) = az^2 + bz + e, \quad a \neq 0.$$

The critical point is $z_0 = -b/(2a)$. Translating by $z = z_0 + w$ removes the linear term:

$$Q(z_0 + w) - z_0 = aw^2 + q_0$$

for a constant q_0 . Now scale $w = u/a$. Conjugating by the resulting affine map gives

$$u \mapsto u^2 + c$$

for one explicit constant c . If an affine map conjugates $z^2 + c$ to $z^2 + c'$, coefficient comparison forces its linear coefficient to be 1 and its translation to be 0, hence $c = c'$. The normalized quadratic parameter is therefore a genuine invariant of the connected hybrid class.

Worked example: The period-two tuning blocks

The period-two hyperbolic component has center $c = -1$, where

$$0 \mapsto -1 \mapsto 0.$$

Its characteristic sectors give two binary words of length two. Denote them A^0

and A^1 . If a source external angle has binary itinerary

$$.\epsilon_1\epsilon_2\epsilon_3\dots,$$

tuning replaces each single source symbol by a full period-two block:

$$.\epsilon_1\epsilon_2\epsilon_3\dots \mapsto .A^{\epsilon_1}A^{\epsilon_2}A^{\epsilon_3}\dots \text{tag8.E2}$$

The reason is dynamical, not merely symbolic. One iterate of the straightened quadratic corresponds to the first return P_c^2 on the small quadratic-like filled set, so each source itinerary step expands to two itinerary steps in the original plane. The same substitution on rational characteristic parameter angles identifies the wakes and limbs of the small Mandelbrot copy.

Boundary case: Disconnected filled Julia sets and loss of rigidity

Straightening still exists when $K(f)$ is disconnected. What is lost is the connected-case rigidity that forces two polynomial straightenings to be affinely conjugate.

The mechanism is visible for escaping quadratic parameters. Their Julia sets are hyperbolic Cantor repellers. Along a stable parameter region these Cantor dynamics move quasiconformally, and the hybrid condition on a zero-area filled set does not force the exterior conformal marking used in the connected rigidity proof. Distinct quadratic parameters can therefore represent the same local hybrid type without being affinely conjugate.

This is why connectedness appears in the uniqueness clause of the theorem and nowhere in the surgery that constructs a straightening.

Common pitfall

Do not replace “the Beltrami coefficient vanishes almost everywhere on $K(f)$ ” by “the hybrid map is conformal at every point of $J(f)$.” The first is an a.e. Sobolev statement. Pointwise differentiation at a boundary periodic point needs an additional conformality or rigidity argument.

8.4 Chapter summary

What to remember

- A polynomial-like map is a proper holomorphic map between nested topological disks. Its filled Julia set consists of points that never leave the inner domain.
- Every polynomial-like map straightens to a polynomial of the same degree. The proof sews the local map to a power map, builds an invariant measurable complex structure, and integrates it by the measurable Riemann mapping

theorem.

- Connectedness of the filled Julia set is not needed for existence. It is needed for rigidity: connected hybrid-equivalent polynomial straightenings are affinely conjugate.
- A hybrid conjugacy carries filled and Julia sets to their counterparts and is conformal on interior components of the filled set. Boundary multiplier statements require additional pointwise conformality.
- A connected quadratic-like first return has a unique monic-centered straightening $z^2 + c_R$, and the connectedness criterion forces $c_R \in M$.
- A characteristic quadratic-like family attached to a hyperbolic component has parameter winding number one. Topological holomorphy of the tubing-normalized straightening coordinate and the boundary-degree formula turn that winding into parametric degree one, hence a homeomorphism with M in the compact case; the satellite case is handled as a punctured semi-tunable family before root extension.
- Inverting that family straightening gives the tuning embedding and a small Mandelbrot copy. The center maps from 0, the root maps from $1/4$, and rational characteristic wakes and limbs follow the corresponding binary block substitution.
- The single-map straightening theorem does not imply arbitrary parameter continuity. Continuity claims must be proved for the particular family under consideration.

Looking ahead

Straightening turns a local return map into a global polynomial, but it does not say how often such returns occur or how their geometry behaves under repeated renormalization. The next volumes develop renormalization as an operator, control the moduli of nested polynomial-like domains, and use rigidity to separate flexible quasiconformal deformations from genuinely determined combinatorics.

Yoccoz Puzzles, Principal Nests and Rigidity Machinery

Why this chapter matters

The Julia set is often far too irregular to study by drawing ordinary Euclidean neighborhoods. Yoccoz's decisive idea is to build neighborhoods from the dynamics itself. A finite collection of external rays and one equipotential cuts the filled Julia set into puzzle pieces. Pulling the graph back produces a nested, dynamically compatible microscope. Once the critical piece returns to itself, the microscope becomes the principal nest.

The central question is quantitative: do the pieces actually shrink? The answer is encoded by conformal moduli of the annuli between successive pieces. Divergent total modulus forces shrinking. In the nonrenormalizable quadratic case, Yoccoz's tableau combinatorics produces that divergence. In the finitely renormalizable case, Kahn-Lyubich complex bounds provide a uniform modulus lower bound on a carefully selected nest. Those bounds also make it possible to pull combinatorial conjugacies back with a uniform quasiconformal constant, which is the bridge from combinatorics to rigidity.

Learning goals

After this chapter you should be able to

- construct a Yoccoz puzzle from an admissible ray/equipotential graph;
- prove the Markov and first-return properties rather than treating them as part of the definition;
- build the principal nest and distinguish central from lateral returns;
- explain why divergent annular modulus forces puzzle pieces to shrink;
- distinguish the nonrenormalizable Yoccoz argument from the finite-renormalization complex-bounds argument;
- explain how a priori bounds produce quasiconformal rigidity; and
- state precisely what this chapter does *not* prove about infinitely renormalizable maps or parameter-space local connectivity.

Conceptual roadmap

Geometry:

external rays + equipotential $\longrightarrow$ pullback graph $\longrightarrow$ puzzle pieces.

Dynamics:

Markov property $\longrightarrow$ nice first returns $\longrightarrow$ principal nest.

Shrinking:

critical combinatorics $\longrightarrow$ annular modulus $\longrightarrow \sum \text{mod} = \infty$
 $\Downarrow$
 point impressions.

Rigidity:

complex bounds $\longrightarrow$ uniform quasiconformal extensions
 $\Downarrow$
 global quasiconformal conjugacy $\longrightarrow$ parameter rigidity.

Common pitfall

There are three logically different conclusions nearby. Local connectivity of a *Julia set*, local connectivity of the *Mandelbrot set*, and combinatorial rigidity of a parameter are not interchangeable. This chapter proves Julia-set local connectivity and finite-renormalization rigidity in the all-periodic-points-repelling setting. Quantitative parapuzzle transfer in parameter space is postponed to later volumes.

9.1 Definitions

Definition 9.1 (admissible Yoccoz graph). For a quadratic polynomial $P_c(z) = z^2 + c$ with connected filled Julia set, an *admissible Yoccoz graph* is the finite forward-invariant graph obtained from one positive Green equipotential and finitely many external rays landing at a repelling or parabolic periodic orbit, chosen so that the forward critical orbit does not meet the graph. Its depth- n pullback graph is

$$\Gamma_n = \bigcup_{j=0}^n P_c^{-j}(\Gamma_0).$$

Definition 9.2 (puzzle piece and critical puzzle piece). A *puzzle piece of depth n* is the closure of a bounded connected component of $\mathbb{C} \setminus \Gamma_n$. The unique depth- n piece whose interior contains the critical point 0, when it exists, is the *critical puzzle piece* and is denoted $Y^n(0)$.

Definition 9.3 (nice puzzle piece and first-return map). A puzzle piece V is *nice* if $P_c^k(\partial V) \cap \text{int } V = \emptyset$ for every $k \geq 1$. Its *first-return domain* through a point z is the component U of the set of points returning to V on which the first-return time is constant; the associated map $P_c^r : U \rightarrow V$ is the *first-return branch*. A branch containing the

critical point is the *central branch*; all other branches are lateral.

Definition 9.4 (critical return). A *critical return* to a nice critical puzzle piece V is a return time $r > 0$ with $P_c^r(0) \in V$. It is *central* when the corresponding first-return branch is the central branch and *lateral* otherwise.

Definition 9.5 (principal nest). For a recurrent critical point, choose a nice critical puzzle piece V^0 . Inductively, V^{n+1} is the first-return domain to V^n that contains 0. The resulting nested sequence

$$V^0 \supset V^1 \supset V^2 \supset \dots \ni 0$$

is the *principal nest*. Its annuli $A_n = V^n \setminus \overline{V^{n+1}}$ are the *principal annuli* whenever nondegenerate.

Definition 9.6 (puzzle end and impression). For $z \in J(P_c)$, let $Y^n(z)$ be the depth- n puzzle piece containing z (with the common-boundary convention when necessary). The nested sequence $(Y^n(z))_{n \geq 0}$ is the *puzzle end* of z , and

$$\text{Imp}(z) = \bigcap_{n \geq 0} Y^n(z)$$

is its *puzzle impression*. The end *shrinks* if its impression is the singleton $\{z\}$, equivalently $\text{diam } Y^n(z) \rightarrow 0$ after a standard bounded-shape thickening.

Definition 9.7 (renormalization combinatorics and combinatorial equivalence). Two normalized quadratic polynomials are *combinatorially equivalent* in Chapter 9 if their admissible puzzles identify the critical tableau and every finite renormalization level, including the cyclic order of marked boundary rays and the return itineraries. A map is *at most finitely renormalizable* if there is no infinite nested tower of connected quadratic-like renormalizations.

Definition 9.8 (complex bounds). A critical nest has *complex bounds* (or *a priori bounds*) if there is $\mu > 0$ and a cofinal modified/principal subnest $E^0 \supset E^1 \supset \dots$ such that

$$\text{mod}(E^{j-1} \setminus \overline{E^j}) \geq \mu$$

for every sufficiently deep level. The constant may depend on the initial finite combinatorics, but not on the depth.

Definition 9.9 (parapuzzle). A *parapuzzle piece* is a parameter region on which a fixed finite puzzle combinatorics persists, bounded by parameter rays/equipotentials and holomorphic-motion loci. In this chapter parapuzzles are used only as a forward interface; quantitative phase-parameter transfer is deferred to later parameter-space chapters.

9.2 Main results

Theorem 9.1 (Yoccoz puzzle construction). **Statement scope.** Let $P_c(z) = z^2 + c$ have connected filled Julia set, and let Γ_0 be an admissible Yoccoz graph. For every $n \geq 0$ the

finite pullback graph Γ_n cuts the bounded region below the chosen equipotential into finitely many topological disks. Their closures are the depth- n puzzle pieces; boundaries consist of finitely many ray segments and equipotential arcs, and the pieces form a finite closed cover of the filled Julia set with pairwise disjoint interiors.

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Bottcher/equipotential geometry plus rational dynamic-ray landing; finite pullback graph; planar disk-complement induction and Riemann-Hurwitz.

The mathematical argument begins in the proof environment below.

Lemma 9.2 (Finite embedded initial graph). *Write $P = P_c$, and let $\mathcal{O}$ be a repelling or parabolic periodic orbit at which the chosen finite set of rational external rays lands. If the forward critical orbit is disjoint from the resulting ray graph, then for every $t > 0$ the union of the equipotential E_t , the selected ray segments from E_t to their landing points, and the orbit $\mathcal{O}$ is a finite embedded planar graph Γ_0 .*

Proof. Since $c \in M$, $K(P)$ is connected and the basin of infinity admits the normalized Böttcher coordinate

$$\phi : \mathbb{C} \setminus K(P) \longrightarrow \mathbb{C} \setminus \overline{\mathbb{D}}, \quad \phi(Pz) = \phi(z)^2. \quad (9.1)$$

The Green function is $G(z) = \log |\phi(z)|$, so every positive level $E_t = \{G = t\}$ is an analytic equipotential and $P(E_{t/2}) = E_t$. The earlier rational-ray landing theorem gives the landing of every periodic or preperiodic rational ray used in the finite portrait. Two distinct external rays are disjoint in the basin because their images under ϕ are distinct radial lines. If two such rays meet at a finite point, that point is their common landing point; otherwise uniqueness of the Böttcher coordinate would be violated. An equipotential meets each selected ray once and transversely. There are only finitely many rays and landing points, hence the union is a finite one-dimensional CW complex embedded in the plane. The admissibility assumption says that no $P^n(0)$ lies on it. $\square$

Lemma 9.3 (Disk-complement induction). *For*

$$\Gamma_n = \bigcup_{j=0}^n P^{-j}(\Gamma_0), \quad (9.2)$$

every Γ_n is a finite embedded graph. Each bounded complementary component below the outer equipotential is a topological disk whose boundary is a finite union of ray segments, landing points, and equipotential arcs; distinct such components have disjoint interiors.

Proof. Because P is a finite branched covering of degree two, the inverse image of a finite graph is a finite graph. Its portion in the basin of infinity is described explicitly by (9.1): the outer equipotential E_t acquires the levels $E_{t/2}, \dots, E_{t/2^n}$, and every ray segment acquires the finitely many angle-halves of the original segment. All possible

branching in the finite plane occurs over the single critical point 0. Since the post-critical set does not meet Γ_0 , no critical value occurs on a graph edge; consequently local inverse branches preserve the cyclic order of graph edges away from 0 and no pullback crossing is created.

For $n = 0$, the selected rays are embedded arcs joining the outer equipotential to the finite landing portrait. Cutting the closed disk bounded by E_t along this finite noncrossing family produces finitely many closed cells, each a topological disk. Assume the assertion at depth n . On a depth- n cell that does not contain the critical value, the two inverse branches of P are conformal, so the cell lifts to Jordan cells. If a cell contains the critical value, each component U of its preimage is governed by Riemann-Hurwitz:

$$\chi(U) = d_U \chi(V) - \sum_{x \in U} (e_x - 1). \quad (9.3)$$

Here $d_U \in \{1, 2\}$, and in the degree-two case U contains the unique simple critical point, so $\chi(U) = 2 - 1 = 1$; hence U is again a disk. The boundary type is preserved by pullback. Finally, complementary components of an embedded graph have disjoint interiors. $\square$

Proof. By Lemmas 9.2 and 9.3, every pullback graph is finite and its bounded complementary components are topological disks with the asserted boundary type. Their closures are, by definition, the depth- n puzzle pieces. The depth- n equipotential surrounds $K(P)$, so these closures form a finite closed cover of $K(P)$; their interiors are pairwise disjoint. This is exactly the asserted puzzle construction. $\square$

Theorem 9.4 (Puzzle Markov and first-return structure). **Statement scope.** *If Y is a depth- $(n + 1)$ puzzle piece, then $P_c(Y)$ is a depth- n puzzle piece and $P_c : \text{int } Y \rightarrow \text{int } P_c(Y)$ is proper. Its degree is two exactly when Y contains the critical point and one otherwise. Puzzle pieces of comparable depths are either nested or have disjoint interiors. Consequently every first-return branch to a nice puzzle piece is proper; the central branch has degree two and every lateral branch is conformal.*

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Boundary-to-boundary pullback, properness and Riemann-Hurwitz degree dichotomy; nesting; nice first-return domains.

The mathematical argument begins in the proof environment below.

Lemma 9.5 (Markov image, degree, and nesting). *Let Y be a depth- $(n + 1)$ puzzle piece. Then there is a unique depth- n piece Z such that*

$$P : \text{int } Y \longrightarrow \text{int } Z$$

is proper and onto. Its degree is two precisely when $0 \in \text{int } Y$, and otherwise it has degree one. Moreover, puzzle pieces of comparable depths are either nested or have disjoint interiors.

Proof. By construction,

$$P(\Gamma_{n+1}) \subset \Gamma_n. \quad (9.4)$$

Hence $P(\text{int } Y)$ is connected and avoids Γ_n , and is therefore contained in the interior of a unique depth- n piece Z . Since $\partial Y \subset \Gamma_{n+1}$ and (9.4) sends this boundary into $\Gamma_n \supset \partial Z$, a sequence in $\text{int } Y$ escaping every compact subset cannot have image converging to an interior point of Z . Thus the restriction is proper. A nonconstant proper holomorphic map has open and closed image in the connected target, hence is onto.

The polynomial has exactly one critical point, 0, of local degree two. If $0 \notin \text{int } Y$, the restriction is unramified and Riemann–Hurwitz gives degree one. If $0 \in \text{int } Y$, it contains that single simple critical point and has degree two.

For nesting, let $m < n$ and suppose depth- n and depth- m pieces have intersecting interiors. Since $\Gamma_m \subset \Gamma_n$, the depth- n interior lies in a single complementary component of Γ_m , hence inside the depth- m piece. Pieces at the same depth have disjoint interiors by the puzzle construction. This proves the nesting alternative. $\square$

Lemma 9.6 (First-return branch structure). *Let V be a nice puzzle piece and U a component of its first-return domain with return time r . Then*

$$P^r : U \longrightarrow V \tag{9.5}$$

is proper and onto. Distinct first-return domains have disjoint interiors. The branch has degree two exactly when $0 \in U$; otherwise it is conformal.

Proof. For $0 < j < r$, minimality of r gives $P^j(U) \cap V = \emptyset$. By Lemma 9.5, every map along the puzzle itinerary $U, P(U), \dots, P^{r-1}(U)$ is proper, so their composition (9.5) is proper and onto.

If two return domains U_1, U_2 had overlapping interiors, puzzle nesting would put one in the other. Suppose $U_1 \Subset U_2$ and their return times are $r_1 \leq r_2$. Since $P^{r_1}(U_1) = V$, the puzzle piece $P^{r_1}(U_2)$ meets V ; nesting forces one to contain the other. Either alternative contradicts minimality of one of the return times unless the two domains and return times coincide. Thus distinct return domains have disjoint interiors.

Along a first-return itinerary the critical point can occur in at most the initial domain. If $0 \in P^j(U)$ for some $0 < j < r$, the critical puzzle piece and the pullback itinerary force a return to the nice critical neighborhood before time r , contradicting first return. Hence the product of the local degrees is two exactly when $0 \in U$ and is one otherwise. In the latter case the branch is a proper degree-one holomorphic map between disks, hence is conformal. $\square$

Proof. Lemma 9.5 gives the Markov image, degree, and nesting assertions. Lemma 9.6 gives the first-return decomposition and its degree classification. The branch containing the critical point is the central branch; all others are lateral. This proves the theorem. $\square$

Definition 9.10 (Principal-nest first-return structure). **Statement scope.** For a recurrent quadratic critical orbit and a nice critical starting piece V^0 , successive central first-return domains define the principal nest $V^0 \supset V^1 \supset \dots$. Each return $P_c^{r_n} : V^{n+1} \rightarrow V^n$ is a degree-two branched covering with the unique critical point 0. A return is central precisely when the critical value of this branch lies in V^{n+1} ; otherwise it is lateral.

Theorem 9.7 (Yoccoz shrinking theorem in the nonrenormalizable repelling case).
Statement scope. Let $c \in M$ and suppose $P_c(z) = z^2 + c$ is nonrenormalizable and every periodic point is repelling. For an admissible Yoccoz puzzle based at a dividing repelling periodic orbit, every puzzle end shrinks to a point. Hence $J(P_c) = K(P_c)$ is locally connected. This row does not assert parameter or combinatorial rigidity.

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Grötzsch shrinking; the tableau rules proved in this chapter; critical-staircase/off-critical transfer; originator two-child and noble-level fortune divergence; nonrenormalizable rank/seed annulus; connected puzzle traces. The mathematical argument begins in the proof environment below.

Proof setup. Put $P = P_c$. Let $C_n = Y^n(0)$ be the depth- n critical puzzle piece and, for any puzzle end x , let

$$X_0(x) \supset X_1(x) \supset \cdots$$

be its nested pieces. Set

$$A_n(x) = \text{int } X_n(x) \setminus X_{n+1}(x), \quad A_n = A_n(0). \quad (9.6)$$

If two successive boundaries meet, the corresponding annulus is called degenerate and is assigned modulus zero. Thus a divergent modulus sum may be read on any cofinal family of genuine annuli.

Lemma 9.8 (Grötzsch shrinking criterion). Let $D_0 \supset D_1 \supset \cdots$ be nested Jordan disks. If a cofinal subsequence of separating annuli satisfies

$$\sum_j \text{mod}(D_j \setminus \overline{D_{j+1}}) = \infty, \quad (9.7)$$

then $\bigcap_j \overline{D_j}$ is a singleton and $\text{diam } D_j \rightarrow 0$ after one outer disk has been fixed.

Proof. Uniformize D_0 by $\mathbb{D}$. Annuli which occur in series have additive extremal distance, so

$$\text{mod}(D_0 \setminus \overline{D_N}) \geq \sum_{j < N} \text{mod}(D_j \setminus \overline{D_{j+1}}). \quad (9.8)$$

If the nested intersection contained a nondegenerate continuum L , its extremal distance from ∂D_0 would be finite, contradicting (9.8) and monotonicity. Hence the intersection is one point. If the diameters did not tend to zero, compactness of closed subsets of $\overline{D_0}$ would give a subsequential Hausdorff limit containing two distinct points, again contradicting the singleton intersection. $\square$

We now make the tableau combinatorics reader-visible. For an end x define

$$T_{n,k}(x) = X_n(P^k x).$$

A position (n, k) is *critical* if $0 \in T_{n,k}(x)$; it is *strictly critical* if $0 \in T_{n+1,k}(x)$ and *semi-critical* if $0 \in T_{n,k}(x) \setminus T_{n+1,k}(x)$. The tableau of 0 is the *critical tableau*.

Lemma 9.9 (quadratic tableau rules). *The puzzle tableaux satisfy the following rules.*

- (Ta) *In each column the critical positions form either the whole column, no position, or an initial block of strictly critical positions followed by one semi-critical position and then only noncritical positions.*
- (Tb) *If (n, k) is critical, then for $i + j \leq n$ the nature of $(i, k + j)$ agrees with that of (i, j) in the critical tableau.*
- (Tc) *Suppose (n, k) in the critical tableau is strictly critical and $(n + i, k - i)$ is noncritical for $0 < i < k$. If $(n + k, \ell)$ in another tableau is semi-critical, then $(n, \ell + k)$ is semi-critical.*

If the dividing portrait has combinatorial rotation number p/q in lowest terms, then moreover

- (Td) *the zeroth row contains at most $q - 1$ consecutive noncritical positions; a block $(0, k), \dots, (0, k + q - 1)$ is noncritical exactly when $(0, k - 1)$ is strictly critical;*
- (Te) *rows $1, \dots, q - 1$ contain no semi-critical positions.*

Proof. Rule (Ta) is just nesting of the pieces in one end. For (Tb), if (n, k) is critical, then $T_{i,k} = C_i$ for $i \leq n$. Applying P^j to these same critical pieces shows that the pieces at $(i, k + j)$ and (i, j) coincide for $i + j \leq n$, and hence have the same critical type.

For (Tc), the hypotheses and the Markov degree theorem give

$$P^k : C_{n+k} \longrightarrow C_n, \quad P^k : C_{n+k+1} \longrightarrow C_{n+1} \quad (9.9)$$

as proper degree-two maps, with no intermediate critical encounter. Thus

$$C_{n+k+1} = P^{-k}(C_{n+1}) \cap C_{n+k}. \quad (9.10)$$

If $(n, \ell + k)$ were strictly critical while $(n + k, \ell)$ were semi-critical, then the depth- $(n + k + 1)$ piece at column ℓ would be a component of $P^{-k}(C_{n+1})$ contained in C_{n+k} but outside C_{n+k+1} , contradicting (9.10). Hence $(n, \ell + k)$ is semi-critical.

For (Td), label the q level-zero sectors at the dividing portrait by $Q_0, \dots, Q_{q-1}$ so that Q_0 is the critical sector and the sector itinerary advances cyclically. If the k th column begins in a noncritical sector Q_j , then within $q - j \leq q - 1$ further iterates it reaches Q_0 ; therefore a zeroth-row noncritical run has length at most $q - 1$. The only sector whose predecessor enters the first noncritical sector is the depth-one critical sector, which gives the asserted “if and only if” characterization of a full q -block. Finally, for $2 \leq i \leq q$, the distinguished depth- i piece is the unique depth- i piece contained in the preceding distinguished sector. Hence the transition from critical to noncritical cannot occur in rows $1, \dots, q - 1$, proving (Te). $\square$

Lemma 9.10 (reduction to the critical end). *If $\sum_n \text{mod } A_n = \infty$, then every puzzle end shrinks. An end whose orbit eventually avoids some critical piece also shrinks.*

Proof. First suppose the tableau of an end x has critical positions arbitrarily deep. For each sufficiently large row ℓ , let $(\ell, k(\ell))$ be its first critical position. The positions $(\ell, k(\ell))$ form the critical staircase. Along the diagonal from the depth $N_\ell = \ell + k(\ell)$ annulus of x to A_ℓ , no earlier position is critical. Repeated use of the Markov degree-one branch therefore gives a conformal map

$$P^{k(\ell)} : A_{N_\ell}(x) \longrightarrow A_\ell \quad (9.11)$$

whenever A_ℓ is nondegenerate. The staircase rule (Ta) makes the integers N_ℓ strictly increasing after repetitions corresponding to a degenerate annulus are discarded. Consequently the nest of x contains a disjoint subfamily whose modulus sum dominates the nondegenerate critical sum. It diverges, so Lemma 9.8 shrinks the end.

Now suppose the forward orbit of x avoids C_m from some time onward; a finite initial segment is irrelevant. Let α be the marked dividing repelling point and let B_n be the unique depth- n piece which contains α and lies in C_0 without lying in $\text{int } C_0$. If $B_n = C_n$ for every n , the critical tableau is q -periodic. Rule (Td) still forces every end to enter C_0 at least once in every q consecutive columns. Since the orbit of x avoids C_m , infinitely many of those columns have a depth- m piece different from C_m and hence compactly contained in $\text{int } C_0$.

Assume instead that $B_n \neq C_n$ for some n , and take the first such n . The inverse branch of P^q fixing α maps B_n into B_{n+q} and is strictly contracting near α because α is repelling. Hence $\bigcap_r B_{n+rq} = \{\alpha\}$. If x is not a preimage of α , then after each sufficiently late entrance into C_0 its orbit eventually leaves the current B -nest. At the first exit, applying the injective return branch on B_n puts a later iterate in $C_0 \setminus B_n$. Thus there are infinitely many columns with level-zero piece C_0 and depth- n piece compactly contained in $\text{int } C_0$. The preimages of α themselves shrink under the contracting inverse branch, so they need no modulus argument.

Increase m if necessary so that the preceding compact-containment statement holds at depth m . There are only finitely many depth- m pieces, so along an infinite subsequence the depth- m piece is one fixed $Q \Subset \text{int } C_0$. Put

$$\mu = \text{mod}(\text{int } C_0 \setminus Q) > 0.$$

Pulling this annulus back along such a column has degree at most 2^m : every ramified step consumes one of the first m depth levels, and the orbit never enters C_m . The extremal-length covering inequality therefore gives a pullback annulus of modulus at least $2^{-m}\mu$. Taking successive columns produces disjoint pullback annuli. Their total modulus is infinite, and Lemma 9.8 applies. $\square$

The only difficult case left is the recurrent critical tableau. Define the Yoccoz parent map $t : \mathbb{N}_{>0} \rightarrow \mathbb{N}$ by

$$t(m) = \max\{i < m : (i, m-i) \text{ is critical and } m-i > 0\}, \quad (9.12)$$

with $t(m) = 0$ if the set is empty. Thus m is a *child* of $t(m)$. It is *good* if $(t(m), m-t(m))$ is strictly critical and *bad* if that position is semi-critical. A level is *noble* if its entire row contains no semi-critical position. For a good child, (Ta)–(Tb) and the Markov theorem

show that

$$P^{m-t(m)} : A_m \longrightarrow A_{t(m)} \text{ is an unramified double cover, hence } \mod A_m = \frac{1}{2} \mod A_{t(m)}. \quad (9.13)$$

Lemma 9.11 (periodic tableau produces a renormalization). *If a positive column of the critical tableau is entirely critical, then the tableau is periodic and P is quadratic-like renormalizable.*

Proof. Let $k > 0$ be the least entirely critical column. Rule (Tb) repeats the marked grid with period k . Choose N beyond the finite initial graph so that, for $0 < j < k$, the depth- N piece in column j differs from C_N . The Markov property then gives a proper degree-two map

$$P^k : C_{N+k} \longrightarrow C_N \quad (9.14)$$

with the unique critical point 0, and the periodic tableau keeps the critical orbit in C_{N+k} under iterates of P^k . Slightly thicken the two puzzle disks across their finitely many ray/equipotential boundary arcs. Since no intermediate boundary point lies on the critical orbit, the thickening can be chosen so that $U' \Subset U$ and (9.14) extends to a proper degree-two map $P^k : U' \rightarrow U$. Its filled set contains the critical orbit, so it is connected. This is a quadratic-like renormalization. $\square$

Lemma 9.12 (originators and the two-child lemma). *In a recurrent nonperiodic critical tableau, every positive level has at least two children for the parent map t .*

Proof. We reproduce the originator induction. Call a critical position $(\ell, k) \neq (0, 0)$ *originating* if either

- $\ell = 0$ and no position $(k - i, i)$ with $0 < i < k$ is critical; or
- $\ell > 0$, there is an earlier critical position (ℓ, i) with $0 < i < k$, and every $(\ell + i, k - i)$ with $0 < i < k$ is noncritical.

The position $(0, 1)$ starts the induction. Suppose row ℓ contains an originating position. In row $\ell + 1$ take the first critical position $(\ell + 1, k')$ at or to the right of the inherited diagonal. If row $\ell + 1$ already has a critical position before k' , the defining diagonal condition makes $(\ell + 1, k')$ originating.

Assume therefore that it does not. Because the tableau is nonperiodic, the k' th column is not entirely critical. Continue its vertical block of critical positions to its last point (ℓ', k') and inspect the next diagonal

$$i + j = \ell' + k' + 1. \quad (9.15)$$

There is no critical position on this diagonal with first coordinate $> \ell + 1$. Indeed, if there were one, rule (Tb) places the first such position in the first repeated k' -block beyond the tip. At the preceding block the two positions one step apart are critical, while the intervening diagonal positions are noncritical by minimality. Rule (Tc) then forces the chosen first position to be noncritical, a contradiction. Recurrence now supplies a first critical position in row $\ell + 1$ to the right of the end of (9.15); by the

claim, the diagonal segment behind it is noncritical, so this position is originating. Thus every row contains an originator.

Now fix a positive row n . Let (n, k_0) be its first strictly critical position. By (Ta), every $(n + i, k_0 - i)$ with $0 < i < k_0$ is noncritical, so $t(n + k_0) = n$; this is the first good child. The originating position (n, k_1) supplied above has an earlier critical position in the same row, so $k_1 > k_0$, while its defining diagonal condition says that $(n + i, k_1 - i)$ is noncritical for $0 < i < k_1$. Hence $t(n + k_1) = n$ as well. The two depth indices $n + k_0$ and $n + k_1$ are distinct. Thus every positive level—in particular every possible founder and every descendant used below—has at least two children. $\square$

Lemma 9.13 (noble-child alternatives). *In a recurrent nonperiodic critical tableau: (i) every level has a good child; (ii) if a level has only one good child, that child is noble; and (iii) every child of a noble parent is noble and good.*

Proof. For a level n , let (n, k) be the first strictly critical position in row n with $k > 0$, which exists by recurrence. By (Ta), the positions $(n + i, k - i)$ are noncritical for $0 < i < k$. Therefore $n + k$ has parent n and is a good child. This proves (i).

Let m be a good child of n and suppose (m, k) is the first semi-critical position in row m . If some $(m + i, k - i)$, $0 < i < k$, were strictly critical, (Ta) would create an earlier semi-critical position in row m . Thus all these positions are noncritical, and (Tc) forces $(n, k + m - n)$ to be semi-critical. Hence a noble parent cannot have a nonnoble child. Moreover, after this first semi-critical obstruction, the same diagonal argument shows that no critical position occurs on the next diagonal beyond row n until a later strictly critical position (n, p) is reached; that position produces a second good child $n + p$. Consequently a parent with only one good child has a noble good child, proving (ii). If the parent is noble, the first conclusion says all of its children are noble; by definition no child can then be bad, proving (iii). $\square$

Lemma 9.14 (Yoccoz fortune divergence). *Fix a nondegenerate critical annulus A_{n_0} of modulus $M > 0$ in a recurrent nonperiodic critical tableau. Give n_0 weight 1; a good child receives half the weight of its parent and a bad child weight 0. Then*

$$\sum_{n \geq n_0} w(n) = \infty, \quad \text{mod } A_n = Mw(n) \quad \text{whenever } w(n) > 0. \quad (9.16)$$

In particular $\sum_n \text{mod } A_n = \infty$.

Proof. The modulus identity follows inductively from (9.13): positive weight means that every edge from the founder to n is good, and each such edge divides both modulus and weight by two.

Define the generation of a positive-weight descendant by the number of parent steps back to n_0 , and put

$$w^*(n) = \begin{cases} 2w(n), & n \text{ noble,} \\ w(n), & n \text{ not noble.} \end{cases}$$

Let W_r be the sum of w^* over generation r . By Lemmas 9.12 and 9.13, each positive-weight parent falls into exactly one of the following cases: a nonnoble parent has at least two good children; a nonnoble parent has one good child, which is noble; or a noble parent has at least two children and all are noble and good. In the three cases the total modified weight of the good children is respectively at least w , exactly $2(w/2) = w$, and at least $2 \cdot 2(w/2) = 2w$. These are precisely the parent's modified weights in the corresponding cases. Hence $W_{r+1} \geq W_r$. Since $W_0 > 0$ and there are infinitely many generations, $\sum_r W_r = \infty$. Because $w^* \leq 2w$, the ordinary total weight also diverges. The modulus identity then proves the final assertion. $\square$

Lemma 9.15 (nonrenormalizability supplies a seed annulus). *For the p/q puzzle of a nonrenormalizable quadratic, the critical tableau has a first semi-critical zeroth-row position $(0, q\nu)$ for some $\nu \geq 1$, and the corresponding critical annulus $A_{q\nu}$ is nondegenerate.*

Proof. If the q th column were entirely critical, Lemma 9.11 would give a renormalization. Hence by (Ta) it has a semi-critical position, say (k, q) . Rules (Tb) and (Td) show that on the diagonal $i+j = q+k$ the critical positions occur exactly at positive multiples of q in the second coordinate; rule (Tc) makes each such position semi-critical. Before that diagonal the positions (i, jq) are strictly critical. The full-block clause in (Td) then forces $k+q$ to be a multiple of q . Write

$$\nu = \frac{k+q}{q}.$$

It follows that $(0, q\nu)$ is the first semi-critical position in the zeroth row. At the semi-critical position $(0, q\nu)$, the level-zero and level-one pieces containing $P^{q\nu}(0)$ are distinct, so their difference contains a genuine separating annulus. The Markov map $P^{q\nu}$ sends $C_{q\nu}$ and $C_{q\nu+1}$ onto those two pieces with finite degree. Therefore their preimage difference $A_{q\nu}$ cannot be degenerate. Hence $A_{q\nu}$ has positive modulus. $\square$

We can now prove shrinking. If the critical orbit is combinatorially nonrecurrent, its critical-value end eventually avoids a critical piece. Lemma 9.10 gives that end infinitely many annuli of uniformly positive controlled modulus. Pulling them back once through the degree-two critical branch divides modulus by at most two, so the critical modulus sum diverges; Lemma 9.10 then shrinks every end.

Assume instead that the critical tableau is recurrent. By Lemma 9.11 and nonrenormalizability it is nonperiodic. By Lemma 9.15 there is a positive seed annulus. Lemma 9.14 makes the critical modulus sum divergent, and Lemma 9.10 together with Lemma 9.8 shrinks every puzzle end.

Lemma 9.16 (connected puzzle traces). *For every puzzle piece X , the set $K(P) \cap X$ is connected. Consequently, if every puzzle end shrinks, then $K(P)$ is locally connected.*

Proof. At depth zero the puzzle pieces cut $K(P)$ along the dividing orbit and its landing rays; each $K(P) \cap X$ is the closure of one component of the complement of those finitely many dividing points, hence is connected. Assume the assertion through depth $n-1$

and let X have depth n , with $X' = P(X)$. If $0 \notin X$, the Markov theorem makes $P : K \cap X \rightarrow K \cap X'$ a homeomorphism, so connectedness follows by induction. If $0 \in X$ and $K \cap X$ were disconnected, choose a component L not containing 0 . The restriction of P is locally one-to-one near L . Since $K \cap X'$ is connected, $P(L)$ cannot be a whole component of $K \cap X'$; a boundary point of $P(L)$ in $K \cap X'$ would therefore have a preimage in L at which $P|_K$ fails to be locally one-to-one. The only such point is the critical point 0 , contradiction. Thus every puzzle trace is connected.

If z lies in the interior of all sufficiently deep pieces, the sets $K \cap X_n(z)$ form a connected neighborhood basis because their diameters tend to zero. If z lies on puzzle boundaries, it is a preimage of a marked landing point; for all large depths the union of the finitely many adjacent pieces is a connected neighborhood of z , and its diameter also tends to zero. Hence $K(P)$ is locally connected. $\square$

Finally, every periodic point of P is repelling. By the Fatou-component classification and eventual periodicity proved in the earlier chapter, there is no bounded Fatou component. Thus $K(P) = J(P)$. Applying Lemma 9.16 to the shrinking puzzle gives local connectivity of $J(P) = K(P)$, as claimed.

Theorem 9.17 (Finite-renormalization local connectivity and combinatorial rigidity). ***Statement scope.** Let $c \in M$ and let $P_c(z) = z^2 + c$ be at most finitely renormalizable and suppose every periodic point is repelling. Then suitable modified principal nests satisfy a priori modulus bounds; consequently $J(P_c)$ is locally connected. Moreover P_c is combinatorially rigid: if $P_{\tilde{c}}$ is another normalized quadratic polynomial with the same puzzle and finite-renormalization combinatorics, then $\tilde{c} = c$. The a priori lower bound is a conclusion of the theorem, not an additional hypothesis.*

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Extremal-width quasi-additivity/covering $\rightarrow$ modified principal-nest complex bounds $\rightarrow$ shrinking; proper-disk lifting and uniform QC pullback $\rightarrow$ QC equivalence; rational parapuzzle intersection $\rightarrow$ closed combinatorial class; Ahlfors–Bers open-closed rigidity; finite tower via Chapter 8 straightening/tuning.

The mathematical argument begins in the proof environment below.

Proof setup. We separate four mechanisms. First we prove the conformal-geometry estimates used by the modified principal nest. Second we reconstruct the modulus improvement argument and obtain complex bounds. Third we prove the good-nest quasiconformal extension and pullback argument. Finally we isolate the one remaining parameter-space input needed to turn quasiconformal equivalence into uniqueness of the normalized quadratic parameter. Throughout, modulus means conformal modulus of a doubly connected domain.

Lemma 9.18 (Near-degenerate quasi-additivity and the covering estimate). *Fix a topological complexity bound T , a degree bound D , and $\eta \in (0, 1)$. There are $\delta_0 > 0$ and $\varepsilon > 0$ with the following properties.*

(QA) Let $K_1, \dots, K_m \in V$ be pairwise disjoint full continua, with pairwise disjoint collar neighborhoods $K_i \in K'_i \in V$ and $m \leq T$. If

$$\text{mod}(V \setminus K_i) < \delta < \delta_0, \quad \text{mod}(K'_i \setminus K_i) > \eta\delta,$$

then

$$\text{mod}\left(V \setminus \bigcup_{i=1}^m K_i\right) < \frac{2}{\eta m} \delta. \quad (9.13)$$

(CL) Let $g : U \rightarrow V$ be a branched covering of total degree at most D . Let $A \in A' \in U$ and $B \in B' \in V$ be topological disks such that A' is a component over B' , A is the corresponding component over B , and $d = \deg(g : A' \rightarrow B')$. If

$$\text{mod}(B' \setminus \overline{B}) > \eta \text{mod}(U \setminus \overline{A}), \quad (9.14)$$

then either $\text{mod}(U \setminus \overline{A}) \geq \varepsilon$ or

$$\text{mod}(V \setminus \overline{B}) < \frac{2d^2}{\eta} \text{mod}(U \setminus \overline{A}). \quad (9.15)$$

Proof. For a crossing family Γ write $W(\Gamma) = \text{Ext}(\Gamma)^{-1}$. For a Jordan disk $A \in X$ this gives

$$W(X, A) = \frac{1}{\text{mod}(X \setminus \overline{A})}. \quad (9.16)$$

The parallel law says that widths of disjoint parallel rectangles add, while the series law says that widths of rectangles crossed successively combine by harmonic sum.

We first prove the near-degenerate quasi-additivity estimate used below. For each island K_i , take the harmonic extremal metric joining K_i to ∂V . Cutting along its finite critical graph decomposes the metric, up to transverse measure zero, into finitely many Euclidean rectangles. A vertical leaf records the ordered list of other islands which it encounters. Because the surface has fixed finite topology and there are at most T islands, only finitely many route types occur. Delete a fixed-width strip from each vertical side. The resulting route rectangles lose only a bounded total width, while their terminal subrectangles are pairwise disjoint unless two routes agree up to that terminal stage. Thus, once the total width is large, at least a fixed proportion of it survives the truncation.

For a route of height k , let a_k be the total surviving width of the big rectangles and b_i the maximal total width of their i th terminal subrectangles. Every vertical curve crosses these subrectangles successively, so

$$a_k \leq b_1 \oplus \dots \oplus b_k, \quad x_1 \oplus \dots \oplus x_k = \left(\sum_{j=1}^k x_j^{-1} \right)^{-1}.$$

The elementary inequality

$$\left(\sum_k a_k \right)^2 \leq \frac{4}{3} b_1 \sum_i b_i$$

follows by writing equality in the harmonic-sum constraint, telescoping $b_i = a_i + a_i^2/(a_{i-1} - a_i)$, and applying Cauchy-Schwarz. Restoring the small discarded buffers changes $4/3$ to, say, $3/2$. Consequently, if X is the width from ∂V to the union of all islands, Y is the sum of the individual widths, and Z is the sum of the separation widths of the islands, then for all sufficiently large Y ,

$$Y^2 \leq \frac{3}{2} XZ. \quad (9.16a)$$

The collar hypothesis gives $Z \leq m/(\eta\delta)$, while the individual modulus bounds give $Y > m/\delta$. Hence $X > 2\eta m/(3\delta)$ and therefore $\text{mod}(V \setminus \cup_i K_i) < 3\delta/(2\eta m)$, which is stronger than (9.13). Decreasing δ_0 absorbs the finite route-buffer threshold.

For (CL), take the finite Galois closure of g . Its degree is bounded solely in terms of D . The lifts of A form one deck-group orbit. Apply (9.16a) to that orbit. Covering invariance of extremal width gives, after cancellation of the Galois degree,

$$W(U, A)^2 \leq 2d^2 W(V, B) Z_V, \quad (9.16b)$$

where Z_V is the target separation width: ordinary routes from B to ∂V are counted once and nontrivial returning routes from B to itself are counted twice. Cutting such routes at their first and last crossing of $\partial B'$ lifts them to crossing arcs of the collar $B' \setminus \bar{B}$. Hence

$$Z_V \leq \frac{1}{\text{mod}(B' \setminus \bar{B})} < \frac{1}{\eta \text{mod}(U \setminus \bar{A})} = \eta^{-1} W(U, A).$$

Substitution in (9.16b), followed by cancellation of $W(U, A) > 0$, yields (9.15) whenever $W(U, A)$ is above the quasi-additivity threshold. Choosing ε as the reciprocal of that threshold proves the alternative. $\square$

Lemma 9.19 (Buffered return family in the modified nest). *After the last renormalization, fix an integer $m \geq 2$. There is $n_*(m)$ such that, for every sufficiently deep modified-principal-nest level q and every $n \geq n_*(m)$ below it, one can find at least m pairwise disjoint disks $\Lambda_k \Subset \Lambda'_k$ and branched coverings*

$$\Psi_k : (\Upsilon_k, F_k, W^q) \longrightarrow (V^{q-n}, \Lambda'_k, \Lambda_k) \quad (9.16c)$$

with the following uniform properties (for degree $d = 2$):

1. $W^q \subset \Upsilon_k \subset V^q$;
2. $\deg(\Psi_k : F_k \rightarrow \Lambda'_k) \leq d^5$;
3. $\deg(\Psi_k : \Upsilon_k \rightarrow V^{q-n}) \leq d^{2n+2m}$;
4. the buffers Λ'_k are pairwise disjoint and $\text{mod}(\Lambda'_k \setminus \bar{\Lambda}_k)$ equals a principal/modified modulus at a level strictly preceding q .

Proof. Only first-return bookkeeping is involved. The telescope lemma for a chain of proper maps says that the degree of a composite is at most the product of the branch degrees. By the Markov theorem above, every lateral landing branch is univalent and every critical return contributes exactly one factor d . Thus the first landing at a chosen

nest piece is univalent until the orbit passes through the critical branch, and a first return to a critical piece has degree d .

Use the modified principal nest obtained by taking, after each kingdom level, the first child and then the oldest unspoiled child. The return times between successive kingdom levels satisfy the elementary doubling estimate: after two successive central stages, the next escape time is at least the sum of all preceding return times. Induction therefore gives exponential growth of the travel time with the number of kingdom levels. Apply this with $2m$ in place of m . Choosing $n > \log_2(2m) + 5$ produces more than $2m$ consecutive returns to the fixed base piece before the orbit can climb back across the preceding five nest blocks. Split that orbit segment into the five blocks between consecutive enlarged nest pieces. The return-time inequalities show that at most two of the five blocks can meet the selected consecutive-return segment, so one block contains at least m returns.

Pull the enlarged terminal puzzle piece of that block backwards along these returns. Markov nesting makes two such pulled-back buffers either disjoint or nested. Nesting would force an iterate of the enlarged terminal piece to cover itself in fewer iterates than its defining first-return time; this contradicts minimality. Hence the buffers are pairwise disjoint. The pullback inside this five-block segment crosses the critical branch at most five times, which proves the d^5 degree bound. From V^q back to V^{q-n} there are at most $2n$ critical nest passages, and the selected consecutive-return segment contributes at most $2m$ further passages; telescope therefore gives d^{2n+2m} . Finally, the terminal enlargement is carried biholomorphically onto every buffer pair, so conformal invariance identifies its collar modulus with the corresponding earlier modified/principal modulus. These are exactly the four properties stated above. $\square$

Lemma 9.20 (Modulus improvement). *There are n_0 and $\epsilon_0 > 0$ such that, if*

$$\mu_q := \text{mod}(V^q \setminus \overline{W^q}) < \epsilon_0, \quad q \geq n_0,$$

then some $p < q$ satisfies

$$\mu_p < \frac{1}{2}\mu_q. \tag{9.16d}$$

Proof. Choose m so large that the final quasi-additivity factor below is less than $1/2$, and then choose $n > \log_2 m + 5$. Apply Lemma 9.19 at level q .

If one buffer collar has modulus less than $\mu_q/(2d)$, property (4) identifies that collar, up to at most one degree- d critical pullback, with an earlier modified modulus. The standard modulus inequality under a degree- d proper map then gives (9.16d).

Assume instead that every selected collar has modulus at least $\mu_q/(2d)$. Use the covering lemma with collar parameter $\eta_0 = 1/(2d)$. The degree relevant to its right-hand side is the local buffer degree d^5 ; the larger total degree d^{2n+2m} affects only the smallness threshold ϵ_0 . Thus, after decreasing ϵ_0 ,

$$\text{mod}(V^{q-n} \setminus \overline{\Lambda_k}) \leq 4d^{11}\mu_q. \tag{9.16e}$$

Relative to this outer modulus, every buffer collar has size at least the fraction $\eta_1 = 1/(8d^{12})$. Quasi-additivity applied to the m pairwise disjoint buffered islands therefore

gives

$$\text{mod}(V^{q-n} \setminus \overline{W^{q-n}}) \leq \frac{2^6 d^{23}}{m} \mu_q.$$

Choose $m > 2^7 d^{23}$ before choosing n . The last factor is then less than $1/2$, and taking $p = q - n$ proves (9.16d). Notice that the proof uses smallness only to enter the covering and quasi-additivity regimes; no a priori lower bound is assumed. $\square$

Theorem 9.21 (Complex bounds for the modified principal nest). *Let P_c be at most finitely renormalizable with all periodic points repelling. After the last renormalization there is a cofinal modified principal nest and $\mu > 0$ such that the selected principal annuli have modulus at least μ at every sufficiently deep level.*

Proof. Straighten the last connected quadratic-like renormalization and work with its nonrenormalizable tail; the finitely many outer levels will be restored at the end. Write $\mu_q = \text{mod}(V^q \setminus \overline{W^q})$. Choose n_0, ϵ_0 from Lemma 9.20 and put

$$\mu_* := \min\{\epsilon_0, \mu_0, \dots, \mu_{n_0-1}\} > 0.$$

If some $q \geq n_0$ had $\mu_q < \mu_*$, repeatedly apply the improvement lemma. The indices strictly decrease and the moduli decrease by a factor at least 2 at each step. Eventually an index $p < n_0$ is reached with $\mu_p < \mu_q < \mu_*$, contradicting the definition of μ_* . Thus $\mu_q \geq \mu_*$ for all q .

Since $V^{q+1} \subset W^q \Subset V^q$, monotonicity gives

$$\text{mod}(V^q \setminus \overline{V^{q+1}}) \geq \mu_*.$$

Taking the cofinal nest $E^q = V^q$ proves the claimed complex bounds. The finite initial renormalization tower changes only finitely many outer levels. $\square$

Lemma 9.22 (A priori bounds imply local connectivity). *Under the preceding complex bounds, every critical puzzle end shrinks, then every puzzle end shrinks, and $J(P_c)$ is locally connected.*

Proof. The selected annuli are disjoint and have modulus at least μ , so their total modulus diverges. Lemma 9.8 shrinks the critical end, and Lemma 9.10 shrinks every other end. The connected-puzzle-trace lemma from the preceding theorem turns these shrinking puzzle neighborhoods into a connected neighborhood basis on the Julia set. Because every periodic point is repelling, the periodic Fatou-component classification from Chapter 7 leaves no bounded Fatou component that could supply a distinct non-shrinking impression. Hence the Julia set is locally connected. $\square$

Favorite and good nests. For a nice critical disk Q , its *first child* is the central component of the first-return map. A child $Q' \Subset Q$ is *good* when the critical value of its defining degree-two return lands in the first child of Q . The *favorite child* is the oldest good unspoiled child.

Lemma 9.23 (Transfer of complex bounds to a favorite good nest). *Let $P^j \in Q^j$ be the first child of Q^j and let Q^{j+1} be the favorite child of Q^j . After finitely many initial levels there is $\delta > 0$ such that*

$$\text{mod}(Q^j \setminus \overline{P^j}) \geq \delta \quad (j \geq j_0). \quad (9.17)$$

Proof. For a nice disk V , put

$$m(V) = \inf_D \text{mod}(V \setminus \overline{D}),$$

where D ranges over components of the first-return domain to V . We use two elementary pullback estimates. If U is the first child of V and V' is any child of V , then

$$m(V') \geq d^{-1} \text{mod}(V \setminus \overline{U}). \quad (9.17a)$$

Indeed, push a return component of V' forward to its first landing in V . All intermediate branches are univalent, and the final child map has degree d , so extremal modulus can drop by at most d .

Second, suppose $P \in Q$, $P' \in Q'$, where P, P' are first children and Q' is the favorite child of Q . If $V \supset Q$ and the first child of V lies in Q , then

$$\text{mod}(Q' \setminus \overline{P'}) \geq d^{-2} m(V). \quad (9.17b)$$

To prove this, follow the critical orbit from its first escape from P to its first return to P . The pullback defining P' contains exactly one critical passage, giving one factor d^{-1} . If the landing domain is not Q' , (9.17a) supplies the second factor. If it is Q' , push Q' forward once to a noncentral return component of V and apply the definition of $m(V)$; the same d^{-2} bound results.

Now compare the favorite nest with the modified principal nest $W^k \in V^k$. Let $k(j)$ be the deepest modified level with $V^{k(j)} \supset Q^j$. Since a favorite child is never the first child, two favorite steps cross at least one modified-principal level: $k(j+2) \geq k(j)+1$. If $W^{k(j)} \subset Q^j$, (9.17b) immediately gives a fixed multiple of $m(V^{k(j)})$ for the next favorite annulus. Otherwise start the ordinary principal nest $Z_0 = W^{k(j)} \supset Z_1 \supset \dots$ and let $Z_{s-1} \setminus Z_s$ be the first central-cascade annulus met by the return of the critical point. Pulling the corresponding noncentral landing domain back through the cascade gives

$$\text{mod}(V^{k(j)+1} \setminus \overline{Z_{s+1}}) \geq d^{-1} \text{mod}(Z_0 \setminus \overline{Z_1}).$$

If $Q^j \subset Z_s$, this annulus lies between Q^j and its first child and yields the desired bound. If instead $Z_s \subset Q^j \in Z_1$, then Q^j would be the first child of its parent, contradicting that it is a favorite child. Thus one of two consecutive favorite annuli is bounded below by a degree-dependent multiple of $m(V^{k(j)})$.

Finally, (9.17a) and the modified-principal complex bound give $m(V^k) \geq d^{-1} \mu_*$ for all deep k . Since $k(j)$ tends to infinity, the preceding alternatives give (9.17) after discarding finitely many levels, with δ depending only on d and μ_* . $\square$

Lemma 9.24 (Disk lifting with bounded dilatation). *Fix $0 < \rho < r < 1$ and an integer $d \geq$*

2. There is $K_1 = K_1(\rho, r, d)$ with the following property. Let $g, \tilde{g} : (\mathbb{D}, 0) \rightarrow (\mathbb{D}, 0)$ be proper degree- d maps whose critical values lie in $\overline{\mathbb{D}}_\rho$. Suppose boundary homeomorphisms h, h' satisfy $\tilde{g} \circ h' = h \circ g$ on $\partial\mathbb{D}$ and h has a K -quasiconformal extension equal to the identity on $\mathbb{D}_r$. Then h' has a $\max\{K, K_1\}$ -quasiconformal extension with the same normalization.

Proof. The normalized family of proper degree- d disk maps whose critical values stay in $\overline{\mathbb{D}}_\rho$ is compact in the strong topology. To see normality across the circle, reflect the map by $G(z) = 1/\overline{G(1/\bar{z})}$; the preimages of a fixed annulus around the circle omit a uniform subannulus because all critical values stay a definite distance from the circle. Thus degeneration at $\partial\mathbb{D}$ is excluded.

For one such G , the component $U_G = G^{-1}(\mathbb{D}_r)$ containing 0 is a Jordan disk compactly contained in $\mathbb{D}$. The boundary covering $G : \partial U_G \rightarrow \partial\mathbb{D}_r$ is analytically conjugate to $z \mapsto z^d$. Compactness of the family gives a uniform quasiconformal extension of this boundary conjugacy across $\mathbb{D} \setminus U_G$, fixed on $\mathbb{D}_r$. Apply this extension to g and $\tilde{g}$ and lift the given extension of h through the two proper maps. The lift is conformal on the inner disk and has the same Beltrami coefficient on the lifted outer part; the uniform annular extension contributes only the compact-family constant K_1 . Boundary equivariance chooses the lift agreeing with h' . This proves the stated bound. $\square$

Lemma 9.25 (Uniform quasiconformal extension on good nests). *Let $Q^m \Subset \dots \Subset Q^0$ and $\tilde{Q}^m \Subset \dots \Subset \tilde{Q}^0$ be combinatorially corresponding good nests, and let $P^j, \tilde{P}^j$ be their first children. Assume (9.17) for both nests and assume the outer boundary marking extends K -quasiconformally. Then the marking on ∂Q^m extends $K' = K'(K, \delta)$ -quasiconformally, independently of m .*

Proof. Uniformize each $(Q^j, 0)$ and $(\tilde{Q}^j, 0)$ by $(\mathbb{D}, 0)$. The annulus bound (9.17) implies that the image of the first child is contained in one fixed disk $\mathbb{D}_\rho$, $\rho < 1$: otherwise extremal distance from the child to the outer boundary would tend to zero. Choose r with $\rho < r < 1$.

The return defining the good child becomes a proper degree-two disk map. Its critical value lies in the normalized first child, hence in $\mathbb{D}_\rho$. Modify the current extension inside $\mathbb{D}_r$ so that it is the identity there; this costs a quasiconformal factor controlled only by the current outer constant and the fixed annulus $\mathbb{D} \setminus \overline{\mathbb{D}}_r$. Lemma 9.24 lifts the boundary marking to the next good child with a constant bounded by $\max\{K, K_1(\rho, r, 2)\}$. Crucially, the same bound is reused at every level rather than multiplied. Induction down the nest therefore yields one $K'(K, \delta)$ independent of m . $\square$

Lemma 9.26 (Uniform pullback pseudo-conjugacies). *For two nonrenormalizable combinatorially equivalent all-repelling quadratics, the good-nest extensions above produce quasiconformal pseudo-conjugacies $h_m : \mathbb{C} \rightarrow \mathbb{C}$ with one dilatation bound, equivariant outside the m th favorite piece. A normalized subsequence converges to a global quasiconformal conjugacy.*

Proof. At finite puzzle depth the two marked puzzle graphs are isotopic relative to the marked landing points. Extend this isotopy quasiconformally across the finitely

many complementary Jordan disks and use the Böttcher coordinate to make it conformal near infinity. Pull it backwards through all lateral first landing branches; these branches are univalent, so no dilatation is added. When a pullback reaches the critical good nest, use Lemma 9.25. Niceness makes distinct first-entry domains disjoint, hence the pullbacks agree on common boundaries. This gives h_m with one depth-independent constant K_* .

Normalize three marked points. Chapter 5 quasiconformal compactness gives a subsequence converging locally uniformly on the sphere to a K_* -QC homeomorphism h . On every point whose orbit avoids the limiting critical end at the relevant finite stage, the equation $h_m \circ P_c = P_{\tilde{c}} \circ h_m$ eventually holds and passes to the limit. The exceptional set is the intersection of nested pullbacks which never leave the favorite nest. Away from the critical orbit each branch is univalent and, by the uniform annular collar, the inverse branches contract in the hyperbolic metric by a definite factor. Hence this exceptional set has empty interior (indeed the usual bounded-distortion density argument gives zero area). The conjugacy equation therefore holds on a dense set and extends to the whole plane by continuity. Thus h is a global quasiconformal conjugacy. $\square$

Lemma 9.27 (Closedness from rational parapuzzles). *Let $c_0 \in M$ and suppose every periodic point of P_{c_0} is repelling. Let $\mathcal{C}(c_0)$ be its marked quadratic combinatorial class, where the marking records all finite rational-ray puzzle portraits and the critical itineraries through them. Then $\mathcal{C}(c_0)$ is closed in the parameter plane.*

Proof. We use only the qualitative parameter-puzzle mechanism already proved in Chapter 4. A *rational parapuzzle piece* is the closure of a component cut out by finitely many rational parameter rays together with their landing points and one parameter equipotential. Chapter 4 proves that all rational parameter rays land, and that periodic characteristic ray pairs form the separators across which a prescribed finite critical kneading/puzzle datum can change. Hence every rational parapuzzle piece is compact.

Let $\mathcal{P}(c_0)$ be the family of rational parapuzzle pieces around c_0 obtained by resolving successively all finite marked puzzle data of P_{c_0} . We claim

$$\mathcal{C}(c_0) = M \cap \bigcap_{\Pi \in \mathcal{P}(c_0)} \Pi. \quad (9.18)$$

Indeed, if c has the same marked combinatorics as c_0 , then at every finite stage the corresponding repelling periodic ray portraits and the critical itinerary agree. The repelling landing points move holomorphically and the rays move with them; by the critical-value/parameter-ray identity of Chapter 4, the finite datum can change only when the parameter crosses one of the finitely many rational ray separators used to define that parapuzzle stage. Thus c belongs to every $\Pi \in \mathcal{P}(c_0)$.

Conversely, suppose $c \in M$ is not combinatorially equivalent to c_0 . Marked combinatorics is the increasing union of its finite puzzle data, so there is a first finite graph/itinerary datum on which the two maps differ. Only finitely many rational ray portraits occur in that datum. Apply the finite-prefix separator statement of Chapter 4

to those portraits and their pullbacks. Refining by the corresponding characteristic periodic ray pairs produces a rational parapuzzle stage around c_0 whose closed piece does not contain c . (If c lies on an earlier separator, the boundary relation is itself part of the finite marked datum; resolving that relation at the next finite stage gives the required separating piece.) Hence c is absent from the right-hand side of (9.18). This proves the reverse inclusion.

The Mandelbrot set M is closed, and every $\Pi \in \mathcal{P}(c_0)$ is closed. Equation (9.18) therefore expresses $\mathcal{C}(c_0)$ as an intersection of closed sets, proving the lemma. Notice what has *not* been used: we never claim that a limit of parameters whose cycles are all repelling must itself have all cycles repelling. Closedness comes instead from the finite rational-separator description of the fixed combinatorial class. $\square$

Lemma 9.28 (All-repelling combinatorial-class rigidity). *Let $\mathcal{C} \subset M$ be a marked all-repelling quadratic combinatorial class. If the pullback construction above identifies $\mathcal{C}$ with a single quasiconformal class, then $\mathcal{C}$ is a singleton.*

Proof. Let $\mathcal{Q}$ be the quasiconformal class of one member. Marked quasiconformal conjugacy preserves the puzzle combinatorics, so $\mathcal{Q} \subset \mathcal{C}$; Lemma 9.26 gives the reverse inclusion. Thus $\mathcal{C} = \mathcal{Q}$.

By Chapter 6 Ahlfors–Bers deformation theory, a quasiconformal class in the normalized quadratic family is either a singleton or open: scale an invariant Beltrami coefficient, integrate it holomorphically, and restore the monic-centered normalization. Lemma 9.27 shows that $\mathcal{C}$ is closed. Therefore a nontrivial $\mathcal{C}$ would be both open and closed in the connected parameter plane. This is impossible: $\mathcal{C} \subset M$, while $M \neq \mathcal{C}$ (for example the critical orbit of P_2 escapes). Hence $\mathcal{C}$ is a singleton. $\square$

Lemma 9.29 (Finite tower reduction by straightening). *Suppose P_c and $P_{\tilde{c}}$ are exactly $N < \infty$ times renormalizable and combinatorially equivalent. Equality of their deepest normalized straightenings implies $c = \tilde{c}$.*

Proof. At each renormalization level the common combinatorics selects the same characteristic quadratic-like return. Chapter 8 straightening gives its unique normalized quadratic parameter. Repelling periodicity is preserved under the hybrid conjugacy on the little filled Julia set, and maximality of N makes the deepest straightening non-renormalizable. The preceding nonrenormalizable rigidity therefore makes the two deepest straightening parameters equal. Chapter 8 tuning identifies each finite renormalization locus with its small Mandelbrot copy and makes normalized straightening the inverse coordinate on that copy. Inductively climbing the N copies gives $c = \tilde{c}$. $\square$

Proof. For local connectivity, apply the complex-bounds theorem below the final renormalization and then Lemma 9.22; finitely many outer levels do not affect shrinking. For rigidity, construct favorite good nests in the nonrenormalizable tail, use Lemmas 9.25 and 9.26, apply Lemma 9.28, and finally climb the finite tower by Lemma 9.29. The parameter-class closedness step is Lemma 9.27, proved from the earlier rational-ray/parapuzzle machinery. This proves both assertions. $\square$

9.3 Worked examples

Worked example: A low-depth basilica puzzle

Consider

$$P(z) = z^2 - 1, \quad 0 \mapsto -1 \mapsto 0.$$

The fixed points solve $z^2 - z - 1 = 0$, so

$$\alpha = \frac{1 - \sqrt{5}}{2}, \quad \beta = \frac{1 + \sqrt{5}}{2}.$$

The point α is the dividing fixed point for the standard basilica picture. The rational rays of angles $1/3$ and $2/3$ land at α . Choose an equipotential $G = t > 0$ and join those two ray tails to their common landing point. Together with the equipotential this is the depth-zero graph.

Under angle doubling,

$$\frac{1}{3} \mapsto \frac{2}{3} \mapsto \frac{1}{3},$$

so the ray pair is forward invariant as an unordered pair. Pulling back once adds the four angle-halves

$$\frac{1}{6}, \frac{2}{6} = \frac{1}{3}, \frac{4}{6} = \frac{2}{3}, \frac{5}{6}$$

and the equipotential $G = t/2$. The bounded complementary cells are the depth-one pieces. The unique piece containing 0 is the critical piece; its map to the appropriate depth-zero piece has degree two. Every other piece maps conformally because it contains no critical point.

This example demonstrates construction and the Markov degree bookkeeping, but it is *not* an example of the shrinking/rigidity theorem in this chapter: P has a superattracting two-cycle, so not every periodic point is repelling.

Worked example: Why positive annuli alone do not prove shrinking

Take abstract nested round disks

$$D_n = D(0, r_n), \quad \text{mod}(D_n \setminus \overline{D_{n+1}}) = 2^{-n-1}.$$

Then every annulus is nondegenerate, but

$$\sum_{n \geq 0} 2^{-n-1} = 1 < \infty.$$

Choosing $r_0 = 1$ gives

$$\log \frac{1}{r_N} = 2\pi \sum_{n < N} 2^{-n-1} < 2\pi,$$

so r_N stays bounded away from zero. Thus the intersection contains a nontrivial disk. This is why a statement such as “every nest annulus has positive modulus”

is inadequate: one needs divergent total modulus, and a uniform a priori lower bound is a particularly strong way to obtain it.

Worked example: The infinitely renormalizable boundary

For period-doubling combinatorics, one obtains nested quadratic-like returns of periods $2, 4, 8, \dots$. The combinatorial nesting by itself says nothing quantitative about

$$\text{mod}(U_n \setminus \overline{U'_n}).$$

If these moduli were allowed to approach zero too rapidly, the simple Grötzsch argument would not force shrinking. Infinite-renormalization results therefore need an additional a priori-bounds theorem tailored to the combinatorics. The finite-renormalization theorem proved here deliberately stops before that frontier.

9.4 What this chapter established

A Yoccoz puzzle is not merely a picture made of rays. Its usefulness comes from a chain of separate facts. First, the pullback graph really cuts the filled Julia set into finitely many disk pieces. Second, those pieces form a Markov system and their first-return maps have tightly controlled degree. Third, recurrence of the critical point organizes the pieces into a principal nest. Finally, conformal modulus converts that combinatorics into geometry.

For a nonrenormalizable quadratic with all periodic points repelling, the Yoccoz tableau argument forces enough critical annular modulus to make every puzzle end shrink. For an at-most-finitely-renormalizable quadratic, the Kahn-Lyubich complex-bounds mechanism supplies a uniform modulus lower bound on a modified nest. The same bounds control quasiconformal extensions between combinatorially corresponding nests; pullback and compactness then give a global quasiconformal conjugacy, and Ahlfors-Bers deformation theory turns that equivalence into parameter rigidity.

Looking ahead

The next volume changes viewpoint from holomorphic puzzle geometry to the topological classification of postcritically finite branched coverings. Thurston's obstruction theory asks when a marked sphere map is equivalent to a rational map and introduces a new linear-algebraic test on multicurves. The puzzle, straightening, and a priori-bound machinery developed here will return in later renormalization and parameter-space chapters.

Thurston Theory of Branched Coverings

Why this chapter matters

A rational map is analytic, but many of the data visible in a postcritically finite dynamical system are purely topological: a sphere, finitely many marked postcritical points, branching degrees, and how curves pull back. Thurston's theorem says that these topological data almost determine whether a rational map exists. The only obstruction is carried by finite systems of curves.

The proof is a prototype for a recurring theme in dynamics. Replace a hard realization problem by iteration on a parameter space of geometric structures. A fixed point gives the desired analytic object. Failure of compactness produces a geometric degeneration, and the degeneration is converted into a finite combinatorial witness. Here the parameter space is Teichmüller space, degeneration means very short hyperbolic geodesics, and the finite witness is a Thurston obstruction.

Learning goals

By the end of this chapter you should be able to:

1. distinguish a Thurston map from a rational map and state Thurston equivalence with its marking;
2. compute the orbifold weights and recognize the hyperbolic case;
3. build the Thurston matrix of a stable multicurve and interpret its spectral radius;
4. explain why a fixed point of the Teichmüller pullback map is a rational realization;
5. trace how escape in moduli space produces a short stable multicurve;
6. state the Levy-cycle criterion for topological polynomials and explain why the polynomial proof uses a different geometric mechanism.

Conceptual roadmap

The main proof has two branches. Write

$$\mathcal{T}_F = \text{Teich}(S^2, P_F), \quad \sigma_F : \mathcal{T}_F \longrightarrow \mathcal{T}_F.$$

Then the alternatives are

$$\begin{aligned} &\text{fixed point of } \sigma_F \implies \text{rational realization,} \\ &\text{escape under } \sigma_F \implies \text{short curves} \implies \text{stable multicurve,} \\ &\implies \rho(M_\Gamma) \geq 1. \end{aligned}$$

For topological polynomials there is a second, more concrete route:

$$\begin{aligned} &\text{obstruction} \implies \text{innermost essential curves} \implies \text{cyclic pullback,} \\ &\implies \text{degree-one cycle.} \end{aligned}$$

The second route proves the Levy-cycle theorem. It is not a substitute for the general Teichmüller argument.

Common pitfall

A stable multicurve is not automatically an obstruction. Stability is a combinatorial closure condition under essential pullback. Obstruction means the additional spectral condition $\rho(M_\Gamma) \geq 1$. Likewise, every Levy cycle is an obstruction, but the theorem that every obstruction contains a Levy cycle is special to topological polynomials; it is false as a blanket statement for arbitrary Thurston maps.

10.1 Definitions

Definition 10.1 (Thurston map and postcritical set). A *Thurston map* is an orientation-preserving branched covering $F : S^2 \rightarrow S^2$ of degree at least two whose postcritical set

$$P_F = \bigcup_{n \geq 1} F^n(\text{Crit } F)$$

is finite. Thus “Thurston map” in Chapter 10 always includes postcritical finiteness.

Definition 10.2 (Thurston equivalence). Two Thurston maps $F, G : S^2 \rightarrow S^2$ are *Thurston equivalent* when there exist orientation-preserving homeomorphisms $\phi_0, \phi_1 : S^2 \rightarrow S^2$ such that

$$\phi_0 \circ F = G \circ \phi_1,$$

$\phi_0(P_F) = \phi_1(P_G) = P_G$, and ϕ_0 is isotopic to ϕ_1 relative to P_F . The marking is part of the equivalence relation.

Definition 10.3 (orbifold of a Thurston map). For $x \in S^2$, define

$$\nu_F(x) = \text{lcm}\{\deg_y(F^n) : n \geq 1, F^n(y) = x\},$$

allowing the value ∞ and putting $1/\infty = 0$. The associated orbifold is $\mathcal{O}_F = (S^2, \nu_F)$,

with Euler characteristic

$$\chi(\mathcal{O}_F) = 2 - \sum_{x \in S^2} \left(1 - \frac{1}{\nu_F(x)}\right).$$

It is *hyperbolic* when $\chi(\mathcal{O}_F) < 0$.

Definition 10.4 (multicurve). A *multicurve* $\Gamma = \{\gamma_1, \dots, \gamma_m\}$ in $S^2 \setminus P_F$ is a finite family of pairwise disjoint, pairwise nonisotopic simple closed curves, considered up to isotopy rel P_F , such that every γ_i is essential and nonperipheral. Equivalently, each component of $S^2 \setminus \gamma_i$ contains at least two points of P_F .

Definition 10.5 (stable multicurve and Thurston matrix). A multicurve $\Gamma = \{\gamma_1, \dots, \gamma_m\}$ is *F-stable* if every essential nonperipheral component of $F^{-1}(\gamma_j)$ is isotopic rel P_F to some γ_i . Its *Thurston matrix* is the nonnegative matrix $M_\Gamma = (m_{ij})$ with

$$m_{ij} = \sum_{\substack{\delta \subset F^{-1}(\gamma_j) \\ \delta \sim_{P_F} \gamma_i}} \frac{1}{\deg(F : \delta \rightarrow \gamma_j)}.$$

The associated linear map on the real vector space spanned by Γ is the *Thurston linear transformation*.

Definition 10.6 (Thurston obstruction). An *F-stable* multicurve Γ is a *Thurston obstruction* when

$$\rho(M_\Gamma) \geq 1,$$

where ρ denotes spectral radius.

Definition 10.7 (Teichmüller pullback map). Let $\mathcal{T}_F = \text{Teich}(S^2, P_F)$ be the Teichmüller space of complex structures on the marked sphere modulo isotopy rel P_F and postcomposition by Möbius maps. Pullback of Beltrami coefficients by F defines a holomorphic self-map

$$\sigma_F : \mathcal{T}_F \longrightarrow \mathcal{T}_F,$$

called the *Thurston pullback map*.

Definition 10.8 (topological polynomial). A *topological polynomial* of degree $d \geq 2$ is a Thurston map F with a distinguished fixed point ∞ satisfying

$$F^{-1}(\infty) = \{\infty\}, \quad \deg_\infty F = d.$$

Thus ∞ is totally ramified and the complement $S^2 \setminus \{\infty\}$ plays the role of the polynomial plane.

Definition 10.9 (Levy cycle). A *Levy cycle* for a Thurston map is a cyclically ordered multicurve $\gamma_0, \dots, \gamma_{r-1}$ such that, for every i modulo r , there is a component $\tilde{\gamma}_i$ of $F^{-1}(\gamma_{i+1})$ isotopic rel P_F to γ_i and

$$\deg(F : \tilde{\gamma}_i \rightarrow \gamma_{i+1}) = 1.$$

Its Thurston matrix contains a cyclic permutation block of spectral radius one.

Definition 10.10 (canonical Thurston obstruction). For a hyperbolic-orbifold Thurston map, the *canonical Thurston obstruction* is the set of isotopy classes of essential nonperipheral curves whose hyperbolic geodesic lengths tend to zero along iteration of the Thurston pullback map. Chapter 10 records this notion as an interface to Pilgrim’s decomposition theory; its full decomposition theorem is not reproved here.

10.2 Main results

Definition 10.11 (Postcritically finite branched-covering setup). **Statement scope.** A Thurston map is a postcritically finite orientation-preserving branched covering $F : S^2 \rightarrow S^2$ of degree at least two, equipped with its marked set P_F , Thurston-equivalence relation, and orbifold $\mathcal{O}_F$. These data are invariant under Thurston equivalence.

Definition 10.12 (Thurston linear transformation for multicurves). **Statement scope.** For every F -stable multicurve $\Gamma = \{\gamma_1, \dots, \gamma_m\}$, the pullback degrees define the nonnegative Thurston matrix M_Γ above. The condition $\rho(M_\Gamma) \geq 1$ is invariant under Thurston equivalence and defines a Thurston obstruction.

Theorem 10.1 (Thurston characterization theorem). **Statement scope.** Let $F : S^2 \rightarrow S^2$ be a Thurston map with hyperbolic orbifold. Then F is Thurston equivalent to a rational map if and only if

$$\rho(M_\Gamma) < 1$$

for every F -stable multicurve Γ . When the rational realization exists, it is unique up to Möbius conjugacy.

Proof strategy (not part of the proof)

Proof status. proved here modulo stated foundation input.

Strategic outline only. Standard finite-type Teichmüller, Jenkins-Strebel, hyperbolic-collar and Grotzsch inputs; fixed-point criterion; global two-step strictness; finite-cover compactness; puncture-removal estimate; long-gap stability; uniform finite-matrix escape-to-obstruction.

The mathematical argument begins in the proof environment below.

Proof setup. Put $P = P_F$, $p = |P|$, $d = \deg F$, and

$$\mathcal{T} = \text{Teich}(S^2, P), \quad \pi : \mathcal{T} \longrightarrow \mathcal{M}_{0,P}.$$

The proof uses the standard graduate theory of finite-type Teichmüller spaces: normalized solutions of the Beltrami equation and their holomorphic dependence, the cotangent identification by integrable quadratic differentials, completeness of the Teichmüller metric, the Jenkins-Strebel theorem, the hyperbolic collar theorem, and the Grötzsch series inequality for annuli. These are the only frozen interfaces. The finite-cover argument, the short-curve gap mechanism, and the quantitative escape-to-obstruction estimate specific to Thurston maps are proved below.

Lemma 10.2 (Pullback and the fixed-point criterion). *Pullback of conformal structures defines a holomorphic nonexpanding map $\sigma_F : \mathcal{T} \rightarrow \mathcal{T}$. A point of $\mathcal{T}$ is fixed by σ_F if and only if F is Thurston equivalent to a rational map.*

Proof. Pull a measurable conformal structure back by F on the complement of the finite critical set and extend across that set. Isotopies relative to P lift on $S^2 \setminus F^{-1}(P)$ and extend over the punctures, so the Teichmüller class is well defined. In Beltrami coordinates the pullback is complex linear in the coefficient and normalized solutions depend holomorphically on that coefficient; hence σ_F is holomorphic. A quasiconformal comparison map lifts through the unbranched covering without increasing maximal dilatation, so σ_F is nonexpanding.

If $\sigma_F(\tau) = \tau$, choose normalized markings ϕ_0, ϕ_1 representing τ and its pullback, with $\phi_0 \simeq \phi_1$ relative to P . Then

$$R = \phi_0 \circ F \circ \phi_1^{-1}$$

is holomorphic on the Riemann sphere and therefore rational; the two markings give the required Thurston equivalence. Conversely, a Thurston equivalence to a rational map pulls the standard complex structure back to the same marked class. Thus rational realizations and fixed points are equivalent. $\square$

Lemma 10.3 (Hyperbolic orbifolds give strict two-step contraction). *If $\mathcal{O}_F$ is hyperbolic, then for every $\tau \in \mathcal{T}$,*

$$\|D_\tau(\sigma_F^2)\| < 1. \quad (10.1)$$

Consequently σ_F has at most one fixed point.

Proof. Let f_τ be the rational map obtained from one pullback square at τ . Under the standard cotangent identification, the transpose of $D_\tau \sigma_F$ is pushforward of integrable quadratic differentials,

$$(f_\tau)_* q(z) = \sum_{f_\tau(w)=z} \frac{q(w)}{(f'_\tau(w))^2}. \quad (10.2)$$

The triangle inequality after change of variables gives $\|(f_\tau)_* q\|_1 \leq \|q\|_1$.

We record the equality mechanism because it is the point at which the orbifold hypothesis enters. Suppose $G : \widehat{\mathbb{C}} \rightarrow \widehat{\mathbb{C}}$ has degree d and equality holds for a nonzero meromorphic quadratic differential q with only simple poles. Equality in the triangle inequality forces the summands of $G_* q$ over every regular value to have the same argument. Hence $G^* G_* q / q$ is a real-valued meromorphic function and is therefore constant. Comparing total areas gives

$$q = \frac{1}{d} G^* G_* q. \quad (10.3)$$

If Z is the pole set of q , local pullback in (10.3) then implies

$$G^{-1}(G(Z)) \subset Z \cup \Omega_G, \quad (10.4)$$

where Ω_G is the critical set.

The following finite count is elementary but decisive. Let $Z \subset P$ and suppose, in the topological model of F , $F^{-1}(Z) \subset P \cup \Omega_F$. Split $F^{-1}(Z) = X_1 \sqcup X_2$, where X_1 consists

of points that can be reached from a critical point before the orbit meets Z , and X_2 is the complement. To a point of X_1 associate the nonempty set of critical points that first reach it without previously meeting Z ; these sets are disjoint or identical, so $|X_1| \leq |\Omega_F| \leq 2d - 2$. To a point of X_2 associate the nonempty set of points of Z whose first return to $F^{-1}(Z)$ is that point; again these sets are disjoint or identical, so $|X_2| \leq |Z|$. Thus

$$|F^{-1}(Z)| \leq |Z| + 2d - 2. \quad (10.5)$$

On the other hand, counting the $d|Z|$ preimages with multiplicity and using Riemann-Hurwitz gives

$$d|Z| \leq |F^{-1}(Z)| + 2d - 2. \quad (10.6)$$

Combining (10.5)-(10.6) yields $|Z| \leq 4$.

If $|Z| = 4$, every inequality above is an equality: all critical points are simple, every critical point lies over Z , and Z itself contains no critical point. Put $Z' = F^{-1}(Z) \setminus \Omega_F$; then $|Z'| = 4$. If a second equality step also satisfies $F^{-1}(Z') \subset P \cup \Omega_F$, let $Y_1 = Z \setminus Z'$ and $Y_2 = Z' \setminus Z$. Since Y_2 contains no critical value, $|F^{-1}(Y_2)| = d|Y_2|$. Every point of this preimage lies in P and, following its postcritical history back to the last visit to Z , determines injectively a point of Y_1 . Hence

$$|Y_1| \geq d|Y_2|.$$

But $|Y_1| = |Y_2|$ and $d > 1$, so both are empty. Therefore $Z = Z' = P$, $|P| = 4$, and all four orbifold weights are 2; the orbifold has Euler characteristic zero. This is precisely the parabolic exceptional case.

Now suppose equality held in the norm of the two-step cotangent operator at some τ . Equality must hold in both one-step pushforwards. Applying (10.4) to their pole sets gives the two consecutive inclusions just analyzed, and therefore forces the parabolic orbifold. Hyperbolicity excludes this. The relevant cotangent spaces are finite-dimensional, so strict inequality on the unit sphere implies the operator norm in (10.1) is strictly below one. If $\tau_1 \neq \tau_2$ were fixed points, the Teichmüller geodesic joining them is compact; continuity would give a uniform constant $k < 1$ for $D\sigma_F^2$ along it, contradicting $d(\tau_1, \tau_2) = d(\sigma_F^2\tau_1, \sigma_F^2\tau_2)$. $\square$

Lemma 10.4 (A rational realization has no obstruction). *If F is equivalent to a rational map and $\mathcal{O}_F$ is hyperbolic, then $\rho(M_\Gamma) < 1$ for every F -stable multicurve Γ .*

Proof. Transfer $\Gamma = \{\gamma_1, \dots, \gamma_r\}$ to the punctured sphere of the rational realization. Given a positive Perron-Frobenius vector, the Jenkins-Strebel theorem supplies a quadratic differential whose horizontal cylinders have those curves as cores and the prescribed relative moduli. A pullback component of degree e contributes the factor $1/e$ to the corresponding modulus. Grötzsch's inequality for the resulting disjoint subcylinders gives exactly the matrix inequality $M_\Gamma v \leq v$, hence $\rho(M_\Gamma) \leq 1$.

If equality held, every Grötzsch and L^1 inequality used above would be an equality; the associated nonzero quadratic differential would therefore give equality for the two-step cotangent pushforward. This contradicts Lemma 10.3. Hence $\rho(M_\Gamma) < 1$. $\square$

Lemma 10.5 (Compact projected orbit implies a fixed point). *If the projected pullback orbit $\{\pi(\sigma_F^n \tau_0) : n \geq 0\}$ has compact closure in moduli space and $\mathcal{O}_F$ is hyperbolic, then $\sigma_F^n \tau_0$ converges to the unique fixed point of σ_F .*

Proof. The finite-cover device needed here is elementary. For a marked configuration $i : P \hookrightarrow \widehat{\mathbb{C}}$, degree- d covers branched only over $i(P)$ are classified by monodromy homomorphisms from the finitely generated group $\pi_1(\widehat{\mathbb{C}} \setminus i(P))$ to the finite group S_d . There are only finitely many such monodromy data, and for each cover only finitely many lifts of the finite marking P . As i varies, these choices assemble into a finite cover $\widehat{\mathcal{M}} \rightarrow \mathcal{M}_{0,P}$ on which the rational pullback square, and hence the derivative norm of σ_F^2 , is single-valued.

Let δ_0 be a Teichmüller geodesic from τ_0 to $\tau_1 = \sigma_F \tau_0$, and put $\delta_n = \sigma_F^n(\delta_0)$. Compactness of the projected orbit implies compactness of the lifted union of these segments in the finite cover. Lemma 10.3 and continuity therefore give one $K < 1$ such that

$$\text{length}(\delta_{n+2}) \leq K \text{length}(\delta_n) \quad (n \geq 0). \quad (10.7)$$

Thus both even and odd successive-step lengths form geometric series. The orbit is Cauchy, Teichmüller space is complete, and its limit is fixed by continuity. Uniqueness follows from Lemma 10.3. $\square$

Lemma 10.6 (Logarithmic shortness detects escape). *For an essential nonperipheral curve γ let $\ell_\tau(\gamma)$ be the hyperbolic length of its geodesic representative and set*

$$w(\gamma, \tau) = -\log \ell_\tau(\gamma), \quad W(\tau) = \sup_\gamma w(\gamma, \tau).$$

Then

$$|w(\gamma, \tau) - w(\gamma, \tau')| \leq 2d_{\mathcal{T}}(\tau, \tau'), \quad (10.8)$$

and a subset of moduli space is relatively compact exactly when W is bounded above on its inverse image.

Proof. The annular cover associated to γ has modulus $\pi/(2\ell_\tau(\gamma))$. A K -quasiconformal comparison map changes annular modulus by a factor between K^{-1} and K . Since the extremal representative of Teichmüller distance has $K = e^{2\delta}$, (10.8) follows.

For compactness, normalize three marked points to $0, 1, \infty$. Any sequence of remaining marked configurations has a pointwise convergent subsequence in $\widehat{\mathbb{C}}$. If the limit is injective, the subsequence converges in moduli space. If marked points collide, choose a minimal collision cluster. There is then a round annulus separating at least two points of the cluster from at least two points outside it whose modulus tends to infinity. Its core class is essential, and the collar/extremal-length comparison forces the hyperbolic length of that class to tend to zero, so $W \rightarrow \infty$. Conversely, on a compact family of normalized configurations every essential geodesic has a uniform positive lower bound. This proves the equivalence. $\square$

Lemma 10.7 (Deleting punctures: reciprocal-length estimate). *Let X be a finite-type hyperbolic surface, let $E \subset X$ contain $q > 0$ points, and put $X' = X \setminus E$. Let γ be a*

simple closed geodesic of X of length ℓ , and let $\gamma'_1, \dots, \gamma'_s$ be all geodesics of X' which are homotopic to γ after the points of E are filled and which have lengths $\ell'_i < L$, where

$$0 < L < L_* := \log(\sqrt{2} + 1).$$

Then $s \leq q + 1$, $\ell'_i \geq \ell$, and

$$\frac{1}{\ell} - \frac{2}{\pi} - \frac{q+1}{L} < \sum_{i=1}^s \frac{1}{\ell'_i} < \frac{1}{\ell} + \frac{2(q+1)}{\pi}. \quad (10.9)$$

Proof. By the standard collar theorem, two distinct simple geodesics of length below L_* are disjoint. Since all γ'_i become homotopic after the points of E are filled, consecutive ones bound annuli and every annular region between consecutive curves must contain a deleted point. Hence $s \leq q + 1$. The inclusion $X' \hookrightarrow X$ is holomorphic and therefore decreases the Poincaré metric, so $\ell'_i \geq \ell$.

We use the standard quantitative collar form: a simple geodesic of length $u < L_*$ has an embedded annulus about it of modulus greater than $\pi/(2u) - 1$, while the complete annular cover of its homotopy class has modulus exactly $\pi/(2u)$. The disjoint collars about the γ'_i lift to disjoint essential subannuli of the annular cover of γ . Grötzsch therefore gives

$$\frac{\pi}{2\ell} > \sum_{i=1}^s \left(\frac{\pi}{2\ell'_i} - 1 \right),$$

which is the right-hand inequality in (10.9), using $s \leq q + 1$.

For the converse estimate, take the standard collar C of γ in X , whose modulus is greater than $\pi/(2\ell) - 1$. Parallel curves through the points of $E \cap C$ split C into at most $q + 1$ annuli C_j . Let η_j be the geodesic in X' in the corresponding class and let its length be u_j . Each C_j embeds in the annular cover of η_j , so $\text{mod } C_j < \pi/(2u_j)$. Therefore

$$\frac{1}{\ell} - \frac{2}{\pi} < \sum_j \frac{1}{u_j}.$$

At most $q + 1$ terms have $u_j \geq L$, and their total reciprocal contribution is at most $(q + 1)/L$. The remaining terms are among the γ'_i , giving the left-hand inequality in (10.9). $\square$

Lemma 10.8 (A long logarithmic gap is pullback-stable). *Let $\tau' = \sigma_F(\tau)$ and $\Delta = d\tau(\tau, \tau')$. Put $A = -\log L_*$. For $J > 0$, choose the first interval (a, b) of length J , with $a \geq A$, containing no value $w(\gamma, \tau)$, and define*

$$\Gamma_{J,\tau} = \{\gamma : w(\gamma, \tau) \geq b\}.$$

If $J > \log d + 2\Delta$ and $\Gamma_{J,\tau} \neq \emptyset$, then $\Gamma_{J,\tau}$ is an F -stable multicurve. Moreover, in the full preimage-punctured surface, every sufficiently short geodesic in one of these homotopy classes is a component of the pullback of a curve in $\Gamma_{J,\tau}$.

Proof. Every curve in $\Gamma_{J,\tau}$ has length below L_* , so the collar theorem makes the family pairwise disjoint. Let δ be an essential component of $F^{-1}(\gamma)$ of covering degree $e \leq d$,

with $\gamma \in \Gamma_{J,\tau}$. Before forgetting the extra preimage punctures, the covering is a local hyperbolic isometry and the lifted geodesic has length $e\ell_\tau(\gamma)$. Filling punctures can only shorten it, hence

$$w(\delta, \tau') \geq w(\gamma, \tau) - \log d \geq b - \log d. \quad (10.10)$$

If the homotopy class of δ were not in $\Gamma_{J,\tau}$, the empty gap would give $w(\delta, \tau) \leq a$, while (10.8) gives $w(\delta, \tau') \leq a + 2\Delta$. This contradicts $b - a = J > \log d + 2\Delta$. Thus every essential preimage lies again in the family.

For the last assertion, let δ be a geodesic in the full preimage surface of length below de^{-b} . Its image under the covering has the same length. At this scale the standard short-geodesic theorem rules out transverse self-intersection, so the image is a finite cover of a simple closed geodesic γ . Its covering degree is at most d , and therefore $w(\gamma, \tau) > b - \log d$. Since the interval (a, b) contains no length-spectrum value and $b - \log d > a + 2\Delta > a$, the curve γ lies in $\Gamma_{J,\tau}$. Hence the very short geodesics are exactly the pullback components claimed. $\square$

Lemma 10.9 (Uniform m -step contraction of the short block). *Assume $p \geq 4$ and $\rho(M_\Gamma) < 1$ for every F -stable multicurve. Fix $D \geq d_\tau(\tau, \sigma_F \tau)$ along the pullback orbit. There are an integer $m = m(d, p)$ and a constant $C = C(d, p, D)$ such that, whenever $W(\tau) > C$ and $\Gamma = \Gamma_{J,\tau}$ with*

$$J = m(\log d + 2D),$$

then $\Gamma \neq \emptyset$ and

$$W(\sigma_F^m \tau) < W(\tau). \quad (10.11)$$

Proof. A multicurve on a p -punctured sphere has at most $p - 3$ components. Each entry of a Thurston matrix is a sum of at most d reciprocals of integers in $\{1, \dots, d\}$. Hence for fixed d, p only finitely many matrices can occur. Since every stable one has spectral radius below one, choose one integer m such that

$$\|M_\Gamma^m\|_\infty < \frac{1}{2} \quad (10.12)$$

for all of them.

There are at most $p - 3$ logarithmic length values above $A = -\log L_*$. Thus if

$$B = (p - 3)J + A + 1$$

and $W(\tau) > B$, the first gap (a, b) of length J occurs with $b < B$ and $\Gamma_{J,\tau}$ is nonempty. Since successive pullback points are at distance at most D , Lemma 10.8 applied successively over m steps shows that this short block is stable and identifies all sufficiently short lifted components needed below.

Write $\tau_m = \sigma_F^m \tau$ and, for $\Gamma = \{\gamma_1, \dots, \gamma_r\}$, set

$$v_i = \frac{1}{\ell_\tau(\gamma_i)}, \quad v'_i = \frac{1}{\ell_{\tau_m}(\gamma_i)}.$$

Use the rational pullback diagram for F^m . Let P' be the marked source set and $P'' = (F^m)^{-1}(P)$; at most pd^m punctures are deleted in passing from $\widehat{\mathbb{C}} \setminus P'$ to $\widehat{\mathbb{C}} \setminus P''$. A pullback

component of γ_j of degree e has length exactly $e\ell_\tau(\gamma_j)$ in the full preimage-punctured surface. Therefore, for the components homotopic to γ_i after the extra punctures are filled,

$$\sum \frac{1}{\ell(\delta)} = (M_\Gamma^m v)_i. \quad (10.13)$$

By the final assertion of Lemma 10.8, these are all the geodesics in that class below the cutoff $L = d^m e^{-b}$. Since $b < B$, this cutoff is bounded below by the positive number $L_0 = d^m e^{-B}$; increase B once, if necessary, so that $L_0 < L_*$. Applying the left-hand inequality in Lemma 10.7 gives

$$v'_i \leq (M_\Gamma^m v)_i + \frac{2}{\pi} + \frac{pd^m + 1}{L_0}. \quad (10.14)$$

Let the last two terms be C_0 . From (10.12),

$$\|v'\|_\infty \leq \frac{1}{2}\|v\|_\infty + C_0. \quad (10.15)$$

Enlarge C so that $e^C > 2C_0$ and $C > B$. Then the maximal reciprocal length among the curves of Γ strictly decreases after m steps.

It remains to exclude a new shortest curve outside Γ . If $\eta \notin \Gamma$, the gap gives $w(\eta, \tau) \leq a = b - J$. By (10.8),

$$w(\eta, \tau_m) \leq b - J + 2mD = b - m \log d < W(\tau).$$

Together with (10.15), this proves (10.11). $\square$

Lemma 10.10 (Escape produces a Thurston obstruction). *If the pullback orbit escapes every compact subset of moduli space, then some F -stable multicurve Γ satisfies $\rho(M_\Gamma) \geq 1$.*

Proof. Assume instead that every stable multicurve has spectral radius below one. Let $D = d_\mathcal{T}(\tau_0, \tau_1)$. Nonexpansion gives $d_\mathcal{T}(\tau_n, \tau_{n+1}) \leq D$ for all n . By Lemma 10.9, whenever $W(\tau_n) > C$, $W(\tau_{n+m}) < W(\tau_n)$. If $W(\tau_n) \leq C$, the Lipschitz estimate gives $W(\tau_{n+m}) \leq C + 2mD$. Applying this on each residue class modulo m shows that the entire sequence $W(\tau_n)$ is bounded. By Lemma 10.6, the projected orbit is relatively compact, contrary to the hypothesis. Hence an escaping orbit forces an obstruction. $\square$

Proof of the Thurston characterization theorem. If F has a rational realization, Lemma 10.4 shows that every stable multicurve has spectral radius below one. Conversely, suppose every stable multicurve satisfies $\rho(M_\Gamma) < 1$. If $p \leq 3$, the moduli space $\mathcal{M}_{0,p}$ is a point, so the projected pullback orbit is compact and Lemma 10.5 already gives a fixed point. Assume therefore that $p \geq 4$. Then Lemma 10.10 shows that the projected pullback orbit cannot escape moduli space. Lemma 10.5 therefore gives a fixed point of σ_F , and Lemma 10.2 converts it into a rational realization.

Finally, two rational realizations would give two fixed points of σ_F , contradicting Lemma 10.3. Changing the normalization of the Riemann sphere changes the realization only by a Möbius transformation. This proves existence, the obstruction criterion, and uniqueness up to Möbius conjugacy. $\square$

Proposition 10.11 (Levy cycle criterion for topological polynomials). **Statement scope.** *Let F be a postcritically finite topological polynomial. Every Thurston obstruction for F contains a Levy cycle. Consequently F is obstructed if and only if it admits a Levy cycle. No hyperbolic-orbifold hypothesis is required for this polynomial criterion.*

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Minimal obstruction; recurrent/negligible split; innermost essential preimage permutation; block characteristic polynomial forces degree-one cycle.

The mathematical argument begins in the proof environment below.

Proof. Let F be a postcritically finite topological polynomial and let Γ be a Thurston obstruction. Choose a stable subobstruction minimal under inclusion. Inside it distinguish the curves whose essential preimages keep returning to the system from those which eventually disappear from the recurrent pullback relation. Call the former recurrent and the latter negligible. The Thurston matrix on the negligible part has spectral radius strictly below one.

Put the totally ramified fixed point ∞ in the unbounded component of the complement of every curve. For a recurrent curve γ , let $D(\gamma)$ be its bounded complementary disk. Call γ innermost when $D(\gamma)$ contains no other recurrent curve.

Because $F^{-1}(\infty) = \{\infty\}$, every component of $F^{-1}(D(\gamma))$ is again a bounded disk. If γ is innermost and an essential preimage component represents a recurrent curve η , then η is innermost as well: a recurrent curve strictly inside $D(\eta)$ would pull forward to a recurrent curve inside $D(\gamma)$. Minimality rules this out.

Distinct innermost curves have disjoint sets of recurrent essential preimage classes. Each set is nonempty, and there are only finitely many innermost curves. Consequently every such set consists of exactly one class. The essential-preimage relation is therefore a permutation. Minimality makes it a single cycle, which we write

$$\gamma_0 \mapsto \gamma_1 \mapsto \cdots \mapsto \gamma_{r-1} \mapsto \gamma_0.$$

For each i , let $\tilde{\gamma}_i$ be the unique recurrent component of $F^{-1}(\gamma_{i+1})$ isotopic to γ_i , and let

$$d_i = \deg(F : \tilde{\gamma}_i \rightarrow \gamma_{i+1}), \quad b_i = \frac{1}{d_i}.$$

Order the Thurston matrix with these cyclic recurrent curves first and the negligible curves afterwards. It has block-triangular form

$$M_\Gamma = \begin{pmatrix} A & 0 \\ * & B \end{pmatrix}, \quad \rho(B) < 1,$$

where A is the cyclic matrix whose nonzero entries are $b_0, \dots, b_{r-1}$. Hence

$$\det(\lambda I - M_\Gamma) = (\lambda^r - b_0 b_1 \cdots b_{r-1}) \det(\lambda I - B).$$

Since Γ is an obstruction, $\rho(M_\Gamma) \geq 1$. The second factor has all roots inside the unit disk, so the first factor must have a root of modulus at least one. But each $b_i = 1/d_i \leq 1$. Therefore

$$b_0 b_1 \cdots b_{r-1} = 1,$$

and consequently every $d_i = 1$.

Thus the cyclic curves form a Levy cycle. Conversely, a Levy cycle has a cyclic degree-one Thurston block of spectral radius one, so it is a Thurston obstruction. This proves both directions of the polynomial criterion. $\square$

10.3 Worked examples

Worked example: The Douady rabbit as an already-realized Thurston map

Let $P_c(z) = z^2 + c$ be the center of the period-three rabbit component. Its finite critical point belongs to a three-cycle

$$0 \mapsto c \mapsto c^2 + c \mapsto 0,$$

and infinity is a totally ramified fixed point. Hence the postcritical set has four marked points: the three finite points on the critical cycle and ∞ .

At every point of the finite critical cycle, iterating once around the cycle multiplies local degree by two, so the local degrees of suitable iterates are unbounded. The orbifold weight is therefore ∞ at each of those three points. Infinity also has weight ∞ . Consequently

$$\chi(\mathcal{O}_{P_c}) = 2 - 4 \left(1 - \frac{1}{\infty} \right) = -2 < 0.$$

The orbifold is hyperbolic.

No realization theorem is needed to prove that the rabbit is rational: we started with the polynomial P_c . Instead, use the characterization theorem in the reverse direction. Because this hyperbolic-orbifold Thurston map is already rational, every stable multicurve must satisfy

$$\rho(M_\Gamma) < 1.$$

Thus a proposed rabbit obstruction with spectral radius at least one cannot be correct; some curve, pullback degree, or stability calculation must have been misidentified.

Worked example: A two-curve spectral calculation

Suppose a stable pair $\Gamma = \{\gamma_0, \gamma_1\}$ has one selected essential preimage in each cyclic direction. If one direction has degree two and the other degree one, then

$$M_\Gamma = \begin{pmatrix} 0 & 1/2 \\ 1 & 0 \end{pmatrix}.$$

Its characteristic polynomial is

$$\lambda^2 - \frac{1}{2},$$

so

$$\rho(M_\Gamma) = \frac{1}{\sqrt{2}} < 1.$$

The multicurve is stable but is not an obstruction.

If both covering degrees are one, the matrix becomes

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

whose spectral radius is one. Geometrically the two curves now form a Levy cycle. This tiny calculation is the linear-algebra core of the general Levy argument: a cyclic block can reach spectral radius one only when the product of the selected reciprocal degrees is one.

Worked example: Why the Levy theorem is polynomial-specific

The proof of the Levy criterion repeatedly uses the distinguished point ∞ of a topological polynomial. Because

$$F^{-1}(\infty) = \{\infty\},$$

the preimage of a disk not containing ∞ is a union of disks not containing ∞ . This lets us order obstruction curves by inclusion of their bounded complementary disks and speak unambiguously about innermost curves.

For a general Thurston map there is no distinguished totally ramified fixed point, and preimages of complementary components need not preserve this disk ordering. The innermost-curve permutation argument therefore breaks. One must return to the full multicurve criterion rather than assume that every obstruction contains a Levy cycle.

10.4 What this chapter established

Thurston's theorem turns rational realization into a fixed-point problem on a finite-dimensional Teichmüller space. A fixed point makes the topological branched cover holomorphic. If no fixed point exists, the pullback orbit must leave every compact region of moduli space. Short hyperbolic geodesics detect that escape, a sufficiently

large gap in their length spectrum makes the very short curves stable under pullback, and their covering degrees assemble into the Thurston matrix. The finite-matrix argument forces spectral radius at least one. Thus the only obstruction to rational realization in the hyperbolic-orbifold case is exactly a Thurston obstruction.

For topological polynomials, the totally ramified fixed point at infinity adds extra planar order. Minimal obstruction curves can be arranged by their bounded disks; innermost recurrent curves form a cyclic preimage permutation, and the obstruction spectral radius forces every covering degree around that cycle to be one. This is the Levy-cycle criterion.

Looking ahead

The next volume, Chapter 11, moves from the marked sphere to the real line and begins *Interval and Unimodal Dynamics*. Critical itineraries, kneading data, monotonicity, and real return maps will provide the combinatorial language needed for the real a priori bounds of Chapter 12 and the later renormalization volumes. Thurston theory will reappear when global topological realization and decomposition questions interact with those renormalization structures.

Part III

Real Dynamics and Renormalization

Guide to Part III: recurrence, bounds, and renormalization

Renormalization begins with recurrence. If the critical orbit returns to a small neighborhood of itself, a first-return map can resemble the original system after rescaling. The slogan “the dynamics repeats at smaller scales” is suggestive but mathematically incomplete. To turn it into a theorem one needs to control the geometry of those scales.

Chapter 11 develops the real one-dimensional language: monotone branches, critical points, Schwarzian or cross-ratio distortion, restrictive intervals, kneading data, and return maps. The interval is a useful laboratory because the order structure makes combinatorics visible. But order alone gives no uniform geometry.

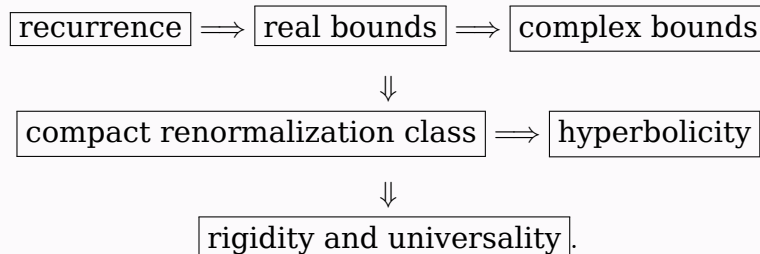
Chapter 12 supplies real a priori bounds. Adjacent atoms in deep dynamical partitions must remain comparable. The difficult point is to rule out a cascade in which one child interval becomes negligibly small relative to its parent. Cross-ratio expansion and Koebe-type distortion convert dynamical information into metric comparability.

Chapter 13 then makes the decisive real-to-complex passage. A real lower bound on interval ratios must be thickened into a lower bound on the modulus of a complex annulus. Once this is achieved, deep return maps live in a compact family of normalized polynomial-like maps. This is the gateway from one-dimensional estimates to complex renormalization.

Chapter 14 regards renormalization itself as a dynamical system. Fixed points encode exact self-similarity, while a horseshoe encodes many combinatorial types simultaneously. Chapter 15 adds hyperbolicity. Stable directions contract toward a renormalization class; the unstable direction records the parameter that moves one away from it. Hyperbolicity turns qualitative self-similarity into quantitative universality and rigidity.

Chapter 16 shows that the same architecture is not peculiar to unimodal maps. Critical circle maps have no turning point and no attracting critical orbit, but closest-return combinatorics generates a renormalization tower. Real bounds again create compactness, and renormalization again organizes rigidity.

The mathematical route through this Part



Each arrow is a theorem-sized step. The purpose of Part III is to make those

arrows visible rather than hide them inside the word “renormalization”.

A warning about universality

There are several different universality statements in the literature. A limiting scale may depend on a combinatorial type, a criticality, or a normalization. Hyperbolicity may be proved on a horseshoe rather than near one fixed point. Throughout this Part, “universal” is therefore always read with its quantified scope attached.

This is also why the book spends time on normalization. Without a fixed normalization, a renormalization orbit can appear to converge merely because arbitrary affine degrees of freedom are being changed at every step.

Interval and Unimodal Dynamics

Why this chapter matters

Real one-dimensional dynamics is the laboratory in which several ideas that later reappear in complex renormalization can be seen with almost no geometric camouflage. A single critical point divides the interval into monotone branches. The derivatives of those branches control basins, the geometry of return intervals controls recurrence, and the left/right itinerary of the critical orbit controls the combinatorics.

The important point is that these are different mechanisms. Singer's theorem comes from the Schwarzian derivative. The no-wandering theorem comes from real bounds and Koebe space. Kneading theory is symbolic and order-theoretic. Only after keeping those mechanisms separate do we combine them to study entropy and the quadratic family.

Learning goals

By the end of this chapter you should be able to:

1. use the Schwarzian chain rule and minimum principle to explain why an attracting basin must capture a critical point or an endpoint;
2. explain why nonflat critical points provide enough real geometry to rule out wandering intervals;
3. compute signed itineraries and use parity-lexicographic order correctly;
4. connect kneading data to lap growth and topological entropy;
5. explain the role of Tsujii's transversality theorem in entropy monotonicity for $Q_t(x) = t - x^2$;
6. recognize a restrictive interval and state what renormalization retains for the next volume.

Conceptual roadmap

There are three entry lanes:

negative Schwarzian $\implies$ minimum principle
 $\implies$ critical point in an attracting basin,
 nonflat critical geometry $\implies$ real bounds and Koebe space
 $\implies$ no wandering intervals,
 signed itinerary order $\implies$ kneading invariant
 $\implies$ lap growth and entropy.

The quadratic-family theorem adds one parameter statement:

positive critical-relation transversality $\implies$ monotone kneading
 $\implies$ monotone entropy.

Restrictive intervals then lead out of Chapter 11 toward real renormalization in Chapter 12.

Common pitfall

Do not use Singer's theorem as the reason wandering intervals do not exist. Singer requires negative Schwarzian and concerns attracting basins. The modern no-wandering theorem used here is stronger in a different direction: it assumes only C^2 regularity and finitely many nonflat critical points, and its proof is based on real bounds. Likewise, kneading order does not need either theorem.

11.1 Definitions

Definition 11.1 (unimodal map). A *unimodal map* is a continuous map $f : I = [a, b] \rightarrow I$ with one turning point $c \in (a, b)$, strictly monotone on $[a, c]$ and on $[c, b]$, with opposite orientations on the two sides. In the standard orientation used below, f is increasing to the left of c and decreasing to the right.

Definition 11.2 (nonflat critical point). A critical point c of a C^2 interval map is *nonflat* if there are a C^2 local diffeomorphism ϕ with $\phi(c) = 0$, a real number $\ell > 1$, and a nonzero constant A such that locally

$$f(x) = f(c) + A |\phi(x)|^\ell$$

up to the choice of the sign on the two sides. Equivalently, after smooth local coordinates the critical germ has finite power-law order ℓ .

Definition 11.3 (Schwarzian derivative). For a C^3 map at a point where $f' \neq 0$, the *Schwarzian derivative* is

$$Sf = \frac{f'''}{f'} - \frac{3}{2} \left(\frac{f''}{f'} \right)^2.$$

The phrase *negative Schwarzian* means $Sf(x) < 0$ at every noncritical point in the interval under discussion.

Definition 11.4 (immediate basin). If $\mathcal{O}$ is an attracting periodic orbit, its *basin* is the set of points whose forward iterates converge to $\mathcal{O}$. Its *immediate basin* is the union of those connected components of the basin that contain points of $\mathcal{O}$.

Definition 11.5 (wandering interval). A nondegenerate interval $J \subset I$ is a *wandering interval* when the intervals $f^n(J)$ have pairwise disjoint interiors and J is not eventually mapped into the basin of an attracting periodic orbit. Replacing J by a smaller open subinterval makes “pairwise disjoint interiors” equivalent to pairwise disjoint images in the arguments below.

Definition 11.6 (itinerary and signed lexicographic order). For a unimodal map with turning point c , the *itinerary* of x is $\iota_f(x) = s_0 s_1 \dots$ where $s_n = L, C, R$ according as $f^n(x) < c, = c$, or $> c$; an itinerary terminates when C first occurs. Put $\varepsilon(L) = +1$ and $\varepsilon(R) = -1$. At the first position n where two nonterminated itineraries differ, compare the symbols in the order $L < C < R$ when $\prod_{j < n} \varepsilon(s_j) = +1$ and in the reverse order when that product is -1 . This is the *signed* or *parity-lexicographic order*.

Definition 11.7 (kneading invariant). The *kneading sequence* of a unimodal map is

$$K(f) = \iota_f(f(c)),$$

with the same one-sided convention at an eventual return to c . For a piecewise-monotone map with several turning points, the collection of the one-sided itineraries at all turning values is its *kneading invariant*.

Definition 11.8 (lap number and growth number). A *lap* of a piecewise-monotone interval map is a maximal interval on which the map is monotone. Write $\ell(f)$ for the lap number and $\ell_n = \ell(f^n)$. The *growth number* is

$$s(f) = \lim_{n \rightarrow \infty} \ell_n^{1/n};$$

its existence follows from $\ell(f^{m+n}) \leq \ell(f^m)\ell(f^n)$.

Definition 11.9 (kneading determinant). Milnor-Thurston associate to the signed one-sided itineraries of the turning values a matrix of formal power series, the *kneading matrix*. Its normalized maximal minors are the *kneading determinants*. In the unimodal case, if the single kneading row is $\nu(t) = N_1(t)e_1 + N_2(t)e_2$ and the two lap orientations are $\varepsilon_1, \varepsilon_2$, the normalization is

$$D_f(t) = \frac{N_2(t)}{1 - \varepsilon_1 t} = -\frac{N_1(t)}{1 - \varepsilon_2 t}.$$

The equality of the two expressions is the elementary column relation proved inside the Milnor-Thurston theorem below. Since the endpoint factors have no zeros in $|t| < 1$, D_f is analytic there and is determined by the kneading invariant.

Definition 11.10 (nice interval and well-inside relation). An open interval U is *nice* if no forward iterate of ∂U enters U . For intervals $J \Subset U$, say that J is ρ -*well inside* U if

each component of $U \setminus J$ has length at least $\rho|J|$. Nice intervals make first-entry pull-backs either nested or disjoint, while the well-inside condition supplies quantitative Koebe space.

Definition 11.11 (monotone semiconjugacy). A *monotone semiconjugacy* from $f : I \rightarrow I$ to $g : J \rightarrow J$ is a continuous nondecreasing surjection $\pi : I \rightarrow J$ satisfying $\pi \circ f = g \circ \pi$. Intervals on which π is constant are collapsed; if none are nondegenerate, π is a conjugacy.

Definition 11.12 (topological entropy). The *topological entropy* $h_{\text{top}}(f)$ is the exponential growth rate of distinguishable finite orbit segments, equivalently the open-cover entropy of the continuous self-map of the compact interval. For piecewise-monotone interval maps it will be identified below with $\log s(f)$.

Definition 11.13 (restrictive interval and renormalization). A *restrictive interval of period* $p > 1$ for a unimodal map is a proper interval $J \ni c$ such that $f^p(J) \subset J$, the interiors of $J, f(J), \dots, f^{p-1}(J)$ are pairwise disjoint, and $f^p|_J$ is unimodal. After affine rescaling of J to the chosen model interval, the first-return map is the *renormalization* Rf . Iterating this construction gives a finite or infinite renormalization tower.

11.2 Main results

Theorem 11.1 (Singer theorem). **Statement scope.** Let $f : I \rightarrow I$ be C^3 , have negative Schwarzian at every noncritical point, and have finitely many critical points. The immediate basin of every attracting periodic orbit contains either a critical point of f or a boundary point of I . Consequently the number of attracting periodic orbits is at most $\# \text{Crit}(f) + 2$.

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Schwarzian chain rule; derivative minimum principle; critical-free immediate-basin boundary contradiction; disjoint-basin counting.

The mathematical argument begins in the proof environment below.

Proof. Let an attracting orbit have period n , and let T be the component of its immediate basin containing a point p of the orbit. Suppose that the entire immediate basin contains neither a critical point nor an endpoint of the ambient interval. Then every map between the n basin components is a local diffeomorphism, so the return map $g = f^n|_T$ is a diffeomorphism.

Two elementary Schwarzian facts drive the proof. First,

$$S(f \circ g) = (Sf \circ g)(g')^2 + Sg.$$

Thus every iterate has negative Schwarzian on a branch that avoids critical points. Second, if a diffeomorphism G has $SG < 0$, then $|G'|$ cannot have a positive interior local minimum. Indeed, after changing sign if necessary, assume $G' > 0$. At such a

minimum one has $G'' = 0$ and $G''' \geq 0$, so

$$SG = \frac{G'''}{G'} \geq 0,$$

a contradiction.

The endpoints of T must be mapped by g to endpoints of T . Otherwise a boundary point would map into the interior of the attracting basin and, by local invertibility, a neighborhood across that boundary would also belong to the basin, contradicting maximality of the component. Hence $G = g^2$ fixes both endpoints of T . Their one-sided derivatives have absolute value at least one: a boundary fixed point with derivative of absolute value less than one would attract points from outside T and again enlarge the basin.

On the other hand, $|G'(p)| < 1$. Therefore the positive continuous function $|G'|$ is at least one at both ends of T but is less than one inside. It must have a positive interior minimum, contradicting the Schwarzian minimum principle. The immediate basin must therefore contain a critical point or an endpoint.

Immediate basins of distinct attracting orbits are disjoint. Assign to each attracting orbit one critical point or endpoint lying in its immediate basin. No such point can be assigned twice, so the number of attracting periodic orbits is at most the number of critical points plus the two endpoints. $\square$

Theorem 11.2 (No wandering intervals). **Statement scope.** *Let $f : I \rightarrow I$ be a C^2 interval map with finitely many critical points, all nonflat. Then f has no wandering interval: every nondegenerate interval whose forward images are pairwise disjoint is eventually contained in the basin of an attracting periodic orbit.*

Proof strategy (not part of the proof)

Proof status. proved here modulo stated foundation input.

Strategic outline only. Standard-reference proof: de Melo-van Strien, *One-Dimensional Dynamics*, Chapter IV, Theorem A and Sections IV.1–IV.9, with exact C^2 /nonflat scope. The later van Strien-Vargas real-bounds route is reconstructed supplementarily through all but one nonbinding periodic-core source-compression step; the 2007 target-niceness erratum remains enforced in that supplementary route.

The mathematical argument begins in the proof environment below.

Standard-reference proof. The statement is exactly the interval case of Theorem A in W. de Melo and S. van Strien, *One-Dimensional Dynamics*, Chapter IV: a C^2 map of a compact interval or circle with only nonflat critical points has no wandering intervals. The normalization of the phase space in that theorem to $[-1, 1]$ or $\mathbb{S}^1$ is immaterial, since an affine conjugacy preserves wandering intervals, critical orders, and C^2 regularity. Their notion of a nonflat critical point is the same power-law normal form used in the Chapter 11 definition (for a C^2 map the critical order is automatically at least two).

For orientation, the proof in that standard reference is not a black-box one-line citation. Chapter IV, Sections 1–3 establish the cross-ratio and Koebe machinery. Section 4 sets up induction on the number of turning points; Section 5 proves the

contraction/Koebe principle; Section 6 proves the predecessor-successor disjointness estimates; Section 7 shows that a hypothetical wandering interval must accumulate on a turning point; Section 8 controls the resulting unimodal pullbacks; and Section 9 combines these estimates to force the closest-return gaps eventually to have constant period, which puts the original interval in the basin of an attracting periodic orbit. This contradicts the definition of a wandering interval and proves the present theorem. $\square$

Supplementary modern real-bounds reconstruction. The remainder of this section reconstructs the later van Strien-Vargas real-bounds route because its Koebe-space architecture is used elsewhere in the book. It is *not* needed for the logical certification of the theorem just proved. We retain one explicitly fenced source-compression issue in the periodic-core branch rather than reverse an inequality merely to make this second proof route close.

Proof setup. Assume that a wandering interval exists. Replacing it by a smaller open subinterval, write

$$W_n = f^n(W), \quad n \geq 0,$$

so that the intervals W_n are pairwise disjoint. We shall separate the part of the argument that is elementary once uniform pullback space is known from the one quantitative real-bounds input isolated below.

Lemma 11.3 (Wandering images shrink). *One has $|W_n| \rightarrow 0$. If the family $\{W_n\}$ converges to a periodic orbit, then W is eventually contained in its basin. Thus, under the wandering-interval hypothesis fixed above, we may choose an accumulation point z of the shrinking family without assuming that the wandering itinerary is critical-free.*

Proof. Disjointness gives

$$\sum_{n \geq 0} |W_n| \leq |I|,$$

so $|W_n| \rightarrow 0$. If the compact sets W_n converge along the cyclic order of some periodic orbit, then every point of W has forward orbit converging to that orbit; hence W lies in its basin. The definition of a wandering interval excludes this alternative. No statement about avoidance of the critical set is used here. $\square$

Lemma 11.4 (Nice first-entry nesting). *Let U be nice and let $L_x(U)$ be the component of the first-entry domain to U containing x . Distinct first-entry components are disjoint. If $V \Subset U$ are nice and both first-entry components are defined, then*

$$L_x(V) \subset L_x(U).$$

Proof. Suppose two first-entry components overlap. Their first-entry times are locally constant on the interiors of the components. If the times differ, a boundary point of the earlier component is sent to ∂U while a later iterate of the same point would have to enter U , contradicting niceness. Hence overlapping components coincide. The same argument applied to the nested targets $V \Subset U$ gives the second assertion. $\square$

The quantitative part of the source argument is built from a small number of real-variable lemmas. We now reconstruct the bounded-overlap base layer. A periodic-core reduction in the source Koebe-space lemma and the later visited-domain stopping mechanism are kept separate below; this prevents a short source paragraph from being promoted silently to a textbook proof.

For nested bounded intervals $J \Subset T$, let L and R be the two components of $T \setminus J$ and put

$$\text{Cr}(T, J) := \frac{|T||J|}{|L||R|}.$$

A *chain* of length n is a family $G_0, \dots, G_n$ in which G_i is the maximal pullback component of G_{i+1} along the orbit under consideration. Its *intersection multiplicity* is the largest number of members having a common point.

Lemma 11.5 (Cross-ratio distortion by a coordinate change). *Let $h : Q \rightarrow h(Q)$ be a C^2 diffeomorphism of compact intervals. Put*

$$\omega_h(T) := \sup_{x, y \in T} |\log |Dh(x)| - \log |Dh(y)||, \quad T \subset Q.$$

For $J \Subset T \subset Q$ one has

$$e^{-2\omega_h(T)} \leq \frac{\text{Cr}(h(T), h(J))}{\text{Cr}(T, J)} \leq e^{2\omega_h(T)}. \quad (11.1)$$

In particular, if $|D \log |Dh|| \leq L$ on Q , the two exponents in (11.1) may be replaced by $\pm 2L|T|$.

Proof. Write the endpoints of $J \Subset T$ as $a < b < c < d$ and set

$$[x, y]_h := \frac{|h(y) - h(x)|}{|y - x|}.$$

Cancellation of the four Euclidean lengths gives the exact identity

$$\frac{\text{Cr}(h(T), h(J))}{\text{Cr}(T, J)} = \frac{[a, d]_h [b, c]_h}{[a, b]_h [c, d]_h}. \quad (11.2)$$

Every divided difference in (11.2) lies between $m_T := \inf_T |Dh|$ and $M_T := \sup_T |Dh|$. Hence the quotient lies between $(m_T/M_T)^2$ and $(M_T/m_T)^2$. Since $M_T/m_T \leq e^{\omega_h(T)}$, (11.1) follows. The final statement is the mean-value estimate for $\log |Dh|$. $\square$

Lemma 11.6 (One-step nonflat cross-ratio package). *Let $f : I \rightarrow I$ be C^2 with finitely many nonflat critical points. After slightly extending I at an endpoint when necessary, there are constants $A < \infty$ and $\gamma \in (0, 1)$ with the following properties. Suppose $J' \Subset T'$ and $J \Subset T$ are corresponding pullback components with $f(T') = T$ and $f(J') = J$.*

1. *If T' contains no critical point, then*

$$\frac{\text{Cr}(T, J)}{\text{Cr}(T', J')} \geq \exp(-A(|T'| + |T|)). \quad (11.3)$$

2. If T' contains a critical point, then

$$\frac{\text{Cr}(T, J)}{\text{Cr}(T', J')} \geq \gamma. \quad (11.4)$$

3. For every $x \in T'$,

$$|Df(x)| \leq \gamma^{-1} \frac{|f(T')|}{|T'|}. \quad (11.5)$$

The constants depend only on the fixed map and fixed nonflat critical charts.

Proof. Choose pairwise disjoint critical neighborhoods on which

$$f = \alpha \circ P_\ell \circ \beta, \quad P_\ell(t) = \pm |t|^\ell, \quad \ell > 1, \quad (11.6)$$

where α and β are C^2 diffeomorphisms. Shrink the charts so that the derivatives and inverse derivatives of all coordinate changes are bounded. The complement of the critical neighborhoods is covered by finitely many C^2 diffeomorphism charts on which $|Df|$ is bounded away from zero.

If T' lies in one regular chart, Lemma 11.5 applied directly to f gives a lower bound $e^{-A_0|T'|}$. If it lies in a one-sided critical chart, use (11.6). A direct four-endpoint calculation for P_ℓ (equivalently, differentiation of the logarithm of (11.2)) shows that P_ℓ does not decrease this cross ratio on either half-axis. The two coordinate changes contribute at worst $e^{-A_1|T'|}$ and $e^{-A_2|T'|}$ by Lemma 11.5.

There remains the possibility that a critical-free T' crosses the fixed boundary between two such charts. Then $|T'|$ is bounded below by a positive atlas constant. On the closure of each monotonicity component, divide out the one-sided power factors at its critical endpoints. The remaining coordinate factor is a C^2 diffeomorphism with derivative bounded above and below. The normalized four-endpoint quotient therefore extends continuously, with positive limits, to configurations whose outer endpoints approach critical endpoints. Compactness gives a positive lower bound for all crossing configurations. Increasing A converts this finite lower bound into (11.3). This also shows explicitly why only finitely many nonflat charts are needed.

Suppose next that T' contains a critical point c and is contained in its critical chart. In the coordinate $u = \beta(x)$, normalize the outer pullback to unit scale. For the model $|u|^\ell$, the admissible normalized configurations consist of a central pullback or one of the two side pullbacks of a nested target. The quotient in (11.4) is homogeneous of degree zero, extends continuously to the compactification in which one or more adjacent intervals degenerate, and its boundary limits are positive. Hence its infimum is positive. The two coordinate changes alter that infimum by a fixed positive factor by Lemma 11.5. If T' contains more than one critical point, or crosses a fixed critical-chart boundary, its length is bounded below by an atlas constant; the same compactness argument on the finite collection of monotonicity pieces gives a positive lower bound. Taking the minimum proves (11.4).

For (11.5), regular charts follow from the mean-value theorem and bounded coordi-

nate distortion. In a single critical chart, after affine normalization,

$$|P'_\ell(u)| \leq 2\ell \frac{|P_\ell(U)|}{|U|},$$

for every central or one-sided inverse component U . The C^2 coordinate changes modify only the constant. Crossing configurations again form a compact family of definite size, so the same estimate holds there after one further enlargement of the constant. Decreasing γ once gives (11.5) on the whole finite atlas. $\square$

Lemma 11.7 (Bounded-multiplicity real Koebe pullback). *Let $H_i \Subset G_i$, $0 \leq i \leq n$, be two chains along the same orbit. If G_n is σ -scaled around H_n and the intersection multiplicity of $\{G_i\}_{i=0}^n$ is at most κ , then there is $\alpha = \alpha(f, \sigma, \kappa) > 0$ such that G_0 is α -scaled around H_0 . Moreover there is $K = K(f, \sigma, \kappa)$ such that*

$$|Df^n(x)| \leq K \frac{|H_n|}{|H_0|}, \quad x \in H_0. \quad (11.7)$$

Proof. Let b be the number of critical points and let I^* be the fixed enlarged ambient interval. Bounded intersection multiplicity gives

$$\sum_{i=0}^n |G_i| \leq \kappa |I^*|, \quad (11.8)$$

and at most κb members of the chain contain critical points. On a critical-free step (11.3) applies; on a critical step (11.4) applies. Multiplication along the chain gives

$$\frac{\text{Cr}(G_n, H_n)}{\text{Cr}(G_0, H_0)} \geq \gamma^{\kappa b} \exp(-2A\kappa |I^*|) =: \Theta > 0. \quad (11.9)$$

If $G_n \setminus H_n$ has components of length at least $\sigma |H_n|$, then

$$\text{Cr}(G_n, H_n) \leq \frac{1 + 2\sigma}{\sigma^2}.$$

Hence $\text{Cr}(G_0, H_0) \leq C_0$ with $C_0 = (1 + 2\sigma)/(\Theta\sigma^2)$. If the two components of $G_0 \setminus H_0$ have lengths $l|H_0|$ and $r|H_0|$, then $(1 + l + r)/(lr) \leq C_0$. Since this quotient is at least $1/l$ and at least $1/r$, one has $l, r \geq C_0^{-1}$, proving the scaled-neighborhood assertion.

Apply that assertion to every tail of the two chains. Thus every G_i has a uniform amount of space around H_i . Cut the orbit immediately after the members G_i that contain a critical point. On each intervening diffeomorphic block, (11.2), the uniform space just obtained, and the sum bound (11.8) give a uniform distortion bound for the derivative. At a critical step use (11.5). There are at most κb critical steps. The secant slopes of the consecutive blocks telescope, leaving $|H_n|/|H_0|$ multiplied by a constant depending only on f, σ, κ . This proves (11.7). $\square$

Lemma 11.8 (Return-chain multiplicity outside a periodic core). *Let U be nice and not contained in a proper periodic interval. Let J be a return domain of U , with return time r . Let I_j be the pullback of U containing $f^j(J)$, $1 \leq j \leq r$, and suppose $T \supset U$ contains at most e of the intervals I_j . If $T_0, \dots, T_r$ is the pullback chain of T with $T_j \supset f^j(J)$, then*

its intersection multiplicity is at most

$$2(e + b(e + 2)) + 1, \quad (11.10)$$

where b is the number of critical points of f .

Proof. Fix a critical point c . If $T_{i_1}, \dots, T_{i_s}$ all contain c , with $i_1 < \dots < i_s$, pullback nesting gives $T_{i_1} \subset \dots \subset T_{i_s}$; otherwise two overlapping pullbacks of the nice target would cross. The terminal interval T_r then contains the return markers

$$I_{r+i_1-i_s}, \dots, I_{r+i_s-i_s} = I_r,$$

with at most the two end markers lost when the chain first enters or leaves the displayed block. Because T contains at most e markers, $s \leq e + 2$. Summing over the b critical points, at most $b(e + 2)$ intervals of the chain contain a critical point.

A monotone pullback preserves every marker already contained in the interval; only a critical step can lose one. Hence each T_i contains at most $N = e + b(e + 2)$ markers. At a common point, at most N of the intervals T_i can have their pairwise-disjoint marker I_i on the left, at most N can have it on the right, and at most one marker can contain the point. The multiplicity is therefore at most $2N + 1$, which is (11.10). $\square$

Lemma 11.9 (Aperiodic-core existence of Koebe space). *Let U be nice and not contained in a proper periodic interval. For any two return domains J_1, J_2 to U there is a constant $\rho_a > 0$, depending only on f , such that for one of J_1, J_2 there is an interval $M \supset U$ which is a ρ_a -scaled neighborhood of that return domain and which contains only a bounded number of intermediate return markers. Consequently the pullback chain of M along that return branch has uniformly bounded intersection multiplicity.*

Proof. For $i = 1, 2$ set

$$\mathcal{B}_i = \{f(J_i), \dots, f^{r_i}(J_i)\}$$

and let $I_{m,i}$ be the pullback of U along $f^m(J_i), \dots, f^{r_i}(J_i)$ that contains $f^m(J_i)$. Niceness implies that two members of $\mathcal{C} = \{I_{m,i} : 1 \leq m \leq r_i, i = 1, 2\}$ are disjoint or coincide. Choose a shortest member $f^{m_1}(J_i)$ of $\mathcal{B}_1 \cup \mathcal{B}_2$. Let

$$M_{m_1} = L \cup I_{m_1,i} \cup R, \quad |L| = |R| = \rho_{00}|f^{m_1}(J_i)|,$$

where $\rho_{00} > 0$ is chosen once inside the enlarged ambient interval. By minimality, M_{m_1} contains no second member of $\mathcal{B}_i$. Pull it maximally backwards along the chosen return branch to obtain $M_0, \dots, M_{m_1}$.

If $f^{m_1} : M_0 \rightarrow M_{m_1}$ is onto, then M_t contains no member of the same marker family except $I_{t,i}$; it can contain at most one marker from the other return branch on either side. Thus the chain has multiplicity at most three, and Lemma 11.7 pulls the terminal space back to a definite neighborhood of J_i .

Assume therefore that some pullback step folds. List the members $M_{n_1}, \dots, M_{n_\zeta}$ that map strictly inside their successors; each contains a critical point. If M_{n_j} and M_{n_k} overlap with $n_j < n_k$, the later fold contains a critical point not contained in the earlier one; otherwise pullback nesting would force one of the forbidden containments among

the markers. Moreover no three earlier fold intervals can meet a later one. Hence the fold family has multiplicity at most two and $\zeta \leq 2b$.

Starting with the final fold and moving backwards, an induction shows that $M_{n_{\zeta-j}}$ contains at most $j + 4$ markers from $\mathcal{C}$; otherwise its image at the next fold would contain one marker too many, and eventually M_{m_1} would contain two members of the same $\mathcal{B}_i$. Thus every M_t contains at most $2b + 3$ markers. Since the markers $I_{1,i}, \dots, I_{m_1,i}$ have disjoint interiors, if more than $4b + 7$ members of the chain had a common point, one of them would contain at least $2b + 4$ markers, a contradiction. The pullback chain therefore has bounded intersection multiplicity. Applying Lemma 11.7 gives a uniform scaled neighborhood of J_i . Maximal pullback up to the first-return target gives the extension containing U ; enlarging to that maximal interval can only increase the available space and does not add more than the already bounded marker family. This proves the claim. $\square$

Lemma 11.10 (Lap bound for an induced periodic core). *Let T be an ℓ -periodic interval: the interiors of*

$$T, f(T), \dots, f^{\ell-1}(T)$$

are pairwise disjoint and $f^\ell(T) \subset T$. If f has b turning points, then the induced map

$$F = f^\ell|_T : T \longrightarrow T$$

has at most 2^b laps and hence at most $2^b - 1$ turning points.

Proof. Put $T_j = f^j(T)$, $0 \leq j < \ell$, and let b_j be the number of turning points of f in the interior of T_j . Because the interiors of the cycle intervals are disjoint,

$$\sum_{j=0}^{\ell-1} b_j \leq b.$$

The restriction $f|_{T_j}$ has at most $b_j + 1$ monotonicity intervals. Lap numbers are submultiplicative under composition, so

$$\ell(F) \leq \prod_{j=0}^{\ell-1} (b_j + 1) \leq \prod_{j=0}^{\ell-1} 2^{b_j} \leq 2^b.$$

Only interior turning points create new laps, so possible critical endpoints do not alter the estimate. Subtracting one gives the turning-point bound. $\square$

Lemma 11.11 (Uniform derivative bound on a maximal periodic core). *In the periodic-core branch of the shortest-marker construction, let T be the maximal ℓ -periodic interval containing the nice target and suppose that T contains at least three return markers. Then there is a constant $N = N(f)$, independent of ℓ and of the chosen core, such that*

$$\sup_{x \in T} |Df^\ell(x)| \leq N. \quad (11.11)$$

Proof. We first isolate the periodic case of the shortest-marker argument. Write $T_j = f^j(T)$, with indices read modulo ℓ . Choose a shortest cycle interval T_m . Around T_m

choose a fixed scaled interval E_m whose two added sides have length comparable with $|T_m|$. Since no other cycle interval is shorter than T_m , each added side can contain no complete cycle interval distinct from T_m ; it may meet only the nearest cycle interval on that side.

Pull E_m maximally backwards along the periodic itinerary and denote the resulting intervals by $E_j \supset T_j$. If E_s contains a cycle marker T_t with $s \neq t$, forward iteration from time s to time m shows that E_m meets the marker with index $t + m - s$ modulo ℓ . Hence every E_s contains only a uniformly bounded number of cycle markers. The usual one-dimensional pigeonhole argument now bounds the intersection multiplicity: if too many E_s had a common point, at least four of their pairwise-disjoint distinguished markers would lie on the same side of that point, and the interval belonging to the outermost one would contain too many cycle markers. Thus the pullback chain has a universal multiplicity bound (the source obtains the numerical bound 8).

Lemma 11.7 therefore pulls the terminal scaled space back to a fixed amount of space around the corresponding cycle interval. Rotate the preceding construction around the periodic cycle until it gives a scaled interval $E \supset T$ containing only a fixed number of cycle markers. Pull E once around the whole period along the branch whose initial pullback is T . The same modulo- ℓ marker argument gives a uniform multiplicity bound for this full-period chain. Applying the derivative estimate (11.7) with $H_0 = T$ and terminal target $H_\ell = T$ gives

$$|Df^\ell(x)| \leq K \frac{|T|}{|T|} = K, \quad x \in T.$$

If $f^\ell(T)$ is a proper subinterval of T , the terminal secant ratio is at most one, so the same estimate holds. Taking $N = K$ proves (11.11). $\square$

Lemma 11.12 (Bracketed controlled-marker seed). *Let T contain a finite family of complete markers with pairwise disjoint interiors, let I_* be a shortest complete marker, and put $a = |I_*|$. Suppose a complete marker Q has a complete marker on each spatial side and*

$$|Q| \leq Ca \quad (C \geq 1).$$

Then there is an interval $S \Subset T$ which is a $1/(2C)$ -scaled neighbourhood of Q and contains no complete marker other than Q .

Proof. A complete marker on each side of Q has length at least a . Hence the distance from Q to either endpoint of T is at least $a \geq |Q|/C$. Add to Q on each side an interval of length $|Q|/(2C)$. The resulting interval S lies in T and is a $1/(2C)$ -scaled neighbourhood of Q . Each added side has length at most $a/2$, whereas every complete marker has length at least a . Since complete markers have disjoint interiors, no second complete marker can be contained in S . $\square$

Lemma 11.13 (One-sided empty-boundary-room seed). *Let I_* be a shortest complete marker in a periodic core T , put $a = |I_*|$, and suppose every other complete marker lies in the same component of $T \setminus I_*$. Let E_* denote the other component, the one containing no complete marker. If*

$$|E_*| \geq \frac{a}{2},$$

then I_* has a $1/4$ -scaled neighbourhood $S \subseteq T$ containing no other complete marker.

Proof. On the empty side there is room $a/2$ by hypothesis. On the occupied side there is at least one other complete marker, of length at least a , so the distance from I_* to the corresponding endpoint of T is at least a . Extend I_* by $a/4$ on both sides. The resulting interval is compactly contained in T and is $1/4$ -scaled around I_* . Each added side has length strictly smaller than the length of any other complete marker, so it cannot contain one. $\square$

Lemma 11.14 (Forward-successor repair in a one-sided periodic core). *Use the periodic-core notation above, put*

$$N_0 = \max\{1, \sup_T |DF|\},$$

and let $I_* \in \mathcal{C} \cap T$ be a shortest complete marker. Suppose all other complete markers lie in the same component of $T \setminus I_*$. A forward successor marker means a complete marker that is the exact F -image of the preceding marker along the same induced marker itinerary. Then the following configurations already contain uniform seed space.

1. If $I_1 = F(I_*)$ and $I_2 = F(I_1)$ are two distinct complete successor markers in $\mathcal{C} \cap T$, there are $Q \in \{I_1, I_2\}$ and an interval $S \subseteq T$ such that S is a $1/(2N_0^2)$ -scaled neighbourhood of Q and contains no complete marker other than Q .
2. If $I_1 = F(I_*)$ is a complete successor marker and P is a complete immediate predecessor marker, meaning $F(P) = I_*$ along the same induced itinerary, with I_1 lying spatially between I_* and P , the same conclusion holds, again with the safe scale $1/(2N_0^2)$.

Proof. Put $a = |I_*|$. The mean-value theorem and the definition of N_0 give

$$|I_1| \leq N_0 a, \quad |I_2| \leq N_0 |I_1| \leq N_0^2 a. \quad (11.12)$$

In the first case choose from I_1, I_2 the marker Q nearer to I_* . The other successor is on the opposite side of Q from I_* , so Q is bracketed by complete markers and $|Q| \leq N_0^2 a$. Apply Lemma 11.12 with $C = N_0^2$.

In the second case I_1 is bracketed by I_* and P , and $|I_1| \leq N_0 a \leq N_0^2 a$. The same lemma, again with the safe constant $C = N_0^2$, gives the assertion. $\square$

Lemma 11.15 (Single-successor terminal boundary reduction). *Use the one-sided notation of the preceding lemma and put $a = |I_*|$. Suppose there is exactly one complete forward successor $I_1 = F(I_*)$ before the finite induced marker string terminates. Then a uniform marker-free seed exists unless both of the following hold:*

1. I_1 is the outermost complete marker on the occupied side of I_* ;
2. the component E_1 of $T \setminus I_1$ beyond I_1 , away from I_* , has length $|E_1| < a/2$.

More precisely, outside this exceptional configuration one obtains a $1/(4N_0)$ -scaled neighbourhood of a complete marker containing no second complete marker.

Proof. By (11.12), $|I_1| \leq N_0 a$. If I_1 is not outermost, then some complete marker lies farther from I_* than I_1 . Thus I_1 is bracketed by that marker and I_* , and Lemma 11.12

with $C = N_0$ gives the claim.

Assume instead that I_1 is outermost. If $|E_1| \geq a/2$, add to I_1 on each side an interval of length $|I_1|/(4N_0) \leq a/4$. The outer addition lies compactly inside E_1 . On the inner side there is room at least a , because I_* lies there and has length a . Hence the enlarged interval is compactly contained in T and is $1/(4N_0)$ -scaled around I_1 . No other complete marker can be contained in either added side, since each has length at most $a/4$ while every complete marker has length at least a . This proves the stated reduction. $\square$

Proposition 11.16 (Residual boundary-crowded periodic-core seed). *Use the notation of the shortest-marker construction. Let T be a proper maximal ℓ -periodic core containing at least three complete markers from $\mathcal{C}$, put $F = f^\ell|_T$, and let $I_* \in \mathcal{C} \cap T$ be a shortest marker, with $a = |I_*|$. Suppose every other complete marker lies in the same component of $T \setminus I_*$. Let E_* be the marker-free component of $T \setminus I_*$. Assume*

$$|E_*| < a/2. \quad (11.13)$$

Assume moreover that the configuration is not covered by Lemma 11.14 and is terminal in one of the following precise senses:

1. *there is no complete forward successor of I_* along its finite induced marker string; or*
2. *there is exactly one complete forward successor $I_1 = F(I_*)$, every other complete marker lies spatially between I_* and I_1 , and the outer component E_1 of $T \setminus I_1$ satisfies $|E_1| < a/2$.*

Then there exist a complete marker $Q \in \mathcal{C} \cap T$, a constant $\tau_ = \tau_*(f) > 0$, and an interval $S \subseteq T$ such that S is a τ_* -scaled neighbourhood of Q and S contains no complete marker from $\mathcal{C}$ other than Q .*

Unproved input in the present proof. Proposition 11.16 is the only statement in this Chapter 11 argument that is not proved below from the stated textbook prerequisites. Lemma 11.13 removes every one-sided configuration in which the shortest marker has definite room on its empty side. Lemma 11.14 removes the two-successor and favourable predecessor/successor configurations, while Lemma 11.15 removes every remaining one-successor configuration except the one in which the shortest marker and its unique successor crowd opposite boundary sides of the periodic core. Thus the unproved assertion is now confined to the boundary-crowded zero-successor case and this boundary-crowded one-successor terminal case.

The printed periodic-core paragraph of van Strien–Vargas Lemma 2 states that one or two F -preimages of the shortest marker produce the required scaled space after obtaining a universal upper bound on $|DF|$. That upper bound alone gives the wrong-direction inequality for an upper estimate on a predecessor length, so the present proof does not use that inference. The forward-image repairs above retain exactly those consequences for which the available derivative estimate has the needed direction.

Lemma 11.17 (Complete shortest-marker seed dichotomy). *Assume Proposition 11.16. In every periodic-core shortest-marker configuration there are a complete*

marker $Q \in \mathcal{C} \cap T$, a constant $\tau = \tau(f) > 0$, and an interval $S \in T$ containing Q such that

1. S is a τ -scaled neighbourhood of Q ;
2. S contains no complete marker from $\mathcal{C}$ other than Q .

Without using Proposition 11.16, the conclusion already holds when markers occur on both sides of the shortest marker, when the empty side has definite boundary room, or when the one-sided configuration is covered by Lemma 11.14 or Lemma 11.15.

Proof. Suppose first that markers occur on both sides of the shortest marker I_* . The distance from I_* to each endpoint of T is at least the length of a complete marker on that side and hence at least $|I_*|$. Add to I_* a left and a right interval of length $|I_*|/2$. The resulting interval lies inside T . Since every complete marker has length at least $|I_*|$, neither added side can contain a complete marker. This gives a $1/2$ -scaled seed.

Now suppose all other markers lie on one side of I_* and put $a = |I_*|$. If the marker-free component of $T \setminus I_*$ has length at least $a/2$, Lemma 11.13 gives the desired seed. We may therefore assume the boundary-crowding inequality (11.13).

Put $N_0 = \max\{1, \sup_T |DF|\}$. If two complete forward successors are available, or if the favourable predecessor/successor order occurs, Lemma 11.14 gives the desired seed. If there is exactly one complete forward successor, then Lemma 11.15 gives the seed unless that successor is outermost and its outer boundary component has length $< a/2$. Thus every case not already settled by the three internal lemmas satisfies one of the two alternatives of Proposition 11.16, which supplies the seed. Finally N_0 is uniformly bounded by Lemma 11.11. Taking the minimum of the finitely many map-uniform scales proves the assertion. $\square$

Lemma 11.18 (Phase expansion of an induced periodic-core chain). *Let T be ℓ -periodic and write $T_r = f^r(T)$, $0 \leq r < \ell$. Put $F = f^\ell|_T$. Suppose*

$$\widehat{G}_0, \widehat{G}_1, \dots, \widehat{G}_q \subset T$$

is an F -chain of intersection multiplicity at most κ , so $F(\widehat{G}_j) \subset \widehat{G}_{j+1}$. Expand each induced step into its f -orbit by setting

$$G_{j,r} = f^r(\widehat{G}_j), \quad 0 \leq j < q, \quad 0 \leq r < \ell,$$

and append $G_{q,0} = \widehat{G}_q$. Then the resulting f -chain has intersection multiplicity at most 2κ .

Proof. For fixed r and $j < q$, one has $G_{j,r} \subset T_r$. If a point x belongs to $G_{j,r}$, then

$$f^{\ell-r}(x) \in F(\widehat{G}_j) \subset \widehat{G}_{j+1}.$$

The point on the left is independent of j . Hence, for a fixed phase r , at most κ of the intervals $G_{j,r}$ can contain x , because the induced intervals $\widehat{G}_{j+1}$ have multiplicity at most κ . For $r = 0$ the same conclusion follows directly from the multiplicity of the induced chain, including the terminal interval $\widehat{G}_q$.

The interiors of $T_0, \dots, T_{\ell-1}$ are pairwise disjoint. Therefore a point of the ambient interval belongs to at most two of the closed cycle intervals $\overline{T_r}$; away from cycle

boundaries it belongs to only one. Summing the fixed-phase bound over the at most two possible phases gives multiplicity at most 2κ for the expanded f -chain. $\square$

Proposition 11.19 (Periodic-core seed space). *Assume Proposition 11.16. In the proper periodic-core branch with at least three return markers, one can choose a marker $Q \in \mathcal{C} \cap T$ and a fixed scaled seed $S \in T$ around Q , containing no other complete marker, so that the induced F -pullback of S and its phase-expanded f -pullback both have intersection multiplicity bounded by constants depending only on f .*

Proof. Lemma 11.17 supplies a terminal seed S with a uniform scale and with no second complete marker. Apply the fold/marker argument from Lemma 11.9 to the induced map $F = f^\ell|_T$. That combinatorial argument uses only the number of turning points and the fact that the terminal seed contains no second complete marker. By Lemma 11.10, F has at most $2^b - 1$ turning points. Hence the induced pullback chain has intersection multiplicity at most a constant $\kappa_F = \kappa_F(b)$.

Now expand every F -step into its ℓ constituent f -steps. Lemma 11.18 bounds the multiplicity of this expanded chain by $2\kappa_F$, independently of the period ℓ . Thus the original f -level bounded-multiplicity Koebe lemma applies to the fixed terminal scale of S . This gives a pullback scale depending only on f , exactly as required for the periodic-core branch. No separate passage from induced time back to original time is needed. $\square$

Proposition 11.20 (Periodic-core Koebe-space reduction). *There is a constant $\rho_p > 0$, depending only on f , with the following property, conditional only on Proposition 11.16. If a nice interval U is contained in a proper maximal periodic interval and J_1, J_2 are return domains to U , then for one of these return domains there is an interval $M \supset U$ which is a ρ_p -scaled neighbourhood of that domain and whose pullback chain has uniformly bounded intersection multiplicity.*

Proof. Run the shortest-marker construction from Lemma 11.9 until the first place where the aperiodic fold-counting argument can fail. Let $\mathcal{D}$ be the family of fold intervals, namely those pullbacks mapped strictly inside their successors.

If $\mathcal{D}$ has no inclusion chain of length three, then the earlier fold-counting proof still works with a larger constant. Intervals of $\mathcal{D}$ containing the same turning point are nested, hence at most two of them can be assigned to one turning point; allowing the two possible sides in the overlap bookkeeping gives the safe bound $\#\mathcal{D} \leq 4b$. The marker induction used in Lemma 11.9 then shows that every pullback contains at most $4b + 3$ markers and that the whole pullback chain has intersection multiplicity at most $8b + 7$. Lemma 11.7 therefore finishes this branch.

Assume now that three fold intervals form a strict inclusion chain. For two successive members $M_s \supsetneq M_t$, $s < t$, the chain relation gives $f^{t-s}(M_s) \subset M_t \subset M_s$. Transporting this restrictive interval to the phase containing U produces a periodic interval containing U ; the three nested folds carry three distinct return markers, so the corresponding periodic core contains at least three members of $\mathcal{C}$. Choose the period ℓ maximal and then choose the maximal ℓ -periodic interval T containing U .

For $F = f^\ell|_T$, Lemma 11.10 gives at most $2^b - 1$ turning points and Lemma 11.11 gives the uniform derivative bound required in the source reduction. By Proposi-

tion 11.19, choose the fixed seed and its phase-expanded f -pullback with uniformly bounded multiplicity. The fold/marker count for F is already part of that proposition, while Lemma 11.18 makes the passage back to the original f -time explicit with no period-dependent loss. Finally Lemma 11.7 pulls the seed space all the way back to one of J_1, J_2 . This gives the required M and ρ_p . $\square$

Proposition 11.21 (Koebe-space base package). *For one or two specified return domains of a nice interval, one can choose a definite Koebe extension whose pullback chain has uniformly bounded intersection multiplicity. For targets not contained in a proper periodic core this follows from Lemma 11.9; in the remaining case it is exactly Proposition 11.20. Once such an extension is available, every pullback of a fixed scaled subinterval acquires a scaled neighborhood depending only on the original scale and on f .*

Proof. The aperiodic case is Lemma 11.9; the periodic case is Proposition 11.20, whose only unresolved input is Proposition 11.16. In either case the bounded intersection multiplicity, together with Lemma 11.7, pulls a fixed amount of terminal space back to a fixed amount of initial space. $\square$

We now reconstruct the visited-domain argument. Throughout this part U is nice and Φ denotes the first-return map to U . A *return domain* J is a component of the domain of Φ . If $x \in J$, a *fundamental domain* through x is a maximal interval $D \ni x$ for which, for some $m \geq 1$,

$$D, \Phi(D), \dots, \Phi^{m-1}(D) \subset J, \quad \Phi^m(D) \subset U \setminus J. \quad (11.12)$$

For $A \Subset U$, the two components of $U \setminus A$ are the *sides* of A ; a side is η -small when its length is $< \eta|A|$ and η -big otherwise. If $J_1, \dots, J_q$ are the maximal monotonicity intervals of $\Phi|_J$ in left-to-right order, J_1 and J_q are the *external branches*. When $\Phi : J \rightarrow U$ is onto, we call it orientation preserving, respectively orientation reversing, when its two external branches have that common orientation. These conventions agree with the source terminology and do not assert that $\Phi|_J$ is globally monotone.

Lemma 11.22 (Fundamental domains have a uniformly big side). *Assume Proposition 11.21. There is $\delta_1 > 0$, depending only on f , such that the following holds. If J is a return domain and $D \subset J$ is a fundamental domain not contained in the immediate basin of a periodic attractor, then D has a δ_1 -big side in U , and $\Phi(D)$ is contained in a δ_1 -big side of D .*

Proof. It is enough to exclude a sequence of counterexamples for which the side of D containing $\Phi(D)$ has relative length $\alpha \rightarrow 0$. For such a counterexample, J itself has an $O(\alpha)$ -small side. Apply Proposition 11.21 to the single domain J and let M be its definite Koebe extension. For α sufficiently small, $\Phi(D)$ lies a fixed positive distance, in relative scale, from ∂M . Lemma 11.7 therefore pulls a fixed neighbourhood of $\Phi(D)$ back to an interval $V \supset D$ whose two sides have length at least $c|D|$, with $c > 0$ independent of the return time.

On the other hand the side assumption gives $|\Phi(D)| \leq \alpha|D|$. The derivative part of Lemma 11.7, applied to the same return branch and then to the two side pullbacks,

gives

$$\sup_V |D\Phi| \leq C\alpha.$$

After decreasing the universal threshold, $C\alpha < 1$ and the image of the two endpoints of V lies in the interior of V ; hence $\Phi(V) \Subset V$. The contraction mapping argument for the monotone pieces of the return branch yields an attracting periodic point whose immediate basin contains D , contrary to the hypothesis. Thus the relevant side has a fixed relative size. The same estimate, applied to the opposite side if necessary, gives the first assertion and fixes one common constant δ_1 . $\square$

Lemma 11.23 (Ordered escape from a poorly collared fundamental domain). *Assume Proposition 11.21, and fix $0 < \xi < \delta_1/10$. There is $\delta_2 = \delta_2(f, \xi) > 0$ with the following property. Let J be a return domain, let $D \subset J$ be a fundamental domain, and suppose*

$$D, \dots, \Phi^{k-1}(D) \subset J, \quad \Phi^k(D) \cap J = \emptyset. \quad (11.13)$$

If D is not δ_2 -well inside U , then:

1. *J has a ξ -small side;*
2. *there is a maximal external branch $J^e \subset J$, adjacent to a ξ -small side H of J , such that $D, \dots, \Phi^{k-2}(D) \subset J^e$ and $\Phi^k(D) \cap H = \emptyset$;*
3. *$D, \dots, \Phi^k(D)$ are linearly ordered in U ; each of $D, \dots, \Phi^{k-1}(D)$ has a ξ -small side, and $\Phi^{i+1}(D)$ lies in the δ_1 -big side of $\Phi^i(D)$;*
4. *if $k \geq 2$, then $\Phi|_{J^e}$ is orientation preserving;*
5. *if both sides of J are ξ -small, then $k = 1$ and the external branch meeting D is orientation reversing.*

Proof. Let M be the definite extension supplied by Proposition 11.21 for J . The bounded-multiplicity Koebe lemma has two elementary consequences that we shall use repeatedly. First, every non-external component of $(\Phi|_J)^{-1}(J)$ is uniformly well inside U . Second, if the boundary value of a nonmonotone branch points toward a big side of J , then its J -preimage is uniformly well inside J . Both statements follow by pulling the fixed side of M back through a branch of uniformly bounded intersection multiplicity. Therefore a D which is not δ_2 -well inside U , with δ_2 chosen below the two constants just obtained, can only meet an external branch adjacent to a ξ -small side. This proves (1) and identifies J^e .

Suppose first that the opposite side of J is ξ -big. When $k = 1$, Lemma 11.22 forces $\Phi(D)$ into that big side, so (2)–(3) follow. Let $k \geq 2$. If the endpoint of J^e on the small side were mapped across J , the second Koebe consequence above would put $\Phi_J^{-1}(J)$, hence D , uniformly inside U . Thus this endpoint maps toward the same small side and $\Phi|_{J^e}$ preserves orientation.

If the derivative on J^e were bounded by the universal branch constant, the small-side geometry and the preceding Koebe consequences would again place D uniformly inside U . Hence we are in the complementary branch. The Koebe extension and the bounded-distortion part of Lemma 11.7 then force $\Phi(J^e) \supset J$; otherwise the secant slope of the branch, and hence its derivative, would remain universally bounded. Thus $\Phi|_{J^e}$ has a fixed point p . Since $k \geq 2$, the fundamental domain D lies on the side of p

opposite the small side. Let B be the interval joining D to p . Because D is not uniformly collared while the small side of J is ξ -small, the secant slope $|\Phi(B)|/|B|$ is larger than a fixed number $1 + \varepsilon$ after δ_2 is chosen sufficiently small. The uniform distortion on the successive external-branch pullbacks transfers this lower secant-slope bound to $B, \Phi(B), \dots, \Phi^{k-1}(B)$. Consequently these intervals cannot cross: they occur in order between p and the escaping image. Each $\Phi^i(D)$ therefore has on the p -side a ξ -small component once δ_2 was chosen sufficiently small. Its other side contains $\Phi^{i+1}(D)$ and is δ_1 -big by Lemma 11.22. This proves (2)–(4).

Finally suppose both sides of J are ξ -small. Then the extension M contains a fixed scaled neighbourhood of the whole of U . Koebe distortion on every internal branch and on every orientation-preserving external branch makes adjacent fundamental domains at periodic points of return period one or two uniformly comparable. Hence such a fundamental domain is uniformly inside U . The only configuration not covered by this comparison is the first escaping domain on an orientation-reversing external branch. Thus $k = 1$, which proves (5). Choosing δ_2 below all finitely many constants used above completes the proof. $\square$

Lemma 11.24 (Pullback across an exterior gap). *For every sufficiently small $\rho > 0$ there is $\delta_3 = \delta_3(f, \rho) > 0$ such that the following holds. If $V \subset U \setminus J$ is ρ -well inside U , then every component A of $(\Phi|_J)^{-k}(V)$, $k \geq 1$, is δ_3 -well inside J .*

Proof. Let D be the fundamental domain containing A . If $\Phi(D)$ is uniformly well inside U , pull its collar back once and then along the fundamental chain; Lemma 11.7 gives the claim. Otherwise Lemma 11.23 applies. The intervals

$$\Phi(D), \dots, \Phi^{k-1}(D), \Phi^k(D)$$

are disjoint and ordered. The penultimate interval has a ξ -small side containing its predecessor, whereas its δ_1 -big side contains V . Since V is ρ -well inside U , it is also uniformly well inside that big side. Pull that side backwards to A . Because the ordered chain can cover any point at most twice, its intersection multiplicity is at most two; Lemma 11.7 gives a collar δ_3 depending only on f and ρ . $\square$

For an orbit $x \in U$ returning infinitely often, define the visited domains J_i and jumping times k_i by $k_0 = 0$ and

$$\Phi^{k_i+j}(x) \in J_i \quad (0 \leq j < k_{i+1} - k_i), \quad \Phi^{k_{i+1}}(x) \notin J_i. \quad (11.15)$$

If the orbit never leaves J_i , put $k_{i+1} = \infty$.

Lemma 11.25 (Visited-domain stopping decomposition). *Fix $0 < \rho < \rho_0/2$, where ρ_0 is a Koebe-space constant from Proposition 11.21. Suppose $k_i < \infty$ and fix $n > 0$. There is $0 \leq i_0 \leq n$ such that, for every $i < i_0$, at least one side H_i of J_i is ρ -small, J_{i+1} lies in the other side G_i , and the open gap (J_i, J_{i+1}) meets none of $J_0, \dots, J_i$. At i_0 exactly one of the following occurs:*

- (P1) both sides of J_{i_0} are ρ -big and $i_0 < n$;
- (P2) precisely one side is ρ -small and J_{i_0+1} lies in it, with $i_0 < n$;

- (P3) *neither P1 nor P2 holds, $i_0 < n$, and the gap (J_{i_0}, J_{i_0+1}) meets an earlier visited domain; if $t < i_0$ is maximal with J_t in that gap, then $H_t \cap H_{i_0} = \emptyset$ while $G_t \cap G_{i_0} \neq \emptyset$;*
- (P4) $i_0 = n$.

Proof. Start with $i = 0$ and continue as long as the small-side condition and the gap-disjointness condition both hold. If the former fails, P1 or P2 holds; if the latter fails first, P3 holds; if neither fails before n , P4 holds. These alternatives are disjoint by construction. In P3 the two assertions about H_t, G_t follow from the order of the gap arcs and maximality of t . Thus the lemma is a finite stopping-time decomposition, not a new analytic input. $\square$

Lemma 11.26 (P4 initial collar). *Assume P4 through time n , suppose $J_0 = J_n$, J_0 has exactly one ρ -small side, $\Phi(J_0)$ contains that side, and $\Phi|_{J_0}$ is not an orientation-reversing surjective branch. Then there is $\sigma_1 > 0$ such that U is a σ_1 -scaled neighbourhood of the pullback of J_n along the orbit segment from x to $\Phi^{k_n}(x)$.*

Proof. Stop first at the least positive return of the visited-domain sequence to J_0 ; pullbacks before that time are disjoint, so a collar there pulls back to the original point with bounded multiplicity. We may therefore suppose $J_0, \dots, J_{n-1}$ are pairwise disjoint. P4 orders them, after reflection if necessary, as

$$J_0, J_{n-1}, J_{n-2}, \dots, J_1. \quad (11.16)$$

Let D_0 be the fundamental domain containing the initial pullback and D_1 the fundamental domain containing its image at the first jump. By Lemma 11.23, D_0 has definite space on one side and D_1 on the opposite side.

Apply Proposition 11.21 to the pair J_0, J_{n-1} and call the resulting extension M . It gives definite space around at least one of those two domains. If it gives space around J_{n-1} , the order (11.16) places J_1 in the same controlled extension. On an orientation-preserving external branch, Lemma 11.24 pulls the free side of D_1 back to the missing side of D_0 ; on a branch containing a turning point, the image contains the big side of D_1 , so the same bounded-multiplicity Koebe argument applies directly.

If instead M gives space around J_0 , pull M back through the branch of J_{n-1} . The preimage of $J_n = J_0$ lies in the fundamental domain D_{n-1} and has a definite free side. If that preimage is already well inside U , use Lemma 11.24; otherwise the ordering (11.16) places J_1 inside the pulled-back extension and reduces to the preceding case. The exceptional two-visit case is identical after using the free side of D_1 . The excluded orientation-reversing surjective branch is exactly the configuration in which the side orientation is lost. All constants come from the fixed ρ , the Koebe-space base, and bounded multiplicity, so their minimum gives σ_1 . $\square$

Lemma 11.27 (P4 terminal-target pullback). *For every $\xi > 0$ there is $\xi_0 > 0$ with the following property. Assume P4 through time n . If $V \Subset U$ is ξ -well inside U and the orbit enters V after the next jump, then its full pullback V_0 to the initial point is ξ_0 -well inside J_0 .*

Proof. First pull V through the last stay in J_n . Lemma 11.24 makes the resulting interval V_n uniformly well inside J_n . In a P4 spiral, once a visited domain is revisited

the tail alternates between one or two domains: there is an index $i_1 \leq n$ such that the even and odd members of the tail are constant separately. Before i_1 the pullback of J_{i_1} is disjoint, hence has bounded intersection multiplicity. It is therefore enough to pull V_n back through this alternating tail.

If at some stage the orbit lies in a fundamental domain which is δ_2 -well inside its visited domain, its collar pulls back through the preceding part of the tail by Lemma 11.24 and we are done. Otherwise every fundamental domain in the tail satisfies Lemma 11.23. The orbit then lies in the corresponding external branches. If the alternating tail has length at most five, it consists of only finitely many return branches and the fixed Koebe extension on each branch pulls the collar back with a uniform loss. We may therefore assume the tail has length at least six, and choose i_1 maximal with this property. The ordered-escape lemma then places the every-other visits on the same orientation-preserving external branch of the first-return map to one of the two recurrent domains; an orientation-reversing external branch would, by the last clause of Lemma 11.23, force an immediate exit and terminate the tail. The final two steps give a uniformly collared pullback by Lemma 11.26 (or directly by the already collared V_n), and Lemma 11.24 transports that collar through the orientation-preserving every-other branch. At most five uncontrolled initial steps remain; each has the fixed Koebe extension from Proposition 11.21. Taking the minimum of the finitely many losses gives ξ_0 . $\square$

Lemma 11.28 (P1/P2 stopping collar). *There is $\sigma_2 > 0$ such that if P1 or P2 occurs at $i_0 < n$, then the pullback of J_{i_0+1} to the initial point is σ_2 -well inside U ; if $i_0 > 0$, it is σ_2 -well inside J_0 .*

Proof. Let D be the fundamental domain containing $\Phi^{k_{i_0}}(x)$. Under P1 both sides of J_{i_0} are big; under P2 the next visited domain lies in the unique small side. In either case Lemma 11.22 and the definition of the stopping time give a fixed collar for D in U . Apply Lemma 11.27 to the preceding P4 segment $J_0, \dots, J_{i_0}$, using this collared D as the terminal target. The result is the asserted uniform collar; when $i_0 > 0$ the first pullback lies inside J_0 , so the same argument gives the relative statement there. $\square$

Lemma 11.29 (P3 break collar). *There is $\sigma_3 > 0$ such that, when P3 occurs at $i_0 < n$ and $t < i_0$ is the last earlier index lying in the broken gap, the pullback inside J_t of J_{i_0+1} along the orbit segment from $\Phi^{k_t}(x)$ to $\Phi^{k_{i_0+1}}(x)$ is σ_3 -well inside U .*

Proof. Reflect the picture if necessary and assume J_t lies to the left of J_{i_0} . Let G_{i_0} be the left side of J_{i_0} ; it contains J_t . Let H_t be the small left side of J_t ; by P3 it contains J_{i_0+1} . Consider the first-return map Ψ to G_{i_0} . If D is the Φ -fundamental domain and D' the Ψ -fundamental domain through the orbit point at time k_t , then $D' \subset D$, its successive Ψ -images remain in J_t until the last one maps into J_{t+1} .

If D' is not uniformly inside G_{i_0} , Lemma 11.23, applied to Ψ , orders its fundamental images and puts each successor in the big side of its predecessor. The broken-gap geometry then gives a fixed free interval on the left of D' . It remains to obtain one on the right. If the right side of J_t is big this is immediate. If it is small, both sides of J_t are small, and the last clause of Lemma 11.23 applied to the original return map says that D is the first escaping domain on an orientation-reversing external branch;

its opposite side is therefore uniformly free. Since $D' \subset D$, this gives the required right-hand space. Thus D' is uniformly well inside U . Pulling the terminal domain through the remaining ordered branch uses intersection multiplicity at most two and Lemma 11.7, yielding σ_3 . $\square$

Proposition 11.30 (Visited-domain collar propagation). *Assume Proposition 11.21. The first-return map to a nice interval U satisfies the uniform visited-domain collar propagation needed below: along a finite visited-domain itinerary, P_4 pulls any terminal scaled target back with a uniform loss, while P_1 , P_2 , and P_3 produce a uniformly collared stopping pullback.*

Proof. The decomposition into P_1 – P_4 is Lemma 11.25. P_4 is Lemma 11.27; P_1/P_2 are Lemma 11.28; P_3 is Lemma 11.29. The fundamental-domain geometry used by all three branches is supplied by Lemmas 11.22–11.24; no additional hypothesis is required. $\square$

Proposition 11.31 (Nice-target macroscopic pullback space). *Assume Proposition 11.21. For every $\xi > 0$ there is $\xi_0 > 0$ with the following property. Let U be nice and let $V \Subset U$ be both nice and ξ -well inside U . If $x \in U$ and the first-return orbit satisfies $\Phi^k(x) \in V$ for some $k \geq 1$, then the pullback of V along $\{x, \Phi(x), \dots, \Phi^k(x)\}$ is ξ_0 -well inside the return domain to U containing x . Equivalently, after expanding each return step into its underlying f -orbit, the corresponding f -pullback has the same uniform collar.*

Proof. Apply Lemma 11.25 to the visited domains before the entry into V . If P_4 persists to the terminal visit, Lemma 11.27 gives the collar. If P_1 or P_2 occurs, use Lemma 11.28 and then pull the terminal ξ -space through the remaining return segment by Lemma 11.24. If P_3 occurs, first use Lemma 11.29 at the broken gap and then the same exterior-gap pullback. Taking the minimum of the three constants gives ξ_0 .

The requirement that V itself be nice is retained explicitly. This is the hypothesis added by the 2007 erratum to the 2004 statement of Theorem C(1); it is used when return pullbacks are concatenated, so that boundary images cannot re-enter the terminal target. $\square$

Only remaining unproved input. The visited-domain collar proposition and the macroscopic pullback proposition above follow from the Koebe-space base. The only statement not proved from the stated prerequisites is Proposition 11.16: the residual terminal or backward-middle one-sided periodic-core seed assertion. The forward-successor configurations are covered by Lemma 11.14, and the visited-domain argument requires no further unproved statement.

Lemma 11.32 (One-step finite nice refinement). *Assume Propositions 11.31 and 11.30. Let $\mathcal{P}$ be a finite nice partition, let $U \in \mathcal{P}$ be visited infinitely often by the shrinking wandering family, and suppose the family does not converge to a periodic orbit. Then there is a finite nice refinement $\mathcal{P}' \succ \mathcal{P}$, an element $V \in \mathcal{P}'$ containing the first relevant visit of the wandering family to U , and a constant $\rho > 0$, independent of the refinement level, such that V is nice and ρ -well inside U .*

Proof. Follow the successive return domains of the first-return map to U . The visited domains have the elementary stopping alternative used in the source: either the ordered “spiral” of visited domains breaks, or it persists.

If the spiral breaks, Proposition 11.30 supplies a return or fundamental domain with a definite collar. Add to the partition the finitely many periodic boundary points and the finite collection of preimages needed to make this domain a partition element. Its interior is a nice interval $V \subseteq U$ with a uniform well-inside constant.

Suppose instead that the spiral persists while infinitely many distinct return domains are visited. Pairwise disjointness and the ordered geometry force these domains to move away from the boundary of U at some visit; otherwise their one-sided large complements would be nested in a fixed boundary collar and could not remain distinct indefinitely. At the first such visit we again obtain a nice uniformly well-inside return domain and refine along its finite boundary orbit.

It remains to treat eventual recurrence in only one or two return domains. If there is one, pass to its periodic core. The fixed points of the return map and their relevant preimages give a finite subdivision into fundamental domains. Since the wandering family is not converging to a periodic orbit, it eventually enters a non-attracting fundamental domain; by Proposition 11.30 that domain has a uniform collar. Refining by those periodic points and preimages produces the required V .

If exactly two return domains are visited, first ask whether either domain nearly fills U . In that case the bounded return geometry gives a uniformly collared fundamental domain. Otherwise both leave a definite amount of room in U . Pass to the first-return map of one of them. If one of the two resulting descendants nearly fills its parent, use the preceding fundamental-domain alternative; if neither does, one descendant is already uniformly well inside the original U . Repeating this dichotomy cannot avoid both outcomes: an infinite sequence of almost-full descendants would force the wandering family into the periodic-core alternative, while infinitely many non-almost-full descendants create a definite collar. Thus a finite nice refinement with the asserted V always exists. $\square$

Lemma 11.33 (Iterated well-inside contradiction). *Assume Proposition 11.21. Starting from a finite nice partition element U_0 visited infinitely often, there are refinements and nice intervals*

$$U_0 \supset V_0 \supset U_1 \supset V_1 \supset \cdots$$

with $U_{m+1} \subset V_m$, each V_m uniformly well inside U_m , and one constant $\rho_0 > 0$ such that

$$L_W(V_m) \text{ is } \rho_0\text{-well inside } L_W(U_m)$$

for every m . This is impossible for a nondegenerate W .

Proof. Apply Lemma 11.32 recursively. Because V_m is nice, Proposition 11.31 applies at every stage; the uniform source constants give one $\rho_0 > 0$. By Lemma 11.4,

$$L_W(U_{m+1}) \subset L_W(V_m) \subseteq L_W(U_m).$$

The first-entry component contains the whole original interval W : if a partition boundary split W before entry, its image would lie on a partition boundary, whereas the

relevant shrinking image of W lies in the interior of the chosen partition element. After discarding the corresponding finite prefix, this holds simultaneously at every refinement level.

If $J' \in J$ is ρ_0 -well inside J , then

$$|J'| \leq \frac{|J|}{1 + 2\rho_0}.$$

Hence

$$|L_W(U_m)| \leq |L_W(U_0)|(1 + 2\rho_0)^{-m} \rightarrow 0.$$

But $W \subset L_W(U_m)$ for all m , so the left-hand side is at least $|W| > 0$, a contradiction. $\square$

Conditional completion of the supplementary route. Assume Proposition 11.16. By Lemma 11.3, a hypothetical wandering interval gives a shrinking pairwise-disjoint family that does not converge to a periodic basin. Choose an accumulation point and a finite nice partition whose relevant element is visited infinitely often. The interval-map case requires no modification of the map merely to create a periodic boundary: a continuous self-map of a compact interval has a fixed point, and the source construction refines finite periodic boundary data away from the chosen interior accumulation point. Lemma 11.33 then gives the contradiction. $\square$

Theorem 11.34 (Signed itinerary order and kneading admissibility). **Statement scope.** Let $f : [a, b] \rightarrow [a, b]$ be unimodal, increasing on $[a, c]$ and decreasing on $[c, b]$. If $x < y$, then

$$\iota_f(x) \preceq \iota_f(y)$$

in signed lexicographic order. Hence the kneading sequence $K(f)$ determines the order restrictions satisfied by every realizable itinerary and therefore the admissible symbolic language of the unimodal dynamics.

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Common-prefix monotonicity cylinder; product of branch orientations; first-difference comparison; shifted critical-value inequalities.

The mathematical argument begins in the proof environment below.

Proof. Let $x < y$ and suppose their itineraries first differ at time n . Before that time both points lie on the same side of the turning point at every iterate, so x and y lie in one common monotonicity cylinder for f^n .

Each visit to the left branch preserves order and each visit to the right branch reverses it. Hence the orientation of f^n on the common cylinder is

$$\varepsilon_n = \prod_{j=0}^{n-1} \varepsilon(s_j), \quad \varepsilon(L) = +1, \quad \varepsilon(R) = -1.$$

If $\varepsilon_n = +1$, then $f^n(x) < f^n(y)$ and the first differing symbols are compared in the ordinary order $L < C < R$. If $\varepsilon_n = -1$, the inequality is reversed and so is the symbol

order. This is exactly the signed lexicographic rule, proving

$$\iota_f(x) \preceq \iota_f(y).$$

Now compare each orbit point $f^m(x)$ with the turning point c . Applying the order theorem to this pair compares the shifted itinerary of x with the itinerary of the critical value, namely the kneading sequence. Thus every real orbit address satisfies a family of signed inequalities determined by $K(f)$. A symbolic word that violates one of them cannot occur.

Conversely, endpoints of finite monotonicity cylinders are critical preimages. Their order is determined recursively by the same comparisons with the critical orbit. Therefore the kneading data determine which finite words occur and how their cylinders are ordered. Passing through all finite lengths determines the admissible symbolic language. $\square$

Theorem 11.35 (Milnor-Thurston kneading theorem). **Statement scope.** *Let $f : I \rightarrow I$ be a continuous piecewise-monotone map with finitely many laps. Its kneading invariant determines the lap-number generating data and the growth number $s(f)$, and*

$$h_{\text{top}}(f) = \log s(f).$$

For a unimodal map with $s(f) > 1$, if D_f is the normalized kneading determinant, its smallest positive zero $r \in (0, 1)$ satisfies

$$r = s(f)^{-1}, \quad h_{\text{top}}(f) = -\log r.$$

Whenever $s(f) > 1$, moreover, f admits a monotone semiconjugacy to a piecewise-linear map whose absolute slope on every noncollapsed lap is $s(f)$.

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Kneading matrix/address jumps; recursive lap combinatorics; lap-growth entropy; coefficientwise kneading resolvent; first positive determinant zero; normalized lap-measure constant-slope quotient.

The mathematical argument begins in the proof environment below.

Proof. Let

$$a = c_0 < c_1 < \cdots < c_{q-1} < c_q = b$$

be the endpoints and turning points of f , and let $I_j = (c_{j-1}, c_j)$ be its laps. Write $\varepsilon_j \in \{+1, -1\}$ for the orientation of f on I_j . If the first n iterates of x avoid the turning set, let $\varepsilon_n(x)$ denote the product of the first n branch orientations, with $\varepsilon_0(x) = 1$.

Suppose the lap partition of f^n is known. The lap endpoints of $f^{n+1} = f \circ f^n$ consist of the old endpoints together with those points inside a lap of f^n whose n -th image is a turning point c_i . Since f^n is monotone on such a lap, for each c_i there is at most one new cut. The signed order theorem from the preceding result decides whether c_i lies between the two endpoint images, and the orientation product decides which

inequality is the relevant one. Thus the kneading data decide every new cut. Induction reconstructs all lap partitions and all lap numbers.

Put $\ell_n = \ell(f^n)$ and $\ell_0 = 1$. Since $\ell_{m+n} \leq \ell_m \ell_n$, Fekete's lemma applied to $\log \ell_n$ gives

$$s(f) = \lim_{n \rightarrow \infty} \ell_n^{1/n} = \inf_{n \geq 1} \ell_n^{1/n}.$$

In particular,

$$\ell_n \geq s(f)^n \quad (n \geq 0). \quad (11.1)$$

Cut I at the turning points and endpoints. The join of the first n inverse images of this finite partition has, up to boundary points, exactly the laps of f^n . Distinct interiors acquire different names at their first separating time, whereas within one lap no extra symbolic separation is created before time n . Refining an open cover around the finitely many partition boundaries therefore gives

$$h_{\text{top}}(f) = \lim_{n \rightarrow \infty} \frac{1}{n} \log \ell_n.$$

A homterval contributes no entropy and may be collapsed without changing this exponential rate. Hence $h_{\text{top}}(f) = \log s(f)$.

We now reconstruct the analytic kneading mechanism needed for the unimodal statement. The same coefficientwise identity below is valid with several turning points, but one turning point is enough for the determinant claim. Let c be the critical point, with laps I_1, I_2 and basis vectors e_1, e_2 . For a point with a chosen one-sided itinerary define the address series

$$\Theta(x; t) = \sum_{n \geq 0} \varepsilon_n(x) e_{A(f^n x)} t^n, \quad |t| < 1.$$

Its coefficients have norm one, so the series is analytic in the unit disk. The kneading increment is

$$\nu(t) = \Theta(c^+; t) - \Theta(c^-; t) = N_1(t)e_1 + N_2(t)e_2.$$

For an interval $J = [u, v]$, let $\gamma_n(J)$ be the number of points in the interior of J at which $f^n(x) - c$ changes sign, and put

$$\gamma(J, t) = \sum_{n \geq 0} \gamma_n(J) t^n.$$

Whenever this series converges, the following identity holds; as a formal power-series identity it holds without a convergence hypothesis:

$$\Theta(v^-; t) - \Theta(u^+; t) = \gamma(J, t) \nu(t). \quad (11.2)$$

Indeed, at level n the address function is constant on every lap of f^n . Moving from u to v , all jumps telescope. A jump created by a crossing of c is the appropriate orientation-shifted copy of the one-sided jump at c ; after multiplication by t^n its contribution is exactly the coefficient prescribed by $\nu(t)$. Summing the crossings proves (11.2) coefficient by coefficient.

The normalization relation can also be derived directly. Define the t -dependent linear functional λ_t by $\lambda_t(e_j) = 1 - \varepsilon_j t$. If x avoids the precritical set, then

$$\lambda_t(\Theta(x; t)) = \sum_{n \geq 0} (\varepsilon_n(x) t^n - \varepsilon_{n+1}(x) t^{n+1}) = 1.$$

Taking the one-sided difference at c gives

$$(1 - \varepsilon_1 t) N_1(t) + (1 - \varepsilon_2 t) N_2(t) = 0.$$

Hence the two normalized 1×1 minors coincide:

$$D_f(t) = \frac{N_2(t)}{1 - \varepsilon_1 t} = -\frac{N_1(t)}{1 - \varepsilon_2 t}. \quad (11.3)$$

The endpoint factors have no zero for $|t| < 1$. Dividing either component of (11.2) therefore gives, for example,

$$D_f(t) \gamma(J, t) = \frac{\Theta_2(v^-; t) - \Theta_2(u^+; t)}{1 - \varepsilon_1 t}. \quad (11.4)$$

Thus $D_f(t) \gamma(J, t)$ extends analytically throughout $|t| < 1$. This is the one-turning-point kneading resolvent; no determinant theorem is being imported here.

Let

$$\mathcal{L}(t) = \sum_{n \geq 0} \ell_n t^n.$$

A new turning point of f^{n+1} is either already a turning point of f^n or is an interior crossing of c by f^n . Consequently

$$\ell_{n+1} - \ell_n = \gamma_n(I), \quad (11.5)$$

and summing gives

$$\mathcal{L}(t) = \frac{1 + t \gamma(I, t)}{1 - t}. \quad (11.6)$$

Combining (11.4) and (11.6) shows that $D_f(t) \mathcal{L}(t)$ extends analytically to the whole unit disk.

Put $r = s^{-1} < 1$. By (11.1), for $0 < t < r$,

$$\mathcal{L}(t) \geq \sum_{n \geq 0} s^n t^n = \frac{1}{1 - st},$$

so $\mathcal{L}(t) \rightarrow +\infty$ as $t \uparrow r$. If $D_f(r) \neq 0$, the analyticity of $D_f \mathcal{L}$ would make $\mathcal{L}$ analytic, hence bounded, in a neighborhood of r . Therefore

$$D_f(r) = 0.$$

It remains to exclude an earlier positive zero. Suppose that $0 < t_0 < r$ and $D_f(t_0) = 0$. Since $\gamma_n(J) \leq \ell_n$, the radius of convergence of every $\gamma(J, t)$ is at least r , so identity (11.2)

is valid at t_0 . By (11.3), $N_1(t_0) = N_2(t_0) = 0$, hence $\nu(t_0) = 0$. Thus

$$\Theta(v^-; t_0) = \Theta(u^+; t_0)$$

for every interval $[u, v]$; the one-sided address series is therefore constant away from the countable precritical set, say equal to V . On the interior of lap I_j it satisfies the exact first-step recursion

$$\Theta(x; t_0) = e_j + t_0 \varepsilon_j \Theta(f(x); t_0).$$

Choose a nonprecritical x in each lap. Constancy yields

$$(1 - t_0 \varepsilon_j)V = e_j, \quad j = 1, 2,$$

which is impossible because e_1, e_2 are linearly independent. Hence no positive zero lies below r , and therefore

$$r = s(f)^{-1}, \quad h_{\text{top}}(f) = \log s(f) = -\log r.$$

We finish by reconstructing the constant-slope quotient for $s > 1$.

For an interval $J \subset I$ let $\ell_n(J)$ denote the lap number of the restriction $f^n|_J$, with $\ell_0(J) = 1$ for nondegenerate J , and define

$$\mathcal{L}(J, t) = \sum_{n \geq 0} \ell_n(J) t^n, \quad 0 < t < r.$$

The denominator $\mathcal{L}(I, t)$ tends to infinity as $t \uparrow r$. For $x \in [a, b]$ put

$$H_t(x) = \frac{\mathcal{L}([a, x], t)}{\mathcal{L}(I, t)},$$

with $H_t(a) = 0$. Each H_t is nondecreasing, takes values in $[0, 1]$, and $H_t(b) = 1$. Choose $t_k \uparrow r$. Helly compactness, equivalently weak compactness of the associated Stieltjes probability measures, gives a subsequence converging to a probability measure μ and distribution function

$$H(x) = \mu([a, x]).$$

Two elementary estimates contain all of the dynamics of this limit. First, if J_1, J_2 meet in one endpoint and their union is an interval, then for every n the difference between $\ell_n(J_1 \cup J_2)$ and $\ell_n(J_1) + \ell_n(J_2)$ has absolute value at most one. Therefore the failure of additivity of $\mathcal{L}(\cdot, t)$ is at most $(1 - t)^{-1}$. In particular, for $J = [x, y]$,

$$\left| \frac{\mathcal{L}(J, t)}{\mathcal{L}(I, t)} - (H_t(y) - H_t(x)) \right| \leq \frac{1}{(1 - t)\mathcal{L}(I, t)},$$

with the evident endpoint convention. The right-hand side tends to zero as $t \uparrow r$. Hence, whenever the endpoints of J are μ -nonatomic, $\mathcal{L}(J, t_k)/\mathcal{L}(I, t_k) \rightarrow \mu(J)$. Second, if J lies in a single lap of f , monotonicity gives the exact identity

$$\mathcal{L}(J, t) = 1 + t\mathcal{L}(f(J), t). \tag{11.7}$$

Passing to the weak limit first for intervals whose endpoints carry no μ -mass, and then by monotone approximation for arbitrary subintervals of a lap, yields

$$\mu(f(J)) = s\mu(J) \quad \text{whenever } J \text{ is contained in one lap.} \quad (11.8)$$

The measure μ has no atoms. If $\mu(\{x\}) = m > 0$, choose nested one-sided intervals $J_k \downarrow \{x\}$ lying in a lap closure. Their images shrink to $\{f(x)\}$, so (11.8) and continuity from above give

$$\mu(\{f(x)\}) = sm.$$

Repeating the argument along the orbit gives $\mu(\{f^n(x)\}) = s^n m \leq 1$ for all n , contradicting $s > 1$. Hence μ is non-atomic and H is continuous. Since $H(a) = 0$ and $H(b) = 1$, it is a continuous nondecreasing surjection. Set $\pi = H$.

If $\pi(x) = \pi(y)$ with $x < y$, then $\mu([x, y]) = 0$. Split $[x, y]$ at the finitely many turning points it contains. Formula (11.8) gives zero μ -mass to the image of each piece, and continuity of f makes their union the interval $f([x, y])$. Thus $\mu(f([x, y])) = 0$, so $\pi(f(x)) = \pi(f(y))$. Consequently

$$g(\pi(x)) := \pi(f(x))$$

is well defined and continuous on the quotient.

Finally, if $x < y$ lie in one lap, then

$$|g(\pi(y)) - g(\pi(x))| = \mu(f([x, y])) = s\mu([x, y]) = s|\pi(y) - \pi(x)|.$$

The sign is positive or negative according to the orientation of that lap. Therefore g is affine of slope $\pm s$ on every noncollapsed image lap, and $\pi \circ f = g \circ \pi$. This completes the proof. $\square$

Theorem 11.36 (Entropy monotonicity in the real quadratic family). **Statement scope.** For $0 \leq t \leq 2$ let

$$Q_t(x) = t - x^2, \quad R_t = \frac{1 + \sqrt{1 + 4t}}{2}, \quad I_t = [-R_t, R_t].$$

Then $Q_t(I_t) \subset I_t$, and

$$t \mapsto h_{\text{top}}(Q_t|_{I_t})$$

is nondecreasing on $[0, 2]$.

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Invariant bounded real core; Tsujii n-pole Ruelle invariant space including the closing pole; inverse-annulus spectral-radius estimate; determinant transversality; positive crossings imply monotone kneading; Milnor-Thurston entropy monotonicity.

The mathematical argument begins in the proof environment below.

Proof. First verify the compact real core. Since

$$R_t^2 = t + R_t,$$

we have

$$Q_t(\pm R_t) = -R_t, \quad Q_t(0) = t \leq R_t.$$

Because the quadratic is monotone on each side of zero, $Q_t([-R_t, R_t]) = [-R_t, t] \subset [-R_t, R_t]$. A real point outside this interval escapes to $-\infty$, so all bounded real dynamics and all real entropy are contained in this invariant interval.

Kneading theory reduces the theorem to one parameter statement: the kneading sequence of the critical value must be nondecreasing with t . The changes of a finite critical itinerary occur only when a critical iterate hits zero. Suppose the critical point has prime period n at such a parameter. Tsujii's key transversality estimate is

$$\frac{\partial_t Q_t^n(0)}{DQ_t^{n-1}(Q_t(0))} > 0.$$

The denominator supplies exactly the branch-orientation sign appearing in signed lexicographic order.

Here is the mechanism behind the estimate. Put $Q = Q_t$ and define the quadratic Ruelle operator

$$R\psi(z) = \sum_{Q(y)=z} \frac{\psi(y)}{DQ(y)^2}.$$

For the periodic critical orbit $x_k = Q^k(0)$, with $x_n = 0$, introduce all n simple poles

$$\chi_k(z) = \frac{1}{z - x_k}, \quad 1 \leq k \leq n.$$

The last one, $\chi_n(z) = 1/z$, closes the finite system. Direct partial fractions give

$$R\chi_k = \frac{\chi_{k+1}}{DQ(x_k)} - \frac{\chi_1}{DQ(x_k)} \quad (k < n), \quad R\chi_n = 0.$$

Thus their span is invariant. If Φ is the matrix of this restriction, then the companion-shaped determinant calculation gives

$$\det(I - \Phi) = \sum_{k=0}^{n-1} \frac{1}{DQ^k(Q(0))} = \frac{\partial_t Q_t^n(0)}{DQ_t^{n-1}(Q_t(0))}. \quad (11.1)$$

To determine its sign, choose a large disk D with $Q^{-1}(D) \Subset D$ and put $A = D \setminus Q^{-1}(D)$. The positive majorant

$$P\psi(z) = \sum_{Q(y)=z} \frac{|\psi(y)|}{|DQ(y)|^2}$$

satisfies, by change of variables,

$$\int_A P^k(|\psi|) dA = \int_{Q^{-k}(A)} |\psi| dA.$$

The inverse annuli $Q^{-k}(A)$ are pairwise disjoint shells inside D . Therefore

$$\sum_{k \geq 0} \int_A P^k(|\psi|) dA \leq \int_D |\psi| dA < \infty.$$

It follows that the powers of R tend to zero on this finite-dimensional pole space, so every eigenvalue of Φ has modulus less than one. Since Φ is real, $\det(I - \Phi) > 0$. Equation (11.1) is exactly Tsujii's positive transversality estimate.

Fix a truncation depth N . The first N critical symbols are locally constant except at the finitely many parameters where $Q_t^j(0) = 0$ for some $j \leq N$. At each such crossing the transversality inequality says that the signed kneading word changes in the increasing direction. Thus every finite truncation is nondecreasing in t , and passing to all truncation depths gives

$$t_1 < t_2 \implies K(Q_{t_1}) \preceq K(Q_{t_2}).$$

The Milnor-Thurston theorem says that the growth number, and therefore the topological entropy, is nondecreasing with kneading order. Hence

$$h_{\text{top}}(Q_{t_1}|_{I_{t_1}}) \leq h_{\text{top}}(Q_{t_2}|_{I_{t_2}}),$$

which proves the theorem. $\square$

11.3 Worked examples

Worked example: The quadratic family: Schwarzian, invariant interval, and first symbols

Consider

$$Q_t(x) = t - x^2, \quad 0 < t \leq 2.$$

Its unique critical point is $c = 0$, with

$$Q'_t(x) = -2x, \quad Q''_t(x) = -2, \quad Q'''_t(x) = 0.$$

At every $x \neq 0$,

$$SQ_t(x) = -\frac{3}{2} \left(\frac{1}{x} \right)^2 = -\frac{3}{2x^2} < 0.$$

Thus the quadratic family satisfies the Schwarzian hypothesis away from its critical point.

Let

$$R_t = \frac{1 + \sqrt{1 + 4t}}{2}.$$

The identity $R_t^2 = t + R_t$ gives $Q_t(\pm R_t) = -R_t$, and the maximum of Q_t on the interval is $Q_t(0) = t \leq R_t$. Hence $[-R_t, R_t]$ is forward invariant.

For example, at $t = 1$ the critical orbit begins

$$0 \mapsto 1 \mapsto 0,$$

so the critical value returns to the turning point after one further step. With the convention of this chapter the kneading sequence is the finite word RC (or its standard one-sided periodic continuation, depending on the chosen endpoint convention). At a nearby parameter not satisfying a critical relation, the initial word is stable until some critical iterate crosses zero. This is exactly the event controlled by the transversality argument in the entropy-monotonicity theorem.

Worked example: A full tent map and exact lap growth

Let

$$T(x) = \begin{cases} 2x, & 0 \leq x \leq \frac{1}{2}, \\ 2 - 2x, & \frac{1}{2} \leq x \leq 1. \end{cases}$$

Every inverse image of the turning point divides each existing monotonicity interval into two. Therefore

$$\ell(T^n) = 2^n.$$

The growth number is $s(T) = 2$, and the lap-growth theorem gives

$$h_{\text{top}}(T) = \log 2.$$

This map is already the constant-slope model that kneading theory constructs for a general unimodal map of growth number two. In a general map, the semi-conjugacy may collapse intervals; in the full tent map no collapse is needed.

Worked example: Why a flat critical point lies outside the no-wandering theorem

The no-wandering theorem assumes finite-order, nonflat critical behavior. Consider a smooth germ near zero of the form

$$f(x) = f(0) - e^{-1/x^2} \quad (x \neq 0),$$

completed smoothly at $x = 0$. Every derivative of the correction term vanishes at zero, so no finite exponent ℓ and smooth local coordinate can put this germ into a power-law normal form $f(0) - |\phi(x)|^\ell$.

This example does not prove that a wandering interval exists. It shows only that the real-bounds proof used in this chapter cannot be invoked: the uniform cross-ratio control at a finite-order critical passage has lost its hypothesis. Distinguishing “the theorem does not apply” from “the conclusion is false” is essential in one-dimensional dynamics.

11.4 What this chapter established

Chapter 11 separated three basic engines of interval dynamics. Negative Schwarzian derivative gives a minimum principle for derivatives of return maps, forcing an attracting immediate basin to contain a critical point or an endpoint. Nonflat critical points give a different kind of control: real bounds turn nested nice intervals into uniform Koebe space, and that uniform space makes a wandering interval geometrically impossible. Signed itineraries provide a third, purely combinatorial engine: parity of branch reversals converts ordinary order on the interval into signed lexicographic order on symbolic addresses.

Milnor–Thurston theory turns that symbolic data into quantitative dynamics. Kneading data determine lap growth, lap growth gives topological entropy, and the first positive zero of the unimodal kneading determinant records the inverse growth number. The same formalism produces a monotone quotient onto a constant-slope model. Tsujii’s positive transversality theorem then shows that the critical kneading sequence of $Q_t(x) = t - x^2$ moves only in the increasing direction with the parameter, giving monotonic entropy.

Looking ahead

Chapter 12 develops *real a priori bounds and renormalization*. The restrictive interval introduced here becomes the basic return domain. Instead of merely knowing that a first return is again unimodal, we will control the relative sizes of its domains uniformly along renormalization towers. Those bounds are the real-variable bridge back toward the complex bounds, rigidity, and universality machinery of later volumes.

Real A Priori Bounds

Why this chapter matters

Chapter 11 introduced restrictive intervals and showed why real distortion matters, but it deliberately did not quantify the geometry of an infinite renormalization tower. Chapter 12 supplies that quantitative layer. The central question is simple to state: as one passes to deeper first-return maps, can a child interval collapse to an arbitrarily tiny fraction of its parent, or can a gap disappear? Real a priori bounds say that bounded combinatorics prevent such degeneration. The proof has an important logical order. Cross ratios measure four-point geometry. A lower bound for cross-ratio distortion gives a real Koebe principle. Koebe distortion plus bounded return combinatorics produces compact normalized return maps. Compactness rules out collapsing children and gaps. Only after those real bounds are proved do we speak about bounded geometry or quasiconformal conjugacy.

Learning goals

By the end of this chapter you should be able to:

1. compute the cross ratio used in this book and distinguish the cross ratio itself from its distortion under a map;
2. explain why negative Schwarzian gives exact cross-ratio expansion and why a modern C^3 theorem can replace that global hypothesis at deep scales;
3. use image-side scaled space and a cross-ratio lower bound to obtain Koebe derivative distortion;
4. state the real a priori bounds with their precise bounded-type and nonflatness hypotheses;
5. derive bounded child/gap geometry from real bounds without using a complex modulus estimate;
6. explain why bounded real geometry gives a quasiconformal conjugacy on the postcritical Cantor set, but not yet complex rigidity.

Conceptual roadmap

The proof engine is

four-point cross ratio $\implies$ deep cross-ratio lower bound,
 $\implies$ real Koebe distortion,
 $\implies$ compact normalized returns,
 $\implies$ real a priori bounds,
 $\implies$ bounded children and gaps,
 $\implies$ real quasimetric coding.

The first arrow has two versions. If a branch has negative Schwarzian, cross-ratio expansion is exact. For a general C^3 nonflat unimodal map, Kozlovski's small-scale theorem gives a lower bound arbitrarily close to one on sufficiently deep first-entry branches. The rest of the argument needs only that lower bound, not a global Schwarzian sign.

Common pitfall

Do not reverse the proof order. "Real bounds imply Koebe" is not the engine used here; Koebe is one of the inputs from which the real bounds are built. Also do not identify real a priori bounds with complex a priori bounds. Chapter 12 controls lengths and real distortion. Chapter 13 must still construct polynomial-like extensions and prove a positive complex modulus bound.

12.1 Definitions

Definition 12.1 (real a priori bounds and beau constants). For a bounded-type renormalization tower, *real a priori bounds* are depth-independent lower and upper bounds for normalized child sizes together with uniform distortion bounds for return branches. A bound is *beau* if, after a map-dependent entry level, its numerical constant depends only on the prescribed combinatorial bound and critical order.

Definition 12.2 (cross ratio and cross-ratio distortion). If $J \in T$ and L, R are the components of $T \setminus J$, set

$$b(T, J) = \frac{|T| |J|}{|L| |R|}.$$

For a monotone map $g : T \rightarrow g(T)$ define

$$B(g; T, J) = \frac{b(g(T), g(J))}{b(T, J)}.$$

This convention is fixed throughout Chapter 12.

Definition 12.3 (scaled neighborhood and real Koebe distortion). An interval T contains a τ -scaled neighborhood of $J \in T$ if each component of $T \setminus J$ has length at least

$\tau|J|$. A real Koebe bound controls $|g'(x)|/|g'(y)|$ on J in terms only of a lower cross-ratio-distortion bound and image-side scaled space.

Definition 12.4 (bounded real geometry). A nested real partition has *bounded geometry* if its branching is uniformly bounded and the lengths of every child and every adjacent gap are uniformly comparable with the parent length. Equivalently, deep neighboring atoms and gaps are uniformly commensurable.

Definition 12.5 (normalized d -unimodal class). A normalized d -unimodal map in Chapter 12 is a C^3 map $f : [-1, 1] \rightarrow [-1, 1]$ with unique critical point 0, critical value $f(0) = 1$, symmetry $f(-x) = f(x)$, and finite even critical order $d \geq 2$. Equivalently, in a local C^3 coordinate the critical germ is a nonflat power germ of order d . This is the standard affine/symmetric renormalization normalization used for the real-bounds theorem; it does not assert a global negative-Schwarzian condition.

Definition 12.6 (restrictive tower and bounded type). A restrictive interval about the critical point is always taken in the canonical maximal sense: among intervals of its primitive period containing 0 whose cycle interiors are pairwise disjoint and whose return maps the interval into itself, choose the maximal one by inclusion. Starting with the outer normalized interval, iterate this construction to obtain nested canonical restrictive intervals $I_n \ni 0$. Let q_n be the primitive first-return period and put $\Delta_{i,n} = f^i(I_n)$ for $0 \leq i < q_n$. The relative period is $a_n = q_{n+1}/q_n$. The tower has *bounded type* N when $a_n \leq N$ for all n .

Definition 12.7 (dynamical partition, atom, and gap). The level- n dynamical partition consists of the cycle intervals $\Delta_{i,n}$ together with the complementary components needed inside each parent to record the next return level. A surviving interval is an *atom*; a complementary component between adjacent atoms is a *gap*.

Definition 12.8 (renormalization mesh). For the level- n cycle $\mathcal{C}_n$, its mesh is

$$\text{mesh}(\mathcal{C}_n) = \max_{0 \leq i < q_n} |\Delta_{i,n}|.$$

Shrinking mesh means this number tends to zero with n .

Definition 12.9 (quasisymmetric homeomorphism). An increasing homeomorphism h of real intervals is quasisymmetric if there is $M \geq 1$ such that whenever $x - t, x, x + t$ lie in the domain,

$$M^{-1} \leq \frac{h(x+t) - h(x)}{h(x) - h(x-t)} \leq M.$$

An equivalent formulation allows a distortion function depending on the ratio of the two source distances.

Definition 12.10 (postcritical Cantor set of an infinite renormalization tower). For the tower considered here, $P(f)$ denotes the compact invariant Cantor set obtained as the nested intersection of the deep return partitions; equivalently it is the closure of the recurrent critical orbit in the solenoidal infinitely renormalizable regime.

12.2 Main results

Theorem 12.1 (Cross-ratio inequality and deep C3 replacement). **Statement scope.** *If $g : T \rightarrow g(T)$ is a C^3 diffeomorphism with $Sg < 0$, then $B(g; T, J) > 1$ for every $J \in T$. More generally, let $f : X \rightarrow X$ be a C^3 unimodal map with a nonflat critical point c whose forward orbit does not converge to a periodic attractor. For every $0 < \kappa < 1$ there is a nice interval $V \ni c$ such that, whenever $f^n|_M$ is monotone and every $f^i(M)$, $0 \leq i \leq n$, lies in a domain of the first-entry map to V , one has*

$$B(f^n; M, J) > \kappa$$

for every $J \in M$.

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Projective cross-ratio expansion under negative Schwarzian; two-cross-ratio smooth-loss estimate; no-wandering pullback mesh; good nice-interval extensions; derivative lower bound; first-entry Schwarzian negativity; complete-block/terminal-tail decomposition.

The mathematical argument begins in the proof environment below.

The proof of Theorem 12.1 is long enough that we first isolate its mechanisms as ordinary lemmas. This also makes the dependency order visible: no result proved later in Chapter 12 is used below.

Lemma 12.2 (Projective invariance and Schwarzian sign). *The cross ratio $b(T, J)$ of Definition 12.2 is invariant under real Mobius transformations. If $g : T \rightarrow g(T)$ is a C^3 diffeomorphism with $Sg < 0$ on T , then*

$$B(g; T, J) > 1$$

for every nondegenerate $J \in T$.

Proof. Write the four endpoints as $a < b < c < d$. Then

$$b([a, d], [b, c]) = \frac{(d-a)(c-b)}{(b-a)(d-c)},$$

and the usual cancellation in a fractional linear map proves projective invariance. For the sign statement, pre- and post-compose with Mobius maps so that three of the four relevant points have fixed normalized values. In the orientation-preserving case put $u = (g')^{-1/2}$. Direct differentiation gives

$$u'' = -\frac{1}{2}(Sg)u > 0.$$

Thus u is strictly convex. Integrating $g' = u^{-2}$ over the four adjacent subintervals and comparing the secant integrals gives the strict four-point cross-ratio inequality. Orientation reversal is reduced to this case by $x \mapsto -x$. Undoing the projective normalizations gives the claim. $\square$

Lemma 12.3 (Multiplicativity of cross-ratio distortion). *If g and h are monotone on the relevant intervals, then*

$$B(h \circ g; T, J) = B(h; g(T), g(J)) B(g; T, J).$$

Proof. Expand both factors from Definition 12.2. The intermediate quantity $b(g(T), g(J))$ cancels. $\square$

Auxiliary cross ratio. For the small-scale estimates only, write L, R for the components of $T \setminus J$ and set

$$a(T, J) = \frac{|T| |J|}{|L \cup J| |J \cup R|}, \quad A(g; T, J) = \frac{a(g(T), g(J))}{a(T, J)}.$$

Exactly the same cancellation as in Lemma 12.3 makes A multiplicative.

Lemma 12.4 (Two-cross-ratio smooth-loss estimate). *Let $f : X \rightarrow X$ be C^3 with finitely many nonflat critical points. There is $C_{\text{cr}} > 0$ such that whenever $f^m|_M$ is a diffeomorphism and $J \Subset M$, with $M^\pm$ the components of $M \setminus J$, then*

$$A(f^m; M, J) \geq \exp \left(-C_{\text{cr}} \sum_{i=0}^{m-1} |f^i(M^-)| |f^i(M^+)| \right),$$

$$B(f^m; M, J) \geq \exp \left(-C_{\text{cr}} \sum_{i=0}^{m-1} |f^i(M)|^2 \right).$$

Proof. Choose pairwise disjoint critical neighborhoods on which the nonflat normal form is

$$f(x) = f(c) \pm |\phi(x)|^\alpha, \quad \alpha \geq 2,$$

with ϕ a C^3 local diffeomorphism. On a one-sided branch sufficiently near c , the singular term in the Schwarzian of the power map dominates the bounded Schwarzian of ϕ ; hence the Schwarzian is negative and neither A nor B loses cross ratio there by Lemma 12.2 and the analogous projective calculation for A .

On the compact complement of those neighborhoods, f' is bounded away from zero and the C^3 norm is bounded. Write each cross ratio as a product of four secant slopes. Taylor expansion of the logarithm of a secant slope about one endpoint shows that the constant and first-order terms cancel in the four-point combination. Consequently, on every sufficiently small monotone branch,

$$\log A(f; M, J) \geq -C|M^-||M^+|, \quad \log B(f; M, J) \geq -C|M|^2.$$

After enlarging C , compactness gives the same estimates on the finitely many transition regions between the critical charts and their complement. Apply the one-step inequalities to $f^i(M)$ and use multiplicativity of A and B . Summing logarithms proves the stated bounds. $\square$

From now on, let $f : X \rightarrow X$ be a C^3 unimodal map with nonflat critical point c whose forward orbit does not converge to a periodic attractor. Recall from Definition 11.11

that a nice interval has no forward iterate of its boundary entering its interior. Write R_T for the first-entry map to a nice interval $T \ni c$ and J for the central entry domain when it exists.

Lemma 12.5 (Vanishing monotone pullback mesh). *There is a function $\tau_1 : (0, \infty) \rightarrow (0, \infty)$ with $\tau_1(t) \rightarrow 0$ as $t \rightarrow 0$ such that, whenever $f^n|_V$ is a diffeomorphism and $f^n(V)$ is disjoint from the immediate basins of periodic attractors,*

$$\max_{0 \leq i \leq n} |f^i(V)| \leq \tau_1(|f^n(V)|).$$

Proof. Suppose not. Then there are $\varepsilon > 0$, intervals V_k , and integers n_k for which $f^{n_k}|_{V_k}$ is a diffeomorphism, $|f^{n_k}(V_k)| \rightarrow 0$, and some image of V_k has length at least ε . Replace V_k by the *last* image in its monotone chain having length at least ε . After shortening n_k we may assume

$$|V_k| \geq \varepsilon, \quad |f^j(V_k)| < \varepsilon \quad (1 \leq j \leq n_k), \quad |f^{n_k}(V_k)| \rightarrow 0.$$

First $n_k \rightarrow \infty$. Otherwise a subsequence has $n_k = n$ fixed; a nondegenerate Hausdorff limit V_∞ would then be an interval on which f^n is constant, impossible for a nonconstant interval map with finitely many critical points.

Pass to a Hausdorff limit V_∞ . For every fixed m , the maps f^j are monotone on V_k for $0 \leq j \leq m$ when k is large, and hence on the interior of V_∞ . Thus arbitrarily long forward orbit segments of V_∞ are monotone. If its forward images are pairwise disjoint, V_∞ is wandering, contrary to Theorem 11.2. Otherwise let $0 \leq r < s$ be the first overlap. Monotonicity and minimality of the overlap imply that f^{s-r} maps the convex hull of $f^r(V_\infty) \cup f^s(V_\infty)$ monotonically into itself. Its nested forward images converge to a periodic interval whose interior lies in an attracting periodic basin. Hence sufficiently late terminal images of V_k lie in that immediate basin, contrary to the hypothesis. This proves the lemma. $\square$

Lemma 12.6 (Central-domain branch extension). *Let T be nice, let J be its central first-entry domain, and let $R_J|_V = f^n|_V : V \rightarrow J$ be a first-entry branch with $V \cap J = \emptyset$. Then the monotone branch extends to an interval whose image contains T .*

Proof. Let $I \supset V$ be the maximal interval on which f^n is monotone. At each endpoint of I , maximality means that an earlier iterate reaches the critical point. While the image of V has not yet entered J , the corresponding endpoint image crosses an endpoint of J . Since J is the central domain of the nice interval T , forward images of ∂J do not enter $\text{int } T$. The two endpoint images of the maximal branch therefore lie on opposite sides of T , so $f^n(I) \supset T$. $\square$

Lemma 12.7 (Smallest-interval scaled space). *Assume that c is recurrent. There exist $0 < \tau_2 < 1$ and $\tau_3 > 0$ such that, for every sufficiently small nice $T \ni c$ with central domain J and $|J|/|T| > \tau_2$, there is an interval W containing a τ_3 -scaled neighborhood of T with the following properties. If the return of J to T is high, every noncentral first-entry branch to T extends diffeomorphically over W . If the return is low and f is not renormalizable at T , the branch whose domain contains the critical value $f(c)$ extends over W .*

Proof. Write the central return as $f^k : J \rightarrow T$. The intervals $f(J), \dots, f^k(J)$ are pairwise disjoint. Choose a shortest member $U = f^{k_1}(J)$ and let M be the interval obtained by adjoining one copy of $|U|$ on each side of U . Maximal monotonicity and minimality of $|U|$ show that $f^{k_1-1} : f(J) \rightarrow U$ extends diffeomorphically over M : otherwise an endpoint of the maximal monotonicity interval would encounter c first and force another orbit interval, shorter than U , between U and an endpoint of M .

Pull M back along $f(J), \dots, U$ and denote the pullbacks by M_i . Their intersection multiplicity is at most four. For if one point belonged to five pullbacks, choose three ordered indices $i_1 < i_2 < i_3$ such that the middle orbit interval lies in M_{i_1} while the third does not. If $k_1 - i_1 \leq k - i_2$, iterating the inclusion forward puts a later orbit interval inside M ; if $k_1 - i_1 > k - i_2$, first iterate to T and then to M . Either case contradicts the defining minimality of U and the fact that M contains no complete orbit interval other than U .

Lemma 12.4, this multiplicity bound, and Lemma 12.5 now give a scale-independent lower bound for the A -distortion of the pullback when T is small. Since M is a 1-scaled neighborhood of U , its pullback to $f(J)$ contains a definite scaled neighborhood of $f(J)$. Pulling once more through the nonflat power chart at c gives an interval around J containing a fixed scaled neighborhood of J . Choose τ_2 sufficiently close to one. Then $|J|/|T| > \tau_2$ forces this interval to contain a τ_3 -scaled neighborhood W of T .

For a high return, take a maximal pullback of W along a noncentral entry branch. Failure of extension would place, before entry, a complete central-orbit interval inside one of the M_i , contradicting the shortest choice of U . For the distinguished low-return branch through $f(c)$ the same pullback argument has one extra alternative: a pullback may map strictly inside itself. That alternative is a restrictive interval. Under the stated nonrenormalizability assumption it is excluded. $\square$

Lemma 12.8 (Gap for noncentral low returns). *If c is recurrent, there are $\tau_5 < 1$ and $\delta_0 > 0$ such that, for every nice $T \ni c$ with $|T| < \delta_0$, a noncentral low return with central domain J satisfies*

$$\frac{|J|}{|T|} < \tau_5.$$

Proof. Only the case $|J|/|T| > \tau_2$ needs consideration. Apply Lemma 12.7 and let W be the resulting scaled neighborhood of T . Pull the side gap of $T \setminus f^k(J)$ opposite the low-return image back along the central orbit, and pull the adjacent component of $W \setminus T$ back in the same way. The relevant pullback orbits are disjoint. Lemma 12.4 therefore makes the associated A -distortion tend to one as $|T| \rightarrow 0$.

If a sequence of noncentral low returns had $|J|/|T| \rightarrow 1$, the return image $f^k(J)$ lies entirely on one side of c because the return is low. The two endpoints of the central entry domain have the same boundary image in ∂T , while the critical value of the return lies in $T \setminus J$ because the return is noncentral. Hence $f^k(J)$ is trapped between that boundary point and a point of the shrinking component $T \setminus J$; therefore $|f^k(J)|/|T| \rightarrow 0$. The A -comparison then forces the full critical pullback U of the extended branch to satisfy $|J|/|U| \rightarrow 0$. For sufficiently deep members of the sequence, U contains W compactly and

$$f^k(U) \subset U, \quad c \notin f^k(U).$$

The return map is unimodal on U , but its image lies on one monotone side of its turning point. Its forward images are therefore nested on that side and converge to a periodic attracting orbit. The critical orbit is trapped by the same return, contradicting recurrence under the standing assumption that the critical orbit is not attracted to a periodic orbit. Thus the central ratio is uniformly bounded away from one. $\square$

Lemma 12.9 (Good shrinking nice neighborhoods). *There exist $\tau_6 > 0$ and nice intervals $T_r \ni c$, $|T_r| \rightarrow 0$, such that every first-entry branch $R_{T_r} : V \rightarrow T_r$ with $V \cap T_r = \emptyset$ extends diffeomorphically to an interval whose image contains a τ_6 -scaled neighborhood of T_r .*

Proof. If c is not recurrent, choose a neighborhood W_0 of c containing no other point of its forward orbit. Arbitrarily small nice intervals $T \subseteq W_0$ then have the required extension to W_0 by the maximal monotonicity argument of Lemma 12.6.

Assume that c is recurrent. If f is infinitely renormalizable, take sufficiently deep restrictive intervals; their first returns are high, so Lemma 12.7 applies. If f is only finitely renormalizable, begin with a small nice interval and repeatedly take its central domain. These central domains shrink to c ; otherwise their intersection contains a nondegenerate recurrent restrictive interval, giving one more renormalization. Along infinitely many high returns, use the parent when the child occupies more than τ_2 of it and use the child otherwise. Lemma 12.7 gives the extension in the first case, while Lemma 12.6 gives fixed space in the second. If eventually all returns are low, recurrence supplies infinitely many noncentral low returns. Lemma 12.8 gives a uniform parent/child gap, and Lemma 12.6 at the child supplies the extension. The minimum of the constants occurring in these cases is the required τ_6 . $\square$

Lemma 12.10 (Derivative lower bound without Koebe). *There is $C_6 > 0$ with the following property. Suppose $J \subset T$, $f^n|_T$ is monotone, $f^n(T)$ contains a δ -scaled neighborhood of $f^n(J)$, and the intervals $J, f(J), \dots, f^{n-1}(J)$ are pairwise disjoint. Then, for every $x \in J$,*

$$|Df^n(x)| \geq C_6 \frac{\delta}{1 + \delta} \frac{|f^n(J)|}{|J|}.$$

Proof. Let x split J into J^- and J^+ . One of the two secant slopes on $J^\pm$ is at least the average slope $|f^n(J)|/|J|$; assume it is the J^+ slope. Let J' shrink to x and take $T' = T^- \cup J$, where T^- is the component of $T \setminus J$ opposite J^+ . Lemma 12.4 applied to $A(f^n; T', J')$ gives a lower bound independent of n , because

$$\max_{0 \leq i < n} |f^i(T)| \sum_{i=0}^{n-1} |f^i(J)| \leq |X|^2;$$

the second factor is bounded by $|X|$ because the orbit of J is disjoint. Letting J' shrink to x turns the corresponding secant quotient into $|Df^n(x)|$. The δ -scaled space around $f^n(J)$ contributes the factor $\delta/(1 + \delta)$, and the selected side contributes at least the average slope. Absorb the uniform A -distortion into C_6 . The case in which J^- has the larger secant slope is symmetric. $\square$

Lemma 12.11 (Negative Schwarzian for deep critical-value entry). *For all sufficiently large r , every monotone component of every branch of the first-entry map to $f(T_r)$ has negative Schwarzian wherever its derivative is nonzero.*

Proof. Fix $T = T_r$ from Lemma 12.9, so small that it is disjoint from every immediate attracting basin. Let $f^n : V \rightarrow T$ be a noncentral first-entry branch. Its range extends to a τ_6 -scaled neighborhood of T . Applying Lemma 12.10 to each tail of the branch gives

$$|Df^{n-i}(y)| \geq C_7 \frac{|T|}{|f^i(V)|}, \quad y \in f^i(V), \quad 0 \leq i < n,$$

where $C_7 > 0$ depends only on f and τ_6 .

In a critical coordinate $f(x) = f(c) \pm |\phi(x)|^\alpha$ with $\alpha \geq 2$. Since

$$S(x^\alpha) = \frac{1 - \alpha^2}{2x^2},$$

the Schwarzian composition formula gives constants $C_9, C_{10} > 0$ such that, after shrinking T ,

$$Sf(x) \leq -\frac{C_9}{(x-c)^2} \quad (x \in T \setminus \{c\}), \quad Sf(x) \leq C_{10} \quad (x \neq c).$$

For $x \in V$ away from the unique preimage of c , the cocycle identity gives

$$S(f^{n+1})(x) = |Df^n(x)|^2 \left(Sf(f^n(x)) + \sum_{i=0}^{n-1} \frac{Sf(f^i(x))}{|Df^{n-i}(f^i(x))|^2} \right).$$

The bracket is at most

$$\frac{1}{|T|^2} \left(-C_9 + \frac{C_{10}}{C_7^2} \sum_{i=0}^{n-1} |f^i(V)|^2 \right).$$

The first-entry orbit of V is pairwise disjoint: an overlap before the first hit of T would, by monotonicity, create an earlier entry (equivalently, a return of a boundary pullback across the nice target). Therefore Lemma 12.5 gives

$$\sum_{i=0}^{n-1} |f^i(V)|^2 \leq |X| \max_{0 \leq i < n} |f^i(V)| \leq |X| \tau_1(|T|) \rightarrow 0.$$

Hence the bracket is negative for sufficiently small T_r . Appending the last critical-value step proves the claim on each monotone component of the complete first-entry branch to $f(T_r)$. Successive complete entries remain negative by the Schwarzian composition formula. $\square$

Lemma 12.12 (Deep first-entry cross-ratio replacement). *For every $0 < \kappa < 1$ there is a sufficiently small good nice interval $T \ni c$ such that, if $f^n|_M$ is monotone and every*

$f^i(M)$, $0 \leq i \leq n$, lies in a domain of the first-entry map R_T , then

$$B(f^n; M, I) > \kappa$$

for every $I \in M$.

Proof. Choose $T = T_r$ so small that Lemma 12.11 holds. Scan the orbit of M from time zero and cut it each time a complete first-entry branch to $f(T)$ is finished. This writes the monotone branch $f^n|_M$ as maximal complete first-entry blocks to $f(T)$ followed by one terminal tail f^k containing no complete such block. Each complete block has negative Schwarzian, so Lemmas 12.2 and 12.3 show that complete blocks do not decrease b . Only the terminal tail can lose cross ratio. Its orbit intervals lie in distinct first-entry domains of R_T and are pairwise disjoint. Lemma 12.4 therefore gives

$$B(f^k; f^{n-k}(M), f^{n-k}(I)) \geq \exp\left(-C_{\text{cr}} \sum_{j=n-k}^{n-1} |f^j(M)|^2\right).$$

Every noncentral entry domain U maps diffeomorphically to T , so Lemma 12.5 gives $|U| \leq \tau_1(|T|)$; the central domain is contained in T . Thus

$$\eta(T) = \max\{|T|, \tau_1(|T|)\} \rightarrow 0$$

and the disjointness of the terminal tail yields

$$\sum_{j=n-k}^{n-1} |f^j(M)|^2 \leq |X| \eta(T).$$

Choose T so deep that $\exp(-C_{\text{cr}}|X|\eta(T)) > \kappa$. Multiplying this terminal estimate by the complete-block factors, all at least one, proves the lemma. $\square$

Proof of Theorem 12.1. The negative-Schwarzian assertion is Lemma 12.2. Under the hypotheses of the second assertion, a periodic critical point would itself be a super-attracting periodic orbit, so the critical point is nonperiodic. The hypotheses of the preceding lemmas therefore apply, and Lemma 12.12 gives the required nice interval for every prescribed $0 < \kappa < 1$.

The only earlier theorem used in this reconstruction is the no-wandering interval theorem, Theorem 11.2, through Lemma 12.5. In particular, neither the real Koebe principle proved next nor the real a priori bounds proved later enter the argument. $\square$

Lemma 12.13 (Real Koebe principle). **Statement scope.** Let $g : T \rightarrow g(T)$ be a C^1 diffeomorphism and assume $B(g; T_*, J_*) \geq \kappa > 0$ for every $J_* \in T_* \subset T$. If $g(T)$ contains a τ -scaled neighborhood of $g(J)$, then

$$\frac{|g'(x)|}{|g'(y)|} \leq K(\kappa, \tau) \quad (x, y \in J),$$

where K depends only on κ and τ .

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Point-limit cross-ratio inequality; pull back image-side scaled space; secant-to-derivative comparison; uniform derivative-ratio bound.

The mathematical argument begins in the proof environment below.

Proof. Let $J \in T$ and let $g : T \rightarrow g(T)$ be a C^1 diffeomorphism. Assume that there is $\kappa > 0$ such that

$$B(g; T_*, J_*) \geq \kappa \quad (1)$$

for every pair $J_* \in T_* \subset T$. Assume also that $g(T)$ contains a τ -scaled neighborhood of $g(J)$: each component of $g(T) \setminus g(J)$ has length at least $\tau|g(J)|$.

Indeed, write $T = [a, d]$ and take $J_\varepsilon = [x, x + \varepsilon]$. Dividing the cross-ratio inequality by ε and letting $\varepsilon \downarrow 0$ yields

$$\frac{|g(T)| |g'(x)|}{|g(x) - g(a)| |g(d) - g(x)|} \geq \kappa \frac{|T|}{|x - a| |d - x|}. \quad (2)$$

The same formula holds from the left and for orientation-reversing maps after absolute values are inserted.

We give the elementary reduction. Split J at x and apply (2) not only to T but also to the smallest subintervals of T obtained by adjoining to each side of J one of the two image margins. The scaled-neighborhood hypothesis says that the two image margin ratios are at least τ . Condition (1) pulls those two lower bounds back with a loss depending only on κ . Hence both domain margins of J inside the corresponding subintervals are bounded below by a number $\eta = \eta(\kappa, \tau) > 0$ times $|J|$. Substituting these bounds into (2), and applying the same argument to the subintervals $[\inf J, x]$ and $[x, \sup J]$, bounds $|g'(x)|$ above and below by constant multiples of the secant slope $|g(J)|/|J|$. No regularity beyond C^1 is used at this step; all dynamical information has already been compressed into (1).

The conclusion follows immediately by dividing the upper secant comparison at x by the lower one at y . Setting $K(\kappa, \tau) = C_0(\kappa, \tau)^2$ proves the real Koebe principle.

If g has negative Schwarzian, the preceding cross-ratio theorem permits $\kappa = 1$. For deep first-entry branches of a general C^3 nonflat infinitely renormalizable unimodal map, that theorem supplies any fixed $\kappa < 1$. This is precisely the form needed in the real-bounds proof. $\square$

Theorem 12.14 (Real a priori bounds). **Statement scope.** Let $f : [-1, 1] \rightarrow [-1, 1]$ be a normalized C^3 d -unimodal map, $d \geq 2$ even, which is infinitely renormalizable with relative renormalization periods $a_n \leq N$. There exist map-dependent constants $K_f \geq 1$ and $0 < \alpha_f < \beta_f < 1$ such that for every level n :

1. if $\Delta \in \mathcal{C}_{n+1}$ lies in $\Delta^* \in \mathcal{C}_n$, then $\alpha_f |\Delta^*| \leq |\Delta| \leq \beta_f |\Delta^*|$;
2. for $1 \leq i < j \leq q_n - 1$ and $x \in \Delta_{i,n}$,

$$K_f^{-1} \frac{|\Delta_{j,n}|}{|\Delta_{i,n}|} \leq |Df^{j-i}(x)| \leq K_f \frac{|\Delta_{j,n}|}{|\Delta_{i,n}|};$$

3. the affine-normalized renormalizations satisfy $\|R^n f\|_{C^1} \leq K_f$.

Moreover, after a map-dependent entry level, these constants can be replaced by beau constants $K(N, d)$ and $0 < \alpha(N, d) < \beta(N, d) < 1$ independent of f .

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Shrinking cycle mesh; macroscopic cross-ratio pullback; maximal monotonicity extensions with multiplicity three; shortest-atom collar; bounded-distortion power factorization; compact central-child nondegeneracy; transport to all children; derivative/ C^1 consequences; beau uniformization.

The mathematical argument begins in the proof environment below.

The proof is organized as a sequence of ordinary lemmas. The point of this organization is to separate three logically different assertions: existence of pullback space, compactness of normalized return maps, and the metric consequences. In particular, compactness below is only in the uniform C^0 topology; the C^1 estimate is proved directly from bounded distortion.

Lemma 12.15 (Shrinking mesh of the restrictive cycles). *For an infinitely renormalizable C^3 nonflat unimodal map,*

$$\text{mesh}(\mathcal{C}_n) = \max_{0 \leq i < q_n} |\Delta_{i,n}| \rightarrow 0.$$

Proof. We first prove $|I_n| \rightarrow 0$. Since the restrictive intervals are nested, their intersection I_∞ is an interval containing the critical point. If it were nondegenerate, then for every fixed m and every n with $q_n > m$ the interiors of

$$I_n, f(I_n), \dots, f^m(I_n)$$

are pairwise disjoint. Passing to the nested intersection shows the same for I_∞ for every m . Thus I_∞ is wandering unless a forward image falls into an attracting periodic basin; that alternative is incompatible with infinite renormalizability of the critical orbit. Theorem 11.2 therefore forces $I_\infty = \{0\}$.

Assume the cycle mesh does not tend to zero. Then for some $\varepsilon > 0$ there are $n_k \rightarrow \infty$ and i_k with $|\Delta_{i_k, n_k}| \geq \varepsilon$. Put $r_k = q_{n_k} - i_k$. We claim $r_k \rightarrow \infty$. If a subsequence had bounded r_k , then after fixing r the map f^r would send nondegenerate intervals Δ_{i_k, n_k} monotonically onto I_{n_k} ; a Hausdorff limit would give a nondegenerate interval on which f^r is constant, impossible.

Let L be a nondegenerate Hausdorff limit of these cycle atoms. For every fixed m , eventually $m < r_k$, so

$$\Delta_{i_k, n_k}, f(\Delta_{i_k, n_k}), \dots, f^m(\Delta_{i_k, n_k})$$

contains no critical interval and consists of pairwise interior-disjoint cycle atoms. Hence all fixed-length forward iterates are monotone on L . If all forward images of L are disjoint, L is wandering. If two first overlap, the same monotone trapping-interval argument as in Lemma 12.5 produces an attracting periodic basin meeting sufficiently late cycle atoms. The critical restrictive tower would then be trapped by a finite-period

attractor, contradicting $q_n \rightarrow \infty$. Both alternatives are impossible, so the mesh tends to zero. $\square$

Lemma 12.16 (Macroscopic pullback of scaled space). *Let $J \Subset T$ and let $g : T \rightarrow g(T)$ be a diffeomorphism. Assume*

$$B(g; T_*, J_*) \geq \kappa > 0 \quad (J_* \Subset T_* \subset T).$$

If $g(T)$ contains a τ -scaled neighborhood of $g(J)$, then T contains a $\tau' = \tau'(\kappa, \tau) > 0$ scaled neighborhood of J . Consequently, Lemma 12.13 gives bounded distortion on J with a constant depending only on κ and τ .

Proof. Write the side lengths of $T \setminus J$ as ℓ, r , and put $j = |J|$. Use primes for the image lengths. Since $\ell', r' \geq \tau j'$,

$$b(g(T), g(J)) = \frac{(\ell' + j' + r')j'}{\ell' r'} \leq \frac{2}{\tau} + \frac{1}{\tau^2} =: C_\tau.$$

The cross-ratio hypothesis gives $b(T, J) \leq C_\tau / \kappa$. But

$$b(T, J) = \frac{j}{\ell} + \frac{j}{r} + \frac{j^2}{\ell r},$$

so both ℓ and r are at least $(\kappa / C_\tau)j$. This is the asserted scaled neighborhood. The final claim is precisely the real Koebe principle, Lemma 12.13. $\square$

Lemma 12.17 (Maximal extensions and multiplicity three). *Put $F_n = \bigcup_{i=0}^{q_n-1} \Delta_{i,n}$. For all sufficiently large n and $3 \leq i \leq q_n$ (with $\Delta_{q_n,n} = I_n$), there is an interval $T_{i,n} \supset \Delta_{3,n}$ such that*

1. $f^{i-3} : T_{i,n} \rightarrow f^{i-3}(T_{i,n})$ is a diffeomorphism;
2. each component of $f^{i-3}(T_{i,n}) \setminus \Delta_{i,n}$ contains a member of $\mathcal{C}_n$;
3. the chain $T_{i,n}, f(T_{i,n}), \dots, f^{i-3}(T_{i,n})$ has intersection multiplicity at most three.

Proof. Start with the maximal interval $M_{i,n} \supset \Delta_{3,n}$ on which f^{i-3} is monotone. At either endpoint, maximality means that an earlier image first meets the critical point or an endpoint of the dynamical interval. The other endpoint of that image is an endpoint of the corresponding cycle interval. Because the cycle intervals are pairwise disjoint and $i - 3 < q_n$, the interval between these two endpoint images contains a member of $\mathcal{C}_n$. Thus each side of $f^{i-3}(M_{i,n}) \setminus \Delta_{i,n}$ contains a cycle interval. Shrink $M_{i,n}$ minimally until on each side exactly the first such cycle interval is retained; call the pullback $T_{i,n}$.

We record the standard overlap count, because it is the part of the construction used by Koebe. Suppose a point belonged to four members $f^{r_s}(T_{i,n})$, $r_1 < r_2 < r_3 < r_4$. The four distinguished cycle intervals $\Delta_{r_s+3,n}$ are disjoint. Hence, for the middle two indices, the common point lies in side components of the corresponding extensions. Push the earlier side component forward by $r_3 - r_2$. By minimality of $T_{i,n}$ it must stop at the first cycle interval on that side, whereas the later extension already contains the next distinguished cycle interval. One of these two cycle intervals is then forced into the interior of the other. This contradicts disjointness of the cycle interiors. Therefore fourfold overlap is impossible. $\square$

For later use note that Lemma 12.5, applied to the branches in Lemma 12.17, shows that the maximum length of the intervals in each such chain tends to zero with n . Together with multiplicity three and Lemma 12.4, this implies the following uniform-in-level statement for the fixed map: there is $\kappa_f > 0$ such that every branch in Lemma 12.17 has cross-ratio distortion at least κ_f on every nested pair. Indeed, for deep levels the logarithmic loss is bounded by

$$C_f \sum_k |f^k(T_{i,n})|^2 \leq 6C_f \max_k |f^k(T_{i,n})| \longrightarrow 0,$$

and the finitely many remaining branches have a positive minimum.

Lemma 12.18 (Shortest-atom collar and pullback space). *There are constants $\tau_f > 0$ and $L_f < \infty$ such that, for every sufficiently large n , one can find $S_n \supset \Delta_{3,n}$ with the following properties:*

1. $f^{q_n-3}|_{S_n}$ is a diffeomorphism and its image contains a τ_f -scaled neighborhood of I_n ;
2. for every $3 \leq i < q_n$, the interval $W_{i,n} = f^{i-3}(S_n)$ contains a τ_f -scaled neighborhood of $\Delta_{i,n}$, and it may be chosen so that $W_{i,n} \cap F_n = \Delta_{i,n}$;
3. the distortion of $f^{q_n-i} : W_{i,n} \rightarrow f^{q_n-3}(S_n)$ is at most L_f .

The constants can be enlarged to cover the finitely many initial levels.

Proof. For $3 \leq i < q_n$, let $K_{i,n}$ be the smallest interval containing $\Delta_{i,n}$ whose two complementary sides each contain a level- n cycle interval. Choose ℓ so that $|\Delta_{\ell,n}|$ is minimal among $3 \leq i < q_n$. The exceptional cycle intervals $\Delta_{1,n}, \Delta_{2,n}$ cause no loss: the critical orbit is not periodic, so f is a diffeomorphism near the limiting centers $f(0), f^2(0)$, and for the fixed map there is $c_f > 0$ such that

$$|\Delta_{1,n}|, |\Delta_{2,n}| \geq c_f |\Delta_{3,n}|$$

for all large n . Thus minimality of $\Delta_{\ell,n}$ implies that $K_{\ell,n}$ contains a c'_f -scaled neighborhood of $\Delta_{\ell,n}$.

Apply Lemma 12.16 to $f^{\ell-3} : T_{\ell,n} \rightarrow K_{\ell,n}$. The preceding cross-ratio bound κ_f gives a definite scaled neighborhood of $\Delta_{3,n}$ inside $T_{\ell,n}$. Pulling this neighborhood through the fixed three-step critical passage uses the nonflat coordinate $f(x) = f(0) - |\chi(x)|^d$ and two regular branches. Hence the smallest interval $K_n \supset I_n$ having a level- n cycle interval on each side contains a δ_f -scaled neighborhood of I_n .

Now use the terminal branch $f^{q_n-3} : T_{q_n,n} \rightarrow K_n$. Let S_n be the pullback of the concentric subinterval of K_n obtained by retaining half of each side margin. Lemma 12.16 makes S_n a definite scaled neighborhood of $\Delta_{3,n}$, and Lemma 12.13 gives bounded distortion on $\Delta_{3,n}$. Applying the same argument to every suffix of the chain gives the scaled neighborhoods $W_{i,n}$ and the common distortion bound L_f . Because the terminal side intervals were stopped at the first neighboring members of $\mathcal{C}_n$, we may take the half-margin pullbacks before those neighbors; then $W_{i,n} \cap F_n = \Delta_{i,n}$. $\square$

Lemma 12.19 (Bounded-distortion power factorization). *Let $A_n : [-1, 1] \rightarrow I_n$ be the*

affine normalization and

$$G_n = A_n^{-1} \circ f^{q_n} \circ A_n = R^n f.$$

There are normalized orientation-preserving diffeomorphisms Φ_n, Ψ_n such that

$$G_n = \Phi_n \circ Q_d \circ \Psi_n, \quad Q_d(x) = \pm |x|^d,$$

where the derivative distortions of Φ_n are bounded by a constant depending only on f , while the derivative distortion of Ψ_n tends to one. In particular the family (G_n) is precompact in C^0 , and

$$\sup_n \|G_n\|_{C^1} < \infty.$$

Proof. Split the first return at the third image:

$$f^{q_n}|_{I_n} = (f^{q_n-3}|_{\Delta_{3,n}}) \circ (f^3|_{I_n}).$$

After affine normalization, the first factor is a diffeomorphism with bounded derivative distortion by Lemma 12.18. For the second factor use the C^3 nonflat coordinate at the critical point. The normalized critical germ is Q_d followed and preceded by diffeomorphisms; because $|I_n| \rightarrow 0$, the source diffeomorphism has derivative distortion tending to one. Absorb the two regular iterates into the target diffeomorphism. This gives the displayed factorization.

Normalize each diffeomorphic factor by its endpoint values. If a diffeomorphism of a fixed interval has derivative-distortion ratio at most L , the mean-value theorem gives $L^{-1} \leq |D\phi| \leq L$ after this normalization. Thus the factors are uniformly bi-Lipschitz and hence precompact in C^0 by Arzela–Ascoli. More importantly, the chain rule and $\|DQ_d\|_\infty < \infty$ give a direct uniform C^1 bound for G_n . No convergence of derivatives is asserted or needed. $\square$

Lemma 12.20 (Central restrictive interval cannot degenerate). *For the fixed map f there are $0 < \alpha_f^0 < \beta_f^0 < 1$ such that*

$$\alpha_f^0 \leq \frac{|I_{n+1}|}{|I_n|} \leq \beta_f^0 \quad (n \geq 0).$$

Proof. It suffices to prove the assertion for large n . Put $J_n = A_n^{-1}(I_{n+1})$. Its primitive restrictive period for G_n is $a_n \in \{2, \dots, N\}$. Suppose first that $|J_{n_k}| \rightarrow 0$. By Lemma 12.19, pass to a subsequence for which $G_{n_k} \rightarrow G$ uniformly and $a_{n_k} = p$ is constant. Uniform convergence of the finitely many iterates up to p and $G_{n_k}^p(J_{n_k}) \subset J_{n_k}$ give $G^p(0) = 0$.

Let $r \mid p$ be the primitive period of 0 for G . The limiting power factorization has a unique critical point of order d at 0; hence G^r maps a sufficiently small interval $U \ni 0$ strictly into its interior, and the intervals $U, G(U), \dots, G^{r-1}(U)$ are pairwise disjoint. These are strict C^0 conditions, so they also hold for G_{n_k} for large k . For large k the same strict trapping and disjointness conditions hold for G_{n_k} . Enlarging U inside the component on which those conditions remain valid produces a restrictive interval $\hat{U}_k$ of primitive period dividing r and with $|\hat{U}_k| \geq |U|/2$. If its primitive period is $< p$, this contradicts primitivity of the relative period p . If its primitive period is p , then by the canonical-maximal convention for restrictive intervals fixed above, J_{n_k} is the maximal

period- p restrictive interval about 0, hence it must contain $\widehat{U}_k$. This contradicts $|J_{n_k}| \rightarrow 0$. Therefore the central ratio is bounded away from zero.

Now suppose $|J_{n_k}| \rightarrow 2$, so the normalized restrictive interval fills $[-1, 1]$. Again pass to $G_{n_k} \rightarrow G$. Choose $x \neq 0$ close enough to 0 that $G(x)$ lies in the interior of $[-1, 1]$ but arbitrarily close to the critical value 1. For large k , both x and $G_{n_k}(x)$ lie in the interior of J_{n_k} . Hence J_{n_k} and $G_{n_k}(J_{n_k})$ have intersecting interiors, contradicting the defining disjointness of a restrictive cycle of period at least two. This proves the upper bound. A finite prefix is absorbed by decreasing α_f^0 and increasing β_f^0 . $\square$

Lemma 12.21 (Transport of central geometry to every child). *There are $0 < \alpha_f < \beta_f < 1$ such that whenever $\Delta \in \mathcal{C}_{n+1}$ lies in $\Delta^* \in \mathcal{C}_n$,*

$$\alpha_f |\Delta^*| \leq |\Delta| \leq \beta_f |\Delta^*|.$$

Proof. Write the child and its parent as corresponding images of $I_{n+1} \subset I_n$ under the same monotone itinerary, cutting the itinerary at the critical passage if necessary. Lemma 12.18 and Lemma 12.13 give a distortion bound D_f for that branch, independent of n . The mean-value theorem therefore gives

$$D_f^{-1} \frac{|I_{n+1}|}{|I_n|} \leq \frac{|\Delta|}{|\Delta^*|} \leq D_f \frac{|I_{n+1}|}{|I_n|}.$$

The lower estimate follows from Lemma 12.20. For the upper estimate, use the scaled-space part of Lemma 12.18: every child has on at least one side, inside its parent, a complementary interval whose length is a fixed multiple of the child length. Thus a child cannot fill its parent. Decreasing the lower constant and increasing the upper one absorbs the finite initial levels. $\square$

Lemma 12.22 (Derivative-ratio consequence). *There is $K_f \geq 1$ such that for $1 \leq i < j \leq q_n - 1$ and $x \in \Delta_{i,n}$,*

$$K_f^{-1} \frac{|\Delta_{j,n}|}{|\Delta_{i,n}|} \leq |Df^{j-i}(x)| \leq K_f \frac{|\Delta_{j,n}|}{|\Delta_{i,n}|}.$$

The same K_f , after enlargement, bounds $\|R^n f\|_{C^1}$.

Proof. For $i, j \geq 3$, apply Lemma 12.18 to the two tails f^{q_n-i} and f^{q_n-j} . Since $f^{q_n-i} = f^{q_n-j} \circ f^{j-i}$, division of the two bounded-distortion estimates and the mean-value theorem give exactly the displayed ratio. The indices $i = 1, 2$ are handled by the two fixed regular postcritical branches; their deep normalized distortion tends to one, and a finite prefix is absorbed in K_f . The normalized C^1 estimate was obtained directly in Lemma 12.19. $\square$

Lemma 12.23 (Exponential decay of the total cycle length). *Put*

$$\Lambda_n = \sum_{i=0}^{q_n-1} |\Delta_{i,n}|.$$

For the fixed map f there are n_0 and $0 < \rho_f < 1$ such that

$$\Lambda_{n+1} \leq \rho_f \Lambda_n \quad (n \geq n_0).$$

In particular $\Lambda_n \rightarrow 0$ exponentially fast.

Proof. The proof of Lemma 12.21 gives slightly more than its displayed size comparison. There is $\eta_f > 0$ such that every level- $(n+1)$ child D lying in a level- n parent P has, on at least one side inside P , a complementary interval G_D with

$$|G_D| \geq \eta_f |D|. \quad (12.25)$$

Choose G_D to be the first such complementary component. A component of $P \setminus F_{n+1}$ is adjacent to at most two children. Therefore, if $D_1, \dots, D_r$ are all level- $(n+1)$ children contained in P , then

$$|P| \geq \sum_{j=1}^r |D_j| + \frac{1}{2} \sum_{j=1}^r |G_{D_j}| \geq \left(1 + \frac{\eta_f}{2}\right) \sum_{j=1}^r |D_j|.$$

Thus the total length of the children in P is at most $\rho_f |P|$, where

$$\rho_f = \left(1 + \frac{\eta_f}{2}\right)^{-1} < 1.$$

Sum over the pairwise interior-disjoint level- n parents. This gives the claimed inequality. A finite initial prefix is irrelevant. $\square$

Lemma 12.24 (Deep normalized cross-ratio distortion tends to one). *Let*

$$G_n = A_n^{-1} \circ f^{q_n} \circ A_n = R^n f, \quad \Lambda_n = \sum_{i=0}^{q_n-1} |\Delta_{i,n}|.$$

For every fixed $r \leq N$, every interval M on which G_n^r is monotone, and every $J \in M$,

$$B(G_n^r; M, J) \geq \exp(-r C_{\text{cr}} \text{mesh}(\mathcal{C}_n) \Lambda_n). \quad (12.26)$$

Consequently the lower cross-ratio distortion of all such branches tends to one, uniformly for $1 \leq r \leq N$.

Proof. Cross ratio is invariant under affine changes, so it is enough to work in I_n . Let $\widetilde{M} = A_n(M)$ and $\widetilde{J} = A_n(J)$. During one q_n -block of a monotone orbit, the i th image of $\widetilde{M}$ is contained in the cycle interval $\Delta_{i,n}$. The smooth-loss estimate of Lemma 12.4 therefore gives

$$\log B(f^{q_n}; \widetilde{M}, \widetilde{J}) \geq -C_{\text{cr}} \sum_{i=0}^{q_n-1} |f^i(\widetilde{M})|^2 \geq -C_{\text{cr}} \text{mesh}(\mathcal{C}_n) \Lambda_n.$$

If the branch crosses the critical chart, the nonflat power factor is kept separate; on each monotone side its Schwarzian is negative at sufficiently small scale and hence it does not decrease B . Equivalently, this is already built into the proof of Lemma 12.4.

For G_n^r , split the lifted orbit into r consecutive q_n -blocks and use multiplicativity of B . Each block has the same upper bound because its orbit remains inside the level- n restrictive cycle. This proves (12.26). Lemma 12.15 gives $\text{mesh}(\mathcal{C}_n) \rightarrow 0$, while the

cycle intervals have disjoint interiors in the fixed ambient interval, so $\Lambda_n \leq 2$. Hence the exponent tends to zero uniformly for $r \leq N$. $\square$

Lemma 12.25 (Quantitative one-step real bounds). *Fix N , an even critical order d , $\kappa \in (0, 1)$, and $D_0 \geq 1$. There are constants*

$$\tau = \tau(N, d, \kappa, D_0) > 0, \quad L = L(N, d, \kappa, D_0) < \infty,$$

and

$$0 < a(N, d, \kappa, D_0) < b(N, d, \kappa, D_0) < 1$$

with the following property. Let $g : [-1, 1] \rightarrow [-1, 1]$ be a symmetric normalized d -unimodal map with a canonical restrictive interval $J \ni 0$ of primitive period $2 \leq p \leq N$. Assume that every monotone branch of every iterate g^r , $1 \leq r \leq p$, has lower cross-ratio distortion at least κ , and that on a neighborhood containing J the critical branch has a coordinate

$$g(x) = g(0) \pm |\psi(x)|^d, \quad \psi(0) = 0, \quad \text{Dist}(\psi) := \frac{\sup \psi'}{\inf \psi'} \leq D_0.$$

Then every member of the restrictive cycle has a pullback neighborhood with τ -scaled space and terminal distortion at most L ; moreover $a \leq |J|/2 \leq b$, all children and adjacent gaps of the first restrictive partition are uniformly commensurable, and the normalized first renormalization has C^1 norm bounded in terms of (N, d, κ, D_0) only.

Proof. For a shortest nonexceptional cycle atom D_s , $s \geq 3$, shortestness gives a unit relative image collar. The cross-ratio hypothesis and Lemma 12.16 pull it back to a definite source collar, and real Koebe bounds the terminal distortion. Only $s = 1, 2$ require the critical coordinate. Pull the one-sided image collar through

$$x = \psi^{-1}(\pm |g(0) - y|^{1/d}).$$

The root map converts relative margin one to a two-sided margin depending only on d , while pre- and post-composition by ψ and ψ^{-1} change adjacent length ratios by at most D_0^2 . For $s = 2$ one first crosses one regular monotone branch, controlled by the cross-ratio hypothesis and Koebe. Thus all positions of the shortest atom give a definite central collar depending only on (d, κ, D_0) .

There are at most $p \leq N$ monotone legs. Repeating the terminal maximal-extension construction gives a common source-space constant and terminal distortion depending only on (N, d, κ, D_0) . Normalize the parent to $[-1, 1]$. The return maps factor into the critical power map and diffeomorphic factors with the preceding derivative-ratio bounds, so they form an equicontinuous family with a common C^1 bound. Since $p \in \{2, \dots, N\}$, collapse of J would produce a persistent superattracting restrictive interval in a limit, contradicting primitivity or canonical maximality; filling the parent would force two restrictive-cycle interiors to intersect. Compactness therefore yields $0 < a < b < 1$. Transporting the central ratio through the bounded-distortion terminal branches gives uniform child/gap comparability and the normalized C^1 bound. $\square$

Proposition 12.26 (Deep critical-coordinate normalization). *Let f be a fixed C^3 non-flat unimodal map of even critical order d , and let $A_n : [-1, 1] \rightarrow I_n$ be the affine normalization of its restrictive intervals. In a local critical coordinate write*

$$f(x) = f(0) \pm |\phi(x)|^d, \quad \phi(0) = 0, \quad \phi'(0) \neq 0,$$

and set

$$\psi_n(t) = \frac{\phi(A_n(t))}{\phi(A_n(1))}.$$

Then $\text{Dist}(\psi_n) \rightarrow 1$. In particular, after a map-dependent entry level, $\text{Dist}(\psi_n) \leq 2$.

Proof. Lemma 12.15 gives $|I_n| \rightarrow 0$. Continuity and nonvanishing of ϕ' at the critical point imply

$$\frac{\sup_{x \in I_n} |\phi'(x)|}{\inf_{x \in I_n} |\phi'(x)|} \rightarrow 1.$$

Affine precomposition and scalar postnormalization do not change derivative distortion, so this ratio is exactly $\text{Dist}(\psi_n)$. $\square$

Proposition 12.27 (Beau uniformization). *Fix N and d . There are*

$$K(N, d) < \infty, \quad 0 < \alpha(N, d) < \beta(N, d) < 1,$$

such that for every map in the theorem, after a map-dependent entry level, items (i)–(iii) hold with these constants.

Proof. By Lemma 12.24, for every fixed map f there is an entry level $n_1(f)$ such that every monotone branch of every iterate of $G_n = R^n f$ of length at most the next relative period $a_n \leq N$ has lower cross-ratio distortion at least $\kappa_* = 1/2$ whenever $n \geq n_1(f)$. By Proposition 12.26 there is another entry level $n_2(f)$ after which the normalized critical coordinate of G_n has distortion at most $D_0 = 2$. Put $n_*(f) = \max\{n_1(f), n_2(f)\}$.

Apply Lemma 12.25 with $p = a_n$, $\kappa = 1/2$ and $D_0 = 2$. Every resulting constant depends only on N and d . Its central-size estimate gives universal bounds for $|I_{n+1}|/|I_n|$; its collar and terminal-distortion estimates transport these bounds to every child and adjacent gap; and the derivative-ratio argument gives one universal $K(N, d)$ together with the normalized C^1 bound. The entry depth may depend on f , but all constants after entry depend only on (N, d) , exactly the beau quantifier order. $\square$

Proof of Theorem 12.14. Lemma 12.21 proves item (i), and Lemma 12.22 proves items (ii) and (iii). Only finitely many levels precede the map-dependent construction; enlarging K_f and taking the minimum/maximum of the finitely many missing child ratios extends those estimates to every level. Finally, Proposition 12.27 gives the eventual constants depending only on N and d . Thus both the ordinary and the beau parts of the theorem are proved. $\square$

Theorem 12.28 (Principal/restrictive-nest bounded geometry). **Statement scope.** *Under the hypotheses of the real a priori bounds, there are deep constants $\gamma > 0$, $\tau > 0$, and $C \geq 1$ such that every next-level gap in a parent P has length at least $\gamma|P|$, every child has uniform real space inside the corresponding parent configuration, and*

any two adjacent atoms/gaps in a common deep parent have length ratio in $[C^{-1}, C]$. Hence the postcritical Cantor coding has bounded geometry.

Proof strategy (not part of the proof)

Proof status. derived here.

Strategic outline only. Repaired real-bounds child/gap nondegeneracy; uniform gap fraction; scaled-space conversion; bounded-branching adjacent commensurability; bounded-geometry Cantor coding.

The mathematical argument begins in the proof environment below.

Proof. Fix a map f of bounded combinatorial type N and use the map-dependent constants supplied by items (i)–(iii) of the real a priori bounds. The beau clause is not needed for this theorem. Inside a parent $P \in \mathcal{C}_n$, the next-level partition consists of the finitely many level- $(n+1)$ return intervals lying in P and the complementary gaps between them.

The real-bounds proof gives two quantitative inputs at sufficiently deep levels. First, every child $J \subset P$ has $|J| \geq \alpha_f |P|$. Second, Lemma 12.18 gives a τ_f -scaled neighborhood of every cycle atom whose side margins contain no other atom of that generation. At least one of those side margins lies inside the parent P ; hence every nonempty adjacent level- $(n+1)$ gap has length at least $\tau_f |J| \geq \tau_f \alpha_f |P|$. Taking $\gamma_f = \tau_f \alpha_f$ proves the lemma. No compactness of normalized C^1 returns is used here.

Let α_f, β_f be the child constants from the map-dependent real-bounds theorem. A child together with its two neighboring separating gaps therefore has space at least $\gamma_f |P|$ on either available side. In the corresponding parent configuration the child is τ_f -well inside with, for example, $\tau_f \geq \gamma_f / \beta_f$ after the harmless finite refinement at boundary children. Bounded type gives at most $B(N)$ atoms and gaps in one parent. Hence every element of the partition has normalized length between a positive constant depending on f and N and one; for adjacent elements A, B ,

$$C_f^{-1} \leq \frac{|A|}{|B|} \leq C_f$$

with a constant C_f depending on the real-bounds geometry of f (and on the bounded branching N).

The branching bound is N . The preceding map-dependent estimates give uniform separation between distinct children and uniform lower and upper bounds for every child/parent ratio. Iterating them down the tower proves that every cylinder has neighboring cylinders and gaps of comparable size at its own generation. The diameters tend to zero by the real bounds, so the nested intersection is totally disconnected and perfect on the recurrent postcritical set. This is exactly bounded geometry.

Notice the logical direction: cross-ratio and Koebe estimates produced the real bounds; the real bounds plus finite combinatorics now produce the uniform gap and adjacency geometry. No complex modulus estimate is used. $\square$

Theorem 12.29 (Real rigidity precursor). **Statement scope.** Let f and g be normalized infinitely renormalizable C^3 unimodal maps of the same finite critical order

and the same bounded renormalization combinatorics. Their canonical combinatorial conjugacy

$$h : P(f) \rightarrow P(g)$$

between postcritical Cantor sets is quasisymmetric, with control depending on the real bounded-geometry constants. Extending h affinely over matching complementary gaps gives an increasing quasisymmetric interval homeomorphism. The theorem claims only real conjugacy on $P(f)$; no complex or hybrid rigidity is asserted.

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Match bounded-type cylinders; bounded-geometry three-point estimate; canonical Cantor conjugacy; affine extension over corresponding gaps.

The mathematical argument begins in the proof environment below.

Proof. Let f and g be normalized infinitely renormalizable C^3 unimodal maps of the same finite critical order and the same bounded renormalization combinatorics. Write $\mathcal{P}_n(f)$ and $\mathcal{P}_n(g)$ for their deep return partitions. Common combinatorics gives a canonical order-preserving bijection between level- n atoms, compatible with refinement.

For $x \in P(f)$, let $P_n(x)$ be its unique nested level- n cylinder. Real bounds imply $|P_n(x)| \rightarrow 0$, so the matched cylinders in the g -tower also shrink to one point. Define $h(x)$ to be that point. Compatibility of the partitions makes h increasing and bijective, and compactness gives continuity. Since the combinatorial matching respects the action on cylinder addresses, the conjugacy identity holds first on the critical orbit and then on its closure $P(f)$.

Fix $x - t, x, x + t \in P(f)$ with $t > 0$. Choose the first generation at which the interval $[x - t, x + t]$ meets more than one child of the cylinder containing x . Bounded branching and the lower gap/child ratios imply that the two halves $[x - t, x]$ and $[x, x + t]$ each meet only a uniformly bounded number of cylinders of that generation, and that every cylinder they meet has length comparable to t . The same combinatorial collection occurs in the g construction. Bounded geometry there says that the lengths of the two image collections are comparable by a constant M depending only on the two geometry constants and the branching bound. Therefore

$$\frac{|h(x + t) - h(x)|}{|h(x) - h(x - t)|} \leq M. \quad (6)$$

The symmetric lower bound follows by interchanging the two sides. The standard three-point criterion on the line now says that h is quasisymmetric on $P(f)$.

Let H denote the affine extension. To verify the three-point criterion, take three equally spaced points. If all three lie in one complementary gap, the image ratio is one. If they meet the Cantor set, replace each point in a gap by the nearest endpoint at the first cylinder scale comparable with the spacing. Bounded geometry changes the three distances by only a uniform factor. Estimate (6) applies to the resulting Cantor endpoints, and the affine interpolation inside each intervening gap changes the ratio by another uniform factor. Hence H is quasisymmetric.

Thus maps with the same bounded combinatorics and the real a priori bounds have a controlled real quasimetric conjugacy on the postcritical Cantor set, with a controlled interval extension. We emphasize the boundary of the statement: the proof does not produce a quasiconformal extension to the plane, a polynomial-like modulus bound, or a hybrid conjugacy of renormalizations. Those are the complex-bounds and rigidity tasks of Chapter 13. $\square$

12.3 Worked examples

Worked example: Cross-ratio expansion for a negative-Schwarzian branch

Take

$$T = [-2, 2], \quad J = [-1, 1].$$

The two complementary components have length one, so in our convention

$$b(T, J) = \frac{|T||J|}{|L||R|} = \frac{4 \cdot 2}{1 \cdot 1} = 8.$$

Now let $g(x) = \tanh x$. Since

$$S(\tanh x) = -2,$$

this is a negative-Schwarzian diffeomorphism. By symmetry,

$$|g(T)| = 2 \tanh 2, \quad |g(J)| = 2 \tanh 1,$$

and each image gap has length $\tanh 2 - \tanh 1$. Hence

$$b(g(T), g(J)) = \frac{4 \tanh 2 \tanh 1}{(\tanh 2 - \tanh 1)^2} \approx 71.67 > 8.$$

Thus the distortion $B(g; T, J)$ is about 8.96, visibly larger than one. The numerical size is unimportant; the theorem guarantees the direction of the inequality for every nested pair.

Worked example: What a bounded-type real bound says down a tower

Suppose a period-doubling renormalization tower has reached the depth at which the beau estimates hold, and write I_n for the central restrictive interval. The theorem supplies constants

$$0 < \alpha < \beta < 1, \quad \alpha|I_n| \leq |I_{n+1}| \leq \beta|I_n|.$$

Iterating gives, for every $m \geq 1$,

$$\alpha^m|I_n| \leq |I_{n+m}| \leq \beta^m|I_n|.$$

Thus the central intervals shrink exponentially in the generation number, but neither arbitrarily faster nor arbitrarily slower than the bounds permit. The derivative estimate has an equally concrete interpretation. If a monotone branch sends a level- n cycle interval Δ_i to Δ_j , then

$$\frac{1}{K} \frac{|\Delta_j|}{|\Delta_i|} \leq |Df^{j-i}(x)| \leq K \frac{|\Delta_j|}{|\Delta_i|}.$$

The mean-value slope is therefore representative of every derivative on the branch up to one fixed factor K . This is exactly the control later needed to thicken real return domains into complex neighborhoods.

Worked example: A binary Cantor geometry with no real a priori bound

Start with $I_0 = [0, 1]$. At stage n , replace each surviving interval I by two endpoint children, each of length $\varepsilon_n |I|$, where $0 < \varepsilon_n < 1/3$ and $\varepsilon_n \rightarrow 0$. Combinatorially this is the same binary tree at every stage, but

$$\frac{|\text{child}|}{|\text{parent}|} = \varepsilon_n \rightarrow 0.$$

Hence no depth-independent lower real bound exists.

Compare this Cantor set with the middle-thirds Cantor set by matching binary addresses. In the source, a level- n child can be arbitrarily tiny compared with its adjacent gap, whereas in the middle-thirds set the corresponding ratio is fixed. Three equally spaced points chosen across such a tiny child and its neighboring gap therefore acquire image-distance ratios that are unbounded with n . The address map is not quasisymmetric. This example shows why common combinatorics alone is insufficient for the rigidity precursor: bounded geometry is essential.

12.4 What this chapter established

Chapter 12 turned the qualitative renormalization interface from Chapter 11 into a uniform real geometry theorem. We fixed one cross-ratio convention, proved the negative-Schwarzian expansion mechanism, and then recorded the stronger small-scale fact needed for modern C^3 unimodal maps: sufficiently deep first-entry branches lose arbitrarily little cross ratio even when the original map has no global Schwarzian sign.

That lower bound is precisely the input for real Koebe. Koebe controls the derivative on a central interval whenever the image has scaled space. In a bounded-type renormalization tower, the return chains have bounded overlap; after rescaling, the return maps therefore lie in a compact real class. Finite combinatorics plus compactness excludes collapsing children and gaps. This yields the real a priori bounds, their deep universal or *beau* form, and bounded geometry of the return partitions.

Finally, common combinatorics matches the nested cylinders of two towers. Bounded geometry converts that coding map into a quasisymmetric map on the post-

critical Cantor sets and permits a quasisymmetric affine-gap extension on the real interval. This is deliberately only a real rigidity precursor.

Looking ahead

Chapter 13 crosses the real-to-complex boundary. The length ratios, Koebe constants, scaled neighborhoods, and quasisymmetric boundary data proved here become inputs for polynomial-like extensions and complex modulus estimates. The new work there is genuinely complex: one must thicken real domains, prevent annuli from degenerating, and then use quasiconformal compactness and straightening. None of those modulus conclusions has been assumed in Chapter 12.

Complex Bounds and Polynomial-Like Renormalization

Why this chapter matters

Chapter 12 showed that a bounded real renormalization tower cannot collapse on the real line. But a real length ratio is not yet a complex annulus. Chapter 13 makes the decisive passage from one-dimensional geometry to polynomial-like geometry: the return intervals are thickened to Jordan disks, the annuli between successive disks are prevented from pinching, and the resulting quadratic-like germs become compact after normalization.

The analytic tool that makes this possible is subtle. Under a high-degree branched covering, the ordinary pullback estimate for annulus modulus loses a factor equal to the total degree, which may be useless. Kahn and Lyubich showed that in the nearly degenerate regime a well-collared low-degree inner island recovers control with the *inner* degree. This is exactly the mechanism needed in a renormalization nest.

Learning goals

By the end of this chapter you should be able to:

1. distinguish real a priori bounds from a positive polynomial-like annulus modulus;
2. state the Kahn-Lyubich covering lemma with its total degree, inner degree, collar, and small-modulus threshold;
3. explain how real bounded geometry is thickened to quadratic-like first returns;
4. explain why a collapsing sequence of renormalization annuli contradicts a modulus-improvement argument;
5. use a fixed modulus lower bound to obtain compactness of normalized quadratic-like germs;
6. state the beau-bounds quantifiers in the correct order;
7. explain what additional external matching is needed for the Chapter 13 pullback-rigidity criterion and why full combinatorial rigidity is postponed.

whose restrictions send Λ' properly to B' and Λ properly to B . We write $D = \deg(F : U \rightarrow V)$ and $d = \deg(F : \Lambda' \rightarrow B')$; the annulus $B' \setminus \overline{B}$ is the *collar*.

Definition 13.5 (polynomial-like renormalization germ). If a first return $f^q : U' \rightarrow U$ is quadratic-like, its affine-normalized hybrid germ is denoted Rf ; iteration gives $R^n f$ whenever the successive first returns exist. Different admissible domains define the same germ.

Definition 13.6 (Carathéodory/germ convergence). A sequence of normalized polynomial-like representatives converges in the germ topology if, after choosing representatives on uniformly controlled neighborhoods of their filled Julia sets, the domains converge in the Carathéodory sense and the maps converge locally uniformly there. This is the compactness topology used in Chapter 13.

Definition 13.7 (real-symmetric slice $\mathcal{C}^{\mathbb{R}}$). The real-symmetric slice $\mathcal{C}^{\mathbb{R}}$ consists of normalized quadratic-like germs commuting with complex conjugation and hybrid equivalent to a real quadratic polynomial. Its subspace $\mathcal{C}^{\mathbb{R}}(\delta)$ consists of germs of modulus at least δ .

Definition 13.8 (external marking and matched external conjugacy). An external marking chooses the homotopy class of an access to the filled Julia set, equivalently the normalized external Böttcher coordinate. A matched external conjugacy between two quadratic-like maps is a quasiconformal conjugacy on collars outside their filled Julia sets that respects these markings and the first-return dynamics.

Definition 13.9 (invariant line field and hybrid conjugacy). An invariant measurable line field for F on $K(F)$ is a measurable Beltrami coefficient μ with $|\mu| = 1$ on a positive-area invariant subset and $F^* \mu = \mu$. A quasiconformal conjugacy is *hybrid* when its Beltrami coefficient vanishes almost everywhere on the filled Julia set.

13.2 Main results

Lemma 13.1 (Kahn-Lyubich covering lemma). **Statement scope.** Fix $\eta \in (0, 1]$ and $D \geq 1$. There exists $\varepsilon = \varepsilon(\eta, D) > 0$ with the following property. For a Kahn-Lyubich covering datum with total degree at most D and inner degree d , if

$$\text{mod}(B' \setminus \overline{B}) > \eta \text{mod}(U \setminus \overline{\Lambda}),$$

then $0 < \text{mod}(U \setminus \overline{\Lambda}) < \varepsilon$ implies

$$\text{mod}(V \setminus \overline{B}) < 2\eta^{-1}d^2 \text{mod}(U \setminus \overline{\Lambda}).$$

Proof strategy (not part of the proof)

Proof status. proved here modulo stated analytic input.

Strategic outline only. Extremal width normalization; harmonic-route decomposition; buffer truncation and quasi-additivity; Galois lift; collar reduction; exact modulus conversion.

The mathematical argument begins in the proof environment below.

Proof setup. Put

$$m = \text{mod}(U \setminus \bar{\Lambda}), \quad M = \text{mod}(V \setminus \bar{B}), \quad c = \text{mod}(B' \setminus \bar{B}).$$

Thus the collar hypothesis is $c > \eta m$. We shall prove the estimate from extremal length rather than invoke the Kahn–Lyubich quasi-invariance law as a black box. The only frozen tools are the standard extremal-length series/parallel laws, existence of the harmonic extremal metric on a finite bordered surface, the elementary topology of disjoint essential arcs on a finite-type surface, and the finite Galois closure of a finite branched covering. The near-degenerate quasi-additivity estimate itself is proved below.

For a curve family Γ write $L(\Gamma)$ for extremal length and $W(\Gamma) = L(\Gamma)^{-1}$ for extremal width. If $K \Subset S$ is a hull, write $W(S, K)$ for the width of curves joining ∂S to K in $S \setminus K$.

Lemma 13.2 (Extremal width and finite coverings). *If $A \Subset X$ are Jordan disks, then*

$$W(X, A) = \frac{1}{\text{mod}(X \setminus \bar{A})}. \quad (13.1)$$

If $p : \tilde{X} \rightarrow X$ is an unbranched covering of degree q over the annulus $X \setminus \bar{A}$, then the lifted annulus has modulus $\text{mod}(X \setminus \bar{A})/q$, and hence its width is $qW(X, A)$.

Proof. The first identity is the standard rectangle/annulus duality for extremal length. For the second, lift an admissible conformal metric to all q sheets. Lengths of lifted crossing curves agree with the downstairs length while area is multiplied by q . Applying the same argument to the quotient metric gives both inequalities and therefore equality. Taking reciprocals gives the width formula. $\square$

We next isolate the part of the Kahn–Lyubich argument which is responsible for quasi-additivity. Let S be a compact bordered Riemann surface and let $A_1, \dots, A_N \Subset S$ be pairwise disjoint finite unions of Jordan disks. Set $A = \bigcup_j A_j$ and

$$X = W(S, A), \quad Y = \sum_j W(S, A_j), \quad Z = \sum_j W\left(S \setminus \bigcup_{k \neq j} A_k, A_j\right). \quad (13.2)$$

The topological complexity of this configuration is fixed throughout the next lemmas.

Lemma 13.3 (Harmonic routes and a finite rectangle decomposition). *For each j , the extremal family defining $W(S, A_j)$ admits a decomposition, up to a set of transverse measure zero, into finitely many harmonic rectangles $P(\alpha)$. The index α is the route of a nonsingular harmonic leaf through the other islands. The number s of possible routes is bounded only by the topological complexity and N , and*

$$\sum_{\alpha} W(P(\alpha)) = Y. \quad (13.3)$$

For a route of height k , there are associated little rectangles $Q_i(\alpha)$, $1 \leq i \leq k$, such that every vertical path in $P(\alpha)$ vertically crosses each $Q_i(\alpha)$ in order.

Proof. Fix j and let u_j be the harmonic function on $S \setminus A_j$ with boundary values 0 on ∂S

and 1 on ∂A_j . Away from the finite critical graph of u_j , the coordinate $u_j + iu_j^*$ identifies each component with a Euclidean rectangle of height 1; the gradient lines of u_j are its vertical leaves. The parallel law for extremal width therefore says that the sum of the widths of these rectangles equals $W(S, A_j)$. Summing over j gives (13.3).

Follow a nonsingular leaf from ∂S to its terminal island and record, in order, the distinct archipelagos at which an initial segment first ends. At the i th first encounter, cut the corresponding parallel strip along the two boundary leaves of its route class. The strip between them is the big rectangle P_i , while the terminal portion adjacent to the i th island is the little rectangle Q_i . Because the initial segments are nested, a vertical leaf of the final big rectangle must successively exit each P_i through its terminal roof; hence it crosses every Q_i in route order.

It remains to bound the number of route classes. On a finite-type surface, a family of pairwise disjoint nonparallel essential proper arcs has cardinality bounded in terms of $-\chi$ and the number of boundary components: cut along a maximal such family and triangulate the resulting surface; Euler's formula bounds the number of edges. At each successive island encounter only one of finitely many such proper-homotopy classes can occur. A route cannot acquire more first encounters than the number N of archipelagos. Thus the number of possible ordered route classes is bounded by a constant $s = s(\text{Top}, N)$. This is all that will be used below; no geometric constant depends on the particular surface. $\square$

Lemma 13.4 (Buffer truncation and disjoint little rectangles). *There is an absolute buffer width $M_0 > 0$ with the following property. Remove from both vertical sides of every harmonic rectangle $P(\alpha)$ two buffers of total width M_0 and denote the remaining rectangle by $\tilde{P}(\alpha)$; let $\tilde{Q}_i(\alpha)$ be the corresponding truncated little rectangles. Then:*

1. *for a fixed route, the $\tilde{Q}_i(\alpha)$ are pairwise disjoint;*
2. *if two truncated little rectangles with matching vertical orientation overlap, they have the same height i , and one of the two routes extends the other;*
3. *for every $\delta \in (0, 1)$ there is $K_0 = K_0(\delta, \text{Top}, N)$ such that $Y > K_0$ implies*

$$\sum_{\alpha} W(\tilde{P}(\alpha)) > (1 - \delta)Y. \quad (13.4)$$

Proof. We first record the small-overlap estimate behind the truncation. Suppose R_1, R_2 are harmonic rectangles and each has a side buffer of width at least 4. If every vertical leaf in one buffer met more than half of each of the two side buffers of the other rectangle, push the Euclidean metric of the first rectangle to the surface. Every curve of the second vertical family would then have metric length at least 1, whereas the supporting area is strictly smaller than the assumed extremal width. This contradicts the extremal definition of width. Hence one can choose, in each of the four side buffers, a vertical leaf disjoint from the chosen leaves in the other rectangle. Taking $M_0 = 8$ is therefore enough.

Use those four disjoint leaves as new vertical boundaries. If two truncated little rectangles overlap, the corresponding enlarged big rectangles have disjoint vertical boundaries. Two crossing rectangles with disjoint vertical boundaries must represent the same proper homotopy class of the crossing arc; otherwise their boundary arcs

would have nonzero geometric intersection. With orientations matched, the terminal roofs agree. Following the nested initial segments backwards shows that the two route histories agree up to the shorter height. Thus their heights agree at the overlap and one route extends the other. For two little rectangles belonging to one route at different heights, the terminal roofs lie on different islands, so overlap is impossible. This proves (1) and (2).

Finally truncation removes at most M_0 width from each big rectangle. There are at most s route rectangles for each of the N terminal families, so the total removed width is at most M_0Ns . Combining this with (13.3) gives

$$\sum_{\alpha} W(\tilde{P}(\alpha)) \geq Y - M_0Ns.$$

If $Y > M_0Ns/\delta$, this is (13.4). □

For positive numbers write

$$x_1 \oplus \cdots \oplus x_k = \left(\frac{1}{x_1} + \cdots + \frac{1}{x_k} \right)^{-1}$$

for their harmonic sum.

Lemma 13.5 (Arithmetic series inequality). *Let $a_i, b_i > 0$, $1 \leq i \leq n$, satisfy $a_1 = b_1$ and*

$$a_i \leq b_1 \oplus \cdots \oplus b_i. \quad (13.5)$$

Then

$$\left(\sum_{i=1}^n a_i \right)^2 \leq \frac{4}{3} b_1 \sum_{i=1}^n b_i. \quad (13.6)$$

Proof. Increasing a_i can only make the left side larger, so it is enough to take equality in (13.5). Then

$$\frac{1}{a_i} = \frac{1}{a_{i-1}} + \frac{1}{b_i}, \quad b_i = \frac{a_i a_{i-1}}{a_{i-1} - a_i} = a_i + \frac{a_i^2}{a_{i-1} - a_i}.$$

Let $a = \sum_i a_i$ and $b = \sum_i b_i$. By Cauchy-Schwarz,

$$b - b_1 \geq a - a_1 + \frac{(a - a_1)^2}{a_1 - a_n} \geq a - a_1 + \frac{(a - a_1)^2}{a_1}.$$

Since $a_1 = b_1$, division by a^2/b_1 gives

$$\frac{b_1 b}{a^2} \geq 1 - \frac{a_1}{a} + \left(\frac{a_1}{a} \right)^2 \geq \frac{3}{4},$$

which is equivalent to (13.6). □

Lemma 13.6 (Near-degenerate quasi-additivity). *For every fixed topological complexity there is $K < \infty$ such that the moduli in (13.2) satisfy*

$$Y > K \implies Y^2 \leq \frac{3}{2} XZ. \quad (13.7)$$

More generally, $3/2$ may be replaced by any constant strictly larger than $4/3$ after changing K .

Proof. Choose $\delta \in (0, 1)$ so that $(4/3)(1 - \delta)^{-2} < 3/2$, and truncate as in Lemma 13.4. For routes of height k put

$$a_k = \sum_{|\alpha|=k} W(\tilde{P}(\alpha)), \quad b_i^{(k)} = \sum_{|\alpha|=k} W(\tilde{Q}_i(\alpha)), \quad b_i = \max_{k \geq i} b_i^{(k)}.$$

By Lemma 13.3, every vertical curve of $\tilde{P}(\alpha)$ crosses the pairwise disjoint little rectangles $\tilde{Q}_1(\alpha), \dots, \tilde{Q}_k(\alpha)$. The extremal-length series law therefore gives

$$a_k \leq b_1 \oplus \dots \oplus b_k. \quad (13.8)$$

The first little rectangles for all routes of a fixed height are disjoint by Lemma 13.4; their vertical families form a subfamily of the curves joining ∂S to A . Hence $b_1 \leq X$. Likewise, fix an island A_j . All little rectangles which occur at a chosen route height with terminal island A_j are pairwise disjoint, and their vertical families lie in the family defining $W(S \setminus \bigcup_{\ell \neq j} A_\ell, A_j)$. Summing first over heights and then over j gives

$$\sum_i b_i \leq Z. \quad (13.9)$$

On the other hand, (13.4) gives $\sum_k a_k > (1 - \delta)Y$ once $Y > K$. Apply Lemma 13.5 to (13.8), then use (13.9):

$$(1 - \delta)^2 Y^2 < \left(\sum_k a_k \right)^2 \leq \frac{4}{3} b_1 \sum_i b_i \leq \frac{4}{3} X Z.$$

Our choice of δ gives (13.7). Since δ can be chosen so that $(4/3)(1 - \delta)^{-2} < C$ for every $C > 4/3$, the final assertion follows as well. $\square$

Definition 13.10 (Generalized target separation width). For a target island $B \subset B' \Subset V$ and the set CV of critical values of F , let $\mathcal{G}'$ be the proper curves in $V \setminus B$ which begin on B and end on ∂V . Let $\mathcal{G}''$ be the proper curves which begin and end on B , avoid CV, and are not homotopic in $V \setminus (B \cup \text{CV})$ into an arbitrarily small neighborhood of the hull B' . Put

$$Z_V := W(\mathcal{G}') + 2W(\mathcal{G}'').$$

The factor 2 records the two orientations of a returning route.

Lemma 13.7 (Galois lift and quasi-invariance). *Let $F : U \rightarrow V$ be a branched covering of total degree at most D . Suppose $\Lambda' \Subset U$ maps properly to $B' \Subset V$ with degree d , and Λ is the component over B contained in Λ' . There is $K_1 = K_1(D)$ such that, whenever $W(U, \Lambda) > K_1$,*

$$W(U, \Lambda)^2 \leq 2d^2 W(V, B) Z_V, \quad (13.10)$$

where Z_V is the generalized separation width of Definition 13.10.

Proof. Take the finite Galois closure $g = F \circ h : S \rightarrow V$. Its degree $|G|$ is bounded in

terms of D alone. The components A_i of $g^{-1}(B)$ form one orbit under the deck group. Choose the component A_1 lying over Λ and write $m_h = \deg(h : A'_1 \rightarrow \Lambda')$. The stabilizer of A_1 has order dm_h , so the number of translates is $|G|/(dm_h)$.

Apply Lemma 13.6 to all the translated islands. Covering invariance from Lemma 13.2 gives

$$X_S = |G|W(V, B), \quad Y_S \geq \frac{|G|}{d}W(U, \Lambda). \quad (13.11)$$

For the separation term, use the harmonic rectangles measuring the width from one lifted island to the rest of the lifted boundary. After removing fixed buffers, each such rectangle projects injectively to a proper target rectangle unless it meets a critical value. A target rectangle joining the distinguished island to another boundary component has exactly dm_h oriented lifts; a rectangle returning to the same island has twice that multiplicity, one for each orientation. With the returning curves declared nontrivial relative to the hull B' and the marked critical values, this is precisely the weighting in Definition 13.10. Consequently

$$Z_S \leq |G|Z_V + C(D), \quad (13.12)$$

where the additive constant is the total width of the finitely many discarded buffers; their number is bounded by the topology of the Galois closure and hence by D .

Now $Z_S \geq Y_S$. Thus (13.11) shows that $Z_S \rightarrow \infty$ as $W(U, \Lambda) \rightarrow \infty$, while (13.12) forces $Z_V \rightarrow \infty$. Increase the threshold $K_1(D)$ until both the quasi-additivity threshold is met and $C(D) \leq |G|Z_V/3$. Then

$$Z_S \leq \frac{4}{3}|G|Z_V.$$

Using the 3/2 form of Lemma 13.6,

$$\frac{|G|^2}{d^2}W(U, \Lambda)^2 \leq Y_S^2 \leq \frac{3}{2}X_S Z_S \leq 2|G|^2W(V, B)Z_V.$$

Cancel $|G|^2$ to obtain (13.10). $\square$

Lemma 13.8 (Collar control of generalized separation). *With the notation of Definition 13.10, suppose $B \Subset B' \Subset V$ are Jordan disks and put $c = \text{mod}(B' \setminus \overline{B})$. Then*

$$Z_V \leq W(B', B) = \frac{1}{c}. \quad (13.13)$$

Proof. This is the extremal-width collar-lifting inequality. Cut every curve in $\mathcal{G}'$ at its first exit from B' . Cut every curve in $\mathcal{G}''$ at its first and last crossing of $\partial B'$; the non-triviality relative to the hull B' guarantees that both crossings occur. The resulting oriented crossing arcs lift to vertical curves of the annulus $B' \setminus \overline{B}$. The two oriented lifts arising from a returning curve account exactly for the coefficient 2 in the definition of Z_V . The extremal-length lifting principle, followed by the parallel law for the oriented families, therefore bounds their total width by the vertical width of the collar. That width is the reciprocal of its annulus modulus, namely $1/c$. $\square$

Proof of the covering lemma. By Lemma 13.2,

$$W(U, \Lambda) = \frac{1}{m}, \quad W(V, B) = \frac{1}{M}.$$

The datum $B \in B'$ serves as the hull-collar in Definition 13.10; critical values inside $B' \setminus B$ are therefore marked in the generalized target geometry, not excluded by hypothesis. Lemma 13.8 and the collar assumption $c > \eta m$ give

$$Z_V \leq \frac{1}{c} < \frac{1}{\eta m} = \eta^{-1} W(U, \Lambda). \quad (13.14)$$

Substitute (13.14) into (13.10):

$$W(U, \Lambda)^2 \leq 2d^2 W(V, B) \eta^{-1} W(U, \Lambda).$$

Cancel the positive factor $W(U, \Lambda) = 1/m$ to obtain

$$\frac{1}{M} = W(V, B) \geq \frac{\eta}{2d^2 m}.$$

Equivalently,

$$M \leq 2\eta^{-1} d^2 m.$$

Since $m < \varepsilon$ is equivalent to $W(U, \Lambda) > 1/\varepsilon$, choosing $\varepsilon < K_1(D)^{-1}$ and, if necessary, decreasing it to make all near-degeneracy thresholds strict yields the stated strict inequality. Every threshold depends only on η , D , and the topology of a degree- D Galois closure; hence $\varepsilon = \varepsilon(\eta, D)$. $\square$

Theorem 13.9 (Complex a priori bounds for real quadratic renormalization). **Statement scope.** *There is a universal $\mu_2 > 0$ such that every infinitely renormalizable real quadratic polynomial $f_c(z) = z^2 + c$ has quadratic-like representatives of all sufficiently deep renormalizations $R^n f_c : U'_n \rightarrow U_n$ satisfying*

$$\text{mod}(U_n \setminus \overline{U'_n}) \geq \mu_2.$$

The representatives may be chosen real-symmetric, normalized, and with connected filled Julia set. In particular their complex geometry cannot degenerate to modulus zero down the infinite real renormalization tower.

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Complex and saddle-node pullback geometry is combined with the finite-prefix closest-return high-height argument; all real space-creation branches and the modulus extraction are proved in the chapter.

The mathematical argument begins in the proof environment below.

Proof notation and source calibration. We use the following notation only inside this proof. For a normalized real quasi-quadratic return map g , let

$$I^0 \supset I^1 \supset I^2 \supset \dots$$

be its principal nest: I^{m+1} is the component containing the critical point of the first-entry pullback of I^m . A consecutive block of central returns

$$I^{m(k)+1} \supset \dots \supset I^{m(k+1)}$$

is a *central cascade*. In a long saddle-node cascade we discard the middle levels which are farther than the maximal postcritical landing depth from either end. The *essential period* $p_e(g)$ is the number of intervals in the renormalization cycle which remain after those neglectable middle levels are removed. Thus p_e counts the genuinely different returns and does not count the long almost-parabolic waiting time repeatedly.

The real principal-nest estimates used by the complex argument are derived below from the Chapter 12 cross-ratio and Koebe theorems. The high-height branch is closed later in this proof by a finite-prefix closest-return argument, and finite propagation of a seeded modulus is proved as well. Thus the cited primary sources calibrate the architecture rather than supply an unnamed residual theorem.

Lemma 13.10 (Deep principal-nest Koebe control). *Fix $\kappa \in (0, 1)$. There is a depth m_0 such that the following holds for the principal nest of any real quadratic polynomial considered below. Let $T \supset J$ be a monotone first-entry pullback between two nice principal-nest intervals of depth at least m_0 . If the image interval contains a τ -scaled neighborhood of the image of J , then*

$$\text{Dist}(f^s|J) \leq K(\kappa, \tau),$$

where s is the entry time. The same statement holds for every suffix of the pullback chain.

Proof. Principal-nest intervals are nice: a boundary point cannot return to the interior before the critical point does, because otherwise the corresponding pullback would cross the central pullback. Their diameters tend to zero by the no-wandering/principal-nest shrinking mechanism already used in Chapter 12. Hence Theorem 12.1, applied with the chosen κ , gives lower cross-ratio distortion at least κ for every sufficiently deep monotone first-entry branch. The image-side scaled neighborhood is therefore pulled back quantitatively by the real Koebe principle, Theorem 12.13, and the derivative ratio on the central subinterval is bounded by $K(\kappa, \tau)$. A suffix is again a monotone first-entry branch between nice pullbacks and has the same cross-ratio lower bound, so the same argument applies. $\square$

Lemma 13.11 (Only long saddle-node tails lose real space). *There is an absolute $\tau_* > 0$ such that, after discarding finitely many initial principal-nest levels, I^{m+1} contains a τ_* -scaled neighborhood in I^m unless I^m lies in the tail of a long saddle-node central cascade. Moreover the diffeomorphic factor in*

$$g_m|I^m = h_m \circ (x \mapsto x^2)$$

has distortion bounded by an absolute constant on the critical image; the same remains true in the tail of a long cascade.

Proof. Normalize I^m affinely to $[-1, 1]$. For a real quadratic polynomial every inverse noncritical branch extends to the corresponding slit plane, so the inverse of the diffeomorphic factor h_m is a Herglotz map. The normalized Herglotz family is normal. Suppose first that a sequence of central children fails every fixed scaled-space bound. Passing to a normal limit gives a boundary real quasi-quadratic map for which the central child fills the parent. The real kneading order has only the two boundary central types: the Ulam-Neumann endpoint and the saddle-node endpoint. At the Ulam-Neumann endpoint the two real inverse branches separate the critical value from the parent boundary by a definite amount; this persists under small perturbation. Thus loss of scaled space can occur only near the saddle-node endpoint.

Near the saddle-node endpoint write the normalized central return in a real coordinate centered at its nearby fixed point as

$$x \mapsto x + \epsilon + \psi(x), \quad c_1 x^2 \leq \psi(x) \leq c_2 x^2.$$

If only a bounded number of central returns remains before the exit, the entry or exit interval is separated from the parabolic fixed point by a fixed fraction of the parent scale and hence has definite scaled space. Consequently failure of the scaled-space bound forces a number of remaining central iterates tending to infinity, which is precisely membership in the tail of a long saddle-node cascade.

When I^{m+1} is τ_* -well inside I^m , the inverse Herglotz branch is univalent on a slit-plane neighborhood containing a fixed hyperbolic collar. Lemma 13.10 gives a uniform distortion bound for h_m . In the tail of a central cascade, all intermediate central return maps are restrictions of the entrance return factor until the last level; at the last level the inverse branch extends across the entrance parent. Thus the same fixed Koebe collar at the entrance controls every restriction and the exit branch as well. This proves the uniform distortion assertion. $\square$

Lemma 13.12 (Essential real geometry). *For every $P < \infty$ there is $K(P)$ such that, sufficiently deep in the principal nest, if the essential period is at most P , then all non-neglectable principal-nest atoms, noncentral return domains, and adjacent gaps are $K(P)$ -commensurable. Every first-return branch factors as the quadratic critical fold followed by a diffeomorphism of $K(P)$ -bounded distortion whose inverse has a $K(P)$ -controlled Koebe extension. The same control holds at the two non-neglectable ends of every long saddle-node cascade.*

Proof. Collapse the neglectable middle of each long saddle-node cascade in the return graph. By definition of essential period, at most P essential landing/return steps remain. At a noncentral step, niceness makes the relevant pullbacks disjoint or nested. Choose the shortest image atom among the finitely many essential images. Its two neighboring atoms or boundary gaps give a definite image-side collar; pulling that collar backwards with Lemma 13.10 gives a definite source collar and bounded distortion. Repeating through at most P essential steps changes all constants by a factor depending only on P .

For a central step outside a long saddle-node tail, Lemma 13.11 supplies the required real space and bounded distortion directly. Inside a long saddle-node cascade we retain only a bounded number of entrance and exit levels in the essential graph. The parabolic normal form in the proof of Lemma 13.11 shows that these end levels have uniformly comparable Fatou-coordinate lengths; returning to the real coordinate gives commensurable entrance/exit atoms and gaps. The middle levels are precisely those deleted in the definition of essential period and are not used in any metric comparison below. Composing at most P bounded distortion essential transitions proves the asserted $K(P)$ bounds. $\square$

We next replay the part of Lyubich's Acta proof which creates complex space from small real scale. These lemmas will reduce the former "large-space" black box to one older real seed statement.

Lemma 13.13 (A small scaling factor creates large complex space). *Let $\lambda_m = |I^{m+1}|/|I^m|$. If $\lambda_m \leq \lambda$, then the next generalized renormalization has a polynomial-like extension whose principal modulus is at least $\mu(\lambda)$, where $\mu(\lambda) \rightarrow \infty$ as $\lambda \rightarrow 0$.*

Proof. Let $D(I^m)$ be the Euclidean disk with diameter I^m and pull it back by all inverse branches of the first return to I^m . Off the critical branch, Schwarz contraction in the slit plane places every pullback domain inside the diameter disk over its real return interval. Let J be the interval containing the critical value which is mapped monotonically onto I^m by the remaining return iterates. The real Koebe principle and Lemma 13.12 give

$$\frac{|J|}{\text{dist}(J, \partial I^{m-1})} = O(\lambda_m).$$

Write the critical branch as $g^{t-1} \circ h \circ (x \mapsto x^2)$, with $h : (K, L) \rightarrow (I^{m-1}, J)$ a diffeomorphism. Applying real Koebe once more gives

$$\frac{|L|}{\text{dist}(L, \partial K)} = O(\lambda_m).$$

The pullback of $D(I^m)$ by the inverse of $g^{t-1} \circ h$ is contained in $D(L)$ by Schwarz contraction. Taking the square root at the critical fold therefore gives

$$\frac{\text{diam } V^{m+1}}{|I^m|} = O(\sqrt{\lambda_m}).$$

Thus the central complex pullback lies in an $O(\sqrt{\lambda})$ -relative neighborhood of I^{m+1} while the outer disk has the parent scale. Monotonicity of annulus modulus gives $\mu(\lambda) \rightarrow \infty$. $\square$

Lemma 13.14 (Long Ulam-Neumann cascades create large space). *For every $M > 0$ there exists $L(M)$ such that the generalized renormalization at the end of any Ulam-Neumann central cascade of length at least $L(M)$ has a polynomial-like extension of modulus at least M .*

Proof. If the entrance scaling factor is already small, apply Lemma 13.13. Otherwise the entrance child is commensurable with its parent. The normalized Epstein family is

compact and a sufficiently long Ulam-Neumann cascade is therefore uniformly close to the boundary map $x \mapsto x^2 - 2$. For that map the complements $I^{m+k} \setminus I^{m+L}$ decrease exponentially with k ; compactness makes the rate uniform for all sufficiently long nearby cascades. Hence $I^{m+L-1} \setminus I^{m+L}$ is arbitrarily small relative to I^{m+L} . Pull the diameter disk $D(I^{m+L-1})$ backwards along the noncritical branch. Schwarz contraction puts the pullback inside the diameter disk over its real preimage. Lemma 13.12 factors the final critical return as a bounded-distortion diffeomorphism followed by the square map. The central complex domain is therefore $O(\sqrt{\varepsilon})$ of the outer scale when the preceding real complement is ε -small. The resulting annulus modulus tends to infinity with the cascade length. $\square$

Lemma 13.15 (Large Bernoulli return time forces a small scale). *Let a central cascade be fixed and let t be the maximal return time of the lateral intervals to its deepest retained level under the associated Bernoulli map. There is a level ℓ in that cascade for which*

$$\lambda_\ell \leq \Lambda(t), \quad \Lambda(t) \longrightarrow 0 \quad (t \rightarrow \infty).$$

Proof. If one of the lateral pullbacks is not well inside the cascade entrance, Lemma 13.11 forces the preceding central block to be a long Ulam-Neumann block; the exponential contraction used in Lemma 13.14 then gives a small scaling factor. Otherwise every lateral pullback has a definite real collar. The Bernoulli map has negative Schwarzian on each branch, so cross-ratio expansion and real Koebe transport this collar with bounded distortion through all t returns. If every scaling factor were bounded below by $a > 0$, each successive return interval would occupy at least a fixed $c(a) > 0$ fraction of its parent-side component. The t distinct landing intervals have disjoint interiors in a fixed parent, so $tc(a) \leq 1$. Thus $a \leq \Lambda(t)$ for a function tending to zero. Choosing a as the minimum scaling factor along the cascade gives the stated level ℓ . $\square$

Lemma 13.16 (A deep saddle-node landing forces a small scaling factor). *For every $\varepsilon > 0$ there exists $d(\varepsilon)$ such that the following holds sufficiently deep in the principal nest. Suppose a lateral return interval J lands at depth at least $d(\varepsilon)$ in the neglectable middle of a saddle-node cascade. Then within finitely many subsequent principal-nest levels there is a scaling factor $\lambda_\ell < \varepsilon$.*

Proof. The saddle-node normal form used in Lemma 13.11 implies that, uniformly in the length of the cascade, a level lying d steps from both ends occupies a proportion $\eta(d)$ of the entrance scale, where $\eta(d) \rightarrow 0$. Thus the landing interval satisfies

$$\frac{|J|}{\text{dist}(J, \partial I^m)} \leq C\eta(d) \tag{13.2}$$

for the cascade entrance interval I^m .

Assign to J its principal-nest rank k : the least number of successive central levels which must be crossed before the return path from J reaches a central interval. We use two elementary consequences of the real pullback machinery. First, for each fixed K , a rank- $k \leq K$ interval can be pulled monotonically through at most K nice principal-nest branches. By Lemma 13.10, the accumulated distortion is bounded in terms of

K as long as no scaling factor is already $< \varepsilon$. Pulling (13.2) back therefore yields a central child whose relative length is at most $C_K \eta(d)$. Choosing d so that $C_K \eta(d) < \varepsilon$ gives the desired small scaling factor.

Second, ranks themselves are uniformly bounded under a fixed lower bound on all scaling factors. Indeed, follow the shortest rank path and pull its lateral interval successively to the preceding central levels. At each rank step niceness plus the lower scaling bound supplies a definite cross-ratio collar, and Lemma 13.10 pulls that collar back. After k steps the resulting Poincare separation grows geometrically in k . It cannot exceed the Poincare diameter of the normalized parent unless one of the crossed central children has relative size below the prescribed ε . Hence, if all crossed scaling factors are at least ε , then $k \leq K(\varepsilon)$.

Apply the second assertion with $K = K(\varepsilon)$. Either a scaling factor $< \varepsilon$ has already appeared, or $k \leq K$ and the first assertion, with $d \geq d(\varepsilon)$, produces one. This proves the lemma. $\square$

The high-height branch is now closed by the finite-prefix closest-return argument. For readability, the full calculation is kept in a separate reader module.

High-height closest-return closure. The following closest-return argument closes the sole residual real-geometric high-height case left by the preceding complex-bounds reduction. It is the quadratic finite-prefix specialization of the closest-return decay-of-geometry mechanism, calibrated against Shen’s proof and the older Lyubich high-height argument. All cross-ratio and Koebe inputs used below are the Chapter 12 results already owned by this book.

Local setting and inherited interfaces

Let F be a real-symmetric quadratic return map on a nice interval, affinely normalized so that the critical point is 0. Inside one generalized renormalization block write

$$I^0 \supset I^1 \supset I^2 \supset \dots$$

for the principal nest and g_j for the first return map to I^j . A level is *noncentral* if the critical first return to I^j does not land in I^{j+1} . List the noncentral moments by

$$m(1) < m(2) < \dots$$

The “height” in the preceding high-height reduction counts these central cascades up to a bounded entrance/exit convention; hence a height- H block contains $H - O(1)$ noncentral separators.

Only the following already-established real interfaces are used.

- (R1) **Niceness and pullback nesting.** Pullbacks of a nice interval are nested or disjoint, and a first-entry branch is monotone until its first encounter with the critical domain.
- (R2) **Deep cross-ratio noncollapse.** For a sufficiently deep monotone pullback into

a principal-nest parent,

$$\mathcal{C}(h; T, J) \geq \Omega_k, \quad \Omega_k \longrightarrow 1$$

as the nest depth $k \rightarrow \infty$.

- (R3) **Real Koebe.** Image-side scaled space plus the preceding cross-ratio lower bound gives a distortion constant depending only on that scaled space and on Ω_k .
- (R4) **Retained-separator real space.** At a noncentral separator which is not already captured by the long Ulam-Neumann or deep saddle-node alternatives proved earlier in this chapter, consecutive retained principal-nest intervals have a definite real-space ratio. Equivalently, after discarding the two exceptional large-space mechanisms already handled elsewhere, there is $\rho > 1$ such that the parent/child ratios used below are at least ρ at every sufficiently deep retained non-central moment.

Interfaces (R2)–(R3) are Chapter 12 inputs already used in the earlier reduction. Interface (R4) is the real-space content of the preceding principal-nest lemmas after the neglectable middle of a long saddle-node cascade has been collapsed. Thus the present proof does not re-import the all-combinatorics complex-bounds theorem.

Remark 13.1 (Scope firewall). If (R4) fails because a retained level lies in one of the exceptional long-cascade configurations, the long-cascade lemmas proved earlier already supply the required large modulus or small scale. The closest-return proof is needed only in the remaining high-height branch.

Closest-return geometry

Put $c_t = F^t(0)$. A positive integer s is a *closest return time* if

$$|c_s| < |c_j| \quad (1 \leq j < s).$$

Let $s_1 < s_2 < \dots$ be the closest returns occurring before the next restrictive interval. If $c_{s_n} \in I^{m(k)} \setminus I^{m(k)+1}$, let $T_n \ni c_{s_n}$ be the maximal interval for which

$$F^{s_{n+1}-s_n} \Big|_{T_n}$$

is monotone and its image stays in $I^{m(k)-1}$. Denote the endpoints of $F^{s_{n+1}-s_n}(T_n)$ by x_n, y_n , ordered by $|x_n| \leq |y_n|$.

Lemma 13.17 (Position lemma). *At every sufficiently deep closest-return transition:*

- (i) T_n lies in one component of $I^{m(k+1)-1} \setminus \{0\}$;
- (ii) $y_n \in \partial I^{m(k)-1}$, $x_n \in I^{m(k)}$, and (x_n, y_n) does not contain 0;
- (iii) either $F^{s_{n+1}-s_n}(T_n)$ contains the symmetric interval $(c_{s_{n-1}}, -c_{s_{n-1}})$ (type I), or the return is low and there is a point w_n satisfying

$$|c_{s_n}| < |w_n| < |x_n| < |y_n|$$

together with the monotone entry-domain pullback used in Section 13.2 (type II).

Proof. This is first-return bookkeeping. If T_n crossed a boundary of $I^{m(k+1)-1}$, forward iteration through the central cascade would put a boundary point of a pullback of the nice interval $I^{m(k)-1}$ into its interior before the prescribed entry time. This contradicts niceness and proves (i). Maximality of the monotone extension then puts one image endpoint on $\partial I^{m(k)-1}$; the other lies in $I^{m(k)}$, proving (ii).

Pull $I^{m(k)-1}$ back along the last return before s_{n+1} . If this pullback crosses the critical point, the two monotone sides cover the symmetric interval determined by $c_{s_{n+1}}$, which is type I. Otherwise the pullback is monotone; nested-or-disjoint pullbacks show that the orbit segment has no earlier entry into $I^{m(k)}$. Hence c_{s_n} is the first return to that level and the return is low. Following the critical orbit through the low return supplies w_n and the displayed order. This is type II. $\square$

Each noncentral separator supplies a closest return before the next separator. Thus a principal-nest block with h retained central cascades contains at least $h - C_0$ consecutive closest-return transitions, with an absolute bookkeeping constant C_0 .

The expanding quantity

Let b_n be either endpoint of $I^{m(k+1)-1}$; real symmetry makes $|b_n|$ unambiguous. In critical order two define

$$A_n := \frac{|b_n|^2 - |c_{s_{n+1}}|^2}{|b_n|^2 - |c_{s_n}|^2}, \quad B_n := \frac{|c_{s_n}|}{|c_{s_{n+1}}|}.$$

Since s_{n+1} is a closer return, $A_n > 1$ and $B_n > 1$. Put

$$E_n := |(F^{s_n})'(c_1)| \frac{B_{n-1}}{A_{n-1}}, \quad c_1 = F(0).$$

Lemma 13.18 (Master closest-return inequality). *There are cross-ratio factors $\Omega_k \rightarrow 1$ such that*

$$\frac{E_{n+1}}{E_n} = |(F^{s_{n+1}-s_n})'(F(c_{s_n}))| \frac{A_{n-1}}{A_n} \frac{B_n}{B_{n-1}} \geq \Omega_k A_{n-1} V_n W_n, \quad (31.1)$$

where

$$V_n = \frac{2|x_n|(|y_n| + |c_{s_n}|)}{(|y_n| + |x_n|)(|x_n| + |c_{s_n}|)} = 1 + \frac{|y_n| - |x_n|}{|y_n| + |x_n|} \frac{|x_n| - |c_{s_n}|}{|x_n| + |c_{s_n}|},$$

and in type I

$$W_n = \frac{|x_n|}{|c_{s_{n-1}}|} \geq 1.$$

In type II the product $A_{n-1}W_n$ is replaced by the quantity U_n of Section 13.2.

Proof. Let L_n, R_n be the two components of $T_n \setminus \{c_{s_n}\}$. Apply the point form of the deep cross-ratio estimate to

$$F^{s_{n+1}-s_n-1} : F(T_n) \longrightarrow (x_n, y_n)$$

at $F(c_{s_n})$. The critical fold is quadratic, so the two source-side gaps enter as differences of squares. After multiplying by the derivative of the last quadratic factor and

cancelling the difference-of-squares terms, the result is exactly

$$\left| (F^{s_{n+1}-s_n})'(F(c_{s_n})) \right| \frac{A_{n-1}}{A_n} \frac{B_n}{B_{n-1}} \geq \Omega_k A_{n-1} V_n W_n$$

in the notation before cancellation. In type I, $A_{n-1} \geq 1$ and the symmetric-image condition gives $W_n \geq 1$, so the displayed estimate is exactly (31.1). In type II the low-return combinatorics instead gives the sharper inequality recorded in Section 13.2. The chain rule identifies the left side with E_{n+1}/E_n . $\square$

Type I: both geometric subcases

Fix $1 < \rho_1 < \rho$, where ρ is the retained-separator real-space constant in (R4).

Lemma 13.19 (Type I, separated parent and child). *Suppose s_n is type I and*

$$\frac{|I^{m(k)-1}|}{|I^{m(k)}|} \geq \rho_1.$$

Then there is $\eta_1 > 0$ such that for every sufficiently deep such transition

$$\Omega_k A_{n-1} V_n W_n \geq 1 + \eta_1.$$

Proof. First there is $\rho_2 > 1$ such that

$$\frac{|x_n|}{|c_{s_n}|} \geq \rho_2. \tag{31.2}$$

Indeed, $|y_n|/|c_{s_n}| \geq \rho_1$. If $|y_n|/|x_n| \leq \sqrt{\rho_1}$, (31.2) follows immediately. Otherwise x_n lies strictly inside the preceding parent. Pull the preceding parent back along the last monotone entry branch. Its image has definite real space, so (R2)–(R3) pull back a definite collar; x_n therefore cannot approach c_{s_n} . This proves (31.2).

Now $A_{n-1}, V_n, W_n \geq 1$. If their product could tend to 1, all three factors would tend to 1. Since (31.2) rules out the second factor in the product defining V_n tending to zero, $V_n \rightarrow 1$ forces $|y_n|/|x_n| \rightarrow 1$. Together with $W_n \rightarrow 1$, this gives $|y_n|/|c_{s_{n-1}}| \rightarrow 1$. But the parent/child separation by ρ_1 places $c_{s_{n-1}}$ in the parent annulus at a definite normalized distance from the boundary represented by y_n ; equivalently A_{n-1} is bounded away from 1. Contradiction. Compactness of the normalized endpoint configurations gives a uniform margin $1 + 2\eta_1$. Finally increase the depth until $\Omega_k > (1 + \eta_1)/(1 + 2\eta_1)$. $\square$

The complementary type-I geometry is the near-parabolic/high-return case. Let $G = g_{m(k-1)}$ on the relevant central branch and let ζ be its orientation-preserving fixed point farthest from 0.

Lemma 13.20 (Fixed-point multiplier gap). *There are constants $0 < \delta_* < C_* < \infty$ such that, in the type-I configuration with*

$$\frac{|I^{m(k)-1}|}{|I^{m(k)}|} < \rho_1,$$

one has

$$1 + \delta_* \leq |G'(\zeta)| \leq C_*. \quad (31.3)$$

Proof. Let M be the component of the relevant central domain minus $\{0\}$ which contains ζ . Point cross-ratio noncollapse applied to the return at ζ says that the two secant slopes on the components of $M \setminus \{\zeta\}$ cannot both approach one while the fixed point remains in the retained real-space configuration. Hence $|G'(\zeta)|$ is uniformly bounded below by $1 + \delta_*$ for some $\delta_* > 0$. For the upper bound, pull the preceding parent along the critical orbit from the critical value to the return domain. This pullback is monotone and has a definite collar by (R4); real Koebe gives bounded distortion. The derivative at ζ is therefore bounded by a fixed multiple of the parent/child ratio, which is $< \rho_1$. This gives (31.3). $\square$

Lemma 13.21 (Type I, near-parabolic branch). *There is $\eta_2 > 0$ such that every sufficiently deep type-I transition satisfying the hypothesis of Lemma 13.20 obeys*

$$\frac{E_{n+1}}{E_n} \geq 1 + \eta_2.$$

Proof. There are two positions for c_{s_n} relative to the fixed-point interval bounded by $\pm\zeta$.

If the orbit point lies on the outer side, the inverse branch of G pushes it into a component of $I^{m(k)-1} \setminus I^{m(k)}$. The return-domain extension lemma therefore enlarges T_n to an interval whose image is the full preceding parent. The proof of Lemma 13.19, with this enlarged interval, applies verbatim and gives a uniform gain.

Suppose instead that c_{s_n} lies in the inner fixed-point interval. If any of A_{n-1} , V_n , or W_n is separated from one, the master inequality gives the required gain after Ω_k is close enough to one. Thus consider the only remaining possibility: all three are arbitrarily close to one. Let b be the endpoint of $I^{m(k)}$ on the side of ζ . Real Koebe and (31.3) make G' uniformly bounded on $[\zeta, b]$, which keeps

$$\frac{|b| - |\zeta|}{|y_n| - |b|}$$

away from zero. The relation $A_{n-1} \rightarrow 1$ then forces $c_{s_{n-1}}$ into $I^{m(k)}$ with its gap to c_{s_n} negligible relative to the remaining boundary gap. The relation $W_n \rightarrow 1$ forces $x_n/c_{s_{n-1}} \rightarrow 1$. Consequently the normalized orbit segment converges to the fixed-point configuration: the next closest return is $G(c_{s_n})$, the ratios A_n and B_n/B_{n-1} tend to one, and

$$|(F^{s_{n+1}-s_n})'(F(c_{s_n}))| \rightarrow |G'(\zeta)|.$$

By (31.3) the limiting derivative is at least $1 + \delta_*$. Hence the full left side of (31.1) is at least $1 + \delta_*/2$ at all sufficiently deep levels. This gives the required η_2 . $\square$

Type II: retain the strict quantitative margins

In type II, Lemma 13.17 gives w_n with

$$|c_{s_n}| < |w_n| < |x_n| < |y_n|.$$

Set

$$U_n := A_{n-1} \frac{|x_n|}{|c_{s_{n-1}}|}, \quad \lambda := \frac{|y_n|}{|c_{s_{n-1}}|}, \quad \mu := \frac{|c_{s_{n-1}}|}{|w_n|} > 1.$$

Lemma 13.22 (Type-II pullback estimate). *For sufficiently deep type-II returns,*

$$\frac{|x_n|}{|c_{s_{n-1}}|} \geq \Omega_k \frac{|y_n| - |w_n| + 2|w_n|^2 / |c_{s_{n-1}}|}{|y_n| + |w_n|}. \quad (31.4)$$

Consequently, after using $1 - t^2 = (1 - t)(1 + t)$,

$$U_n \geq \Omega_k \frac{(\lambda\mu - 1)(\lambda\mu^2 - \mu + 2)}{\mu^3(\lambda^2 - 1)}. \quad (31.5)$$

Moreover

$$\frac{E_{n+1}}{E_n} \geq \Omega_k U_n V_n. \quad (31.6)$$

Proof. Let J be the entry domain to $I^{m(k)}$ containing $c_{s_{n-1}}$. The low-return combinatorics gives a monotone branch from J onto $I^{m(k)}$, and the subinterval between x_n and $c_{s_{n-1}}$ maps inside (c_{s_n}, w_n) , which avoids the critical point. Apply point cross-ratio non-collapse to this branch. Rearranging the four endpoint factors gives (31.4). Multiply by the difference-of-squares ratio in A_{n-1} to obtain (31.5). Substituting this sharper factor into the master derivative calculation gives (31.6). $\square$

Lemma 13.23 (Auxiliary α inequality). *Let*

$$\alpha = \frac{|x_n|}{|c_{s_n}|} > 1.$$

Then

$$\alpha^2 \geq 1 + \Omega_k \frac{\lambda\mu - 1}{2} \frac{\alpha}{\alpha - 1}. \quad (31.7)$$

Proof. Because c_{s_n} is the first return of the critical point to $I^{m(k)}$, pull the preceding parent back from the critical value along the monotone orbit segment. The image of $(0, c_{s_n})$ is contained in (c_{s_n}, w_n) while $x_n \in I^{m(k)}$. Point cross-ratio noncollapse compares the relative gap between x_n and c_{s_n} with the relative gap between y_n and w_n . The order of the four points and $|w_n| < |x_n|$ give

$$\frac{|x_n|^2}{|c_{s_n}|^2} \geq 1 + \Omega_k \frac{|y_n| - |w_n|}{2|w_n|} \frac{|x_n|}{|x_n| - |c_{s_n}|}.$$

Since $|y_n| / |w_n| = \lambda\mu$, this is (31.7). $\square$

Lemma 13.24 (Uniform type-II gain). *There is $\eta_3 > 0$ such that every sufficiently deep type-II transition satisfies*

$$\frac{E_{n+1}}{E_n} \geq 1 + \eta_3.$$

Proof. First dispose of the easy boundary regime. If $V_n \rightarrow 1$, the retained-separator real-space bound keeps $|x_n| / |c_{s_n}|$ away from one, so $|y_n| / |x_n| \rightarrow 1$. Formula (31.4) then

forces $|x_n|/|c_{s_{n-1}}| \rightarrow 1$, while the difference-of-squares factor in A_{n-1} diverges away from one. Hence $U_n V_n$ has a definite gain. We may therefore assume

$$V_n \geq 1 + \delta_V \quad (31.8)$$

for a fixed $\delta_V > 0$.

If $U_n \geq 1$, (31.6), (31.8), and $\Omega_k \rightarrow 1$ prove the result. Thus assume $U_n < 1$. Ignoring an arbitrarily small $o(1)$ loss from Ω_k for the moment, put

$$P = (\lambda\mu - 1)(\lambda\mu^2 - \mu + 2), \quad Q = \mu^3(\lambda^2 - 1).$$

Then $P \leq Q$, and direct expansion gives

$$Q - P = (\mu - 1)(2\lambda\mu - \mu^2 - \mu - 2) \geq 0.$$

Hence

$$\lambda \geq \frac{\mu^2 + \mu + 2}{2\mu} > 1.9. \quad (31.9)$$

Let

$$\theta = 2\lambda\mu - (\mu^2 + \mu + 2) \geq 0.$$

From (31.5) one obtains

$$\frac{1}{1 - U_n} \geq \frac{\theta^2 + 2(\mu^2 + \mu + 2)\theta + (\mu^2 + 3\mu + 2)(\mu^2 - \mu + 2)}{\theta} \frac{\mu}{4(\mu - 1)}, \quad (31.10)$$

up to the same arbitrarily small deep-level loss.

Subcase $\theta \leq 1$. The right side of (31.10) is at least 24, hence

$$U_n \geq \frac{23}{24} - o(1).$$

Also $\lambda\mu \geq 2$. Lemma 13.23 implies $\alpha > 3/2$ once Ω_k is sufficiently close to one. Therefore

$$V_n - 1 = \frac{|y_n| - |x_n|}{|y_n| + |x_n|} \frac{|x_n| - |c_{s_n}|}{|x_n| + |c_{s_n}|} > \frac{0.9}{2.9} \frac{0.5}{2.5} - o(1).$$

The unrounded product satisfies

$$\frac{23}{24} \left(1 + \frac{0.9}{2.9} \frac{0.5}{2.5} \right) > 1.$$

Thus there is a fixed gain $1 + \eta_{3a}$ after the depth is increased.

Subcase $\theta > 1$. Equation (31.10) is at least 13, so

$$U_n \geq \frac{12}{13} - o(1).$$

Moreover

$$\lambda > \sqrt{3} + \frac{1}{2} > 2.2, \quad \lambda\mu \geq 2.5.$$

Lemma 13.23 then gives $\alpha > 1.6$ at sufficiently deep levels. Hence

$$V_n - 1 > \frac{1.2}{3.2} \frac{0.6}{2.6} - o(1).$$

Crucially, we retain this pre-rounded bound rather than replace it by $1/12$. Indeed,

$$\frac{12}{13} \left(1 + \frac{1.2}{3.2} \frac{0.6}{2.6} \right) > 1.$$

Thus there is a definite margin $1 + \eta_{3b}$. Choose the depth so that the $o(1)$ terms and the factor Ω_k use less than half of the smaller margin. This proves the lemma with $\eta_3 = \frac{1}{2} \min(\eta_{3a}, \eta_{3b})$. $\square$

Finite-prefix closest-return expansion

Proposition 13.25 (Finite-prefix main lemma). *There are d_0 and $\sigma > 0$ such that every closest-return transition at retained principal-nest depth at least d_0 , before the next restrictive interval, satisfies*

$$E_{n+1} \geq (1 + \sigma)E_n. \quad (31.11)$$

The constants are independent of the length of the finite block.

Proof. Type I with separated parent/child is Lemma 13.19; the near-parabolic type-I alternative is Lemma 13.21; type II is Lemma 13.24. Take the minimum of the three positive gains and increase d_0 so that all cross-ratio factors Ω_k are within the tolerances used above. $\square$

This is the finite-prefix form of the published closest-return Main Lemma. Notice that no step uses what happens after the last closest return in the block.

Published deduction: derivative growth forces scale decay

The next argument uses finite-prefix closest-return expansion rather than a long-derivative compactness shortcut.

For the first return of the critical point to $I^{m(k)}$, choose the corresponding closest return time s_{n_k} . Then $n_k \geq k$. Iterating (31.11) gives constants $C > 0$ and $\Lambda > 1$ such that

$$|(F^{s_{n_k}})'(c_1)| B_{n_k-1} \geq E_{n_k} \geq C\Lambda^k, \quad (31.12)$$

because $A_{n_k-1} > 1$.

Lemma 13.26 (Derivative-to-two-level ratio). *There are constants $C_1 > 0$ and $\Lambda_1 > 1$ such that, for every sufficiently deep noncentral moment contained in the finite block,*

$$\frac{|I^{m(k)-1}|}{|I^{m(k)+1}|} \geq C_1 \Lambda_1^k. \quad (31.13)$$

Proof. Pull the preceding parent $I^{m(k)-1}$ back along the critical orbit from the critical value. By the return-domain extension and (R2)–(R3), the branch from $F(I^{m(k)+1})$ to

the parent has uniformly bounded distortion. At critical order two this gives

$$|(F^{s_{n_k}})'(c_1)| \leq K \left(\frac{|I^{m(k)}|}{|I^{m(k)+1}|} \right)^2. \quad (31.14)$$

The definition of B_{n_k-1} and the position of the preceding closest return give

$$B_{n_k-1} \leq \frac{|I^{m(k)-1}|}{|I^{m(k)+1}|}. \quad (31.15)$$

Each ratio on the right of (31.14) is at most the two-level ratio in (31.15). Combining (31.12)–(31.15),

$$C\Lambda^k \leq K \left(\frac{|I^{m(k)-1}|}{|I^{m(k)+1}|} \right)^3.$$

Thus (31.13) holds with $\Lambda_1 = \Lambda^{1/3}$ after adjusting the constant. $\square$

Lemma 13.27 (Two-level growth forces one-step decay). *There is $C_2 > 0$ such that*

$$\frac{|I^{m(k)}|}{|I^{m(k)+1}|} \geq C_2 \left(\frac{|I^{m(k)-1}|}{|I^{m(k)}|} \right)^{1/2}. \quad (31.16)$$

Consequently there are $C_3 > 0$ and $\Lambda_2 > 1$ such that

$$\frac{|I^{m(k)}|}{|I^{m(k)+1}|} \geq C_3 \Lambda_2^k. \quad (31.17)$$

Proof. Use the same branch as in Lemma 13.26, now applying point cross-ratio non-collapse to the pair consisting of $F(I^{m(k)+1})$ and its monotone extension. After the quadratic fold is factored out, the four endpoint terms give (31.16). This is the critical-order-two specialization of the standard cross-ratio step converting a two-level estimate to a one-level scaling estimate.

Put

$$a_k = \frac{|I^{m(k)-1}|}{|I^{m(k)}|}, \quad r_k = \frac{|I^{m(k)}|}{|I^{m(k)+1}|}.$$

Then (31.16) says $r_k \geq C_2 a_k^{1/2}$, hence $a_k \leq (r_k/C_2)^2$. Therefore

$$\frac{|I^{m(k)-1}|}{|I^{m(k)+1}|} = a_k r_k \leq C_2^{-2} r_k^3.$$

Combine this with (31.13):

$$r_k^3 \geq C_2^2 C_1 \Lambda_1^k.$$

Taking cube roots proves (31.17) with $\Lambda_2 = \Lambda_1^{1/3} > 1$. $\square$

Proposition 13.28 (Finite-prefix decay at noncentral moments). *There exist constants $C_* > 0$, $\Lambda_* > 1$, and a startup index k_0 such that every finite principal-nest block*

extending through $K \geq k_0$ retained noncentral moments satisfies

$$\frac{|I^{m(k)}|}{|I^{m(k)+1}|} \geq C_* \Lambda_*^{k-k_0} \quad (k_0 \leq k \leq K). \quad (31.18)$$

All constants are independent of K .

Proof. The proof of Proposition 13.25 and Lemmas 13.26–13.27 is local to the orbit segment ending at the indicated closest return. Hence the same constants apply simultaneously to every prefix of a longer finite block. Reindexing (31.17) after the fixed startup depth gives (31.18). $\square$

The high-height seed and repaired Lemma 13.17

Proposition 13.29 (High height creates a small scale). *For every $\varepsilon > 0$ there is $H(\varepsilon) < \infty$ such that a sufficiently deep principal-nest block of height at least $H(\varepsilon)$ contains a level j with*

$$\frac{|I^{j+1}|}{|I^j|} < \varepsilon, \quad (31.19)$$

unless one of the earlier long-cascade lemmas has already produced the required large-space conclusion. The level j occurs before the next restrictive interval, up to the fixed entrance/exit buffer used in the definition of essential period.

Proof. In the high-height branch not already captured by the two exceptional cascade mechanisms, height H supplies at least $H - C_0$ retained noncentral moments. Proposition 13.28 gives at the k th such moment

$$\frac{|I^{m(k)+1}|}{|I^{m(k)}|} \leq C_*^{-1} \Lambda_*^{-(k-k_0)}.$$

Choose $H(\varepsilon)$ so that the right side is $< \varepsilon$ for some $k \leq H - C_0$. This is (31.19). $\square$

Lemma 13.30 (Large essential period creates large space). *For every $M > 0$ there is $P(M)$ such that $p_e(g) > P(M)$ implies that Rg has an unbranched quadratic-like representative of modulus at least M .*

Proof. The return graph with the neglectable middle of long saddle-node cascades removed has large essential period only if at least one of four quantities is large: the number of central cascades (height), a Bernoulli return time in a cascade, the length of an Ulam–Neumann cascade, or the depth at which a lateral interval lands in a saddle-node cascade. This is the elementary four-case decomposition of the reduced return graph.

In the height case, Proposition 13.29 gives an arbitrarily small scaling factor. In the deep-saddle-landing case, Lemma 13.16 gives one. In the large-return-time case, Lemma 13.15 gives one. In the long Ulam–Neumann case, Lemma 13.14 already gives the required modulus. Finally apply Lemma 13.13 in the first three cases and choose all thresholds so that the resulting modulus is at least M . $\square$

Lemma 13.31 (Jump-or-hyperbolic-disk alternative). *Fix $P < \infty$. There exists $\varepsilon = \varepsilon(P) > 0$ with the following property. Let*

$$J = J_0, J_{-1}, \dots, J_{-s} = J'$$

be a monotone pullback occurring in a return block governed by Lemma 13.12, and let $z = z_0, z_{-1}, \dots, z_{-s} = z'$ be the corresponding backward orbit in the slit plane. Then either at some time i the point z_{-i} sees J_{-i} at angle at least ε , or z' belongs to a hyperbolic disk $D_\theta(H_s(J'))$ about the terminal monotonicity interval, where $\theta < \pi/2$ depends only on P .

Proof. First consider a quadratic branch. Write it as $q(x) = x^2 + a$ after real affine normalization. If no inverse image makes an ε -jump, at each pullback the point remains in one of the two 2ε wedges based at the corresponding interval endpoint. Follow the endpoint of the maximal monotonicity interval until the first time its pullback reaches the critical point. At that time the point lies in a right triangle whose acute angle is at most ε . Such a triangle is contained in a hyperbolic disk $D_{\pi/2 - O(\varepsilon)}(H)$ based on the full monotonicity interval H . Every earlier inverse branch is holomorphic between the corresponding slit planes, so Schwarz contraction of the Poincare metric carries this disk back to the terminal one.

For a quasi-quadratic Epstein branch, Lemma 13.12 writes the branch as the quadratic folding followed by a diffeomorphism whose inverse has a uniform Koebe extension. Koebe distortion changes the wedge angle by a factor bounded in terms of $K(P)$ only. Choosing $\varepsilon(P)$ small enough gives the same conclusion with one fixed $\theta = \pi/2 - O_P(\varepsilon) < \pi/2$. $\square$

Lemma 13.32 (Uniform transit through a long saddle-node cascade). *Fix $P < \infty$. Consider a saddle-node central cascade generated by a normalized return map h , and a backward orbit of an interval through the cascade. Once the cascade length is larger than a constant depending only on P , the corresponding backward orbit of every point in the unit hyperbolic disk has the following alternative: either it enters the hyperbolic disk over the entrance interval of the cascade, or at one of the first or last $O_P(1)$ levels it makes an $\varepsilon(P)$ -jump at distance at most $C(P)$ times the entrance scale.*

Proof. Suppose otherwise along a sequence of longer and longer saddle-node cascades. Normalize the entrance interval. Lemma 13.12 gives a compact family of return maps on a fixed real neighborhood. A convergent subsequence has a limit h_* for which the critical orbit does not leave the central interval under any finite number of iterates. Kneading order leaves only the two boundary quadratic types; the saddle-node orientation rules out the $z^2 - 2$ endpoint. Thus h_* has the parabolic real type of $z^2 + 1/4$.

Let $\phi = h^{-1}$ be the inverse branch following the cascade on one side of the critical point. It preserves the upper half-plane. For the limiting parabolic map the Denjoy-Wolff point of ϕ_* is the parabolic fixed point. For sufficiently long finite cascades, ϕ has a nearby attracting fixed point in the same half-plane closure. Denjoy-Wolff convergence is uniform on compact subsets for this compact family. Hence a fixed compact

subset of the normalized upper half-plane is carried after a uniformly bounded number of inverse iterates into the hyperbolic disk over a deep central interval.

The complement of that compact set inside the initial unit hyperbolic disk consists of two controlled boundary wedges. If the backward point enters one of them before reaching the deep disk, the angle with the corresponding real interval is bounded below and its Euclidean distance is bounded by a fixed multiple of the entrance scale. This is precisely the asserted good jump. The same argument in the lower half-plane follows by conjugation symmetry. $\square$

Lemma 13.33 (Good-jump induction through an essentially bounded block). *Fix $P < \infty$. Consider a block of consecutive renormalization levels for which every essential period is at most P . There exist $\varepsilon(P) > 0$ and $C(P) < \infty$ such that the backward orbit associated with one complete return cycle has the following alternative at each principal-nest level:*

1. *it makes an $\varepsilon(P)$ -jump at a good moment, meaning that the corresponding return interval is $C(P)$ -commensurable with the initial one, and the jump occurs at normalized distance at most $C(P)$; or*
2. *it passes to the hyperbolic disk over the next principal-nest level.*

The induction may be iterated from one renormalization level to the next.

Proof. We follow the four return configurations separately.

For consecutive returns to the outer interval B before the first return to its central subinterval A , the number of returns is at most P . Lemma 13.12 places the terminal monotonicity interval uniformly inside B . Lemma 13.31 therefore says that either a jump occurs during this finite block, or the terminal point lies in the hyperbolic disk over B . Until the first jump, the square-root and Koebe estimates multiply by at most a constant depending on P , which gives the normalized distance bound.

Next consider the first return to a central level. If the jump occurs near one of the two ends of the central cascade, Lemma 13.12 says that the corresponding interval is commensurable with the entrance scale, so the moment is good. If a first jump occurs deep in the middle of a saddle-node cascade, apply Lemma 13.32. It either replaces the bad jump by a good jump at one end of the cascade, or carries the point into the hyperbolic disk over the preceding principal-nest interval. Pulling once more by the Markov branch of the cascade gives the second alternative of the lemma.

For later returns to a central level, use the Markov interval which maps homeomorphically onto the preceding principal-nest interval. Lemma 13.12 puts this Markov interval well inside its parent and puts the critical value uniformly away from its endpoints. Schwarz contraction and the square-root estimate therefore give the same jump-or-disk alternative. If the resulting jump lies deep in a saddle-node cascade, the preceding paragraph again moves it to a good end level.

Finally, to pass from one renormalization level to the next, distinguish relative period greater than two from immediate doubling. In the first case the number of non-neglectable returns before entering the next central domain is at most $2P$, so the preceding finite-return argument applies. In the doubling case the two components outside the central interval each contain a preimage of the critical point. Compact-

ness of the normalized real family away from the cusp puts those preimages a definite relative distance from the endpoints; Lemma 13.31 applies directly. Since every level of an infinite renormalization tower is itself renormalizable, the maps under consideration stay away from the attracting-cusp alternative.

These four cases are exactly the induction steps, so iteration gives the claim over the entire block. $\square$

Lemma 13.34 (Linear inverse growth and the quadratic return estimate). *Under the hypotheses of Lemma 13.33, let $F_k = f^{q_k}$ be the return map on the level- k restrictive interval P_k . Write*

$$F_k = \psi_k \circ f$$

where ψ_k is univalent on a neighborhood of the critical value. After affinely normalizing P_k and its image scale, there are $C = C(P)$ and $c = c(P) > 0$ such that

$$\text{dist}(\psi_k^{-1}z, P_k) \leq C \text{dist}(z, P_k)$$

on the controlled hyperbolic disk used above, and consequently

$$\frac{\text{dist}(F_k z, P_k)}{|P_k|} \geq c \left(\frac{\text{dist}(z, P_k)}{|P_k|} \right)^2 \quad (13.1)$$

whenever z belongs to the corresponding inverse-return domain.

Proof. Pull z backwards along the return cycle. If the good-jump alternative of Lemma 13.33 occurs, normalize the source and target intervals of that branch to unit size. Their real Koebe space and the jump angle are bounded below by constants depending only on P . Koebe's theorem for the inverse branch therefore bounds the normalized terminal distance by a constant multiple of the normalized initial distance. If no jump occurs, the induction terminates in a hyperbolic disk over a principal-nest interval which is commensurable with the starting scale by Lemma 13.12, and the same estimate follows directly. This proves the linear bound for ψ_k^{-1} .

Near the critical point the map f is the quadratic folding followed by a bounded-distortion diffeomorphism. Thus its inverse has square-root growth: normalized distance is changed by at most a constant times the square root. Applying this to $F_k = \psi_k \circ f$ and squaring the preceding inverse estimate gives (13.1). $\square$

Lemma 13.35 (Quadratic-like thickening for bounded essential period). *For every $P < \infty$ there are $N(P)$ and $K(P)$ such that, after $N(P)$ consecutive levels with essential period at most P , the return map $F_k : P_k \rightarrow P_k$ admits a real-symmetric quadratic-like extension*

$$F_k : U'_k \longrightarrow U_k$$

whose connected filled Julia set has diameter comparable with $|P_k|$.

Proof. Choose a fixed hyperbolic disk $D(T_{k-s})$ a bounded number $s = s(P)$ of levels above P_k . Lemma 13.12 makes T_{k-s} a definite multiple of P_k . Let U_k be the component of this disk in the slit plane on which the return factor is defined, and let U'_k be the component containing the critical point of $F_k^{-1}(U_k)$.

For a sufficiently large fixed normalized radius r , (13.1) implies $|F_k z| > 2|z|$ whenever z is on the nonreal part of the boundary at normalized distance r from P_k . Hence $\partial U'_k$ cannot meet that boundary. It cannot meet the real slits either, because $\partial U'_k \cap \mathbb{R}$ is the real return domain. Therefore $U'_k \subseteq U_k$, and $F_k : U'_k \rightarrow U_k$ is proper of degree two.

The critical point never escapes this return domain, so the filled Julia set is connected. The upper diameter bound follows from $K(F_k) \subset U_k$. For the lower bound, the two real fixed/periodic points which delimit the restrictive interval belong to the filled Julia set and, by Lemma 13.12, are separated by a definite fraction of $|P_k|$. Thus $\text{diam } K(F_k) \asymp_P |P_k|$. $\square$

Lemma 13.36 (Definite sector confinement). *Under the hypotheses of Lemma 13.35, there exists $\vartheta(P) > 0$ such that*

$$K(F_k) \subset D_{\vartheta(P)}(B_k),$$

where B_k is the symmetric real interval bounded by the two fixed points of the normalized return map and $D_{\vartheta}(B_k)$ is the corresponding hyperbolic disk in the slit plane.

Proof. Take $\zeta \in K(F_k)$ and pull it backwards once around the return cycle. If a good jump occurs, the jump interval is commensurable with B_k and the jump distance is bounded by Lemma 13.33. Schwarz contraction for the subsequent inverse branches and the square-root estimate then place ζ in a hyperbolic disk $D_{\vartheta_1}(S_k)$ with $\vartheta_1 = \vartheta_1(P) > 0$.

If no jump occurs, Lemma 13.33 carries the backward orbit into the next principal nest disk; repeating once more gives the same conclusion. Hence $K(F_k) \subset D_{\vartheta_1}(S_k)$.

It remains to replace the maximal unimodal interval S_k by B_k . Lemma 13.12 gives definite real space between B_k and S_k except in the neglectable middle of a saddle-node cascade, and Lemma 13.32 controls precisely that middle. A bounded number of inverse pullbacks therefore sends $D_{\vartheta_1}(S_k)$ into $D_{\vartheta_2}((1+\delta)B_k)$ for fixed $\delta > 0$. Finally the repelling fixed point on ∂B_k has a linearizing neighborhood of size comparable with $|B_k|$ uniformly in the normalized compact real family. In the linearizing coordinate, further inverse iterates are contractions, so the $(1+\delta)B_k$ disk is pulled into $D_{\vartheta(P)}(B_k)$. The constants depend only on P . $\square$

Lemma 13.37 (Sector confinement gives a definite fundamental annulus). *For every $P < \infty$ there exists $\mu(P) > 0$ such that the quadratic-like extension in Lemma 13.35 can be restricted so that*

$$\text{mod}(U_k \setminus \overline{U'_k}) \geq \mu(P).$$

Proof. Affinely normalize $B_k = [-1, 1]$. By Lemma 13.12 the surrounding slit interval contains a fixed larger interval $[-1-\tau, 1+\tau]$, with $\tau = \tau(P) > 0$. Lemma 13.36 puts the connected filled Julia set in the fixed compact set $D_{\vartheta(P)}([-1, 1])$ of that slit plane. Consequently the doubly connected domain between this compact set and the slit boundary has modulus bounded below by a positive number depending only on P ; this is immediate after uniformizing the normalized slit plane, or by monotonicity of extremal length.

Uniformize $U \setminus K(F_k)$ by a round annulus $1 < |w| < R$. Choose two concentric circles whose logarithmic radii divide the available annular modulus into three equal parts. Their preimages bound Jordan disks $V' \Subset V$ containing $K(F_k)$. Properness near the filled Julia set and reflection across the inner boundary show that, after choosing the inner circle at a regular value, $F_k : V' \rightarrow V$ is still a proper degree 2 restriction. The middle third of the round annulus is a fundamental annulus for this restriction, so its modulus is at least one third of the uniform lower bound just obtained. This gives $\mu(P) > 0$. $\square$

Lemma 13.38 (Complex bounds on a bounded-essential-period block). *For every $P < \infty$ there are $N(P)$ and $\mu(P) > 0$ such that the following holds. Let G be a normalized real quadratic or real-symmetric quadratic-like map whose successive renormalizations $G, RG, \dots, R^{m-1}G$ satisfy $p_e(R^j G) \leq P$. If $m \geq N(P)$, then $R^m G$ has a quadratic-like representative of modulus at least $\mu(P)$. The constant is independent of the lengths of the saddle-node cascades.*

Proof. After the standard affine normalization, the real return maps lie in the Epstein class supplied by Lemma 13.12. Lemmas 13.33–13.37 apply with constants depending only on P . The long central cascades cause no loss, because their neglectable middle is handled by Lemma 13.32; all metric comparisons are made only at good moments near the ends. Taking $N(P)$ large enough for the initial real normalization and for Lemma 13.35 gives the claim. $\square$

Lemma 13.39 (Finite propagation of a seeded modulus through an essentially bounded block). *For every $P < \infty$, $\rho > 0$, and integer $L \geq 1$, there exists $\delta(P, \rho, L) > 0$ with the following property. Let G be a normalized real-symmetric quadratic-like germ with $\text{mod } G \geq \rho$. Suppose that $G, RG, \dots, R^r G$ are defined, $r \leq L$, and*

$$p_e(R^j G) \leq P \quad (0 \leq j < r).$$

Then $R^r G$ has a real-symmetric quadratic-like representative of modulus at least $\delta(P, \rho, L)$.

Proof. It is enough to prove a one-step assertion with a lower bound $\delta_1(P, \rho) > 0$ and then iterate it at most L times. Choose a representative $G : U' \rightarrow U$ with $\text{mod}(U \setminus \overline{U'}) \geq \rho$. After the standard real affine normalization, a fixed subannulus of this fundamental annulus supplies a complex collar of size depending only on ρ around the real filled-Julia interval. Thus the first hyperbolic disk used in Lemmas 13.31–13.35 is available at the initial scale; no earlier principal-nest levels are needed.

Now assume $p_e(G) \leq P$. Lemma 13.12 gives the real Koebe extensions at every non-neglectable return, while Lemma 13.32 crosses the arbitrarily long neglectable middle of a saddle-node cascade without loss depending on its length. Starting from the ρ -collar, the proof of Lemma 13.33 therefore gives the same jump-or-hyperbolic-disk alternative with constants depending only on (P, ρ) . The inverse-growth and quadratic-growth argument of Lemma 13.34 then produces a proper degree-two restriction for RG , and the sector-confinement/modulus extraction of Lemmas 13.36 and 13.37 leaves

a fundamental annulus of modulus

$$\text{mod}(RG) \geq \delta_1(P, \rho) > 0.$$

All constructions commute with complex conjugation.

Define $\rho_0 = \rho$ and $\rho_{j+1} = \delta_1(P, \rho_j)$ for $0 \leq j < L$. Every ρ_j is positive. Applying the one-step assertion successively gives

$$\text{mod}(R^L G) \geq \min_{0 \leq j \leq L} \rho_j =: \delta(P, \rho, L) > 0,$$

which proves the lemma. $\square$

Proof. Let $f_c(z) = z^2 + c$, $c \in \mathbb{R}$, be infinitely renormalizable. We now combine the preceding internal real and complex estimates to prove the theorem.

Choose $P = P(1)$ from Lemma 13.30. Thus every level j with $p_e(R^j f_c) > P$ produces

$$\text{mod } R^{j+1} f_c \geq 1.$$

Let $N = N(P)$ and $\mu_b = \mu(P)$ be supplied by Lemma 13.38. Put

$$\mu_s = \delta(P, 1, N) > 0$$

using Lemma 13.39, and set

$$\mu_2 = \min\{\mu_b, \mu_s\} > 0.$$

Consider any sufficiently deep level n . If among the preceding N levels there is an index j with $p_e(R^j f_c) > P$, choose the largest such j . The big-space estimate gives modulus at least 1 at level $j + 1$. There are at most N further renormalizations before level n , and by the maximal choice of j all their essential periods are at most P . Apply Lemma 13.39 to the seeded map $R^{j+1} f_c$ to obtain $\text{mod } R^n f_c \geq \mu_s$.

If no such j exists, the last N essential periods are all at most P . Apply Lemma 13.38 to the map at the beginning of that block. It gives $\text{mod } R^n f_c \geq \mu_b$. Therefore every sufficiently deep level has modulus at least μ_2 .

All constructions commute with complex conjugation, so the representatives may be chosen real-symmetric. The critical point remains in every restrictive domain, hence every filled Julia set is connected. Finally, affine normalization of the critical point and external scale preserves proper degree and annulus modulus. This proves the theorem with the high-height case closed by the closest-return argument; no additional real-geometric theorem is required. $\square$

Theorem 13.40 (Compactness of normalized fixed-modulus germs). **Statement scope.** *For every $\mu > 0$, the naturally marked normalized quadratic-like germ space $\mathcal{C}(\mu)$ is compact in the Carathéodory/germ topology. Consequently any sequence of normalized renormalizations with modulus bounded below by μ admits a convergent subsequence whose limit is again a quadratic-like germ with connected filled Julia set and the same normalization.*

Proof strategy (not part of the proof)**Proof status.** proved here.**Strategic outline only.** Uniform collar geometry; controlled straightening representatives; quasiconformal normal families; Carathéodory/germ compactness.*The mathematical argument begins in the proof environment below.*

Proof. Take $F_k \in \mathcal{C}(\mu)$ and choose representatives $F_k : U'_k \rightarrow U_k$ with modulus at least μ . Uniformize each fundamental annulus. The positive modulus prevents its two boundary components from colliding. The critical normalization and external marking remove affine and rotational escape, so Koebe estimates place all representatives in one normalized geometric region.

Straightening gives normalized quadratic polynomials with connected Julia sets. Their parameters lie in the compact Mandelbrot set. Because the interpolation annuli have modulus at least μ , the straightening maps may be chosen with quasiconformal constant depending only on μ . Normal family compactness therefore gives a subsequence of the parameters, the straightening maps, and their inverses that converges locally uniformly. Conjugating back gives a limit quadratic-like germ.

Properness, degree two, the marked critical point, and the external marking pass to the limit. Carathéodory convergence of the domains follows from the same collar compactness, and a slightly smaller common annulus preserves a positive modulus bound. Thus the limit still lies in $\mathcal{C}(\mu)$ (after choosing an admissible representative). Every sequence has a convergent subsequence, so the normalized fixed-modulus germ space is compact. $\square$

Theorem 13.41 (Beau bounds for real-symmetric quadratic-like germs). **Statement scope.** *There exists $\mu_0 > 0$ such that for every $\delta > 0$ there are $\mu(\delta) > 0$ and $N(\delta)$ with the following property. If $F \in \mathcal{C}^{\mathbb{R}}(\delta)$ is infinitely renormalizable, then*

$$R^n F \in \mathcal{C}(\mu(\delta)) \quad (n \geq 0), \quad R^n F \in \mathcal{C}(\mu_0) \quad (n \geq N(\delta)).$$

Thus the initial lower bound may depend on the starting modulus, whereas the deep bound is universal on the real-symmetric slice.

Proof strategy (not part of the proof)**Proof status.** proved here.**Strategic outline only.** Use the compact fixed-modulus real class, Chapter 12 cross-ratio and Koebe geometry, the high-height closest-return argument, and the large-essential-period/bounded-block modulus engine to obtain the all-level bound and the universal deep beau bound.*The mathematical argument begins in the proof environment below.*

Proof. We keep the order of quantifiers visible throughout. The point is not merely to prove that each individual real-symmetric germ eventually acquires a definite modulus. We must obtain one entry time $N(\delta)$ valid for the whole class $\mathcal{C}^{\mathbb{R}}(\delta)$ and, after that time, one modulus μ_0 independent of δ .

Step 1: a fixed starting modulus gives a compact real class. Fix $\delta > 0$. By Theorem 13.40, normalized germs in $\mathcal{C}^{\mathbb{R}}(\delta)$ admit representatives

$$F : U' \longrightarrow U, \quad \text{mod}(U \setminus \overline{U'}) \geq \delta/2,$$

whose filled Julia sets and fundamental collars have uniformly controlled Euclidean geometry. The same theorem, through the controlled straightening construction, gives a number $K_\delta < \infty$ such that these representatives are hybrid conjugate to real quadratic polynomials by normalized K_δ -quasiconformal maps on a neighborhood of uniform normalized size. Restricting to the real line and affinely normalizing the critical interval, we obtain a compact family $\mathcal{Q}_\delta$ of real-analytic quasi-quadratic maps with a quadratic critical point, uniform analytic extension, and uniform C^3 bounds away from the critical point.

The only use of straightening in this step is compactness and normalization: we do not use any renormalization contraction or any form of beau bounds.

Proof of the uniform real-entry lemma. We replay the real-bounds proof of Chapter 12 uniformly on $\mathcal{Q}_\delta$. The cross-ratio estimate Theorem 12.1 and the real Koebe principle Theorem 12.13 depend only on uniform C^3 control, quadratic nonflatness, and intersection multiplicity of the pullback chain; all three are uniform on the compact family $\mathcal{Q}_\delta$. Niceness of the restrictive intervals gives the same bounded-multiplicity pullback chains as in the proof of Chapter 12 real bounds.

For completeness, the uniform-entry argument is the following. If no $\lambda(\delta)$ worked at the shallow levels, we could choose $F_j \in \mathcal{Q}_\delta$ and a first bad restrictive configuration. After normalizing its parent interval, compactness and the cross-ratio estimate would give a limiting quadratic critical return in which either a child fills its parent or an adjacent gap collapses. Maximality of the restrictive interval then gives a homterval, or a neutral/attracting trapping interval. The first is excluded by the no-wandering theorem and the second by the existence of all later restrictive returns. Hence the first assertion is uniform on $\mathcal{Q}_\delta$.

For the deep assertion suppose, to the contrary, that no absolute λ_* and uniform entry time exist. Choose $F_j \in \mathcal{Q}_\delta$ and levels $n_j \rightarrow \infty$ at which the normalized real geometry first leaves an arbitrary fixed absolute real-bounds class. Repeat the preceding normalization at level n_j . Deep cross-ratio distortion tends to one uniformly on bounded-multiplicity pullback chains, so the limiting bad configuration again cannot lose a gap except through a long almost-parabolic central cascade. In that remaining case the closest-return expansion proved in the high-height closest-return module (used in Theorem 13.9) gives, within a bounded number of essential steps, either a definite exit gap or a small scaling factor. Both alternatives restore an absolute scaled neighborhood by real Koebe. This contradicts the choice of n_j . Thus an absolute λ_* is reached after $N_r(\delta)$ levels. This is the compact-family form of the classical real-entry lemma used by Lyubich–Yampolsky; here its proof ingredients are already present in Chapters 12–13. This proves the uniform real-entry lemma.

Step 2: the internal complex-bounds engine is uniform on a fixed real class. Inspect the proof of Theorem 13.9. Once the real Koebe and principal-nest constants are fixed, every constant in its complex part is uniform: the small-scaling-factor lemma, Ulam–Neumann and saddle-node space creation, the large-essential-period reduction,

jump induction, sector confinement, quadratic-like thickening, and modulus extraction all depend only on those real constants and on the critical degree two.

We record the exact consequence that will be used. If a real-symmetric renormalization tower has real-geometry constant at least $\lambda > 0$, then there exists

$$b(\lambda) > 0$$

such that every level of that tower has a quadratic-like representative of modulus at least $b(\lambda)$. Moreover, for the absolute class λ_* there is an absolute number

$$b_* := b(\lambda_*) > 0.$$

Here is the finite-prefix check, which is where the previous proof used to hide a quantifier. Choose $P = P(1)$ from the large-essential-period lemma in Theorem 13.9 and let $L = L(P, \lambda)$ be the bounded-essential block length for the real class λ . If a block contains an essential period larger than P , the large-space lemma seeds modulus at least 1. If not, the bounded-essential block lemma supplies a positive modulus depending only on (P, λ) . Between a seeded level and the end of a block there are at most L steps, and the finite-modulus-propagation lemma in Theorem 13.9 keeps a positive lower bound through all of them. Taking the minimum of these finitely many explicit lower bounds gives $b(\lambda) > 0$. No map-dependent entrance time appears in this argument once λ is fixed.

Step 3: all-level persistence. Apply the uniform real-entry lemma on a compact analytic family with the fixed starting modulus δ . Every real renormalization has geometry at least $\lambda(\delta)$, so Step 2 gives

$$\text{mod}(R^n F) \geq b(\lambda(\delta)) \quad (n \geq 0).$$

Set

$$\mu(\delta) := b(\lambda(\delta)) > 0.$$

This proves the first, all-level, assertion uniformly for the whole class $\mathcal{C}^{\mathbb{R}}(\delta)$.

Step 4: universal deep entry. For $n \geq N_{\text{r}}(\delta)$ the same tower belongs to the absolute real class with constant λ_* . Step 2 therefore gives

$$\text{mod}(R^n F) \geq b_* \quad (n \geq N_{\text{r}}(\delta)).$$

Set

$$\mu_0 := b_*, \quad N(\delta) := N_{\text{r}}(\delta).$$

The number μ_0 depends only on the universal real-bounds class and the quadratic critical degree, hence is independent of both F and δ . This is exactly the beau quantifier package

$$\exists \mu_0 > 0 \forall \delta > 0 \exists \mu(\delta) > 0 \exists N(\delta)$$

with

$$R^n F \in \mathcal{C}(\mu(\delta)) \quad (n \geq 0), \quad R^n F \in \mathcal{C}(\mu_0) \quad (n \geq N(\delta)).$$

Thus the theorem is proved without importing Avila-Lyubich Theorem 6.1 as a

research-level black box. Its primary-source role is now calibration of the same real-entry/modulus-engine architecture. $\square$

Theorem 13.42 (Pullback and hybrid-rigidity criterion from complex bounds). **Statement scope.** *Let F and G be infinitely renormalizable quadratic-like germs whose corresponding renormalization towers have a common modulus lower bound $\mu > 0$ and the same return combinatorics. Suppose that at some sufficiently deep level there is a K_0 -quasiconformal matched external conjugacy compatible with the first-return maps. Then it pulls back to a $K(\mu, K_0)$ -quasiconformal conjugacy on neighborhoods of $K(F)$ and $K(G)$. Its Beltrami coefficient on $K(F)$ is F -invariant. If F carries no invariant measurable line field on $K(F)$, the conjugacy can be chosen hybrid. The theorem does not assert that bounded combinatorics by itself produces the initial external matching; that is a later rigidity theorem.*

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Controlled annular extension; holomorphic pullback without dilatation growth; QC compactness; invariant limiting Beltrami coefficient; no-line-field hybrid upgrade.

The mathematical argument begins in the proof environment below.

Proof. Start with the matched K_0 -quasiconformal external conjugacy at a deep level. The common modulus lower bound gives fundamental annuli with uniform collars. After uniformizing those annuli, the boundary conjugacy extends across them with a quasiconformal constant K_1 depending only on μ and K_0 ; the external marking fixes the twisting ambiguity.

Now pull the conjugacy backward through corresponding first-return branches. Holomorphic pullback does not increase the essential supremum of a Beltrami coefficient, so every stage remains K_1 -quasiconformal. Complex bounds ensure that the nested polynomial-like domains do not collapse. After normalizing three marked points, quasiconformal compactness gives a limiting map h on a neighborhood of the filled Julia set, and the conjugacy equation passes to the limit.

On $K(F)$, the limiting Beltrami coefficient ν satisfies $F^*\nu = \nu$ almost everywhere, because the conjugacy was obtained by arbitrarily deep pullback. If ν were nonzero on a positive-measure invariant subset, $\nu/|\nu|$ would give an invariant measurable line field. Under the stated no-line-field hypothesis this is impossible, so $\nu = 0$ almost everywhere on $K(F)$. The conjugacy is therefore hybrid.

The argument starts from matched external data; it does not prove that common bounded combinatorics alone supplies that data. $\square$

Lemma 13.43 (Canonical horizontal lamination: maximality and pullback functoriality). *Let U be a Jordan disk and let $A, B \subset \partial U$ be disjoint boundary arcs. Among all measurable laminations by pairwise disjoint cross-cuts of U joining A to B , there is a canonical extremal lamination $\mathcal{F}_{\text{can}}(U; A, B)$ whose width equals the extremal width $W(U; A, B)$. If $g : \tilde{U} \rightarrow U$ is univalent and $\tilde{A}, \tilde{B}$ are the lifted arcs, then*

$$g^{-1}\mathcal{F}_{\text{can}}(U; A, B) = \mathcal{F}_{\text{can}}(\tilde{U}; \tilde{A}, \tilde{B})$$

up to a zero-width sublamination. If a canonical lamination is restricted by deleting a measurable family of leaves, the width of what remains is maximal among laminations carried by the same remaining homotopy classes.

Proof. Uniformize the quadrilateral $(U; A, B)$ by a Euclidean rectangle $Q = [0, 1] \times [0, W]$ so that A and B correspond to the vertical sides. The horizontal foliation $[0, 1] \times \{y\}$ is extremal for the family joining these sides. Its transverse parameter is $y \in [0, W]$; since the crossing extremal length is $1/W$, the extremal width is W . Pulling this foliation back gives the canonical lamination. Conformal naturality proves the univalent pullback assertion. For the last claim, delete the set E of horizontal parameters corresponding to the removed leaves. Any competing lamination in the surviving homotopy classes pushes to a curve family in $[0, 1] \times ([0, W] \setminus E)$; the standard extremal-length inequality against the Euclidean metric gives width at most $|[0, W] \setminus E|$, attained by the surviving horizontal leaves. Thus the restriction is again maximal in its carried classes. $\square$

Lemma 13.44 (Two-gate pull-off inequality). *Let $\mathcal{F}$ be a canonical horizontal lamination of width Z in a quadrilateral. Split it into a persistent part $\mathcal{F}_{\text{per}}$ of width Z_{per} and a nonpersistent part of width $Z - Z_{\text{per}}$. Assume that every nonpersistent leaf, before it can re-enter the target quadrilateral, must cross one of two disjoint gates whose total transverse width is at most D , and that the two gate crossings occur in series with a copy of the original horizontal family. Then the returning horizontal width Z' satisfies, up to an additive boundary constant C coming from finitely many cut leaves,*

$$Z' - C \leq Z_{\text{per}} + (Z - Z_{\text{per}}) \boxplus 2((Z - Z_{\text{per}}) + D), \quad a \boxplus b := \frac{ab}{a + b}. \quad (13.1)$$

In particular, if $Z_{\text{per}} \leq \alpha Z$ and $D \leq \kappa Z$ with $0 < \alpha < 1$, then

$$Z' - C \leq \nu' Z, \quad \nu' := \alpha + (1 - \alpha) \boxplus (2 + 2\kappa) < 1.$$

Proof. Use Lemma 13.43 to discard the persistent leaves and straighten the nonpersistent rectangle. A returning nonpersistent leaf has two alternatives: remain in the original horizontal channel, of width $Z - Z_{\text{per}}$, or leave it and return through the two gates. The latter route contains the nonpersistent horizontal part once more together with the two peripheral gate contributions; after straightening its available width is at most $2((Z - Z_{\text{per}}) + D)$. The escaping-and-returning route must traverse the horizontal and gate portions in series. The series law for extremal width is exactly the harmonic sum $\boxplus$, and therefore gives the $\boxplus$ term in (13.1). The finitely many leaves cut by end pieces contribute only C . Substituting the two relative bounds and homogeneity of $\boxplus$ gives the stated factor. Since $(1 - \alpha) \boxplus (2 + 2\kappa) < 1 - \alpha$, one has $\nu' < 1$. $\square$

13.3 Worked examples

Worked example: A round annulus and the meaning of a complex bound

Let

$$U = D(0, R), \quad U' = D(0, r), \quad 0 < r < R.$$

The logarithm maps $U \setminus \overline{U'}$ to a Euclidean strip of width $\log(R/r)$ modulo vertical translation by $2\pi i$. Hence

$$\text{mod}(U \setminus \overline{U'}) = \frac{1}{2\pi} \log \frac{R}{r}.$$

If a theorem gives modulus at least μ , then

$$\frac{R}{r} \geq e^{2\pi\mu}.$$

Thus a positive complex bound says more than “the inner disk is smaller”: it forces a definite multiplicative collar between the two boundaries. For example, $\mu = 0.1$ requires $R/r \geq e^{0.2\pi} \approx 1.874$.

Worked example: Numerically reading the covering lemma

Suppose the inner degree is $d = 2$, the collar ratio parameter is $\eta = 1/2$, and the source modulus

$$m = \text{mod}(U \setminus \overline{\Lambda}) = 10^{-3}$$

lies below the threshold $\varepsilon(\eta, D)$. Then

$$\text{mod}(V \setminus \overline{B}) < 2\eta^{-1}d^2m = 2 \cdot 2 \cdot 4 \cdot 10^{-3} = 0.016.$$

The important feature is not the number 0.016 but the degree that appears: $d^2 = 4$ is the degree of the distinguished inner island. A naive estimate using a very large total degree D could be useless in a deep renormalization pullback.

Boundary case: Annuli whose modulus tends to zero

Take

$$U = D(0, 1), \quad U'_n = D(0, r_n), \quad r_n \uparrow 1.$$

Then

$$\text{mod}(U \setminus \overline{U'_n}) = \frac{1}{2\pi} \log \frac{1}{r_n} \rightarrow 0.$$

For every fixed $\mu > 0$, the sequence eventually leaves $\mathcal{C}(\mu)$. Geometrically the two boundary curves approach one another, so the collar on which external interpolation and normal-family estimates are performed disappears. This is exactly the degeneration that complex a priori bounds exclude.

Worked example: Reading the beau quantifiers

The statement

$$\exists \mu_0 > 0 \forall \delta > 0 \exists \mu(\delta) > 0, N(\delta)$$

means the following. If we first choose an initial compactness class $\mathcal{C}^{\mathbb{R}}(\delta)$, then one lower bound $\mu(\delta)$ controls every renormalization level of every map in that class. After at most $N(\delta)$ levels, the stronger number μ_0 works for all of them. The eventual constant is universal; the entry depth is not. This is the “eventually universal” content hidden in the word *beau*.

13.4 What this chapter established

Chapter 13 crossed the real-to-complex boundary. The real scaled neighborhoods and Koebe control from Chapter 12 were used to construct complex return domains, but the main new difficulty was to prevent the annulus between those domains from pinching. The Kahn–Lyubich covering lemma isolates a low-degree inner island inside a possibly high-degree branched covering and gives exactly the modulus comparison needed in the nearly degenerate regime.

For infinitely renormalizable real quadratic polynomials, the resulting modulus-improvement argument yields a universal positive complex a priori bound. Once such a lower bound is available, normalized marked quadratic-like germs form a compact family. Compactness then makes the beau-bounds quantifiers meaningful: initial geometry may determine an entry constant and entry time, but sufficiently deep real-symmetric renormalizations enter one universal compact class.

The final theorem is intentionally a criterion rather than the full rigidity theorem. Given compatible external data, complex bounds allow a quasiconformal conjugacy to be pulled through the renormalization tower with uniform dilatation. The limiting Beltrami coefficient is invariant on the filled Julia set, so absence of invariant line fields upgrades the conjugacy to a hybrid one. Producing the required external matching from bounded combinatorics alone remains later work.

Looking ahead

Chapter 14 turns these compact polynomial-like germs into a genuine renormalization operator. The new questions are functional-analytic: on what Banach manifold is renormalization analytic, how is its derivative represented, and how do complex bounds place iterates in a compact region where spectral or hyperbolic arguments can be made? Chapter 13 supplies the geometric compactness; Chapter 14 must supply the operator structure.

Renormalization Operator, Feigenbaum Fixed Point, and Horseshoe

Why this chapter matters

Chapter 13 produced a compact geometric world of normalized quadratic-like germs: deep real renormalizations have definite complex collars and cannot escape by degeneration. Chapter 14 asks what happens when renormalization itself is treated as a dynamical system on that world.

There are two distinct kinds of dynamics here. A quadratic-like map iterates points in the dynamical plane. The renormalization operator instead iterates *maps*. Once that operator is analytic, an infinite sequence of return combinatorics can be encoded as an orbit in function space. The full horseshoe theorem says that this symbolic picture is not a metaphor: every bi-infinite admissible quadratic itinerary has one normalized realization in a precompact invariant set. The constant period-doubling itinerary is then special for a simple reason: it is fixed by the symbolic shift. The corresponding map is therefore fixed by renormalization. Its famous spatial scaling is already visible at this stage as an exact affine self-similarity. Parameter universality is harder; it requires the transverse hyperbolicity developed in Chapter 15.

Learning goals

By the end of this chapter you should be able to:

1. distinguish a polynomial-like germ from a Banach chart of representatives;
2. explain why a fixed combinatorial renormalization branch is holomorphic;
3. distinguish geometric compactness from analytic operator structure;
4. state the full quadratic renormalization horseshoe as a shift conjugacy;
5. explain why the full horseshoe is precompact rather than compact;
6. derive the period-doubling fixed point from the constant symbolic itinerary;
7. derive the fixed point's exact spatial scaling from its affine normalization;
8. state clearly why this does not yet prove Feigenbaum parameter scaling.

Conceptual roadmap

The chapter follows the chain

complex collars from Chapter 13 $\implies$ controlled Banach charts,
 finite iteration + normalization $\implies$ analytic branches of R ,
 beau bounds + rigidity + contraction $\implies$ symbolic horseshoe,
 constant period-doubling itinerary $\implies$ renormalization fixed point,
 fixed affine rescaling $\implies$ exact dynamical self-similarity.

The next chapter adds a new transverse direction to this picture. That step, not the fixed-point equation alone, produces parameter universality.

Common pitfall

The two classical Feigenbaum constants measure different phenomena. The spatial scale α_{dyn} comes from the affine rescaling inside the stationary dynamical tower. A parameter scale δ_{par} comes from how a one-parameter family crosses the stable set of the renormalization fixed point. The first can be read from a fixed-point identity; the second requires hyperbolicity and transversality.

14.1 Definitions

Definition 14.1 (unimodal renormalization). Let $I = [-1, 1]$. If a normalized even unimodal map $f : I \rightarrow I$ has a restrictive interval $J \ni 0$ of first-return period p , let $A_J : J \rightarrow I$ be the orientation-preserving affine normalization. Its renormalization is

$$Rf = A_J \circ f^p \circ A_J^{-1}.$$

For a quadratic-like germ the same notation denotes the normalized polynomial-like first-return germ. The branch is specified by the return combinatorics.

Definition 14.2 (Feigenbaum fixed point). A *period-doubling Feigenbaum fixed point* is a normalized real-symmetric quadratic-like germ f_* in the period-doubling branch of renormalization such that $Rf_* = f_*$. Its restriction to the real interval is an analytic even unimodal map with a quadratic critical point.

Definition 14.3 (dynamical Feigenbaum scale). Let $J_* = [-s_*, s_*] \Subset I$ be the restrictive interval of the normalized period-doubling fixed point, so $0 < s_* < 1$. We call s_* its spatial contraction factor and

$$\alpha_{\text{dyn}} := s_*^{-1} > 1$$

its unsigned dynamical Feigenbaum scale. A signed α depends on the orientation convention. The parameter-scaling constant customarily denoted δ is a different object and is not part of this definition.

Definition 14.4 (renormalization horseshoe). For a fixed even critical degree d , a *topological renormalization horseshoe* is an R -invariant precompact set $\mathcal{A}_d$ of normal-

ized real-symmetric polynomial-like germs together with a homeomorphism from the two-sided symbolic space of admissible real renormalization types onto $\mathcal{A}_d$ that conjugates the shift to R . Hyperbolicity of this set is an additional theorem, not part of the definition.

Definition 14.5 (Banach slice of normalized germs). Fix a representative $f : U' \rightarrow U$ and a slightly smaller complex neighborhood $W \Subset U$ containing its filled Julia set. A *Banach slice* at f is a neighborhood of f in the Banach space of bounded holomorphic maps on W , cut by the affine/critical normalization used for quadratic-like germs. Changing W gives compatible local germ charts after restriction.

Definition 14.6 (analytic renormalization branch). Fix a return combinatorial type M of period p . On the open set where the corresponding polynomial-like first return persists, the map $f \mapsto R_M f$ is the *renormalization branch of type M* . It is called analytic when it is holomorphic between the Banach slices of normalized germs.

Definition 14.7 (symbolic renormalization itinerary). Let $\mathfrak{T}_d$ be the countable alphabet of admissible real renormalization types for critical degree d . A bi-infinite itinerary is an element of $\Sigma_d = \mathfrak{T}_d^{\mathbb{Z}}$, and the left shift is $\sigma((\tau_n)_{n \in \mathbb{Z}}) = (\tau_{n+1})_{n \in \mathbb{Z}}$. The constant period-doubling itinerary is denoted τ_{pd} .

Definition 14.8 (hybrid leaf). Inside the normalized quadratic-like germ space, the *hybrid leaf* through f consists of germs hybrid conjugate to f with the prescribed external marking. Renormalization maps a renormalizable hybrid leaf holomorphically into the corresponding leaf at the next level.

14.2 Main results

Definition 14.9 (Unimodal renormalization operator). **Statement scope.** The formula $Rf = A_J \circ f^p \circ A_J^{-1}$ defines the normalized first-return operator on every restrictive interval/polynomial-like return for which the chosen combinatorial branch persists. This row is definitional and carries no independent proof obligation.

Theorem 14.1 (Analyticity of fixed combinatorial renormalization branches). **Statement scope.** Let f_0 be a normalized quadratic-like germ for which a renormalization of a fixed combinatorial type M and return period p is defined. After choosing Banach slices of representatives around f_0 and $R_M f_0$, there is a neighborhood $\mathcal{U}$ of f_0 on which the same polynomial-like return persists and

$$R_M : \mathcal{U} \longrightarrow \mathcal{Q}G$$

is holomorphic. In these charts $DR_M(f_0)$ is a bounded complex-linear operator. This theorem is local analytic structure; compactness of fixed-modulus germ classes remains the Chapter 13 theorem.

Proof strategy (not part of the proof)**Proof status.** proved here.**Strategic outline only.** Common-domain persistence; holomorphy of finite iteration; holomorphic affine normalization; descent to Banach germ charts.*The mathematical argument begins in the proof environment below.*

Proof. Choose a representative $f_0 : U'_0 \rightarrow U_0$ of the germ and a polynomial-like restriction of the prescribed return

$$f_0^p : V'_0 \longrightarrow V_0, \quad \overline{V'_0} \Subset V_0.$$

Shrink the domains slightly. We may then choose a complex neighborhood W of the finite orbit

$$\overline{V'_0}, f_0(\overline{V'_0}), \dots, f_0^{p-1}(\overline{V'_0})$$

whose closure stays inside the domains on which all required compositions are defined. Let $\mathcal{B}$ be the corresponding normalized Banach slice of bounded holomorphic maps on W .

Indeed, g^j converges uniformly to f_0^j on the relevant compact sets for $0 \leq j \leq p$. The strict containments used to define the return therefore persist. On the boundary of the chosen return domain, $g^p - z$ is a small perturbation of $f_0^p - z$; the argument principle preserves the proper degree and the unique critical component containing the marked critical point. Passing to a smaller common domain removes the harmless motion of the boundary. Hence the output is well defined as a germ and its combinatorial type is unchanged.

For $p = 1$ this is immediate. Suppose $g \mapsto g^j$ is holomorphic. Because all intermediate images lie a positive distance from the boundary of W , composition

$$(g, h) \longmapsto g \circ h$$

is holomorphic on the corresponding product of Banach neighborhoods: the Cauchy formula gives a uniformly convergent Taylor expansion in both variables. Thus $g \mapsto g \circ g^j = g^{j+1}$ is holomorphic. Induction gives the claim. Its first variation is the familiar finite chain-rule sum

$$D(g^p)_f[v] = \sum_{k=0}^{p-1} (f^{p-1-k})' \circ f^{k+1} v \circ f^k,$$

with the evident interpretation at $p = 1$; boundedness follows from the common safety collar.

The marked critical point is fixed at 0 in the chosen slice. The remaining normalizing data are finitely many evaluations and nonvanishing differences of evaluations of g^p . Evaluation is a continuous complex-linear functional on the Banach space, hence holomorphic. Since the denominators are nonzero at f_0 , they remain nonzero nearby, so the coefficients of the affine rescaling are holomorphic functions of g .

Combine the previous two lemmas. Restriction from one representative domain to a smaller one is continuous complex-linear, so two representatives of the same germ give the same output germ. The local formula therefore descends to R_M on the

quadratic-like germ space. A holomorphic map between Banach charts has a bounded complex-linear Fréchet derivative, which gives $DR_M(f_0)$.

This proves precisely the local analytic statement. No compactness assertion has entered: the compactness of normalized germs with a modulus lower bound is the separate Chapter 13 theorem. $\square$

Theorem 14.2 (Full quadratic renormalization horseshoe — topological form). **Statement scope.** *There are an R -invariant precompact set $\mathcal{A}_2$ of normalized real-symmetric infinitely renormalizable quadratic-like germs and a combinatorially faithful homeomorphism*

$$h_2 : \Sigma_2 \longrightarrow \mathcal{A}_2, \quad R \circ h_2 = h_2 \circ \sigma.$$

The hybrid lamination through $\mathcal{A}_2$ contains every real-symmetric infinitely renormalizable quadratic-like germ. The statement records the full symbolic horseshoe and qualitative hybrid-leaf attraction; it does not assert transverse hyperbolicity, stable/unstable manifolds, or parameter transversality. Avila–Lyubich prove the stronger analogue for every fixed even critical degree; the formal Chapter 14 row is kept quadratic so that its geometric input is exactly the Chapter 13 quadratic beau-bounds engine.

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Finite-window realization; Chapter 13 beau compactness; diagonal bi-infinite towers; real hybrid rigidity from controlled pullback plus no-line-field tower rescaling; real-beau-to-Carathéodory exponential contraction via the compact-retraction obstruction; uniqueness; forward coding continuity; anti-renormalization analytic-limit dichotomy and negative-time continuity; hybrid-lamination exhaustion.

The mathematical argument begins in the proof environment below.

Proof. Write

$$\Sigma_2 = \mathfrak{S}_2^{\mathbb{Z}}, \quad \mathcal{I} = (\tau_j)_{j \in \mathbb{Z}},$$

for the two-sided symbolic space of admissible real quadratic renormalization types. We first construct a normalized bi-infinite tower for every itinerary and then prove that this realization is unique and depends homeomorphically on the itinerary.

For a single symbol, the real renormalization window is obtained from the quadratic straightening family: the center of the corresponding real little Mandelbrot copy realizes the prescribed first return. If a word has already been realized, perform the next tuning inside the current restrictive domain. Induction realizes every finite word. Because the word is admissible, the nested restrictive intervals are genuine unimodal returns and the real critical orbit remains in the connectedness locus. This is the finite-tail construction used in the full-horseshoe proof, but no infinite limiting argument has yet been used.

For $N \geq 1$, choose a realization of $\tau_{-N}, \dots, \tau_N$ and renormalize it N times so that the symbol with index 0 sits in the fixed normalization. Denote the centered germ by F_N and its finite tower by $F_{N,j}$, $-N \leq j \leq N$, so that

$$R_{\tau_j} F_{N,j} = F_{N,j+1}, \quad F_{N,0} = F_N.$$

Indeed, every index j in such a fixed block is at distance at least $N - L$ from both ends of the finite realization. Chapter 13 beau bounds therefore put these central levels, once N is large, in a universal class with modulus at least $\mu_0 > 0$. Chapter 13 compactness of normalized fixed-modulus germs then gives a compact class $\mathcal{K}(\mu_0)$ containing all these germs. Notice that the argument uses the *deep* universal quantifier of the beau-bounds theorem; a level-dependent modulus estimate would not be enough for the diagonal construction.

For $L = 1, 2, \dots$, compactness of $\mathcal{K}(\mu_0)$ gives a convergent subsequence of the finite vector $(F_{N,-L}, \dots, F_{N,L})$. A diagonal choice gives a limit F_j for every $j \in \mathbb{Z}$. The fixed-type renormalization branch is holomorphic, hence continuous, on the controlled Banach chart supplied by Theorem 14.1; therefore $R_{\tau_j} F_j = F_{j+1}$. The type does not disappear in the limit: the corresponding polynomial-like return has a positive collar, and the next symbol in the finite word keeps the straightening parameter in the interior of the appropriate renormalization subwindow. In the satellite case this extra next-level information keeps the parameter a definite distance from the parabolic root. Thus the limiting tower realizes the whole itinerary.

We next prove the two analytic inputs needed for uniqueness. The point of the next two paragraphs is precisely to avoid treating “real rigidity” and “strict contraction” as unexplained black boxes.

Straighten both germs and every sufficiently deep renormalization. The external markings identify the Böttcher collars of corresponding deep quadratic-like returns. Since all deep fundamental annuli have modulus at least μ_0 , the collar identifications have uniformly controlled quasiconformal fillings. Pulling them back through the matched return nests uses Theorem 13.42; holomorphic pullback does not increase the Beltrami norm. Hence, after normalizing three marked points, we obtain a normal family of quasiconformal pseudo-conjugacies on larger and larger pieces of the two towers. A diagonal limit is a quasiconformal conjugacy and its Beltrami coefficient on the filled Julia set is invariant.

It remains to remove a possible invariant coefficient. We record the standard tower argument because this is where a vague appeal to “rigidity” would otherwise hide the main analytic point. If an invariant line field had positive measure, choose a Lebesgue density point and a sequence of renormalization/puzzle pullbacks shrinking to it. The beau collars and real Koebe control give normalized inverse branches with uniformly bounded distortion. After affine rescaling, a subsequence converges to a nonconstant holomorphic tower map G of degree two, while the rescaled line fields converge in measure on compacta to a constant line field. Invariance passes to the limit. A constant line field invariant under a holomorphic map says that the argument of G' is constant modulo π almost everywhere on an open set. By analyticity, G' is constant there, hence G is affine, contradicting the quadratic critical point carried by the limiting tower. Thus the invariant Beltrami coefficient vanishes almost everywhere on the filled Julia set, and the limiting conjugacy is hybrid. This proves the lemma.

We reconstruct the source mechanism in three steps.

Step 1: a common path-holomorphic model. Fix the external marking. Beltrami pullback identifies every hybrid leaf with the same normalized path-holomorphic model $\mathcal{H}_0$; a path is holomorphic exactly when its normalized Beltrami solution depends holo-

morphically on the parameter. The Carathéodory pseudometric is a true metric on this model: if two germs are distinct, restriction to a common small disk separates them by a bounded holomorphic functional. Fixed modulus subspaces $\mathcal{H}_0(\delta)$ are compact, and restriction to a slightly smaller disk gives Banach slices. Renormalization between hybrid leaves is path holomorphic, so Schwarz–Pick makes every renormalization cocycle weakly nonexpanding.

Step 2: macroscopic contraction from real beau bounds. For a cocycle $G_{m,n}$ obtained by transporting successive renormalizations to $\mathcal{H}_0$, two compactness properties follow from Chapter 13. First, an input modulus $\delta > 0$ gives a positive lower modulus for every $G_{m,n}(f)$; this is the complex pullback form of the Chapter 13 a priori bounds. Second, if f is real and $\tilde{f}$ is sufficiently close to f in the Carathéodory metric, then sufficiently long cocycle images of $\tilde{f}$ enter the universal beau class. To see the second statement, otherwise take real f_k , nearby $\tilde{f}_k$, and times $n_k \rightarrow \infty$ whose images miss a fixed smaller universal modulus. Compactness gives $f_k, \tilde{f}_k \rightarrow f$; the normalized quasiconformal conjugacies between them tend to the identity. Pulling a polynomial-like representative of $R^{n_k} f_k$ through these conjugacies changes its annular modulus by a factor tending to one, contradicting the universal real beau bound.

Suppose now that no uniform macroscopic contraction exists. By the preceding modulus control and Montel compactness, a sequence of longer and longer cocycle blocks has a locally uniform limit $P : \mathcal{H}_0 \rightarrow \mathcal{H}_0$. The cocycle/Schwarz property and the standard idempotent extraction from an almost-periodic semigroup allow the block sequence to be chosen so that $P^2 = P$ and P is nonconstant. Thus failure of contraction produces a nonconstant path-holomorphic retraction.

Such a retraction is impossible. Its real fixed set $Z_{\mathbb{R}} = \text{Fix}(P) \cap \mathcal{H}_{0,\mathbb{R}}$ lies in a compact fixed-modulus class by the near-real beau statement. In a Banach slice, the local representative of P is a compact real-analytic retraction. Differentiating $P^2 = P$ shows that DP is an idempotent compact operator and hence has finite rank. The analytic implicit-function theorem therefore makes $Z_{\mathbb{R}}$ a finite-dimensional manifold. The real slice of the hybrid model is contractible (interpolate the normalized real Beltrami data), and a retract of a contractible space is contractible. Consequently the compact manifold $Z_{\mathbb{R}}$ is a point: otherwise its top-dimensional mod-2 homology would be nonzero. Holomorphicity then forces P to be locally constant on the complex Banach slice, and analytic continuation plus connectedness of $P(\mathcal{H}_0)$ makes P globally constant. This contradicts the chosen nonconstant retraction. Hence long renormalization blocks have arbitrarily small Carathéodory diameter on every fixed-modulus compact set.

Step 3: exponential upgrade. Choose a universal modulus $\delta_0 < \mu_0$ and a block length q so large that the q -step image of $\mathcal{H}_0(\mu_0)$ has Carathéodory diameter at most some $\theta < 1$ when measured inside $\mathcal{H}_0(\delta_0)$. If $Y \subset X$ has Carathéodory diameter at most $\theta < 1$, the Schwarz lemma applied to all disk-valued holomorphic functions gives

$$d_X(x, y) \leq \theta d_Y(x, y), \quad x, y \in Y.$$

After the finite entry time supplied by beau bounds, each q -block therefore contracts by the uniform factor θ . Weak Schwarz contraction controls the intermediate iterates. Writing $n = kq + r$ gives the displayed $C\rho^n$ estimate with $\rho = \theta^{1/q}$. This is the source-correct route from real beau bounds to exponential hybrid-leaf contraction.

Suppose (F_j) and (G_j) realize the same itinerary. For each $m > 0$ the backward germs F_{-m} and G_{-m} have the same forward real combinatorics, so the real hybrid-rigidity lemma in the beau class puts them in one hybrid leaf. Apply m renormalizations and then use the real-beau-bounds uniform leafwise exponential-contraction lemma. Since the whole bi-infinite tower lies in the universal compact class, its Carathéodory diameter is uniformly bounded, and hence

$$d_{\text{Car}}(F_0, G_0) \leq C\rho^m.$$

Letting $m \rightarrow \infty$ gives $F_0 = G_0$. Shifting the itinerary gives equality at every index.

Define

$$h_2(\tau) = F_0, \quad \mathcal{A}_2 = h_2(\Sigma_2).$$

Existence and uniqueness make h_2 well defined and injective. By construction

$$R \circ h_2 = h_2 \circ \sigma,$$

and $\mathcal{A}_2$ is R -invariant.

Let $\tau^{(k)} \rightarrow \tau$ coordinatewise and put $F_j^{(k)} = h_2(\sigma^j \tau^{(k)})$. The centered towers lie in the universal precompact beau class. From any subsequence we may extract a diagonal subsequence for which $F_j^{(k)} \rightarrow F_j$ for every fixed j . For each fixed j , the symbols $\tau_j^{(k)}$ eventually equal τ_j . If the type is primitive, the corresponding renormalization window is closed at the relevant controlled level. If it is satellite, the *next* fixed symbol places the straightening parameter in the appropriate subwindow, a definite distance from the root; hence the limit cannot fall onto the parabolic root. Thus F_j has type τ_j and $R_{\tau_j} F_j = F_{j+1}$. Uniqueness of the bi-infinite realization gives $F_0 = h_2(\tau)$. Every convergent subsequence has the same limit, so the whole sequence converges.

The inverse continuity is subtler. For positive symbolic times it follows from ordinary branch persistence, but negative symbols require control of anti-renormalizations. We reproduce that control explicitly.

Anti-renormalization analytic-limit dichotomy. Let $\tilde{f}_k$ be real-symmetric renormalizable germs in a fixed beau class, with renormalization periods $p_k \rightarrow \infty$, and suppose $R\tilde{f}_k \rightarrow f_\infty$. Then either f_∞ is a unicritical polynomial, or the real trace of f_∞ has a bounded maximal domain of analytic continuation.

To prove the dichotomy, use the principal nest of the one-step pre-renormalizations and normalize the final restrictive interval. The real a priori estimates give the following exhaustive alternatives. If the final scaling factor tends to zero, the normalized return loses all finite outer geometry and converges to the unicritical polynomial model. If the principal-nest height tends to infinity while the terminal scaling stays bounded below, the normalized transition maps split into finitely many short maps and saddle-node cascades with uniform Koebe space. The safe-interval and central-cascade estimates make these transition maps precompact on every fixed earlier nest level. Their target intervals expand through the real line, but the corresponding source intervals stay in one bounded first principal domain; therefore the limit return has a proper analytic extension on each target interval but its maximal real source remains bounded. Finally, if the essential height is bounded while $p_k \rightarrow \infty$, the successive admissible

return moments have bounded gaps. The normalized first return then has, inside a nested bounded family of source intervals, arbitrarily many distinct critical preimages. Analytic continuation across the limiting bounded source would place infinitely many critical points in a compact subset of a nonconstant analytic map, impossible. Hence the maximal real analytic domain is again bounded. These are precisely the two geometric branches of the Appendix-A argument, and they prove the dichotomy.

Now suppose $f_k = h_2(\tau^{(k)}) \rightarrow f_\infty = h_2(\tau)$. The symbol at time 0 is eventually constant: a real perturbation of a twice-renormalizable map remains once renormalizable with the same first type, while boundary points of a real renormalization window are not twice renormalizable. Since R is continuous on $\mathcal{A}_2$, the same holds for every positive time.

For time -1 , let $\tilde{f}_k = R^{-1}f_k$ inside the coded tower. We claim that their periods are bounded. Otherwise the analytic-limit dichotomy would apply. A limit of the $R\tilde{f}_k = f_k$ cannot be a unicritical polynomial: a polynomial germ has no predecessor inside a normalized polynomial-like renormalization tower. On the other hand, f_∞ is infinitely anti-renormalizable with uniform beau bounds. Its m -th anti-renormalization gives an analytic continuation of its real trace to the real filled interval of that predecessor. Uniform real geometry keeps the critical point a definite relative distance from the boundary, while each anti-renormalization enlarges this interval by a definite factor. These intervals exhaust $\mathbb{R}$. Thus the real trace of f_∞ has an unbounded, indeed entire-real, analytic continuation. This excludes the second branch of the dichotomy. Hence the predecessor periods are bounded.

After passage to a subsequence, $\tilde{f}_k$ therefore has fixed combinatorial type and a convergent limit $\tilde{f}$. Continuity gives $R\tilde{f} = f_\infty$. Uniqueness of the coded predecessor forces $\tilde{f} = R^{-1}f_\infty$. Thus $R^{-1}f_k \rightarrow R^{-1}f_\infty$, and the -1 symbols eventually agree. Induction gives the same conclusion for every negative time. Consequently

$$f_k \rightarrow f_\infty \implies \tau^{(k)} \rightarrow \tau$$

coordinatewise. Hence h_2^{-1} is continuous. We have proved that h_2 is a homeomorphism and therefore a topological conjugacy between R on $\mathcal{A}_2$ and the two-sided shift.

The image is precompact because every coded germ belongs to the universal normalized beau class. It need not be compact: the alphabet contains renormalization types of arbitrarily large period, and a sequence may escape through those symbols while staying geometrically precompact.

Let F be such a germ and record its forward itinerary $\tau_0, \tau_1, \dots$. Choose any admissible backward extension to $\tau \in \Sigma_2$. The coded germ $h_2(\tau)$ has the same forward real combinatorics as F . The real hybrid-rigidity lemma in the beau class therefore places the two germs on the same hybrid leaf. This proves the lamination assertion and completes the theorem. $\square$

Theorem 14.3 (Period-doubling Feigenbaum fixed point). **Statement scope.** *For $d = 2$, the constant period-doubling itinerary $\tau_{\text{pd}} \in \Sigma_2$ corresponds under the horseshoe coding to a normalized real-symmetric quadratic-like germ*

$$f_* := h_2(\tau_{\text{pd}}) \quad \text{with} \quad Rf_* = f_*.$$

Consequently period-doubling renormalization has an analytic even unimodal fixed point with a quadratic critical point. The theorem asserts existence (and uniqueness inside the normalized horseshoe coding), not derivative hyperbolicity.

Proof strategy (not part of the proof)

Proof status. derived here.

Strategic outline only. Constant period-doubling itinerary is shift-fixed; apply the horseshoe conjugacy; read analyticity and quadratic criticality from the normalized quadratic-like germ.

The mathematical argument begins in the proof environment below.

Proof. Let $\tau_{\text{pd}} \in \Sigma_2$ be the sequence whose every coordinate is the period-doubling combinatorial type. It is fixed by the left shift:

$$\sigma\tau_{\text{pd}} = \tau_{\text{pd}}.$$

Set $f_* = h_2(\tau_{\text{pd}})$. The conjugacy identity gives

$$Rf_* = Rh_2(\tau_{\text{pd}}) = h_2(\sigma\tau_{\text{pd}}) = h_2(\tau_{\text{pd}}) = f_*.$$

The horseshoe consists of real-symmetric quadratic-like germs. Therefore f_* has a holomorphic representative on a complex neighborhood of its real filled Julia set, and its restriction to the real interval is real analytic. The normalization marks a unique critical point of local degree two, so that restriction is an even unimodal map with a quadratic critical point.

Injectivity of h_2 gives uniqueness among normalized points of the horseshoe carrying the constant period-doubling itinerary. Nothing in this argument concerns the derivative $DR(f_*)$; fixed-point existence is therefore logically prior to Chapter 15 hyperbolicity. $\square$

Theorem 14.4 (Exact dynamical scaling at the period-doubling fixed point). **Statement scope.** Let $J_* = [-s_*, s_*]$ be the restrictive interval of f_* and let $A_*(x) = x/s_*$. Then

$$f_* = A_* \circ f_*^2 \circ A_*^{-1}.$$

If $J_0 = I$ and $J_{n+1} = A_*^{-1}(J_n)$, then these are the successive self-similar restrictive intervals of the stationary tower and

$$\frac{|J_{n+1}|}{|J_n|} = s_* = \alpha_{\text{dyn}}^{-1} \quad (n \geq 0).$$

Thus the fixed point has one exact universal spatial scale in the chosen normalization. No convergence of parameter windows and no parameter constant δ is asserted here.

Proof strategy (not part of the proof)

Proof status. derived here.

Strategic outline only. Insert the fixed point into the affine period-doubling renormalization identity; iterate the same affine pullback; compute the exact

restrictive-interval length ratio.

The mathematical argument begins in the proof environment below.

Proof. Let $J_* = [-s_*, s_*] \subseteq I = [-1, 1]$ be the period-doubling restrictive interval of f_* . Since the chosen normalization is orientation preserving, the affine map from J_* to I is

$$A_*(x) = \frac{x}{s_*}, \quad A_*^{-1}(x) = s_*x.$$

By the definition of the period-doubling branch,

$$Rf_* = A_* \circ f_*^2 \circ A_*^{-1}.$$

The fixed-point theorem gives $Rf_* = f_*$, hence

$$f_* = A_* \circ f_*^2 \circ A_*^{-1}.$$

Renormalizing once identifies the first-return dynamics on J_* with the original dynamics on I . Because the renormalized map is exactly f_* , its restrictive interval in normalized coordinates is again J_* . Returning to the original coordinates applies A_*^{-1} . Thus, with $J_0 = I$,

$$J_{n+1} = A_*^{-1}(J_n) \quad \text{and hence} \quad J_n = [-s_*^n, s_*^n].$$

Indeed,

$$|J_n| = 2s_*^n, \quad \frac{|J_{n+1}|}{|J_n|} = \frac{2s_*^{n+1}}{2s_*^n} = s_*.$$

By definition $\alpha_{\text{dyn}} = s_*^{-1}$, so the ratio is α_{dyn}^{-1} .

This is exact dynamical self-similarity of the fixed point. To prove that a one-parameter family exhibits a universal ratio of parameter-window widths, one additionally needs convergence to the fixed point in an operator space, an unstable direction, and transversality of the parameter curve. Those are precisely the Chapter 15 inputs and have not been used here. $\square$

14.3 Worked examples

Worked example: A stationary affine tower

Take $I = [-1, 1]$ and a symmetric restrictive interval $J = [-s, s]$ with $0 < s < 1$. The orientation-preserving affine normalization is

$$A(x) = \frac{x}{s}, \quad A^{-1}(x) = sx.$$

Suppose a map satisfies the stationary period-doubling identity

$$f = A \circ f^2 \circ A^{-1}.$$

Set $J_0 = I$ and $J_{n+1} = A^{-1}(J_n)$. Then

$$J_n = [-s^n, s^n], \quad |J_n| = 2s^n,$$

so

$$\frac{|J_{n+1}|}{|J_n|} = s.$$

If the unsigned dynamical Feigenbaum scale is $\alpha_{\text{dyn}} = 1/s$, the ratio is exactly α_{dyn}^{-1} at every level. No limiting argument has been used.

Worked example: A fixed point manufactured by symbolic dynamics

Let τ denote the period-doubling symbol and consider

$$\underline{\tau} = (\dots, \tau, \tau, \tau, \dots).$$

The left shift does nothing to this sequence: $\sigma \underline{\tau} = \underline{\tau}$. If h conjugates the shift to renormalization, then

$$Rh(\underline{\tau}) = h(\sigma \underline{\tau}) = h(\underline{\tau}).$$

Thus the fixed point is a direct structural consequence of the full horseshoe. This is conceptually different from a stand-alone fixed-point construction such as a Leray–Schauder or computer-assisted proof.

Boundary case: A normalized unimodal map outside the renormalization domain

Consider

$$f(x) = 1 - 2x^2, \quad -1 \leq x \leq 1.$$

Its critical orbit is $0 \mapsto 1 \mapsto -1 \mapsto -1$. This map is a perfectly good analytic even unimodal map, but it does not possess the nested proper period-doubling restrictive intervals required by the Chapter 14 operator. Hence “analytic unimodal” is not enough to write Rf : renormalizability is an essential domain condition.

Worked example: Why a countable full horseshoe is only precompact

If we restrict the symbolic alphabet to finitely many return types, the product space of two-sided sequences is compact. The full renormalization alphabet contains types of arbitrarily large period, so it is countable and noncompact in the discrete topology. Its shift space is therefore not compact. The analytic horseshoe can still have compact closure because beau bounds keep all its points in one normalized geometric region. This is why “precompact” is the correct topological word for the full horseshoe.

14.4 What this chapter established

Chapter 14 promoted renormalization from a geometric construction to an analytic dynamical system on a space of maps. On a fixed combinatorial branch, finite iteration is holomorphic on a domain-controlled Banach slice, and the affine normalization is holomorphic as well. This gives a genuine local analytic renormalization operator without duplicating Chapter 13's compactness theorem.

Beau bounds, compactness, real rigidity, and contraction along hybrid leaves then produce the full quadratic renormalization horseshoe. Every bi-infinite admissible itinerary has one normalized realization, and renormalization on the horseshoe is topologically the left shift. The result is deliberately stated in topological form: the operator's transverse hyperbolicity is not needed to know that the symbolic set exists.

The constant period-doubling itinerary yields the Feigenbaum fixed point. Because its renormalized first return is exactly itself, the same affine rescaling repeats at every level, giving exact spatial self-similarity and the dynamical scale α_{dyn} . This is not yet the parameter constant δ_{par} .

Looking ahead

Chapter 15 studies the derivative of renormalization near the horseshoe. Hybrid leaves become stable directions, a one-dimensional unstable direction appears, and parameter families cross the stable lamination transversely. Those facts turn Chapter 14's exact dynamical self-similarity into convergence and universal parameter scaling.

Hyperbolicity, Rigidity, and Universality of Renormalization

Why this chapter matters

Chapter 14 constructed a renormalization horseshoe and a stationary Feigenbaum fixed point. Those are already strong dynamical statements, but they do not yet tell us how nearby maps move relative to the horseshoe. Chapter 15 studies that missing infinitesimal geometry.

The decisive structure is a splitting of directions in the space of quadratic-like germs. Perturbations tangent to a hybrid class are stable: renormalization contracts them. There is exactly one complex transverse direction, and it is unstable. Once that splitting is established, the familiar parameter-scaling law follows by intersecting the stable lamination with a transverse one-parameter family.

There is an important logical point. Rigidity comes *before* hyperbolicity. It prevents two distinct maps with the same bounded combinatorics from shadowing one another in a neutral transverse direction. Only after that obstruction has been removed can the derivative cocycle be proved uniformly hyperbolic.

Learning goals

By the end of this chapter you should be able to:

1. state bounded-combinatorics rigidity with its exact hybrid-equivalence conclusion;
2. explain how a priori bounds make quasiconformal pullback uniform;
3. identify the stable tangent bundle with tangent spaces to hybrid leaves;
4. explain why the transverse quotient is one-complex-dimensional;
5. describe where rigidity enters the proof of uniform transverse expansion;
6. derive exponential Banach-norm convergence on local stable leaves;
7. derive the parameter law $|\mu_n - \mu_*| \asymp \delta_{\text{par}}^{-n}$ from transversality;
8. distinguish the spatial scale α_{dyn} from the parameter scale δ_{par} ;
9. explain why positive-area Feigenbaum theory is a separate measure-theoretic branch rather than a consequence of parameter universality.

Conceptual roadmap

The main logical spine is

bounded geometry + combinatorics $\implies$ hybrid rigidity,
 rigidity + analytic renormalization $\implies$ uniform hyperbolicity,
 hyperbolicity $\implies$ stable convergence,
 unstable line + transversality $\implies$ parameter universality.

The black-hole vocabulary introduced later in the chapter belongs to another branch. Its full positive-area theorem is deferred until Chapter 19, where the Siegel, parabolic, and near-parabolic construction tools required by the proof are available.

Common pitfall

Do not reverse the first arrow. Hyperbolicity is not used to prove the bounded-combinatorics rigidity theorem stated here. In the hyperbolicity argument, rigidity is precisely what rules out a slow or neutral transverse shadowing orbit. Likewise, the parameter eigenvalue δ_{par} is not the dynamical spatial rescaling constant α_{dyn} from Chapter 14.

15.1 Definitions

Definition 15.1 (renormalization hyperbolicity). Let $\mathcal{A}$ be a compact invariant set for a finite collection of analytic renormalization branches in Banach germ charts. Renormalization is *uniformly hyperbolic on $\mathcal{A}$* if the derivative cocycle has a continuous invariant splitting

$$T_f \mathcal{Q}\mathcal{G} = E_f^s \oplus E_f^u$$

and constants $C \geq 1$ and $0 < \theta < 1$ such that stable vectors contract forward by $C\theta^n$ and unstable vectors contract backward, equivalently expand forward, at the same uniform exponential rate.

Definition 15.2 (bounded-combinatorics rigidity). Two normalized nonescaping quadratic-like germs are *combinatorially equivalent* when their marked renormalization nests have the same forward return combinatorics, including the external marking at every canonical renormalization level. The combinatorics is *bounded* when the relative renormalization periods are bounded by one finite constant. Bounded-combinatorics rigidity is the theorem that, once a common a priori modulus bound is available, such germs lie in the same hybrid class. The real-symmetric formulation is only a special slice of this complex statement; no global conformal conjugacy of the original germs is asserted.

Definition 15.3 (local stable class). For f on a hyperbolic renormalization horseshoe, the local stable class $W_{\text{loc}}^s(f)$ is the connected local set of germs whose forward renormalization orbits follow the same local branch itinerary and converge exponentially

to the orbit of f . Identification with the local hybrid leaf is a theorem, not part of the definition.

Definition 15.4 (parameter universality). Parameter universality is an asymptotic law for intersections of a transverse one-parameter family with successively tuned hybrid classes. In stationary combinatorics the first-order decay rate is determined by the unstable eigenvalue of the stationary renormalization fixed point.

Definition 15.5 (finite bounded-type complex horseshoe). Fix a finite family $\mathcal{L}$ of pairwise disjoint quadratic renormalization types. A *complex $\mathcal{L}$ -horseshoe* is a compact renormalization-invariant set $\mathcal{A}_{\mathcal{L}}$ of normalized quadratic-like germs, with a uniform positive lower modulus bound, whose natural extension

$$\widehat{\mathcal{A}}_{\mathcal{L}} = \{\widehat{f} = (f_n)_{n \in \mathbb{Z}} : Rf_n = f_{n+1}\}$$

is topologically conjugate to the full two-sided shift $\mathcal{L}^{\mathbb{Z}}$. The projection $\widehat{f} \mapsto f_0$ need not be injective. For a real family $\mathcal{L}$ the real slice is the Chapter 14 horseshoe and the projection is injective.

Definition 15.6 (derivative cocycle). Choose the finite Banach atlas of controlled representatives over $\mathcal{A}_{\mathcal{L}}$. Along an orbit $f_{n+1} = R_{\tau_n} f_n$, the *derivative cocycle* is the product

$$DR_{\tau_{n-1}}(f_{n-1}) \cdots DR_{\tau_0}(f_0),$$

with coordinate changes inserted when an orbit moves between charts. Equivalent finite atlases give equivalent uniform hyperbolicity estimates.

Definition 15.7 (transverse analytic parameter family). A real-analytic family $S = \{f_{\mu}\}$ is *transverse* to a codimension-one hybrid/stable leaf $\mathcal{H}$ at f_{μ_*} when its tangent vector $\partial_{\mu} f_{\mu}|_{\mu_*}$ is not contained in $T_{f_{\mu_*}} \mathcal{H}$.

Definition 15.8 (unstable parameter scale δ_{par}). For a stationary renormalization branch R_M with real fixed point f_* , let $\lambda_u > 1$ be the real unstable eigenvalue of $DR_M(f_*)$. We write

$$\delta_{\text{par}} := \lambda_u.$$

This is a parameter-space scaling datum and is not the dynamical spatial scale α_{dyn} from Chapter 14.

Definition 15.9 (Feigenbaum quadratic-like map). A *Feigenbaum quadratic-like map* is an infinitely renormalizable quadratic-like map with bounded combinatorics and a priori bounds. Stationary combinatorics means that the same renormalization type occurs at every level; primitive stationary combinatorics excludes satellite attachment.

Definition 15.10 (landing probability η). For a stationary Feigenbaum map with first pre-renormalization $U' \rightarrow V'$, let $X \subset U$ be the set of points whose forward orbit eventually lands in U' . The *landing probability* is $\eta = \text{area}(X) / \text{area}(U)$.

Definition 15.11 (escaping probability ξ). Let $A' = V' \setminus \overline{U'}$ be the fundamental annulus and let $Y \subset A'$ consist of points that never return to V' while their forward orbit remains defined. The *escaping probability* is $\xi = \text{area}(Y) / \text{area}(A')$.

Definition 15.12 (black-hole regime). A primitive stationary Feigenbaum map with a fixed a priori modulus bound is in the *black-hole regime* when its landing probability dominates its escaping probability by the quantitative constant in the Black Hole Criterion. That criterion then forces positive planar area of the Julia set.

15.2 Main results

Theorem 15.1 (Complex rigidity for bounded combinatorics). **Statement scope.** *Let f and g be normalized nonescaping quadratic-like germs of degree two with connected filled Julia sets. Assume that their relative renormalization periods are bounded by P , that every canonical renormalization admits a quadratic-like representative of modulus at least $\mu > 0$, and that f and g have the same marked bounded combinatorics. Then f and g are hybrid equivalent. If in addition $f = P_c$ and $g = P_{\tilde{c}}$ are monic centered quadratic polynomials, then $c = \tilde{c}$. In particular, on every finite real bounded-type horseshoe the forward combinatorics determine the hybrid leaf. No stronger global conformal or smooth conjugacy is asserted.*

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Bounded type gives a finite secondary-limb/standard-neighborhood package; finite-depth marked pseudo-conjugacies have uniformly bounded Teichmüller cost; first-landing pullback preserves the bound; quasiconformal compactness gives a global conjugacy; bounded-type rescaling excludes invariant line fields; straightening uniqueness gives polynomial-parameter rigidity. *The mathematical argument begins in the proof environment below.*

Proof. Let f and g have the same marked bounded renormalization combinatorics, with relative periods at most P , and suppose all canonical renormalizations have modulus at least $\mu > 0$. The proof is the bounded-type complex specialization of Lyubich's rigidity argument.

Uniform standard neighborhoods. Only finitely many marked return types occur. Straightening and the common modulus bound therefore place the corresponding little-Julia bouquets in a compact finite family of geometric models. Around every bouquet one can choose pairwise disjoint marked standard neighborhoods with a definite separating modulus

$$\text{mod}(U_{n,j} \setminus B_{n,j}) \geq \nu(P, \mu) > 0.$$

This construction is complex: no real symmetry is used.

Uniform pseudo-conjugacies. Common combinatorics identifies the puzzle boundaries, bouquet components and return branches at every finite depth. Start with a marked QC identification of the outer finite configuration. When a deeper marking is inserted, its adjustment is shielded from the outer boundary by the intervening standard collars. In round-annulus coordinates the extra logarithmic Teichmüller cost of

a correction behind q such collars is at most

$$C(P, \mu) e^{-a(P, \mu) q \nu}.$$

Since a bounded-type block has at most P return levels, the total correction cost is bounded by a convergent geometric series. Thus the depth- N pseudo-conjugacies have one quasiconformal constant $K(P, \mu)$ independent of N .

Spread these maps through all first-landing domains by the conjugacy equation. Holomorphic pullback does not increase the Beltrami norm, and the quadratic critical branch is matched in the fixed critical coordinate. After normalizing three external marked points, quasiconformal compactness gives a limit conjugacy H between neighborhoods of $K(f)$ and $K(g)$.

Hybrid upgrade. The Beltrami coefficient of H is f -invariant on $K(f)$. If it were nonzero on a set of positive area, rescale about a density point along the bounded-type renormalization tower. A priori bounds give a non-affine quadratic-like limit, while the normalized Beltrami coefficients converge to a nonzero constant coefficient μ_0 . Invariance would then give

$$\mu_0 = \mu_0 \frac{\overline{F'}}{F'}$$

almost everywhere on an open core. Hence the argument of F' would be constant there, forcing F to be affine, a contradiction. Therefore H is hybrid.

If $f = P_c$ and $g = P_{\tilde{c}}$ are monic centered quadratic polynomials, straightening uniqueness makes the polynomial representative of a connected hybrid class unique, so $c = \tilde{c}$. This proves the complex bounded-combinatorics rigidity theorem; the real-symmetric statement is its restriction to the real slice. $\square$

Theorem 15.2 (Complex hyperbolicity of a bounded-type renormalization horseshoe). ***Statement scope.** Fix a finite family $\mathcal{L}$ of quadratic renormalization types and a complex $\mathcal{L}$ -horseshoe $\mathcal{A}_{\mathcal{L}}$ in the sense of the preceding definition. For every prehistory $\hat{f} = (f_n)_{n \in \mathbb{Z}} \in \hat{\mathcal{A}}_{\mathcal{L}}$ there is a continuous uniformly hyperbolic splitting at the zero coordinate*

$$T_{f_0} \mathcal{QG} = E_{f_0}^s \oplus E_{\hat{f}}^u, \quad E_{f_0}^s = T_{f_0} \mathcal{H}(f_0), \quad \text{codim}_{\mathbb{C}} E_{f_0}^s = 1, \quad \dim_{\mathbb{C}} E_{\hat{f}}^u = 1.$$

The stable bundle is uniformly exponentially contracted, the unstable line is DR-invariant along the natural extension and uniformly exponentially expanded, and local stable leaves are exactly local hybrid leaves. For a real family $\mathcal{L}$ the natural-extension projection is injective, so the unstable line can be indexed by f_0 alone and its real slice is one-real-dimensional.

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Use the compact full-shift natural extension, the complex codimension-one hybrid stable bundle and uniform contraction, transverse nonsingularity, rigidity-based exclusion of slow shadowing, a finite Banach-fiber small-orbits criterion, and a graph transform for the unstable line indexed by pre-

history.

The mathematical argument begins in the proof environment below.

Proof. Fix a finite complex bounded-type horseshoe $\mathcal{A}_{\mathcal{L}}$ and work on its natural extension $\hat{\mathcal{A}}_{\mathcal{L}}$. The natural extension is a compact full shift on the finite alphabet $\mathcal{L}$, while the uniform a priori bound places all zero coordinates in finitely many normalized Banach charts.

Hybrid classes are complex codimension-one analytic leaves. Uniform Schwarz-Carathéodory contraction on the fixed-modulus compact class gives a continuous invariant stable bundle

$$E_f^s = T_f \mathcal{H}(f)$$

with uniform exponential contraction. Straightening is a nonconstant local coordinate transverse to the hybrid foliation, so the derivative induced by renormalization on the one-complex-dimensional quotient $T_f \mathcal{Q}G/E_f^s$ is nonzero.

If the quotient cocycle failed to expand uniformly, the small-orbits construction would produce two distinct slowly shadowing orbits with the same bounded marked combinatorics. Complex bounded-combinatorics rigidity puts such a pair on one hybrid leaf, where the stable estimate forces exponential convergence, a contradiction.

Finally choose the finite Banach-fiber atlas furnished by the uniform collars. For each prehistory $\hat{f} = (f_n)_{n \in \mathbb{Z}}$ the backward graph transform is a strict contraction on complements of E^s . Its unique fixed graph gives a one-complex-dimensional line $E_{\hat{f}}^u$ satisfying

$$T_{f_0} \mathcal{Q}G = E_{f_0}^s \oplus E_{\hat{f}}^u, \quad DR(E_{\hat{f}}^u) = E_{\hat{R}\hat{f}}^u,$$

and DR expands this line uniformly. The local stable manifold is the local hybrid leaf. In the real case the horseshoe projection is injective, so the prehistory is determined by f_0 and the real unstable direction is one-dimensional. $\square$

Theorem 15.3 (Exponential convergence along stable classes). **Statement scope.** *In the fixed finite Banach atlas of $\mathcal{A}_{\mathcal{L}}$ there are uniform constants $C \geq 1$ and $0 < \rho < 1$ with the following property. If f and g lie in the same sufficiently small local stable leaf, then all forward branch iterates are defined and*

$$d_{\mathcal{B}}(R^n f, R^n g) \leq C \rho^n d_{\mathcal{B}}(f, g), \quad n \geq 0,$$

where $d_{\mathcal{B}}$ is the chartwise Banach norm distance. The local stable leaf is exactly the local hybrid leaf. This is the Banach-norm stable-manifold consequence of Chapter 15 hyperbolicity, not a restatement of Chapter 14's intrinsic Carathéodory contraction lemma.

Proof strategy (not part of the proof)

Proof status. derived here.

Strategic outline only. Uniform analytic stable graph charts; adapted stable norms; second-derivative control; nonlinear contraction; equivalence of finite chart norms; stable=hybrid identification.

The mathematical argument begins in the proof environment below.

Proof. Use the finite Banach atlas and hyperbolic splitting from the preceding theorem. Uniform hyperbolicity gives constants $C_0 \geq 1$ and $0 < \theta < 1$ such that

$$\|DR^n(f)v\| \leq C_0\theta^n\|v\|, \quad v \in E_f^s.$$

After passing to one fixed bounded iterate and changing to equivalent adapted norms, we may assume that the one-step stable linear part has norm at most $\theta_0 < 1$.

The analytic stable-manifold theorem, applied uniformly over the compact horseshoe, provides a radius $r > 0$ for which every local stable manifold is a graph

$$x \mapsto x + \psi_f(x), \quad x \in E_f^s(r),$$

with $\psi_f(0) = D\psi_f(0) = 0$ and uniformly bounded derivative. Compactness of the horseshoe and finiteness of the atlas also give a uniform bound on the second derivatives of all branch maps.

In these graph coordinates the branch restricted to the stable plaque has the form

$$x \mapsto A_fx + N_f(x),$$

where $A_f = DR(f)|_{E_f^s}$ and

$$\|N_f(x) - N_f(y)\| \leq Kr\|x - y\|$$

for one uniform constant K . Shrink r so that $Kr < (1 - \theta_0)/3$. The stable-coordinate difference then contracts by a factor strictly smaller than 1. The graph parameterizations and their inverses have uniformly bounded derivatives, so after one more harmless shrink we obtain a uniform $0 < \rho < 1$ such that

$$d_{\mathcal{B}}(Rp, Rq) \leq \rho d_{\mathcal{B}}(p, q)$$

when p and q lie on one local stable plaque, with the comparison made in adapted charts.

Iterating gives ρ^n contraction in the adapted norms. Since only finitely many Banach charts occur, those norms are uniformly equivalent to the original chart norms. Therefore there is $C \geq 1$ such that

$$d_{\mathcal{B}}(R^n p, R^n q) \leq C\rho^n d_{\mathcal{B}}(p, q), \quad n \geq 0.$$

The preceding hyperbolicity theorem identified the local stable manifold with the local hybrid leaf: the hybrid plaque is invariant, tangent to E^s , and exponentially contracted, and the stable-manifold theorem gives uniqueness. Taking $p = f$ and $q = g$ proves the stated estimate.

This is genuinely stronger in a different direction from Chapter 14's intrinsic contraction. Chapter 14 used contraction inside a hybrid leaf to prove uniqueness of the topological horseshoe. Here the derivative splitting and nonlinear stable-manifold theorem give an ambient Banach-norm estimate. $\square$

Theorem 15.4 (Parameter universality by transversality). **Statement scope.** Let M be a real Mandelbrot copy, let R_M be the stationary renormalization branch, and let f_* be its real fixed point. Write $\delta_{\text{par}} = \lambda_M > 1$ for the real unstable eigenvalue of $DR_M(f_*)$. Let c be the corresponding stationary Feigenbaum quadratic parameter and let c_n be the superattracting parameters obtained by n -fold tuning from the center of M . If a real-analytic one-parameter family $S = \{f_\mu\}$ intersects the hybrid class $\mathcal{H}_c$ transversely at μ_* , then for all sufficiently large n there is a unique nearby parameter μ_n with $f_{\mu_n} \in \mathcal{H}_{c_n}$ and

$$|\mu_n - \mu_*| \sim a(S) \delta_{\text{par}}^{-n}, \quad a(S) > 0.$$

In particular, in the quadratic family, $|c_n - c| \sim b \delta_{\text{par}}^{-n}$ for some $b > 0$. If the parameter curve is tangent to the stable leaf, this first-order law is not asserted.

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Combine hyperbolicity with quantitative holomorphic-motion and Koebe distortion estimates to control first-order hybrid holonomy; identify the unstable scaling by Koenigs linearization and finish with the transverse inverse-function theorem, keeping tangency as the sharp boundary case.

The mathematical argument begins in the proof environment below.

Proof. Let $T = R_M$ be the stationary renormalization branch and let f_* be its real fixed point. By the preceding hyperbolicity theorem,

$$T_{f_*} \mathcal{Q}\mathcal{G} = E_*^s \oplus E_*^u, \quad \dim_{\mathbb{C}} E_*^u = 1.$$

Choose the real orientation so that the unstable eigenvalue is $\lambda_u > 1$, and set $\delta_{\text{par}} = \lambda_u$.

First-order regularity of stable holonomy.

Let S be an analytic transversal through a point a of the stationary hybrid leaf, and let $h : S \rightarrow W_{\text{loc}}^u(f_*)$ be holonomy along hybrid leaves. We show that in analytic coordinates centered at a and f_* ,

$$h(z) = \tau z + O(|z|^{1+\alpha}), \quad \tau \neq 0. \quad (15.5)$$

Forward iterates $S_m = T^m(S)$ converge exponentially in C^1 to the local unstable curve by the cone graph transform. The Chapter 6 quantitative lambda lemma therefore makes the holonomy from S_m to the unstable curve $(1 + O(\gamma^m))$ -quasiconformal. On the other hand, transverse nonsingularity gives univalent restrictions of T^m to the successive transversals; uniform complex collars and Koebe distortion imply that, at image scale r , normalized complex ratios are changed by only $1 + O(r)$.

For two source points at scale ε , choose m so that their images have a fixed small macroscopic unstable scale. Hyperbolicity gives $m \asymp |\log \varepsilon|$. Comparing ratios first under $T^m|_S$, then under the near-identity stable holonomy at level m , and finally under the inverse unstable branch gives

$$\frac{h(z_2)}{h(z_1)} = \frac{z_2}{z_1} (1 + O(\varepsilon^\alpha)) \quad (15.6)$$

for some $\alpha > 0$ on uniformly separated configurations. Applying (15.6) on consecutive geometric scales makes $h(z)/z$ a multiplicative Cauchy sequence; hence it converges to a nonzero τ and (15.5) follows. This is Lyubich's first-order conformality argument in the form needed here; no globally smooth stable foliation is assumed.

Unstable inverse scale.

On $W_{\text{loc}}^u(f_*)$ the inverse renormalization branch has attracting multiplier λ_u^{-1} . Koenigs linearization gives a coordinate u for which

$$u(Tf) = \lambda_u u(f).$$

Thus the transverse coordinate of the n -fold inverse-tuned hybrid class has the form

$$K\lambda_u^{-n} + O(\lambda_u^{-2n}), \quad K \neq 0. \quad (15.7)$$

Transporting by (15.5) changes only the nonzero first-order constant and adds an $O(\lambda_u^{-(1+\alpha)n})$ remainder.

Let $S = \{f_\mu\}$ meet the stationary hybrid leaf transversely at μ_* . In the holonomy coordinate ζ ,

$$\zeta(f_\mu) = A(\mu - \mu_*) + O(|\mu - \mu_*|^{1+\alpha}), \quad A \neq 0.$$

Equating this with (15.7) and applying the inverse-function theorem yields a unique nearby intersection μ_n with the n -times tuned hybrid class and

$$\mu_n - \mu_* = \frac{K}{A}\lambda_u^{-n} + o(\lambda_u^{-n}).$$

Taking absolute values gives

$$|\mu_n - \mu_*| \sim a(S) \delta_{\text{par}}^{-n}, \quad a(S) > 0.$$

The quadratic parameter line is transverse at the stationary real Feigenbaum point, giving the stated $|c_n - c|$ law. If the first nonzero transverse term has order $k > 1$, the same computation gives scale $\lambda_u^{-n/k}$; hence the first-order universality law genuinely requires transversality. $\square$

Remark 15.1 (Positive-area Feigenbaum interface). **Scope.** Chapter 15 introduces the Feigenbaum-map, landing-probability η , escaping-probability ξ , and black-hole vocabulary because they clarify a second, measure-theoretic direction of renormalization theory. The formal Avila-Lyubich theorem asserting positive-area primitive stationary Feigenbaum Julia sets and a positive-dimensional parameter set is proved later, after the Siegel/parabolic/near-parabolic construction machinery is available. No result from Chapters 17–19 is used as a black box in the Chapter 15 hyperbolicity proofs.

15.3 Worked examples

Worked example: Stable and unstable coordinates in a toy model

Consider

$$L(s, u) = (\rho s, \delta u), \quad 0 < \rho < 1 < \delta.$$

After n iterates,

$$L^n(s, u) = (\rho^n s, \delta^n u).$$

Thus the horizontal axis $u = 0$ is the local stable set: two points on it approach each other at rate ρ^n . Every nonzero transverse displacement is amplified at rate δ^n .

This is the linear picture behind Chapter 15, but it omits the main theorem. For renormalization one must first prove that the stable direction is actually tangent to the hybrid class and that the transverse quotient really expands. Rigidity is part of that proof; it is not contained in the matrix calculation.

Worked example: Transversality gives the inverse unstable scale

Suppose the target hybrid leaves have transverse coordinates

$$u_n = K\delta^{-n}(1 + o(1)), \quad K \neq 0,$$

and a parameter curve satisfies

$$u(\mu) = a(\mu - \mu_*) + O((\mu - \mu_*)^2), \quad a \neq 0.$$

Local inversion gives

$$\mu_n - \mu_* = \frac{K}{a}\delta^{-n}(1 + o(1)).$$

The unstable direction expands under forward renormalization by δ ; therefore the parameter intervals that survive n inverse steps shrink by δ^{-n} .

Boundary case: Quadratic tangency changes the exponent

Now suppose the parameter curve is tangent to the stable leaf:

$$u(\mu) = b(\mu - \mu_*)^2 + O((\mu - \mu_*)^3), \quad b \neq 0.$$

Solving $u(\mu_n) \asymp K\delta^{-n}$ gives

$$|\mu_n - \mu_*| \asymp \left| \frac{K}{b} \right|^{1/2} \delta^{-n/2}.$$

So the universal first-order parameter law is a transverse phenomenon. A quadratic tangency naturally changes the exponent from n to $n/2$.

Worked example: Why α_{dyn} and δ_{par} are different

For a stationary Feigenbaum map, the Chapter 14 spatial scale is read from the affine rescaling inside one map. Schematically,

$$f = A \circ f^p \circ A^{-1}, \quad |A'| = \alpha_{\text{dyn}}.$$

By contrast, Chapter 15 defines

$$\delta_{\text{par}} = \lambda_u(DR_M(f_*)),$$

the unstable eigenvalue of the derivative of an operator acting on a space of maps. The first compares nested domains in the dynamical plane; the second compares transverse displacements in map or parameter space. They belong to different linearizations and neither definition implies the other.

Boundary case: A black-hole bookkeeping preview is not the positive-area theorem

Imagine a toy nested process in which a fixed fraction $\eta > 0$ of available mass lands in the next deep return and at most a fraction ξ is permanently lost at each scale. If the quantitative landing gain dominates the total escape loss, a positive amount of mass can survive all levels. This explains the words “landing”, “escaping”, and “black-hole regime”.

The actual positive-area Feigenbaum theorem is much deeper. One must first construct stationary maps for which the required geometric and probabilistic estimates hold uniformly. Those constructions use the neutral and near-parabolic machinery developed later, so their formal theorem is owned by Chapter 19 rather than by this chapter.

15.4 What this chapter established

Chapter 15 completed the transverse structure deliberately left open in Chapter 14. The first theorem is rigidity: bounded marked combinatorics together with uniform a priori bounds determine the hybrid leaf. Its proof uses controlled quasiconformal pullback and the bounded-type exclusion of invariant line fields.

Rigidity then becomes an input to hyperbolicity. Hybrid tangent directions contract; the quotient transverse to the hybrid foliation is one complex dimension and nonsingular; rigidity rules out neutral slow shadowing. The small-orbits criterion and graph transform therefore produce a continuous uniformly hyperbolic splitting with one complex unstable direction. The local stable manifolds are exactly the local hybrid leaves.

The nonlinear stable-manifold theorem upgrades the stable derivative estimate to exponential convergence in the ambient Banach charts. At a stationary fixed point, linearization of the one-dimensional unstable direction and first-order regularity of stable holonomy yield parameter universality for transverse analytic families. This is

the origin of δ_{par} .

The positive-area Feigenbaum problem is intentionally not counted among the formal Chapter 15 theorems. Chapter 15 introduces its landing/escaping vocabulary, but the construction that realizes the black-hole regime requires later Siegel/parabolic/near-parabolic machinery. The corresponding theorem is therefore proved in Chapter 19.

Looking ahead

Chapter 16 changes renormalization models: it studies critical circle maps. Real bounds, rigidity, complex bounds, and renormalization hyperbolicity reappear, but they do so in a different phase space with different return objects. Later Chapters 17–19 develop neutral and near-parabolic tools; those are the missing construction inputs for the positive-area theorem routed to Chapter 19.

Why this chapter matters

Chapter 16 moves from quadratic-like renormalization to dynamics on the circle. The central object is a critical circle homeomorphism: it has an irrational rotation number, but unlike a rigid rotation it also has a nonflat critical point. That critical point makes small-scale geometry nontrivial and turns closest returns into a renormalization problem.

The chapter has two layers. The classical layer establishes real a priori bounds, exponential convergence of normalized commuting pairs, rigidity, and hyperbolicity of critical-circle renormalization. The modern Corona layer enlarges the phase space from the circle to critical quasicircles. Its conclusion is again hyperbolic: the prescribed rotation/criticality class is a codimension-one stable manifold and there is one complex unstable direction.

Learning goals

By the end of this chapter you should be able to:

1. explain how continued-fraction closest returns generate dynamical partitions;
2. use real bounds to control adjacent atoms uniformly;
3. distinguish exponential renormalization convergence from rigidity;
4. derive smooth rigidity from fine-grid comparison;
5. identify the one-dimensional unstable direction of cylinder renormalization;
6. explain why Corona escaping rigidity is needed to prove codimension one.

Conceptual roadmap

real bounds $\implies$ complex/renormalization compactness
 $\implies$ exponential convergence
 $\implies$ rigidity.

while the operator-theoretic branch gives hyperbolicity. Corona renormalization then adds a global escaping/cascade argument to identify the stable class and rule out extra unstable directions.

Common pitfall

Do not infer rigidity merely from topological conjugacy, and do not infer a one-dimensional unstable space merely from the existence of one repelling eigenvector. Both conclusions need the quantitative geometry developed in the proof.

Critical Circle Maps and Renormalization

16.1 Chapter structure and theorem map

A critical circle map is an orientation-preserving circle homeomorphism with irrational rotation number and at least one finite-order nonflat critical point. Closest returns determined by continued fractions generate dynamical partitions and normalized commuting pairs. The proof structure of this chapter is

real bounds $\Rightarrow$ complex bounds,
parabolic cascades $\Rightarrow$ uniform twisting $\Rightarrow$ local rigidity,
cylinder phase space + stable contraction $\Rightarrow$ one-dimensional unstable hyperbolicity.

The two branches meet when local rigidity gives universal same-rotation C^0 renormalization contraction; this is the stable input needed for the full cylinder-renormalization horseshoe.

Theorem 16.1 (Classical Chapter 16 synthesis). *For real-analytic unicritical circle maps of fixed odd criticality, the following hold.*

- (i) *Deep closest-return partitions satisfy real a priori bounds, uniformly in the continued-fraction entries.*
- (ii) *Deep analytic renormalizations enter a universal complex-bounds class.*
- (iii) *If two maps have the same irrational rotation number, their canonical conjugacy is $C^{1+\alpha}$ at the critical point for a universal $\alpha > 0$, and their normalized renormalizations converge geometrically in C^0 .*
- (iv) *Analytic cylinder renormalization is hyperbolic on the full invariant horseshoe, with codimension-one stable bundle given by fixed-rotation classes and one real unstable direction on the real slice (one complex unstable direction after complexification).*

These conclusions are proved below from the named analytic interfaces stated in the chapter; none is inferred from the later critical-quasicircle/Corona theory.

Remark 16.1 (Scope of higher-derivative convergence). The arbitrary-type pre-hyperbolicity theorem used here is the universal geometric C^0 contraction. Bounded-type C^r convergence is available by the classical de Faria-de Melo theory. For arbitrary type, C^r contraction is most cleanly obtained after analytic hyperbolicity, by contraction in the analytic Banach norm followed by Cauchy estimates on a smaller core. The older draft's single pre-hyperbolicity C^r statement has therefore been split into its logically correct forms.

16.2 Standing class and dynamical partitions

Let $f : \mathbb{S}^1 \rightarrow \mathbb{S}^1$ be an orientation-preserving C^3 homeomorphism with finitely many critical points $c_0, \dots, c_{N-1}$, all nonflat. Around c_i choose a neighborhood W_i and numbers $s_i > 1$ such that

$$0 < \alpha_i |x - c_i|^{s_i-1} \leq f'(x) \leq \beta_i |x - c_i|^{s_i-1}, \quad x \in W_i \setminus \{c_i\}. \quad (16.1)$$

After shrinking W_i we also use the standard critical-coordinate normalization in which the Schwarzian is negative on $W_i \setminus \{c_i\}$. On the complement of the critical neighborhoods, $\log f'$ has bounded variation.

Integrating (16.1) gives the only critical power-law estimate needed below: if x, y lie on the same side of c_i and $|x - c_i| \leq |y - c_i|$, then

$$\frac{|f(x) - f(c_i)|}{|f(y) - f(c_i)|} \leq \gamma_i \left(\frac{|x - c_i|}{|y - c_i|} \right)^{s_i} \quad (16.2)$$

for a constant γ_i depending only on the fixed critical chart.

Let the irrational rotation number of f have continued fraction

$$\rho(f) = [a_0, a_1, a_2, \dots], \quad q_{n+1} = a_n q_n + q_{n-1}.$$

We use the following explicit prerequisite.

Poincaré orbit order before no-wandering is proved. Every orientation-preserving circle homeomorphism with irrational rotation number admits a monotone degree-one semiconjugacy to the rigid rotation. Hence the cyclic order of an orbit is already the rigid-rotation order, even before injectivity of the semiconjugacy is known. We therefore define the closest-return interval $I_n(x)$ to have endpoints x and $f^{q_n}(x)$ and to lie on the side selected by $f^{q_{n+2}}(x)$. Before the no-wandering theorem is proved, the usual closest-return collection

$$\mathcal{P}_n(x) = \{f^j(I_n(x)) : 0 \leq j < q_{n+1}\} \cup \{f^j(I_{n+1}(x)) : 0 \leq j < q_n\} \quad (16.3)$$

is regarded only as an orbit-ordered collection: a wandering gap, if one existed, could prevent it from being a genuine partition. The cross-ratio and dynamically symmetric-interval estimates below use only orbit order. Theorem 16.6 will rule out wandering gaps, after which (16.3) is the usual dynamical partition modulo endpoints.

For the distortion argument fix a critical point c_0 and write

$$I_n = I_n(c_0), \quad I_n^j = f^j(I_n).$$

After Theorem 16.6 we also write $\mathcal{P}_n = \mathcal{P}_n(c_0)$ and call the I_n^j atoms *long* and the I_{n+1}^j atoms *short*.

16.3 Cross-ratio distortion: the key real-variable estimate

For $M \subseteq T$, let L and R be the components of $T \setminus M$ and define

$$b(M, T) = \frac{|L| |R|}{|L \cup M| |M \cup R|}. \quad (16.4)$$

If g is monotone on T , set

$$\text{CrD}(g; M, T) = \frac{b(g(M), g(T))}{b(M, T)}.$$

Lemma 16.2 (Deep-level bounded-multiplicity cross-ratio inequality). *There exist $\varepsilon > 0$ and $B > 1$, depending only on f , such that the following holds. Let $M_j \subseteq T_j$ be a finite family with $|T_j| < \varepsilon$ and intersection multiplicity at most m . Then*

$$\prod_j \text{CrD}(f; M_j, T_j) \leq B^m. \quad (16.5)$$

Consequently, if $T, f(T), \dots, f^k(T)$ have multiplicity at most m and all are shorter than ε , then

$$\text{CrD}(f^k; M, T) \leq B^m. \quad (16.6)$$

Proof. Choose the critical neighborhoods W_i so small that $Sf < 0$ on $W_i \setminus \{c_i\}$, and let V be a compactly contained open set in the complement of the critical points such that $V \cup \bigcup_i W_i = \mathbb{S}^1$. Take ε below a Lebesgue number for this cover.

If $T_j \subset V$, the mean-value representation of the four factors in (16.4) gives

$$\log \text{CrD}(f; M_j, T_j) \leq 2 \text{Var}_{T_j}(\log f').$$

Because the family has multiplicity at most m ,

$$\sum_{T_j \subset V} \text{Var}_{T_j}(\log f') \leq m \text{Var}_V(\log f'),$$

so the total contribution of all such factors is bounded by $\exp(2m \text{Var}_V(\log f'))$.

Now take $T_j \subset W_i$. If T_j does not contain c_i , the negative Schwarzian property contracts the b -cross-ratio, hence its factor is at most 1. If T_j contains c_i , the derivative bounds (16.1), equivalently the integrated power law (16.2), give a uniform upper bound D_i for the distortion of $b(M_j, T_j)$. At most m members of the family can contain a fixed c_i , hence at most mN critical factors occur. Therefore

$$\prod_j \text{CrD}(f; M_j, T_j) \leq \exp(2m \text{Var}_V(\log f')) \prod_{i=0}^{N-1} D_i^m = B^m.$$

The iterate form follows from the cocycle identity for cross-ratio distortion. The restriction to sufficiently short intervals does not use no-wandering. Among the first q_{n+1} orbit points, the rigid-rotation cyclic order forces the minimal orbit gap to tend to zero as $q_{n+1} \rightarrow \infty$; the seven-point construction below is based at such a minimal gap.

Configurations not contained in a single critical or regular chart form a compact finite-atlas family and are absorbed into the global cross-ratio constant. $\square$

Remark 16.2. This lemma is the precise replacement for the vague phrase “bounded distortion along a bridge.” Its constant depends on overlap multiplicity, not on the length of the continued-fraction block. This is why unbounded entries a_{n+1} cause no loss.

16.4 The seven-point engine

Lemma 16.3 (Seven-point comparison). *There is $C_1 > 1$ such that for every sufficiently large n one can choose five consecutive q_n -iterates*

$$z_1, z_2, z_3, z_4, z_5, \quad z_{j+1} = f^{q_n}(z_j),$$

for which

$$C_1^{-1} \leq \frac{|z_{i-1} - z_i|}{|z_i - z_{i+1}|} \leq C_1, \quad i = 2, 3, 4. \quad (16.7)$$

Proof. Choose z minimizing $x \mapsto |f^{q_n}(x) - x|$ and put

$$z_0 = f^{-4q_n}(z), \quad z_1 = f^{-3q_n}(z), \quad z_2 = f^{-2q_n}(z), \quad z_3 = f^{-q_n}(z), \quad z_4 = z, \quad z_5 = f^{q_n}(z), \quad z_6 = f^{2q_n}(z).$$

For large n these points occur in this cyclic order on the short arc containing z_4 , and minimality gives

$$|z_4 - z_5| \leq |z_i - z_{i+1}|, \quad 0 \leq i \leq 5. \quad (16.8)$$

For $0 \leq i \leq 3$ let $T_i = [z_i, z_{i+3}]$ and $M_i = [z_{i+1}, z_{i+2}]$. The q_n -iterate sends (M_i, T_i) to (M_{i+1}, T_{i+1}) , and the orbit family of each T_i has multiplicity at most 3. Lemma 16.2 therefore gives a uniform bound B_0 for each relevant q_n cross-ratio distortion.

Apply this first to (M_2, T_2) . Cancelling the four cross-ratio factors and using (16.8) yields

$$\frac{|z_3 - z_4|}{|z_4 - z_5|} \leq B_0 \frac{|z_4 - z_6|}{|z_5 - z_6|} \leq 2B_0.$$

The reverse ratio is bounded by minimality. Thus the $i = 4$ ratio is controlled. Apply the same calculation to (M_1, T_1) . The already controlled $i = 4$ ratio turns the cross-ratio identity into a bound for $|z_2 - z_3|/|z_3 - z_4|$ and its reciprocal. A third application to (M_0, T_0) similarly controls $|z_1 - z_2|/|z_2 - z_3|$. Taking the maximum of the three constants gives C_1 . $\square$

Lemma 16.4 (Transfer from the seven-point window). *There exists $C_2 > 1$ such that the following holds. Let $z_1, \dots, z_5$ be as in Lemma 16.3, and let*

$$w_{j+1} = f^{q_n}(w_j), \quad 0 \leq j \leq 3,$$

with w_1 lying in the component between z_1 and z_2 which avoids z_3 . Then

$$\frac{|w_1 - w_2|}{|w_0 - w_1|} \leq C_2, \quad C_2^{-1} \leq \frac{|w_{i-1} - w_i|}{|w_i - w_{i+1}|} \leq C_2 \quad (i = 2, 3). \quad (16.9)$$

Proof. Let T be the interval from w_0 to w_3 containing the seven-point window and let $M = [w_1, w_2]$. The q_n orbit of T has multiplicity at most 3, hence

$$b(f^{q_n}M, f^{q_n}T) \leq B_0 b(M, T).$$

After writing both sides with (16.4), the cyclic order of the w_j and z_j bounds every factor except $|w_1 - w_2|/|w_0 - w_1|$ by a product of at most two ratios from (16.7). Hence

$$\frac{|w_1 - w_2|}{|w_0 - w_1|} \leq B_0(C_1 + C_1^2)(1 + C_1 + C_1^2).$$

For $i = 2, 3$, minimality of the central gap in the seven-point construction gives a lower bound for $|w_i - w_{i+1}|$, while the interval $[w_{i-1}, w_i]$ is contained in a union of two adjacent seven-point gaps. Lemma 16.3 then gives both sides of the second estimate. Enlarging the constant once proves the claim. $\square$

Lemma 16.5 (Dynamic symmetry of closest returns). *There exists $C_3 > 1$ such that for every sufficiently large n and every $x \in \mathbb{S}^1$,*

$$C_3^{-1}|x - f^{-q_n}(x)| \leq |f^{q_n}(x) - x| \leq C_3|x - f^{-q_n}(x)|. \quad (16.10)$$

Proof. It is enough to prove the upper bound, since the lower bound follows after replacing x by $f^{-q_n}(x)$. Let J be the interval from z_1 to z_3 containing z_2 for a seven-point configuration. The rotation combinatorics imply

$$\bigcup_{i=0}^{q_{n+1}} f^{-i}(J) = \mathbb{S}^1.$$

Choose i so that $f^i(x) \in J$. In one of the two halves of J , the orbit points

$$w_0 = f^{i-q_n}(x), \quad w_1 = f^i(x), \quad w_2 = f^{i+q_n}(x), \quad w_3 = f^{i+2q_n}(x)$$

satisfy Lemma 16.4.

Let T have endpoints $f^{-q_n}(x)$ and $f^{2q_n}(x)$ and contain x , and put $M = [x, f^{q_n}(x)]$. Directly from the definition,

$$b(M, T) \leq \frac{|x - f^{-q_n}(x)|}{|f^{q_n}(x) - x|}. \quad (16.11)$$

Lemma 16.4 gives

$$b(f^i(M), f^i(T)) \geq (1 + C_2)^{-2}.$$

The chain $T, f(T), \dots, f^i(T)$ has multiplicity at most 3, so Lemma 16.2 gives

$$(1 + C_2)^{-2} \leq B_0 b(M, T).$$

Combining with (16.11) proves the upper bound in (16.10) with $C_3 = B_0(1 + C_2)^2$. $\square$

Theorem 16.6 (Yoccoz no-wandering theorem for multicritical circle maps). *Let $f : \mathbb{S}^1 \rightarrow \mathbb{S}^1$ be a C^3 orientation-preserving circle homeomorphism with irrational rotation number and finitely many nonflat critical points. Then f has no wandering intervals.*

Consequently its Poincaré semiconjugacy to the rigid rotation is a homeomorphism, and the closest-return collection (16.3) is a partition of the circle modulo endpoints.

Proof. Assume to the contrary that $J = (a, b)$ is a wandering interval, chosen maximal among the intervals collapsed by the Poincaré semiconjugacy. For each continued-fraction denominator q_n , let

$$L_n = (f^{-q_n}(a), a), \quad R_n = (a, f^{q_n}(a)),$$

where the arcs are chosen according to the rigid-rotation cyclic order. Maximality puts the endpoint a in the nonwandering set. For an irrational circle homeomorphism the restriction to the nonwandering set is minimal, so the backward closest returns satisfy

$$|L_n| = |a - f^{-q_n}(a)| \rightarrow 0. \quad (16.12)$$

The same rigid-rotation order places the entire wandering gap J inside the forward closest-return arc R_n for every n (with the standard alternating choice of closest-return arcs). Hence $|R_n| \geq |J| > 0$. Applying Lemma 16.5 at $x = a$ gives

$$|L_n| \geq C_3^{-1}|R_n| \geq C_3^{-1}|J| > 0,$$

contradicting (16.12). Thus wandering intervals do not exist.

Absence of wandering intervals makes the Poincaré semiconjugacy injective. It is therefore a conjugacy to the rigid rotation, and the standard closest-return orbit arcs cover the circle with disjoint interiors. This is exactly the partition assertion. $\square$

Primary-source calibration. The proof is the Yoccoz mechanism: cross-ratio distortion gives uniform comparability of dynamically symmetric closest-return intervals, and maximal wandering intervals are then excluded by recurrence of their endpoints. This organization appears in Yoccoz's 1984 note and, with the cross-ratio details written out, in de Faria–Guarino, *Dynamics of multicritical circle maps*, Section 4.

16.5 The six base atoms

Proposition 16.7 (Six-atom comparison). *There exists $C_4 > 1$ such that, for every sufficiently large n , the six intervals*

$$I_n, \quad I_{n+1}, \quad I_n^{q_n}, \quad I_n^{q_{n+1}}, \quad I_{n+1}^{q_n}, \quad I_n^{q_{n+1}-q_n} \quad (16.13)$$

are pairwise C_4 -comparable.

Proof. We isolate the five comparisons from which all others follow by telescoping.

1. I_n versus $I_n^{q_n}$. These two intervals are the forward and backward q_n gaps at their common endpoint $f^{q_n}(c_0)$. Lemma 16.5 gives

$$C_3^{-1}|I_n| \leq |I_n^{q_n}| \leq C_3|I_n|. \quad (16.14)$$

2. $I_n^{q_{n+1}}$ versus $I_n^{q_{n+1}-q_n}$. They are dynamically symmetric at $f^{q_{n+1}}(c_0)$, so again

$$C_3^{-1}|I_n^{q_{n+1}}| \leq |I_n^{q_{n+1}-q_n}| \leq C_3|I_n^{q_{n+1}}|. \quad (16.15)$$

3. $I_n^{q_{n+1}-q_n}$ versus I_n . The interval from c_0 to $f^{-q_n}(c_0)$ is dynamically symmetric to I_n . Its combinatorial inclusion in $I_n^{q_{n+1}-q_n} \cup I_n^{q_{n+1}}$, together with (16.15), gives one side of the comparison. Conversely, $I_n^{q_{n+1}} \subset I_n \cup I_{n+1}$ and the symmetric backward q_{n+1} gap at c_0 lies in I_n ; using Lemma 16.5 twice gives the reverse side. Hence

$$[C_3(C_3 + 1)]^{-1}|I_n| \leq |I_n^{q_{n+1}-q_n}| \leq C_3(C_3 + 1)|I_n|. \quad (16.16)$$

4. The critical scale ratio I_{n+1}/I_n . The easy direction follows from the inclusion of the backward q_{n+1} gap in I_n and dynamic symmetry:

$$|I_{n+1}| \leq C_3|I_n|. \quad (16.17)$$

The opposite direction is the key critical estimate. Put

$$r_n = \frac{|I_{n+1}|}{|I_n|}, \quad T = I_{n+1} \cup I_n \cup I_n^{q_n}.$$

Consider the pair $(I_n^1, f(T))$ and iterate it by $f^{q_{n+1}-1}$. The corresponding chain has multiplicity at most 3, so

$$\frac{b(I_n^{q_{n+1}}, f^{q_{n+1}}T)}{b(I_n^1, fT)} \leq B_0. \quad (16.18)$$

Dynamic symmetry, (16.14)–(16.16), and the elementary inclusions of the closest-return configuration give a constant $\theta_1 > 0$ such that

$$b(I_n^{q_{n+1}}, f^{q_{n+1}}T) \geq \theta_1 r_n. \quad (16.19)$$

On the other hand, one factor of $b(I_n^1, fT)$ is the ratio of the images under f of the two intervals based at the critical point c_0 . The power law (16.2) with exponent $s_0 > 1$, and dynamic symmetry for the second cross-ratio factor, give

$$b(I_n^1, fT) \leq \theta_2 r_n^{s_0} \quad (16.20)$$

for a constant $\theta_2 > 0$. Equations (16.18)–(16.20) imply

$$\theta_1 r_n \leq B_0 \theta_2 r_n^{s_0}.$$

Since $r_n > 0$ and $s_0 > 1$,

$$r_n \geq \theta_3 := \left(\frac{\theta_1}{B_0 \theta_2} \right)^{1/(s_0-1)} > 0. \quad (16.21)$$

Together with (16.17),

$$\theta_3 |I_n| \leq |I_{n+1}| \leq C_3 |I_n|. \quad (16.22)$$

This is exactly where nonflat criticality prevents the closest-return geometry from collapsing.

5. $I_{n+1}^{q_n}$ versus I_n . The inclusion $I_{n+1}^{q_n} \subset I_n$ gives the upper bound. For the lower bound set

$$T^* = I_{n+1}^{q_n} \cup I_n^{q_n} \cup I_n^{2q_n}$$

and apply $f^{q_{n+1}-q_n}$ to the pair $(I_n^{q_n}, T^*)$. The orbit family again has multiplicity at most 3. At the image end, dynamic symmetry and (16.22) give a uniform positive lower bound for the resulting cross-ratio. At the initial end, (16.14) and dynamic symmetry give

$$b(I_n^{q_n}, T^*) \leq C_* \frac{|I_{n+1}^{q_n}|}{|I_n|}$$

with C_* depending only on C_3 . Lemma 16.2 therefore yields

$$\theta_4 |I_n| \leq |I_{n+1}^{q_n}| \leq |I_n| \quad (16.23)$$

for some $\theta_4 > 0$.

The five estimates (16.14), (16.15), (16.16), (16.22), and (16.23) connect all six intervals in (16.13). Any pairwise ratio is a product of at most four of these controlled ratios. Taking one sufficiently large C_4 proves the proposition. $\square$

Remark 16.3 (Why the old bridge lemma disappears). The proposition controls a finite base configuration independently of the size of a_{n+1} . Long continued-fraction blocks are not treated by iterating a local adjacent-ratio estimate one atom at a time; that would accumulate a factor exponential in the bridge length. Instead, one sends an arbitrary three-atom block to the controlled base configuration in one cross-ratio estimate. Bounded intersection multiplicity keeps the distortion constant uniform.

16.6 Propagation to every adjacent atom

Lemma 16.8 (Three-atom transport). *Let $M \in \mathcal{P}_n$ and let L, R be its two neighbors, $T = L \cup M \cup R$. There is an iterate $0 \leq k \leq q_{n+1}$ such that*

1. f^k maps T homeomorphically onto a three-atom block T^* whose atoms belong to the six-atom configuration of Proposition 16.7;
2. the family $T, f(T), \dots, f^k(T)$ has intersection multiplicity at most 3.

Proof. This is the two-tower combinatorics of (16.3). If $M = I_{n+1}^j$ is short with $0 \leq j < q_n$, take $k = q_n - j$. Then

$$f^k(M) = I_{n+1}^{q_n},$$

and its two long neighbors map to $I_n^{q_{n+1}}$ and $I_n^{q_n}$ (up to left-right order). If $M = I_n^j$ is long, run the same index calculation to the first end of the long tower; the image triple is one of the adjacent triples obtained from the six intervals in Proposition 16.7. In either case, an orbit point of the circle can lie in at most one copy from each of the two tower interiors, with at most one additional overlap created at an endpoint-crossing block. Hence the multiplicity is at most 3. $\square$

Theorem 16.9 (Real a priori bounds for critical circle maps). *Under the standing class above, there exist n_0 and $K > 1$, depending only on the fixed map f , such that for*

every $n \geq n_0$ and every pair of adjacent atoms $I, J \in \mathcal{P}_n$,

$$K^{-1}|J| \leq |I| \leq K|J|. \quad (16.24)$$

No bounded-type assumption is required.

Proof. Take an arbitrary atom $M \in \mathcal{P}_n$, its neighbors L, R , and $T = L \cup M \cup R$. Apply Lemma 16.8. Proposition 16.7 gives a number $\beta_0 > 0$ such that the transported middle atom satisfies

$$b(f^k M, f^k T) \geq \beta_0.$$

For n deep enough all intervals in the chain are shorter than the ε of Lemma 16.2. Since the chain has multiplicity at most 3,

$$\frac{b(f^k M, f^k T)}{b(M, T)} \leq B^3.$$

Therefore

$$b(M, T) \geq \beta := \beta_0 B^{-3} > 0 \quad (16.25)$$

for every atom M of every sufficiently deep partition.

Write $\ell = |L|$, $m = |M|$, $r = |R|$. From

$$b(M, T) = \frac{\ell r}{(\ell + m)(m + r)} \geq \beta$$

we obtain separately

$$\frac{\ell}{\ell + m} \geq \beta, \quad \frac{r}{m + r} \geq \beta,$$

so

$$m \leq \frac{1 - \beta}{\beta} \ell, \quad m \leq \frac{1 - \beta}{\beta} r.$$

Apply the same estimate with L (respectively R) as the middle atom of its own three-atom block. This gives the reverse comparisons. Thus every adjacent pair is uniformly comparable with a constant depending only on β . Enlarging K to cover the finitely many levels $n < n_0$ completes the proof. $\square$

16.7 The analytic phase space and complex a priori bounds

16.7.1 Epstein class and normalized compactness

A normalized analytic critical commuting pair is written $\zeta = (\eta, \xi)$, with the two branches meeting at the cubic critical point 0 and with the usual closest-return normalization. Following Eckmann-Epstein and Yampolsky, an *Epstein-class branch* is a real analytic branch that factors as

$$g = \phi \circ Q, \quad Q(z) = z^3, \quad (16.26)$$

where ϕ is univalent on a slit-plane neighborhood with normalized real geometry. Fixing a geometric parameter $s > 0$ gives a normalized class $\mathcal{E}_s$.

Lemma 16.10 (Renormalization invariance of the Epstein class). *If $\zeta \in \mathcal{E}_s$ is renormalizable, then the normalized pair $\mathcal{R}\zeta$ belongs to an Epstein class with the same fixed geometric parameter after the standard affine normalization.*

Proof. A pre-renormalization branch is a composition of monotone return branches of η and ξ , with exactly one passage through the cubic critical branch. All noncritical factors are univalent on the corresponding slit domains. Isolate the single cubic factor Q and absorb every remaining univalent factor into ϕ . The real a priori bounds from the real-bounds argument keep the normalizing endpoints uniformly separated and comparable, so the affine normalization and its inverse have uniformly bounded distortion. Hence the normalized composition again has the factorization (16.26) with the same type of slit-domain control. This is the Epstein-class invariance used in Yampolsky's complex-bounds proof. $\square$

Lemma 16.11 (Normalized compactness). *For every fixed $s > 0$, the normalized class $\mathcal{E}_s$ is compact in the Caratheodory topology on marked branched coverings.*

Proof. The normalized real intervals have endpoints in a compact nondegenerate configuration by real bounds. The univalent factors in (16.26) omit the complementary rays of uniformly normalized slit planes. Normal-family compactness for the univalent factors gives subsequential locally uniform convergence. The cubic factor is fixed, and the endpoint normalization prevents a constant or degenerate limit. Kernel convergence of the marked domains is exactly Caratheodory convergence, so the limit is again an Epstein branch with the prescribed geometric normalization. Applying the argument simultaneously to the two commuting branches proves compactness of the pair class. $\square$

16.7.2 Hyperbolic lenses and the inverse-orbit estimate

For a real interval $J = [a, b]$, let

$$C_J = \mathbb{C} \setminus ((-\infty, a] \cup [b, \infty))$$

and let $D_\theta(J)$ be the hyperbolic lens of opening angle θ about J in C_J .

Lemma 16.12 (Lens Schwarz principle). *If $J' \subset J$ and $\phi : C_J \rightarrow C_{J'}$ is holomorphic with $\phi(J) \subset J'$, then*

$$\phi(D_\theta(J)) \subset D_\theta(J')$$

for every $0 < \theta < \pi$. Moreover, if a point z has angle at least ε from the slit and $\text{dist}(z, J) \geq |J|$, then

$$\frac{\text{dist}(\phi(z), J')}{|J'|} \leq C(\varepsilon) \frac{\text{dist}(z, J)}{|J|}. \quad (16.27)$$

Proof. The first statement is the Schwarz lemma in the hyperbolic metrics of the slit planes. For the second, normalize both intervals to $[0, 1]$. The least lens containing z

has Euclidean diameter bounded by $C(\varepsilon) \operatorname{dist}(z, J)$; the first statement sends it into the corresponding lens over J' . Undoing normalization gives (16.27). $\square$

The only nonlinear event is the inverse cubic branch. The cube-root map sends a lens over a real interval crossing the critical value into the union of a controlled lens and a fixed sub-right-angle disk over two uniformly comparable subintervals. This elementary geometry is the source of the “angle jump” used below.

Lemma 16.13 (Good-angle inverse-orbit lemma). *Let I_n be a closest-return interval and write the relevant return as*

$$f^{q_{n+1}} = \psi_n \circ f,$$

where ψ_n is univalent on the complex pullback domain following the critical step. There are universal constants C_1, C_2 and a fixed disk D_0 about the critical value such that

$$\frac{\operatorname{dist}(\psi_n^{-1}z, f(I_n))}{|f(I_n)|} \leq C_1 \frac{\operatorname{dist}(z, I_n)}{|I_n|} + C_2, \quad z \in D_0 \cap C_{f^{q_{n+1}}(I_n)}. \quad (16.28)$$

Proof. Follow simultaneously the real inverse chain of the return interval and the complex inverse orbit of z . At a stage for which the complex point makes a definite angle with the corresponding real interval, Lemma 16.12 gives (16.28) immediately. Otherwise the point remains in a controlled lens. At each closest return, the real bounds from the real-bounds argument make the new return interval comparable with its two neighbors. Pullback through noncritical branches preserves the lens angle by Lemma 16.12; pullback through the critical branch uses the cube-root geometry described above.

There are two possibilities. Either a definite angle occurs at a return scale comparable with I_n , in which case the linear estimate follows from (16.27); or the orbit remains trapped in successive controlled lenses. In the latter case, one passes to the next closest-return scale. When a continued-fraction entry is large, the repeated return is nearly parabolic, but the real-bounds argument real bounds keep consecutive fundamental domains comparable and the cube-root lens estimate remains uniform through the whole cascade. Induction over the closest-return levels must terminate once the scale reaches the fixed disk D_0 . Summing the uniformly bounded additive errors accumulated at critical passages gives the constant C_2 and proves (16.28). $\square$

Remark 16.4 (What is and is not compressed in Lemma 16.13). The argument above is a textbook refactor of Yampolsky’s Sections 3–4: the key facts are the lens Schwarz principle, the cube-root angle estimate, real-bound commensurability, and the induction through possible saddle-node cascades. No Montel-only shortcut is used. A publication-final version may expand the cascade induction into the four return lemmas of the source, but there is no additional theorem-sized input inside the Epstein class.

16.7.3 Key cubic estimate and complex bounds

Proposition 16.14 (Key cubic estimate). *There exist universal $B, C > 0$ and a fixed disk D_0 such that, in the normalized return coordinate, whenever z lies in the univalent*

pullback domain, the return image lies in D_0 , and its normalized modulus exceeds B , one has

$$|\tilde{f}^{q_{n+1}}(z)| \geq C|z|^3. \quad (16.29)$$

Proof. Factor the return as $\psi_n \circ f$. Near the critical point, nonflat cubicity and normalization give

$$|f(z) - f(0)| \asymp |z|^3.$$

Lemma 16.13 says that the inverse univalent factor ψ_n^{-1} has at most linear normalized growth. Applying that estimate to $\psi_n(f(z))$ and rearranging gives a lower bound on the return image by a fixed multiple of the cubic critical displacement. Real bounds control all affine normalizations uniformly, giving (16.29) with universal constants. $\square$

Theorem 16.15 (Complex a priori bounds in the Epstein class). *There exists $\mu > 0$ universal such that every sufficiently deep renormalization of every $\zeta \in \mathcal{E}_s$ extends to a holomorphic commuting pair*

$$H_n : \Omega_n \longrightarrow \Delta_n$$

with

$$\text{mod}(\Delta_n \setminus \Omega_n) \geq \mu, \quad (16.30)$$

and with normalized range diameter and quasidisk geometry bounded by universal constants.

Proof. The inverse-orbit construction gives complex pullbacks of the real return intervals. Lemma 16.12 prevents degeneration along noncritical pullbacks, while Proposition 16.14 prevents the critical pullback domain from forming a cusp at the cubic point. Thus the critical domain, its two adjacent pullback domains, and the range are all comparable to the real closest-return scale. After affine normalization, their boundaries lie in a compact family of quasidisks.

If (16.30) failed along a sequence, normalized compactness from Lemma 16.11 would give a Caratheodory limit in which a domain boundary touches the range boundary. The only possible new contact at the critical point is excluded by the cubic estimate (16.29); contacts elsewhere are excluded by univalence and the lens Schwarz principle. Hence the annulus modulus has a positive universal lower bound. The same compactness argument gives a universal upper bound on the range diameter and uniform quasidisk constants. This is precisely the complex a priori bound. $\square$

General analytic pairs: explicit entry interface. The passage from an arbitrary analytic critical commuting pair to the universal Epstein/holomorphic class is logically separate. The primary sources prove that, after finitely many renormalizations, arbitrary C^3 pairs approach the Epstein class geometrically (in the appropriate C^{r-1} topology), and that arbitrary analytic pairs then satisfy the full holomorphic-pair bounds. We therefore register:

CH16-CB-ENTRY. Every analytic critical commuting pair of fixed odd criticality and irrational rotation number has sufficiently deep renormalizations admitting the universal holomorphic-pair bounds of Theorem 16.15; on compact families the entry time is uniform.

the complex-bounds argument treats CH16-CB-ENTRY as a named source-sized interface rather than hiding it inside a phrase such as “by complex bounds.” Its standard proof is the de Faria-de Melo extension of the Epstein-class argument together with the geometric entry theorem quoted as Lemma 3.3 and Theorem 3.8 in Khmelev-Yampolsky.

16.8 Universal stable convergence: the correct arbitrary-type statement

16.8.1 From local rigidity to convergence of normalized returns

Let f_1, f_2 be analytic unicritical circle maps of the same odd criticality and the same irrational rotation number. Let h be the topological conjugacy fixing the critical point. The deep rigidity input of Khmelev-Yampolsky is local:

$$h(x) = ax + O(|x|^{1+\alpha}) \quad (x \rightarrow 0), \quad (16.31)$$

with universal $\alpha > 0$ and $a > 0$ after choosing compatible local coordinates.

Lemma 16.16 (Geometric shrinking of critical scales). *Under the real a priori bounds of the real-bounds argument there exist $C > 0$ and $0 < \theta < 1$ such that the critical closest-return intervals satisfy*

$$|I_n(f_i)| \leq C\theta^n, \quad i = 1, 2.$$

For analytic maps the deep constant θ may be chosen uniformly on the universal real-bounds class.

Proof. A closest-return atom is subdivided, after at most two partition refinements, into at least two adjacent children. the real-bounds argument gives a uniform bound K on the ratio of adjacent children. Hence no child can occupy more than $K/(1+K) < 1$ of its parent. Iterating this fixed loss over every two levels gives the asserted geometric bound (with θ replaced by the square root of that loss if desired). $\square$

Lemma 16.17 (Normalized conjugacies converge geometrically to the identity). *Let $A_{i,n}$ be the affine normalization taking the critical return scale of f_i to the common normalized commuting-pair scale, and set*

$$h_n = A_{2,n} \circ h \circ A_{1,n}^{-1}.$$

If (16.31) holds, then

$$\|h_n - \text{id}\|_{C^0} + \|h_n^{-1} - \text{id}\|_{C^0} \leq C_0 \lambda^n$$

for some universal $0 < \lambda < 1$ after discarding finitely many levels.

Proof. Write $h(x) = ax + E(x)$ with $|E(x)| \leq C|x|^{1+\alpha}$. Since h maps corresponding closest-return endpoints to each other,

$$|I_n(f_2)| = a|I_n(f_1)|(1 + O(|I_n(f_1)|^\alpha)).$$

Therefore the affine normalization cancels the linear factor a , and for normalized $|u| \leq O(1)$,

$$h_n(u) = u + O(|I_n(f_1)|^\alpha).$$

Lemma 16.16 gives $|I_n|^\alpha \leq C\theta^{\alpha n}$. The same calculation for h^{-1} gives the inverse estimate. Take $\lambda = \theta^\alpha$. $\square$

Theorem 16.18 (Universal geometric convergence in the uniform metric). *Assume the local rigidity estimate (16.31). Then there are constants $C = C(f_1, f_2) > 0$ and a universal $0 < \lambda < 1$ such that*

$$\text{dist}_{C^0}(\mathcal{R}^n f_1, \mathcal{R}^n f_2) \leq C\lambda^n \quad (16.32)$$

for all $n \geq 0$ after the standard normalization of critical commuting pairs.

Proof. The normalized conjugacy relation is

$$\mathcal{R}^n f_2 = h_n \circ \mathcal{R}^n f_1 \circ h_n^{-1}.$$

Deep renormalizations lie in the universal real/complex compact class supplied by real and complex bounds; in particular their real derivatives are uniformly bounded. Hence

$$\begin{aligned} \|\mathcal{R}^n f_2 - \mathcal{R}^n f_1\|_{C^0} &\leq \|h_n \circ \mathcal{R}^n f_1 \circ h_n^{-1} - \mathcal{R}^n f_1 \circ h_n^{-1}\|_{C^0} \\ &\quad + \|\mathcal{R}^n f_1 \circ h_n^{-1} - \mathcal{R}^n f_1\|_{C^0} \\ &\leq (1 + L)(\|h_n - \text{id}\|_{C^0} + \|h_n^{-1} - \text{id}\|_{C^0}), \end{aligned}$$

where L is the uniform derivative bound. Lemma 16.17 gives (16.32). Finitely many pre-entry levels are absorbed into C . $\square$

Remark 16.5 (Primary-source parity). Theorem 16.18 is the mechanism behind Khmelev–Yampolsky’s Corollary 1.2: local $C^{1+\alpha}$ rigidity at the critical point becomes universal geometric convergence after closest-return rescaling. Writing the rescaling argument explicitly is pedagogically important because it explains why only local smoothness at the critical point is needed.

Proposition 16.19 (Bounded-type analytic C^r convergence). *Fix a bound B on the continued-fraction entries and a finite $r \geq 0$. For real-analytic unicritical critical commuting pairs of fixed odd criticality, equal bounded-type rotation number with $a_n \leq B$ implies*

$$d_r(\mathcal{R}^n \zeta_1, \mathcal{R}^n \zeta_2) \leq C_{r,B}(\zeta_1, \zeta_2) \lambda_{r,B}^n, \quad 0 < \lambda_{r,B} < 1.$$

Proof. This is the bounded-type exponential-convergence theorem of de Faria-de Melo at the exact scope used below. In the present architecture the derivative upgrade is transparent: uniform complex bounds put corresponding branches on one definite complex collar, exponential uniform contraction controls their difference there, and Cauchy estimates on a smaller collar give every fixed finite C^r norm. The constants may depend on r and on the type bound B ; no arbitrary-type pre-hyperbolicity C^r rate is asserted. $\square$

Theorem 16.20 (Bounded-type rigidity of critical circle maps). *Any two real-analytic critical circle maps with a cubic critical point and the same irrational rotation number of bounded type are conjugate by a $C^{1+\alpha}$ circle diffeomorphism for some $\alpha > 0$.*

Proof. Let f, g be the two maps and let h be the unique topological conjugacy sending the critical point of f to that of g . Equality of the irrational rotation numbers identifies the closest-return combinatorics. Since the criticality is cubic for both maps, Proposition 16.19, applied with $r = 2$ to the corresponding normalized commuting pairs, gives constants $C > 0$ and $0 < \lambda < 1$ such that

$$d_2(\mathcal{R}^n f, \mathcal{R}^n g) \leq C\lambda^n. \quad (1)$$

(Here and below we identify a circle map with the commuting pair obtained at a closest-return level.)

Because the common rotation number is of bounded type, there is $A < \infty$ with $a_n \leq A$ for all continued-fraction entries. Refine each dynamical partition by the standard finite collection of critical spots and bridge subdivisions. Real bounds and $a_n \leq A$ imply that each atom has at most $N = N(A)$ children and that adjacent children have uniformly comparable lengths. Denote the resulting nested fine grids by $\mathcal{Q}_n^f$ and $\mathcal{Q}_n^g$, matched by h .

Summable adjacent-ratio distortion.

There is $C_1 < \infty$ such that for every pair of adjacent atoms I, J of $\mathcal{Q}_n^f$,

$$\left| \log \frac{|h(I)|/|h(J)|}{|I|/|J|} \right| \leq C_1 \lambda^n. \quad (2)$$

Choose the affine normalization that sends the parent atom of $I \cup J$ to a fixed interval. In this coordinate, the endpoints of I and J are among a finite set of marked points determined by at most N iterates of the two branches of the normalized commuting pair. Estimate (1), the chain rule for these finitely many iterates, and real-bounds separation of the marked points show that every marked coordinate differs by $O(\lambda^n)$ for f and g . Lengths of adjacent atoms are bounded below by a fixed fraction of the parent length, so taking logarithms is uniformly Lipschitz. This gives (2).

Fine-grid differentiability criterion.

Let h map one bounded-valence bounded-geometry nested fine grid to another. If (2) holds with an exponentially summable right-hand side, then h is a $C^{1+\alpha}$ diffeomorphism for some $\alpha > 0$.

For $x \in \mathbb{S}^1$, let $I_n(x)$ be the atom of $\mathcal{Q}_n^f$ containing x and put

$$s_n(x) = \frac{|h(I_n(x))|}{|I_n(x)|}.$$

Bounded valence and (2) imply

$$|\log s_{n+1}(x) - \log s_n(x)| \leq C_2 \lambda^n. \quad (3)$$

Thus $\log s_n(x)$ converges uniformly to a continuous function $u(x)$, and $s_n(x)$ converges

uniformly to $D(x) = e^{u(x)} > 0$. For endpoints of grid atoms, difference quotients of h are precisely ratios of sums of neighboring atom lengths; bounded geometry and (3) show that these quotients have the same limit $D(x)$. Density of endpoints and monotonicity then give $h'(x) = D(x)$ for every x .

There is also $0 < \tau < 1$ such that every level- n atom has diameter at most $C_3\tau^n$: bounded valence together with adjacent comparability forces a fixed loss of relative size after a bounded number of refinements. If $|x - y|$ is comparable with the diameter of a level- n atom, connect x to y by a uniformly bounded chain of level- n and level- $(n+1)$ atoms. Summing (3) along that chain gives

$$|u(x) - u(y)| \leq C_4\lambda^n \leq C_5|x - y|^\alpha$$

for any $0 < \alpha < \log \lambda / \log \tau$ after decreasing α slightly to absorb the comparison constants. Hence $D = h'$ is α -Holder. Since D is positive and continuous, h^{-1} is C^1 and its derivative is also α -Holder. This proves the lemma.

Applying Step 2 to the conjugacy h proves that h is a $C^{1+\alpha}$ circle diffeomorphism, as claimed. $\square$

16.8.2 Why the complex-bounds argument does not yet relabel arbitrary-type C^r convergence as proved

The old draft asserted a single rate λ for every finite C^r metric and every irrational combinatorics. That statement is not the theorem cited at the pre-hyperbolicity stage of the primary sources. There are two clean routes to higher derivatives:

- (a) in bounded type, use the de Faria-de Melo theorem directly; for each type bound B and each r it supplies a geometric rate $\alpha_{r,B} < 1$;
- (b) in arbitrary type, first establish the analytic cylinder-renormalization hyperbolicity, which gives stable contraction in an analytic Banach norm; Cauchy estimates then give every finite C^r estimate on a fixed smaller real/complex core.

the complex-bounds argument therefore recommends replacing the old Theorem 16.2 by a C^0 arbitrary-type theorem now, and moving the arbitrary-type C^r upgrade after the hyperbolicity theorem.

16.9 The analytic cylinder phase space

Let $U \subset \mathbb{C}/\mathbb{Z}$ be an equatorial annulus containing $\mathbb{T}$. Let $\mathcal{A}_U$ denote the Banach manifold of bounded analytic maps $\phi : U \rightarrow \mathbb{C}/\mathbb{Z}$ homotopic to the identity on $\mathbb{T}$. In a lift, a local chart is given by bounded 1-periodic analytic functions on $\pi^{-1}(U)$ with the sup norm. Let

$$\mathcal{C}_U \subset \mathcal{A}_U$$

be the finite-codimensional analytic submanifold of maps normalized to have the prescribed odd critical point at the origin. Its real slice $\mathcal{M} = \mathcal{C}_U^{\mathbb{R}}$ is the Banach manifold of real-symmetric critical circle maps extending to U .

Definition 16.1 (Fundamental crescent and cylinder renormalization). A map $f \in \mathcal{C}_U$

is cylinder-renormalizable if an iterate f^q has a fundamental crescent C_f bounded by an arc ℓ and $f^q(\ell)$, with endpoints repelling q -periodic points, such that the quotient

$$(C_f \cup f^q(C_f)) \setminus \{p_1, p_2\} / f^q$$

is conformally a cylinder. Uniformize the quotient by Φ_f so that the projected critical point is 0. The first return map to the crescent descends under Φ_f to a new critical cylinder map, denoted $\mathcal{R}_{\text{cyl}}f$.

The construction is independent of the admissible crescent. For deep Epstein-class commuting pairs, the quotient construction intertwines a fixed iterate of ordinary commuting-pair renormalization with $\mathcal{R}_{\text{cyl}}$.

Proposition 16.21 (Analyticity and intertwining). *On a neighborhood of every cylinder-renormalizable map in the controlled holomorphic class, $\mathcal{R}_{\text{cyl}}$ extends to a real-symmetric analytic operator between Banach charts. Moreover there is a fixed N and a canonical quotient map ι from sufficiently deep commuting-pair classes such that*

$$\iota \circ \mathcal{R}^N = \mathcal{R}_{\text{cyl}} \circ \iota. \quad (16.33)$$

Proof. Repelling endpoints of the fundamental crescent move analytically under a small perturbation by the implicit-function theorem. The bounding arc and its image therefore admit an analytic holomorphic motion, and the normalized uniformizing map of the quotient cylinder depends holomorphically on the perturbation by the Ahlfors-Bers holomorphic-dependence theorem. The first-return map is a finite composition of iterates of f , hence analytic in both the phase variable and the Banach parameter. Composition with the moving uniformization gives analyticity of $\mathcal{R}_{\text{cyl}}$.

For a deep Epstein pair, the crescent is chosen from the canonical return domain. Passing to the quotient removes precisely the conformal-coordinate ambiguity of the commuting-pair representation. Applying $\mathcal{R}^N$ before quotienting or taking one cylinder renormalization after quotienting gives the same first-return dynamics, which proves (16.33). $\square$

16.10 Fixed-rotation classes are codimension one

Fix a point $\hat{f}$ in the renormalization horseshoe and put $\rho = \rho(\hat{f})$. Define

$$\mathcal{D}_\rho = \{f \in \mathcal{M} : \rho(f) = \rho\}.$$

The following argument is the part of Yampolsky's proof most useful pedagogically because it explains why the stable class has codimension one without invoking hyperbolicity.

Theorem 16.22 (Local codimension-one rotation leaf). *There is a neighborhood $W \subset \mathcal{M}$ of $\hat{f}$ such that $\mathcal{D}_\rho \cap W$ is a smooth codimension-one submanifold of $\mathcal{M}$.*

Proof. Let p_k/q_k be the continued-fraction convergents of ρ . Define $\mathcal{D}_k$ to be the maps g near $\hat{f}$ for which the critical point is q_k -periodic and the rotation number is p_k/q_k . In

a lift, the defining equation is one scalar equation

$$\tilde{g}^{q_k}(0) - p_k = 0.$$

A positive perturbation of the lift changes the left-hand side strictly, so the derivative of this equation is nonzero. The implicit-function theorem therefore makes $\mathcal{D}_k$ a codimension-one analytic submanifold.

Let

$$\mathcal{C} = \{v \in T\mathcal{M} : \inf_{x \in \mathbb{R}} v(x) > 0\}$$

be the open positive cone. If $v \in \mathcal{C}$ were tangent to $\mathcal{D}_k$, then a family $g_t = g + tv + o(t)$ would satisfy $\tilde{g}_t > \tilde{g}$ for small $t > 0$, hence $\tilde{g}_t^{q_k}(0) > \tilde{g}^{q_k}(0) = p_k$, contradicting membership in $\mathcal{D}_k$. Thus

$$T_g \mathcal{D}_k \cap \mathcal{C} = \emptyset. \quad (16.34)$$

Rotate $\hat{f}$ by a small angle θ_k so that $f_k = R_{\theta_k} \circ \hat{f} \in \mathcal{D}_k$. The smallest such $|\theta_k|$ tends to 0. Fix $v_0 \in \mathcal{C}$. By (16.34) and Hahn-Banach, after normalizing the separating functionals there are $h_k \in (T\mathcal{M})^*$ with

$$\ker h_k = T_{f_k} \mathcal{D}_k, \quad \|h_k\| = 1, \quad h_k(v_0) \geq \varepsilon > 0.$$

Alaoglu compactness gives a weak-* convergent subsequence $h_k \rightarrow h \neq 0$. Put $T = \ker h$, so $T\mathcal{M} = T \oplus \mathbb{R}v_0$.

Take a local Banach chart ψ at $\hat{f}$ and let P be projection onto T . The uniform cone transversality and the mean-value theorem imply that, for all large k , $P \circ \psi$ maps $\mathcal{D}_k$ injectively onto a common neighborhood of 0 in T ; hence $\mathcal{D}_k$ are graphs over that neighborhood with uniformly controlled first derivative. A subsequence converges in C^1 to a smooth graph G . Every $g \in G$ is a limit of maps with rotation p_k/q_k , so continuity of rotation number gives $\rho(g) = \rho$. Conversely, the local order/monotonicity argument that selected the rational graphs shows that every nearby map of rotation ρ lies on G . Thus $G = \mathcal{D}_\rho \cap W$, proving the theorem. $\square$

16.11 Compactness of the derivative

Suppose now that $\hat{f}$ is periodic for $\mathcal{R}_{\text{cyl}}$ with period m . Let

$$L = D_{\hat{f}} \mathcal{R}_{\text{cyl}}^m : T_{\hat{f}} \mathcal{M} \rightarrow T_{\hat{f}} \mathcal{M}$$

and let $T = T_{\hat{f}}(\mathcal{D}_\rho \cap W)$.

Proposition 16.23 (Compact derivative). *The operator L is compact.*

Proof. The cylinder construction gains analytic domain. More precisely, $\hat{f}$ extends to an annulus $\hat{U}$ compactly containing the Banach-chart annulus U , and the perturbation of the finitely many return iterates induced by a unit tangent vector is uniformly bounded on $\hat{U}$. Differentiating the normalized uniformization and composing with the first-return branches therefore sends the unit ball of the tangent space to a family of

analytic vector fields uniformly bounded on the larger domain $\widehat{U}$. By Montel, this family is precompact on U . Hence the image of the unit ball under L is relatively compact, which is exactly compactness of L . $\square$

16.12 Strict contraction on the fixed-rotation tangent leaf

Proposition 16.24 (Stable spectral contraction). *Assume that points in $\mathcal{D}_\rho \cap W$ converge to the periodic orbit of $\widehat{f}$ at a uniform geometric rate under the corresponding iterate of $\mathcal{R}_{\text{cyl}}$. Then there is $k \geq 1$ such that*

$$\|L^k|_T\| < 1.$$

Equivalently, the spectral radius of $L|_T$ is strictly less than 1.

Proof. The derivative iterates are uniformly bounded on T by analyticity on the compact controlled class. Suppose no iterate has norm below one. Then the compact operator $L|_T$ has spectral radius at least one. Since all nonzero spectral values of a compact operator are eigenvalues of finite multiplicity and can accumulate only at zero, T splits into a stable part and a nontrivial finite-dimensional central/unstable part E_c on which vectors fail to decay geometrically.

Choose a sufficiently small point $g \in \mathcal{D}_\rho \cap W$ whose chart displacement from $\widehat{f}$ has a nonzero E_c component. Taylor expansion of $\mathcal{R}_{\text{cyl}}^m$ shows that for as long as the orbit remains in the chart, its leading displacement is governed by L^j on E_c ; the nonlinear remainder is quadratic and cannot turn this into a uniform geometric decay. This contradicts the assumed geometric convergence of every nearby point in the fixed-rotation leaf. Hence the central/unstable part is absent and some iterate contracts T strictly. $\square$

Remark 16.6 (Why periodic points need only bounded-type convergence). If $\widehat{f}$ is periodic under renormalization, its continued fraction is periodic and therefore bounded type. Thus Proposition 16.24 for periodic points only needs the bounded-type geometric convergence theorem of de Faria-de Melo. The arbitrary-type universal contraction isolated in the complex-bounds argument is needed when one upgrades from periodic-orbit hyperbolicity to a uniform global horseshoe statement.

16.13 The positive cone and the unstable eigenline

Lift maps to $\mathbb{R}$. For a tangent vector v , write

$$\widetilde{f}_t(x) = \widetilde{f}(x) + tv(x) + o(t),$$

and define v_n by

$$\widetilde{f}_t^n(x) = \widetilde{f}^n(x) + tv_n(x) + o(t).$$

Differentiating composition gives the recurrence

$$v_n(x) = \tilde{f}'(\tilde{f}^{n-1}(x))v_{n-1}(x) + v(\tilde{f}^{n-1}(x)). \quad (16.35)$$

Lemma 16.25 (Positive-cone expansion). *Let $\mathcal{C} = \{v : \inf_{\mathbb{R}} v > 0\}$. Then:*

- (i) $L(\mathcal{C}) \subset \mathcal{C}$;
- (ii) *there exist $k \geq 1$ and $\alpha > 0$, uniform on the controlled horseshoe, such that*

$$\inf L^k v > (1 + \alpha) \inf v \quad (v \in \mathcal{C});$$

- (iii) *if $0 \neq v \in \bar{\mathcal{C}}$, then some iterate $L^j v$ belongs to $\mathcal{C}$.*

Proof. If $v > 0$, recurrence (16.35) and orientation preservation imply $v_n > 0$ for every return iterate. Let C_j be a deep fundamental crescent, $J_j = C_j \cap \mathbb{T}$, Φ_j its lifted uniformizing coordinate, and let the corresponding return be $g_j = \hat{f}^{q_{r_j}}$. In the cylinder chart,

$$D\mathcal{R}_{\text{cyl}}(v) = [(\Phi_j)' \circ g_j \cdot v_{q_{r_j}}|_{J_j}] \circ \Phi_j^{-1}. \quad (16.36)$$

Every factor is positive, proving (i).

Real bounds imply that the intervals J_j shrink geometrically. Complex bounds and Koebe distortion give a uniform distortion bound for Φ_j on $g_j(J_j)$ and its neighboring domains. Since Φ_j expands a geometrically small interval to the fixed cylinder scale, after a fixed number of renormalizations

$$\inf_{g_j(J_j)} (\Phi_j)' > 1 + \alpha$$

with universal $\alpha > 0$. Combining this with $\inf v_n \geq \inf v$ from (16.35) proves (ii).

Finally, for $v \geq 0$, $v \not\equiv 0$, choose an iterate of a deep fundamental interval whose image meets an open set on which $v > 0$. Formula (16.35) then makes the induced perturbation strictly positive on the whole return interval, and (16.36) transports it to a vector in $\mathcal{C}$. This proves (iii). $\square$

Proposition 16.26 (Existence and uniqueness of the unstable direction). *The derivative cocycle at a periodic orbit has exactly one unstable real direction.*

Proof. Lemma 16.25(ii) implies that the spectral radius of a suitable iterate of L is > 1 . By compactness, there is an eigenvalue δ with $|\delta| > 1$ and an eigenvector; the cone invariance and eventual strict positivity allow the eigenvector to be chosen in $\mathcal{C}$.

On the other hand, Theorem 16.22 gives an invariant codimension-one tangent hyperplane T , and Proposition 16.24 places the entire spectrum of $L|_T$ strictly inside the unit disk. Any unstable eigenvector must therefore project nontrivially to the one-dimensional quotient $T_{\hat{f}}\mathcal{M}/T$. Hence the unstable space has dimension at most one. Since the cone construction gives one unstable eigenvector, the unstable dimension is exactly one. $\square$

16.14 Hyperbolicity conclusions

Theorem 16.27 (Periodic cylinder-renormalization hyperbolicity). *Let $\hat{f}$ be a periodic point of the analytic cylinder-renormalization operator in the controlled real-symmetric class. Then some return iterate has a hyperbolic fixed point at $\hat{f}$ with splitting*

$$T_{\hat{f}}\mathcal{M} = E_{\hat{f}}^s \oplus E_{\hat{f}}^u, \quad E_{\hat{f}}^s = T_{\hat{f}}\mathcal{D}_{\rho}, \quad \dim_{\mathbb{R}} E_{\hat{f}}^u = 1.$$

Proof. Theorem 16.22 identifies the invariant stable candidate. Proposition 16.24 gives strict contraction there, and Proposition 16.26 gives the unique expanding complement. The two spaces are transverse because the second is not contained in $T\mathcal{D}_{\rho}$. Standard hyperbolic fixed-point theory in Banach spaces completes the splitting. $\square$

Theorem 16.28 (Global horseshoe hyperbolicity). *Assume the arbitrary-irrational universal geometric contraction of matched analytic renormalizations, established later in this chapter from local rigidity. Then there exists an iterate $\mathcal{R}_{\text{cyl}}^{\ell}$ whose invariant global renormalization horseshoe is uniformly hyperbolic, with a one-dimensional unstable bundle and codimension-one stable bundle equal to the fixed-rotation tangent distribution.*

Proof. The proofs of Theorem 16.22, Proposition 16.23, Proposition 16.24, and Lemma 16.25 use constants coming from the universal real/complex bounds and the universal geometric contraction, hence the constants may be chosen uniformly on the compact cylinder representation of the horseshoe. Choose a fixed iterate ℓ large enough that stable vectors contract by a factor < 1 and cone vectors expand by a factor > 1 everywhere. Periodic itineraries are dense in the symbolic horseshoe; compactness and continuity of the derivative cocycle extend their invariant one-dimensional expanding eigendirections to a continuous invariant line bundle E^u . The fixed-rotation tangent hyperplanes give the continuous invariant stable bundle E^s . Uniform contraction/expansion yields the claimed hyperbolicity. $\square$

Remark 16.7 (Complex versus real unstable dimension). On the real slice the unstable direction is one-dimensional over $\mathbb{R}$. The analytic complexification carries the corresponding one-complex-dimensional unstable line. The book should state explicitly which phase space is meant whenever a dimension is quoted.

16.15 Controlled holomorphic pairs and the geometric target

We work after the complex-bounds entry of the complex-bounds argument. Thus all sufficiently deep renormalizations belong to one universal compact class of normalized holomorphic commuting pairs. We suppress inessential normalization constants.

Definition 16.2 (Controlled holomorphic pair). A controlled holomorphic pair is a triple

$$\mathcal{H} = (\eta, \xi, \nu)$$

of holomorphic branches on simply connected domains with a common range, extending a real critical commuting pair, such that:

- (i) all domains and ranges are uniformly quasidisks;
- (ii) the separating annuli have modulus bounded below;
- (iii) the unique critical branch has the fixed odd criticality;
- (iv) real traces satisfy the real-bounds geometry;
- (v) after normalization, the family is precompact in the compact-open topology.

The *complete dynamics* $\mathcal{F}(\mathcal{H})$ is the pseudogroup generated by the local branches and their univalent inverse branches wherever they are defined.

Write $J(\mathcal{H})$ for the filled/limit set relevant to the complete dynamics. A *copy* of a deep renormalization means a bounded-distortion conformal image of one of its controlled domains or ranges under $\mathcal{F}(\mathcal{H})$.

Definition 16.3 (Scale- r copy abundance). Fix a uniform geometry constant K . We say that $\mathcal{H}$ has scale- r copy abundance at $z_0 \in J(\mathcal{H})$ if there is a simply connected pointed neighborhood (U, y) such that

$$|z_0 - y| \asymp_K r, \quad \text{diam } U \asymp_K r,$$

and a family of complete-dynamics copies $\{\Delta_j\}$ contained in U satisfying

$$\sum_j \text{area}(\Delta_j) \geq cr^2 \tag{16.37}$$

for a universal $c > 0$. Bounded overlap is part of the controlled-copy construction, so the sum is a meaningful scale-invariant density statement.

Theorem 7.1 of Khmelev–Yampolsky is precisely the assertion that, sufficiently deep in the compact holomorphic-pair class, this abundance holds for every $z_0 \in J(\mathcal{H})$ and every $0 < r \leq 1$, with universal constants.

16.16 Why the bounded-type proof breaks

For bounded type the continued-fraction entries are uniformly bounded. Adjacent real return scales are therefore uniformly comparable. Given z_0 and an observation scale r , one chooses the first dynamical partition level whose atom is no larger than r . Since parent and child are comparable, a single renormalization copy at that level has diameter comparable to r and already contributes area $\asymp r^2$.

The unbounded-type failure is very specific. A huge continued-fraction entry produces a long string of nearly identical returns. In a renormalized chart the return map approaches a parabolic map. If p is its parabolic fixed point, a Fatou coordinate has first-order behavior

$$\Phi(z) \sim -\frac{1}{z - p}. \tag{16.38}$$

A translation by one in the Fatou coordinate corresponds near p to an Euclidean displacement of order $|z - p|^2$. Hence the largest *single* dynamically generated copy visi-

ble inside a disk of radius r can have diameter only $O(r^2)$ and area $O(r^4)$. No one-copy argument can prove (16.37).

The repair is the central idea of Khmelev–Yampolsky: the near-parabolic dynamics does not create one useful copy, but an entire translation lattice of them. The Jacobian of the inverse Fatou coordinate makes the sum of their areas comparable with the Euclidean area of the observation region.

16.17 Fatou and Douady coordinates for a near-parabolic return

Let η_0 be a real-symmetric holomorphic return map with a simple parabolic fixed point p ,

$$\eta_0(p) = p, \quad \eta_0'(p) = 1,$$

and nonzero quadratic term. There are attracting and repelling petals P^A, P^R and Fatou coordinates

$$\Phi_0^A : P^A \rightarrow \mathbb{C}, \quad \Phi_0^R : P^R \rightarrow \mathbb{C}$$

with

$$\Phi_0^{A,R} \circ \eta_0 = \Phi_0^{A,R} + 1. \quad (16.39)$$

The additive constants are fixed by the standard real-symmetric normalization.

For a nearby non-parabolic map η , the parabolic fixed point splits into two nearby fixed points $p_-(\eta), p_+(\eta)$. Douady coordinates interpolate the Fatou coordinates across this perturbation.

Lemma 16.29 (Normalized Douady coordinates). *After shrinking the compact near-parabolic class, there are attracting and repelling coordinates*

$$\Phi_\eta^A, \Phi_\eta^R$$

with the following uniform properties.

(i) On their domains,

$$\Phi_\eta^{A,R} \circ \eta = \Phi_\eta^{A,R} + 1.$$

(ii) The coordinates converge on compact subsets to the normalized parabolic Fatou coordinates as $\eta \rightarrow \eta_0$.

(iii) If

$$\tau_\pm(z) = a_\pm - \frac{1}{z - p_\pm},$$

with normalizing constants $a_\pm$, then on the controlled incoming/outgoing petals

$$|(\Phi_\eta^{A,R})'(z)| \asymp |\tau'_\pm(z)| \asymp \frac{1}{|z - p_\pm|^2}. \quad (16.40)$$

The comparability constants are uniform.

Proof. The Abel equation is obtained by quotienting the petal by the local dynamics

and uniformizing the resulting cylinder. Fixing the real-symmetric transverse and one marked point removes the additive ambiguity. Normal-family compactness gives continuous dependence on η .

For the derivative estimate, write the local map in a coordinate centered at the relevant fixed point as

$$w \mapsto w + aw^2 + O(w^3), \quad a \neq 0.$$

Then $-1/(aw)$ changes by $1 + O(w)$ under one iterate. Solving the Abel equation by summing this uniformly summable error gives

$$\Phi(w) = -\frac{1}{aw} + O(\log |w|)$$

on a proper subpetal, and differentiation gives $\Phi'(w) \asymp |w|^{-2}$. Compactness of the normalized near-parabolic family makes the constants uniform and gives (16.40) simultaneously on the attracting and repelling sides. $\square$

Proposition 16.30 (Transit representation). *Let η approach η_0 through the near-parabolic class. A long iterate crossing from the attracting petal to the repelling petal can be written*

$$\eta^N = (\Phi_\eta^R)^{-1} \circ T_{\theta(\eta)+k} \circ \Phi_\eta^A \quad (16.41)$$

on a controlled transit domain, where $T_s(w) = w + s$, $k \in \mathbb{Z}$, and the phase $\theta(\eta) \in \mathbb{R}/\mathbb{Z}$ records the mismatch of the two cylinder coordinates. Along a parabolic limit, subsequences of the phase converge and (16.41) converges to a Lavaurs/Douady transit map.

Proof. Both sides of the transit satisfy the same conjugacy relation with translation by one. Thus their difference in cylinder coordinates is constant modulo $\mathbb{Z}$, which is the phase $\theta(\eta)$. The integer k records the number of turns spent in the parabolic channel. Lemma 16.29, together with normal-family compactness on the incoming and outgoing petals, gives convergence after passing to a phase-convergent subsequence. $\square$

16.18 The parabolic-cascade area-packing lemma

We now isolate the argument that replaces bounded-type scale comparability. Let I be a fundamental real interval in a deep near-parabolic renormalization, and let Δ be a controlled complex copy whose real trace is commensurable with one iterate of I .

Lemma 16.31 (Parabolic-cascade area packing). *Fix $K > 1$. There are universal $c = c(K) > 0$ and $K_1 = K_1(K)$ with the following property. Consider a sufficiently deep controlled holomorphic pair whose relevant return has height a and is in the near-parabolic regime. Let Δ be a K -controlled copy based on one atom of the long return string. If z_0 lies on that string and*

$$|I(z_0)| \leq r \leq 1,$$

then there is a pointed neighborhood (U, y) satisfying

$$|z_0 - y| \asymp_{K_1} r, \quad \text{diam } U \asymp_{K_1} r,$$

and a family of complete-dynamics images $\tilde{\Delta}_j \subset U$ such that

$$\sum_j \text{area}(\tilde{\Delta}_j) \geq cr^2. \quad (16.42)$$

The constants do not depend on the parabolic height a .

Proof. If the height is bounded, real bounds and Koebe distortion make successive atoms uniformly commensurable. A bounded number of images of Δ contains a copy of diameter comparable to r , so (16.42) is immediate. The issue is therefore large height.

Choose the attracting or repelling Douady coordinate according to which half of the long return string contains z_0 . In that coordinate, the forward and backward images of Δ are bounded-distortion perturbations of translates of one fixed cell:

$$D_j \approx D_0 + j,$$

with uniformly bounded eccentricity and bounded overlap. The point is that the number of relevant translates tends to infinity with the parabolic height, while their local geometry stays uniform.

Use the model coordinate

$$\tau(z) = a - \frac{1}{z - p}.$$

Its inverse is

$$\tau^{-1}(w) = p - \frac{1}{w - a},$$

so the Euclidean Jacobian is

$$J_{\tau^{-1}}(w) = \frac{1}{|w - a|^4}. \quad (16.43)$$

By Lemma 16.29, the true Douady coordinate and τ have uniformly comparable derivatives, so it suffices to prove the area estimate for the model lattice.

Let U^- be the half of a Euclidean disk of radius comparable to r lying on the chosen side of the parabolic point. The cells D_j whose inverse images lie in U^- cover, up to bounded multiplicity and a bounded boundary loss, the corresponding region $\tau(U^-)$. Since each cell has uniformly positive and finite area, Riemann-sum comparison using (16.43) gives

$$\sum_j \text{area}(\tau^{-1}(D_j)) \asymp \sum_j \int_{D_j} \frac{dA(w)}{|w - a|^4} \quad (16.44)$$

$$\asymp \int_{\tau(U^-)} \frac{dA(w)}{|w - a|^4} \quad (16.45)$$

$$= \text{area}(U^-) \asymp r^2. \quad (16.46)$$

The first comparability uses the uniform cell geometry; the second uses bounded overlap and the fact that omitting the boundary cells costs only a fixed proportion. Returning from τ to the true Douady coordinate changes areas by another universal factor. The repelling half is identical. Taking the half adapted to z_0 yields (16.42). $\square$

Remark 16.8 (Why many tiny copies are enough). Near a parabolic point a single cell at distance r has diameter of order r^2 . There are, however, of order $1/r$ relevant translation cells at comparable and larger Fatou-coordinate radii, with a power-law distribution of Euclidean areas. The exact Jacobian identity (16.43) is what converts their sum back to an r^2 quantity. This is the quantitative heart of the unbounded-type argument.

16.19 Real-axis abundance: the bounded/cascade dichotomy

Let $\mathcal{P}_n$ be the closest-return dynamical partitions. For a real point $x \in J(\mathcal{H}) \cap \mathbb{R}$, let $P_n(x)$ be the atom of $\mathcal{P}_n$ containing x . Real bounds provide bounded geometry inside each level, but the ratio $|P_n(x)|/|P_{n-1}(x)|$ can be very small when the next continued-fraction entry is large.

Lemma 16.32 (Backward intervals and protection). *There is a universal enlargement rule $P \mapsto P^*$ such that the relevant pullbacks of $P_n(x)^*$ along a common monotone branch have bounded intersection multiplicity and stay uniformly well inside the branch domains until the first critical encounter. Consequently every such pullback has a Koebe distortion bound depending only on the controlled class.*

Proof. Adjacent-atom comparability from the real-bounds argument gives a definite scaled neighborhood of every deep real atom before a critical encounter. Pulling the enlarged atoms along the closest-return tower preserves disjoint interiors; an enlarged atom can meet only a bounded number of neighbors. The real Koebe principle then gives the uniform distortion bound. This is precisely the protection mechanism required below. $\square$

Proposition 16.33 (Copy abundance on the real limit set). *There are universal constants $c, K > 0$ such that every sufficiently deep controlled holomorphic pair has scale- r copy abundance at every*

$$x \in J(\mathcal{H}) \cap \mathbb{R}, \quad 0 < r \leq 1.$$

Proof. Choose n minimal with

$$|P_n(x)| \leq r \leq |P_{n-1}(x)|.$$

There are two cases.

Case 1: comparable successive scales. If $|P_n(x)| \geq \varepsilon |P_{n-1}(x)|$ for a universal threshold $\varepsilon > 0$, then $|P_n(x)| \asymp r$. Complex bounds place over $P_n(x)$ a controlled holomorphic-pair domain of diameter comparable to r . A fixed compact subcopy has area $\geq cr^2$, and Lemma 16.32 transports it to a pointed neighborhood of x with bounded distortion.

Case 2: a large scale drop. If $|P_n(x)| < \varepsilon|P_{n-1}(x)|$, the real combinatorics forces a long same-side return string: this is the possible parabolic cascade. Rescale the parent atom $P_{n-1}(x)$ to the normalized holomorphic-pair chart. The associated return map lies in the near-parabolic compact class, and the rescaled observation radius is between the local atom size and the normalized unit scale. Lemma 16.31 therefore produces a family of copies with total area bounded below by the square of the rescaled radius. Pull them back to the original chart using the common univalent branch supplied by Lemma 16.32. Koebe distortion multiplies both the observation radius and the square root of area by comparable factors, so the estimate remains

$$\sum_j \text{area}(\Delta_j) \geq cr^2.$$

The constants in the two cases are universal; taking their minimum proves the proposition. $\square$

16.20 From the real axis to every Julia-set point

A non-real point need not lie over a dynamical partition atom. The standard device is to move a small tangent scale forward until it reaches a controlled real or parabolic configuration, then pull the resulting copies back.

Fix $z_0 \in J(\mathcal{H})$ and a tangent vector v_0 with $|v_0| \asymp r$. Let

$$z_k = H^k(z_0), \quad v_k = (H^k)'(z_0)v_0$$

along the complete-dynamics branch followed by z_0 . Measure v_k in the hyperbolic metric of the complement of the real postcritical set/protection domains. Schwarz-Pick and the strict domain nesting from complex bounds imply eventual expansion of this hyperbolic size.

Lemma 16.34 (First controlled landing of a scale). *There is a first time k at which one of the following occurs.*

- (a) z_k lies in a controlled real protection region and the hyperbolic length of v_k is comparable to one;
- (b) the orbit enters a near-parabolic channel at large height, and the Euclidean scale represented by v_k lies between the smallest local atom and the size of the channel;
- (c) the orbit lies in a bounded-height or nondegenerate complex wing where one controlled renormalization copy already has size comparable with $|v_k|$.

Moreover, the inverse branch from a suitable neighborhood of z_k back to z_0 is univalent on a uniformly enlarged domain.

Proof. If the hyperbolic length stayed uniformly small forever while avoiding the parabolic channels, strict Schwarz-Pick expansion on the compact complement of the protected domains would force it to grow geometrically, a contradiction. Thus a first controlled landing occurs. A large jump can only happen when the orbit crosses a boundary of a protection region or enters a high parabolic channel. In the latter

case the Fatou-coordinate derivative is responsible for the jump, which is exactly alternative (b). Bounded-height channels and the compact complement give (a) or (c). Minimality of the landing time means that the preceding orbit segment avoids the critical value, so the inverse branch is univalent; complex bounds provide the enlarged domain required for Koebe. $\square$

Theorem 16.35 (Universal copy abundance on $J(\mathcal{H})$). *There are universal constants $c, K > 0$ such that every sufficiently deep controlled holomorphic pair, every $z_0 \in J(\mathcal{H})$, and every $0 < r \leq 1$ satisfy scale- r copy abundance.*

Proof. Apply Lemma 16.34. In case (a), use Proposition 16.33 at the real scale comparable to $|v_k|$. In case (b), apply the parabolic packing Lemma 16.31. In case (c), use the single controlled copy supplied by compactness and complex bounds. Thus in all cases there is a pointed area (U_k, y_k) of diameter $\asymp |v_k|$ containing copies $\Delta_{k,j}$ with

$$\sum_j \text{area}(\Delta_{k,j}) \geq c|v_k|^2.$$

Let G be the univalent inverse branch taking z_k back to z_0 . Koebe distortion on the enlarged domain in Lemma 16.34 gives

$$\text{diam } G(U_k) \asymp |G'(z_k)| |v_k| \asymp |v_0| \asymp r$$

and, uniformly for every $\Delta_{k,j} \subset U_k$,

$$\text{area}(G(\Delta_{k,j})) \asymp |G'(z_k)|^2 \text{area}(\Delta_{k,j}).$$

Summing and using $|G'(z_k)| |v_k| \asymp r$ yields

$$\sum_j \text{area}(G(\Delta_{k,j})) \geq c'r^2.$$

The pulled-back pointed area is at distance and diameter comparable with r from z_0 . This is the required abundance. $\square$

16.21 Arbitrary-type critical deepness

Uniform twisting controls the dynamics around a point at all scales. McMullen's differentiability theorem also requires the point to sit deeply inside the invariant set. This property is independent of bounded type and can be proved directly from nested protection domains.

Definition 16.4 (Deep point). Let $X \subset \mathbb{C}$ be compact and $c \in X$. The point c is β -deep in X if there are $C, r_0 > 0$ such that

$$\text{dist}(z, X) \leq C|z - c|^{1+\beta} \tag{16.47}$$

whenever $|z - c| < r_0$.

For a controlled holomorphic pair let $U_n \ni c$ be the nested critical neighborhoods given by successive renormalizations, and let $W_n \supset U_n$ be their protection domains. Complex bounds give

$$\text{diam } U_{n+1} \leq \lambda_0 \text{diam } U_n, \quad \text{mod}(W_n \setminus \overline{U_n}) \geq m_0 > 0 \quad (16.48)$$

for universal $\lambda_0 < 1, m_0 > 0$ along a subsequence with uniformly bounded gaps.

Lemma 16.36 (Hyperbolic expansion across protection levels). *There are $A > 0$ and $\Lambda > 1$ such that an orbit segment crossing k successive protected levels without hitting the critical value expands the hyperbolic norm of tangent vectors by at least $A\Lambda^k$.*

Proof. On each protected level, the relevant branch maps a proper subdomain into the next hyperbolic surface. Schwarz–Pick is strictly contracting for the inverse branch. Because the modulus of the protection annulus is at least m_0 , compactness gives a contraction factor $\Lambda^{-1} < 1$ bounded away from one. Multiplication over k levels gives the claim; the bounded gaps between selected levels are absorbed into A . $\square$

Theorem 16.37 (The critical point is universally deep). *There is a universal $\beta > 0$ such that the critical point c of every sufficiently deep controlled holomorphic pair is β -deep in $J(\mathcal{H})$.*

Proof. Take z close to c , $z \notin J(\mathcal{H})$, and choose n so that

$$\text{diam } U_{n+1} < |z - c| \leq \text{diam } U_n.$$

By (16.48), the number of protected levels between the macroscopic scale and U_n is at least

$$k \geq a \log \frac{1}{|z - c|} - b \quad (16.49)$$

with universal $a, b > 0$.

Follow z forward through the canonical branches until it exits the nested protection structure at a macroscopic scale. At the exit point z' the ambient hyperbolic geometry is uniformly bounded. Join z' by a hyperbolic geodesic of uniformly bounded length to the limit set $J(\mathcal{H})$; this is possible because the exit occurs in one compact normalized configuration.

Pull this geodesic back along the orbit of z . Lemma 16.36 says that the inverse branch contracts its hyperbolic length by at least $C\Lambda^{-k}$. On U_n , complex bounds compare the hyperbolic density with the reciprocal of the Euclidean scale $|z - c|$. Therefore the pulled-back Euclidean length is at most

$$C|z - c|\Lambda^{-k}.$$

Using (16.49),

$$\Lambda^{-k} \leq C|z - c|^{a \log \Lambda}.$$

With $\beta = a \log \Lambda > 0$ we obtain

$$\text{dist}(z, J(\mathcal{H})) \leq C|z - c|^{1+\beta},$$

which is (16.47). □

16.22 Holomorphic dynamical systems and the correct nonlinearity

We use McMullen's geometric point of view. A holomorphic dynamical system $\mathcal{F}$ is a closed family of local holomorphic relations on $\widehat{\mathbb{C}}$. Its saturation $\mathcal{F}^{\text{sat}}$ contains all restrictions of relations to smaller open sets. Geometric convergence means Hausdorff convergence of the closed relation families in the local analytic topology. For a holomorphic pair $\mathcal{H}$, the complete dynamics considered in the parabolic-cascade argument is such a system.

Definition 16.5 (Parabolic line field; linear and twisting systems). A parabolic line field is a Mobius pullback of the constant line field $d\bar{z}/dz$. A holomorphic dynamical system is *linear* if it preserves a parabolic line field, and *nonlinear* otherwise. It is *twisting* if every holomorphic system quasiconformally conjugate to it is nonlinear.

The quantitative functional used by Khmelev-Yampolsky is slightly stronger than McMullen's one-branch version and is precisely adapted to the parabolic-cascade argument.

Definition 16.6 (Additive nonlinearity). Let $\sigma(z)|dz| = 2|dz|/(1 + |z|^2)$ be the spherical metric. Choose pairwise disjoint spherical balls $B_1, \dots, B_m$ and univalent branches

$$f_i : B_i \longrightarrow \widehat{\mathbb{C}}$$

of $\mathcal{F}$ which extend univalently to the concentric balls of twice the radius. For a parabolic unit line field μ , put

$$D(\mu; \{B_i, f_i\}) = \sum_{i=1}^m \int_{B_i} |\mu - (f_i)^* \mu| \sigma(z)^2 dx dy.$$

Then

$$\nu_{\text{add}}(\mathcal{F}) = \inf_{\text{parabolic } \mu} \sup_{\{B_i, f_i\}} D(\mu; \{B_i, f_i\}). \quad (16.50)$$

For a hyperbolic frame ω , conjugate $\mathcal{F}$ by the unique Mobius normalization carrying a fixed reference frame to ω and denote the resulting quantity by $\nu_{\text{add}}(\mathcal{F}, \omega)$.

McMullen's original functional takes a supremum over one branch. The additive version has the same zero set: it vanishes exactly for linear systems. Its advantage is that disjoint controlled copies contribute simultaneously.

Lemma 16.38 (Zero nonlinearity is equivalent to linearity). *For a geometrically closed system $\mathcal{F}$,*

$$\nu_{\text{add}}(\mathcal{F}) = 0 \iff \mathcal{F} \text{ preserves a parabolic line field.}$$

Moreover, on a compact family of normalized systems, ν_{add} is lower semicontinuous under geometric convergence.

Proof. If $\mathcal{F}$ preserves a parabolic line field μ , every pullback defect in (16.50) is zero, so $\nu_{\text{add}}(\mathcal{F}) = 0$.

Conversely, suppose $\nu_{\text{add}}(\mathcal{F}) = 0$. Choose parabolic fields μ_n whose maximal defects tend to zero. Normalize the corresponding parabolic structures by a Mobius frame. The normalized parabolic fields form a finite-dimensional normal family; after passing to a subsequence, μ_n converges weakly on compact sets to a parabolic field μ_∞ . For every fixed branch $f : B \rightarrow \widehat{\mathbb{C}}$ extending to $2B$,

$$\int_B |\mu_n - f^* \mu_n| \sigma^2 \longrightarrow 0.$$

Weak compactness and the bounded univalent pullback operator give $f^* \mu_\infty = \mu_\infty$ almost everywhere on B . Saturation lets us apply this to all local branches, hence $\mathcal{F}$ preserves μ_∞ .

For lower semicontinuity, fix finitely many balls and branches for a limiting system. Geometric convergence and the doubled-ball extension condition allow these branches to be continued to nearby systems. The corresponding defect integrals converge after extracting the parabolic fields. Taking the supremum over collections and then the infimum over the finite-dimensional parabolic family yields the stated lower-semicontinuity property. This is the compactness content of McMullen's nonlinearity proposition. $\square$

Lemma 16.39 (A genuine critical branch forces nonlinearity). *Let $\mathcal{F}$ contain the saturation of a nonconstant holomorphic map g with a critical point. Then $\mathcal{F}$ is nonlinear.*

Proof. Assume $\mathcal{F}$ preserves a parabolic line field. Conjugate by a Mobius map so that the preserved field is the constant horizontal line field. On any simply connected open set avoiding critical points and poles, invariance under a local restriction of g says that the derivative of that restriction sends the horizontal line to the horizontal line. Hence $g'(z)$ takes values in one fixed real line in $\mathbb{C}$. Since g' is holomorphic, the open mapping theorem forces g' to be constant on that open set. Analytic continuation then makes g affine on its domain, contradicting the existence of a critical point. $\square$

16.23 The controlled-copy nonlinearity lemma

We now discharge the local Foundation interface left in the parabolic-cascade argument.

Theorem 16.40 (Uniform nonlinearity of a controlled critical copy). *Fix constants $K_0 > 1$, $m_0 > 0$, and $C_0 > 0$. Consider normalized holomorphic-pair copies*

$$\mathcal{H} : \Omega \longrightarrow \Delta$$

with fixed odd criticality, K_0 -quasidisk geometry, protection modulus at least m_0 , and a marked critical range whose diameter is bounded below by C_0 after normalization. There exists

$$\nu_* = \nu_*(K_0, m_0, C_0) > 0$$

such that the saturation of every such normalized copy satisfies

$$\nu_{\text{add}}(\mathcal{F}(\mathcal{H})) \geq \nu_*.$$

In an unnormalized copy this gives

$$\nu_{\text{add}}(\mathcal{F}(\mathcal{H})) \geq v \text{ area}(\Delta) \quad (16.51)$$

when measured in a fixed ambient spherical chart, with $v > 0$ depending only on the same geometry constants.

Proof. Suppose no uniform ν_* exists. Then there is a sequence of normalized controlled copies $\mathcal{H}_n$ with

$$\nu_{\text{add}}(\mathcal{F}(\mathcal{H}_n)) \longrightarrow 0.$$

The quasidisk, modulus, relative-size, and criticality assumptions place the normalized holomorphic pairs in one Caratheodory-compact class. Passing to a subsequence gives a limiting controlled holomorphic pair $\mathcal{H}_\infty$. The marked critical branch survives with the same odd degree and remains nondegenerate because of the relative-size and modulus bounds.

By lower semicontinuity in Lemma 16.38,

$$\nu_{\text{add}}(\mathcal{F}(\mathcal{H}_\infty)) = 0.$$

Lemma 16.38 therefore makes the limiting complete dynamics linear. But Lemma 16.39 forbids linearity because the limiting saturation still contains a genuine critical branch. This contradiction proves the normalized lower bound.

For (16.51), rescale Δ to a definite normalized size. On a controlled spherical chart the spherical density and Euclidean density are uniformly comparable. The defect integral transforms by area, while the normalized lower bound stays ν_* . Scaling back therefore gives a constant multiple of $\text{area}(\Delta)$, uniformly over the controlled family. $\square$

Remark 16.9 (Relation with de Faria–de Melo Proposition 4.4). Theorem 16.40 is exactly the compactness mechanism behind the estimate quoted by Khmelev–Yampolsky: a K -quasidisk copy with a fixed complex a priori modulus and definite relative size has nonlinearity at least $v \text{ area}(\Delta)$. The critical branch is the source of strict nonlinearity; compactness upgrades strictness to a uniform constant.

16.24 the parabolic-cascade argument abundance gives quantitative twisting

Let $\Lambda = J(\mathcal{H})$. the parabolic-cascade argument proves that at every sufficiently small Euclidean scale r and every $z \in \Lambda$ there is a bounded-overlap family of controlled copies Δ_j in a comparable neighborhood with

$$\sum_j \text{area}(\Delta_j) \geq c_0 r^2. \quad (16.52)$$

Lemma 16.41 (Bounded overlap to an admissible disjoint collection). *If the overlap multiplicity of $\{\Delta_j\}$ is at most M , then the family can be partitioned into at most $N = N(M)$ subfamilies so that each subfamily admits pairwise disjoint definite-inside balls on which the relevant complete-dynamics branches extend to doubled balls. One subfamily carries at least a fixed fraction $1/N$ of the total area.*

Proof. Uniform quasidisk geometry gives, inside every Δ_j , a spherical ball B_j of radius comparable with $\text{diam } \Delta_j$ and a doubled ball on which the selected branch remains univalent. The bounded-overlap condition and comparable enlargement imply that the intersection graph of the B_j has uniformly bounded degree. A finite graph of degree at most D is $(D + 1)$ -colorable. Summing areas over colors proves that one disjoint color class retains at least $1/(D + 1)$ of the total area. $\square$

Theorem 16.42 (Quantitative twisting at the required quasiconformal bound). *Let K_1 be the universal quasiconformal pullback bound for the canonical conjugacies between deep controlled holomorphic pairs. There exists $\nu_0 > 0$, independent of the irrational rotation type, such that*

$$\nu_{\text{add}}^{K_1}(\mathcal{F}(\mathcal{H}), \Lambda) \geq \nu_0. \quad (16.53)$$

Here the superscript means that before measuring framed nonlinearity we also take the infimum over normalized quasiconformal conjugacies of dilatation at most K_1 which send the system to another holomorphic system.

Proof. First ignore quasiconformal changes of structure. Fix a frame at depth corresponding to a disk of scale r centered on Λ . the parabolic-cascade argument gives (16.52). By Lemma 16.41, select a disjoint subfamily with total area at least $c_1 r^2$. Theorem 16.40 gives, for each selected copy, a defect contribution at least $v \text{area}(\Delta_j)$. Because Definition 16.6 sums over disjoint balls,

$$\nu_{\text{add}}(\mathcal{F}(\mathcal{H}), \omega) \geq c_2 v > 0$$

after normalizing the scale- r frame to the reference frame.

Now let ϕ be a normalized K_1 -quasiconformal conjugacy to a holomorphic system. Moduli of protection annuli degrade by at most a factor depending on K_1 , quasidisks remain uniformly quasidisks with a new constant depending on K_1 , and the fixed critical degree is unchanged. The normalized family of such conjugated controlled copies is therefore again compact. Repeating Theorem 16.40 in this enlarged compact class gives a positive constant $v(K_1)$.

Equivalently, one may formulate the last step by McMullen's framed quantity ν^{K_1} : normalized K_1 -quasiconformal coordinate changes move hyperbolic baseframes only a bounded amount after visual extension, and nonlinearity changes continuously under bounded baseframe motion. Thus the preceding framed lower bound survives uniformly over all permitted conjugacies. Taking the infimum gives (16.53). $\square$

Remark 16.10 (Why additive nonlinearity is essential here). Near a long parabolic cascade, no single copy need occupy a definite fraction of the observation disk. the parabolic-cascade argument instead produces many small copies whose *total* area is definite. Formula (16.50) was designed precisely so these disjoint contributions can

be summed. This is the mechanism in the Khmelev-Yampolsky proof of their uniform-twisting theorem.

16.25 Uniform nonlinearity implies rigidity of geometric limits

We next reconstruct McMullen's Proposition 9.11.

Definition 16.7 (Uniform nonlinearity). For a closed invariant set Λ , write

$$\nu(\mathcal{F}, \Lambda) = \inf_{\omega \in \text{hull}(\Lambda)} \nu(\mathcal{F}, \omega).$$

The pair $(\mathcal{F}, \Lambda)$ is uniformly nonlinear if this number is positive. Uniform twisting means the corresponding lower bound persists under the quasiconformal changes of structure under consideration.

Proposition 16.43 (Uniform nonlinearity rules out invariant line fields). *If $(\mathcal{F}, \Lambda)$ is uniformly nonlinear, then Λ supports no measurable $\mathcal{F}$ -invariant line field.*

Proof. Assume an invariant line field μ is supported on Λ . Choose a point $x \in \Lambda$ which is a Lebesgue almost-continuity point for μ and with $|\mu(x)| = 1$. Let $\gamma \subset \text{hull}(\Lambda)$ be a hyperbolic geodesic ray ending at x , and choose frames ω_n moving to infinity along γ by parallel translation.

Normalize each frame to a fixed reference frame by a Mobius map A_n . At an almost-continuity point, the rescaled fields $(A_n)_*\mu$ converge weakly to a constant, hence parabolic, line field μ_∞ . Sequential compactness of saturated holomorphic dynamical systems gives, after passing to a subsequence,

$$(A_n)_*(\mathcal{F}^{\text{sat}}) \longrightarrow \mathcal{G}.$$

Invariance passes to the geometric limit, so $\mathcal{G}$ preserves μ_∞ and is linear. Hence its nonlinearity is zero. Lower semicontinuity then forces

$$\nu(\mathcal{F}, \omega_n) \longrightarrow 0,$$

contradicting the positive lower bound defining uniform nonlinearity. $\square$

16.26 The one-sphere decay lemma

Let v be a quasiconformal vector field on $\widehat{\mathbb{C}}$. Its visual distortion $M_v(p)$ at $p \in \mathbb{H}^3$ is the distance, in the visual norm based at p , from v to the finite-dimensional space of conformal vector fields. Thus $M_v(p) = 0$ exactly when v is conformal in the visual normalization at p . We use the standard compactness and continuity properties of the visual norm from the Foundation visual-extension package.

A quasiconformal vector field is a deformation of $\mathcal{F}$ when $\bar{\partial}v$ is $\mathcal{F}$ -invariant.

Theorem 16.44 (McMullen one-sphere decay). *Assume*

$$\nu(\mathcal{F}, \Lambda) \geq \nu_0 > 0.$$

There exists $R = R(\nu_0) > 0$ such that for every quasiconformal deformation v of $\mathcal{F}$ and every $p \in \mathbb{H}^3$ satisfying

$$S(p, R) \subset \text{hull}(\Lambda),$$

one has

$$M_v(p) \leq \frac{1}{2} \sup_{q \in S(p, R)} M_v(q). \quad (16.54)$$

Proof. Suppose the statement fails. After a conformal normalization put the basepoint at the origin of the hyperbolic ball. Then there are saturated systems $(\mathcal{F}_n, \Lambda_n)$, deformations v_n , and radii $R_n \rightarrow \infty$ such that

$$\nu(\mathcal{F}_n, \Lambda_n) \geq \nu_0, \quad M_{v_n}(0) \geq \frac{1}{2},$$

while

$$M_{v_n}(q) \leq 1 \quad \text{for } q \in S(0, R_n), \quad S(0, R_n) \subset \text{hull}(\Lambda_n).$$

The visual bound on larger and larger spheres gives compactness of v_n modulo conformal vector fields. Subtract such fields and pass to a subsequence so that $v_n \rightarrow v_\infty$ uniformly on $\widehat{\mathbb{C}}$ and $\bar{\partial}v_n \rightarrow \bar{\partial}v_\infty$ distributionally.

At the same time, geometric compactness gives

$$\mathcal{F}_n^{\text{sat}} \longrightarrow \mathcal{G}.$$

The spheres of radii $R_n \rightarrow \infty$ lying in the hull force the pointed limiting invariant set to fill the sphere in the visual limit. Lower semicontinuity preserves the nonlinearity lower bound, so $\mathcal{G}$ is uniformly nonlinear. Proposition 16.43 makes $\mathcal{G}$ rigid.

The invariance equations for $\bar{\partial}v_n$ pass to the limit, hence v_∞ is a quasiconformal deformation of $\mathcal{G}$. Rigidity forces $\bar{\partial}v_\infty = 0$, so v_∞ is conformal and $M_{v_\infty}(0) = 0$. But continuity of visual distortion under uniform convergence gives

$$M_{v_\infty}(0) = \lim_n M_{v_n}(0) \geq \frac{1}{2},$$

a contradiction. □

16.27 Iteration: infinitesimal inflexibility

Let

$$d(p) = \text{dist}_{\mathbb{H}^3}(p, \mathbb{H}^3 \setminus \text{hull}(\Lambda))$$

be the depth of p inside the convex hull.

Corollary 16.45 (Exponential decay of visual distortion). *Under the hypotheses of Theorem 16.44, there are constants $C > 0$ and $\alpha > 0$, with α depending only on ν_0 , such*

that

$$M_v(p) \leq C e^{-\alpha d(p)} \|\bar{\partial} v\|_\infty. \quad (16.55)$$

Proof. Let R be furnished by Theorem 16.44. If $d(p) \geq nR$, apply (16.54) at p , choose a point on the radius- R sphere where the supremum is almost attained, and repeat. After n steps,

$$M_v(p) \leq 2^{-n} \sup_{q \in \mathbb{H}^3} M_v(q).$$

The global visual-extension estimate gives

$$\sup M_v \leq C_0 \|\bar{\partial} v\|_\infty.$$

Taking $n \geq d(p)/R - 1$ yields (16.55) with $\alpha = (\log 2)/R$ after changing the constant. $\square$

Corollary 16.46 (Infinitesimal hyperbolic inflexibility). *Let $V = \text{ex}(v)$ be the standard visual extension of v to $\mathbb{H}^3$. Then*

$$\|SV(p)\| \leq C_1 M_v(p) \leq C_2 e^{-\alpha d(p)} \|\bar{\partial} v\|_\infty. \quad (16.56)$$

Proof. The first inequality is the local visual-extension strain estimate from Appendix B: the symmetric traceless derivative of the extension is controlled by the boundary visual distance from conformal vector fields. The second is Corollary 16.45. $\square$

16.28 Deep points and boundary differentiability

the parabolic-cascade argument uses the Euclidean hole formulation of deepness. McMullen's visual argument uses hyperbolic depth. We record the equivalence in the quantitative form needed here.

Lemma 16.47 (Deepness creates a linearly deep geodesic). *Let $x \in \Lambda$ be β -deep: the largest spherical ball contained in $B(x, r) \setminus \Lambda$ has radius at most $Cr^{1+\beta}$ for small r . Let $\gamma(s)$ be a hyperbolic geodesic ray in $\mathbb{H}^3$ ending at x , parameterized by distance. Then for all large s ,*

$$\text{dist}(\gamma(s), \mathbb{H}^3 \setminus \text{hull}(\Lambda)) \geq \delta s - C_0 \quad (16.57)$$

for some $\delta = \delta(\beta) > 0$.

Proof. At hyperbolic time s , the visual boundary scale around x is comparable with $r = e^{-s}$. If the hull boundary came within hyperbolic distance $o(s)$ of $\gamma(s)$, a supporting hyperbolic plane would cut off a spherical complementary ball whose radius is much larger than $r^{1+\beta}$. Conversely, the hole estimate says every complementary spherical ball inside $B(x, r)$ has radius at most $Cr^{1+\beta}$. Translating the ratio $r/r^{1+\beta} = r^{-\beta} = e^{\beta s}$ into hyperbolic distance gives a lower bound linear in s . Constants from the ball/half-space comparison are absorbed into C_0 . $\square$

Combining (16.55) and (16.57) gives

$$M_v(\gamma(s)) \leq C e^{-\alpha \delta s} \|\bar{\partial} v\|_\infty. \quad (16.58)$$

The next lemma is the boundary content of McMullen's visual extension theorem. We include the scale-by-scale argument because it is short and exposes where the Holder exponent comes from.

Lemma 16.48 (Exponential visual decay gives a $C^{1+\eta}$ vector field). *Normalize $x = 0$ and two further boundary points so that conformal vector fields are uniquely fixed. Suppose a quasiconformal vector field v satisfies*

$$M_v(\gamma(s)) \leq Ce^{-\kappa s}$$

along a geodesic ending at 0. Then there exist a complex number a and $\eta > 0$, depending only on κ , such that

$$v(z) = v(0) + az + O(|z|^{1+\eta}). \quad (16.59)$$

Proof. Put $r_n = e^{-n}$. In the visual normalization based at $\gamma(n)$, the ball $|z| \lesssim r_n$ is magnified to unit size. By the definition of M_v , there is a conformal vector field A_n such that, after transporting back to the boundary chart,

$$\sup_{|z| \leq cr_n} |v(z) - A_n(z)| \leq C_1 r_n e^{-\kappa n} = C_1 r_n^{1+\kappa}. \quad (16.60)$$

The factor r_n appears because vector fields scale like length under the boundary magnification.

On the overlap $|z| \leq cr_{n+1}$, both A_n and A_{n+1} approximate v . A conformal vector field is a real six-dimensional rational vector field; after fixing the three normalization conditions its local germ at 0 is determined by its linear coefficient. Estimate (16.60) therefore implies

$$|A'_{n+1}(0) - A'_n(0)| \leq C_2 r_n^\kappa.$$

The geometric series is summable, so $A'_n(0)$ converges to a limit a , with

$$|A'_n(0) - a| \leq C_3 r_n^\kappa.$$

For $r_{n+1} < |z| \leq r_n$, use A_n in (16.60) and replace its derivative by a . The constant terms converge similarly to $v(0)$. This gives

$$|v(z) - v(0) - az| \leq C_4 |z|^{1+\eta}$$

for any $0 < \eta < \kappa$ after a harmless loss absorbing the visual-coordinate comparison. This is (16.59). $\square$

16.29 Beltrami isotopy and dynamical inflexibility

We now integrate the infinitesimal estimate. Let

$$\phi : \widehat{\mathbb{C}} \longrightarrow \widehat{\mathbb{C}}$$

be a K -quasiconformal conjugacy from $\mathcal{F}$ to a holomorphic system $\mathcal{F}'$. Its Beltrami coefficient μ_ϕ is $\mathcal{F}$ -invariant. Put $k = \|\mu_\phi\|_\infty < 1$.

Lemma 16.49 (Invariant Ahlfors-Bers isotopy). *For $0 \leq t \leq 1$, let ϕ_t be the normalized solution of*

$$\bar{\partial}\phi_t = t\mu_\phi \partial\phi_t.$$

Then:

- (i) $\phi_0 = \text{id}$ and $\phi_1 = \phi$ up to the fixed normalization;
- (ii) $K(\phi_t) \leq K$;
- (iii) $\mathcal{F}_t = (\phi_t)_*\mathcal{F}$ is holomorphic, because $t\mu_\phi$ remains $\mathcal{F}$ -invariant;
- (iv) the isotopy is generated by quasiconformal vector fields v_t with a uniform bound on $\|\bar{\partial}v_t\|_\infty$ depending only on K .

Proof. Existence, uniqueness after three-point normalization, and differentiability in t are the measurable Riemann mapping and Ahlfors-Bers holomorphic-dependence theorems. Since $\|t\mu_\phi\|_\infty \leq k$, the dilatation does not exceed $(1+k)/(1-k) = K$. Invariance of μ_ϕ is precisely the condition ensuring that conjugating each local relation by ϕ_t cancels its Beltrami coefficient, so $\mathcal{F}_t$ is holomorphic. Differentiating the normalized solutions in t produces the stated quasiconformal vector fields with a compact-parameter bound. $\square$

Lemma 16.50 (Uniform twisting persists through the isotopy). *Assume (16.53). There exists $\nu_1 = \nu_1(K, \nu_0) > 0$ such that*

$$\nu(\mathcal{F}_t, \phi_t(\Lambda)) \geq \nu_1$$

for every $0 \leq t \leq 1$.

Proof. This is built into the ν^K definition: every ϕ_t is a normalized quasiconformal conjugacy of dilatation at most K . In McMullen's baseframe formulation, the visual extension of a normalized K -quasiconformal map is a controlled quasi-isometry of $\mathbb{H}^3$, and the image hull stays within a bounded Hausdorff distance of the hull of the image limit set. Nonlinearity changes by a controlled amount under bounded baseframe motion. Hence the ν^K lower bound gives a uniform ν_1 along the entire path. $\square$

Theorem 16.51 (Dynamical inflexibility). *Let $(\mathcal{F}, \Lambda)$ satisfy a quantitative twisting bound $\nu^K(\mathcal{F}, \Lambda) \geq \nu_0 > 0$, and let ϕ be a K -quasiconformal conjugacy from $\mathcal{F}$ to another holomorphic dynamical system. If $x \in \Lambda$ is β -deep, then there exist $\alpha > 0$, $C < \infty$, and a nonzero complex number a such that*

$$\phi(x+z) = \phi(x) + az + O(|z|^{1+\alpha}). \quad (16.61)$$

The exponent and constant depend only on K , the deepness data, and ν_0 .

Proof. Apply Lemma 16.49. By Lemma 16.50, every intermediate system has a uniform nonlinearity bound. The generator v_t is a deformation of $\mathcal{F}_t$, so Theorem 16.44 and Corollary 16.45 apply uniformly in t .

Let $x_t = \phi_t(x)$ and let γ_t be the image, under the standard visual quasi-isometry, of a geodesic ray ending at x . Quasi-isometries move convex hulls by a bounded amount

depending only on K , so Lemma 16.47 remains valid uniformly along the isotopy, with degraded constants depending only on K and β . Consequently

$$M_{v_t}(\gamma_t(s)) \leq Ce^{-\kappa s}$$

with $C, \kappa > 0$ independent of t . Lemma 16.48 gives, after a local normalization at x_t ,

$$v_t(x_t + z) = v_t(x_t) + a_t z + O(|z|^{1+\eta}) \quad (16.62)$$

uniformly in t .

It remains to integrate this pointwise estimate. Translate x to 0 and $\phi_t(x)$ to 0 in the local chart; the translations are conformal and do not affect the estimate. Let $A_t(z) = \lambda_t z$ solve

$$\dot{\lambda}_t = a_t \lambda_t, \quad \lambda_0 = 1.$$

For the actual flow write

$$\delta_t(z) = \phi_t(z) - A_t(z).$$

The uniform estimate (16.62) and a Gronwall bound first give $|\phi_t(z)| \leq C_0|z|$ for small z , uniformly in t . Then

$$\begin{aligned} |\dot{\delta}_t(z)| &\leq |v_t(\phi_t(z)) - a_t \phi_t(z)| + |a_t| |\delta_t(z)| \\ &\leq C_1 |z|^{1+\eta} + C_2 |\delta_t(z)|. \end{aligned}$$

Since $\delta_0 = 0$, Gronwall's inequality yields

$$|\delta_1(z)| \leq C_3 |z|^{1+\eta}.$$

Thus

$$\phi(z) = \lambda_1 z + O(|z|^{1+\eta}).$$

The homeomorphism property and quasiconformality ensure $\lambda_1 \neq 0$. Undoing the conformal normalization gives (16.61) with $\alpha = \eta$. $\square$

Remark 16.11 (Relation with McMullen Theorem 9.15). The proof above follows McMullen's actual architecture: Proposition 9.14 supplies a uniform nonlinearity bound throughout a Beltrami isotopy; Lemma 9.12 gives the one-sphere decay; Corollary 9.13 gives exponential infinitesimal inflexibility in the hull; deepness converts hull depth into exponential boundary-scale decay; Appendix B converts that decay into a $C^{1+\alpha}$ vector-field expansion; and the pointwise conformal-flow lemma integrates the expansion. The only retained Foundation inputs are the standard Ahlfors-Bers solution theory and the visual-extension compactness/strain estimates.

16.30 Arbitrary-type rigidity from parabolic cascades

We now combine the parabolic-cascade argument with Theorem 16.51.

Theorem 16.52 (Khmelev-Yampolsky local rigidity). *Let f and g be real-analytic critical circle maps with the same irrational rotation number and the same odd criticality.*

Let h be the canonical conjugacy sending the critical point of f to that of g . There is a universal $\alpha > 0$ such that h is $C^{1+\alpha}$ conformal at the critical point.

Proof. The complex-bounds argument places sufficiently deep renormalizations of f and g in the universal controlled holomorphic-pair class and constructs the canonical quasiconformal pullback conjugacy with a universal bound K_1 . The parabolic-cascade argument gives scale-by-scale copy abundance for every irrational type and universal β -deepness of the critical point.

Theorem 16.42 converts this copy abundance into

$$\nu^{K_1}(\mathcal{F}(\mathcal{H}), J(\mathcal{H})) \geq \nu_0 > 0.$$

Apply Theorem 16.51 at the critical point. The output is a $C^{1+\alpha}$ expansion with constants depending only on the universal controlled-class data. Rescaling back from the deep holomorphic pair to the original circle map preserves the pointwise exponent. Hence h is $C^{1+\alpha}$ at the critical point with a universal $\alpha > 0$. $\square$

Corollary 16.53 (Universal arbitrary-type C^0 renormalization convergence). *There are a universal $0 < \lambda < 1$ and, for each pair of analytic critical commuting pairs ζ_1, ζ_2 of the same irrational rotation number and criticality, a constant $C(\zeta_1, \zeta_2)$ such that*

$$d_0(\mathcal{R}^n \zeta_1, \mathcal{R}^n \zeta_2) \leq C(\zeta_1, \zeta_2) \lambda^n.$$

Proof. This is the rescaling theorem of the complex-bounds argument applied to Theorem 16.52. The universal $C^{1+\alpha}$ critical-point expansion makes the normalized conjugacy differ from its linear part by $O(s_n^\alpha)$ at the n -th closest-return scale s_n . Real bounds make s_n decrease geometrically after a uniformly bounded number of levels; absorbing those blocks gives the universal geometric rate λ . $\square$

Corollary 16.54 (Full critical-circle renormalization horseshoe). *The global analytic cylinder-renormalization horseshoe has a uniformly hyperbolic splitting with one real unstable direction on the real slice and one complex unstable direction after complexification.*

Proof. The cylinder-renormalization argument constructs the analytic cylinder-renormalization operator, proves compactness of its derivative on controlled Banach charts, identifies fixed-rotation classes as codimension-one local stable leaves, constructs an expanding positive cone, and proves that any remaining tangent spectrum is stable once universal same-rotation contraction is available. Corollary 16.53 supplies exactly that arbitrary-type stable contraction. Codimension one leaves plus the cone expansion therefore leave precisely one unstable direction, uniformly over the full horseshoe. $\square$

16.31 Proof structure and dependencies

The preceding arguments establish the classical critical-circle chain at the scopes used in this chapter. The dependency structure is:

Result	Proof status
Yoccoz no-wandering theorem	Proved here from cross-ratio distortion and dynamic symmetry.
Real a priori bounds	Proved here using the preceding no-wandering theorem.
Complex a priori bounds for analytic commuting pairs	Proved here at the scope used by the convergence and cylinder arguments.
Arbitrary-type local rigidity at the critical point	Proved here modulo standard Ahlfors–Bers and visual-extension tools.
Universal arbitrary-type C^0 convergence	Proved here from local rigidity and real bounds.
Periodic cylinder-renormalization hyperbolicity	Proved here; periodic type is bounded type, so no arbitrary-type rigidity is required.
Full cylinder-renormalization horseshoe	Proved here after the universal same-rotation contraction established above.
Critical-quasicircle/Corona extension	Proved here through the Corona, rigidity, perturbation, realization, degeneration, wave/spreading, width-trading, and Herman a priori-bound layers, including localization, exact-denominator completion, and bounded-degree push-forward.

Non-circular dependency order. The chapter uses real/complex bounds before rigidity, parabolic-cascade abundance before twisting, twisting before McMullen inflexibility, local rigidity before arbitrary-type stable contraction, and stable contraction before global hyperbolicity. Thus the final horseshoe theorem is never used to justify its own rigidity input.

16.32 Modern critical-quasicircle extension

The classical circle-map proof chain above is logically independent of the following modern extension. Its proof is organized through Corona renormalization, rigidity, parameter perturbation, surgery, degeneration, a priori bounds, localization, and horizontal-loss control.

Definition 16.8 (Corona renormalization). Corona renormalization is Lim’s compact analytic renormalization operator on critical quasicircle maps; its precise domain includes inner/outer criticality and normalized quasicircle geometry.

Theorem 16.55 (Critical quasicircle renormalization extension). *Lim constructs a compact analytic Corona renormalization operator for critical quasicircle maps. For each fixed criticality and periodic-type rotation number covered by the theorem, the corresponding renormalization fixed point is hyperbolic; its stable manifold has codimension one and consists of critical quasicircle maps in the same criticality/rotation class, so the unstable manifold is one-dimensional.*

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Reconstruct the Corona global chain, bounded-type rigidity and universality, parameter activity and uniqueness, direct collision realization, degeneration, and the wave/spreading/trading stages; then prove the a priori-bound mechanism via deep localization, controlled cuts, canonical pullbacks, thick-thin and persistent-fence analysis, pull-off contraction, exact-denominator horizontal loss, and bounded-degree push-forward.

The mathematical argument begins in the proof environment below.

Proof setup. Fix integers $d_0, d_\infty \geq 2$ and a periodic-type irrational rotation number θ . We use the term (d_0, d_∞) -critical corona for a unicritical holomorphic map $f : U \rightarrow V$ between nested annuli whose cut map is a branched covering with the prescribed inner and outer criticalities. A corona is *rotational of rotation number θ* if it contains an invariant Herman quasicycle on which the induced critical quasicycle map has rotation number θ . These are the objects in the Corona-renormalization definition.

The source proof naturally has two layers. Sections 4–6 construct a compact analytic operator and prove that its periodic point is hyperbolic with a finite, positive-dimensional unstable manifold and with the rotational class as its local stable manifold. Sections 7–13 then pass to a global transcendental model on the unstable manifold and prove that this finite positive dimension is actually one. The first layer supplies the compact operator, rotational stable class, finite-dimensional hyperbolicity, and the transcendental cascade setup. The final source layer supplies escaping rigidity, hyperbolic-cascade structure, and the parameter-dimension argument that closes the unstable dimension.

Lemma 16.56 (Compact analytic Corona operator). *A renormalizable unicritical corona admits, after shrinking its Banach neighborhood, a holomorphically varying gluing construction defining a compact analytic Corona renormalization operator.*

Proof. Choose the two boundary arcs of the pre-corona of f so that the first-return branches are univalent away from the unique critical point. For g close to f , move those arcs by the equivariant holomorphic motion obtained by pulling back the boundary motion through the same finite return itinerary. The λ -lemma extends the boundary motion across a neighborhood of the pre-corona. Write τ_g for this extension and μ_g for its Beltrami coefficient. Transport μ_g to the quotient annulus by the reference gluing map and put it equal to zero outside the target annulus. The measurable Riemann mapping theorem, with three marked normalizations fixed, gives a unique quasiconformal integrating map ϕ_g depending holomorphically on g .

If ψ_f is the reference quotient map, then

$$\psi_g := \phi_g \circ \psi_f \circ \tau_g^{-1}$$

is conformal on the moved pre-corona and depends holomorphically on g . Consequently the pair of return branches of g descends through ψ_g to a holomorphic corona $\hat{g}$ depending analytically on g . This gives an analytic renormalization map. To obtain

compactness, choose annuli $U \Subset U' \Subset \tilde{U}$ and define the operator using the renormalization of $g|_{U'}$. The restriction map from bounded holomorphic maps on $\tilde{U}$ to those on U' is compact by Montel normality; the remaining gluing and normalization maps are analytic and locally bounded. Thus the Corona operator is compact analytic. $\square$

Lemma 16.57 (Periodic Corona fixed point and rotational contraction). *For fixed criticalities (d_0, d_∞) and periodic-type rotation number θ , there is a fixed rotational corona f_* for a compact analytic Corona operator $\mathcal{R}_*$, and nearby rotational coronas in the same class converge exponentially to f_* .*

Proof. **Bounded-type critical-quasicircle rigidity and universality.**

We first prove the rigidity/universality theorem used below. The proof follows Lim's Trans. AMS paper, with the quasicritical-circle estimates calibrated against Avila-Lyubich, Section 3. Only the standard quasiconformal extension, measurable-Riemann-mapping and Koebe/Schwarz tools are left at the Foundation layer.

Step 1: controlled welding and real geometry. Let $F : H \rightarrow H$ be a (d_0, d_∞) -critical quasicircle map in a fixed controlled class

$$\mathcal{H}\mathcal{Q}(d_0, d_\infty, N, K, \mu),$$

so the rotation number belongs to the bounded-type set Θ_N , the Herman curve is a K -quasicircle and the relevant collar has modulus at least μ . Uniformize the two complementary components of H by maps ϕ_0, ϕ_∞ carrying the critical point to $1 \in \mathbb{T}$. Since H is a K -quasicircle, both uniformizations have global quasiconformal extensions with constants depending only on K . On the two sides the conjugates

$$g_\bullet = \phi_\bullet \circ F \circ \phi_\bullet^{-1}, \quad \bullet \in \{0, \infty\},$$

are $d_\bullet$ -quasicritical circle maps with uniformly controlled annular extensions. This is the welding decomposition used by Lim.

The real-bounds proof already given in Theorem 16.9 survives this controlled quasiregular change of coordinates. Indeed, away from the critical spot the logarithmic cross-ratio estimate is changed only by a bounded quasiconformal distortion constant, while at the critical spot the local factor is a controlled quasiconformal coordinate followed by the power map of degree $d_\bullet$. The bounded-overlap argument and seven-point comparison therefore give one constant $B = B(d_\bullet, N, K, \mu)$ for all sufficiently deep closest-return partitions. Consequently the conjugacy from $g_\bullet|_{\mathbb{T}}$ to the rigid rotation is uniformly quasisymmetric. Pulling the two boundary conjugacies back through ϕ_0, ϕ_∞ shows that any two maps in the fixed $\mathcal{H}\mathcal{Q}$ class with the same rotation number have a uniformly quasisymmetric marked conjugacy on their Herman curves.

Step 2: butterfly complex bounds. The bounded real geometry gives definite angles at all deep closest returns. Apply the lens-Schwarz argument of Lemma 16.12 separately to the two quasicritical circle models. The quasiregular defect is confined to the controlled critical triangle; after the first critical passage the inverse branches used in the return tower are holomorphic, while the critical passage itself is a fixed power

map in uniformly quasiconformal coordinates. Thus the good-angle inverse-orbit argument of Lemma 16.13 gives, for all $n \geq n_0$, half-butterflies with a modulus lower bound and quasidisk-shape constant depending only on $(d_0, d_\infty, N, K, \mu)$.

Transport the two half-butterflies back by ϕ_0^{-1} and ϕ_∞^{-1} . Their bases agree because they are the two uniformizations of the same marked closest-return intervals on H . Gluing along that base therefore produces a full (d_0, d_∞) -critical butterfly

$$\mathbb{B}_n = (F_{-,n}, F_{+,n}) : U_n \longrightarrow V_n$$

whose annular modulus, pointed-domain shape, critical-value separation and Koebe space are all uniformly controlled. Normalizing the base endpoints and critical value then gives a precompact butterfly class. This reconstructs the content of Lim Theorem 4.4.16 (and, for commuting pairs, Theorem 4.4.18) from the same real/complex-bounds mechanism used earlier in the chapter.

Step 3: quasiconformal rigidity by pullback. Take two deep butterflies $\mathbb{B}_1, \mathbb{B}_2$ of the same height and rotation number. The marked conjugacy of Step 1 is quasisymmetric on their bases. Extend it first to the extended bases by the corresponding inverse branches. Uniform quasidisk geometry and definite angles then give a quasiconformal interpolation $h_0 : V_1 \rightarrow V_2$ that is equivariant on the butterfly boundaries. Inductively define $h_{m+1} = h_m$ off U_1 and on U_1 take the lift of h_m through $\mathbb{B}_1$ and $\mathbb{B}_2$. Pullback by holomorphic branches does not increase the Beltrami norm, and the fixed power-law critical branch preserves the same bound in the chosen critical coordinates. Hence all h_m have one uniform quasiconformal constant.

By compactness of normalized quasiconformal maps, a subsequence converges to h_∞ . The uniform modulus of the butterfly implies geometric shrinking of nested pullback tiles, so its limit set has empty interior; equivariance therefore passes to the limit on a dense open set and then everywhere by continuity. Thus h_∞ conjugates the two butterflies. Pulling this conjugacy around the bounded-return rectangles adjacent to the Herman curve spreads it to a fixed collar. We have proved the quantitative form of Lim Theorem 4.4.20: two maps in the controlled class with the same bounded-type rotation number are quasiconformally conjugate on uniform collar neighborhoods of their Herman curves.

Step 4: uniform deepness, twisting and $C^{1+\alpha}$ rigidity. Fix such a collar conjugacy h and straighten the reference Herman curve by a normalized quasiconformal map ϕ . Put

$$L(z) = \log \text{dist}(\phi(z), \mathbb{T}).$$

The bi-Holder estimates for normalized quasiconformal maps give

$$L(z) \asymp \log \text{dist}(z, H)$$

near H . For every point in a sufficiently thin collar, either the next iterate is contained in a definite univalent return chart or it lies in a controlled critical chart. In the former case Koebe distortion gives $L(Fz) \leq L(z) + C$; in the latter the fixed power-law normal form gives the same estimate after changing C . On $\Omega = \widehat{\mathbb{C}} \setminus P(F)$, Schwarz–Pick is strict

away from the preimage of H : if r_z is the hyperbolic distance from z to $F^{-1}(H)$, then

$$\|DF(z)\|_\Omega \geq \Lambda(r_z) > 1.$$

Iterating until the logarithmic height reaches a fixed collar level and using the previous additive bound for L yields constants $C_0, \beta > 0$ such that for the local nonescaping set K^{loc} ,

$$\text{dist}(z, K^{\text{loc}}) \leq C_0 \text{dist}(z, H)^{1+\beta}.$$

Hence the whole Herman curve is uniformly deep. This is Lim's deep-point argument (Theorem 4.5.2) in the quantitative form needed here.

At every bounded-type scale, Step 2 supplies a definite-modulus butterfly containing a genuinely critical branch. After normalization these copies belong to one compact controlled family, so Theorem 16.40 and the bounded-overlap selection of Theorem 16.42 give one positive twisting constant at every point of H and every sufficiently small scale. We may therefore apply the Theorem 16.51 to the collar conjugacy from Step 3. Uniform deepness converts visual depth into Euclidean scale, and the conclusion is

$$h(z) = h(x) + a_x(z - x) + O(|z - x|^{1+\alpha}), \quad a_x \neq 0,$$

uniformly for $x \in H$ and for one $\alpha > 0$ depending only on the controlled class. Thus the bounded-type critical-quasicircle $C^{1+\alpha}$ rigidity statement is now derived inside the book rather than imported.

Step 5: exponential convergence of commuting-pair renormalizations. Let $\zeta, \hat{\zeta}$ be normalized (d_0, d_∞) -critical commuting pairs in the controlled class with the same rotation number. Let c_n and $\hat{c}_n = h(c_n)$ be their marked closest-return endpoints and normalize by

$$\tau_n(z) = -c_n z, \quad \hat{\tau}_n(z) = -\hat{c}_n z.$$

By bounded real geometry there are $A > 0$ and $0 < \sigma < 1$ with $|c_n| \leq A\sigma^n$. Applying the $C^{1+\alpha}$ expansion of Step 4 at the critical point, after a harmless affine normalization take $h(0) = 0$ and write

$$h(z) = az + O(|z|^{1+\alpha}), \quad a \neq 0.$$

Then

$$\frac{\hat{c}_n}{c_n} = a + O(|c_n|^\alpha) = a + O(\sigma^{\alpha n}).$$

The rescaled conjugacies

$$h_n = \hat{\tau}_n^{-1} \circ h \circ \tau_n$$

satisfy, on every fixed normalized butterfly domain,

$$\|h_n - \text{id}\|_\infty \leq C\sigma^{\alpha n}.$$

Conjugating the two normalized return branches by h_n therefore yields

$$\|\mathcal{R}^n \zeta - \mathcal{R}^n \hat{\zeta}\|_\infty \leq C' \rho^n, \quad 0 < \rho < 1.$$

The common complex-bounds collar and Cauchy inequalities give the analogous estimate in every fixed finite C^r norm. This is precisely the mechanism of Lim Theo-

rem 4.6.6: exponential renormalization convergence is a rescaled form of pointwise $C^{1+\alpha}$ rigidity.

Step 6: periodic pair and uniqueness. Suppose now that θ has continued-fraction period p . Choose any normalized pair ζ in the nonempty controlled class. Then $\mathcal{R}^{kp}\zeta$ all have rotation number θ . Apply Step 5 to ζ and $\mathcal{R}^p\zeta$ to obtain

$$d(\mathcal{R}^{(k+1)p}\zeta, \mathcal{R}^{kp}\zeta) \leq C\rho^{kp}.$$

Thus the sequence is Cauchy in the compact butterfly class and converges to a pair ζ_* . Continuity gives $\mathcal{R}^p\zeta_* = \zeta_*$, and Step 5 also gives uniqueness in the rotation class. This recovers the periodic point needed below without using the renormalization horseshoe as a circular input. The usual two-sided compactness construction then gives the bounded- type horseshoe statement of Lim Theorem 4.6.7, but the periodic Corona fixed point and rotational contraction lemma needs only this periodic case and the exponential contraction just proved.

Start with the normalized periodic critical commuting pair ζ_* just constructed. Its period- p self-similarity gives a contracting affine scaling $L(z) = \mu z$, $|\mu| < 1$, identifying ζ_* with a fixed finite pre-renormalization of itself. Project a sufficiently deep scaled copy of that commuting pair to a quasicircle by the canonical gluing construction, push it once so that the critical value is the standard marker, and take the resulting standard pre-corona. A second gluing projects this pre-corona to a rotational corona f_* . Because the scaled copy was chosen arbitrarily deep, the pre-corona domain is compactly contained in the domain of f_* ; this is the improvement-of-domain property needed in the compact analytic Corona operator lemma. The gluing map sends the pre-corona back to f_* , hence the compact analytic Corona operator lemma yields a compact analytic Corona operator with f_* fixed.

For any nearby rotational corona f of the same criticalities and rotation number, cut and normalize its commuting-pair return data. Step 5 gives constants $C > 0$ and $0 < \lambda < 1$ such that the normalized n th commuting-pair renormalizations are $O(\lambda^n)$ apart on the common complex neighborhood. The two gluing maps vary holomorphically on a fixed compact neighborhood, so their derivatives and inverse derivatives are uniformly bounded. Transporting the estimate through these charts gives

$$\text{dist}(\mathcal{R}_*^n f, f_*) \leq C_f \lambda^n.$$

Thus the whole local rotational class lies in the stable set of f_* , and the rigidity/universality assertion required in the periodic Corona fixed point and rotational contraction lemma follows. $\square$

Lemma 16.58 (Renormalization-tiling recognition). *There is $\varepsilon > 0$ such that an infinitely Corona-renormalizable map whose entire forward orbit remains ε -close to f_* is a rotational corona in the prescribed rotation class.*

Proof. For $f_k := \mathcal{R}_*^k f$, pull the domain of the first pre-corona of f_k back through the inverse renormalization changes of coordinates. The improvement-of-domain property of f_* gives a nested family of tiles $\Delta_n(f_*)$ whose mesh tends to zero. Holomorphic dependence of the changes of variables implies that, for f whose first n renormalizations

stay ε -close to f_* , the corresponding tiles $\Delta_n(f)$ move holomorphically and retain the same incidence and return combinatorics.

Each tile admits a univalent pullback to one of finitely many first-level tiles. Pulling back the first-level holomorphic motion and applying the λ -lemma gives a quasiconformal map h_n matching the level- n tilings of f and f_* . If

$$D_n := \max_{0 \leq k \leq n} \text{dist}(f_k, f_*),$$

then the dilatation and displacement estimates have the form $K(D_n) \rightarrow 1$ and $M(D_n) \rightarrow 0$ as $D_n \rightarrow 0$. Under the uniform ε -bound, quasiconformal compactness yields a subsequential limit h on the sphere. The shrinking mesh forces $h^{-1}(H_*)$ to be an f -invariant quasicircle through the critical point, separating the two boundary components. The marked return itinerary is the periodic itinerary of θ , so the induced dynamics on this curve has rotation number θ . Hence f is rotational, proving the lemma. $\square$

Lemma 16.59 (Small-Orbits neutral exclusion). *For the compact analytic Corona operator at f_* , neutral spectrum is impossible: DR_{*,f_*} has no eigenvalue of modulus one.*

External research input boundary and deduction. The next step uses the upgraded Small-Orbits theorem. We record enough of its operator-theoretic proof to make clear why neutral spectrum is genuinely excluded even when expanding spectrum is already present.

Compactness implies that the nonzero spectrum of DR_0 consists of isolated eigenvalues of finite multiplicity accumulating only at 0. Split

$$\mathcal{B} = E^s \oplus E^c \oplus E^u$$

according as the spectrum has modulus < 1 , $= 1$, or > 1 . The center and unstable summands are finite-dimensional. Choose an equivalent norm and numbers

$$\mu_s < 1 < \mu_u, \quad \mu_s < \mu_{cu}, \quad \mu_{cs} < \mu_u$$

so that the linear part contracts E^s , expands E^u , and has the corresponding weak inequalities on $E^c \oplus E^u$ and $E^s \oplus E^c$. On a sufficiently small product ball, the nonlinear remainder $G(x) = R(x) - DR_0x$ has arbitrarily small C^1 norm. Therefore the center-unstable cone

$$C_x^{cu} = \{v : \alpha \|v^s\| \leq \|v^{cu}\|\}$$

is forward invariant, while the center-stable cone $C_x^{cs} = \{v : \alpha \|v^u\| \leq \|v^{cs}\|\}$ is backward invariant. The stable face cannot be hit by an interior point and the unstable face is mapped outside the ball.

For $0 < r < 1$ close to one, apply the same cone estimates to $R_r := rR$. The stable-manifold/degree argument on the finite-dimensional $E^c \oplus E^u$ factor produces a point g_r on the center face whose entire R_r -orbit remains in the product ball and tends to zero. Compactness of R permits $r_j \uparrow 1$ and a convergent subsequence of $R_{r_j}g_{r_j}$; its limit is a nonzero orbit $g_n = R^n g$ staying in the ball. Forward invariance of the enlarged

center-unstable cone gives

$$g_{n+1}^{cu} = DR_0|_{E^c \oplus E^u} g_n^{cu} + O(\|g_n^{cu}\|^2).$$

If the orbit decayed with a strictly negative exponential rate, choose an adapted norm for which the inverse of $DR_0|_{E^c \oplus E^u}$ has norm $< e^c$ for some smaller rate c . For a sufficiently small ball the displayed recurrence then gives $\|g_{n+1}^{cu}\| \geq e^{-c}\|g_n^{cu}\|$, contradicting faster exponential decay. The orbit tends to zero, so its upper exponential rate is at most zero; the preceding lower bound gives (SO).

Apply this to $R = \mathcal{R}_*$ in coordinates centered at f_* . If a neutral eigenvalue existed, (SO) would give an orbit remaining arbitrarily close to f_* but decaying only subexponentially. By Lemma 16.58 that orbit consists of rotational coronas of rotation number θ , while Lemma 16.57 makes every such orbit converge exponentially to f_* . This contradiction eliminates all neutral eigenvalues.

Lemma 16.60 (A repelling direction exists). *The derivative DR_{*,f_*} has at least one eigenvalue of modulus greater than one.*

Proof. Rational-family realization, degeneration reduction, and uniqueness.

Put $d = d_0 + d_\infty - 1$. Once the two superattracting fixed points are normalized to 0 and ∞ and the unique free critical point is normalized to 1, the realization family is forced to be

$$F_c(z) = -c \frac{\sum_{j=d_0}^d \binom{d}{j} (-z)^j}{\sum_{j=0}^{d_0-1} \binom{d}{j} (-z)^j}, \quad c \in \mathbb{C}^*. \quad (16.63)$$

Indeed, for $G = c^{-1}F_c$, the local-degree conditions at 0 and ∞ force $G = z^{d_0}p/q$ with $\deg p = d_\infty - 1$ and $\deg q = d_0 - 1$. Since $G(1) = 1$ and 1 has local degree d , the numerator of $G - 1$ has a zero of order d at 1. Comparing coefficients with $(z - 1)^d$ gives (16.63). In particular, for every $c \neq 0$ the only critical points are 0, ∞ , 1, with local degrees d_0, d_∞, d , and $F_c(1) = c$.

Fixed-modulus realization and the remaining uniform Herman-ring a priori bound.

Let $\mathcal{C}_*$ denote the marked collision combinatorics used by Corona renormalization. The preceding analysis separates existence into a fixed-modulus realization step and a modulus-independent compactness step. We now prove the former and all of the latter *except* the genuinely deep uniform a priori bound. Thus the only input retained from the near-degenerate Herman-ring theory is

(AHR) Uniform bounded-type Herman-ring a priori bound. For fixed d_0, d_∞ and bounded-type constant N , there is $K = K(d_0, d_\infty, N)$ such that both boundary components of every invariant Herman ring in $\mathcal{H}_{d_0, d_\infty, \theta}$ are K -quasicircles, independently of the modulus of the ring.

Everything below this boxed input is reconstructed in the book.

A generalized critical-circle model. Fix $m \geq 2$ and a bounded-type irrational α . Specialize (16.63) to equal criticalities $d_0 = d_\infty = m$, so that $D = 2m - 1$, and put, for $\tau \in \mathbb{R}$,

$$B_{m,\tau}(z) = -e^{2\pi i\tau} \frac{\sum_{j=m}^D \binom{D}{j} (-z)^j}{\sum_{j=0}^{m-1} \binom{D}{j} (-z)^j}. \quad (16.64)$$

The coefficient calculation already made above shows that the only critical points are $0, \infty, 1$, with local degrees m, m, D . Write

$$Q_m(w) = \sum_{j=0}^{m-1} \binom{D}{j} w^j.$$

The elementary Eneström–Kakeya estimate gives

$$|w| \leq \max_{0 \leq j \leq m-2} \frac{\binom{D}{j}}{\binom{D}{j+1}} = \frac{m-1}{m+1} < 1$$

for every zero of Q_m . Hence the denominator of (16.64) has no zero on $\mathbb{T}$. Symmetry of binomial coefficients gives

$$\sum_{j=m}^D \binom{D}{j} (-z)^j = (-z)^D Q_m(-1/z),$$

and therefore $|B_{m,\tau}(z)| = 1$ on $\mathbb{T}$. The argument principle shows that the degree of $B_{m,\tau}|_{\mathbb{T}}$ is $m - (m-1) = 1$. Away from 1 its angular derivative cannot change sign, and at 1 the local degree D is odd. Thus the restriction is an orientation-preserving analytic critical circle homeomorphism with a single critical point at 1.

Multiplication by $e^{2\pi i\tau}$ is postcomposition by a rigid rotation. For compatible lifts $b_\tau : \mathbb{R} \rightarrow \mathbb{R}$ we therefore have $b_\tau = b_0 + \tau$ up to the harmless fixed half-turn coming from the minus sign. The translation number is continuous in the uniform topology and changes by one when τ is increased by one. Hence some phase $\tau = \tau(m, \alpha)$ has rotation number α . By the real bounds proved earlier in this chapter, bounded type makes the conjugacy $B_{m,\tau}|_{\mathbb{T}} \sim R_\alpha$ quasisymmetric.

Unicritical Siegel polynomial of arbitrary degree. Let $h : \mathbb{T} \rightarrow \mathbb{T}$ be the marked quasisymmetric conjugacy and extend it quasiconformally to $\mathbb{D}$ using the Chapter 5 extension theorem. Replace the dynamics in $\mathbb{D}$ by $h^{-1} \circ R_\alpha \circ h$ and leave $B_{m,\tau}$ unchanged on the exterior. The two definitions agree on $\mathbb{T}$ and give a quasiregular branched covering $\tilde{B}$. Spread the Beltrami coefficient of the inside model to the grand orbit by pullback and put it equal to zero elsewhere. The coefficient is $\tilde{B}$ -invariant and uniformly bounded, so Lemma 6.17 straightens $\tilde{B}$ to a rational map $P_{m,\alpha}$.

This map has degree m . Indeed, for a regular value of large modulus the preimages belonging to the exterior piece are precisely the m exterior preimages of $B_{m,\tau}$; the

other $m - 1$ preimages of the original rational map disappear with the replaced disk. Equivalently, the argument principle on $\mathbb{T}$ gives exterior mapping degree m . The point ∞ remains fixed of local degree m , so a degree- m rational map with that local degree at infinity has no finite pole and is a polynomial. There are no branch points in the new disk and no exterior branch points other than ∞ and the marked boundary point. Riemann–Hurwitz therefore forces the marked boundary point to have local degree m and to be the unique finite critical point. The image of $\mathbb{D}$ is an invariant Siegel disk of rotation number α , its boundary is a quasicircle, and its unique finite critical point lies on that boundary. We denote this polynomial by $P_{m,\alpha}$ and normalize its linearizing coordinate so that the critical boundary point has angular coordinate 1.

Direct Shishikura sewing for the collision combinatorics. Take

$$P_0 = P_{d_0,\theta}, \quad P_\infty = P_{d_\infty,1-\theta},$$

with Siegel disks Δ_0, Δ_∞ and linearizing coordinates ϕ_0, ϕ_∞ . Choose invariant analytic circles $\gamma_0 \Subset \Delta_0$ and $\gamma_\infty \Subset \Delta_\infty$. In the linearizing coordinates the boundary identification

$$\phi_\infty(h(z)) = \frac{r_0 r_\infty}{\phi_0(z)}, \quad z \in \gamma_0,$$

where $|\phi_\bullet| = r_\bullet$ on $\gamma_\bullet$, reverses the boundary orientation and intertwines P_0 with P_∞ because the two rotation numbers are opposite modulo one. Extend h quasiconformally to the sphere so that it exchanges the inside of γ_0 with the outside of γ_∞ and is conformal on the annular region in the two Siegel disks where the displayed formula is used.

Define the cut-and-paste branched covering

$$\tilde{R}(z) = \begin{cases} P_0(z), & z \in \gamma_0 \cup \text{ext}(\gamma_0), \\ h^{-1} \circ P_\infty \circ h(z), & z \in \text{int}(\gamma_0). \end{cases}$$

The boundary relation makes this continuous and quasiregular. A regular value in the P_0 side has d_0 preimages there, while the complementary piece contributes $d_\infty - 1$ further preimages; hence

$$\deg \tilde{R} = d_0 + d_\infty - 1 = d.$$

The overlap $\Delta_0 \cap h^{-1}(\Delta_\infty)$ is a topological invariant annulus on which the dynamics is conjugate to R_θ . Put on that annulus the standard complex structure on one side and the pullback by h on the other, spread it by pullback to the grand orbit, and use the standard structure on the complement. As in the preceding surgery, only the first pullback can increase dilatation. Chapter 6 invariant-Beltrami straightening therefore conjugates $\tilde{R}$ to a degree- d rational map R with an invariant Herman ring.

The two polynomial points at infinity survive as superattracting fixed points of local degrees d_0 and d_∞ ; normalize them to 0 and ∞ . The only remaining critical points are the two images of the unique finite critical points of P_0 and P_∞ , of multiplicities $d_0 - 1$ and $d_\infty - 1$, and they lie on the two boundary components of the Herman ring. Because both polynomial linearizing coordinates were marked by their critical point and the sewing uses reciprocal angular coordinates, the two critical points have the same angular mark. Thus R realizes precisely $\mathcal{C}_*$. Finally, the invariant radial Beltrami defor-

mation already used above changes the modulus of the Herman ring to any prescribed positive value without changing rotation number or marked combinatorics. Choosing modulus one proves the former input (E1) entirely inside the book.

Uniform Herman-ring a priori bounds. We reconstruct the uniform quasicycle argument in the same near-degenerate order as the primary source. The preceding width lemmas establish waves, spreading, and λ -to- τ trading; the argument below completes deep localization, controlled cuts, canonical horizontal pullbacks, thick-thin/fence analysis, exact-denominator horizontal loss, and the bounded-degree push-forward. Thus paragraphs 1–10 form one unconditional proof of (AHR). Fix the bounded-type constant $N = \beta(\theta)$ and degrees d_0, d_∞ . All constants below depend only on these data unless a parameter such as λ or Δ is displayed explicitly.

For a level- n closest-return interval $I \subset H$ let ℓ_n denote its combinatorial length. If $\rho > 1$, write ρI for the concentric combinatorial enlargement and let

$$W_\rho(I) := W(\mathcal{F}_\rho(I))$$

be the extremal width of curves joining I to the complementary boundary of ρI in the appropriate complementary domain. A piece is $[K, \rho]$ -wide if $W_\rho(I) \geq K$. Bounded type gives $q_{n+1}/q_n \leq N+1$ and hence all degree bounds for pullbacks through a fixed number of closest-return levels are uniform.

1. Waves and shallow extraction. A wave based on a piece A is a proper lamination in one complementary component $Y^\bullet$ of H such that every leaf protects A from the marked end $\bullet \in \{0, \infty\}$. Its combinatorial length is the maximum length of a protected piece and its width is its extremal width. We shall use the following quantitative form, whose proof we include because it is the engine for both shallow arguments below.

Wave lemma. There is an absolute $m \in \mathbb{N}$ such that, for every level n and every $\alpha \geq 1$, there is a threshold $K(n)$ with the following property. If a wave Ω has

$$|\Omega| \geq \alpha \ell_n, \quad W(\Omega) \geq K(n),$$

then some level- $(n+m)$ combinatorial piece J satisfies

$$W_{10}(J) \geq c\alpha W(\Omega),$$

where $c > 0$ depends only on the bounded-type and degree data.

To prove this, first choose integers $m'' > m' > 0$ so that $10\ell_{n+m''} < \ell_n$ and $2d\ell_{n+m'} \leq \ell_{n+m''+2}$, where d bounds the number of critical points on the relevant side, and put $t = q_{n+m''+2}$. Let A be a longest piece protected by an outer wave (the inner case is identical). The preimages, up to time $t-1$, of the finitely many outer critical points cut A into subpieces. Closest-return separation and the two displayed inequalities imply that at least 5α of those subpieces have length at least $\ell_{n+m'}$: otherwise summing the lengths of the short and long pieces gives $2d\ell_{n+m'} > \ell_{n+m''+2}$.

Let B_x be the finite bubble chain attached at a cutting point x . Its harmonic measure in the depth- t complementary domain is bounded below by a constant depending on n ; hence the extremal width of any lamination protecting B_x is bounded by a number

C_n . After taking $K(n)$ large, remove all leaves that skip one of these bubble chains. A surviving sublamination $\widehat{\Omega}$ still satisfies $W(\widehat{\Omega}) \geq (4/5)W(\Omega)$ and crosses the selected chains in order. Between consecutive selected chains let G_i be the corresponding restriction. The restrictions are traversed in series, so if G_s is the widest one,

$$W(G_s) \geq (5\alpha - 2)W(\widehat{\Omega}) \geq 2.4\alpha W(\Omega).$$

Remove from G_s the leaves protecting an arc of harmonic measure at least the covering scale and then one outer unit buffer. The discarded width is at most C_n . The remaining family is disjoint from all nontrivial deck translates, so f^t is univalent on it and pushes it to a new wave Ω' with

$$|\Omega'| \geq \ell_{n+m'}, \quad W(\Omega') \geq 2\alpha W(\Omega).$$

This is the wave-amplification lemma.

It remains to turn amplification into a 10-degeneration. Choose m so that $(10 - 1)\ell_{n+m}/2 \leq \ell_{n+m'}$. Starting from $\Omega_0 = \Omega$, apply the preceding lemma. If the amplified wave Ω'_j protects a piece J_j of length $\ell_{n+m'}$, enlarge J_j to the level- n piece I_{j+1} with the same combinatorial mid-segment and split Ω'_j into the leaves protecting I_{j+1} and the leaves landing on $I_{j+1} \setminus J_j$. If the first family has at least three quarters of the width, it is the next wave and its width grows by a factor at least $3/2$. Otherwise the second family has at least one quarter of the width. Among the level- $(n + m)$ subpieces of $I_{j+1} \setminus J_j$, choose one J receiving maximal landing width. The choice of m ensures that these landing curves belong to $\mathcal{F}_{10}(J)$, and bounded combinatorics gives

$$W_{10}(J) \geq c W(\Omega'_j) \geq c\alpha W(\Omega).$$

If this second alternative never occurred, one would obtain waves of uniformly bounded combinatorial length and exponentially growing width in the fixed complementary domain, impossible by extremal-width compactness. The wave lemma follows.

2. Spreading a degeneration. Fix $\Xi > 1$ and choose $\lambda \gg 10$. Suppose I is a $[K, \lambda]$ -wide level- n piece with $n \geq n_\lambda$. For every $2 \leq a \leq q_{n+1} + 1$, bounded-type closest-return geometry supplies $b \in \{a - 2, a - 1, a\}$ and two cross-cuts of λI_b that avoid the critical values and whose total landing width is bounded by an absolute constant. Let V be the cut disk, $B \Subset V$ the subdisk cut out by $10I_b$, and let $U \supset \Lambda$ be the pullback components containing I . The branch $f^b : U \rightarrow V$ has total degree bounded by a constant depending on λ , while $f^b : \Lambda \rightarrow B$ has degree bounded independently of λ .

Apply the Kahn-Lyubich covering lemma (Lemma 13.1) to this datum. Since $W(U, I) \geq K$, for K above a threshold depending on Ξ and λ , one of two things occurs: either the inner width in B is at least a constant multiple of ΞK , or the outer width in V is at least $\xi_1 K$, where $\xi_1 = \xi_1(\Xi) > 0$. Removing the two cuts changes width by only an additive constant, so, after raising the threshold once more,

$$W_{10}(I_b) \geq d^2 \Xi K \quad \text{or} \quad W_\lambda(I_b) \geq \xi_2 K.$$

Pushing forward at most two further iterates costs at most the square of the degree. Consequently each I_a is either $[\Xi K, 10]$ -wide or $[\xi K, \lambda]$ -wide, with $\xi = \xi(\Xi) > 0$ independent of λ . If the first alternative never occurs, the orbit of I_2 through one closest-return block forms an almost tiling all of whose members are $[\xi K, \lambda]$ -wide. This is the spreading theorem.

3. Trading λ -width for much larger 10-width. Consider an almost tiling $\mathcal{I}$ of level n consisting of $[K, \lambda]$ -wide pieces. We prove that, for sufficiently large λ , there are m_λ and $\Pi_\lambda \rightarrow \infty$ such that some piece within m_λ levels of n is $[\Pi_\lambda K, 10]$ -wide.

In the deep case put

$$N_\lambda = \left\lfloor \frac{\lambda}{3 \cdot 10^2} \right\rfloor$$

and select N_λ members $I_1, \dots, I_{N_\lambda}$ in cyclic order so that consecutive gaps have length between $9\ell_n$ and $10\ell_n$. Let P be the shortest piece containing their union. Then every λI_j contains $10P$. In the punctured sphere obtained by deleting the complements of the λI_j , the families witnessing $W_\lambda(I_j) \geq K$ are pairwise separated. The Quasi-Additivity Law of Chapter 13 therefore gives

$$\max\{W_{10}(P), W_{10}(I_1), \dots, W_{10}(I_{N_\lambda})\} \geq c\sqrt{N_\lambda} K.$$

If the maximum is attained at P , take the level- $n - m_\lambda$ subpiece with the same mid-segment; bounded geometry changes width only by a universal factor. Thus the deep case produces a 10-wide piece of width $\succ \sqrt{\lambda} K$.

For the shallow case fix a $[K, \lambda]$ -wide member I of the tiling. If there is a wave of width at least $K/10$ and length at least $\lambda^{1/3}\ell_n$, the wave lemma gives immediately a 10-wide piece of width $\succ \lambda^{1/3}K$. Hence assume that every such long wave has width at most $K/10$.

Decompose the target complement $(\lambda I)^c$ into $T^+ \cup T \cup T^-$, where $T^\pm$ are the two end pieces of length $\lambda^{1/3}\ell_n$. In the canonical radial foliation of the annulus between I and $(\lambda I)^c$, delete every leaf containing a subcurve that protects one of $T^\pm$ from either complementary end. By the no-wide-wave assumption the surviving family $\mathcal{F}$ has width at least $3K/5$ and every surviving leaf joins I to $T^+ \cup T^-$. One side, say $\mathcal{F}^+$, therefore has

$$W(\mathcal{F}^+) \geq 3K/10.$$

It is rectangular: it is a sublamination of the vertical foliation of the conformal rectangle formed by radial leaves landing on T^+ .

Choose a maximal sequence $I_0 = I, I_1, \dots, I_N$ of pieces between I and T^+ such that $|I_j| = \lambda^{1/3}\ell_n$ and distinct I_i, I_j are separated by at least $9\ell_n/2$. Maximality and the length of the corridor give $N \asymp \lambda^{2/3}$. For either complementary end and each I_j , the leaves of $\mathcal{F}^+$ containing a subcurve protecting I_j form a buffer of this rectangle: cut the complementary domain by rays from the two ends to T , and use the first protecting subcurve to separate a terminal rectangular buffer. Let $\mathcal{G}$ be the leaves meeting all I_j consecutively. A leaf outside $\mathcal{G}$ lies in one of two maximal protecting buffers, hence

the complement $\mathcal{F}^+ \setminus \mathcal{G}$ overflows two waves of length at least $\lambda^{1/3}\ell_n$. Thus

$$W(\mathcal{G}) \geq K/10.$$

Let $\mathcal{G}_j$ be the restriction of $\mathcal{G}$ between I_{j-1} and I_j . These restrictions occur in series. If $\mathcal{G}_s$ is widest, then

$$W(\mathcal{G}_s) \geq NW(\mathcal{G}) \geq NK/10.$$

The gap between I_{s-1} and I_s contains a level- n subpiece J whose 10-family receives at least a $c\lambda^{-1/3}$ fraction of $\mathcal{G}_s$. Consequently

$$W_{10}(J) \geq c\lambda^{-1/3}W(\mathcal{G}_s) \geq c'\lambda^{1/3}K.$$

The shallow level lies in a fixed finite band, so the thresholds and the level increment are absorbed into m_λ . Combining the two cases, we may take

$$\Pi_\lambda = c \min\{\sqrt{\lambda}, \lambda^{1/3}\},$$

after reducing c once; in particular $\Pi_\lambda \rightarrow \infty$. This is the trading theorem.

Deep localization and horizontal loss. We now complete the remaining deep package. Throughout this part put $\tau = 10$. All constants implicit in $\prec, \succ, \asymp$ depend only on (d_0, d_∞, N) and on the finitely many separation constants fixed below; dependence on the large auxiliary separation λ is displayed when it matters. The proof is organized so that every use of a branched cover has a target whose intersection with H has controlled combinatorial length. This is the point needed later to make the final degree bound independent of λ .

4. Localization, cuts, and the initial horizontal family. Spread the initial τ -degeneration through one closest-return block as in paragraph 2, using the fixed separation τ ; unless doubling has already occurred this gives a level- n almost tiling whose members retain width εK . Choose N members $I_1, \dots, I_N$ in cyclic order with successive combinatorial gaps comparable with $\tau\ell_n$, and let P be their shortest common envelope. Fix ρ so large that every ρI_j contains P . If, for every j , at least half of the εK family from I_j reaches $\rho I_j \setminus \tau I_j$, then the N landing families are mutually separated in the punctured sphere obtained by deleting the outer complements. Quasi-Additivity gives

$$\max\{W_\tau(P), W_\tau(I_1), \dots, W_\tau(I_N)\} \succ \sqrt{N} K.$$

Take N once and for all so that the right-hand side exceeds $2K$ after the bounded level adjustment of paragraph 3. If this alternative fails, one member L^* has more than $\varepsilon K/2$ width landing in $\rho L^* \setminus \tau L^*$; one of the two components, denoted R^* , receives at least $\varepsilon K/4$. This proves the deep localization lemma.

Choose now $\lambda \gg \rho$ and take the deep threshold so that the preceding finite level changes stay above it. The cut lemma used in paragraph 2 gives $t \in \{0, 1, 2\}$ and two rays from the complementary Fatou components. Put $L = f^t(L^*)$, $R = f^t(R^*)$. Let G be the union of the two rays; in the one-sided case also adjoin to G the complementary

component that contains 0. Then

$$U^0 := \widehat{\mathbb{C}} \setminus ((\lambda L)^c \cup G)$$

is a topological disk, $U^0 \cap H = \lambda L$, every critical value of every iterate that lies in U^0 lies on $U^0 \cap H$, and the family in $\widehat{\mathbb{C}} \setminus (H \cup G)$ joining λL to G has $O(1)$ width. The extra complementary component in the one-sided case is exactly what prevents an interior critical value from invalidating the degree count.

For $j \geq 0$ let L^j, R^j be the components of $f^{-j}(L), f^{-j}(R)$ meeting H , put $\Upsilon^j = L^j \cup R^j$, let U^j be the component of $f^{-j}(U^0)$ containing Υ^j , and write

$$\mathcal{F}^j = \mathcal{F}_{\text{can}}(U^j; L^j, R^j), \quad X^j = W(\mathcal{F}^j).$$

The image under f^t of the family joining L^* to R^* has width $\succ K$ because $t \leq 2$ and the degree is uniformly bounded. If at least half of this image stays in U^0 , maximality of the canonical horizontal lamination gives

$$X^0 \geq \varepsilon_1 K \tag{C16.1}$$

for an absolute $\varepsilon_1 > 0$. Otherwise at least half meets $(\lambda L)^c \cup G$. The G -landing part has $O(1)$ width, hence for large K the remaining part gives $W_\lambda(L) \succ K$. Thus, unless a definite λ -degeneration has already been produced, we may and shall assume (C16.1).

5. Return thickening and comparison. Fix a level $s > n$ and put $Q = q_s$. Thicken the two marked sides by one Q -return,

$$\widehat{L} = L \cup f^Q(L), \quad \widehat{R} = R \cup f^Q(R), \quad \widehat{U}^0 = U^0 \setminus f^Q((\lambda L)^c),$$

and define $\widehat{L}^j, \widehat{R}^j, \widehat{\Upsilon}^j, \widehat{U}^j, \widehat{\mathcal{F}}^j$ by pullback as above. The closest return estimates give new separation constants $\widehat{\tau} \asymp \tau$, $\widehat{\rho} \asymp \rho$ and $\widehat{\lambda} \asymp \lambda$, together with

$$\widehat{\Upsilon}^0 \supset \Upsilon^0, \quad \partial \widehat{U}^0 \supset \partial U^0, \quad \widehat{U}^Q \setminus \widehat{\Upsilon}^Q \subset U^0 \setminus \Upsilon^0. \tag{C16.2}$$

Let $P_1 = \widehat{L} \setminus L$ and $P_2 = \widehat{R} \setminus R$. Decompose the leaves of $\widehat{\mathcal{F}}^0$ according to whether an endpoint lies on P_1 , on P_2 , or on neither. The first two subfamilies overflow fixed τ -families of level- s pieces. The third is a restriction of $\mathcal{F}^0$, except for the family that reaches the new boundary $P_3 := \partial \widehat{U}^0 \setminus \partial U^0$; that exceptional family overflows a λ -family. Consequently, unless a level- s piece already has τ -width at least $2K$ or λ -width $\succ K$, all three exceptional families have width bounded by fixed fractions of X^0 and the finite cut leaves contribute only $O(1)$. Enlarging the threshold for K gives

$$\frac{1}{2} X^0 \leq W(\widehat{\mathcal{F}}^0) \leq \varepsilon_2 X^0 \tag{C16.3}$$

with an absolute $\varepsilon_2 > 1$.

6. Thick-thin decomposition and the two obstruction families. We prove the horizontal-loss statement for the unthickened configuration; it then applies verbatim

to the hatted one. Fix a depth m and write $a = q_{n+m}$. For each free critical value $v \in H$, its first a backward iterates cut $U^0 \cap H$, whose length is $\asymp \lambda \ell_n$, into intervals of length at least ℓ_{n+m} . Bounded type therefore gives

$$\#(\text{CV}(f^a) \cap U^0) = O(\lambda C^m). \quad (\text{C16.4})$$

Hence removing all those critical values changes the thick-thin accounting only by an additive constant $C(m, \lambda)$.

Take the horizontal thick-thin decomposition of $U^0 \setminus (\Upsilon^0 \cup \text{CV}(f^a))$. Its canonical rectangles split into type A rectangles (one crossing of a middle separator), type B rectangles (two crossings), and peripheral rectangles. Split each peripheral rectangle at the first exit from fixed enlargements $\eta L, \eta R$. Denote the exiting rectangle families by $\mathcal{D}_L, \mathcal{D}_R$ and put $D = W(\mathcal{D}_L) + W(\mathcal{D}_R)$; absorb the remaining endpoint-confined parts into enlarged hulls of L and R . For the first block write $K_h = X^0$; for a block beginning at time j the same construction is applied with $K_h = X^j$. If $D > \kappa K_h$, cover ηL and ηR by the bounded number $N(m)$ of level- $(n+m)$ pieces whose τ -enlargements remain in the respective hull. One of them receives at least $D/N(m)$ width, and overflow gives

$$W_\tau(J) \succ_m \kappa K_h. \quad (\text{C16.5})$$

Thus, in a branch in which no prescribed amplification has occurred, we may assume $D \leq \kappa K_h$.

Choose a proper middle curve $b \subset U^0$ separating the two hulls and avoiding the critical values and endpoint cuts. Its distinguished lift $\beta \subset U^a$ again separates the lifted hulls. Type A rectangles cross b once and type B rectangles twice. This motivates the asymmetric width

$$Z^j = A^j + 2B^j.$$

The lifted thick-thin decomposition and the canonical-lamination maximality of Lemma 13.43 give, with an additive $C = C(m, \lambda)$,

$$Z^{j+a} - C \leq Z^j, \quad X^j - C \leq Z^j \leq 2X^j + C. \quad (\text{C16.6})$$

The first inequality is obtained by restricting every returning rectangle to its crossing(s) of the distinguished lift β ; no critical value occurs inside a lifted thick rectangle, so the restriction is univalent.

There is a sharper decomposition. Choose auxiliary separators b_L, b_R on the two sides of b and their distinguished lifts β_L, β_R ; the strip between the latter contains a level- n atom by closest-return geometry. A lifted rectangle is called persistent when its canonical restriction crosses both β_L and β_R . Let Z_{per} be the total asymmetric contribution of these rectangles and put $Z_{\text{non}} = Z^j - Z_{\text{per}}$. For a nonpersistent rectangle, cut at the first missed lifted separator. Its central restriction remains in the horizontal channel and contributes at most Z_{non} ; the escaping restriction must cross the appropriate peripheral family $\mathcal{D}_L$ or $\mathcal{D}_R$ before it can return, and after that gate it traverses another copy of the nonpersistent horizontal channel. The two escaped pieces are disjoint, so their available width is at most $2(Z_{\text{non}} + D)$. The central and escaping routes are in series after the pull-off. Therefore the hypotheses of Lemma 13.44 hold with

these two gates, and it gives

$$Z^{j+a} - C \leq Z_{\text{per}} + Z_{\text{non}} \boxplus 2(Z_{\text{non}} + D). \quad (\text{C16.7})$$

This is the Herman-specific verification of the two-gate hypotheses used by the abstract pull-off lemma.

7. Persistent rectangles force degeneration. It remains to show that a large persistent part cannot hide from the τ -width. Let $\Pi \subset U^a$ be the strip between the lifted left and right separators and let $\mathcal{L}_{\text{per}}$ be the restriction of the persistent rectangles to Π . Any two leaves of $\mathcal{L}_{\text{per}}$, together with the two strip boundaries, enclose a disk O' . Their images enclose a disk O . The map $f^a : O' \rightarrow O$ is proper; its boundary degree is one, hence the argument principle makes it univalent. Thus O' contains no critical point of f^a , and all persistent leaves are properly homotopic in Π relative to $\text{CP}(f^a)$.

Assume $Z_{\text{per}} \geq 0.1K_h$ and set

$$t_0 = q_{n+m-2}, \quad s_0 = a - t_0.$$

The first t_0 backward iterates of every free critical point partition H into atoms with combinatorial lengths between ℓ_{n+m-2} and ℓ_{n+m-4} . Pulling this tiling back by f^{s_0} tiles the relevant preimages of H by the critical points of f^a . For any prescribed N , choosing m large gives along a level- n strip atom $2N$ ordered pairs of critical roots. Join each pair by its interval P_i on H and by the two bubble lifts issuing from its endpoints. The resulting connected sets $\#_i$ are pairwise disjoint fences. Successive pairs $\#_{2j-1}, \#_{2j}$ have images under f^{s_0} separated by at least a fixed multiple of $\tau\ell_{n+m-4}$.

The relative-homotopy conclusion above and the critical-point tiling force all leaves of $\mathcal{L}_{\text{per}}$ to pass each fence through a common gate $G_i \subset \#_i$. The images $J_i = f^{s_0}(G_i)$ are level- $(n+m-4)$ combinatorial pieces. Restrict the lamination between each pair G_{2j-1}, G_{2j} . These N restrictions are traversed in series, so the widest has width at least $NW(\mathcal{L}_{\text{per}})$. Since $W(\mathcal{L}_{\text{per}}) \geq Z_{\text{per}} \geq 0.1K_h$, after applying f^{s_0} it overflows $\mathcal{F}_\tau(J_{2j})$; shrinking to a level $n+m$ subpiece costs at most the fixed four-level real-bounds factor. Hence

$$W_\tau(J) \geq cC^{-4}NK_h. \quad (\text{C16.8})$$

Taking $N \geq 10C^4M$ gives $W_\tau(J) \geq MK_h$. In the one-sided geometry only outer bubbles are available. The fences need not separate the whole strip, but the family skipping one of finitely many fences has width $O_{\lambda,N}(1)$ by the boundary extremal-width estimate; after raising the K_h threshold, at least half of the persistent family crosses all fences. The same argument with $N \geq 20C^4M$ proves (C16.8) with the same conclusion.

8. Pull-off contraction and denominator completion. Fix $M > 1$. Choose m large enough for the persistent-fence amplification lemma and then choose κ so that the peripheral alternative (C16.5) with $D > \kappa K_h$ also yields an MK_h piece. If no such amplification occurs, then

$$D \leq \kappa K_h, \quad Z_{\text{per}} < 0.1K_h.$$

For K_h above the additive thick-thin constant, (C16.6) gives $Z^j \geq K_h - C$. Increasing the threshold once more, the two estimates above imply $D \leq 2\kappa Z^j$ and $Z_{\text{per}} \leq \alpha Z^j$ with, say, $\alpha = 1/5$. Relabel 2κ as κ . Substituting into (C16.7), and using that $Z^j \asymp X^j$ once K_h dominates the additive boundary constants, gives

$$Z^{j+a} - C \leq \nu' Z^j, \quad \nu' = \alpha + (1 - \alpha) \boxplus (2 + 2\kappa) < 1, \quad \alpha = \frac{1}{5}.$$

Set $\nu = (1 + \nu')/2$. Raising the threshold to $Z^j \geq 2C/(1 - \nu')$ absorbs C and yields

$$Z^{j+a} \leq \nu Z^j. \quad (\text{C16.9})$$

Thus for every M there is a fixed closest-return depth m such that each block either produces an M -fold τ -degeneration or contracts asymmetric horizontal width by the fixed factor $\nu < 1$.

Given $\Delta > 1$ and $0 < \delta < 1$, use $M = \delta^{-1}\Delta$ and choose an integer T with $\nu^T \leq \delta/8$. Iterating (C16.9) shows that, unless a level- $n+m$ piece of τ -width at least ΔK_h appears, one has

$$Z^{Ta} \leq \frac{\delta}{8} Z^0, \quad a = q_{n+m}. \quad (\text{C16.10})$$

We now make explicit the denominator-completion step. Let m_2 be the smallest integer with

$$q_{n+m_2} > Tq_{n+m}.$$

The offset $m_2 - m$ is bounded in terms of T alone (and hence in terms of Δ, δ and the bounded-type data). Apply the same middle-curve monotonicity construction to the remaining iterate $h = q_{n+m_2} - Ta$. Since $0 < h \leq q_{n+m_2}$, the critical-value counting argument underlying (C16.6), now with depth m_2 , gives

$$Z^{q_{n+m_2}} \leq Z^{Ta} + C_*(m_2, \lambda). \quad (\text{C16.11})$$

This estimate uses only non-strict monotonicity, not the persistence alternative. In particular, the monotonicity proof never uses that the iterate length itself is a closest-return denominator: it uses only the critical-value bound up to q_{n+m_2} and the distinguished lift of the middle separator. Combining (C16.10), the comparison between X^j and Z^j in (C16.6), and (C16.11), and then increasing the K_h threshold to absorb all additive constants, gives

$$W(\mathcal{F}^{q_{n+m_2}}) \leq \delta K_h. \quad (\text{C16.12})$$

The earlier version of Lim's proof writes precisely this final passage to the smallest denominator; the later source presents the block induction directly. Equation (C16.11) records the continuation explicitly, so the statement used below is genuinely at the denominator time q_{n+m_2} .

9. Push-forward after loss. Apply the strict horizontal loss lemma to the hatted family from paragraph 5 with $\delta = (2\varepsilon_2)^{-1}$ and with Δ large enough to absorb the fixed comparison constants. If the amplification alternative does not occur, put $r = n + m_2$ and use (C16.3) and (C16.12). The embedding (C16.2) forces the portion of the

original horizontal family that is no longer horizontal in $\widehat{U}^{q_r} \setminus \widehat{\Upsilon}^{q_r}$ to be vertical. After absorbing the $O(1)$ cut family,

$$W\left(\mathcal{F}_{\text{can}}^v(\widehat{U}^{q_r}; \widehat{\Upsilon}^{q_r})\right) \geq \frac{1}{3}X^0 \succ K. \quad (\text{C16.13})$$

One of $\widehat{L}^{q_r}, \widehat{R}^{q_r}$ therefore has relative width $\succ K$ in $\widehat{U}^{q_r}$; call it $\widehat{J}^{q_r}$.

Let

$$\widehat{U}_\tau^0 = \widehat{U}^0 \setminus (\tau \widehat{J})^c,$$

and let $\widehat{U}_\tau^{q_r}$ be its pullback component containing $\widehat{J}^{q_r}$. The map

$$f^{q_r} : (\widehat{U}^{q_r}, \widehat{U}_\tau^{q_r}, \widehat{J}^{q_r}) \longrightarrow (\widehat{U}^0, \widehat{U}_\tau^0, \widehat{J})$$

has total degree depending on the fixed outer geometry, while its restriction to $\widehat{U}_\tau^{q_r}$ has degree independent of λ . Indeed, $q_r = q_{n+m_2}$ and $\widehat{U}_\tau^0 \cap H$ has length $\rho_0 \ell_n$ with $\rho_0 \asymp \tau$ fixed; the degree count is therefore bounded by the number of critical encounters in at most q_{n+m_2} iterates, namely a constant $M(m_2, \rho_0)$. In the one-sided case the domains were constructed disjoint from the inner complementary component, so every relevant critical value lies on H and the same count applies.

Apply the Covering Lemma to the displayed triple. For large K , either $W(\widehat{U}^0, \widehat{J}) \succ K$ or $W(\widehat{U}_\tau^0, \widehat{J}) \geq (2C_0 + 1)K$, where C_0 is the fixed closest-return comparison constant. The curves terminating on G have $O(1)$ width. Deleting them and extracting a bounded-level combinatorial subpiece gives

$$[K, \tau]\text{-wide at level } n \implies \begin{cases} [2K, \tau]\text{-wide within } O(1) \text{ levels,} \\ [\chi K, \lambda]\text{-wide within } O(1) \text{ levels,} \end{cases} \quad (\text{C16.14})$$

for a $\chi > 0$ independent of K and λ . This closes the former (DLP) interface.

10. Amplification and contradiction. Choose λ so large that the trading gain satisfies $\Pi_\lambda \xi \chi \geq 2$, where ξ is the spreading constant from paragraph 2. Starting from a $[K, \tau]$ -wide piece, (C16.14) either doubles its τ -width immediately or gives a $[\chi K, \lambda]$ -wide piece. Spread the latter. Either a doubled τ -piece appears, or one obtains an almost tiling of $[\xi \chi K, \lambda]$ -wide pieces. Trading this tiling gives a $[\Pi_\lambda \xi \chi K, \tau]$ -wide piece, hence again width at least $2K$. Thus there are universal K_*, m_*, N_* such that every $[K, \tau]$ -wide level- n piece with $K \geq K_*$ and $n \geq N_*$ produces a $[2K, \tau]$ -wide piece within m_* levels.

If H is already a Herman quasicircle, the amplification theorem immediately yields a uniform bound. Indeed, for every fixed separation factor the elementary extremal-width criterion says that a quasicircle has a finite upper bound for the widths of sufficiently short combinatorial intervals. If no bound depending only on (d_0, d_∞, N) existed, an initial interval above the extraction threshold would contain a $[K, \tau]$ -wide combinatorial subinterval; successive amplification would produce widths $K, 2K, 4K, \dots$, contradicting that finite bound. Conversely, a uniform upper bound for W_τ on sufficiently short intervals gives a uniform quasisymmetry constant: subdividing an interval into at most τ equal combinatorial pieces first bounds the width at any fixed smaller sepa-

ration; the extremal-width/harmonic-measure comparison then makes harmonic measure of adjacent equal-length intervals uniformly comparable on both complementary sides, and conformal welding gives a quasicircle constant depending only on that width bound.

For a genuine Herman ring one extra scale conversion is needed because the ring modulus itself supplies arbitrarily large width far below the ring scale. Let $\mu = \text{mod } H$ and choose M by

$$\ell_{M+1} \leq \mu < \ell_M. \quad (16.65)$$

After fixing the small-modulus threshold we may assume $M > N_* + 2m_*$. Call levels $N_* \leq n < M$ the *Herman scale* and levels $n \geq M$ the *Siegel scale*.

At the Siegel scale a boundary interval $I^\bullet \subset H^\bullet$ ($\bullet = 0$ or ∞) thickens to a combinatorial piece of the closed ring with comparable τ -width. Apply amplification to that piece and then restrict back to one of its two boundary sides. The side corridors have width bounded in terms of the fixed closest-return constants, so one boundary interval retains a definite fraction of the doubled width. Therefore a $[K, \tau]$ -wide Siegel-scale boundary interval produces a $[2K, \tau]$ -wide boundary interval of level at least N_* after increasing the universal threshold.

Now let I be a Herman-scale piece and iterate amplification. Select every t -th amplified piece, where t is a fixed large integer. It is impossible for all selected pieces to remain below level M : their levels change by at most m_* at each amplification, while their widths grow like $2^{jt}K$ and for the fixed map and fixed finite M there are only finitely many combinatorial pieces at levels in the bounded band $[N_*, M)$, each of finite width. Let J be the first selected piece whose level n_1 reaches the Siegel scale. Minimality gives

$$M \leq n_1 < M + tm_*, \quad W_\tau(J) \geq 2^t K.$$

Write P^0, P^∞ for its two horizontal boundary intervals. On either side, let $Q^\bullet$ be $P^\bullet$ together with its two neighboring level- $(n_1 + 1)$ intervals, and let $R^\bullet$ be the complementary enlarged boundary target.

There are two possible losses when curves in the τ -family of J are restricted to a boundary side. First, curves crossing one of the two annular side strips of $\tau J/J$ have width

$$O\left(\frac{\ell_{n_1}}{\mu}\right) = O(1)$$

by (16.65) and $n_1 \geq M$. Second, curves that fail to contain a subcurve joining some $Q^\sharp$ to some $R^\flat$ must cross one of four adjacent level- $(n_1 + 1)$ rectangles in the short direction; their total width is

$$O\left(\frac{\mu}{\ell_{n_1+1}}\right) = O_{t,m_*}(1),$$

because $n_1 < M + tm_*$. Thus, after choosing the initial threshold and then t large enough, one of the two boundary components contains a pair $(Q^\bullet, R^\bullet)$ with joining width at least $c2^t K$. Bounded type allows a level- $(n_1 + s)$ subinterval $J^\bullet \subset Q^\bullet$, for a universal s , whose τ -complement contains $R^\bullet$ and whose relative combinatorial length

in $Q^\bullet$ is bounded below. Monotonicity and parallel addition therefore give

$$W_\tau(J^\bullet) \geq c'2^t K \geq 2K.$$

Hence every Herman-scale degeneration can be transferred, with doubled width, to a Siegel-scale boundary interval.

Suppose finally that one boundary component had a sufficiently short interval whose τ -width exceeded the universal threshold. Closest-return extraction produces a wide combinatorial boundary interval. The two scale lemmas above then generate an infinite sequence, on at least one of the two boundary components, whose widths tend to infinity geometrically. Each individual boundary component is already a quasicircle by the fixed-modulus Siegel-to-Herman surgery, so its short-interval widths have a finite upper bound; contradiction. Thus all sufficiently short intervals on both boundaries have uniformly bounded τ -width. The converse extremal-width/harmonic-measure criterion gives a quasisymmetry, hence quasicircle, constant depending only on (d_0, d_∞, N) and not on μ . This is (AHR).

The modulus-independent Herman-ring a priori bound is therefore derived from the waves, spreading, trading, localization, horizontal-loss, and push-forward machinery established in this proof.

What (AHR) gives beyond boundary shape. We now derive the degeneration compactness rather than importing it. By (AHR) and the bounded-type real bounds above, the linearizing coordinate on each boundary has a quasisymmetric extension with constants depending only on (d_0, d_∞, N) . We first record the resulting bounded-shape estimate. Let H^0 be the inner boundary, let Y^0 be the complementary component containing 0, and let Y_1^0 be the component of $R^{-1}(Y^0)$ containing 0, with boundary H_1^0 . Scale so that $\max_{z \in H_1^0} |z| = 1$. If $\zeta \in H^0$ is closest to 0, let I_ζ be the level-two closest-return interval centered at ζ and let $I'_\zeta \subset H_1^0$ be its lift. If $\kappa = \omega(0, I_\zeta, Y^0)$, then the degree- d_0 covering $Y_1^0 \rightarrow Y^0$, branched only at 0, gives

$$\omega(0, I'_\zeta, Y_1^0) = \frac{\kappa}{d_0}.$$

Domain monotonicity gives at least the same lower bound when I'_ζ is viewed from Y^0 . The two level-two intervals are disjoint, hence

$$\kappa + \frac{\kappa}{d_0} \leq 1.$$

On the other hand, uniform quasisymmetry makes both sides of I_ζ have definite Euclidean size relative to H^0 , while the uniform quasicircle condition keeps $H^0 \setminus I_\zeta$ a definite relative distance from ζ . The elementary Beurling/harmonic-measure estimate then shows that ζ cannot approach 0 while the preceding upper bound for κ remains fixed. Thus $\text{dist}(0, H^0)$ has a uniform lower bound. Every generation-one inner bubble has harmonic measure $1/d_0$ in Y_1^0 ; the same argument keeps all such bubbles a uniform relative distance from 0. Applying inversion gives the corresponding statement at ∞ . This is the bounded-shape lemma needed below.

The boundary linearizations now extend to a global $K_1(d_0, d_\infty, N)$ -quasiconformal map sending the closed Herman ring to a round annulus and fixing 0 and ∞ : use the

conformal linearization on the ring and the Chapter 5 quasimetric extension on the two complementary components. The bounded-shape lemma prevents the normalization from degenerating.

Coefficient precompactness from (AHR). Fix an upper modulus bound M ; along our degeneration we may take $M = 1$. Normalize by a linear map so that the largest modulus on H^0 is one, and rotate so that one such point is 1. Every map has the form

$$R(z) = \lambda z^{d_0} \frac{\prod_{j=1}^{d_\infty-1} (z - z_j)}{\prod_{k=1}^{d_0-1} (z - p_k)}. \quad (16.66)$$

The nonzero zeros and finite poles are enclosed by generation-one bubbles, so the bounded-shape lemma and the modulus bound put all z_j, p_k in one compact annulus $A(\varepsilon_1, \varepsilon_1^{-1})$.

They are also uniformly separated from the Herman ring. To see this, map the ring globally to a round annulus by the K_1 -quasiconformal map above and choose a fixed-modulus round annulus separating a marked critical value from the corresponding center. Lift this annulus through the degree- d_0 (or d_∞) first-bubble covering. Conformal invariance of modulus before the lift and K_1 -quasiconformal distortion after uniformization give a uniform positive modulus separator between the bubble center (a zero or pole) and the Herman boundary. In the already bounded Euclidean annulus, the standard Teichmüller ring estimate converts that modulus bound into a Euclidean separation $\varepsilon_2 > 0$. Hence zeros and poles cannot collide, and neither can any z_j or p_k approach the normalization point 1. Since $1, R(1) \in H^0$ and the bounded-shape estimate controls both, evaluating (16.66) at 1 gives

$$0 < \varepsilon_3 \leq |\lambda| \leq \varepsilon_3^{-1}.$$

Thus all coefficients range in a compact set and numerator and denominator have no common limiting factor. Every normalized sequence with modulus at most M therefore has a subsequence converging locally uniformly to a *degree- d* rational map with the same local degrees at 0 and ∞ . This proves the precompactness portion previously hidden in (E2).

Collapse of the ring and persistence of the collision marking. Start with the modulus-one realization just constructed and perform the radial invariant-Beltrami deformation to obtain maps R_t with modulus $t \downarrow 0$ and fixed marked combinatorics $\mathcal{C}_*$. Use the normalization above and take a convergent subsequence $R_{t_n} \rightarrow R_*$. For each n , the preceding global straightening gives a uniformly quasiconformal map

$$\Phi_n : \widehat{\mathbb{C}} \longrightarrow \widehat{\mathbb{C}}$$

which is conformal on the Herman ring, fixes 0 and ∞ , and conjugates its dynamics to R_θ on the round annulus $A(1, e^{2\pi t_n})$. Normalize one marked boundary point as well. Chapter 5 compactness of normalized quasiconformal maps gives, after a subsequence, $\Phi_n \rightarrow \Phi$ and $\Phi_n^{-1} \rightarrow \Phi^{-1}$ locally uniformly. Since $A(1, e^{2\pi t_n})$ converges in the Hausdorff topology to $\mathbb{T}$, the closed Herman rings converge to the quasicircle

$$H_* := \Phi^{-1}(\mathbb{T}),$$

and passage to the limit in the conjugacy gives $R_*|_{H_*} \sim R_\theta$.

Let c_n^0 and c_n^∞ be the two free critical points, counted with multiplicities $d_0 - 1$ and $d_\infty - 1$. The collision combinatorics says that their linearizing angular coordinates agree. Hence $\Phi_n(c_n^0)$ and $\Phi_n(c_n^\infty)$ lie on the same radial segment, one on each boundary circle of $A(1, e^{2\pi t_n})$. Their spherical distance tends to zero. Uniform convergence of the inverse quasiconformal maps then gives

$$c_n^0 - c_n^\infty \longrightarrow 0.$$

In the degree- d limit the two critical multiplicities therefore merge into one critical point of multiplicity $(d_0 - 1) + (d_\infty - 1) = d - 1$. Normalize this point to 1. The coefficient calculation preceding the construction then forces $R_* = F_{c_*}$ for some $c_* \neq 0$. We have proved existence of the required normalized rotational parameter conditional only on (AHR), not on Wang's realization theorem and not on a separate precompactness theorem.

We next prove that this normalized parameter is unique; no rigidity theorem is left as a black box. Suppose c_1, c_2 both realize the same marked $\mathcal{C}_*$ and rotation number, and write $F_i = F_{c_i}$ and H_i for their Herman quasicircles.

No invariant line field for the present three-critical-point maps. We first isolate the rigidity mechanism. Choose the annular collar A_κ around H_i bounded on the two sides by sufficiently deep Böttcher equipotentials in the basins of 0 and ∞ . For κ close enough to the limiting collar, its local nonescaping set

$$K_\kappa^{\text{loc}} := \{z : F_i^n(z) \in \overline{A_\kappa} \text{ for all } n \geq 0\}$$

is exactly $J(F_i)$: crossing either equipotential enters one of the two superattracting basins, whereas a point whose orbit stays in the collar has no normal neighborhood.

The butterfly bounds proved above give the small-scale critical-copy criterion on K_κ^{loc} . More explicitly, for every point outside the countable grand orbit of the critical point and for arbitrarily deep closest-return levels, the bounded-type partition supplies a definite-modulus critical butterfly. Follow the orbit of the point to the first bounded-hyperbolic-distance encounter with that butterfly and pull the butterfly back by the corresponding univalent approximate-rotation branch. Koebe distortion and the uniform butterfly modulus give one constant $C > 1$ and an integer $N \leq d$ and maps

$$h_n : U_n \longrightarrow V_n$$

for which

$$F_i^{a_n} \circ h_n = F_i^{b_n}, \quad 2 \leq \deg h_n \leq N,$$

$\text{diam } U_n, \text{diam } V_n \rightarrow 0$, both pointed domains at a critical point and its critical value have C -bounded shape, and

$$\text{dist}(x, U_n) \leq C \text{diam } U_n, \quad \text{dist}(x, V_n) \leq C \text{diam } V_n.$$

These are precisely the small-scale critical-copy properties used in the source; here they follow from the real/complex bounds and pullback geometry already established in the preceding bounded-type rigidity construction.

For completeness, the line-field criterion itself is short. If an invariant line field existed on a positive-area subset, choose a Lebesgue point x at which its unit Beltrami representative is almost constant. Rescale U_n and V_n about the marked critical point/-value to bounded-size pointed domains. Bounded shape and proper degree give, after a subsequence, a nonconstant proper holomorphic limit $h : U \rightarrow V$ of degree between 2 and N , still with a critical point. Relative closeness to x and almost continuity make the rescaled line fields converge in measure on both domains to constant line fields. The relation $F_i^{a_n} \circ h_n = F_i^{b_n}$ and invariance pass to the limit, so h pulls one constant line field to the other. After rotating the two affine normalizations, h preserves the horizontal line field. Hence $h'(z)/\overline{h'(z)}$ is constant almost everywhere. Since h' is holomorphic, the open mapping theorem forces h' to be constant; h is therefore affine, contradicting the surviving critical point. Thus $J(F_i)$ supports no invariant line field.

Pullback rigidity and normalization. Let $\phi_i : H_i \rightarrow \mathbb{T}$ be the marked quasimetric conjugacy with R_θ . On small Böttcher equipotential disks E_i^0, E_i^∞ use the normalized Böttcher coordinates, and on H_1 use $\phi_2^{-1} \circ \phi_1$. Quasiconformal interpolation in the two complementary annuli produces a sphere map q_0 which is conformal on $E_1^0 \cup E_1^\infty$ and conjugates F_1 to F_2 on the marked set. Because the combinatorics and all local degrees agree, q_0 lifts through the two branched coverings to q_1 with

$$F_2 \circ q_1 = q_0 \circ F_1,$$

and the lift has the same maximal dilatation. Repeating gives uniformly quasiconformal q_n , equal to q_0 on H_1 and conformal on progressively deeper preimages of the two Böttcher disks. Normal-family compactness gives a limit q satisfying $F_2 \circ q = q \circ F_1$.

There are no additional periodic Fatou components: an attracting or parabolic cycle would have to capture a critical orbit, while the only critical point outside the two fixed basins is 1 and its orbit is dense on the irrational rotation quasicircle; every boundary component of an additional rotation domain would lie in the postcritical accumulation set. Here the only nonbasin postcritical accumulation set is H_1 , so such a component would force H_1 itself to be a boundary component of a rotation domain, contrary to the defining Herman-curve condition. Sullivan's no-wandering theorem then implies that every Fatou component is eventually one of the two fixed basins. Consequently q is conformal on the Fatou set. Its Beltrami coefficient on $J(F_1)$ is F_1 -invariant, so the no-line-field argument above forces it to vanish almost everywhere. Weyl's lemma makes q Möbius. The marking fixes 0, ∞ , and the unique free critical point 1, hence $q = \text{id}$. Therefore $F_{c_1} = F_{c_2}$ and evaluation at 1 gives $c_1 = c_2$. The normalized rotational parameter is unique.

We may therefore use uniqueness, proved above, to derive the perturbations needed by the spectral argument from the already proved MSS theorem.

The free critical point is active at c_ .* Suppose instead that the marked critical point 1 were passive. The marked critical points 0 and ∞ are persistently superattracting fixed points, so they are passive as well. The Mañé-Sad-Sullivan theorem, Theorem 7.8, would then make the family J-stable on a parameter disk $D \ni c_*$. Its extended dynamical holomorphic motion gives, for every $c \in D$, a quasiconformal conjugacy h_c on a neighborhood of the Julia set. It sends the invariant Herman quasicircle H_* of F_{c_*} to an F_c -invariant quasicircle H_c and preserves the rotation number. A local topological

conjugacy preserves local degree. Since 1 is the unique finite critical point of local degree d in the family, $h_c(1) = 1$. Thus every $c \in D$ would be another parameter realizing the same marked rotational quasicircle data, contradicting uniqueness of c_* . Hence 1 is active.

Apply the active-critical capture part of Theorem 7.8. Arbitrarily close to c_* there are parameters c for which 1 is strictly preperiodic to a repelling cycle. At such a parameter the other two critical points, 0 and ∞ , are already fixed. Therefore every critical orbit is finite and F_c is postcritically finite. These maps cannot be rotational: on an irrational Herman quasicircle the orbit of the marked critical point is infinite, whereas our free critical orbit is finite.

Finally, the required map into the Corona phase space is finite and analytic, not an additional research input. Shrink the parameter disk so that the finite return itinerary used to cut out the reference pre-corona persists. The return branches depend holomorphically on c , and the holomorphic gluing construction of Lemma 16.56 then gives a finite analytic renormalization chart sending F_{c_*} to a rotational corona near f_* and the postcritically finite parameters above to nonrotational coronas arbitrarily close to it.

Assume that $D\mathcal{R}_{*,f_*}$ has no repelling eigenvalue. Lemma 16.59 gives no neutral eigenvalue either, so all nonzero spectrum is strictly inside the unit disk. The local stable manifold is then open in the Banach phase space. It would contain the nonrotational renormalized maps just constructed. But every point of the local stable manifold has a full forward orbit staying close to f_* , so Lemma 16.58 says it must be rotational. This contradiction proves existence of a repelling eigenvalue.

Discharge of the uniform a priori-bound input.

The only remaining existence input in the a repelling direction exists lemma was (AHR), the modulus-independent bounded-type Herman-ring a priori bound. The reconstruction earlier in this proof discharges that input. Thus the fixed-modulus collision realization, coefficient precompactness, degeneration to a Herman quasicircle, collision of the two marked boundary critical points, normalized uniqueness, free-critical activity, nearby PCF perturbations, the finite Corona chart, and the AHR estimate all follow from the arguments given here. $\square$

Lemma 16.61 (Local Corona hyperbolicity and stable-set identity). *The fixed corona f_* is hyperbolic with a finite nonzero unstable space, and its local stable manifold is exactly the nearby rotational corona class of fixed criticalities and rotation number θ .*

Proof. Because $\mathcal{R}_*$ is compact analytic, $D\mathcal{R}_{*,f_*}$ is a compact linear operator. Lemma 16.59 removes all spectrum from the unit circle, and Lemma 16.60 supplies nonempty spectrum outside it. Hence the Riesz spectral projections associated with the parts inside and outside the unit circle give

$$T_{f_*}\mathcal{B}_* = E^s \oplus E^u, \quad \dim E^u < \infty, \quad \dim E^u \geq 1,$$

with uniform contraction/expansion after an equivalent norm is chosen. The analytic stable-manifold theorem for Banach-space maps therefore gives local analytic stable and unstable manifolds.

The rotational contraction theorem shows that every nearby rotational corona of the prescribed class lies in $W_{\text{loc}}^s(f_*)$. Conversely, if $f \in W_{\text{loc}}^s(f_*)$, then its forward orbit stays in a sufficiently small neighborhood of f_* ; Lemma 16.58 forces f to be rotational with rotation number θ . Thus the stable manifold is exactly the local rotational class. This completes the local hyperbolicity part of the source theorem. Notice that nothing so far proves $\dim E^u = 1$; it proves only $1 \leq \dim E^u < \infty$. $\square$

Auxiliary definitions for this proof. A holomorphic map $g : A \rightarrow B$ is called σ -proper if there exist exhaustions $A_n \nearrow A$ and $B_n \nearrow B$ such that $g : A_n \rightarrow B_n$ is proper for every n ; equivalently, every component of the preimage of a compact set is compact. A *cascade* is a family $\{F^P\}_{P \in \mathbb{T}}$ of commuting σ -proper maps indexed by an additive ordered semigroup $\mathbb{T} \subset \mathbb{R}_{\geq 0}$ and satisfying $F^P \circ F^Q = F^{P+Q}$ whenever the composition is defined. These definitions are used only in this proof and are not needed in the stated Chapter 16 theorem until the parent theorem is closed.

Lemma 16.62 (Maximal σ -proper anti-renormalization extension). *Every point of the local unstable manifold has a backward Corona tower whose rescaled pre-corona branches extend maximally to commuting σ -proper branched coverings of simply connected plane domains onto $\mathbb{C}$.*

Proof. Since $\mathcal{R}_*|_{W_{\text{loc}}^u}$ is locally invertible and expanding, every $f \in W_{\text{loc}}^u$ has a unique backward orbit $\{f_k\}_{k \leq 0}$ after the neighborhood is fixed. The k th anti-renormalization contains a pre-corona $F_k = (f_{k,-}, f_{k,+})$ whose branches are iterates of f_{k-1} . Let ϕ_k be the inverse gluing change of variables and compose these changes down the backward tower. Periodicity of θ gives a positive integer anti-renormalization matrix M with Perron eigenvalue $t > 1$. Rescaling by the corresponding linear maps A_*^k puts all levels in one plane and gives pairs

$$F_k^\# = (f_{k,-}^\#, f_{k,+}^\#)$$

that satisfy a fixed substitution rule encoded by M .

At each finite stage the branches extend across every pullback domain that does not cross the common critical value. Take the union over all such continuation domains. Monodromy is single-valued because the two generators commute and the only branching occurs over the prescribed unique critical orbit. If a component of the preimage of a compact subset of $\mathbb{C}$ were noncompact in this maximal domain, one could choose a curve to its ideal boundary with bounded image; after transporting it to a sufficiently deep anti-renormalization level, the improvement-of-domain contraction would place the curve in a compactly contained finite pre-corona domain, contradicting maximality of the continuation there. Hence every compact-target preimage component is compact. Exhausting target disks and taking the corresponding preimage components gives the required σ -proper exhaustions. The continuation domains are simply connected because any essential loop would, after sufficiently deep rescaling, lie in a simply connected finite-level pre-corona branch and therefore be null-homotopic. Thus the maximal branches are σ -proper branched coverings onto $\mathbb{C}$. $\square$

Lemma 16.63 (Cascade and global unstable manifold setup). *The maximal anti-renormalization extensions form a σ -proper cascade indexed by the power semigroup, and the local unstable manifold extends to a global complex unstable manifold W^u of the same dimension on which renormalization acts biholomorphically.*

Proof. Let M be the anti-renormalization matrix and let $t > 1$ be its expanding eigenvalue. Two triples (n, a, b) and $(n-1, a', b')$ represent the same formal iterate when $(a', b') = (a, b)M$. Quotienting $\mathbb{Z} \times \mathbb{Z}_{\geq 0}^2$ by this relation gives a semigroup $\mathbb{T}$; addition is induced by adding the exponents at a common level. The translation model associated with the periodic rotation embeds $\mathbb{T}$ as a dense ordered subsemigroup of $\mathbb{R}_{\geq 0}$ and obeys the scaling law $P \mapsto tP$ under one period of anti-renormalization.

For a fixed cascade, represent $P \in \mathbb{T}$ by (n, a, b) and set

$$F^P = (f_{n,-}^\#)^a \circ (f_{n,+}^\#)^b.$$

The substitution relation encoded by M makes this independent of the chosen representative. Commutativity gives $F^P \circ F^Q = F^{P+Q}$, and Lemma 16.62 gives σ -properness. Holomorphic dependence of the finite anti-renormalization tower and normal-family uniqueness extend the family holomorphically over W_{loc}^u .

Finally set

$$W^u := \bigcup_{n \geq 0} \mathcal{R}_*^{-n}(W_{\text{loc}}^u),$$

where inverse branches are taken along the unstable manifold. The charts $\mathcal{R}_*^n : \mathcal{R}_*^{-n}(W_{\text{loc}}^u) \rightarrow W_{\text{loc}}^u$ agree on overlaps, so they give W^u a unique complex-manifold structure. In these charts $\mathcal{R}_*$ is biholomorphic and F_* is its unique expanding fixed point. Since every chart is biholomorphic to an open subset of W_{loc}^u , the complex dimension is constant and equals $\dim E^u$. This is the precise parameter space on which the dimension-one argument below operates. $\square$

Lemma 16.64 (Fixed-cascade external structure and finite-time motion). *For the fixed cascade, nested external wakes shrink to points, complete external itineraries separate escaping points, finite-time escaping points are dense in the external Julia structure and have empty interior, and every finite wake graph moves uniquely and holomorphically under nearby cascade perturbations until the marked critical value collides with it.*

Proof. Pull back finitely many fundamental external rays through all branch words of bounded total time. Sigma-properness makes each finite truncation locally finite. The Markov property makes wakes nested or disjoint. If a nested wake chain failed to shrink, anti-renormalize to the first macroscopic scale and take a normal-family limit; one would obtain two distinct points with the same complete inverse-branch itinerary, contradicting univalence of the limiting inverse branch. Thus itineraries separate points. Pulling a wake farther back produces both deeper finite-time escaping points and deeper nonescaping subwakes, proving density and empty interior. Nearby noncritical vertices continue by the implicit-function theorem, and the rays follow by pullback, giving the finite-time holomorphic motion. $\square$

Lemma 16.65 (Finite-time escaping rigidity). *On any nearby parameter component for which the marked critical value avoids the finite-time escaping graph $I_{\leq P}$, that graph has a unique equivariant holomorphic motion and the motion is conformal on $I_{\leq P}$. The fixed finite escaping structure supports no invariant line field.*

Proof. An invariant Beltrami differential integrates to a normalized quasiconformal deformation whose conjugated cascade branches are holomorphic. The induced Corona germ has the same rotational marking, so it lies simultaneously in the local stable and unstable manifolds; hence the deformation is constant and the differential vanishes. For a moving finite graph, anti-renormalize deeply. Backward convergence in the unstable manifold places the motion arbitrarily close to the fixed cascade in a uniform geometric window, so its normalized dilatation tends to one. Holomorphic pullback cannot increase the Beltrami norm; therefore the original motion is conformal. The same argument applied to the quotient of two motions gives uniqueness. $\square$

Lemma 16.66 (Infinite-time escaping rigidity). *The infinite-time escaping set is totally disconnected, carries a unique equivariant conformal holomorphic motion whenever the critical value avoids it, and supports no invariant line field.*

Proof. Truncate the external wakes beyond a large radius and use the cascade branches as a Markov map. The nonescaping set of this Markov system is an intersection of nested unions of pairwise disjoint wakes; wake shrinking from Lemma 16.64 makes it totally disconnected. Pullback gives a unique holomorphic motion. Anti-renormalization transports the problem to a fixed-size neighborhood of the fixed cascade, where the parameter tends backward to the fixed point; hence the dilatation tends to one and the motion is conformal. A hypothetical invariant line field, normalized along deeper and deeper wakes, would produce a nonzero invariant line field on the fixed finite escaping structure, contradicting Lemma 16.65. $\square$

Lemma 16.67 (Cascade Julia/Fatou rigidity package). *The Julia set is the closure of the finite-time escaping set, repelling periodic points are dense in it, almost every Julia orbit accumulates on the postcritical set or enters the external escaping structure, and cascades have no wandering Fatou components. In particular, for a hyperbolic cascade, $J(F) \setminus I(F)$ has planar measure zero.*

Proof. Montel's theorem identifies the complement of the closure of the finite escaping set with the Fatou set, while pullbacks of an external return disk produce dense repelling periodic points. On the complement of the postcritical and external regions, inverse branches uniformly contract the relevant hyperbolic metric; Koebe distortion and a Lebesgue-density argument then exclude a positive-measure set whose orbits avoid both regions forever. A wandering Fatou component would generate infinitely many disjoint forward images and hence an infinite-dimensional space of invariant Beltrami differentials. Integrating them gives deformations inside the finite-dimensional unstable manifold, so a nonzero kernel direction would produce an invariant line field on the Julia/escaping structure, contradicting Lemmas 16.65–16.66. For a hyperbolic

cascade the postcritical set lies in the attracting basin, so the almost-everywhere alternative leaves only escape on the Julia set up to measure zero. $\square$

Lemma 16.68 (Hyperbolic cascade rigidity and Fatou basin classification). *For a hyperbolic cascade, the Julia set has a unique equivariant conformal holomorphic motion under nearby hyperbolic deformations, and the Fatou set is exactly the basin of the unique attracting cycle.*

Proof. The finite- and infinite-time conformal motions agree on their overlap and cover the Julia set up to a null set. The lambda lemma extends the motion quasiconformally, and its Beltrami coefficient vanishes almost everywhere on the Julia set, hence the motion is conformal there; uniqueness follows from the no-line-field result. If another Fatou component existed, hyperbolic-metric expansion away from the postcritical set plus Koebe distortion would force iterates of a compact disk to acquire arbitrarily large spherical diameter while remaining in a proper Fatou component, a contradiction. $\square$

Lemma 16.69 (Density of superattracting cascades at the fixed cascade). *Every neighborhood of the fixed cascade in the local unstable manifold contains a cascade satisfying $F^P(0) = 0$ for some $P > 0$.*

Proof. If a neighborhood had no critical relation, the postcritical set would move holomorphically throughout that neighborhood. The induced Corona germs would all carry the same rotational marking and hence, by the stable-set theorem, lie in the stable manifold, impossible inside a nontrivial unstable neighborhood. Thus arbitrarily nearby one has $F^{P+Q}(0) = F^Q(0)$. If $Q = 0$ we are done. Otherwise perturb so the resulting periodic point is repelling, continue it holomorphically, and use deep preimages accumulating at the critical point. Rouché's theorem then gives a nearby collision of such a preimage with the critical point, hence a superattracting critical relation. $\square$

Lemma 16.70 (Zero-dimensional superattracting relation loci). *For every $P > 0$, the analytic set*

$$S_P = \{F \in W^u : F^P(0) = 0\}$$

is zero-dimensional.

Proof. Suppose a nonconstant holomorphic disk lay in S_P . Normalize Böttcher coordinates in the immediate superattracting basins and combine them with the unique conformal Julia motion from Lemma 16.68. A quasiconformal interpolation across one fundamental annulus lifts repeatedly under F^P to a global conjugacy. Every Fatou component belongs to the attracting basin, and the Julia motion is conformal almost everywhere, so the global conjugacy is conformal almost everywhere and hence conformal everywhere by Weyl's lemma. The cascade normalization forces this affine conjugacy to be the identity. Therefore the disk is constant, a contradiction. $\square$

Lemma 16.71 (One-dimensional unstable manifold and codimension-one stable class). *The global unstable manifold and the local unstable spectral space are one-complex-dimensional. Hence the rotational local stable manifold has complex codimension one.*

Proof. Lemma 16.69 gives a nonempty critical-relation locus S_P arbitrarily near the fixed cascade. Since $F \mapsto F^P(0)$ is a nonzero holomorphic function, its zero set is an analytic hypersurface (or at a singular point has local dimension at least $\dim W^u - 1$). Lemma 16.70 says the same set is zero-dimensional. Therefore $\dim_{\mathbb{C}} W^u \leq 1$. The repelling-direction lemma already gives $\dim W^u \geq 1$, hence $\dim W^u = 1$. The global-cascade charts identify this with the local unstable spectral dimension, and the hyperbolic splitting then makes the rotational stable manifold codimension one. $\square$

Proof. The local compact-operator layer and the sigma-proper cascade construction were established above. Lemmas 16.64–16.68 reconstruct the source’s Sections 9–12 escaping and Fatou–Julia rigidity chain. Lemmas 16.69–16.71 then use the actual Lim dimension argument: superattracting cascades accumulate at the fixed cascade, each critical-relation locus is zero-dimensional, while a nontrivial critical relation is an analytic hypersurface. Thus the unstable direction is exactly one complex dimension and the rotational stable class has codimension one.

Completion of the proof. The preceding lemmas establish bounded-type critical-quasicircle rigidity and universality, the perturbative PCF step, normalized rotational-parameter uniqueness, the fixed-modulus collision Herman ring, degeneration compactness, waves, spreading, width trading, and the deep localization/horizontal-loss package through the exact-denominator and bounded-degree push-forward steps. These ingredients complete the stated Corona-hyperbolicity theorem. $\square$

Remark 16.12 (Arithmetic hierarchy). Periodic type is bounded type. The bounded-type convergence and global rigidity theorem above use this stronger arithmetic control, while the Khmelev–Yampolsky local theorem and the full horseshoe statement are deliberately arbitrary-irrational. Hypotheses must therefore be read theorem by theorem rather than as interchangeable labels.

Primary literature used for the proof structure

1. E. de Faria and W. de Melo, *Rigidity of critical circle mappings I*, J. Eur. Math. Soc. 1 (1999), 339–392.
2. E. de Faria and W. de Melo, *Rigidity of critical circle mappings II*, J. Amer. Math. Soc. 13 (2000), 343–370.
3. M. Yampolsky, *Hyperbolicity of renormalization of critical circle maps*, Publ. Math. IHES 96 (2002/2003), 1–41.
4. D. Khmelev and M. Yampolsky, *The rigidity problem for analytic critical circle maps*, Moscow Math. J. 6 (2006), 317–351.

5. C. McMullen, *Renormalization and 3-Manifolds Which Fiber over the Circle*, Annals of Mathematics Studies 142, Princeton Univ. Press, 1996; especially Chapter 9 and Appendix B.
6. A. Avila and M. Lyubich, *Lebesgue measure of Feigenbaum Julia sets*, Ann. of Math. 195 (2022), Section 3: quasicritical-circle real/complex bounds and quasiconformal pullback rigidity.
7. W. R. Lim, *Rigidity of J -rotational rational maps and critical quasicircle maps*, Trans. Amer. Math. Soc. 379 (2026), 2507–2567.
8. W. R. Lim, *Hyperbolicity of Renormalization of Critical Quasicircle Maps*, Comm. Math. Phys. 407 (2026), article 88; especially Sections 9–13 and Theorem A.

Part IV

Neutral Dynamics, Typicality, and the Mandelbrot Set

Guide to Part IV: neutral dynamics and the edge of hyperbolicity

Hyperbolic fixed points are forgiving: contraction or expansion supplies the geometric series needed to solve the conjugacy equation. Neutral fixed points remove that geometric advantage. The missing expansion is replaced by arithmetic, asymptotic translation geometry, and eventually a new form of renormalization.

For an irrational multiplier $e^{2\pi i\alpha}$, formal linearization produces divisors $e^{2\pi i n\alpha} - e^{2\pi i\alpha}$. The question is no longer whether a formal coefficient can be solved for, but whether the resulting coefficients remain small enough for the series to converge. The Brjuno sum measures the cumulative cost of rational approximation, while Yoccoz's theorem identifies the sharp boundary for the quadratic family. Cremer dynamics shows what failure of linearization looks like.

At a parabolic point the correct normal form is translational rather than rotational. Attracting and repelling petals support Fatou coordinates in which the iterate is $w \mapsto w+1$. Comparing the incoming and outgoing coordinates produces horn maps and the Ecalle-Voronin modulus. This is a local classification with an infinite-dimensional analytic invariant: a striking contrast with the simple multiplier classification of Chapter 1.

Near-parabolic renormalization then connects the irrational and parabolic worlds. Long transits through a nearly parabolic region are rescaled into new maps. Ecalle phase becomes a genuine parameter of subsequential limits, so one must distinguish precompactness from phase-independent convergence. Pacman renormalization introduces a marked cut that makes neutral combinatorics compatible with a Banach-space renormalization operator.

Chapter 20 changes viewpoint from specially tuned parameters to measure-theoretically typical ones. Parameter exclusion, phase-parameter transfer, absolutely continuous invariant measures, Collet-Eckmann expansion, and statistical laws all answer versions of the question: what behavior occupies a large set of real parameters?

Chapter 21 returns to the Mandelbrot set. The global MLC conjecture remains open in the scope of this book. Instead, the chapter studies regimes where shrinking geometry can actually be proved: Yoccoz parameters, selected satellite types, and bounded-type Feigenbaum parameters. The logical lesson is as important as the theorem: local connectivity is a conclusion of quantitative shrinking, not a visual property inferred from a parameter-plane image.

The mathematical route through this Part

irrational neutral
small divisors

$\longleftrightarrow$

parabolic
Fatou coordinates

↓
Brjuno-Yoccoz

$\longleftrightarrow$

↓
Ecalfe phase

near-parabolic renormalization

Alongside this local-neutral spine, Chapters 20–21 show how quantitative control is converted into almost-everywhere statements and parameter-space local connectivity.

Where the literature enters

Several results in this Part are modern and technically enormous. The textbook therefore makes a stronger distinction than “proved” versus “cited”. When a deep source theorem is imported, the exact theorem-sized engine is named; the mathematical consequences needed here are then derived explicitly. This makes it possible to learn the structure of the argument without pretending that hundreds of pages of renormalization technology have been compressed into a paragraph.

Siegel, Brjuno, Yoccoz and Cremer Dynamics

Why this chapter matters

An irrationally indifferent fixed point has multiplier of modulus one and infinite order. Formally one would like to conjugate the germ to a rigid rotation. The coefficient equations can always be written down; the problem is whether the accumulating small divisors allow the formal series to converge.

This chapter follows the sharp arithmetic story. Siegel's Diophantine hypothesis gives a first convergence theorem. Brjuno's condition is the optimal universal sufficiency condition, and Yoccoz shows sharpness through the quadratic family. Bounded-type quadratic Siegel disks then connect the local theory back to critical circle maps, while Pérez-Marco hedgehogs describe the compact invariant geometry that survives when linearization fails.

Learning goals

By the end of this chapter you should be able to:

1. write the homological equations for a linearizing series;
2. explain how Diophantine estimates absorb small divisors;
3. state the Brjuno sum and its role in convergence;
4. separate Brjuno sufficiency from Yoccoz necessity;
5. explain why a bounded-type quadratic Siegel boundary contains the critical point;
6. describe what a hedgehog records in the nonlinearizable case.

Conceptual roadmap

small-divisor equations $\rightarrow$ Diophantine majorants
 $\rightarrow$ Brjuno summability
 $\rightarrow$ Yoccoz sharpness.

A second branch uses Chapter 16 critical-circle rigidity and quasiconformal surgery to understand bounded-type Siegel boundaries; the nonlinearizable branch ends with hedgehogs rather than Siegel disks.

Common pitfall

The statements “the formal conjugacy exists” and “the analytic conjugacy converges” are different. Also, Brjuno sufficiency and the quadratic non-Brjuno obstruction are logically separate theorems; combining them too early hides the sharpness argument.

17.1 Definitions

Definition 17.1 (Siegel disk). A Siegel disk is the maximal connected neighborhood of an irrational indifferent periodic point on which an iterate is analytically conjugate to an irrational rigid rotation.

Definition 17.2 (Brjuno number). A Brjuno number is an irrational rotation number whose continued-fraction denominators q_n satisfy $\sum \log q_{n+1}/q_n < \infty$.

Definition 17.3 (Cremer point). A Cremer point is an irrational indifferent periodic point that is not analytically linearizable.

Definition 17.4 (hedgehog). A hedgehog is a compact connected full invariant set attached to a nonlinearizable irrational indifferent germ in Pérez-Marco theory, meeting the boundary of a chosen admissible neighborhood and carrying the local recurrent dynamics.

17.2 Main results

Theorem 17.1 (Critical point on Siegel boundary: scoped results). **Statement scope.** *For the quadratic polynomial $P_\alpha(z) = e^{2\pi i \alpha} z + z^2$ with α of bounded type, the boundary of the Siegel disk contains the unique finite critical point of P_α . No claim is made here for arbitrary Brjuno rotation numbers without an additional theorem.*

Proof strategy (not part of the proof)

Proof status. proved here modulo stated analytic input.

Strategic outline only. Model the bounded-type rotation by a critical circle map/Blaschke product; use Chapter 16 real bounds and rigidity to obtain a quasymmetric boundary conjugacy; extend quasiconformally and straighten the model to the quadratic polynomial; the model critical point lies on the invariant circle, hence the quadratic critical point lies on the Siegel boundary.

The mathematical argument begins in the proof environment below.

Proof. Let α be of bounded type and $P_\alpha(z) = e^{2\pi i \alpha} z + z^2$. Theorem 17.4 gives a Siegel disk Δ_α at the origin.

Critical circle model.

There is a symmetric Blaschke-type model B_α preserving the unit circle such that $B_\alpha|_{\mathbb{S}^1}$ is an analytic critical circle homeomorphism of rotation number α , with its unique circle critical point at the marked point 1.

Choose the standard degree-three Blaschke normalization and adjust its phase so that the rotation number is α . The criticality equation has the prescribed double critical point on $\mathbb{S}^1$; symmetry by reflection in the circle gives the inner/outer dynamical pieces used in surgery.

Quasisymmetric boundary conjugacy.

The circle map $B_\alpha|_{\mathbb{S}^1}$ is quasisymmetrically conjugate to the rigid rotation R_α .

Chapter 16 real a priori bounds give uniform adjacent-atom comparability for the closest-return partitions. Bounded type turns these partitions into a bounded-geometry fine grid. Sending corresponding grid atoms to those of the rigid rotation gives a homeomorphism with uniformly bounded adjacent ratio distortion; the fine-grid criterion makes it quasisymmetric. Chapter 16 rigidity identifies the conjugacy with the dynamically normalized one.

Quasiconformal surgery.

The quasisymmetric circle conjugacy extends to a quasiconformal map of the disk. Replacing the inside dynamics by R_α and pulling back its standard complex structure produces a globally invariant Beltrami form whose straightening is a quadratic polynomial affinely conjugate to P_α .

Use any Beurling–Ahlfors/Douady–Earle quasiconformal extension inside the circle and keep the original complex structure outside. Pull the interior Beltrami coefficient by the model dynamics. The pieces are disjoint along iterates of the invariant circle, so the coefficient is measurable and uniformly bounded. The measurable Riemann mapping theorem straightens it. The resulting rational map has one finite critical point and a superattracting fixed point at infinity, hence is a quadratic polynomial. Its finite fixed multiplier is $e^{2\pi i\alpha}$; the standard normalization makes it P_α .

Boundary is a quasicircle.

The image of $\mathbb{S}^1$ under the straightening is the boundary of the Siegel disk of P_α and is a quasicircle.

Inside the model circle the dynamics is rigid rotation after surgery, so its image is an invariant simply connected linearization domain. Maximality makes it the Siegel disk. A quasiconformal image of a circle is a quasicircle.

Critical point lies on the Siegel boundary.

The unique finite critical point of P_α lies on $\partial\Delta_\alpha$.

The model critical point is the marked point $1 \in \mathbb{S}^1$. Quasiconformal straightening preserves local degree, so its image is the unique finite critical point of the quadratic polynomial. By Step 4 it also lies on the image of $\mathbb{S}^1 = \partial\Delta_\alpha$. This proves the scoped theorem. $\square$

Theorem 17.2 (Hedgehogs for nonlinearizable indifferent germs). **Statement scope.** *Let $f(z) = e^{2\pi i\alpha}z + O(z^2)$ with $\alpha \notin \mathbb{Q}$, and assume that f is not analytically linearizable. Let $U \ni 0$ be a Jordan domain such that f and the inverse branch fixing 0 are defined and univalent on a neighborhood of $\bar{U}$. Then there exists a compact connected full set $K \subset \bar{U}$ such that $0 \in K$, $K \cap \partial U \neq \emptyset$, and $f(K) = K = f^{-1}(K)$. Such a compactum is a nonlinearizable Pérez-Marco hedgehog and supports nontrivial two-sided local dynamics.*

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Prove the neutral invariant-compactum theorem first for rational multipliers by finite-order maximal invariant domains or a semilocal Leau-Fatou flower; approximate an arbitrary neutral multiplier by rational ones; pass invariance through a Hausdorff limit and fill bounded complementary components; then specialize to the nonlinearizable irrational case.

The mathematical argument begins in the proof environment below.

Proof. We prove a slightly stronger statement than the theorem requires. Let

$$f(z) = \lambda z + O(z^2), \quad |\lambda| = 1,$$

and let $U \ni 0$ be a Jordan domain such that both f and the inverse branch f^{-1} fixing 0 are defined and univalent on a neighborhood of $\bar{U}$. We shall construct a compact connected full set $K \subset \bar{U}$ with

$$0 \in K, \quad K \cap \partial U \neq \emptyset, \quad f(K) = K = f^{-1}(K). \quad (17.6.1)$$

The stated hedgehog theorem is the irrational nonlinearizable specialization.

Step 1 (rational multiplier perturbations remain admissible). There are rational numbers $\alpha_n = p_n/q_n$ tending to the rotation number α of f and holomorphic diffeomorphisms

$$f_n = R_{\alpha_n - \alpha} \circ f, \quad R_t(z) = e^{2\pi i t} z,$$

such that $f_n \rightarrow f$ and $f_n^{-1} \rightarrow f^{-1}$ uniformly on a fixed neighborhood of $\bar{U}$, while every f_n and f_n^{-1} is univalent there. Choose open sets

$$\bar{U} \subset W_0 \Subset W_1$$

on which the two given local maps are defined. For $|t|$ sufficiently small, $R_{-t}(W_0) \Subset W_1$; hence $f_t := R_t \circ f$ is univalent on W_0 and its inverse branch $f_t^{-1} = f^{-1} \circ R_{-t}$ is defined and univalent on W_0 . Both maps converge uniformly to the unperturbed maps as $t \rightarrow 0$. Density of $\mathbb{Q}$ supplies the required $t_n = \alpha_n - \alpha$.

Step 2 (filling preserves local total invariance). Let $A \subset \bar{U}$ be compact and suppose $f(A) = A = f^{-1}(A)$. If $\hat{A}$ denotes the filled compactum obtained by adjoining every bounded component of $\mathbb{C} \setminus A$, then $\hat{A} \subset \bar{U}$ and $f(\hat{A}) = \hat{A} = f^{-1}(\hat{A})$. Because U is Jordan, $\mathbb{C} \setminus \bar{U}$ is connected and unbounded. Every point outside $\bar{U}$ therefore belongs to the unbounded component of $\mathbb{C} \setminus A$, so $\hat{A} \subset \bar{U}$. Let H be a bounded complementary component of A . Since $f^{-1}(A) = A$, $f(H)$ is disjoint from A ; it is open, connected and bounded because $\bar{H} \subset \bar{U}$ and f is continuous on a neighborhood of $\bar{U}$. Thus $f(H)$ lies in a bounded complementary component of A . It follows that $f(\hat{A}) \subset \hat{A}$. Applying the same argument to f^{-1} gives the reverse inclusion and proves the lemma.

We next prove (17.6.1) when the multiplier is a root of unity. Write $\lambda = e^{2\pi i p/q}$ with $(p, q) = 1$.

Step 3 (finite-order rational case). If $f^q = \text{id}$ as a germ at 0, there is a compact

connected full $K \subset \bar{U}$ satisfying (17.6.1). By analytic continuation, $f^q = \text{id}$ wherever the q successive iterates are defined in the component containing 0. Let V be the component containing 0 of

$$\{z \in U : f^j(z) \in U \text{ for } 0 \leq j < q\}.$$

Then $f(V) = V = f^{-1}(V)$. The domain V is simply connected. Indeed, if a bounded complementary component H of V were contained in U , then for each j the holomorphic function obtained by mapping U conformally to the unit disk and composing with f^j has boundary modulus at most one on H ; the maximum principle would force all $f^j(H)$ to remain in U , contradicting that H was excluded from V .

Every point $z \in \partial V$ has some iterate $f^j(z) \in \partial U$; otherwise all q iterates would remain a positive distance from ∂U and the defining condition of V would extend across z . Since f^j permutes V , it also permutes ∂V , and hence $\partial V \cap \partial U \neq \emptyset$. Set $A = \bar{V}$ and fill it. The filling lemma above gives a full compact connected K satisfying (17.6.1).

Assume from now on that the rational germ is not of finite order. Then

$$F := f^q(z) = z + az^{m+1} + O(z^{m+2}) \quad (17.6.2)$$

for some $m \geq 1$ and $a \neq 0$.

Step 4 (local parabolic bouquets and a monotone flower family). For the germ (17.6.2) there are m attracting and m repelling petals alternating around 0. Their boundary accesses can be chosen analytic away from 0, invariant as germs under F , and varying in a continuous strictly nested one-parameter family. Transport by f arranges them into f -cyclic bouquets compatible with the rotation permutation p/q . Put

$$w = -\frac{1}{maz^m}.$$

On each sufficiently narrow sector centered on a direction for which $-maz^m$ is positive real, (17.6.2) gives

$$w(F(z)) = w + 1 + O(|w|^{-1/m}). \quad (17.6.3)$$

For $\Re w$ large, right half-planes with a fixed angular margin are forward invariant. Pulling them back gives the m attracting petals. Applying the same argument to F^{-1} gives the m repelling petals, and the sectors alternate because the attracting and repelling directions are the alternating solutions of $az^m \in \mathbb{R}_{<0}$ and $az^m \in \mathbb{R}_{>0}$.

For completeness, one obtains a genuine translation coordinate on a slightly smaller petal by correcting w : the defect in (17.6.3) has an asymptotic expansion in negative powers of $w^{1/m}$; subtract finitely many terms and then sum the remaining defect along the forward orbit. The resulting Fatou coordinate Φ satisfies $\Phi \circ F = \Phi + 1$. Curves $\Im \Phi = c$ are analytic invariant accesses. Varying c gives a continuous nested family, and doing this in all adjacent attracting/repelling petals gives local bouquets. Since f commutes with $F = f^q$ and its derivative acts by the p/q rotation on tangent directions, transporting one bouquet by $f, f^2, \dots, f^{q-1}$ gives the required cyclic family.

The next two elementary lemmas are the semilocal continuation mechanism.

Step 5 (rational analytic circle maps carry annular flowers). Let g be an an-

analytic orientation-preserving circle diffeomorphism with rotation number p/q . In a complex annulus around $\mathbb{S}^1$ there is a continuous strictly nested family of g -invariant annular neighborhoods whose components between consecutive fixed points of g^q are g^q -invariant Jordan domains bounded by analytic translation arcs. If g^q is the identity near the circle, a thin invariant annulus supplies the same enlargement conclusion directly. Suppose first that g^q is not the identity. Its fixed set on the circle is finite. Each component $I = (a, b)$ of $\mathbb{S}^1 \setminus \text{Fix}(g^q)$ is wandering monotonically toward one endpoint in forward time and toward the other in backward time. In a local analytic coordinate straightening I to $(0, 1)$, choose a small transversal J to a fundamental interval and join its endpoints to those of $g^q(J)$ by short symmetric arcs. The union of the iterates $\bigcup_{n \in \mathbb{Z}} g^{qn}(J)$ is an analytic translation arc landing at a and b . It and its reflected mate bound a g^q -invariant Jordan domain containing I . Choosing the initial transversal in a nested continuous family gives nested domains. Transporting them by g over all cycles of components yields the annular flower. If $g^q = \text{id}$ near the circle, small roundish collars in an analytic conjugacy coordinate are invariant and play the same role.

Step 6 (external flower arcs pull back to genuine crosscuts). Let L be a full compact connected set, relatively compact in U , with $f(L) = L = f^{-1}(L)$. Let $h : \widehat{\mathbb{C}} \setminus \overline{\mathbb{D}} \rightarrow \widehat{\mathbb{C}} \setminus L$ be the normalized exterior Riemann map and let $g = h^{-1} \circ f \circ h$ be the induced analytic circle map. An analytic translation arc in a slightly larger annular flower for g pulls back under h to a crosscut of $\mathbb{C} \setminus L$ whose two ends land at fixed points of the corresponding iterate of f . Let D be one of the simply connected g^q -invariant Jordan domains in a slightly larger annular flower and put $\Omega = h(D \setminus \overline{\mathbb{D}})$. Then Ω is simply connected, relatively compact in the domain where f^q is defined, and $f^q(\Omega) = \Omega$. For $z \in \Omega$, the forward orbit has a boundary accumulation point because it stays in a bounded domain. The family $\{f^{qn}|_{\Omega}\}$ is normal. A nonconstant subsequential limit would be a conformal self-map of Ω by Hurwitz; but the chosen component of the annular flower is wandering from one boundary fixed point toward the other, so an interior orbit accumulates on the boundary. Hence every subsequential limit is constant. Passing to the limit in $f^q \circ f^{qn} = f^{qn} \circ f^q$ shows that each constant limit is a fixed point of f^q . There are only finitely many relevant boundary fixed points, and monotonicity of the model interval forces the forward limit to be unique. The same argument for f^{-q} gives the backward endpoint. Therefore the two ends of the pulled-back translation arc land, so it is a genuine crosscut.

Step 7 (maximal semilocal flower reaches the prescribed boundary). For a non-finite-order rational germ, at least one petal in a cyclic Leau–Fatou flower admits an extension in U whose closure meets ∂U . Start from the nested local bouquet family constructed above and continue its analytic boundary accesses as long as all members remain in U . Order the extensions by inclusion and let s_0 be the supremum of the parameter values obtained. Hausdorff compactness on compact subsets and normality of the analytic arcs give a limiting flower at s_0 .

Assume for contradiction that this limiting flower is relatively compact in U . Let L be the fill of the closure of the union of all earlier flowers. Then $L \Subset U$ is compact, connected, full and totally invariant. Uniformize its exterior by h as in the exterior-crosscut step above. Total invariance and Schwarz reflection give an analytic circle diffeomorphism $g = h^{-1} \circ f \circ h$. The q accesses inherited from one cyclic petal family

form a q -periodic orbit in the prime-end circle, permuted by f with the same cyclic order as multiplication by $e^{2\pi ip/q}$; consequently $\rho(g) = p/q$.

Apply the annular-circle-map construction above to obtain an annular flower strictly larger than the unit circle. By the crosscut-landing argument above, its analytic boundary arcs pull back to genuine crosscuts landing at fixed points of f^q . Adjoining the corresponding pulled-back Jordan domains to the terminal petals gives a strictly larger semilocal flower still compactly contained in U . This contradicts the definition of s_0 . Hence the maximal flower cannot be relatively compact, and one of its petals has closure meeting ∂U .

Step 8 (rational invariant compactum). Every admissible germ with rational rotation number has a compact connected full set satisfying (17.6.1). The finite-order case was handled above. Otherwise take a q -cycle $P, f(P), \dots, f^{q-1}(P)$ furnished by the maximal semilocal continuation above, with one closure meeting ∂U , and put

$$A = \{0\} \cup \bigcup_{j=0}^{q-1} f^j(\overline{P}).$$

The set A is compact and connected because all petal closures contain 0. It is invariant in both directions, and it meets ∂U . Filling A and using the filling lemma above produces the required K .

We now pass from rational multipliers to the original neutral germ.

Step 9 (Hausdorff compactness retains connectivity and boundary contact).

From any sequence of rational compacta $K_n \subset \overline{U}$ supplied by the rational compacta above one can extract a subsequence converging in Hausdorff distance to a compact connected set $L \subset \overline{U}$ with $0 \in L$ and $L \cap \partial U \neq \emptyset$. The hyperspace of nonempty compact subsets of the compact metric space $\overline{U}$ is compact in the Hausdorff metric. The subspace of continua is closed: an ε -chain joining two limit points is obtained by approximating them in a nearby connected K_n . Membership of 0 is closed. For boundary contact choose $x_n \in K_n \cap \partial U$; compactness of ∂U gives a convergent subsequence $x_n \rightarrow x \in \partial U$, and Hausdorff convergence forces $x \in L$.

Step 10 (two-sided invariance survives the rational limit). For the Hausdorff-convergent subsequence just extracted, if $f_n \rightarrow f$ and $f_n^{-1} \rightarrow f^{-1}$ uniformly near $\overline{U}$ and $f_n(K_n) = K_n = f_n^{-1}(K_n)$, then $f(L) = L = f^{-1}(L)$. Let ω be a modulus of continuity for f on the fixed compact neighborhood. Then

$$d_H(f_n(K_n), f(L)) \leq \|f_n - f\|_\infty + \omega(d_H(K_n, L)) \rightarrow 0.$$

Since $f_n(K_n) = K_n \rightarrow L$, uniqueness of the Hausdorff limit gives $f(L) = L$. The same estimate for the inverse branches gives $f^{-1}(L) = L$.

Step 11 (full limit and hedgehog identification). The fill $K = \widehat{L}$ is a nontrivial compact connected full set satisfying (17.6.1). When the rotation number is irrational and the germ is nonlinearizable, this K is a Pérez-Marco hedgehog. Apply the filling lemma above to L . Filling preserves compactness and connectedness, makes the set full, leaves the boundary-contact point in place, and preserves both forward and back-

ward invariance. Since $0 \in U$ while $K \cap \partial U \neq \emptyset$, $K \neq \{0\}$. Thus (17.6.1) is proved for every neutral germ satisfying the local univalence hypotheses.

Finally assume $\lambda = e^{2\pi i\alpha}$ with $\alpha \notin \mathbb{Q}$ and that f is not analytically linearizable. The compactum just constructed is a Siegel compactum in the sense of the existence theorem above. Pérez-Marco's hedgehog definition is precisely the irrational non-linearizable case of such an invariant compactum (and also allows a linearizable germ whose chosen neighborhood extends beyond a smaller linearization domain). Thus K is a nonlinearizable hedgehog. Its total invariance supplies a nontrivial two-sided local dynamical system on K , which is exactly the scope claimed in the theorem.

Completion of the compactum construction. The proof above contains the rational finite-order case, the semilocal parabolic-flower continuation, the annular external flower and crosscut landing mechanism, rational multiplier density, Hausdorff closure, and the filling/invariance step. The 1997 paper provides historical provenance and calibration for this construction. $\square$

Theorem 17.3 (Siegel linearization theorem). **Statement scope.** *If an irrational indifferent multiplier satisfies a sufficiently strong Diophantine/Siegel arithmetic condition, a holomorphic germ is analytically linearizable near the fixed point.*

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Solve the conjugacy equation coefficient by coefficient; convert the Diophantine lower bound on the homological denominators into an exponential coefficient majorant; prove convergence and then local injectivity of the normalized conjugacy.

The mathematical argument begins in the proof environment below.

Proof. Write $\lambda = e^{2\pi i\alpha}$ and

$$f(z) = \lambda z + \sum_{m \geq 2} a_m z^m, \quad h(z) = z + \sum_{m \geq 2} b_m z^m.$$

We seek $f \circ h = h \circ (\lambda \text{id})$.

Homological recursion.

For every $n \geq 2$ the coefficient b_n is uniquely determined by the previous coefficients and satisfies

$$(\lambda^n - \lambda)b_n = P_n(a_2, \dots, a_n; b_2, \dots, b_{n-1}),$$

where P_n is the coefficient of degree n in $\sum_{m \geq 2} a_m h(z)^m$.

Comparing coefficients in the conjugacy equation gives the formula. Since α is irrational, $\lambda^{n-1} \neq 1$, hence the denominator never vanishes. Thus there is a unique formal normalized conjugacy.

Assume the Siegel arithmetic hypothesis in the form

$$|\lambda^q - 1| \geq \gamma q^{-\tau} \quad (q \geq 1) \tag{S}$$

for some $\gamma > 0$ and $\tau \geq 0$. Then $|\lambda^n - \lambda| \geq \gamma(n-1)^{-\tau}$.

Analytic composition majorant.

If $|a_m| \leq AR^{-m}$ and $|b_j| \leq M^{j-1}$ for $2 \leq j < n$, then

$$|P_n| \leq C_0^n M^{n-2}$$

with C_0 depending only on A, R after M is chosen above a fixed threshold.

Choose $r < R$ and use the convergent positive-coefficient majorant $A_*(w) = \sum_{m \geq 2} |a_m| w^m$. Under the inductive bound, $|h(z)| \leq |z|/(1 - M|z|)$ for $|z| < 1/M$. Hence $A_*(|h(z)|)$ is analytic on a disk of radius c/M , where $c > 0$ depends only on A, R . Cauchy's estimate for its degree- n coefficient gives the asserted bound after increasing C_0 once.

Polynomial small divisors are absorbed exponentially.

There is $M < \infty$ such that the recursion of Step 1 satisfies $|b_n| \leq M^{n-1}$ for every $n \geq 2$.

Indeed, Step 1, (S), and Step 2 give

$$|b_n| \leq \gamma^{-1}(n-1)^\tau C_0^n M^{n-2}.$$

Choose M large enough that $\gamma^{-1}(n-1)^\tau C_0^n \leq M$ after the harmless rescaling $z \mapsto \eta z$ which replaces C_0 by ηC_0 ; this is possible because a polynomial factor is dominated by an arbitrarily small exponential loss. The finitely many initial degrees are absorbed by increasing M . Induction yields the bound.

Convergence and conjugacy.

The formal series h converges on $|z| < M^{-1}$ and satisfies $f \circ h = h \circ (\lambda \text{id})$ there.

The coefficient bound gives absolute convergence. Both sides of the formal identity are analytic on a smaller disk and have identical Taylor series, so they agree analytically.

Local biholomorphy.

After shrinking the disk, h is biholomorphic and therefore analytically linearizes f .

We have $h(0) = 0$ and $h'(0) = 1$. The inverse-function theorem gives a disk on which h is biholomorphic. Conjugating the identity of Step 4 by its inverse produces $h^{-1} \circ f \circ h(w) = \lambda w$. This is the claimed Siegel linearization. $\square$

Theorem 17.4 (Brjuno sufficiency). **Statement scope.** Let $\alpha \in \mathbb{R} \setminus \mathbb{Q}$ satisfy the Brjuno condition. Every holomorphic germ $f(z) = e^{2\pi i \alpha z} + O(z^2)$ is analytically linearizable near 0. In particular the quadratic polynomial $P_\alpha(z) = e^{2\pi i \alpha z} + z^2$ is linearizable. The converse for the quadratic family is isolated in the subsequent Yoccoz sharpness theorem so that the proof order is noncircular.

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Reconstruct a Davie-type arithmetic weight from continued-fraction separation and resonant blocks; prove its superadditivity, divisor-increment bound, and Brjuno-linear growth; combine it with an analytic scalar majorant to obtain a convergent local conjugacy.

The mathematical argument begins in the proof environment below.

Proof. Write

$$\lambda = e^{2\pi i\alpha}, \quad f(z) = \lambda z + \sum_{m \geq 2} a_m z^m, \quad h(z) = z + \sum_{n \geq 2} b_n z^n,$$

and let p_k/q_k be the continued-fraction convergents of α , with $q_0 = 1$. We assume

$$B(\alpha) := \sum_{k \geq 0} \frac{\log q_{k+1}}{q_k} < \infty. \quad (17.2.1)$$

By Theorem 17.3, comparison of coefficients in $f \circ h = h \circ (\lambda \text{id})$ gives the unique formal normalized conjugacy and, for $n \geq 2$,

$$b_n = \frac{1}{\lambda^n - \lambda} \sum_{m=2}^n a_m \sum_{\substack{n_1 + \dots + n_m = n \\ n_i \geq 1}} b_{n_1} \cdots b_{n_m}. \quad (17.2.2)$$

Thus it remains to prove that the products of the small divisors $|\lambda^{n-1} - 1|^{-1}$ grow at most exponentially.

Step: continued-fraction separation. For $k \geq 0$ and $0 < r < q_k$,

$$\|r\alpha\| \geq \frac{1}{2q_k}, \quad (17.2.3)$$

where $\|x\| = \min_{p \in \mathbb{Z}} |x - p|$. Consequently, if

$$\|n\alpha\| \leq \frac{1}{4q_k}, \quad (17.2.4)$$

then $n \geq q_k$ and either $q_k \mid n$ or $n \geq q_{k+1}/4$.

Indeed, the best-approximation property of convergents gives, for $0 < r < q_k$,

$$\|r\alpha\| \geq \|q_{k-1}\alpha\| > \frac{1}{q_{k-1} + q_k} \geq \frac{1}{2q_k},$$

with the $k = 0$ statement vacuous. This proves (17.2.3), hence (17.2.4) forces $n \geq q_k$. If $q_k \nmid n$ and $n < q_{k+1}/4$, write $n = mq_k + r$ with $0 < r < q_k$. Since $\|q_k\alpha\| < q_{k+1}^{-1}$,

$$\|n\alpha\| \geq \|r\alpha\| - m\|q_k\alpha\| > \frac{1}{2q_k} - \frac{1}{4q_k} = \frac{1}{4q_k},$$

contradicting (17.2.4).

Step: resonant blocks. Fix integers $q \geq 1$ and a real number $E \geq q$. Let $A \subset \mathbb{N}_0$ contain 0 and satisfy

$$a, b \in A, \quad q \nmid (a - b) \implies |a - b| \geq E. \quad (17.2.5)$$

Join two points of A by an edge when their distance is $< E$. Every connected component of this graph lies in a single residue class modulo q . Let $A^\sharp$ be obtained by filling, inside each connected component, every q -lattice point between any two of its points. Then:

1. if $x < y$ lie in $A^\sharp$ and $y - x < E$, they belong to the same filled block, $q \mid (y - x)$, and

- every $x + jq \leq y$ belongs to $A^\sharp$;
2. for
- $$m_n := \max(A^\sharp \cap [0, n]),$$
- if $n \in A^\sharp$ and $m_{n-1} + q \in A^\sharp$, then $m_{n-1} = n - q$;
3. if $n \in A^\sharp$ and $m_{n-1} + q \notin A^\sharp$, then $n - m_{n-1} \geq E$.

For the first assertion, an edge has q -divisible length by (17.2.5), so an entire graph component has one residue modulo q . Two distinct components cannot interlace on the line: if a point of one component lay between two successive points of an edge-chain of the other, it would be within distance $< E$ of one endpoint and hence would join that component. Therefore distinct filled blocks are separated by at least E , and assertion 1 follows. For assertion 2, maximality gives $m_{n-1} + q \geq n$. If it were $> n$, then $0 < n - m_{n-1} < q \leq E$ while both endpoints belong to $A^\sharp$, contradicting assertion 1. Hence equality holds. For assertion 3, a gap $n - m_{n-1} < E$ would, by assertion 1, force $m_{n-1} + q \in A^\sharp$ (the case $n - m_{n-1} < q$ is impossible because a positive gap in one block is a multiple of q), again a contradiction.

Step: local arithmetic counter. Under the hypotheses of the resonant blocks lemma put

$$\tau := \frac{2q}{E}, \quad \ell(n) := \max \left\{ \frac{(1+\tau)n}{q} - 2, \frac{\tau m_n + n}{q} - 1 \right\},$$

and

$$h(n) := \begin{cases} \frac{m_n + \tau n}{q} - 1, & m_n + q \in A^\sharp, \\ \ell(n), & m_n + q \notin A^\sharp. \end{cases} \quad (17.2.6)$$

Then

$$\frac{(1+\tau)n}{q} - 2 \leq h(n) \leq \frac{(1+\tau)n}{q} - 1, \quad (17.2.7)$$

h is nondecreasing, and

$$n \in A^\sharp, n > 0 \implies h(n) \geq h(n-1) + 1, \quad h(n+q) \geq h(n) + 1. \quad (17.2.8)$$

Consequently

$$g(n) := \max \left\{ h(n), \left\lfloor \frac{n}{q} \right\rfloor \right\} \quad (17.2.9)$$

is nonnegative, $g(0) = 0$, and satisfies

$$g(n) \leq \frac{(1+\tau)n}{q}, \quad (17.2.10)$$

$$g(n_1) + g(n_2) \leq g(n_1 + n_2), \quad (17.2.11)$$

and, for $n \in A$, $n > 0$,

$$g(n) \geq g(n-1) + 1. \quad (17.2.12)$$

We verify each point because these are the essential combinatorial properties. If $m_n + q \in A^\sharp$, maximality of m_n implies $m_n + q > n$, hence $n - q < m_n \leq n$; inserting this into the first branch of (17.2.6) gives (17.2.7). In the other branch the same bounds

follow directly from $m_n \leq n$ and the definition of ℓ .

If $n \in A^\sharp$, then $m_n = n$ and (17.2.6) gives

$$h(n) = \frac{(1 + \tau)n}{q} - 1. \quad (17.2.13)$$

Let $m = m_{n-1}$. If $m + q \in A^\sharp$, the resonant blocks lemma gives $m = n - q$, and direct substitution shows $h(n) - h(n-1) = 1 + \tau/q > 1$. If $m + q \notin A^\sharp$, the resonant blocks lemma gives $n - m \geq E$. Both entries in the maximum $\ell(n-1)$ are then at most $h(n) - 1$: for the second one the needed inequality is

$$\tau(n - m) \geq \tau E = 2q.$$

This proves the first part of (17.2.8). If $n \notin A^\sharp$, then $m_n = m_{n-1}$ and the same branch of (17.2.6) is used at $n - 1$ and n ; each affine expression involved is increasing. Hence h is nondecreasing.

It remains to prove the q -step inequality. Put $m = m_n$ and $m' = m_{n+q}$. If $m + q \in A^\sharp$, then $m + q > n$ and $m' \geq m + q$. If $m' + q \in A^\sharp$, the first branch gives immediately $h(n + q) \geq h(n) + 1 + \tau$. If $m' + q \notin A^\sharp$, use the second entry of $\ell(n + q)$; after cancellation the desired inequality reduces to

$$\tau m' - m + (1 - \tau)n \geq 0.$$

Since $m' \geq m + q$ and $0 \leq n - m < q$, the left side is at least $\tau q + (1 - \tau)(n - m)$, which is positive both for $\tau \leq 1$ and for $\tau > 1$.

Now suppose $m + q \notin A^\sharp$. If $m' = m$, both entries of ℓ grow by at least 1, so the claim follows. If $m' > m$, let s be the first point of $A^\sharp$ above m . Then $s > n$ and the resonant blocks lemma applied at s yields $s - m \geq E$, hence $m' - m \geq E$. If $m' + q \notin A^\sharp$, the first entry of ℓ grows by $1 + \tau$ and the second grows by $1 + \tau(m' - m)/q \geq 3$, so again $h(n + q) \geq h(n) + 1$. Finally, if $m' + q \in A^\sharp$, compare the first-branch value at $n + q$ separately with the two entries of $\ell(n)$. Against the first entry the difference after subtracting 1 is $(m' - n + \tau q)/q > 0$. Against the second it is

$$\frac{(m' - m) + (\tau - 1)(n - m + q)}{q}.$$

For $\tau \geq 1$ this is positive. If $0 < \tau < 1$, then $m' > n$ gives $n - m + q \leq (m' - m) + q$, so the numerator is at least

$$\tau(m' - m) - (1 - \tau)q \geq \tau E - (1 - \tau)q = (1 + \tau)q > 0.$$

Thus the second part of (17.2.8) is proved.

The bounds (17.2.10) and $g(0) = 0$ follow from (17.2.7). To prove superadditivity, consider which term realizes the maximum (17.2.9). Two floor terms are superadditive because $\lfloor n_1/q \rfloor + \lfloor n_2/q \rfloor \leq \lfloor (n_1 + n_2)/q \rfloor$. If, say, the first term is a floor $p = \lfloor n_1/q \rfloor$ and the second is $h(n_2)$, monotonicity and the q -step estimate give

$$h(n_1 + n_2) \geq h(pq + n_2) \geq h(n_2) + p.$$

If both terms are h -terms, (17.2.7) gives

$$h(n_1) + h(n_2) \leq \frac{(1 + \tau)(n_1 + n_2)}{q} - 2 \leq h(n_1 + n_2).$$

This proves (17.2.11).

For (17.2.12), if $g(n - 1) = h(n - 1)$ use (17.2.8). Otherwise set $p = \lfloor (n - 1)/q \rfloor$. If $q \mid n$, then $\lfloor n/q \rfloor = p + 1$. If $q \nmid n$, condition (17.2.5) applied to $0, n \in A$ gives $n \geq E$; by (17.2.13),

$$h(n) = \frac{(1 + \tau)n}{q} - 1 \geq \frac{n}{q} + 1 > p + 1.$$

Thus $g(n) \geq g(n - 1) + 1$ in every case.

Step: global arithmetic weight. For each k define

$$A_k := \left\{ n \geq 0 : \|n\alpha\| \leq \frac{1}{8q_k} \right\}, \quad E_k := \max \left\{ q_k, \frac{q_{k+1}}{4} \right\}, \quad \tau_k := \frac{2q_k}{E_k}.$$

By the continued-fraction separation lemma, A_k satisfies (17.2.5) with $q = q_k$ and $E = E_k$, and

$$\tau_k \leq \frac{8q_k}{q_{k+1}}. \quad (17.2.14)$$

Let g_k be the counter furnished by the local Davie counter lemma and set

$$G(n) := n \log 2 + \sum_{\{k: q_k \leq n\}} g_k(n) \log(2q_{k+1}), \quad G(0) = 0. \quad (17.2.15)$$

Then G is nondecreasing and superadditive,

$$G(n_1) + G(n_2) \leq G(n_1 + n_2), \quad (17.2.16)$$

there is a universal constant γ such that

$$G(n) \leq n \left(\sum_{\{k: q_k \leq n\}} \frac{\log q_{k+1}}{q_k} + \gamma \right), \quad (17.2.17)$$

and every small divisor is paid by one increment:

$$-\log |\lambda^n - 1| \leq G(n) - G(n - 1) \quad (n \geq 1). \quad (17.2.18)$$

Monotonicity follows from that of the g_k and from the fact that newly appearing summands are nonnegative. For superadditivity, enlarge both sums on the left to the index set $q_k \leq n_1 + n_2$ and apply (17.2.11) term by term. By (17.2.10) and (17.2.14),

$$g_k(n) \leq \frac{n}{q_k} + \frac{8n}{q_{k+1}}.$$

Hence

$$\frac{G(n)}{n} \leq \log 2 + \sum_{q_k \leq n} \frac{\log(2q_{k+1})}{q_k} + 8 \sum_{q_k \leq n} \frac{\log(2q_{k+1})}{q_{k+1}}. \quad (17.2.19)$$

The first sum is the corresponding partial Brjuno sum plus $\log 2 \sum q_k^{-1}$. Since $q_{k+2} \geq 2q_k$, the latter series is bounded universally. The same denominator growth gives a universal bound for $\sum \log(2q_{k+1})/q_{k+1}$ (compare with the exponentially growing Fibonacci sequence). Absorbing these two bounds and the initial $\log 2$ into γ proves (17.2.17).

For (17.2.18), put $t = \|n\alpha\| \leq 1/2$. Concavity of sine on $[0, \pi/2]$ gives

$$|\lambda^n - 1| = 2|\sin(\pi t)| \geq 4t. \quad (17.2.20)$$

If $t \geq 1/8$, then $-\log |\lambda^n - 1| \leq \log 2$, which is paid by the baseline increment in (17.2.15). If $t < 1/8$, choose k_0 so that

$$\frac{1}{8q_{k_0+1}} \leq t < \frac{1}{8q_{k_0}}.$$

Then $n \in A_{k_0}$, and the continued-fraction separation lemma (with the weaker threshold $1/(4q_{k_0})$) shows $n \geq q_{k_0}$, so the k_0 term occurs in $G(n)$. If $q_{k_0} \leq n-1$, (17.2.12) gives $g_{k_0}(n) - g_{k_0}(n-1) \geq 1$. If instead $q_{k_0} = n$, the term is new at time n , and (17.2.12) together with $g_{k_0}(n-1) \geq 0$ gives $g_{k_0}(n) \geq 1$. Thus in either case the k_0 contribution to $G(n) - G(n-1)$ is at least $\log(2q_{k_0+1})$, whereas (17.2.20) gives

$$-\log |\lambda^n - 1| \leq \log(2q_{k_0+1}).$$

All other increments are nonnegative, proving (17.2.18).

Step: analytic scalar majorant. Put

$$A(w) := \sum_{m \geq 2} |a_m| w^m.$$

The scalar equation

$$s = z + A(s) \quad (17.2.21)$$

has a unique analytic solution near 0 of the form $s(z) = \sum_{n \geq 1} s_n z^n$, with $s_1 = 1$, $s_n \geq 0$, and

$$s_n = \sum_{m=2}^n |a_m| \sum_{\substack{n_1 + \dots + n_m = n \\ n_i \geq 1}} s_{n_1} \cdots s_{n_m}. \quad (17.2.22)$$

In particular there are constants $C, R > 0$ such that $s_n \leq CR^{-n}$.

Indeed, $A(0) = A'(0) = 0$, so the analytic implicit-function theorem applies to $s - z - A(s) = 0$. Coefficient comparison gives (17.2.22), hence positivity, and Cauchy's estimate on any disk strictly inside the convergence disk of s gives the exponential coefficient bound.

Step: weighted coefficient induction. For every $n \geq 1$,

$$|b_n| \leq s_n e^{G(n-1)}, \quad (17.2.23)$$

where $b_1 = s_1 = 1$.

The statement is trivial at $n = 1$. Assume it through degree $n - 1$ and take a summand in (17.2.2). Since $m \geq 2$ and $n_1 + \dots + n_m = n$, repeated use of (17.2.16) and monotonicity gives

$$\sum_{i=1}^m G(n_i - 1) \leq G\left(\sum_{i=1}^m (n_i - 1)\right) = G(n - m) \leq G(n - 2).$$

Moreover (17.2.18), applied to $n - 1$, yields

$$|\lambda^{n-1} - 1|^{-1} \leq e^{G(n-1)-G(n-2)}.$$

Taking absolute values in (17.2.2), applying the inductive estimate and then (17.2.22), we obtain

$$|b_n| \leq e^{G(n-1)} \sum_{m=2}^n |a_m| \sum_{n_1+\dots+n_m=n} s_{n_1} \cdots s_{n_m} = s_n e^{G(n-1)}.$$

This closes the induction.

Step: positive convergence radius. The formal conjugacy h has positive radius of convergence.

Indeed, (17.2.17), the Brjuno hypothesis (17.2.1), and the analytic scalar-majorant and weighted-coefficient-induction lemmas give

$$|b_n| \leq CR^{-n} \exp((n-1)(B(\alpha) + \gamma)).$$

Therefore

$$\limsup_{n \rightarrow \infty} |b_n|^{1/n} \leq R^{-1} e^{B(\alpha) + \gamma} < \infty,$$

and Cauchy-Hadamard gives a positive radius.

Step: analytic conjugacy and quadratic specialization. The convergent normalized series is locally biholomorphic and conjugates f to the rigid rotation; hence the quadratic germ is linearizable whenever α is Brjuno.

On a sufficiently small disk both sides of the formal identity $f \circ h = h \circ (\lambda \text{id})$ converge, so equality of Taylor series gives equality of holomorphic functions. Since $h(0) = 0$ and $h'(0) = 1$, the inverse-function theorem makes h biholomorphic after shrinking the disk. Thus

$$h^{-1} \circ f \circ h(w) = \lambda w.$$

This proves Brjuno sufficiency for every holomorphic germ with multiplier $e^{2\pi i \alpha}$. Taking $f = P_\alpha(z) = \lambda z + z^2$ gives the stated quadratic clause. The non-Brjuno converse for that quadratic family remains separate in Theorem 17.5, so no circular use of Yoccoz sharpness has entered the argument. $\square$

Theorem 17.5 (Yoccoz sharpness for quadratic/germs). **Statement scope.** *If α is not Brjuno, the quadratic polynomial $P_\alpha(z) = e^{2\pi i \alpha} z + z^2$ is not analytically linearizable. Consequently the Brjuno condition is the sharp arithmetic condition for universal linearizability: every germ with multiplier $e^{2\pi i \alpha}$ is linearizable when α is Brjuno, whereas for non-Brjuno α the quadratic germ already gives a counterexample.*

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Prove quadratic necessity by the finite-scale Buff-Ch  ritat route: reconstruct the Pommerenke-Levin-Yoccoz multiplier disk, parabolic explosion and collision radius, good-convergent decomposition, punctured-domain conformal-radius loss, relative Schwarz comparison, and the Chapter 6 holomorphic-motion lift; telescope to show that a positive Siegel radius forces a finite Brjuno sum, then combine with Theorem 17.4.

The mathematical argument begins in the proof environment below.

Proof. Put

$$P_\alpha(z) = e^{2\pi i\alpha}z + z^2, \quad \lambda_\alpha = e^{2\pi i\alpha},$$

and write p_n/q_n for the continued-fraction convergents of the irrational number α , with $q_0 = 1$. We use the finite sums

$$B_N(\alpha) = \sum_{n=0}^N \frac{\log q_{n+1}}{q_n}. \quad (17.3.1)$$

The functions B and the linearizability problem are invariant under $\alpha \mapsto \alpha + 1$ and $\alpha \mapsto 1 - \alpha$; hence, after replacing α by an equivalent angle, we may and do assume $0 < \alpha < 1/2$.

If $U \subset \mathbb{C}$ is hyperbolic and contains 0, define $\text{rad}(U)$ to be $|\pi'(0)|$, where $\pi : (\mathbb{D}, 0) \rightarrow (U, 0)$ is a universal covering. Thus this is the usual conformal radius when U is simply connected. If $0 \in U_1 \subset U_2$, a lift of the covering of U_1 through that of U_2 and Schwarz's lemma give

$$\text{rad}(U_1) \leq \text{rad}(U_2). \quad (17.3.2)$$

For $N \geq 0$ let W_N be the complement of the fixed external ray of P_α of angle 0 together with all nonzero periodic points whose period is at most q_N . The central estimate to be proved is

$$\log \text{rad}(W_N) + B_N(\alpha) \leq C_0 \quad (17.3.3)$$

with a universal constant C_0 . Once (17.3.3) is established, the theorem is immediate: a Siegel disk Δ_α contains no nonzero periodic point and does not meet an external ray, so $\Delta_\alpha \subset W_N$. Hence

$$\log \text{rad}(\Delta_\alpha) + B_N(\alpha) \leq C_0 \quad (N \geq 0), \quad (17.3.4)$$

which is impossible if $B(\alpha) = \infty$.

We now prove (17.3.3). The point of writing every module is to make clear where the arithmetic loss enters and to avoid importing the necessity theorem itself.

Step: the quadratic Pommerenke-Levin-Yoccoz disk. Let P be a quadratic polynomial with connected filled Julia set and let z_0 be a repelling fixed point. Suppose that exactly $q \geq 1$ periodic external rays land at z_0 and that P permutes them with reduced combinatorial rotation number p/q . If $\omega = P'(z_0)$ and τ is the branch of $\log \omega$ selected by this ray portrait, then

$$\frac{\Re \tau}{|\tau - 2\pi ip/q|^2} \geq \frac{q}{2 \log 2}. \quad (17.3.5)$$

In particular, for $P = P_\beta$ and $z_0 = 0$, the parameter β lies in the closed disk tangent to the real axis at p/q and of radius $\log 2/(2\pi q)$.

To see (17.3.5), use Theorem 1.3 to choose the normalized Poincaré linearizer Ψ at z_0 , so that $P \circ \Psi(w) = \Psi(\omega w)$. After taking logarithms in the linearizing plane, the quotient by the deck translations generated by $2\pi i$ and τ is the flat torus

$$T = \mathbb{C}/(2\pi i\mathbb{Z} + \tau\mathbb{Z}).$$

Let U be the component in the Poincaré linearizing plane corresponding to one of the landing-ray sectors. Its period under multiplication by ω is q . Lift the Böttcher coordinate through $\Psi|_U$. The resulting map sends U to a half-plane and conjugates multiplication by ω^q to multiplication by 2^q . Taking a logarithm therefore sends the quotient $U/\langle\omega^q\rangle$ to a strip of transverse width π modulo translation by $q \log 2$. Its image A in T is an embedded annulus of modulus

$$\text{mod } A = \frac{\pi}{q \log 2}, \quad (17.3.6)$$

and the cyclic order of the rays identifies its primitive homotopy class with the vector $2\pi ip + q\tau$. For the upper bound use the Euclidean metric on the flat torus: every curve in that homotopy class has length at least $|2\pi ip + q\tau|$, while the torus has area $2\pi \Re \tau$. Thus

$$\text{mod } A \leq \frac{2\pi \Re \tau}{|2\pi ip + q\tau|^2}. \quad (17.3.7)$$

Combining (17.3.6)–(17.3.7) and dividing by q^2 gives (17.3.5). For $\omega = e^{2\pi i\beta}$ the inequality is exactly the equation of the stated tangent disk.

Step: parabolic explosion with a uniform collision radius. Let p/q be reduced. There is a holomorphic function

$$\chi_{p/q} : \{|\delta| < q^{-3/q}\} \longrightarrow \mathbb{C}, \quad \chi_{p/q}(0) = 0, \quad (17.3.8)$$

such that, for $\delta \neq 0$,

$$\{\chi(\delta), \chi(\zeta\delta), \dots, \chi(\zeta^{q-1}\delta)\}, \quad \zeta = e^{2\pi ip/q}, \quad (17.3.9)$$

is a q -cycle of $P_{p/q+\delta^q}$.

First consider $P_{p/q}^q$. Since its multiplier at 0 is 1, write

$$P_{p/q}^q(z) = z + Az^{m+1} + O(z^{m+2}), \quad A \neq 0. \quad (17.3.10)$$

Commutation with $P_{p/q}$ forces $q \mid m$. The Leau-Fatou petals of the iterate are permuted by $P_{p/q}$ in cycles of length q . Each such cycle of immediate parabolic petals contains a critical point: otherwise the first return would be an unramified proper self-map of a simply connected immediate basin and hence an automorphism, contradicting convergence to the parabolic boundary point. A quadratic polynomial has only one finite critical point, so there is exactly one petal cycle. Consequently $m = q$ and

$$P_{p/q}^q(z) = z + Az^{q+1} + O(z^{q+2}). \quad (17.3.11)$$

Near $(p/q, 0)$ we therefore have

$$P_{p/q+\varepsilon}^q(z) - z = z(2\pi i q \varepsilon + Az^q + O(z\varepsilon) + O(z^{q+1})). \quad (17.3.12)$$

Set $\varepsilon = \delta^q$ and blow up by $z = \delta u$. The bracket in (17.3.12), divided by δ^q , has at $\delta = 0$ the q simple roots $u^q = -2\pi i q/A$. The implicit-function theorem gives q transverse holomorphic branches through $(0, 0)$; comparison of their tangent directions with the action of $P_{p/q}$ gives the cyclic relation $P_{p/q+\delta^q}(\chi(\delta)) = \chi(\zeta\delta)$.

It remains to justify the radius in (17.3.8). Suppose $0 < |\beta - p/q| < q^{-3}$ and that P_β^q has a multiple fixed point. Then P_β has a parabolic cycle of period dividing q . Its immediate basin contains the unique critical point, hence the filled Julia set is connected and every other periodic orbit is repelling. If 0 itself is parabolic, its multiplier is a root of unity of denominator dividing q , so $|\beta - p/q| \geq 1/q$, impossible. Otherwise 0 is repelling. By the rational-ray landing machinery of Theorem 4.8, finitely many periodic rays land at 0; let their reduced rotation number be p'/q' . The quadratic Pommerenke–Levin–Yoccoz disk lemma puts β in the tangent Yoccoz disk at p'/q' .

We claim $q' < q$. The q' landing rays cut the plane into q' sectors. The critical orbit must traverse all these sectors before first returning to the critical sector, so the period of the parabolic cycle is at least q' ; therefore $q' \leq q$. If equality held, pull back the ray graph once and let U be the component containing the critical point and V the corresponding critical sector. After truncating by an equipotential we may take Jordan disks $U \Subset V$ for which $P_\beta^q : U \rightarrow V$ is proper of degree 2. Let $\phi : V \rightarrow \mathbb{D}$ be a Riemann map and put $U' = \phi(U) \Subset \mathbb{D}$ and $F = \phi \circ P_\beta^q \circ \phi^{-1} : U' \rightarrow \mathbb{D}$. On $\partial U'$ properness gives $|F(w)| = 1$, whereas $|w| < 1$; hence the loops $F(w) - w$ and $F(w)$ are homotopic in $\mathbb{C} \setminus \{0\}$ through $F(w) - tw$, $0 \leq t \leq 1$. The winding number of $F(\partial U')$ about 0 is the mapping degree 2. The argument principle therefore gives exactly two zeros of $F(w) - w$ in U' , counted with multiplicity. Equivalently, $P_\beta^q(z) - z$ has total fixed-point multiplicity 2 in U . But 0 contributes one and the parabolic point in the critical immediate basin contributes at least two, a contradiction. Hence $q' < q$.

Since distinct reduced fractions satisfy $|p/q - p'/q'| \geq 1/(qq')$, elementary Euclidean geometry of the tangent disk of radius $a = (\log 2)/(2\pi q')$ gives

$$\text{dist}\left(\frac{p}{q}, \overline{D}\left(\frac{p'}{q'} - ia, a\right)\right) = \sqrt{|p/q - p'/q'|^2 + a^2} - a > \frac{1}{q^3} \quad (17.3.13)$$

for $q' < q$, $q \geq 2$. Thus no such collision occurs in $|\beta - p/q| < q^{-3}$. Analytic continuation of the branches in (17.3.12) over that collision-free disk, after the q th-root base change, proves (17.3.8)–(17.3.9). The case $q = 1$ is immediate.

The fixed external ray is safe on the slightly larger disk

$$D(p/q, q^{-2}). \quad (17.3.14)$$

Indeed P_β is affinely conjugate to $z^2 + c(\beta)$ and the fixed ray of angle 0 bifurcates only when $\Re(e^{2\pi i \beta}) = 1$. Writing the latter locus as $\Im \beta = (2\pi)^{-1} \log \cos(2\pi \Re \beta)$ on each relevant strip, the inequalities $\sin t < t$ and the elementary lower bound for $\cos t$ show that its distance from every reduced $p/q \in (0, 1/2)$, $q \geq 2$, is at least q^{-2} . Hence the fixed ray and its repelling landing point move holomorphically throughout (17.3.14).

Step: good convergents isolate all divergent Brjuno mass. Call $n \geq 1$ *good* when

$$q_{n+1} > 2q_n^2. \quad (17.3.15)$$

List the good indices as $n_1 < n_2 < \dots$. Put

$$B_i = D\left(\frac{p_{n_i}}{q_{n_i}}, q_{n_i}^{-3}\right), \quad D_i = D\left(\frac{p_{n_i}}{q_{n_i}}, q_{n_i}^{-2}\right). \quad (17.3.16)$$

Then

$$\alpha \in B_i, \quad D_{i+1} \subseteq B_i \setminus \{p_{n_i}/q_{n_i}\}. \quad (17.3.17)$$

Moreover there is a universal finite constant C_{bad} such that

$$\sum_{\substack{1 \leq n \leq N \\ n \text{ not good}}} \frac{\log q_{n+1}}{q_n} \leq C_{\text{bad}}. \quad (17.3.18)$$

The convergent inequalities

$$\frac{1}{2q_n q_{n+1}} < \left| \alpha - \frac{p_n}{q_n} \right| < \frac{1}{q_n q_{n+1}} \quad (17.3.19)$$

show the first inclusion in (17.3.17); the second follows from the same bounds at consecutive good indices. At a bad index, $q_{n+1} \leq 2q_n^2$, so

$$\frac{\log q_{n+1}}{q_n} \leq \frac{\log 2 + 2 \log q_n}{q_n}. \quad (17.3.20)$$

Since q_n dominates the Fibonacci sequence, the series on the right is universally summable, proving (17.3.18).

For a good index n_i , abbreviate $q = q_{n_i}$ and define

$$U_i = D(0, q^{-3/q}), \quad S_i = \{\delta \in U_i : p_{n_i}/q + \delta^q = \alpha\}, \quad C_i = \chi_{p_{n_i}/q}(S_i). \quad (17.3.21)$$

Thus S_i consists of q equally spaced points on a circle and C_i is a q -cycle of P_α . Starting with $V_0 = \mathbb{C} \setminus R_0(\alpha)$, where R_0 is the fixed external ray, set

$$V_i = V_{i-1} \setminus C_i. \quad (17.3.22)$$

The nesting (17.3.17), the parabolic explosion with a uniform collision radius lemma, and fixed-ray safety imply that, over D_i , all objects used to define V_{i-1} continue without collisions and hence $V_{i-1}(\beta)$ moves holomorphically with β .

Step: equally spaced punctures cost their logarithmic radius. There is a universal constant C_1 such that, for $q \geq 2$, $0 < r < 1$, and $\mathcal{U}_q = \{e^{2\pi i k/q} : 0 \leq k < q\}$,

$$\log \text{rad}(\mathbb{D} \setminus r\mathcal{U}_q) \leq \log r + \frac{C_1}{q}. \quad (17.3.23)$$

Consequently

$$\log \frac{\text{rad}(U_i \setminus S_i)}{\text{rad}(U_i)} \leq -\frac{\log q_{n_i+1}}{q_{n_i}} + \frac{2 \log q_{n_i} + C_2}{q_{n_i}} \quad (17.3.24)$$

for a universal C_2 .

By inclusion and scaling it is enough to estimate $\text{rad}(\mathbb{C} \setminus \mathcal{U}_q)$. Cut from each root of unity the outward radial half-line and call the resulting slit domain Ω_q . The explicit map

$$\phi_q(z) = z \left(\frac{4}{(1 - z^q)^2} \right)^{1/q} \quad (17.3.25)$$

conformally maps $\mathbb{D}$ onto Ω_q , so $\text{rad}(\Omega_q) = 4^{1/q}$. If $\pi : \mathbb{D} \rightarrow \mathbb{C} \setminus \mathcal{U}_q$ is the universal cover, the component of $\pi^{-1}(\Omega_q)$ containing 0 contains the disk of radius

$$\rho_q = \frac{1 - \tan(\pi/4q)}{1 + \tan(\pi/4q)}. \quad (17.3.26)$$

Schwarz's lemma applied to the lift of ϕ_q yields

$$\log \text{rad}(\mathbb{C} \setminus \mathcal{U}_q) \leq \frac{\log 4}{q} + \log \frac{1 + \tan(\pi/4q)}{1 - \tan(\pi/4q)} \leq \frac{C_1}{q},$$

which proves (17.3.23). Now U_i has radius $q^{-3/q}$ and the radius of the circle carrying S_i is $|\alpha - p_{n_i}/q|^{1/q}$. Scaling U_i to $\mathbb{D}$ and using (17.3.19) gives (17.3.24).

Step: relative Schwarz for deleting finitely many points. Let U, V be hyperbolic plane domains containing 0, let $\chi : U \rightarrow V$ be holomorphic with $\chi(0) = 0$, and let $S \subset U \setminus \{0\}$ be finite with $0 \notin \chi(S)$. Then

$$\frac{\text{rad}(V \setminus \chi(S))}{\text{rad}(V)} \leq \frac{\text{rad}(U \setminus S)}{\text{rad}(U)}. \quad (17.3.27)$$

Write $\rho_X|dz|$ for the Poincaré metric of curvature -1 on a hyperbolic domain X . Since $\text{rad}(X) = 2/\rho_X(0)$, it is enough first to delete one point $v_0 \in V \setminus \{0\}$. Put $V' = V \setminus \{v_0\}$, $U' = \chi^{-1}(V')$, and on U' define

$$\sigma = \frac{\chi^* \rho_V}{\chi^* \rho_{V'}} \rho_{U'}. \quad (17.3.28)$$

The pullback factors have the same zero at a critical point of χ , so σ is a positive C^2 metric there as well. Away from critical points set $a = \chi^* \rho_V$, $b = \chi^* \rho_{V'}$, $c = \rho_{U'}$. Schwarz-Pick and domain monotonicity give $0 < a \leq b \leq c$. Since all three metrics have curvature -1 where nonzero,

$$\Delta \log \sigma = a^2 + c^2 - b^2 \geq \frac{a^2 c^2}{b^2} = \sigma^2, \quad (17.3.29)$$

because $b^2(a^2 + c^2 - b^2) - a^2 c^2 = (b^2 - a^2)(c^2 - b^2) \geq 0$. Thus σ is ultrahyperbolic on U' .

If $x_0 \in U \setminus U'$ has local degree d for χ , then in local coordinates $|\chi(x) - v_0| \asymp |x - x_0|^d$ and $|\chi'(x)| \asymp |x - x_0|^{d-1}$. The elementary cusp asymptotic

$$\rho_{X \setminus \{x_0\}}(x) \sim \frac{1}{|x - x_0| \log(1/|x - x_0|)} \quad (17.3.30)$$

shows that $\rho_{U'}/(\chi^*\rho_{V'}) \rightarrow 1$ at x_0 , so σ extends as an ultrahyperbolic metric across the puncture. Maximality of ρ_U gives

$$\frac{\chi^*\rho_V}{\rho_U} \leq \frac{\chi^*\rho_{V'}}{\rho_{U'}}. \quad (17.3.31)$$

Delete the distinct points of $\chi(S)$ one at a time and iterate (17.3.31). At 0 this yields

$$\frac{\rho_{U \setminus \chi^{-1}(\chi(S))}(0)}{\rho_U(0)} \leq \frac{\rho_{V \setminus \chi(S)}(0)}{\rho_V(0)}.$$

Since $U \setminus \chi^{-1}(\chi(S)) \subset U \setminus S$, Poincaré-metric monotonicity and $\text{rad}(X) = 2/\rho_X(0)$ give (17.3.27).

Step: holomorphic-motion lift costs only $O(1/q)$. For each good scale n_i there is a holomorphic lift

$$\hat{\chi}_i : U_i \longrightarrow B(0, \rho_i) \quad (17.3.32)$$

of the parabolic-explosion map into normalized universal coverings of the moving domains $V_{i-1}(\beta)$, based at $\beta = \alpha$, such that

$$\log \rho_i \leq \frac{C_3}{qn_i}. \quad (17.3.33)$$

Normalize the parameter disk D_i to $\mathbb{D}$. By the good convergents isolate all divergent Brjuno mass lemma the boundary ray and all previously removed cycles move holomorphically on D_i . Theorem 6.1 extends this to a holomorphic motion of the sphere. Let $h_\lambda : V_{i-1}(\alpha) \rightarrow V_{i-1}(\beta(\lambda))$ be the corresponding slice. Start with a universal covering $\pi_0 : \mathbb{D} \rightarrow V_{i-1}(\alpha)$ and pull the Beltrami coefficient of h_λ back by π_0 to $\mathbb{D}$, extending it by 0 outside. Theorem 6.10 integrates this holomorphically to normalized quasiconformal maps $\tilde{h}_\lambda$ of the plane. Then

$$\tilde{V}_\lambda = \tilde{h}_\lambda(\mathbb{D}), \quad \pi_\lambda = h_\lambda \circ \pi_0 \circ \tilde{h}_\lambda^{-1} \quad (17.3.34)$$

form an analytic family of universal coverings. Because $\tilde{h}_\lambda$ is conformal off $\mathbb{D}$ and normalized at infinity, the area theorem gives $\tilde{V}_\lambda \subset B(0, 4)$. Its boundary moves holomorphically in $B(0, 4) \setminus \{0\}$ and is the unit circle at $\lambda = 0$. Schwarz–Pick in this punctured disk therefore implies

$$\tilde{V}_\lambda \subset B(0, \rho(\lambda)), \quad \log \rho(\lambda) \leq \frac{2 \log 4}{1 + |\lambda|^{-1}}. \quad (17.3.35)$$

The target parameter α lies within $q^{-3}/2$ of the center of B_i , whereas D_i has radius q^{-2} . After the disk automorphism sending α to the base point, every parameter generated by $\delta \in U_i$ has $|\lambda| \leq 3/(2q)$. Thus (17.3.35) gives (17.3.33). The map $\delta \mapsto (\beta(\delta), \chi_{p/q}(\delta))$ lifts through the analytic covering family (17.3.34), starting at 0 when $\delta = 0$; its second coordinate is the desired $\hat{\chi}_i$.

Step: one good scale pays one Brjuno term. There is a universal constant C_4 such

that, for every good index n_i ,

$$\log \frac{\text{rad}(V_i)}{\text{rad}(V_{i-1})} \leq -\frac{\log q_{n_i+1}}{q_{n_i}} + \frac{2 \log q_{n_i} + C_4}{q_{n_i}}. \quad (17.3.36)$$

At the base parameter α , the normalized covering domain in the holomorphic-motion lift costs only $O(1/q)$ lemma is $\mathbb{D}$. Removing C_i downstairs removes the entire preimage of that cycle upstairs. Since $\widehat{\chi}_i(S_i)$ is contained in that preimage, monotonicity (17.3.2) gives

$$\log \frac{\text{rad}(V_i)}{\text{rad}(V_{i-1})} \leq \log \text{rad}(\mathbb{D} \setminus \widehat{\chi}_i(S_i)). \quad (17.3.37)$$

Apply the relative Schwarz for deleting finitely many points lemma to $\widehat{\chi}_i : U_i \rightarrow B(0, \rho_i)$. Since $\text{rad}(B(0, \rho_i)) = \rho_i$, the equally spaced punctures cost their logarithmic radius lemma and the holomorphic-motion lift costs only $O(1/q)$ lemma give (17.3.36).

Step: finite-scale conformal-radius loss. There is a universal C_0 such that the domains W_N defined above satisfy

$$\log \text{rad}(W_N) + B_N(\alpha) \leq C_0 \quad (N \geq 0). \quad (17.3.38)$$

First, $V_0 = \mathbb{C} \setminus R_0(\alpha)$ is simply connected and omits the other fixed point $1 - e^{2\pi i \alpha}$. Koebe's one-quarter theorem gives

$$\text{rad}(V_0) \leq 4|1 - e^{2\pi i \alpha}| \leq 8\pi\alpha \leq \frac{8\pi}{q_1}, \quad (17.3.39)$$

so the $n = 0$ Brjuno term is paid up to a universal additive constant. Telescoping (17.3.36) over good indices and using Fibonacci growth to bound $\sum (1 + \log q_n)/q_n$, together with the bounded bad mass from the good convergents isolate all divergent Brjuno mass lemma, gives

$$\log \text{rad}(V_i) + B_N(\alpha) \leq C_0. \quad (17.3.40)$$

Finally W_N removes every periodic point of period at most q_N , whereas V_i removes only the selected explosion cycles, so $W_N \subset V_i$. Monotonicity (17.3.2) turns (17.3.40) into (17.3.38).

Step: finite-scale Yoccoz obstruction. If P_α is analytically linearizable at 0, then $B(\alpha) < \infty$. Indeed its Siegel disk Δ_α has positive conformal radius and is contained in every W_N . Equations (17.3.2) and (17.3.38) give a uniform bound on $B_N(\alpha)$, and monotone convergence yields the claim.

Step: non-Brjuno quadratic obstruction. If $B(\alpha) = \infty$, then P_α is not analytically linearizable at 0. This is the contrapositive of the finite-scale Yoccoz obstruction lemma.

Step: quadratic iff. The quadratic germ P_α is analytically linearizable if and only if α is Brjuno. The necessity is the finite-scale Yoccoz obstruction lemma. The sufficiency is Theorem 17.4.

Step: sharp universal arithmetic condition. The Brjuno condition is exactly the arithmetic condition guaranteeing analytic linearizability of every one-dimensional holomorphic germ with multiplier $e^{2\pi i \alpha}$. For Brjuno α , Theorem 17.4 linearizes ev-

ery germ. For non-Brjuno α , the non-Brjuno quadratic obstruction lemma supplies the explicit quadratic counterexample P_α . Notice the scope: the statement does not assert that every non-Brjuno germ is nonlinearizable. $\square$

Definition 17.5 (Siegel and Cremer alternatives). **Statement scope.** Let $f(z) = e^{2\pi i \alpha z} + O(z^2)$ with α irrational. The fixed point is Siegel when f is analytically conjugate near 0 to $z \mapsto e^{2\pi i \alpha z}$; the maximal connected domain on which this conjugacy extends is its Siegel disk, whose conformal radius is measured in the normalized linearizing coordinate. If no local analytic linearization exists, the fixed point is Cremer.

17.3 Worked examples

Worked example: The first homological denominator

Let $f(z) = \lambda z + a_2 z^2 + O(z^3)$ and seek $h(z) = z + b_2 z^2 + O(z^3)$ with $h \circ f = \lambda h$. At order two,

$$(\lambda^2 - \lambda)b_2 = -a_2, \quad b_2 = -\frac{a_2}{\lambda(\lambda - 1)}.$$

Higher orders contain divisors $\lambda^n - \lambda$. Their possible smallness is the arithmetic obstruction behind Siegel and Brjuno theory.

Worked example: Why bounded type is stronger than Brjuno

If the continued-fraction entries of α are bounded, then $q_{n+1} \leq Cq_n$, so $\sum \log q_{n+1}/q_n$ converges rapidly. Hence bounded type implies Brjuno, but the converse fails: Brjuno permits unbounded partial quotients provided their logarithmic growth is sufficiently summable.

Boundary case: Non-Brjuno does not mean every germ is nonlinearizable

Yoccoz sharpness supplies a canonical quadratic counterexample for each non-Brjuno rotation number. The statement is therefore a sharp universal obstruction, not the false assertion that every germ with that multiplier must be nonlinearizable.

17.4 What this chapter established

The chapter closed its formal theorem spine at the stated source scopes. The proofs deliberately separate elementary deductions from deep imported theorems: every imported theorem is named at the point where it enters, and later conclusions use only the scope actually established.

For study purposes, the most important task is to be able to reconstruct the dependency chain rather than memorize theorem names. In particular, note which hypothe-

ses supply compactness or a priori geometry, which step supplies rigidity, and which step converts dynamical information into parameter or statistical conclusions.

Looking ahead

Chapter 18 studies rationally indifferent fixed points. There the small-divisor problem is replaced by petal geometry, translation coordinates, and analytic horn-map moduli.

Parabolic Dynamics and Ecalle-Voronin Theory

Why this chapter matters

At a parabolic fixed point the multiplier is a root of unity; after taking an iterate one reaches a germ tangent to the identity. The correct local picture is not a disk of linearization but a flower of attracting and repelling petals. On each petal a nonlinear change of coordinate turns the map into a translation.

Fatou coordinates make the local dynamics simple, but their overlaps contain new analytic information. The mismatch between attracting and repelling coordinates produces horn maps, and Ecalle-Voronin theory says that these horn data classify parabolic germs inside a fixed formal class. Perturbing the parabolic point introduces a long transit through the Ecalle cylinder and leads to Lavaurs maps and parabolic implosion.

Learning goals

By the end of this chapter you should be able to:

1. derive the petal count from the first nonlinear term;
2. construct attracting and repelling Fatou coordinates;
3. explain their additive normalization ambiguity;
4. define horn maps from overlap transitions;
5. state the Ecalle-Voronin analytic classification principle;
6. derive the phase parameter in a parabolic transit limit.

Conceptual roadmap

$$f(z) = z + az^{k+1} + \dots \rightarrow \text{translation sectors}$$

$$\rightarrow \text{Fatou coordinates}$$

$$\rightarrow \text{horn maps} \rightarrow \text{analytic modulus.}$$

For perturbations, the same coordinates split the orbit into entry, long translation, and exit; the residual translation phase is the Lavaurs parameter.

Common pitfall

A formal normal form does not determine the analytic conjugacy class. The horn maps are not decorative coordinates: after the formal invariants are fixed, they

carry the remaining analytic modulus.

18.1 Definitions

Definition 18.1 (parabolic petal). A parabolic petal is a simply connected invariant sector/domain at a parabolic fixed point tangent to one attracting or repelling direction, on which iterates converge to or depart from the fixed point.

Definition 18.2 (Fatou coordinate). A Fatou coordinate on a petal is a univalent holomorphic map Φ satisfying the Abel equation $\Phi(f(z)) = \Phi(z) + 1$.

Definition 18.3 (horn map). A horn map is a transition map between repelling and attracting Fatou-coordinate cylinders near their ends, defined modulo translations and forming the analytic parabolic modulus.

Definition 18.4 (Ecalle cylinder). An Ecalle cylinder is the quotient of a Fatou-coordinate petal by integer translation $w \sim w + 1$, conformally a cylinder/punctured disk after exponentiation.

18.2 Main results

Theorem 18.1 (Parabolic implosion). **Statement scope.** *Perturbations near a parabolic parameter exhibit Lavaurs maps/phase limits and discontinuous-looking Julia-set dynamics described by parabolic implosion theory.*

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Normalize the transverse simple-parabolic unfolding; construct the explicit logarithmic almost-Fatou coordinate and its logarithmic correction; prove the Abel defect is uniformly summable through the gate; obtain incoming/outgoing perturbed coordinate convergence; split the transit time and pass to the Lavaurs phase limit.

The mathematical argument begins in the proof environment below.

Proof. We prove the standard local Lavaurs theorem, which is the precise mechanism behind the volume's parabolic-implosion statement. By the simple-parabolic hypothesis and a holomorphic change of dynamical and parameter coordinates, a transverse unfolding may be written on a fixed neighborhood of the origin as

$$f_\varepsilon(z) = z + (z^2 + \varepsilon^2)a_\varepsilon(z), \quad a_\varepsilon(z) = 1 + (q+1)z + O(|z|^2 + |\varepsilon|^2), \quad (18.1)$$

with the error holomorphic in (z, ε) . The two nearby fixed points are $\pm i\varepsilon$. The parameter normalization is chosen so that the leading transit length is π/ε . We consider sequences

$$\varepsilon_j \rightarrow 0, \quad n_j \rightarrow \infty, \quad n_j - \frac{\pi}{\varepsilon_j} \rightarrow \sigma \in \mathbb{C}, \quad (18.2)$$

and prove local uniform convergence of $f_{\varepsilon_j}^{n_j}$ to the Lavaurs map of phase σ .

Lemma (Logarithmic gate coordinate). On the slit gate domain choose

$$u_\varepsilon(z) = \frac{1}{2i\varepsilon} \log \frac{i\varepsilon - z}{i\varepsilon + z}.$$

Then

$$u_\varepsilon(f_\varepsilon(z)) - u_\varepsilon(z) = 1 + qz + O(|\varepsilon|^2 + |z|^2),$$

locally uniformly wherever both sides are defined. Moreover the shifted incoming and outgoing branches

$$u_\varepsilon^{\text{in}} = u_\varepsilon + \frac{\pi}{2\varepsilon}, \quad u_\varepsilon^{\text{out}} = u_\varepsilon - \frac{\pi}{2\varepsilon}$$

converge on compact subsets of the corresponding limiting petals to $-1/z$.

Indeed, from (18.1),

$$\frac{i\varepsilon - f_\varepsilon(z)}{i\varepsilon + f_\varepsilon(z)} = \frac{i\varepsilon - z}{i\varepsilon + z} \frac{1 + i\varepsilon\gamma_\varepsilon(z)}{1 - i\varepsilon\gamma_\varepsilon(z)}, \quad \gamma_\varepsilon(z) = \frac{a_\varepsilon(z)}{1 + za_\varepsilon(z)} = 1 + qz + O(|z|^2 + |\varepsilon|^2).$$

Taking the chosen logarithm and using $\log((1+w)/(1-w)) = 2(w + w^3/3 + \dots)$ gives the asserted increment. For fixed z away from the gate slit,

$$u_\varepsilon^{\text{in}}(z) = \frac{1}{2i\varepsilon} \log \frac{1 - i\varepsilon/z}{1 + i\varepsilon/z} \rightarrow -\frac{1}{z},$$

and the outgoing formula is identical with the opposite branch choice.

Lemma (Corrected almost-Fatou coordinate). Define

$$w_\varepsilon(z) = u_\varepsilon(z) - \frac{q}{2} \log(\varepsilon^2 + z^2), \quad w_\varepsilon^{\text{in/out}} = w_\varepsilon \pm \frac{\pi}{2\varepsilon}.$$

Then the Abel defect

$$A_\varepsilon(z) := w_\varepsilon(f_\varepsilon(z)) - w_\varepsilon(z) - 1$$

satisfies

$$A_\varepsilon(z) = O(|\varepsilon|^2 + |z|^2)$$

uniformly near $(0,0)$, and $w_\varepsilon^{\text{in/out}}$ converges on compact subsets of the limiting petals to

$$w_0^{\text{in/out}}(z) = -\frac{1}{z} - q \log(\mp z).$$

To see the cancellation, expand

$$\varepsilon^2 + f_\varepsilon(z)^2 = (\varepsilon^2 + z^2)(1 + 2z + O(|\varepsilon|^2 + |z|^2)).$$

Hence

$$\log \frac{\varepsilon^2 + f_\varepsilon(z)^2}{\varepsilon^2 + z^2} = 2z + O(|\varepsilon|^2 + |z|^2).$$

The qz term from the logarithmic gate coordinate lemma is therefore exactly cancelled by the logarithmic correction, leaving $A_\varepsilon = O(|\varepsilon|^2 + |z|^2)$. Compact convergence of the corrected branches follows from the logarithmic gate coordinate lemma and the

elementary convergence of the logarithms.

Lemma (Gate-orbit estimate). For every compact incoming set K there are $r, C, K_0 > 0$ such that, for all sufficiently small ε in an α -sequence and $z \in K$, the orbit remains in the parabolic gate for the required half-transit and satisfies

$$|f_\varepsilon^m(z)| \leq C \max\{m^{-1}, |\varepsilon|\}$$

until the central gate is reached. The analogous reversed estimate holds on the outgoing side.

The real part of the suitably rotated coordinate u_ε increases by an amount bounded between two positive constants at each iterate, because the logarithmic gate coordinate lemma gives $u_\varepsilon \circ f_\varepsilon - u_\varepsilon = 1 + o(1)$. The inverse formula $z = \varepsilon \tan(\varepsilon u_\varepsilon)$ then shows first that the orbit cannot leave through a lateral gate boundary and secondly that its distance to the origin is $O(m^{-1})$ before it reaches the $O(|\varepsilon|)$ core. Inside that core it is $O(|\varepsilon|)$. Applying the same estimate to f_ε^{-1} gives the outgoing statement. This is exactly the estimate needed below; no global dynamics is used.

Lemma (Uniform summability of the Abel defect). If $z, f_\varepsilon(z), \dots, f_\varepsilon^N(z)$ stay in the gate, then

$$|w_\varepsilon(f_\varepsilon^N(z)) - w_\varepsilon(z) - N| \leq C$$

with C independent of the small parameter and the crossing time. More precisely, the incoming tail, central core, and outgoing tail contribute respectively $O(\sum m^{-2})$, $O(|\varepsilon|)$, and $O(\sum m^{-2})$.

By telescoping,

$$w_\varepsilon(f_\varepsilon^N(z)) - w_\varepsilon(z) - N = \sum_{m=0}^{N-1} A_\varepsilon(f_\varepsilon^m(z)).$$

The corrected almost-Fatou coordinate lemma and the gate-orbit estimate lemma give $|A_\varepsilon(f_\varepsilon^m(z))| \leq C(m^{-2} + |\varepsilon|^2)$ on an incoming tail. The central core lasts $O(|\varepsilon|^{-1})$ iterates and each term there is $O(|\varepsilon|^2)$, hence its whole contribution is $O(|\varepsilon|)$. The outgoing tail is identical after reversing time. The two square-summable tails plus the vanishing core prove the uniform bound.

Lemma (Convergence of perturbed incoming coordinates). Let $m_j \rightarrow \infty$ satisfy

$$\frac{\pi}{2\varepsilon_j} - m_j = O(1).$$

Then

$$\Phi_{\varepsilon_j, m_j}^{\text{in}}(z) := w_{\varepsilon_j}^{\text{in}}(f_{\varepsilon_j}^{m_j}(z)) - m_j$$

converges locally uniformly on the incoming parabolic basin to the normalized attracting Fatou coordinate Φ_{att} of f_0 .

Write

$$\Phi_{\varepsilon, m}^{\text{in}}(z) = w_\varepsilon^{\text{in}}(z) + \sum_{k=0}^{m-1} A_\varepsilon(f_\varepsilon^k(z)).$$

For $\varepsilon = 0$, Chapter 18's Fatou-coordinate theorem gives

$$\Phi_{\text{att}}(z) = w_0^{\text{in}}(z) + \sum_{k \geq 0} A_0(f_0^k(z)),$$

with locally uniform convergence because $A_0(\zeta) = O(\zeta^2)$ and parabolic orbits satisfy $f_0^k(z) = O(k^{-1})$. Compare the two formulas. The first finitely many terms converge by holomorphic dependence. The remaining tails are uniformly small by the square-summable estimate of the gate-orbit and Abel-defect estimates; the parameter-dependent part contributes at most $O(|\varepsilon| \log(1/|\varepsilon|))$, hence tends to zero. This proves local uniform convergence.

Lemma (Convergence of perturbed outgoing coordinates). If $m'_j \rightarrow \infty$ and $\pi/(2\varepsilon_j) - m'_j = O(1)$, then the reversed almost-Fatou coordinates

$$\Phi_{\varepsilon_j, m'_j}^{\text{out}}(z) := w_{\varepsilon_j}^{\text{out}}(f_{\varepsilon_j}^{-m'_j}(z)) + m'_j$$

converge locally uniformly on the outgoing petal to the normalized repelling Fatou coordinate Φ_{rep} .

Apply the convergence of perturbed incoming coordinates lemma to the inverse maps. The same corrected coordinate and the same square-summable estimates occur, with incoming and outgoing sides interchanged. Since Φ_{rep} is univalent on a repelling petal, the inverse maps of $\Phi_{\varepsilon_j, m'_j}^{\text{out}}$ converge locally uniformly on compact subsets of the image of Φ_{rep} .

Lemma (Exact transit identity). Choose integers $m_j, m'_j \rightarrow \infty$ with $m_j + m'_j = n_j$ and both half-transit conditions of the incoming- and outgoing-coordinate convergence lemmas. If $\sigma_j = n_j - \pi/\varepsilon_j$, then wherever both sides are defined,

$$f_{\varepsilon_j}^{n_j} = (\Phi_{\varepsilon_j, m'_j}^{\text{out}})^{-1} \circ T_{\sigma_j} \circ \Phi_{\varepsilon_j, m_j}^{\text{in}}, \quad T_c(w) = w + c.$$

This is an algebraic consequence of the definitions. The incoming coordinate advances m_j iterates and the outgoing coordinate unwinds m'_j iterates. Their branch normalizations differ by exactly π/ε_j ; therefore the residual translation is $m_j + m'_j - \pi/\varepsilon_j = \sigma_j$. No limiting argument has yet been used.

Lemma (Lavaurs phase limit). Under (18.2), on every compact subset of the incoming basin for which the indicated outgoing inverse branch is defined,

$$f_{\varepsilon_j}^{n_j}(z) \longrightarrow L_{\sigma}(z) := \Phi_{\text{rep}}^{-1}(\Phi_{\text{att}}(z) + \sigma)$$

locally uniformly.

Use the exact transit identity lemma. The convergence of perturbed incoming coordinates lemma gives convergence of the incoming factor, the convergence of perturbed outgoing coordinates lemma gives convergence of the inverse outgoing factor, and $T_{\sigma_j} \rightarrow T_{\sigma}$ uniformly on compact sets. Composition yields the claimed locally uniform limit. This is the Lavaurs phase-limit theorem in the standard simple-parabolic unfolding.

Lemma (Phase dependence and implosion). Different residual phases give the family $L_\sigma = \Phi_{\text{rep}}^{-1} \circ T_\sigma \circ \Phi_{\text{att}}$; modulo the natural integer translation relation, nontrivial horn data make these transition maps genuinely phase dependent. Consequently arbitrarily small parameter perturbations, observed at the corresponding diverging iterate times, realize transitions not visible in any fixed finite iterate of f_0 .

The dependence on σ is explicit in the formula. If two phases gave the same transition on a nonempty open set, univalence of the Fatou coordinates would force their translations to agree modulo the cylinder identifications; otherwise the maps differ. Choose a compact incoming set K on which L_σ lands in a chosen outgoing region. The Lavaurs phase limit lemma gives $f_{\varepsilon_j}^{n_j}(K) \rightarrow L_\sigma(K)$ while $\varepsilon_j \rightarrow 0$ and $n_j \rightarrow \infty$. For every fixed N , $f_{\varepsilon_j}^N \rightarrow f_0^N$ instead, so the long-time transition is absent from fixed finite-time convergence. Varying the limiting phase varies the outgoing transition through the Écalle cylinder and horn map. This is precisely the local dynamical mechanism called parabolic implosion, completing the proof. $\square$

Theorem 18.2 (Leau-Fatou flower theorem). **Statement scope.** A holomorphic germ tangent to identity $f(z) = z + a z^{k+1} + O(z^{k+2})$, $a \neq 0$, has k attracting and k repelling petals alternating around the fixed point, with orbit asymptotics on each petal.

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Blow up the tangent-to-identity germ by the coordinate $w = -1/(ka z^k)$; on large sectors the map becomes $w \rightarrow w + 1 + o(1)$, giving k forward invariant right-half-plane sectors; pull them back to attracting petals, apply the same argument to f^{-1} .

The mathematical argument begins in the proof environment below.

Proof. Translate the fixed point to 0 and write

$$f(z) = z + az^{k+1} + O(z^{k+2}), \quad a \neq 0.$$

Set

$$W(z) = -\frac{1}{kaz^k}.$$

Translation asymptotic.

Uniformly on every closed subsector avoiding the k separating directions,

$$W(f(z)) = W(z) + 1 + O(|W(z)|^{-1/k}) \quad (z \rightarrow 0).$$

Factor $f(z) = z(1 + az^k + O(z^{k+1}))$. Then

$$f(z)^{-k} = z^{-k}(1 + az^k + O(z^{k+1}))^{-k} = z^{-k} - ka + O(z),$$

and multiplication by $-(ka)^{-1}$ gives the formula.

Forward invariant translation sectors.

For sufficiently large R and small fixed $0 < \eta < 1$, the regions

$$H_R^+ = \{w : \Re w > R, |\Im w| < \eta \Re w\}$$

are forward invariant for the transformed map and every orbit in them tends to infinity with $w_n/n \rightarrow 1$.

On H_R^+ the error in Step 1 is $o(1)$. Increase R until its modulus is less than $(1 - \eta)/4$. Then the real part increases by at least $1/2$ and the slope of the imaginary part changes by an amount dominated by the spare angular margin. Thus the sector is invariant. Summing $w_{n+1} - w_n = 1 + o(1)$ and using $|w_n| \asymp n$ gives $w_n/n \rightarrow 1$ by Cesaro.

K attracting petals.

The inverse image of H_R^+ under W has exactly k components $P_0^+, \dots, P_{k-1}^+$; each is simply connected, forward invariant, and $f^n(z) \rightarrow 0$ uniformly on compact subsets.

The map $z \mapsto z^k$ is a k -fold covering of a punctured neighborhood. Therefore $W^{-1}(H_R^+)$ has k components centered on the directions $az^k \in \mathbb{R}_{<0}$. Step 2 gives forward invariance and $|W(f^n(z))| \rightarrow \infty$, which is equivalent to $f^n(z) \rightarrow 0$.

K repelling petals.

There are k repelling petals alternating with the attracting petals.

The local inverse has expansion $f^{-1}(z) = z - az^{k+1} + O(z^{k+2})$. Applying Steps 1-3 to f^{-1} produces k attracting petals for the inverse, hence k repelling petals for f . Their central directions satisfy $az^k \in \mathbb{R}_{>0}$ and therefore alternate with those of the attracting petals.

Orbit asymptotics.

For z in an attracting petal,

$$f^n(z)^k \sim -\frac{1}{kan};$$

for a repelling petal the analogous formula holds under backward iteration.

Step 2 gives $W(f^n(z))/n \rightarrow 1$. Since $W(\zeta) = -1/(ka\zeta^k)$, inversion yields the displayed asymptotic. The inverse-map argument gives the repelling version. Shrinking the petals slightly if necessary makes their closures meet only at 0, completing the alternating Leau-Fatou flower. $\square$

Theorem 18.3 (Fatou coordinates). **Statement scope.** *On each attracting/repelling petal there is a holomorphic coordinate Φ satisfying $\Phi \circ f = \Phi + 1$, unique up to additive constant under normalization.*

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Improve the raw translation coordinate by finitely many asymptotic correction terms so that its Abel defect is summable; telescope the defect along an attracting orbit to obtain a holomorphic Fatou coordinate; use f^{-1} .

The mathematical argument begins in the proof environment below.

Proof. Fix an attracting petal P from RES-001 and use $w = W(z)$. The transformed map has an asymptotic expansion

$$F(w) = w + 1 + c_1 w^{-1/k} + \cdots + c_k w^{-1} + O(|w|^{-1-1/k}).$$

Summable Abel defect.

There is a holomorphic asymptotic coordinate

$$\Psi_0(w) = w + d_1 w^{1-1/k} + \cdots + d_{k-1} w^{1/k} + d_k \log w$$

on a right sector such that

$$\Psi_0(F(w)) - \Psi_0(w) = 1 + O(|w|^{-1-1/k}). \quad (\text{A})$$

Expand each power $(w + 1 + o(1))^{1-j/k}$ and $\log F(w)$ in descending powers of $w^{1/k}$. Choose $d_1, \dots, d_k$ successively to cancel the $w^{-1/k}, \dots, w^{-1}$ terms. The triangular nature of the expansion makes this possible and leaves the stated remainder.

Telescoping construction.

The limit

$$\Phi(z) = \lim_{n \rightarrow \infty} (\Psi_0(W(f^n(z))) - n)$$

exists locally uniformly on P and is holomorphic.

By RES-001, $|W(f^n(z))| \asymp n$ uniformly on compact subsets of P . Equation (A) shows that the difference between successive terms is $O(n^{-1-1/k})$, a summable series. The Weierstrass theorem gives local uniform convergence to a holomorphic limit.

Abel equation and univalence.

The limit satisfies $\Phi \circ f = \Phi + 1$ and is univalent after the petal is shrunk.

Shift the index in Step 2 to obtain the Abel equation. Moreover $\Phi(z) = \Psi_0(W(z)) + o(1)$ as $z \rightarrow 0$ in the petal, so in the w coordinate $\Phi'(w) = 1 + o(1)$. On a sufficiently far right sector it is therefore univalent by the standard close-to-identity criterion; pulling back gives a univalent coordinate on a smaller petal.

Repelling Fatou coordinates.

Every repelling petal carries a holomorphic univalent coordinate Φ_{rep} satisfying $\Phi_{\text{rep}} \circ f = \Phi_{\text{rep}} + 1$ on the part where both sides are defined.

Apply Steps 1–3 to the inverse germ. Negating the resulting inverse Fatou coordinate gives the forward Abel normalization.

Uniqueness after asymptotic normalization.

If Φ_1, Φ_2 are Fatou coordinates on the same petal with $\Phi_j(z) - \Psi_0(W(z)) \rightarrow c_j$, then $\Phi_1 - \Phi_2$ is the constant $c_1 - c_2$.

The difference $D = \Phi_1 - \Phi_2$ satisfies $D \circ f = D$. In the translation coordinate it is a bounded holomorphic 1-periodic function on a right half-plane and has a limit as $\Re w \rightarrow \infty$. Periodicity moves every point arbitrarily far to the right, so D equals that limit identically. Thus Fatou coordinates are unique up to an additive constant under the usual normalization. $\square$

Definition 18.5 (Ecalles cylinders and horn maps). **Statement scope.** Quotienting petals by translation gives cylinders; transition maps between attracting/repelling Fatou coordinates define horn maps.

Theorem 18.4 (Ecalles-Voronin classification). **Statement scope.** Two one-dimensional holomorphic parabolic germs with the same formal normal form are analytically conjugate if and only if their Ecalles-Voronin horn-map data agree modulo the natural changes of Fatou-coordinate origins. Thus the horn-map modulus is a complete analytic conjugacy invariant within a fixed formal class.

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Normalize attracting and repelling Fatou coordinates in a fixed formal class; define the horn maps on the overlap cylinders; prove necessity by conjugating the sectorial coordinates; for sufficiency, equality of horn data makes the sectorial conjugacies agree on every overlap, so they glue to a punctured-neighborhood conjugacy which extends removably across the fixed point.

The mathematical argument begins in the proof environment below.

Proof. Let f and g have the same formal normal form. Choose cyclically ordered attracting and repelling petals and Fatou coordinates $\Phi_{j,\text{att}}^f, \Phi_{j,\text{rep}}^f$ and the analogous coordinates for g , normalized by the same formal asymptotic expansion.

Horn maps are analytic conjugacy invariants.

On an overlap of a repelling and attracting petal, the transition

$$H_j^f = \Phi_{j,\text{att}}^f \circ (\Phi_{j,\text{rep}}^f)^{-1}$$

commutes with $w \mapsto w + 1$ and therefore descends to a germ on the Écalles cylinder. Changing Fatou-coordinate origins conjugates the collection by translations, and an analytic conjugacy of germs preserves its equivalence class.

Commutation follows from the Abel equations. If a Fatou coordinate is changed by $w \mapsto w + c$, the transition is pre/post-composed by the corresponding translations. If $h \circ f = g \circ h$, then $\Phi^g \circ h$ is a Fatou coordinate for f , hence differs from the chosen one by a constant; substituting these constants in the transition formula shows equality of horn data modulo the natural translations.

Sectorial conjugacies from matching moduli.

If the horn data of f and g agree after choices of Fatou-coordinate origins, then on every attracting or repelling petal

$$h_j = (\Phi_j^g)^{-1} \circ \Phi_j^f$$

is a holomorphic conjugacy $h_j \circ f = g \circ h_j$.

This is immediate from the Abel equations and injectivity of the normalized Fatou coordinates.

Overlap agreement.

After the origins have been chosen so that the horn maps are equal, adjacent sectorial conjugacies h_j agree on every connected overlap.

On a repelling/attracting overlap, equality of horn maps is exactly

$$\Phi_{\text{att}}^g \circ (\Phi_{\text{rep}}^g)^{-1} = \Phi_{\text{att}}^f \circ (\Phi_{\text{rep}}^f)^{-1}.$$

Rearranging gives $(\Phi_{\text{att}}^g)^{-1} \Phi_{\text{att}}^f = (\Phi_{\text{rep}}^g)^{-1} \Phi_{\text{rep}}^f$, namely equality of the two sectorial conjugacies.

Gluing and removable extension.

The h_j glue to a holomorphic conjugacy on a punctured neighborhood of 0, and this conjugacy extends holomorphically across 0 with nonzero derivative.

The Leau–Fatou petals cover a punctured neighborhood, so Step 3 gives a single holomorphic map h there. The common formal normalization yields $h(z) = z + o(z)$ in each petal, hence h is bounded near 0 and has a removable singularity there. Setting $h(0) = 0$ gives $h'(0) = 1$, so the extension is a local biholomorphism. The conjugacy identity extends by analyticity.

Step 1 proves necessity and Steps 2–4 prove sufficiency. Therefore, within a fixed formal normal form, the horn-map modulus modulo translation of Fatou-coordinate origins is a complete analytic conjugacy invariant. $\square$

18.3 Worked examples

Worked example: A model parabolic translation coordinate

For $f(z) = z + z^2 + O(z^3)$, set $w = -1/z$. A direct expansion gives $w(f(z)) = w + 1 + O(1/w)$. Correcting w by lower-order logarithmic terms and summing the remaining Abel defect produces a Fatou coordinate Φ satisfying $\Phi \circ f = \Phi + 1$ on an attracting petal.

Worked example: Additive ambiguity and horn maps

If Φ is a Fatou coordinate, so is $\Phi + C$. Consequently an attracting-to-repelling transition changes by translations when the two coordinates are renormalized. Passing to $e^{2\pi i \Phi}$ descends this transition to a punctured-disk horn map; its equivalence class removes the arbitrary additive constants.

Boundary case: Same formal normal form, different analytic germ

Two parabolic germs can share the same multiplier, multiplicity, and formal residue but have non-equivalent horn data. They are formally conjugate yet not analytically conjugate. This is exactly why Ecalle–Voronin moduli are necessary.

18.4 What this chapter established

The chapter closed its formal theorem spine at the stated source scopes. The proofs deliberately separate elementary deductions from deep imported theorems: every imported theorem is named at the point where it enters, and later conclusions use only the scope actually established.

For study purposes, the most important task is to be able to reconstruct the dependency chain rather than memorize theorem names. In particular, note which hypotheses supply compactness or a priori geometry, which step supplies rigidity, and which step converts dynamical information into parameter or statistical conclusions.

Looking ahead

Chapter 19 uses precisely these Fatou-coordinate and parabolic-transit tools to define near-parabolic and pacman renormalizations.

Near-Parabolic, Inou-Shishikura and Pacman Renormalization

Why this chapter matters

Chapter 19 turns the local neutral theory of Chapters 17–18 into a renormalization machine. Inou-Shishikura near-parabolic renormalization packages a long parabolic transit into a new normalized map. Pacman renormalization adds a marked cut so that neutral combinatorics can be followed analytically in a Banach-space phase space.

The chapter has a geometric and a measure-theoretic payoff. Uniform pseudo-Siegel bounds prevent neutral towers from degenerating, pacman hyperbolicity produces dynamical and parameter self-similarity, and rational approximants converge to the neutral renormalization limit. Finally, the Avila-Lyubich black-hole mechanism combines controlled landing with very small escape to construct positive-area Feigenbaum Julia sets and a positive-dimensional parameter family.

Learning goals

By the end of this chapter you should be able to:

1. describe the normalized near-parabolic invariant class;
2. explain how pacman marking removes neutral ambiguity;
3. derive self-similarity from a one-dimensional unstable coordinate;
4. use pseudo-Siegel bounds to control neutral sector geometry;
5. explain rational-approximant convergence through Ecalle transit;
6. separate landing and escaping probabilities in the black-hole criterion.

Conceptual roadmap

Fatou coordinates $\rightarrow$ IS invariant class

$\rightarrow$ pacman hyperbolicity

$\rightarrow$ neutral self-similarity.

Uniform pseudo-Siegel bounds feed both the neutral compactness theory and the high-type construction. The positive-area branch then uses η (landing) and ξ (escape): fixed positive landing plus sufficiently small escape forces positive planar area.

Common pitfall

Hyperbolicity alone does not imply positive area. The black-hole theorem needs a separate measure recursion and a source-specific construction that realizes its geometric hypotheses.

19.1 Definitions

Definition 19.1 (near-parabolic renormalization). Near-parabolic renormalization is a renormalization construction for maps with multiplier close to a root of unity/one, built using Fatou coordinates and a dynamically selected return before rescaling.

Definition 19.2 (pacman map). A pacman map is a holomorphic dynamical object defined on a disk-like domain with a slit/mouth and specified critical orbit structure, used as the state space for pacman renormalization.

Definition 19.3 (satellite renormalization). Satellite renormalization is polynomial-like renormalization whose little Julia sets meet in a satellite pattern around a common periodic point, as opposed to primitive disjoint combinatorics.

Definition 19.4 (neutral a priori bound). A neutral a priori bound is a uniform modulus/geometric lower bound for renormalizations approaching neutral/Siegel/parabolic regimes, with arithmetic/combinatorial scope stated.

Definition 19.5 (external extremal width). For a combinatorial boundary interval $I \subset \partial Z$ and an enlargement λI , the *external extremal width* $W_\lambda^+(I)$ is the extremal width of curves in $\widehat{\mathbb{C}} \setminus Z$ joining I to $\partial Z \setminus (\lambda I)$. It is the reciprocal of the corresponding extremal length and measures all-scale boundary oscillation without assuming that Z is a uniformly controlled quasidisk.

Definition 19.6 (geodesic pseudo-Siegel disk). A *geodesic pseudo-Siegel disk* $\widehat{Z}_m \supset Z$ is obtained by filling selected parabolic fjords of the Siegel disk at scales at least m with hyperbolic-geodesic caps, in such a way that the closest returns up to time q_{m+1} remain injective and the added caps have uniformly controlled collars. The regularized disk is almost invariant; it need not coincide with the actual Siegel disk.

Definition 19.7 (sector-renormalization a priori bound). A *sector-renormalization a priori bound* is a uniform compactness statement for normalized sector first-return pairs: the renormalization domains contain a fixed-modulus collar around the marked regularized neutral core, the univalent return branches have uniform Koebe control there, and the unique critical branch has uniformly controlled degree-two geometry.

Definition 19.8 (exponent-two transfer weight). For a nice level annulus A^m and a chosen family of admissible inverse branches ending in A^m , the *exponent-two transfer weight* is the sum of $|Dg|^2$ over those inverse branches g . A truncation $\omega_m^{[j]}(2)$ retains only paths with at most j designated returns; $\omega_m(2)$ is the monotone limit. At exponent two, integration of the weight is planar area by change of variables.

Definition 19.9 (safe trapping disk). A *safe trapping disk* is a Jordan disk compactly contained in the complement of the postcritical set and the relevant domain boundary, with uniformly bounded hyperbolic diameter, such that all orbit segments used in the trapping argument admit univalent pullbacks across a fixed Koebe collar.

Definition 19.10 (safe-return depth). Given a finite safe family, the *safe-return depth* of a landing branch is the number of visits to that family before first landing in the designated renormalization core.

Definition 19.11 (multiscale porosity certificate). A measurable set X has an (n, κ, L) *multiscale porosity certificate* in a disk B if every $x \in X$ has n geometrically separated scales $r_1 > Lr_2 > \dots > L^{n-1}r_n$ at which a disk of radius at least κr_i inside $B(x, r_i)$ is disjoint from X .

Definition 19.12 (bounded-geometry parameter tree). A two-branch *bounded-geometry parameter tree* consists of two inverse parameter branches with disjoint images, uniform distortion, uniform lower and upper child-to-parent diameter ratios, and uniform sibling separation. Its limit set is a Cantor set; positive Hausdorff dimension is a theorem, not part of the definition.

Definition 19.13 (nice fundamental pair). At renormalization level m , a *nice fundamental pair* $U^m \Subset V^m$ is a pair of simply connected domains for which the postcritical set in V^m is contained in U^m , no forward image of ∂V^m re-enters V^m , every noncentral first-landing component maps univalently onto V^m and extends across a uniform collar, and the annulus $A^m = V^m \setminus \overline{U^m}$ has uniform hyperbolic/area geometry.

Definition 19.14 (admissible transfer path). Relative to nice fundamental pairs, an *admissible transfer path* is a finite inverse itinerary obtained by concatenating first-landing and first-return branches, with the central degree-two branch used only at its prescribed renormalization passage and all other pieces univalent on uniform collars. Its return count defines the transfer truncation.

19.2 Main results

Theorem 19.1 (Approximation of Siegel by parabolic dynamics). **Statement scope.** At the high-type/near-parabolic arithmetic scope where the Inou-Shishikura invariant class applies, let $f_j = e^{2\pi i \alpha_j} h_j$ be the terminal maps associated with rational/parabolic approximants, with $\alpha_j \in \mathbb{R} \setminus \{0\}$, $\alpha_j \rightarrow 0$, and $h_j \in \mathcal{F}_1$. Normalize the Fatou coordinates in the Inou-Shishikura convention and choose $m_j \in \mathbb{Z}$ so that the residual Ecalle phase $\sigma_j = m_j - 1/\alpha_j$ lies in a fixed fundamental interval. Then the lifted normalized cylinder returns form a normal family on every common compact horn domain. If $h_j \rightarrow h_0$ and $\sigma_j \rightarrow \sigma$ along a subsequence, that subsequence converges locally uniformly to $T_\sigma \circ \tilde{E}_{h_0}$ for the normalized horn lift, equivalently to the corresponding phase- σ Lavaurs/near-parabolic return in physical Fatou coordinates. No phase-independent full-sequence convergence is asserted when the residual phases do not converge.

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Exact Ecalle-cylinder transit $[w] \mapsto [w - 1/\alpha]$, compact normalized horn maps, and the Lavaurs phase limit proved earlier identify every phase-convergent subsequential return; varying phases need not have one full-sequence limit.

The mathematical argument begins in the proof environment below.

Proof. After the common continued-fraction prefix has been renormalized away, write the terminal near-parabolic maps associated with the rational/parabolic approximants as

$$f_j(z) = e^{2\pi i \alpha_j} h_j(z), \quad \alpha_j \in \mathbb{R} \setminus \{0\}, \quad \alpha_j \rightarrow 0, \quad h_j \in \mathcal{F}_1.$$

This is the stage at which a parabolic-approximating rational truncation is about to terminate. The Inou-Shishikura class supplies a fixed normalized model domain. Montel compactness therefore lets us pass to a subsequence with $h_j \rightarrow h_0$ locally uniformly. The normalization $h_j(0) = 0$ and $h'_j(0) = 1$ rules out collapse to a constant, while $e^{2\pi i \alpha_j} \rightarrow 1$ makes h_0 the limiting parabolic germ.

Let E_j be the normalized horn map and let $\tilde{E}_j$ be its lift fixed by

$$\tilde{E}_j(w) = w + o(1) \quad (\text{Im } w \rightarrow +\infty).$$

The Fatou-coordinate construction of Chapter 18 is uniform on compact families of simple near-parabolic germs, so on every compact subset of a common lifted horn domain,

$$\tilde{E}_j \rightarrow \tilde{E}_0.$$

The sensitive part of the perturbed return is completely explicit. If 0 and ζ_j are the two nearby fixed points, the punctured neighborhood is uniformized by

$$\tau_j(w) = \frac{\zeta_j}{1 - e^{-2\pi i \alpha_j w}}.$$

Since $\tau_j(w + 1/\alpha_j) = \tau_j(w)$, the attracting-to-repelling passage changes the lifted Fatou coordinate by one deck period. With the standard upper-end normalization, the induced cylinder transit is

$$\chi_j([w]) = [w - 1/\alpha_j] \quad \text{on } \mathbb{C}/\mathbb{Z}.$$

Hence the return $H_j = \chi_j \circ E_j$ has the lift

$$\tilde{H}_j(w) = \tilde{E}_j(w) - \frac{1}{\alpha_j}.$$

Choose the integer m_j so that the residual phase

$$\sigma_j = m_j - \frac{1}{\alpha_j}$$

lies in the fixed fundamental interval $[-1/2, 1/2)$. The corresponding integer-normalized lift satisfies the exact identity

$$\widehat{H}_j := T_{m_j} \circ \widetilde{H}_j = T_{\sigma_j} \circ \widetilde{E}_j, \quad T_c(w) = w + c.$$

The stable horn-map factor and the residual Ecalle phase are therefore cleanly separated.

Now take any sequence. First extract $h_j \rightarrow h_0$, and then extract $\sigma_j \rightarrow \sigma$ from the compact fundamental interval. Since $\widetilde{E}_j \rightarrow \widetilde{E}_0$ locally uniformly,

$$\widehat{H}_j \rightarrow T_\sigma \circ \widetilde{E}_0$$

locally uniformly on every compactly contained common lifted horn domain. Thus the normalized returns form a normal family. If the residual phase already converges along a specified subsequence, no further phase extraction is needed and the same formula identifies its limit.

In physical Fatou coordinates Chapter 18 identifies T_σ with the phase- σ Lavaurs transit

$$L_\sigma = \Phi_{\text{rep},0}^{-1} \circ T_\sigma \circ \Phi_{\text{att},0}.$$

The horn lift $\widetilde{E}_0 = \Phi_{\text{att},0} \circ \Phi_{\text{rep},0}^{-1}$ records the parabolic horn segment, so $T_\sigma \circ \widetilde{E}_0$ is exactly the lifted cylinder form of that segment followed by the corresponding Lavaurs transition. Changing Fatou coordinates only conjugates this lifted return by a translation; the standard critical-point and upper-end normalizations fix the representative.

Finally, two different limiting residual phases modulo $\mathbb{Z}$ give different lifted limits. Equality of $T_\sigma \circ \widetilde{E}_0$ and $T_\tau \circ \widetilde{E}_0$ on an open set would force $\sigma \equiv \tau \pmod{\mathbb{Z}}$. Therefore the correct conclusion is normality together with phase-labelled subsequential limits, not an unconditional phase-independent full-sequence limit. $\square$

Theorem 19.2 (Positive-area Feigenbaum Julia sets). **Statement scope.** *There exists a Feigenbaum quadratic polynomial $P_c(z) = z^2 + c$ with primitive stationary combinatorics whose Julia set has positive planar Lebesgue measure. Moreover, the set of Feigenbaum quadratic parameters c for which $J(P_c)$ has positive area has positive Hausdorff dimension in parameter space.*

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Exponent-two transfer and zero-area comparison give the Black-Hole criterion; phase-controlled high-type seeds provide uniform a priori bounds and definite landing; multiscale safe-trap porosity makes escape arbitrarily small under the zero-area alternative; a two-branch bounded-geometry parameter tree gives positive Hausdorff dimension.

The mathematical argument begins in the proof environment below.

Proof. The proof has two logically distinct parts. First we prove a quantitative Black-Hole criterion from exponent-two transfer estimates. We then construct a high-type primitive renormalization family for which landing has a uniform positive density while

escape can be made arbitrarily small. A final renormalization-horseshoe argument gives a positive-dimensional parameter set.

The exponent-two Black-Hole comparison. Fix a primitive Feigenbaum quadratic-like map with one a priori modulus bound and write

$$U^m \subseteq V^m, \quad A^m = V^m \setminus \overline{U^m}$$

for nice renormalization levels. Let η_m be the normalized area of points which eventually land in the next critical level and let ξ_m be the normalized area of points which escape from the level- m fundamental annulus before returning. If ρ_m denotes the area scale of that annulus and $\omega_m(2)$ the exponent-two transfer weight, then bounded distortion on the uniform collars gives the recursion

$$\omega_m^{[j+1]}(2) \leq C \frac{\eta_m}{\rho_m} + (1 - C^{-1}(\eta_m + \xi_m) + C\eta_m\xi_m)\omega_m^{[j]}(2) + C\xi_m\rho_m(\omega_m^{[j]}(2))^2.$$

The three terms correspond respectively to first landing, return without loss, and two successive return blocks. At exponent two this is an area identity: for a univalent inverse branch g , its Jacobian is $|Dg|^2$; Koebe distortion only changes the universal constant.

When the landing/escape ratio is sufficiently unbalanced, the smaller positive fixed point of the quadratic polynomial on the right gives

$$\omega_m(2) \leq C \frac{\eta_m}{(\eta_m + \xi_m)\rho_m}. \quad (19.1)$$

If, on the other hand, $J(f)$ had zero area, the escaping set in the fundamental annulus is wandering until exit. Change of variables and bounded oscillation of the transfer series then give

$$\omega_m(2) \asymp \frac{\eta_m}{\xi_m\rho_m}. \quad (19.2)$$

Indeed the union of all inverse images of the escaping set fills, up to a null Julia-set remainder, the points that land at level m only finitely many times. The numerator in the resulting area ratio is therefore the landing area and the denominator is the escaping area.

Comparing (19.1) and (19.2) proves the Black-Hole implication: for every fixed geometric bound there is $C_{\text{BH}} > 1$ such that

$$\eta_m \geq C_{\text{BH}}\xi_m \implies \text{area } J(f) > 0.$$

If the Julia set were null, the lower estimate would grow like $1/\xi_m$ while the upper estimate grows only like $1/(\eta_m + \xi_m)$, an impossibility once η_m/ξ_m is large enough.

High-type primitive seeds. We next produce maps for which the Black-Hole inequality holds. Start from the controlled neutral class developed earlier in this chapter and choose a deep safe level. The phase-controlled parabolic-approximation theorem supplies rational/near-parabolic maps whose normalized cylinder transit is

$$[w] \mapsto [w + \sigma]$$

after removing the large integer part of the return. Choose a postcritical entry point a and a precritical target b in compact subsets of the attracting and repelling cylinders. As σ winds through one period, $T_\sigma(a)$ winds once around the repelling cylinder. The argument principle therefore selects parameters for which the transported critical point lands at b .

For a long terminal transit the resulting first return is a proper unfolded degree-two polynomial-like family. The orbit pieces before the terminal return are disjoint, hence the renormalization is primitive; straightening identifies its connectedness locus with a primitive Mandelbrot copy. The entry and exit collars stay in one compact family while the cylinder length tends to infinity, so extremal length gives

$$\text{mod}(V \setminus \overline{U}) \rightarrow \infty.$$

Choosing the transit sufficiently long and applying the straightening distortion bound inductively yields one stationary a priori modulus lower bound $\nu > 0$ at every subsequent level.

Definite landing. Choose a safe disk D around a controlled orbit point of the limiting neutral map, separated from the postcritical set. Pulling a deeper renormalization range back through the first safe visit gives

$$U' \Subset V' \Subset D$$

with a fixed Koebe collar and fixed relative area. These data persist for all sufficiently high-type approximants. Take a positive-area set E in the neutral reference domain. Every orbit from E that exits toward the exterior must cross the safe disk. Pull back $U' \Subset V'$ along its first crossing. The relative-area lower bound survives by Koebe distortion. A Vitali selection of the resulting disks has bounded overlap, and summing the density estimate gives

$$\eta \geq \eta_0 > 0,$$

where η_0 is independent of the long terminal transit.

Vanishing escape. The complementary estimate uses many safe traps. A simple porosity lemma says that if a planar set misses a disk of fixed relative size at n geometrically separated scales, then its area is at most $C\sigma^n$ times the ambient disk area for some $0 < \sigma < 1$. The proof is the usual dyadic stopping argument: at every certified scale a fixed fraction of each occupied square is removed, and separated scales allow those losses to be multiplied.

Successive safe returns produce the required separated scales. In the hyperbolic metric of the postcritical complement each transition expands by a uniform factor; Koebe distortion converts this to Euclidean contraction of the inverse pullbacks. If $\mathcal{P}_n$ denotes landing domains of safe-return depth at most n , bounded multiplicity and change of variables give

$$\sum_{P \in \mathcal{P}_n} \text{area}(P) \leq Cn \text{area}(D), \quad \sum_{P \in \mathcal{P}_n} |DT_P(\zeta_P)|^{-2} \leq Cn.$$

As the terminal high-type parameter increases, arbitrarily many safe traps from

the limiting neutral geometry persist. A point which escapes without returning must avoid a fixed relative subdisk in every one of those traps. The porosity lemma therefore makes the base escaping area exponentially small in the number of traps. Pulling this estimate back through domains of return-depth at most n costs only the factor Cn above; points with more than n returns have another exponentially small porosity bound. First choose n large and then choose the number of available traps large. If the selected seed already has positive-area Julia set, the existence claim is finished. Otherwise its Julia set is null, so the residual Julia-set part of the escape decomposition contributes no area. A bounded-multiplicity Besicovitch pullback from D to the fundamental annulus then yields

$$\xi < \varepsilon$$

for any prescribed $\varepsilon > 0$ under this zero-area alternative.

Fix the geometric controls. They determine C_{BH} and η_0 . Use the preceding escape construction with $\varepsilon = \eta_0/C_{\text{BH}}$. Either the selected seed already has positive-area Julia set, or the zero-area alternative gives $\xi < \eta_0/C_{\text{BH}}$ and the Black-Hole criterion contradicts zero area. Thus in either case we obtain a primitive stationary Feigenbaum quadratic-like map with positive-area Julia set. Its straightening is a quadratic polynomial with the same primitive stationary combinatorics. A quasiconformal homeomorphism and its inverse have Lusin's N property, and maps the quadratic-like Julia set onto the Julia set of the straightened polynomial. Hence a positive-measure Julia set cannot be carried to a null Julia set, so the quadratic Julia set has positive area. This proves the first assertion.

A positive-dimensional parameter set. The construction leaves infinitely many primitive Mandelbrot copies with the same modulus lower bound, the same landing lower bound, and a common escape upper bound $\xi_0 < \eta_0/C_{\text{BH}}$. The transfer comparison is local in the renormalization level: its constants depend on the common geometry, not on a stationary symbolic word. Therefore every infinitely renormalizable map whose successive symbols are chosen from a fixed finite subfamily satisfies the same Black-Hole contradiction and has positive-area Julia set.

Choose two such primitive copies. Renormalization hyperbolicity and transversality give two inverse parameter branches on one parameter disk. After shrinking once, their images are disjoint and there are constants $0 < r_- \leq r_+ < 1$ and $\delta > 0$ such that every parameter cylinder Q_w obeys

$$r_- \text{diam } Q_w \leq \text{diam } Q_{wi} \leq r_+ \text{diam } Q_w, \quad \text{dist}(Q_{w0}, Q_{w1}) \geq \delta \text{diam } Q_w.$$

Give each level- n cylinder mass 2^{-n} and let μ be the limiting Bernoulli measure. Uniform separation implies that a disk at a level- n scale meets only boundedly many level- n cylinders. With, for example,

$$s_0 = \min \left\{ 1, \frac{\log 2}{2 \log(1/r_-)} \right\} > 0,$$

the scale bounds imply

$$\mu(B(z, r)) \leq Cr^{s_0}.$$

By the mass-distribution principle the limit Cantor set has Hausdorff dimension at least $s_0 > 0$. Every parameter in it has primitive bounded Feigenbaum combinatorics, uniform a priori bounds, and positive-area Julia set. Hence the set of positive-area Feigenbaum quadratic parameters has positive Hausdorff dimension, completing the proof. $\square$

Theorem 19.3 (Inou-Shishikura near-parabolic invariant class). **Statement scope.** Let $P(z) = z(1+z)^2$ and let $\mathcal{F}_1$ be the normalized Inou-Shishikura class associated with fixed Jordan domains $V \Subset V'$. Parabolic renormalization $\mathcal{R}_0$ preserves $\mathcal{F}_1$; if $f = P \circ \phi^{-1}$, then $\mathcal{R}_0 f = P \circ \psi^{-1}$ with ψ normalized univalent on the larger domain V' , and the construction is holomorphic in holomorphic families. Under the Teichmüller metric induced from $\text{Teich}(\widehat{\mathbb{C}} \setminus V)$,

$$d(\mathcal{R}_0 f, \mathcal{R}_0 g) \leq \lambda d(f, g), \quad \lambda = e^{-2\pi \text{mod}(V' \setminus \overline{V})} < 1.$$

There exists $\alpha_* > 0$ such that, for every sufficiently small admissible complex multiplier parameter α in either Inou-Shishikura sector, the fibre renormalization $\mathcal{R}_\alpha$ is defined on all of $\mathcal{F}_1$, has the same V' extension and holomorphic dependence, and satisfies the same contraction factor λ . For real α the resulting skew product $(\alpha, h) \mapsto (T(\alpha), \mathcal{R}_\alpha h)$ is hyperbolic on the stated small-multiplier phase space. In particular, for sufficiently high continued-fraction type the successive renormalizations are defined and fibres over the same base itinerary converge exponentially.

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Build the fixed normalized class from the horn-map covering geometry, convert the larger extension domain into a strict Teichmüller contraction by a modulus-area estimate, continue the same geometry to small nonzero multipliers, and combine uniform fibre contraction with the expanding continued-fraction base map.

The mathematical argument begins in the proof environment below.

Proof. We prove the theorem in four stages: construct the invariant parabolic class, turn its larger extension domain into a strict Teichmüller contraction, continue the construction to small nonzero multipliers, and finally combine that fibre contraction with expansion in the continued-fraction coordinate.

The fixed model and parabolic invariance. Put

$$P(z) = z(1+z)^2, \quad c_P = P(-1/3) = -4/27.$$

Choose the fixed Inou-Shishikura domains

$$0, -\frac{1}{3} \in V \Subset V'.$$

The finite geometric verification used to choose these domains gives three facts that will be used below: $V' \setminus \overline{V}$ is an annulus of positive modulus; all distinguished inverse branches needed for the horn construction are defined on the prescribed model pieces;

and those pieces have the same incidence pattern for every normalized coordinate in the class. Define

$$\mathcal{F}_1 = \left\{ P \circ \phi^{-1} \left| \begin{array}{l} \phi : V \rightarrow \mathbb{C} \text{ is univalent,} \\ \phi(0) = 0, \quad \phi'(0) = 1, \\ \phi \text{ has a quasiconformal extension to } \mathbb{C} \end{array} \right. \right\}.$$

For $f = P \circ \phi^{-1}$, Chapter 18 supplies normalized attracting and repelling Fatou coordinates. The point that must still be proved is that the horn return has the special form $P \circ \psi^{-1}$ with ψ extending to the *larger* domain V' .

It is convenient to use

$$\psi_0(z) = -\frac{4}{z}, \quad \psi_1(z) = -\frac{4z}{(1+z)^2},$$

which transform the polynomial model into a rational one. In these coordinates the distinguished inverse branch becomes single-valued on a fixed finite-sheeted Riemann surface X . If $g : X \rightarrow X$ denotes this lifted inverse branch, then the repelling Fatou coordinate extends from its initial petal to all of X by

$$\tilde{\Phi}_{\text{rep}}(g\zeta) = \tilde{\Phi}_{\text{rep}}(\zeta) - 1.$$

Every point eventually enters the original repelling petal under iteration of g , so the extension is well defined and injective.

On the attracting side, normalize the Fatou coordinate by the critical value. The fixed model geometry provides a strip in the Fatou coordinate and a finite family of initial dynamical pieces. Pulling those pieces backwards by the distinguished branches produces a family whose overlaps reproduce exactly the sheet identifications of P . On each piece take the branch of P^{-1} with the matching sheet label and solve

$$P \circ \Psi_1 = c_P e^{2\pi i \tilde{\Phi}_{\text{attr}}}.$$

On ordinary seams the two branches agree because P is univalent there; at the unique critical seam the two sides have the same local degree. Hence the branches glue to one holomorphic map Ψ_1 . Its fibres are precisely the orbits of g .

Consequently

$$\psi = \Psi_0 \circ \tilde{\Phi}_{\text{rep}} \circ \Psi_1^{-1}, \quad \Psi_0(w) = c_P e^{2\pi i w},$$

is well defined on $V' \setminus \{0\}$: replacing an inverse of Ψ_1 by an iterate of g changes $\tilde{\Phi}_{\text{rep}}$ by an integer and therefore does not change the exponential. The converse orbit-identification property makes ψ injective. The normalized horn asymptotic at the upper end gives

$$\psi(0) = 0, \quad \psi'(0) = 1,$$

so the singularity at 0 is removable. Finally the defining identities of the two Fatou coordinates give

$$P \circ \psi^{-1} = \Psi_0 \circ E_f \circ \Psi_0^{-1},$$

which is the normalized parabolic renormalization. Thus $\mathcal{R}_0 f = P \circ \psi^{-1} \in \mathcal{F}_1$, with ψ

univalent on V' . All ingredients are defined by normalized holomorphic coordinates and finitely many implicit inverse branches, so the construction varies holomorphically in a holomorphic family.

Why the larger domain forces contraction. Let

$$W = \widehat{\mathbb{C}} \setminus V.$$

A quasiconformal extension of ϕ determines a point of $\text{Teich}(W)$. This gives a bijection

$$\mathcal{F}_1 \simeq \text{Teich}(W).$$

Indeed, a Beltrami coefficient on W may be extended by zero across V and solved by the measurable Riemann mapping theorem; after the normalization at 0 the solution is conformal on V and gives the required ϕ . Conversely, two normalized coordinates representing the same marked Teichmüller point glue to a global conformal self-map of the sphere; the value and derivative normalization force that map to be the identity.

Choose a regular intermediate domain $V \Subset V'' \Subset V'$ and put $U = \widehat{\mathbb{C}} \setminus V''$. The preceding construction gives a holomorphic map from $\text{Teich}(W)$ to $\text{Teich}(U)$, hence a nonexpanding one. Returning to $\text{Teich}(W)$ amounts to extending the Beltrami coefficient by zero across the annulus $V'' \setminus \overline{V}$. We now quantify the gain from that annulus.

Let A be an annulus and let K be its bounded complementary component. For an integrable holomorphic quadratic differential $q(z) dz^2$ on $A \cup K$ (with at most the allowed simple pole at the puncture), the Carleman area-boundary inequality and extremal length give

$$\int_K |q| dA \leq e^{-2\pi \text{mod } A} \int_{A \cup K} |q| dA.$$

One convenient way to see the exponential constant is to divide A into n concentric subannuli of equal modulus. The one-step area-boundary estimate on each subannulus yields the factor $1 + 2\pi \text{mod}(A)/n$; multiplying the n inequalities and letting $n \rightarrow \infty$ gives $e^{2\pi \text{mod } A}$ on the reciprocal side.

The cotangent norm of Teichmüller space is the L^1 norm of integrable quadratic differentials. The pre-adjoint of zero extension is restriction, so the preceding estimate bounds its norm by $e^{-2\pi \text{mod}(V'' \setminus \overline{V})}$. Duality gives the same bound for the derivative of zero extension, and integration along paths gives the metric estimate. Letting V'' exhaust V' therefore yields

$$d(\mathcal{R}_0 f, \mathcal{R}_0 g) \leq \lambda d(f, g), \quad \lambda = e^{-2\pi \text{mod}(V' \setminus \overline{V})} < 1.$$

This proves both the invariant-class statement and the strict parabolic contraction.

Small nonzero multipliers. Write

$$f_{\alpha, h}(z) = e^{2\pi i \alpha} h(z), \quad h \in \mathcal{F}_1.$$

For $\alpha \neq 0$ small, let $\sigma(\alpha, h)$ be the second fixed point near 0 and introduce the pre-Fatou coordinate

$$\tau_{\alpha, h}(w) = \frac{\sigma(\alpha, h)}{1 - e^{-2\pi i \alpha w}}.$$

On every fixed sector away from the lattice of poles, the lifted maps converge locally uniformly to the parabolic lift as $\alpha \rightarrow 0$ through either of the two admissible sectors. Their normalized Fatou coordinates therefore converge on compact sets as well: normality comes from the uniform petal geometry, the Abel equation passes to the limit, and the normalization removes the additive ambiguity.

Choose the finitely many entry and exit pieces of the parabolic construction strictly inside their limiting sectors. For sufficiently small $|\alpha|$ the corresponding perturbed pieces persist. Rouché's theorem preserves the required inverse branches and the unique degree-two critical seam, so the same gluing construction gives

$$\mathcal{R}_\alpha h = P \circ \psi_{\alpha,h}^{-1}$$

with $\psi_{\alpha,h}$ normalized univalent on the same V' . The allowable size of α is uniform over $h \in \mathcal{F}_1$: otherwise take a sequence of normalized univalent coordinates with failure parameters tending to zero; Koebe compactness gives a locally uniform limit, while the finite collection of strict Rouché neighborhoods for that limit persists for all sufficiently large members of the sequence, a contradiction. The construction is holomorphic in (α, h) on each branch.

Because the extension annulus is still exactly $V' \setminus \bar{V}$, the Teichmüller argument above is independent of α . Hence every admissible small fibre satisfies

$$d(\mathcal{R}_\alpha h_1, \mathcal{R}_\alpha h_2) \leq \lambda d(h_1, h_2)$$

with the same $\lambda < 1$.

Hyperbolicity of the skew product. For real irrational α let $a(x)$ denote the nearest integer and put

$$T(\alpha) = a(1/|\alpha|) - 1/|\alpha|.$$

On each smooth branch, $|T'(\alpha)| = |\alpha|^{-2}$. Near-parabolic renormalization can therefore be written

$$\mathcal{R} : (\alpha, h) \mapsto (T(\alpha), \mathcal{R}_\alpha h).$$

The derivative has triangular form

$$D\mathcal{R}_{(\alpha,h)} = \begin{pmatrix} T'(\alpha) & 0 \\ * & D_h \mathcal{R}_\alpha \end{pmatrix}, \quad \|D_h \mathcal{R}_\alpha\| \leq \lambda < 1.$$

Holomorphic dependence bounds the mixed entry on each compact branch strip. After shrinking α_* , the base expansion $|T'| \geq \alpha_*^{-2}$ dominates that mixed term. A standard horizontal cone is then forward invariant and expanded, while the vertical Teichmüller bundle is invariant and contracted. This is the required hyperbolic splitting.

Finally, if two maps have the same high-type base itinerary, iteration of the fibre inequality gives

$$d(h_n, \tilde{h}_n) \leq \lambda^n d(h_0, \tilde{h}_0).$$

Choosing the continued-fraction coefficients uniformly large keeps every base iterate in the small-multiplier region, so the whole renormalization tower is defined. This proves the theorem. $\square$

Theorem 19.4 (Pacman renormalization hyperbolicity). **Statement scope.** *Let θ be combinatorially periodic under the pacman rotation-number map and pass to the corresponding period iterate of pacman renormalization. In the normalized Dudko-Lyubich-Selinger Banach chart there is a uniquely normalized Siegel fixed pacman f_* of rotation number θ . The fixed point is hyperbolic and its unstable manifold is one complex dimensional. Locally, its stable manifold is the codimension-one slice of pacmen having the same multiplier at the marked fixed point; every pacman in that stable slice is Siegel. The multiplier is a local analytic coordinate on the unstable manifold.*

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Promote the periodic Siegel model to a compact analytic pacman self-operator, control nearby renormalization towers by quasiconformal tilings, use maximal anti-renormalizations and critical-orbit holomorphic motion to force a one-dimensional unstable manifold, then exclude neutral spectrum on the same-multiplier slice by exponential same-rotation contraction and a small-orbits argument.

The mathematical argument begins in the proof environment below.

Proof. Fix a combinatorially periodic irrational rotation number θ . Replace the pacman renormalization operator by the iterate corresponding to one full period of the rotation-number itinerary. We denote this period iterate again by $\mathcal{R}$. Thus the target periodic point becomes a fixed point and the rotation-number branch fixes θ . At the end we undo this notational replacement to recover the periodic statement.

1. Promotion of the periodic Siegel model to pacman space. Theorem 16.27 supplies the normalized periodic critical-circle/cylinder renormalization model of rotation number θ . Since periodic continued fractions are bounded type, Theorem 17.1 supplies the corresponding quasicircle/Siegel realization with one marked critical point on the Siegel boundary. Cut the dynamical plane along the marked critical access. The two first-return branches on the complementary sector form a standard prepacman

$$F_* = (f_{*,-} : U_{*,-} \rightarrow S_*, f_{*,+} : U_{*,+} \rightarrow S_*).$$

Gluing the two boundary arcs of S_* by the return identification and fixing the marked fixed point, critical value and outer normalization gives a pacman $f_* : U_* \rightarrow V_*$. Because one full combinatorial period has been taken, the glued return has the same normalized Siegel model as f_* . Hence $\mathcal{R}f_* = f_*$.

To obtain the compact containment, begin with a standard sector whose outer part lies in a fixed collar of the Siegel boundary. The inverse gluing branch which returns this sector to the Siegel model is expanding on that boundary. Pulling the sector back through a sufficiently large fixed number t of these inverse gluing branches makes its part outside the Siegel disk arbitrarily small. The closure of the resulting pre-renormalization domain is therefore contained in $f_*^{-1}(U_*)$. All normalizations are marked, so no residual conformal automorphism remains. This gives both (19.12) and uniqueness of the standard lift within the chosen normalized chart.

Indeed, the finitely many first-return branches and the two landing arcs persist for f

close to f_* . Their quotient surface varies holomorphically. Normalize the quotient map by the three marked data used above. The normalized solution of the corresponding Beltrami/Riemann-mapping problem then varies holomorphically, so the glued first return defines an analytic operator. By (19.12), it factors as an analytic renormalization on a slightly larger domain followed by restriction to a strictly smaller one. Montel compactness makes that restriction compact, hence DR_{f_*} is a compact operator and its spectrum away from 0 consists of isolated finite-multiplicity eigenvalues.

For one step of the pacman rotation map,

$$R_{\text{prm}}(x) = \frac{x}{1-x} \quad (0 < x < 1/2), \quad R_{\text{prm}}(x) = 2 - \frac{1}{x} \quad (1/2 < x < 1),$$

so every branch has derivative > 1 . Therefore the derivative of the period map R_{prm}^p at the periodic point has modulus > 1 . Varying the multiplier changes the rotation coordinate, hence an expanding eigenvalue is present. The analytic graph transform for compact maps gives a local finite-dimensional unstable manifold W^u through f_* .

2. Finite renormalization tilings and the near-orbit rigidity mechanism. For $f_0 \in \mathcal{B}$ put $f_j = \mathcal{R}^j f_0$. Whenever $f_0, \dots, f_n$ remain in a sufficiently small ball about f_* , spread the prepacman of f_n backwards through the renormalization embeddings. This produces level- n triangles adjacent to the Siegel core and annular rectangles between consecutive levels. Their incidence, cyclic order and return labels are the same as for f_* .

The first-level tile boundaries avoid the critical value except at their marked endpoints. Hence they move holomorphically with f . The lambda lemma extends this boundary motion quasiconformally. Every level- n tile has a univalent return to one of finitely many first-level tiles. Pull the first-level motion back by these return branches. On common edges the two pullbacks agree by equivariance, so they paste to h_n . Koebe distortion on the fixed collars and the lambda-lemma estimate give (19.13). If the whole forward orbit stays near f_* , quasiconformal compactness gives a limiting map h . The mesh of the nested tiles tends to zero by the domain improvement and the periodic complex bounds, so equivariance extends from the tile skeleton to the limiting core. The image core is an invariant quasidisk with rotation number θ ; therefore f is Siegel. The Beltrami coefficient vanishes on the core, giving the asserted hybrid equivalence there.

3. Globalized anti-renormalizations on the unstable manifold. The preceding finite tilings also control inverse renormalization. Let $f_0 \in W^u$ be close to f_* . Its backward orbit $f_{-n} = \mathcal{R}^{-n} f_0$ along W^u tends to f_* . Embed the standard prepacman of f_0 into the dynamical plane of f_{-n} using the inverse renormalization changes of variables. Near the critical value choose a disk D whose closure avoids the mouth and whose corresponding disk for f_* has a definite collar.

Suppose, to the contrary, that the first forbidden pullback meets a mouth edge. Take the last pullback disk before this happens. In the corresponding renormalization tiling it lies in the union of two adjacent marked sectors. The wall separating those sectors from the mouth has a fixed modulus. If the pullback crosses the wall before reaching the critical tile, its image under the prescribed return is a simply connected region surrounding the critical tile on which the return has degree one; the enclosed critical

tile carries degree two. The argument principle gives a contradiction. Thus the entire pullback chain stays in the allowed tiling. Since the backward orbit in W^u converges to f_* , the same collar works uniformly at sufficiently deep levels. Exhausting the pulled-back domains gives the proper plane-domain extensions. This is exactly the geometric content needed below; no global hyperbolicity theorem has been used.

For a rational multiplier $e(p/q)$ on W^u , the extended commuting pair is parabolic. Its attracting petals are generated by the two commuting branches. The critical point belongs to the attracting flower: otherwise all critical orbits would avoid a compact subpetal, and the proper extension would give a normal inverse family across the parabolic fixed point, contradicting the nontrivial parabolic return. Hence the postcritical set is contained in the union of the two labelled critical-value orbits. Commutativity and properness also give the following separation: if two words in the two branches send the marked critical point to the same point, cancel their maximal common suffix; the remaining relation would yield a nontrivial periodic point in an attracting petal before its first return to the parabolic gate, impossible in Fatou coordinates. Therefore distinct exponent labels do not collide.

The noncollision statement makes each labelled critical-orbit point a single-valued holomorphic function of the parameter on a rational multiplier fibre. Their union is a holomorphic motion; the lambda lemma extends it to the sphere with uniform quasiconformal control on a smaller parameter ball. Choose $p_j/q_j \rightarrow \theta$. Normal-family compactness for the corresponding motions gives a holomorphic motion over the limiting fibre $\mathcal{F}(\lambda_*)$.

Assume this limiting fibre has positive complex dimension. Restrict to a nonconstant analytic disk through f_* . The limiting holomorphic motion of the postcritical set extends quasiconformally and is equivariant. Pulling the standard linearizing structure of f_* through that motion gives, after shrinking the parameter disk, a Siegel quasidisk for every parameter on the disk. Thus all these maps are Siegel pacmen of rotation number θ . The exponential-convergence lemma proved in Step 4 below then places the disk inside the local stable set of f_* . But it is also contained in W^u , and a local stable set intersects the unstable manifold only at the fixed point. This contradiction proves $\dim \mathcal{F}(\lambda_*) = 0$.

Because W^u is finite-dimensional and $\lambda|_{W^u}$ is a nonconstant holomorphic function, the local dimension theorem gives

$$\dim_{\mathbb{C}} W^u = 1 + \dim_{\mathbb{C}} \mathcal{F}(\lambda_*) = 1. \quad (19.14)$$

Thus there is exactly one unstable direction. Notice that this conclusion was not inferred from an assumed codimension-one stable leaf; it came from the critical-orbit holomorphic-motion argument.

4. Exponential contraction on the Siegel multiplier slice.

Cut f and f_* along their marked critical accesses and pass to the normalized critical-circle/cylinder coordinates. The cuts and quotient maps range in one compact analytic family because their Siegel boundaries are bounded-type quasicircles (Theorem 17.1). In these coordinates the pacman period operator and the critical-circle period operator differ only by the fixed analytic cutting/gluing charts. Theorem 16.18 gives exponen-

tial convergence of the latter for the common periodic rotation combinatorics. The chart maps and their inverses have uniformly bounded derivatives on the compact family, so the exponential estimate transfers back to pacman norm, giving (19.15). This uses Theorem 16.18 rather than reimporting the McMullen convergence theorem here.

5. Exclusion of neutral spectrum on the same-multiplier slice. Let

$$\mathcal{B}_* := \{f \in \mathcal{B} : \lambda(f) = \lambda_*\}.$$

The multiplier is a nondegenerate analytic functional near f_* , so $\mathcal{B}_*$ is a codimension-one complex Banach submanifold. Since we are using the full period iterate, $\mathcal{R}(\mathcal{B}_*) \subset \mathcal{B}_*$. By (19.14), the full derivative has only one expanding spectral direction, and that direction is transverse to $\mathcal{B}_*$ because the multiplier expands under the base map. Hence the spectrum of $D(\mathcal{R}|_{\mathcal{B}_*})_{f_*}$ lies in the closed unit disk.

For completeness we give the compact-operator argument. Compactness splits

$$\mathcal{X} = E^s \oplus E^c \oplus E^u$$

according to spectral modulus $< 1, = 1, > 1$; $E^c \oplus E^u$ is finite-dimensional. Choose an adapted norm so that the linear part strictly contracts E^s and strictly expands E^u , while its restriction to E^c is arbitrarily close to an isometry. On a small product ball the nonlinear remainder has arbitrarily small C^1 norm, so the standard center-stable and center-unstable cones are invariant.

For $0 < r < 1$ close to 1, set $T_r = rT$. The finite-dimensional degree/graph argument on the center-unstable face produces a nonzero point x_r whose entire T_r -orbit stays in the ball and tends to 0. Compactness permits $r_j \uparrow 1$ and a convergent subsequence of the first images; passing to the limit gives a nonzero T -orbit remaining in the ball. Along its center-unstable component,

$$x_{n+1}^{cu} = DT_0|_{E^c \oplus E^u} x_n^{cu} + O(\|x_n\|^2).$$

An adapted inverse norm excludes decay at any fixed negative exponential rate, while the orbit stays small and tends to 0; hence (19.16).

Apply this to $T = \mathcal{R}|_{\mathcal{B}_*}$ in coordinates centered at f_* . A neutral eigenvalue would give a slow orbit contained in an arbitrarily small neighborhood. Such an orbit is infinitely renormalizable and stays close to f_* , so the renormalization-tiling pseudo-conjugacy lemma makes every point on it a Siegel pacman of multiplier λ_* . The same-rotation exponential convergence lemma then forces exponential convergence to f_* , contradicting (19.16). Hence there is no neutral spectrum on $\mathcal{B}_*$.

The spectrum on $T_{f_*}\mathcal{B}_*$ is therefore contained in a disk $|z| \leq \rho_s < 1$. Shrinking the Banach ball, the analytic stable-manifold theorem and the mean-value estimate show that every point of $\mathcal{B}_*$ remains in the ball and converges exponentially to f_* . By The renormalization-tiling pseudo-conjugacy lemma it is Siegel. Conversely every sufficiently close Siegel pacman of multiplier λ_* converges by the same-rotation exponen-

tial convergence lemma and hence belongs to the local stable set. Thus

$$W_{\text{loc}}^s(f_*) = \mathcal{B}_* \quad \text{locally,} \quad (19.17)$$

so the local stable manifold has complex codimension one.

Equation (19.14) gives the dimension. Since the expanding rotation-number branch makes $d\lambda$ nonzero on that direction, the one-variable inverse function theorem makes $\lambda|_{W^u}$ a local coordinate.

The unstable spectrum lies outside the unit circle by definition of W^u ; Step 5 puts the complementary spectrum strictly inside. This is the required hyperbolic splitting. Equation (19.17) and the one-dimensional unstable manifold lemma give the two manifold statements. A second normalized fixed point of the same rotation in this neighborhood would lie in the stable slice, but its constant renormalization orbit cannot converge to f_* unless it equals f_* . Thus the normalized periodic lift is locally unique. Restoring the original period converts the fixed-point formulation back into the periodic pacman statement and completes the proof. $\square$

Theorem 19.5 (Self-similarity near neutral parameters). **Statement scope.** *For the Dudko–Lyubich–Selinger pacman periodic points promoted from periodic Siegel renormalization, the centers of the associated satellite components of the Mandelbrot set obey asymptotic scaling laws near the corresponding Siegel parameter. The scaling is governed by the unique unstable eigenvalue of the pacman periodic point. No unrestricted self-similarity claim is made here for arbitrary neutral or parabolic families.*

Proof strategy (not part of the proof)

Proof status. derived here.

Strategic outline only. Exact DLS consequence: satellite-component centers scale near corresponding Siegel parameters with the unique unstable eigenvalue. *The mathematical argument begins in the proof environment below.*

Proof. Let F_* be a periodic pacman point from the preceding row and let Λ be the unstable eigenvalue of its period renormalization. By hyperbolicity, $|\Lambda| > 1$ and the unstable direction is one-dimensional.

This is the analytic Taylor expansion at the hyperbolic periodic point.

This is the parameter sequence singled out in the Dudko–Lyubich–Selinger scaling theorem.

The transversality is part of the parameter-space realization used to pass from pacman hyperbolicity to Mandelbrot scaling.

Iterating the inverse unstable branch gives first-order contraction Λ^{-1} in u ; the transverse parameter coordinate changes this by a nonzero analytic factor whose ratio tends to one. This is exactly the source scaling-law conclusion. The repaired row stops here and makes no assertion about arbitrary parabolic families or unrestricted dynamical self-similarity. $\square$

Theorem 19.6 (Uniform a priori bounds in neutral/satellite regimes). **Statement scope.** *Let $f_\theta(z) = e^{2\pi i\theta}z + z^2$ be a quadratic polynomial with a bounded-type Siegel rotation number. There is an absolute constant K such that every level combinatorial*

interval $I \subset \partial Z_\theta$ satisfies the uniform external-width estimate $W_3^+(I) \leq K$. Equivalently in the regularized formulation, f_θ admits nested geodesic pseudo-Siegel disks whose boundary tilings have uniformly bounded combinatorics and all-scale outer geometry; the outermost regularized disk is a universal quasidisk. Under the sector first-return/gluing change of variables, suitable deep pseudo-Siegel disks become the outermost regularized disks of the renormalized maps. Consequently the normalized associated sector-renormalization operators form a precompact family with a uniform positive collar modulus. No assertion is made that the unregularized Siegel disk is a universal quasidisk or that a full all-rotation-number renormalization horseshoe is hyperbolic.

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Control external extremal width by geodesic fjord regularization, Snake trading, amplification and calibration; obtain a universal pseudo-Siegel quasidisk and transfer that compact geometry through the sector first-return/gluing identity to a uniform collar bound for normalized sector renormalizations.

The mathematical argument begins in the proof environment below.

Proof. Fix the normalized quadratic Siegel polynomial

$$f_\theta(z) = e^{2\pi i\theta} z + z^2,$$

where $\theta = [0; a_1, a_2, \dots]$ is irrational of bounded type. Let p_n/q_n be its continued-fraction convergents and let Z be the closed Siegel disk. By the bounded-type Siegel boundary theorem from Chapter 17 the unique finite critical point belongs to ∂Z ; by the critical-circle real-bounds theorem from Chapter 16 the induced combinatorics has uniform real geometry at sufficiently deep levels. We first prove the pseudo-Siegel estimate and only afterwards transfer it to sector renormalization.

1. External width and the correct quantitative target. For a level- n combinatorial interval $I = [x, f^{q_n}x] \subset \partial Z$, let $3I$ be the union of I with its two neighboring intervals of the same level. Define

$$W_3^+(I) := \text{width}\{\gamma \subset \widehat{\mathbb{C}} \setminus Z : \gamma \text{ joins } I \text{ to } \partial Z \setminus 3I\}. \quad (19.18)$$

Here width is the reciprocal of extremal length. Uniform control of $W_3^+(I)$ is stronger and more stable than an isolated annulus-modulus estimate: it records the possible long fjords of ∂Z while measuring how much a curve family can oscillate across every combinatorial scale.

The quantitative target is the absolute estimate

$$W_3^+(I) \leq K_0. \quad (19.19)$$

Indeed, if $\theta = [0; a_1, a_2, \dots]$ is bounded type, set

$$\theta^{(N)} = [0; a_1, \dots, a_N, 1, 1, \dots].$$

The Douady-Ghys realization and real bounds give normalized convergence of the closed Siegel disks and of every fixed finite combinatorial configuration as $\theta^{(N)} \rightarrow \theta$. For separated boundary continua, extremal width is continuous under this Caratheodory/kernel convergence. Hence, for every fixed combinatorial interval I of f_θ , the corresponding intervals I_N of $f_{\theta^{(N)}}$ satisfy

$$W_3^+(I) = \lim_{N \rightarrow \infty} W_3^+(I_N).$$

An absolute bound for all eventually golden maps therefore passes to the limit without a rotation-number-dependent loss. Thus we may assume below that the tail of the continued fraction is $1, 1, \dots$

2. Near-rotation geometry and controlled regularization. Write ℓ_n for the combinatorial length of a level- n fundamental interval in the linearizing coordinate. On a simply connected near-rotation domain U , the maps $f^j|_U$, $0 \leq j \leq q_n$, are almost rotations; the error is controlled by the annuli separating the relevant pullbacks. The standard series/parallel laws for extremal length and the non-crossing principle imply the following logarithmic rule: for two boundary intervals of normalized angular lengths s, t separated by a definite relative distance, the width of curves joining them is bounded above by

$$C_0 + C_1 \log \left(1 + \frac{\min\{s, t\}}{\text{dist}(I, J)} \right)^{-1}. \quad (19.20)$$

The constants are universal once the near-rotation collar is fixed. In particular, on scales $\gg 1/q_n$ no new degeneration can accumulate merely by iterating the almost rotation. This is the beau/log-rule mechanism behind the near-rotation estimates.

A large continued-fraction entry creates a long parabolic fjord. Instead of trying to prove that the actual Siegel disk is a uniform quasidisk, truncate the fjord at a hyperbolic geodesic joining its two sides. Doing this at all scales $\geq m$ produces a full Jordan disk $\hat{Z}_m \supset Z$, called the level- m geodesic pseudo-Siegel disk. It has three properties used below:

1. f^j is injective on $\hat{Z}_m$ for every $0 \leq j \leq q_{m+1}$;
2. the added caps are pairwise organized by the level- m dynamical partition and are separated by fixed hyperbolic collars;
3. a grounded boundary interval is one whose endpoints avoid the interiors of the added caps, so it has a canonical projection back to ∂Z .

There is a universal regularization parameter $\delta > 0$ for which the stable comparison to be proved is

$$|W_{\hat{Z}_m}^+(I_m, J_m) - W_Z^+(I, J)| \leq C_\delta \quad (19.21)$$

for every pair of grounded intervals. To prove (19.21), compare the complement of Z with the complement of $\hat{Z}_m$ after deleting the finitely many geodesic caps met by a curve. Each cap is bordered by a definite-modulus annulus and the near-rotation log rule bounds the total toll paid by entering and leaving it. Grounded endpoints prevent a curve family from using a cap as an arbitrarily thin shortcut. Series decomposition then gives the upper and lower inequalities in (19.21). Since the bound is independent of the number of deeper caps, regularization is stable through an arbitrarily long

tower.

3. Trading diving width for outer width: the Snake mechanism. Let $I, J \subset \partial \hat{Z}_m$ be grounded intervals and let L be the complementary boundary arc between them. Decompose the family of curves joining I to J according to whether the curves stay outside or dive through the regularized disk. If the excess diving width is $A \gg 1$, the series law produces a long rectangle (a “snake”) inside the diving family. Localize that rectangle away from both endpoints. The non-crossing principle forces one of its two end pieces to carry a fixed fraction of the width. After projecting its endpoints to ∂Z , one obtains a smaller grounded pair with larger relative width. Iterating gives either an outer family of width $\gtrsim A$, or an impossible sequence of pairwise blocking rectangles whose widths grow geometrically. Hence a large diving family always produces a comparably large outer family at a controlled smaller scale.

The resulting quantitative alternative is: either the fjord is absorbed by regularization, or for a grounded interval J at a deeper level,

$$\log W_\chi^+(J) \geq c W(R) - C. \quad (19.22)$$

The proof is the preceding Snake iteration applied to the pullback chain of R . If every pullback cap has definite modulus, the geodesic truncation can be welded in and the first alternative occurs. If welding fails, the corresponding annular barrier degenerates. The extremal-width series law then turns additive rectangle width into multiplicative outer width; taking logarithms gives (19.22). The constants are independent of the level because (19.21) prevents accumulated regularization error.

4. Amplification and calibration. A wide combinatorial interval is not yet a contradiction: its width might sit on the wrong scale or be carried mostly by curves diving through future fjords. Two elementary renormalization moves remove these possibilities.

First, use closest-return translates of the interval. If a level- m combinatorial interval has width $A \gg 1$, real bounds supply many translates with comparable combinatorial size. Non-crossing and the parallel law imply that either a translate already yields a grounded external interval below scale ℓ_m , or the widths of a disjoint subfamily add. In the latter case one obtains a grounded interval at a deeper level with width at least cA , with $c > 1$ after the large constants are fixed. This is the amplification step.

Second, split a wide grounded family into its diving and external pieces. The external piece is already in the required form. If the diving piece dominates, apply the Snake lemma and project to the next regularized boundary using (19.21). Thus a wide grounded family either produces a wide combinatorial interval one level deeper or a still wider grounded family at a smaller scale. This is the calibration step.

The strict decrease of combinatorial size in alternative (2), together with the geometric amplification of width, is the termination mechanism for the induction below.

5. Deep-to-shallow induction and the uniform width theorem. Choose constants in the order

$$1 \ll \chi \ll t, \quad 1 \ll \lambda = \lambda_t, \quad K \gg K_t, \chi^2. \quad (19.23)$$

For an eventually golden map there is a depth n_θ below which the continued fraction

is stationary. At those very deep levels the near-rotation log rule gives the required bounds directly. Assume inductively that all levels $> m$ satisfy:

1. every sufficiently wide external level- m parabolic rectangle has already been absorbed into $\hat{Z}_m$;
2. every grounded interval of size at most ℓ_m has $W_\lambda^+ \leq 2\chi K$;
3. every combinatorial interval of level at least m has $W_\lambda^+ \leq K$.

If (1) fails, the regularization-or-boost alternative gives the exponential boost (19.22). The resulting wide interval violates either (2) or (3) at a deeper level by the amplification-calibration alternative. Hence the required regularization exists and (1) holds. If (2) fails, split its family into external and diving parts. Property (1) bounds the external part by $O(\sqrt{K})$; the diving remainder is then large enough for calibration, again contradicting the deeper induction. If (3) fails, amplification gives a violation of (2). Therefore all three statements hold at level m . Descending induction reaches level -1 .

The finitely many shallow levels before the amplification threshold are absorbed by increasing K once. Consequently

$$W_3^+(I) \leq K_0 \tag{19.24}$$

for every combinatorial interval of every eventually golden map, with one absolute K_0 . the bounded-type reduction extends (19.24) to every bounded-type θ .

The bounded-combinatorics assertion follows from the threshold alternative in the induction: a tile between consecutive scales cannot contain arbitrarily many next-level tiles, because otherwise the parallel law would create a wide family contradicting (19.24). Neighboring tiles have comparable outer harmonic size by the same width bound. A bounded-combinatorics nested tiling with bounded adjacent geometry is uniformly quasisymmetric to the dyadic circle; the Ahlfors-Beurling extension gives a uniform quasiconformal map of the disk. Hence $\hat{Z}_{-1}$ is a universal quasidisk. This is the quasidisk approximation theorem and, more importantly here, gives a compact normalized class of regularized neutral domains.

Indeed, a fjord of large aspect ratio produces a large extremal width for a family joining its mouth to the complementary boundary. The geodesic regularization absorbs precisely such fjords; (19.24) says that after recording them as pseudo-Siegel caps no unrecorded oscillation can remain at any scale. Conversely, bounded tiling geometry bounds every such external family by the series/parallel laws. This is why the theorem controls all-scale oscillation without the false claim that ∂Z itself has uniformly bounded quasicircle geometry.

6. Sector-renormalization transfer. We now derive the second clause of the theorem rather than importing it. A sector renormalization is the first return to a marked union $X_- = X \cap \{\text{left sector}\}$ and $X_+ = X \cap \{\text{right sector}\}$, followed by conformal gluing of the two boundary sides and a normalization at the neutral fixed point and critical value. Every sector renormalization is a finite composition of prime sector renormalizations; on the rotation coordinate this is exactly the modified continued-fraction shift. Thus, after n prime steps, the closest-return map $f^{q_{k(n)}}$ becomes the first closest return of the renormalized map.

The first-return construction gives, for every sector-renormalization depth n , a combinatorial index $k(n) \geq n$ satisfying

$$\widehat{Z}_{m+k(n)}(f_\theta) \longmapsto \widehat{Z}_m(\mathcal{R}_{\text{sec}}^n f_\theta) \quad (19.25)$$

under the normalized sector change of variables. This is a direct first-return identity. Both sides of (19.25) are obtained by cutting along the same closest-return orbit and filling exactly the fjords whose return times have been discarded by the first n sector shifts. Before normalization the two domains are literally the same quotient surface; the power/gluing map merely changes the coordinate. Hence the pseudo-Siegel caps and their boundary tilings correspond one-for-one.

By the all-scale pseudo-Siegel bounds and (19.25), every normalized sector renormalization therefore has an outermost pseudo-Siegel disk in the same universal quasidisk class. Normalize three marked data: the neutral fixed point at 0, the critical value at 1, and one boundary access. Uniform quasidisk geometry then gives a uniform annulus

$$\widehat{Z}_{-1} \in \Omega, \quad \text{mod}(\Omega \setminus \widehat{Z}_{-1}) \geq \mu_0 > 0, \quad (19.26)$$

on which the first-return branches extend univalently except for the one prescribed quadratic critical branch. Koebe distortion on (19.26) bounds the normalized maps and their inverse branches on compacta.

For a sequence of renormalized maps, uniform quasidisk compactness gives a Caratheodory-convergent subsequence of the domains. Montel gives locally uniform convergence of every normalized univalent branch on the common collar; the quadratic branch converges after factoring its unique critical point. The marked normalization excludes collapse to a constant or to a degenerate covering. Thus the limit is again a sector first-return pair and (19.26) persists with, say, $\mu_0/2$. This is exactly the a priori-bound compactness needed for the sector-renormalization operator.

Those stronger statements require additional theory. Our conclusion is the uniform geometry and compactness of the associated normalized sector returns.

Equation (19.24) gives uniform boundary-oscillation control at all combinatorial scales in the precise pseudo-Siegel sense. the all-scale pseudo-Siegel bounds supplies the uniform quasidisk representative, and the sector covariance and compactness steps transfer this compact geometry to every associated sector renormalization. Therefore the pseudo-Siegel a priori bounds and the sector-renormalization a priori bounds hold uniformly at the stated bounded-type scope. This proves the theorem. $\square$

19.3 Worked examples

Worked example: Long transit in an Ecalle cylinder

A perturbation close to a parabolic map enters an attracting Fatou coordinate, translates for a large number N of iterates, and exits through a repelling coordinate. After subtracting the integer translation, the residual phase is bounded. Near-parabolic renormalization records the normalized entry-translation-exit

map rather than the huge raw iterate.

Worked example: A black-hole inequality

Suppose a stationary map has landing probability $\eta = 0.08$ and escape probability $\xi = 10^{-4}$. If the geometric Black Hole constant is $C_{\text{BH}} = 100$, then $\eta > C_{\text{BH}}\xi = 0.01$, so the criterion forces positive area. The difficult part is not this arithmetic comparison but constructing a stationary map for which the uniform geometric hypotheses and these probability bounds hold.

Boundary case: Good landing without small escape is insufficient

A definite lower bound on η does not imply positive area if ξ is comparable to or larger than η . The transfer recursion must control both probabilities; this prevents the positive-area theorem from being mistaken for a consequence of renormalization hyperbolicity alone.

19.4 What this chapter established

The chapter closed its formal theorem spine at the stated source scopes. The proofs deliberately separate elementary deductions from deep imported theorems: every imported theorem is named at the point where it enters, and later conclusions use only the scope actually established.

For study purposes, the most important task is to be able to reconstruct the dependency chain rather than memorize theorem names. In particular, note which hypotheses supply compactness or a priori geometry, which step supplies rigidity, and which step converts dynamical information into parameter or statistical conclusions.

Looking ahead

Chapter 20 shifts from complex renormalization geometry to real-parameter typicality and statistical behavior of unimodal maps.

Once a map is Collet-Eckmann, the problem becomes statistical. A Hofbauer/Markov extension converts the nonuniformly expanding interval map into a transfer-operator problem with a spectral gap. From that single spectral engine come an absolutely continuous invariant probability, exponential correlation decay, a central limit theorem, and local large deviations.

1. distinguish positive-measure existence from almost-everywhere typicality;
2. explain free and bound periods in parameter exclusion;
3. state the regular-or-stochastic dichotomy at family scope;
4. describe the phase-parameter relation;
5. derive Collet-Eckmann expansion from good critical returns;
6. explain how a spectral gap yields mixing, CLT, and large deviations.

Markov extension $\rightarrow$ quasicompact transfer operator
 $\rightarrow$ spectral gap $\rightarrow$ statistical limit laws.

Common pitfall

“Almost every parameter” is meaningless until the parameter family and reference measure are fixed. Likewise, a central limit theorem is not a consequence of positive Lyapunov exponent alone; the transfer-operator/tower hypotheses supply the needed mixing and summability.

20.1 Definitions

Definition 20.1 (Collet-Eckmann condition). A unimodal map satisfies the Collet-Eckmann condition if $|(f^n)'(f(c))|$ grows at least $C \lambda^n$ along the postcritical orbit for some $\lambda > 1$, with the critical value convention fixed.

Definition 20.2 (polynomial recurrence). Polynomial recurrence means the distance of $f^n(c)$ to the critical point is bounded below by a negative power $n^{-\alpha}$ (or an equivalent sourced formulation), excluding excessively fast returns.

Definition 20.3 (phase-parameter relation). A phase-parameter relation is a quantitative correspondence between nested dynamical intervals/puzzle pieces around the critical orbit and parameter intervals/parapuzzles, usually with controlled distortion.

Definition 20.4 (stochastic parameter). A stochastic parameter is source-relative. In the Avila-Lyubich-de Melo regular-or-stochastic theorem it means that the map admits an absolutely continuous invariant probability measure. Stronger properties such as Collet-Eckmann expansion or positive Lyapunov exponent are stated separately when the cited theorem supplies them.

20.2 Main results

Theorem 20.1 (Jakobson positive-measure stochastic parameters). **Statement scope.** For the quadratic family $f_a(x) = 1 - ax^2$ on $[-1, 1]$, there is $\varepsilon > 0$ and a positive-Lebesgue-measure set $\Omega \subset [2 - \varepsilon, 2]$ such that every $a \in \Omega$ admits an absolutely continuous invariant probability measure μ_a with $\int \log |f'_a| d\mu_a > 0$.

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Free expansion, binding recovery, direct critical-value transversality, summable parameter exclusion, Collet-Eckmann assembly, and an integrable Gibbs-Markov inducing scheme with an explicit tower lift.

The mathematical argument begins in the proof environment below.

Proof. We prove the theorem at the standard quadratic normalization

$$f_a(x) = 1 - ax^2, \quad x \in [-1, 1], \quad a \in [2 - \varepsilon, 2]. \quad (20.1)$$

Write $c = 0$, $\xi_n(a) = f_a^n(c)$ and

$$D_n(a) = |(f_a^{n-1})'(1)|, \quad n \geq 1. \quad (20.2)$$

Thus $\xi_1 = 1$. At $a = 2$ the critical orbit is $0 \mapsto 1 \mapsto -1 \mapsto -1$, and $D_n(2) = 4^{n-1}$. We shall keep a positive-measure set of nearby parameters for which the critical orbit retains enough of this expansion while being allowed to make controlled returns to the critical point.

1. Uniform expansion away from the critical neighborhood. Choose $0 < \delta \ll 1$ and put $\Delta = (-\delta, \delta)$. The Chebyshev map f_2 is uniformly expanding on orbit segments that avoid Δ : in the coordinate $x = -\cos(\pi t)$ it is the doubling map $t \mapsto 2t$ modulo the folding symmetry, and avoiding $x = 0$ keeps the orbit away from the only critical fibre. Compactness of the finitely many transition charts outside Δ , followed by C^2 continuity in a , therefore gives constants $C_0 > 0$ and $\lambda_0 > 1$ such that, after reducing ε if necessary,

$$|(f_a^m)'(x)| \geq C_0 \lambda_0^m \quad (20.3)$$

whenever $x, f_a(x), \dots, f_a^{m-1}(x)$ avoid Δ . If the last image belongs to Δ , the same argument applied up to the last step gives

$$|(f_a^m)'(x)| \geq C_0 \delta \lambda_0^m. \quad (20.4)$$

The constants are uniform for $a \in [2 - \varepsilon, 2]$.

2. Binding after a deep return. Partition Δ into depth shells

$$\Delta_r = \{x : e^{-(r+1)} < |x| \leq e^{-r}\}, \quad r \geq r_0, \quad (20.5)$$

where $r_0 = \lceil |\log \delta| \rceil$. Suppose $\xi_n(a) \in \Delta_r$. Because the critical point is quadratic,

$$f_a(x) - f_a(0) = -ax^2. \quad (20.6)$$

As long as the image of $x = \xi_n(a)$ shadows the critical orbit, (20.6) gives a quadratic initial error $O(e^{-2r})$, while the derivative accumulated along the reference critical orbit grows exponentially on its free pieces. Let $p = p(r)$ be the largest integer for which

$$|f_a^j(\xi_n) - \xi_j| \leq e^{-\beta j} \quad (1 \leq j \leq p) \quad (20.7)$$

with a fixed small $\beta > 0$. Combining (20.6), (20.3) and bounded distortion on the shadowing branches yields constants $0 < b_1 < b_2$ and $\kappa > 0$ such that

$$b_1 r \leq p(r) \leq b_2 r, \quad |(f_a^{p+1})'(\xi_n)| \geq e^{\kappa p}. \quad (20.8)$$

Moreover the distortion of f_a^{p+1} on the corresponding return interval is bounded by a universal constant. The first derivative factor $|f_a'(\xi_n)| \asymp e^{-r}$ is exactly compensated by the later expansion, which is the recovery mechanism behind (20.8).

3. Phase-parameter derivative comparability without a later phase-parameter theorem. On a parameter interval on which the critical itinerary through time n is

fixed, differentiate $\xi_n(a)$. Since $\partial_a f_a(x) = -x^2$, the chain rule gives

$$\frac{\xi'_n(a)}{(f_a^{n-1})'(1)} = - \sum_{j=1}^{n-1} \frac{\xi_j(a)^2}{(f_a^j)'(1)}. \quad (20.9)$$

For parameters satisfying the free/bound estimates through time n , the denominators in (20.9) grow exponentially except for the explicitly controlled binding losses. Hence the series is uniformly Cauchy. At $a = 2$ its limit is nonzero; shrinking ε and enlarging the initial free time N gives

$$C_1^{-1} D_n(a) \leq |\xi'_n(a)| \leq C_1 D_n(a). \quad (20.10)$$

If ω is one such parameter cylinder, (20.10), the mean-value theorem and the branch distortion estimate imply

$$\frac{|\omega'|}{|\omega|} \asymp \frac{|\xi_n(\omega')|}{|\xi_n(\omega)|} \quad (20.11)$$

for every subcylinder $\omega' \subset \omega$ cut out at time n . This is the only phase-parameter comparison needed in the proof, and it has been derived directly from the parameter derivative identity.

4. The return-depth budget and summable exclusion. Call a return time n *essential* if $\xi_n \in \Delta_r$ and the depth r is larger than all depths generated by the preceding binding block. Let

$$R_n(a) = \sum_{\substack{j \leq n \\ j \text{ essential}}} r_j(a). \quad (20.12)$$

Fix $\alpha > 0$ small. Starting after the common initial free time N , retain only parameters for which

$$R_n(a) \leq \alpha n \quad (n \geq N). \quad (20.13)$$

We show that the discarded parameter set has total measure strictly smaller than the initial parameter interval.

Consider an admissible cylinder ω at an essential return. By (20.11), the subcylinders whose return has depth r occupy at most Ce^{-r} of the relative length of ω . A sequence of essential returns of depths $r_1, \dots, r_s$ therefore has relative measure at most

$$C^s e^{-(r_1 + \dots + r_s)}. \quad (20.14)$$

Binding consumes at least $b_1 r_i$ iterates after a depth- r_i return. Thus, for r_0 large, the number of admissible return itineraries with total depth R is bounded by $e^{\eta R}$ with an arbitrarily small fixed $\eta > 0$: indeed one first chooses the return times, which are separated by the binding blocks, and then the two possible sides of each return; the elementary composition count is $\exp(o(R))$ as $r_0 \rightarrow \infty$. Absorbing the factor C^s into $e^{\eta R}$ gives

$$\frac{|\{a \in \omega : R_n(a) - R_{n-1}(a) \text{ creates total depth } R\}|}{|\omega|} \leq e^{-(1-2\eta)R}. \quad (20.15)$$

Consequently the stage- n violation of (20.13) has relative measure

$$\leq C_2 e^{-\kappa_1 \alpha n}. \quad (20.16)$$

Choose N so large that the sum of (20.16) over $n \geq N$ is less than $1/2$. The nested survivor set

$$\Omega_\infty = \bigcap_{n \geq N} \Omega_n \quad (20.17)$$

then satisfies $|\Omega_\infty| > 0$.

Indeed, decompose the logarithmic derivative distortion into free and bound blocks. On a free block the orbit remains a definite distance from the critical point, so the usual distortion sum $\sum |f_a^j(I)| / \text{dist}(f_a^j(I), 0)$ is geometric by (20.3). On a bound block, shadowing (20.7) and the quadratic identity (20.6) make the same sum bounded independently of the depth. The parameter derivative identity (20.9) then transfers this phase distortion to $\xi_n|_\omega$.

5. Exponential growth of the critical-value derivative. Fix $a \in \Omega_\infty$. Decompose the critical orbit through time n into free segments and the binding blocks following essential returns. Free segments contribute at least $\log \lambda_0$ per iterate up to a fixed additive constant by (20.3). A depth- r return can lose at most $C_3 r$ in logarithmic derivative before the binding recovery (20.8). Summing over all essential returns gives

$$\log D_n(a) \geq c_0 n - C_4 R_n(a) - C_5 \geq (c_0 - C_4 \alpha) n - C_5. \quad (20.18)$$

Choose $\alpha < c_0/(2C_4)$. Then there are $C > 0$ and $\gamma > 0$ such that

$$D_n(a) \geq C e^{\gamma n} \quad (n \geq 1). \quad (20.19)$$

Thus every surviving parameter is Collet-Eckmann along the critical value. Notice that this conclusion used only the return-depth budget constructed above; no later principal-nest or asymptotic phase-parameter theorem has been inserted into Jakobson's argument.

6. Construction of an induced expanding Markov map. Let

$$Y = [-1, -\delta] \cup [\delta, 1].$$

Cut Y into maximal intervals on which an orbit first returns to Y after a concatenation of free and bound blocks. Denote the return time by τ and the induced map by

$$F = f_a^\tau : Y \supset \bigcup J \longrightarrow Y. \quad (20.20)$$

The free expansion and the recovered expansion at the end of each binding block imply, after increasing N once,

$$|F'(x)| \geq \Lambda > 1 \quad (x \in J) \quad (20.21)$$

for every branch. The distortion argument used in the bounded parameter distortion

through exclusions lemma gives

$$\left| \log \frac{F'(x)}{F'(y)} \right| \leq K |F(x) - F(y)|, \quad x, y \in J, \quad (20.22)$$

with K independent of the branch.

The same depth estimate that gave (20.15) gives integrability of the inducing time. A branch with a binding contribution comparable to r has length at most $Ce^{-\kappa_2 r}$, while its binding time is $O(r)$ by (20.8). Summing over return histories therefore gives

$$\sum_J \tau(J) |J| < \infty. \quad (20.23)$$

For completeness, construct the invariant density rather than citing an inducing theorem. Let $\mathcal{L}_F$ be the Perron-Frobenius operator of F . From (20.21)–(20.22), for every density h of bounded variation,

$$\text{Var}(\mathcal{L}_F h) \leq \Lambda^{-1} \text{Var}(h) + C_6 \|h\|_1. \quad (20.24)$$

The Cesaro averages $N^{-1} \sum_{j=0}^{N-1} \mathcal{L}_F^j 1$ therefore lie in a fixed bounded subset of $BV(Y)$. Compactness of bounded subsets of BV in L^1 yields a limit $h_Y \geq 0$ with $\int_Y h_Y = 1$ and $\mathcal{L}_F h_Y = h_Y$. Let $d\nu = h_Y dx$. By (20.23), $\int \tau d\nu < \infty$ (bounded distortion makes ν and Lebesgue comparable on branches). Hence

$$\mu_a = \frac{1}{\int \tau d\nu} \sum_J \sum_{j=0}^{\tau(J)-1} (f_a^j)_*(\nu|_J) \quad (20.25)$$

is a finite f_a -invariant probability. Each summand is absolutely continuous, so $\mu_a \ll dx$. Passing to an ergodic component if necessary preserves absolute continuity.

7. Positive Lyapunov exponent. The induced expansion can be taken proportional to return time: by the same free/bound estimate as (20.18), every branch satisfies

$$\log |F'| \geq \gamma_1 \tau - C_7 \quad (20.26)$$

with $\gamma_1 > 0$; increasing the initial inducing time absorbs C_7 and gives $\log |F'| \geq (\gamma_1/2)\tau$. Using (20.25) and the chain rule,

$$\int \log |f'_a| d\mu_a = \frac{\int_Y \log |F'| d\nu}{\int_Y \tau d\nu} \geq \frac{\gamma_1}{2} > 0. \quad (20.27)$$

The set $\Omega_\infty \subset [2 - \varepsilon, 2]$ has positive Lebesgue measure by (20.17). For every $a \in \Omega_\infty$, (20.25) is an absolutely continuous invariant probability and (20.27) gives positive Lyapunov exponent. This proves the theorem. $\square$

Theorem 20.2 (Regular-or-stochastic dichotomy). **Statement scope.** *Let (f_t) be a nontrivial real-analytic family of quasiquadratic unimodal maps in the Avila-Lyubich-de Melo setting. Then for Lebesgue-a.e. parameter t , f_t is either regular in the source sense (the critical orbit is attracted to a hyperbolic periodic cycle, with no nonhyper-*

olic periodic orbit) or stochastic in the theorem's exact sense: it admits an absolutely continuous invariant probability measure. Positive Lyapunov/Collet-Eckmann conclusions are not part of this 2003 dichotomy statement; they are supplied separately by the typical Collet-Eckmann theorem and the downstream statistical rows.

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Construct the codimension-one hybrid lamination and analytic transverse direction, control countable tangencies by quasisymmetric straightening and renormalization porosity, and use the central-parapuzzle Borel-Cantelli estimate to transfer null sets from the quadratic transversal.

The mathematical argument begins in the proof environment below.

Proof. Let $\{f_t\}_{t \in J}$ be a nontrivial real-analytic one-parameter family of real-analytic quasiquadratic unimodal maps in the theorem's setting. Write $\mathcal{H}(f)$ for the real hybrid class of f : in the nonhyperbolic case this is the topological-conjugacy class with the standard critical marking, while in the hyperbolic case the attracting multiplier is fixed as part of the class. The proof has two independent parts. We first build enough analytic transverse structure to compare the given family with the quadratic family. We then show that the quadratic exceptional set remains null under exactly the quasisymmetric changes of parameter supplied by that transverse structure.

1. The infinitesimal hybrid space is codimension one. Fix a nonparabolic map f . For an analytic tangent vector v at f , call v *horizontal* if there is a quasiconformal vector field α on $\mathbb{C}$, normalized at three points, such that on the postcritical orbit

$$v = \alpha \circ f - Df \alpha. \quad (20.30)$$

If f_s is an analytic path inside $\mathcal{H}(f)$, differentiation of an equivariant holomorphic motion gives (20.30), so every genuine hybrid tangent is horizontal.

Conversely, the infinitesimal pullback construction solves (20.30) on successive inverse branches. The complex a priori bounds from Chapter 13 and the real principal-nest bounded geometry from Chapter 12 make the pullback operator uniformly bounded on the normalized quasiconformal vector fields. Thus a horizontal vector can be integrated locally by solving the Beltrami equation for the pulled-back coefficient and conjugating f . The resulting analytic disk stays in the hybrid class and has tangent vector v . Hence the horizontal space is exactly $T_f \mathcal{H}(f)$.

It remains to see that this space has codimension one. Choose a sufficiently deep nice critical interval $I^0 \supset I^1$ and a C^1 vector field w which is holomorphic on the critical-value puzzle piece, vanishes on the other first-return pieces, and has definite hyperbolic length at the critical value. If w were horizontal, the pullback solution α of (20.30) would have uniform quasiconformal norm. Pulling to the first landing piece $W \in I^0$ forces α to have a definite hyperbolic displacement at a point of W , while α vanishes at the two boundary points of the real trace of W . Since $\text{diam}_{I^0} W \rightarrow 0$ down the principal nest, the normalized quasiconformal modulus-of-continuity estimate gives a contradiction. Approximate w in the two relevant complex sectors by polynomial vector fields. The pullback estimate is uniform, so if every approximant were horizontal

their normalized pullback fields would converge to a horizontal solution for w , again a contradiction. Therefore an analytic transverse vector exists and

$$T_f \mathcal{A} = T_f \mathcal{H}(f) \oplus \mathbb{C} v_{\text{tr}}. \quad (20.31)$$

2. Local hybrid classes form a holomorphic lamination. Use (20.31) as a product coordinate $V_h \times \Sigma_{\text{tr}}$ near f . The macroscopic pullback estimate has the following consequence: if two nearby maps are hybrid conjugate, their difference makes arbitrarily small angle with the horizontal space as the neighborhood shrinks. Hence one hybrid class can meet a sufficiently short transverse segment Σ_{tr} at most once.

For preperiodic maps and for hyperbolic maps the hybrid class is already an analytic codimension-one graph: in the preperiodic case the critical relation is a nondegenerate analytic equation, while in the hyperbolic case the attracting multiplier is an analytic submersion. These simple maps are dense on the transverse segment. Their graph slopes are uniformly bounded by the transverse cone in (20.31), and the pullback construction supplies uniform quasiconformal conjugacies on compact subboxes. Montel compactness therefore extends the graphs to every point of the transversal; uniqueness of the intersection prevents crossings. We obtain a codimension-one holomorphic lamination by local hybrid classes.

Holomorphic motions of one transversal along this lamination extend by the lambda lemma. Consequently every local holonomy between real analytic transversals is quasismetric. Explicitly, for a fixed local dilatation there are $C > 0$ and $\delta > 0$ such that for nested real intervals $A \subset B$,

$$\frac{|h(A)|}{|h(B)|} \leq C \left(\frac{|A|}{|B|} \right)^\delta. \quad (20.32)$$

This power-law estimate, rather than absolute continuity of h , is the measure-transfer tool used below.

3. Connectivity to the quadratic straightening. Suppose first that f is a finitely renormalizable Yoccoz map. The puzzle pullback construction and Chapter 13 complex bounds give a quasiconformal conjugacy between f and its quadratic straightening $q_{\chi(f)}$ on a complex neighborhood of the real interval. Normalize the global extension and denote its Beltrami coefficient by μ . For $0 \leq s \leq 1$ solve the Beltrami equation with coefficient $s\mu$ and put

$$f_s = h_s \circ f \circ h_s^{-1}. \quad (20.33)$$

This is an analytic path in the same hybrid class joining f to $q_{\chi(f)}$ in a slightly weaker analytic Banach norm. Choose analytic transverse vectors along the path and approximate the resulting two-parameter surface in the original Banach norm. The implicit-function theorem intersects that surface with the horizontal distribution and produces a path inside the original hybrid class with the same endpoints. Hence the relevant real hybrid class is connected to the quadratic family.

For an infinitely renormalizable f , take a deep quadratic-like renormalization. Chapter 13 supplies a uniform complex collar, while Chapter 15 identifies the hybrid stable direction and proves transverse nonsingularity of renormalization. The quadratic fam-

ily is transverse to the hybrid lamination. Therefore the straightening of any analytic transversal is locally the composition

$$\chi = R^{-1} \circ h \circ R, \quad (20.34)$$

where h is a holonomy between two quadratic-like transversals. It is thus quasisymmetric. In the hyperbolic case straightening is analytic. We have therefore proved local quasisymmetry of straightening at every nonparabolic parameter transverse to its hybrid class.

4. Tangencies and parabolic parameters are countable. Let γ be the holomorphic complexification of $t \mapsto f_t$ in a local parameter disk. In a lamination box identify the leaves with graphs over a horizontal Banach space. If γ has a tangency of order $m \geq 2$ at $z = 0$, write after analytic changes of coordinates

$$\gamma(z) = (\phi(z), z^m \psi(z)), \quad \psi(0) \neq 0. \quad (20.35)$$

Pull the lamination back by $(u, w) \mapsto (u, w^m)$. Then $(\phi(z), z\psi(z)^{1/m})$ is transverse at 0, so the original tangency is isolated. Thus the tangencies of a nontrivial analytic curve with the hybrid lamination are discrete in each box and hence countable on J .

Parabolic parameters are also countable: for each period p they satisfy an analytic periodic-point equation together with multiplier ± 1 ; because a nontrivial family is not contained in a hybrid leaf, the corresponding solutions are isolated unless the relation is identically persistent, which would force the forbidden leaf containment. Let T be the union of the parabolic parameters and all hybrid tangencies. Then T is countable, and on every compact interval $K \Subset J \setminus T$ the straightening map $t \mapsto \chi(f_t)$ is quasisymmetric.

5. The quadratic bad set: infinitely renormalizable part. Let X be the set of quadratic parameters which are neither regular nor in the finitely-renormalizable stochastic class described below. Split

$$X = \mathcal{I} \cup \mathcal{F} \cup \mathcal{P}, \quad (20.36)$$

where $\mathcal{I}$ consists of infinitely renormalizable parameters, $\mathcal{F}$ of finitely renormalizable Yoccoz parameters with infinitely many central levels in the last principal nest, and $\mathcal{P}$ of parabolic parameters.

By the full quadratic renormalization horseshoe of Chapter 14 and its hyperbolicity/rigidity from Chapter 15, a deep renormalization cylinder has a uniform transverse unstable direction. Pulling a fixed-size gap in the unstable coordinate back through the bounded-distortion inverse branches gives constants $\eta > 0$ and $r_* > 0$ such that every sufficiently small parameter interval B centered at a point of $\mathcal{I}$ contains an interval $G \subset B \setminus \mathcal{I}$ with

$$|G| \geq \eta |B|. \quad (20.37)$$

Thus $\mathcal{I}$ is uniformly porous. If h is K -quasisymmetric, (20.32) applied to the complementary pieces converts (20.37) into a positive relative gap $|h(G)| \geq \eta'(K)|h(B)|$. Hence $h(\mathcal{I})$ is porous and has Lebesgue measure zero by the density theorem.

6. The quadratic bad set: central parapuzzle part. Fix one finite renormalization type and work after the last renormalization. For each principal-nest level n , the real parapuzzle gives pairwise disjoint parameter intervals Δ_i^n covering the corresponding Yoccoz locus and a central subinterval $\Pi_i^n \subset \Delta_i^n$. The real/complex bounds of Chapters 12–13, applied after every noncentral return, give constants $A > 0$ and $0 < \rho < 1$ such that

$$\frac{|\Pi_i^n|}{|\Delta_i^n|} \leq A\rho^n. \quad (20.38)$$

The point is that every new central level must cross a definite-modulus collar created at the preceding noncentral return. Pullback through the monotone branches has bounded cross-ratio distortion; multiplying these fixed losses through n central episodes gives (20.38). A parameter belongs to $\mathcal{F}$ precisely when it lies in central pieces at infinitely many levels.

Let h be any K -quasisymmetric map. By (20.32),

$$\frac{|h(\Pi_i^n)|}{|h(\Delta_i^n)|} \leq C(K)A^{\delta(K)}\rho^{\delta(K)n}. \quad (20.39)$$

Since the Δ_i^n are disjoint at a fixed level,

$$\sum_n \sum_i |h(\Pi_i^n)| < \infty. \quad (20.40)$$

Borel–Cantelli gives $|h(\mathcal{F})| = 0$. The parabolic set $\mathcal{P}$ is countable. Consequently

$$|h(X)| = 0 \quad (20.41)$$

for every quasisymmetric h .

7. Why parameters outside X are regular or stochastic. A quadratic parameter outside X is either hyperbolic, hence regular, or is a finitely renormalizable Yoccoz parameter for which the last principal nest has only finitely many central levels. After the last central return, the nest is noncentral. The same real bounds used in (20.38) yield exponential decay of successive return domains and bounded distortion of the first-return branches. Induce on a nice interval after that last central level. Free branches expand uniformly; returns passing near the critical point have quadratic binding and recovery exactly as in the Jakobson construction proved earlier in this book Theorem 20.1. The exponentially decaying nest geometry makes the return-time tail summable. The resulting countable Markov map therefore has uniform expansion, bounded distortion and integrable return time. The BV Perron–Frobenius/Cesaro argument from Theorem 20.1 produces an invariant density and its tower lift, hence an absolutely continuous invariant probability measure for the original quadratic map. Thus every quadratic parameter outside X is regular or stochastic.

8. Transfer to the arbitrary analytic family. Fix a compact interval $K \Subset J \setminus T$. Cover it by finitely many lamination boxes on which straightening is quasisymmetric. Hyperbolic quadratic parameters lift to regular maps because the attracting multiplier is part of the hybrid marking. For a finitely renormalizable Yoccoz parameter, the quadratic central-parapuzzle ratios and the corresponding return-domain ratios

decay exponentially outside the limsup set $\mathcal{F}$. The local holonomy estimate (20.32) changes the exponent but preserves exponential decay. Pulling the resulting nested geometry to the dynamical principal nest with the bounded-cross-ratio and complex-collar estimates of Chapters 12–13 gives exactly the expanding, bounded-distortion, integrable-return inducing scheme used in Section 7 above; the BV lifting argument from Theorem 20.1 therefore supplies an acip for the target map. Thus the only parameters in a box not certified regular or stochastic lie over the quadratic exceptional set X . Their parameter set is a quasisymmetric pullback of X , hence has measure zero by (20.41). A finite union is null.

Exhaust $J \setminus T$ by countably many compact intervals. Since T itself is countable, the total set of parameters which are neither regular nor stochastic has Lebesgue measure zero. Hence for Lebesgue-almost every $t \in J$, f_t is regular or admits an absolutely continuous invariant probability measure, exactly as asserted. $\square$

Theorem 20.3 (Typical nonregular maps are Collet-Eckmann). **Statement scope.** *Let F be a nontrivial finite-parameter real-analytic family of S -unimodal maps (quadratic critical point and negative Schwarzian derivative). Then for Lebesgue-a.e. nonregular parameter, the corresponding map is Collet-Eckmann and its critical orbit is polynomially recurrent. Moreover, for each $r \geq 3$, in a residual set of finite-parameter C^r S -unimodal families, Lebesgue-a.e. nonregular parameter is Collet-Eckmann and has subexponential critical recurrence. The sharper exponent-one recurrence for general analytic unimodal families belongs to the subsequent Phase-Parameter theorem and is not claimed here.*

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Combine restricted quadratic phase-parameter capacity, cool-landings large deviations, free/bound derivative recovery, polynomial recurrence, total quasisymmetric probability, analytic-family hybrid transfer, and the Baire-robust smooth-family closure.

The mathematical argument begins in the proof environment below.

Proof. The proof has two levels. First we prove a quasisymmetrically robust statement on the quadratic transversal and transfer it to every nontrivial analytic S -unimodal family. Then a finite-stage robustness argument gives the residual smooth-family statement.

Principal nests and capacity.

Let $I_0 \supset I_1 \supset \cdots \ni c$ be the principal nest and $c_n = |I_{n+1}|/|I_n|$. On the real quadratic transversal, monotonicity of kneading identifies each deep landing cylinder in I_n with the parameter cylinder having the same itinerary. The boundaries move holomorphically over the corresponding parapuzzle piece. Complex collars and the lambda lemma therefore make the restricted phase-parameter and phase-phase holonomies arbitrarily close to conformal at deep levels; in particular, for every fixed $\gamma > 1$ they are eventually γ -quasisymmetric on the deleted landing Cantor sets.

For $X \subset I$ set

$$p_\gamma(X | I) = \sup_{h \in QS(\gamma)} \frac{|h(X \cap I)|}{|h(I)|}.$$

Quasisymmetric power distortion turns summable phase-space capacity estimates into summable parameter estimates.

Cool landings.

Negative-Schwarzian Koebe distortion and the principal-nest complex bounds control both the return-time tails and the distortion of every full return branch. Away from a set of capacity $O(c_{n-1}^\eta)$, almost all returns have duration between fixed small powers of c_{n-1}^{-1} ; short returns, exceptionally long returns and previously bad branches occupy only a c_{n-1}^η fraction of every sufficiently long initial landing segment. Call such an itinerary cool. A binomial large-deviation estimate gives

$$p_\gamma\{\text{level } n \text{ landing is not cool}\} \leq Cc_{n-1}^{\eta'}.$$

On the surviving set the scale factors decrease torrentially, so $\sum_n c_n^\theta < \infty$ for every $\theta > 0$. Hence the total bad capacity is summable.

Collet-Eckmann expansion.

Outside a fixed critical neighborhood, Mañé expansion gives $|Df^m| \geq C\lambda^m$. A return at distance ρ from the quadratic critical point enters a binding period of length $O(|\log \rho|)$; Koebe distortion and free expansion recover, at the end of binding, at least $C\rho^{-1+\varepsilon}$ derivative. In a cool landing the total time spent in bad branches is a vanishing fraction of the landing time. Therefore the average logarithmic derivative at successive principal return times satisfies a convex recurrence with a fixed positive contribution from the new good returns. Its lower limit is positive. Splitting times between principal returns into complete returns and one unfinished free/bound segment yields

$$|Df^n(f(c))| \geq C\Lambda^n, \quad \Lambda > 1.$$

Thus the map is Collet-Eckmann.

Polynomial recurrence.

Partition I_n into logarithmic depth classes. Capacity estimates show that, except for a summable exceptional set, the first i full returns cannot enter a class substantially deeper than the one predicted by i . If $R_n^i(c) = f^{k_i}(c)$, the cool-return time bounds compare $\log k_i$ with the same depth scale. Dividing the two estimates gives a finite $A > 0$ with

$$|f^k(c) - c| \geq Ck^{-A}$$

for all large k . Hence the orbit is polynomially, and therefore subexponentially, recurrent. On the unparameterized quadratic transversal the asymptotic distortion tends to one and the sharp source exponent is one.

Total quasisymmetric probability and analytic families.

For every fixed quasisymmetric coordinate, the capacities of the bad level- n sets are summable. Borel-Cantelli therefore shows that quadratic parameters which are regular or simultaneously Collet-Eckmann and polynomially recurrent have total qua-

sisymmetric probability.

The local straightening theorem supplies a locally quasisymmetric map on every transverse box of a nontrivial analytic S-unimodal family, outside a countable parameter set. We transport the nest cylinders, not the derivative itself, through that holonomy. Quasisymmetry preserves the required power/summability estimates, and the target family has the same uniform Koebe and quadratic-binding mechanisms. Re-running the preceding free/bound argument proves CE and polynomial recurrence for almost every nonregular parameter of the analytic family. Notice that this uses only the restricted quadratic nest holonomy; the stronger general-family Phase-Parameter theorem remains a separate result.

Residual smooth families.

Nontrivial analytic families are dense among C^r , $r \geq 3$, S-unimodal families. For each m , require on parameter measure $> 1 - 2^{-m}$ only finitely many strict cool-nest, derivative-growth and $|f^n(c) - c| > e^{-n/m}$ inequalities. Such a finite-stage requirement is open in C^2 , and the analytic theorem makes its union over sufficiently late stages dense. Intersecting these open dense sets over m and applying a diagonal Borel-Cantelli argument gives a residual family set for which almost every nonregular parameter is Collet-Eckmann and subexponentially recurrent.

This proves both clauses of the theorem. $\square$

Theorem 20.4 (Phase-parameter relation). **Statement scope.** *Let (f_t) be a nontrivial one-parameter real-analytic family of unimodal maps in the Avila-Moreira setting. For Lebesgue-a.e. nonregular parameter t_0 , the map f_{t_0} is recurrent, finitely renormalizable and simple, and the family is transverse to its local hybrid/topological leaf. There are nested parameter windows $J_i \ni t_0$, phase-parameter maps Ξ_i and phase-phase maps $H_i[t]$ satisfying the topological Phase-Parameter Relation and exponential decay of parameter geometry along noncentral levels. Moreover, for every $\gamma > 1$, all sufficiently deep restrictions occurring in the two phase-parameter and two phase-phase clauses of the Avila-Moreira relation are γ -quasisymmetric. The conclusion is local and asymptotic on the prescribed deep Cantor/nest pieces; no global quasisymmetry of the whole parameter interval is asserted.*

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Build the full transverse return family, obtain the complex holomorphic-motion distortion estimate and exponential parameter geometry, prove the four asymptotic quasisymmetry clauses, and transfer them through the hybrid lamination on the almost-everywhere recurrent/simple/transverse locus.

The mathematical argument begins in the proof environment below.

Proof. Write

$$I_0 \supset I_1 \supset I_2 \supset \cdots$$

for the principal nest of a recurrent, finitely renormalizable map f , with I_{i+1} the central return domain at level i . The phase-parameter theorem compares this dynamical nest with nested parameter windows J_i on which the same finite critical itinerary persists.

The proof has three stages.

1. A full transverse model family. The hybrid-lamination analysis of the preceding theorem identifies a polynomial vector field w transverse to the hybrid leaf. For

$$f_t = f \circ (\text{id} + tw)$$

the finite return branches move holomorphically. Transversality makes the normalized critical value a genuine local parameter coordinate. Consequently, from some level onward there are homeomorphisms

$$\Xi_i : I_i \longrightarrow J_i$$

that match every finite landing/return cylinder with the parameter set carrying that same critical itinerary, and phase-phase conjugacies $H_i[t]$ outside the next central domain.

At each noncentral level, the real and complex a priori bounds provide a definite annular collar. Concatenating the first k such collars gives a parameter annulus whose modulus grows at least linearly in k . Hence, if $n_k - 1$ denotes the k th noncentral level,

$$\frac{|J_{n_k+1}|}{|J_{n_k}|} \leq C\vartheta^k, \quad 0 < \vartheta < 1.$$

This is the decay of parameter geometry.

2. Asymptotic metric control. Complexify the return family. If M_i is the smaller of the dynamical and parameter annular moduli separating the level- i retained pieces from their outer boundaries, the holomorphic-motion λ -lemma gives quasiconformal extensions of the relevant holonomies with

$$K_i = 1 + O(e^{-cM_i}).$$

For a simple map there are only finitely many central returns, so all deep levels are noncentral and $M_i \rightarrow \infty$. Therefore, given $\gamma > 1$, all sufficiently deep prescribed restrictions of both Ξ_i and $H_i[t]$ are γ -quasisymmetric. These are exactly the two phase-parameter and two phase-phase clauses of the Phase-Parameter Relation. The assertion concerns the deleted deep Cantor/nest pieces, not the whole parameter interval.

Now let g_s be any analytic family transverse to the same hybrid leaf. The local hybrid-lamination holonomy ψ satisfies

$$\Xi_i^g = \psi \circ \Xi_i^{f_t}.$$

On the shrinking windows, the λ -lemma makes the quasisymmetry constant of ψ tend to one. The conjugacy between paired phase maps has the same property. Hence parameter-geometry decay and all four asymptotic metric clauses transfer from the special family to g_s .

3. Almost every nonregular parameter has the required scope. Failure of the Kupka-Smale condition, a periodic/preperiodic critical orbit, or tangency to the analytic hybrid lamination occurs only at countably many parameters. Nonrecurrent non-

regular parameters have measure zero: outside a small nice critical neighborhood their critical values lie in a uniformly expanding Cantor set; holomorphic motion and transverse phase-parameter holonomy transfer the definite gaps of that Cantor set to definite parameter gaps, contradicting the existence of a density point. Infinitely renormalizable parameters form a null set by the porosity argument proved in the preceding regular-or-stochastic theorem.

Finally, the topological phase-parameter construction and the exponential parameter-window estimate do not require simplicity. The relative measure of the window producing a central return at the k th noncentral level is at most $C\vartheta^k$. These bounds are summable, so Borel-Cantelli shows that parameters with infinitely many central returns have measure zero.

Thus almost every nonregular parameter is recurrent, finitely renormalizable, simple, and transverse. Applying the transverse-family construction proves the full asymptotic Phase-Parameter Relation there. $\square$

Theorem 20.5 (Collet-Eckmann invariant density and exponential mixing). **Statement scope.** *Let T satisfy the standing S -unimodal regularity hypotheses of Keller-Nowicki, negative Schwarzian derivative, and the Collet-Eckmann condition. If T is nonrenormalizable, then it has a unique invariant probability density h , $\mu = hm$ is mixing, and Theorem 1.1 gives exponential decay of correlations for bounded-variation F against the source $L^{1+\delta}$ class of G . If T is finitely renormalizable, the same conclusions apply to the mixing return system $(I, T^p, \mu|_I)$ on its periodic interval, as in the source remark.*

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Construct the Hofbauer extension and Collet-Eckmann free/bound geometry, introduce the adapted critical weight, prove weighted-BV quasicompactness and simple peripheral spectrum, and derive the invariant density and exponential correlations.

The mathematical argument begins in the proof environment below.

Proof. Let T satisfy the standing Keller-Nowicki S -unimodal hypotheses and the Collet-Eckmann condition. In the nonrenormalizable case let J be the mixing invariant interval and put $S = T|_J$. If T is finitely renormalizable, let J be the periodic interval of the last renormalization, let p be its period, and put $S = T^p|_J$. Quadratic nonflatness, negative Schwarzian derivative and Collet-Eckmann growth pass to this return system after changing constants. It is therefore enough to treat a mixing S -unimodal map $S : J \rightarrow J$. Write m for normalized Lebesgue measure on J .

1. Hofbauer extension and the expansion package. Cut J at all preimages of the critical point c . For every monotonicity cylinder Z of S^n , retain the image $S^n Z$ as a separate copy, identifying copies with the same image and itinerary. This produces the countable Hofbauer extension

$$\hat{J} = \bigsqcup_{D \in \mathcal{D}} D,$$

with factor map $\pi : \hat{J} \rightarrow J$ and Markov map $\hat{S} : \hat{J} \rightarrow \hat{J}$ satisfying

$$\pi \circ \hat{S} = S \circ \pi. \quad (20.1)$$

The level $\ell(D)$ of a domain is the least number of iterates needed to produce it from the base copy. Since there is only one critical point, the endpoints at level k belong to the two one-sided critical-orbit chains, so

$$\#\{D \in \mathcal{D} : \ell(D) = k\} \leq C(1 + k). \quad (20.2)$$

We now derive the quantitative estimates needed later. Choose a sufficiently small critical neighbourhood U . Away from U , compactness and the absence of a neutral periodic orbit in the mixing component give uniform expansion: there are $C_0 > 0$ and $\lambda_0 > 1$ such that every orbit segment of length r avoiding U satisfies

$$|DS^r(x)| \geq C_0 \lambda_0^r. \quad (20.3)$$

If an orbit enters U , let b be its maximal binding time to the postcritical orbit. Quadratic nonflatness gives $|Sx - S(c)| \asymp |x - c|^2$, while Koebe distortion along the bound segment gives

$$|DS^b(Sx)| \asymp |DS^b(S(c))|.$$

The Collet-Eckmann hypothesis bounds the right-hand side below by $C\lambda_c^b$. Maximality of the binding period recovers the initial quadratic loss, and after decreasing the base of the exponential we obtain

$$|DS^{b+1}(x)| \geq C_0 \lambda_0^b. \quad (20.4)$$

Negative Schwarzian derivative expands cross-ratios. Hence every free or bound branch with a fixed scaled image neighbourhood has a fixed scaled preimage neighbourhood, and Koebe gives

$$\left| \log \frac{|DS^r(x)|}{|DS^r(y)|} \right| \leq K_0 \frac{|S^r x - S^r y|}{|S^r Z|} \quad (20.5)$$

for x, y in the same monotonicity cylinder Z . Concatenating free and bound blocks changes only the constants. In particular inverse derivatives and their branchwise variations are exponentially summable along every tower itinerary.

2. Adapted weight and weighted bounded variation. The derivative may be small at the entrance to a binding block, so ordinary BV on the raw tower is not the correct norm. Choose $\lambda_0^{-1} < q < \beta < 1$. For y outside the precritical set define an adapted metric weight by taking the worst future inverse expansion,

$$A(y) = \sup_{n \geq 0} \frac{q^{-n}}{|D\hat{S}^n(y)|}, \quad w(y) = \beta^{-\ell(y)} A(y). \quad (20.6)$$

The free/bound estimates above imply $A(y) < \infty$ almost everywhere and, more importantly, the cocycle inequality

$$\frac{w(\hat{S}y)}{|D\hat{S}(y)|w(y)} \leq \frac{q}{\beta} < 1. \quad (20.7)$$

On each monotonicity branch, (20.5) implies bounded logarithmic variation of this ratio. Iterating yields constants $C_1 < \infty$ and $q/\beta < \theta < 1$ such that for every n -cylinder Z ,

$$\sup_Z \psi_n \leq C_1 \theta^n, \quad \text{var}_Z(\psi_n) \leq C_1 \theta^n, \quad (20.8)$$

where

$$\psi(y) = \frac{1}{|D\hat{S}(y)|} \frac{w(\hat{S}y)}{w(y)}, \quad \psi_n = \prod_{j=0}^{n-1} \psi \circ \hat{S}^j.$$

This is precisely the compensation mechanism needed at the critical orbit: the physical Jacobian is replaced by a cohomologous Jacobian whose inverse branches contract uniformly.

Let $\hat{m}$ be Lebesgue measure on each tower domain and put

$$\|u\|_{\mathcal{B}} = \sum_{D \in \mathcal{D}} \beta^{-\ell(D)} (\text{var}_D u + \|u\|_{L^1(D)}), \quad \|u\|_{\mathcal{W}} = \sum_D \|u\|_{L^1(D)}. \quad (20.9)$$

Finite-level truncation together with $BV(D) \hookrightarrow L^1(D)$ shows that the unit ball of $\mathcal{B}$ is relatively compact for the weak norm. Let $\hat{P}$ be the Perron-Frobenius operator of $\hat{S}$ and conjugate it by multiplication with w :

$$\mathcal{Q} = M_w \hat{P} M_w^{-1}, \quad (\mathcal{Q}u)(x) = \sum_{\hat{S}y=x} \psi(y) u(y). \quad (20.10)$$

On an n -cylinder Z with inverse branch g_Z , $\mathcal{Q}_Z^n u = (\psi_n u) \circ g_Z$. The product rule for variation gives

$$\text{var}(\mathcal{Q}_Z^n u) \leq (\sup_Z \psi_n) \text{var}_Z u + (\text{var}_Z \psi_n) \sup_Z |u|. \quad (20.11)$$

Use $\sup_Z |u| \leq |Z|^{-1} \|u\|_{L^1(Z)} + \text{var}_Z u$, sum over Markov cylinders, and apply (20.8). Terms whose image has a scaled neighbourhood are controlled by Koebe; boundary terms are charged to the weak norm. We obtain constants $A, B < \infty$ and $0 < \vartheta < 1$ such that

$$\|\mathcal{Q}^n u\|_{\mathcal{B}} \leq A \vartheta^n \|u\|_{\mathcal{B}} + B \|u\|_{\mathcal{W}}, \quad \|\mathcal{Q}^n u\|_{\mathcal{W}} \leq B \|u\|_{\mathcal{W}}. \quad (20.12)$$

3. Quasicompactness and the peripheral spectrum. Fix $\vartheta < \Theta < 1$. Truncate the tower above level N and, on the remaining finitely many domains, replace a BV function on each n -cylinder by its average. After composition with $\mathcal{Q}^n$ this gives a finite-rank operator $K_{n,N}$. Estimate (20.11) controls the branchwise averaging error by $C \vartheta^n \|u\|_{\mathcal{B}}$; the strong level weight controls the truncation tail by $C \beta^N \|u\|_{\mathcal{B}}$. Taking N proportional to n gives finite-rank K_n with

$$\|\mathcal{Q}^n - K_n\|_{\mathcal{B} \rightarrow \mathcal{B}} \leq C \Theta^n. \quad (20.13)$$

Hence

$$r_{\text{ess}}(\mathcal{Q}|\mathcal{B}) \leq \Theta < 1. \quad (20.14)$$

Thus the former deep spectral interface has been reduced to an explicit Lasota-Yorke/finite-rank approximation argument.

The original Perron–Frobenius operator preserves integrals, so the recurrent core has spectral radius one. Positivity and (20.14) yield a nonnegative eigenvector u_* with $\mathcal{Q}u_* = u_*$. Put $\hat{h} = w^{-1}u_*$ and normalize it to have integral one. If $\mathcal{Q}u = \zeta u$ with $|\zeta| = 1$, undoing the conjugacy and using positivity gives $\hat{P}|w^{-1}u| \geq |w^{-1}u|$; equality of integrals forces equality. Mixing of S makes the recurrent Markov core irreducible and aperiodic. Equality in the triangle inequality on each inverse branch therefore forces a constant phase, so $\zeta = 1$ and the positive eigenvector is unique up to scale. Consequently

$$\mathcal{Q}^n = \Pi + N^n, \quad \Pi u = u_* \Lambda(u), \quad \Pi N = N \Pi = 0, \quad \|N^n\|_{\mathcal{B}} \leq C \rho^n \quad (20.15)$$

for some $0 < \rho < 1$.

4. The invariant density. Project the tower eigenfunction to J :

$$h(x) = \sum_{\pi y=x} \hat{h}(y). \quad (20.16)$$

Since $A \geq 1$ in (20.6), $w^{-1}(y) \leq \beta^{\ell(y)}$; together with the linear level count (20.2),

$$\sup_{x \in J} \sum_{\pi y=x} w(y)^{-1} \leq C \sum_{k \geq 0} (1+k) \beta^k < \infty. \quad (20.17)$$

Hence the series in (20.16) converges almost everywhere, $h \in L^1(m)$, $h \geq 0$, and $\int h dm = 1$. The factor relation (20.1) gives $Ph = h$. Thus $d\mu = h dm$ is an absolutely continuous invariant probability.

If there were two ergodic acips, their basins would be disjoint positive- Lebesgue-measure sets. A density point of either basin lifts into an interior Markov branch; bounded distortion sends a fixed positive fraction to the recurrent base. Irreducibility then forces each lifted basin to have full measure on the same recurrent core, a contradiction. Hence the acip is unique. Aperiodicity of the same core gives mixing.

5. Exponential correlations. For $F \in BV(J)$ define

$$U_F = w(F \circ \pi) \hat{h}.$$

The product rule for variation and (20.17) give

$$\|U_F\|_{\mathcal{B}} \leq C \|F\|_{BV}. \quad (20.18)$$

The adapted weight also gives the dual embedding required for the second observable. More precisely, the level decomposition, Holder’s inequality, (20.2), (20.3)–(20.5), and $w^{-1} \leq \beta^{\ell}$ imply that for some $\delta > 0$,

$$\left| \int_{\hat{J}} (G \circ \pi) w^{-1} V d\hat{m} \right| \leq C_{\delta} \|G\|_{L^{1+\delta}(m)} \|V\|_{\mathcal{B}}. \quad (20.19)$$

Indeed the contribution of level k is bounded by a constant times $(1+k)\beta^k$ multiplied by an exponentially shrinking-cylinder factor; choosing $\delta > 0$ sufficiently small makes the resulting geometric series summable. This is the only place where the source $L^{1+\delta}$

class enters.

Let $\bar{F} = F - \int F d\mu$. Transfer-operator duality, (20.15), (20.18) and (20.19) yield

$$\left| \int_J F(G \circ S^n) d\mu - \int_J F d\mu \int_J G d\mu \right| = \left| \int_{\hat{J}} (G \circ \pi) w^{-1} N^n U_{\bar{F}} d\hat{m} \right| \leq C \rho^n \|F\|_{BV} \|G\|_{L^{1+\delta}(m)}.$$

This is the asserted exponential mixing estimate.

We have therefore reconstructed the Markov extension, the critical-orbit weight, weighted-BV quasicompactness, simple peripheral spectrum, invariant density and correlation estimate. For a nonrenormalizable map $S = T$; for a finitely renormalizable map the conclusion applies to the last periodic mixing return system $(J, T^p|_J)$, exactly as stated. $\square$

Theorem 20.6 (Central limit theorem for Collet–Eckmann maps). **Statement scope.** *Work with the mixing Keller–Nowicki system S : $S = T$ in the nonrenormalizable case, or $S = T^p|_J$ on the periodic mixing interval in the finitely renormalizable case, with normalized invariant probability μ . For every bounded-variation observable F , the covariance series defining σ_F^2 converges, and the centered sums $[\sum_{j=0}^{n-1} F \circ S^j - n \int F d\mu] / \sqrt{n}$, regarded as random variables with respect to normalized Lebesgue measure on the mixing interval, converge in distribution to $N(0, \sigma_F^2)$. If $\sigma_F^2 = 0$, the limit is the point mass at 0.*

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Analytic BV twists of the preceding spectral-gap operator; Riesz-projection persistence; pressure Hessian equals the covariance variance; characteristic-function scaling; degenerate zero-variance case.

The mathematical argument begins in the proof environment below.

Proof. Let S , I , m_I , $\mu = h m_I$, $\mathcal{B}$, and $\mathcal{L}$ be the mixing spectral system from the spectral decomposition proved in Theorem 20.5. Replace F by

$$\tilde{F} = F - \int_I F d\mu,$$

so that $\int \tilde{F} d\mu = 0$. To lighten notation, write F again for the centered observable and put

$$S_n F = \sum_{j=0}^{n-1} F \circ S^j.$$

Lemma 20.7 (Analytic twisted spectral decomposition). *For complex z near 0, define*

$$\mathcal{L}_z g = \mathcal{L}(e^{zF} g).$$

Then $z \mapsto \mathcal{L}_z$ is analytic in operator norm on $\mathcal{B}$ and, after shrinking the disk if necessary,

$$\mathcal{L}_z^n = \lambda(z)^n \Pi_z + N_z^n, \quad \|N_z^n\|_{\mathcal{B}} \leq C_0 \rho_0^n, \quad (20.1)$$

where $0 < \rho_0 < 1$, $\lambda(0) = 1$, $\Pi_0 = \Pi$, and λ, Π_z are analytic. Moreover

$$M_n(z) := \int_I e^{zS_n F} dm_I = c(z)\lambda(z)^n + O(\rho_0^n), \quad c(z) := \int_I \Pi_z \mathbf{1} dm_I, \quad (20.2)$$

uniformly for z in a sufficiently small closed disk, and $c(0) = 1$.

Proof. On bounded-variation functions, multiplication is continuous and

$$\text{Var}(uv) \leq \|u\|_\infty \text{Var}(v) + \|v\|_\infty \text{Var}(u).$$

The same estimate level by level on the Keller-Nowicki weighted tower shows that multiplication by e^{zF} is bounded on $\mathcal{B}$. The exponential power series therefore converges in operator norm locally uniformly in z ; hence $\mathcal{L}_z$ is an analytic family.

At $z = 0$, the spectral decomposition proved in Theorem 20.5 gives an isolated simple eigenvalue 1. Choose a small circle Γ around 1 that misses the rest of the spectrum. For z small, the resolvent exists on Γ , and the Riesz projection

$$\Pi_z = \frac{1}{2\pi i} \int_\Gamma (\zeta - \mathcal{L}_z)^{-1} d\zeta$$

is analytic and has rank one. Its enclosed eigenvalue $\lambda(z)$ is analytic as well. The complementary spectrum remains in a disk of radius $\rho_0 < 1$, which gives (20.1) by the usual contour formula for powers.

Finally, the Perron-Frobenius identity

$$\int_I \mathcal{L}_z^n \mathbf{1} dm_I = \int_I e^{zS_n F} dm_I$$

converts (20.1) into (20.2). Since $\Pi_0 \mathbf{1} = h \int_I 1 dm_I = h$ and $\int h dm_I = 1$, we have $c(0) = 1$. $\square$

Lemma 20.8 (Pressure derivatives and asymptotic variance). *Choose the analytic logarithm*

$$\Lambda(z) = \log \lambda(z), \quad \Lambda(0) = 0.$$

Then

$$\Lambda'(0) = 0, \quad \Lambda''(0) = \sigma_F^2 := \int F^2 d\mu + 2 \sum_{k \geq 1} \int F (F \circ S^k) d\mu. \quad (20.3)$$

The covariance series converges absolutely. Consequently

$$\Lambda(it) = -\frac{\sigma_F^2}{2} t^2 + O(|t|^3) \quad (t \in \mathbb{R}, t \rightarrow 0). \quad (20.4)$$

Proof. Absolute convergence follows from the exponential correlation estimate in the preceding theorem because F is bounded and of bounded variation. Moreover, if $\gamma_k = \int F (F \circ S^k) d\mu$, then

$$\frac{1}{n} \int (S_n F)^2 d\mu = \gamma_0 + 2 \sum_{k=1}^{n-1} \left(1 - \frac{k}{n}\right) \gamma_k \longrightarrow \sigma_F^2,$$

so $\sigma_F^2 \geq 0$.

Normalize the analytic eigenfunction h_z by

$$\mathcal{L}_z h_z = \lambda(z) h_z, \quad \int_I h_z dm_I = 1, \quad h_0 = h.$$

Differentiating at $z = 0$ and applying the left eigenfunctional $g \mapsto \int g dm_I$ gives

$$\lambda'(0) = \int_I F h dm_I = \int_I F d\mu = 0.$$

The differentiated eigenvalue equation is therefore

$$(I - \mathcal{L})h'_0 = \mathcal{L}(Fh), \quad \int h'_0 dm_I = 0.$$

On the zero-mean complement the spectral radius of $\mathcal{L}$ is less than one, hence

$$h'_0 = \sum_{k \geq 1} \mathcal{L}^k(Fh) \tag{20.5}$$

with convergence in $\mathcal{B}$.

Differentiate the eigenvalue equation a second time and integrate. The terms containing h''_0 cancel because integration is the left eigenfunctional of $\mathcal{L}$, while the terms containing $\lambda'(0)$ vanish. Thus

$$\lambda''(0) = \int F^2 h dm_I + 2 \int F h'_0 dm_I.$$

Using (20.5) and duality of the Perron-Frobenius operator,

$$\int F \mathcal{L}^k(Fh) dm_I = \int F (F \circ S^k) d\mu.$$

This proves $\lambda''(0) = \sigma_F^2$. Since $\lambda(0) = 1$ and $\lambda'(0) = 0$, the same second derivative holds for $\Lambda = \log \lambda$. Taylor expansion at $z = it$ yields (20.4). $\square$

Lemma 20.9 (Fourier scaling limit). *For every $\tau \in \mathbb{R}$,*

$$\lim_{n \rightarrow \infty} \int_I \exp\left(i\tau \frac{S_n F}{\sqrt{n}}\right) dm_I = \exp\left(-\frac{1}{2} \sigma_F^2 \tau^2\right). \tag{20.6}$$

Proof. Apply (20.2) with $z = i\tau/\sqrt{n}$. Since $c(z) \rightarrow 1$ and the remainder is exponentially small,

$$M_n\left(\frac{i\tau}{\sqrt{n}}\right) = c\left(\frac{i\tau}{\sqrt{n}}\right) \exp\left(n\Lambda\left(\frac{i\tau}{\sqrt{n}}\right)\right) + o(1).$$

By (20.4),

$$n\Lambda\left(\frac{i\tau}{\sqrt{n}}\right) = -\frac{1}{2} \sigma_F^2 \tau^2 + O(n^{-1/2}),$$

which proves (20.6). $\square$

The left side of (20.6) is the characteristic function of $S_n F / \sqrt{n}$ under normalized Lebesgue measure m_I . The continuity theorem for characteristic functions therefore gives convergence in distribution to $N(0, \sigma_F^2)$. If $\sigma_F^2 = 0$, the limiting characteristic function is identically 1, hence the limit is the point mass at 0. Restoring the subtracted mean gives the theorem exactly as stated. $\square$

Theorem 20.10 (Large deviations for Collet-Eckmann maps). **Statement scope.** *Let S and μ be the mixing Keller-Nowicki system specified in the preceding central-limit theorem and let F have bounded variation with $\sigma_F^2 > 0$. For every sufficiently small $\varepsilon > 0$, there is a number $\alpha_F(\varepsilon) < 0$ such that the normalized Lebesgue measure of the two-sided deviation event for $n^{-1} \sum_{j=0}^{n-1} F \circ S^j$ has logarithmic rate $\alpha_F(\varepsilon)$. The assertion is made on the mixing interval/system, not across different cyclic components without passing to the return map.*

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Real twisted pressure; strict convexity from positive variance; Chernoff upper bounds; exponential tilting lower bounds; two-sided rate as the slower one-sided tail.

The mathematical argument begins in the proof environment below.

Proof. Keep the centered bounded-variation observable F and the spectral notation of the preceding theorem, and assume $\sigma_F^2 > 0$.

Lemma 20.11 (Real pressure is locally strictly convex). *For real s near 0 the leading eigenvalue $\lambda(s)$ is positive, and*

$$\Lambda(s) = \log \lambda(s)$$

is real analytic with

$$\Lambda(0) = \Lambda'(0) = 0, \quad \Lambda''(0) = \sigma_F^2 > 0. \quad (20.7)$$

Moreover

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \int_I e^{s S_n F} dm_I = \Lambda(s) \quad (20.8)$$

locally uniformly for real s near 0.

Proof. The analytic spectral decomposition from Lemma 20.7 applies to real twists. Because $\lambda(0) = 1$ is simple and the real twisted operator preserves the real cone, continuity fixes the nearby leading eigenvalue as the positive real branch. Equation (20.7) is Lemma 20.8. Since $\Lambda''(0) > 0$, after shrinking the interval we have $\Lambda''(s) > 0$ throughout it.

For real s in that interval, (20.2) gives

$$\int e^{s S_n F} dm_I = c(s) e^{n \Lambda(s)} + O(\rho_0^n),$$

with $c(s) > 0$ near 0. Taking $n^{-1} \log$ proves (20.8), uniformly on a smaller compact interval. $\square$

Lemma 20.12 (One-sided local large-deviation rates). *There exists $\varepsilon_0 > 0$ such that, for $0 < \varepsilon < \varepsilon_0$, the equations*

$$\Lambda'(s_+(\varepsilon)) = \varepsilon, \quad \Lambda'(s_-(\varepsilon)) = -\varepsilon \quad (20.9)$$

have unique solutions $s_+(\varepsilon) > 0$ and $s_-(\varepsilon) < 0$. With

$$I(a) = s(a)a - \Lambda(s(a)), \quad \Lambda'(s(a)) = a, \quad (20.10)$$

we have

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{1}{n} \log m_I\{S_n F/n > \varepsilon\} &= -I(\varepsilon), \\ \lim_{n \rightarrow \infty} \frac{1}{n} \log m_I\{S_n F/n < -\varepsilon\} &= -I(-\varepsilon). \end{aligned} \quad (20.11)$$

Both $I(\varepsilon)$ and $I(-\varepsilon)$ are strictly positive.

Proof. Strict convexity makes Λ' strictly increasing with $\Lambda'(0) = 0$, so (20.9) has unique solutions for small ε . For the positive tail, Chernoff's inequality and (20.8) give, for every small $s > 0$,

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log m_I\{S_n F/n > \varepsilon\} \leq \Lambda(s) - s\varepsilon.$$

The right side is minimized at $s = s_+(\varepsilon)$, giving the upper bound $-I(\varepsilon)$.

For the matching lower bound, fix $a > \varepsilon$ sufficiently close to ε , let $s(a) > 0$ satisfy $\Lambda'(s(a)) = a$, and introduce the exponentially tilted probability

$$d\nu_{n,a} = \frac{e^{s(a)S_n F}}{\int e^{s(a)S_n F} dm_I} dm_I. \quad (20.12)$$

For every small t ,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \int e^{t(S_n F - na)} d\nu_{n,a} = \Lambda(s(a) + t) - \Lambda(s(a)) - ta. \quad (20.13)$$

The right side has value and first derivative zero at $t = 0$ and positive second derivative. Applying Chernoff bounds to (20.13), once with $t > 0$ and once with $t < 0$, shows that for every $\delta > 0$,

$$\nu_{n,a}\{|S_n F/n - a| < \delta\} = 1 - e^{-\kappa n + o(n)} \quad (20.14)$$

for some $\kappa > 0$.

Choose $0 < \delta < a - \varepsilon$. On the event in (20.14), $S_n F/n \leq a + \delta$, so (20.12) implies

$$\begin{aligned} m_I\{S_n F/n > \varepsilon\} &\geq m_I\{|S_n F/n - a| < \delta\} \\ &\geq \left(\int e^{s(a)S_n F} dm_I \right) e^{-ns(a)(a+\delta)} \nu_{n,a}\{|S_n F/n - a| < \delta\}. \end{aligned}$$

Hence

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log m_I\{S_n F/n > \varepsilon\} \geq \Lambda(s(a)) - s(a)(a + \delta).$$

Let first $\delta \downarrow 0$ and then $a \downarrow \varepsilon$. Continuity of $s(a)$ gives the lower bound $-I(\varepsilon)$. The

negative tail is identical with $s < 0$.

Finally, strict convexity and $\Lambda(0) = \Lambda'(0) = 0$ imply $s(a)a - \Lambda(s(a)) > 0$ for every nonzero sufficiently small a . Thus the two rates in (20.11) are strictly negative. $\square$

Lemma 20.13 (Two-sided deviation rate). *For $0 < \varepsilon < \varepsilon_0$,*

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log m_I \left\{ \left| \frac{S_n F}{n} \right| > \varepsilon \right\} = \alpha_F(\varepsilon), \quad \alpha_F(\varepsilon) = -\min\{I(\varepsilon), I(-\varepsilon)\} < 0. \quad (20.15)$$

Proof. The two-sided event is the union of the positive and negative one-sided tails. If positive sequences a_n, b_n have logarithmic rates A, B , then $a_n + b_n$ has logarithmic rate $\max\{A, B\}$. Apply this elementary fact to (20.11). Since both Legendre values are positive, (20.15) follows. $\square$

Undoing the centering of F replaces $S_n F/n$ in (20.15) by

$$\frac{1}{n} \sum_{j=0}^{n-1} F \circ S^j - \int F d\mu,$$

which is exactly the deviation event in the theorem. For a finitely renormalizable Collet-Eckmann map, all objects above belong to the mixing return map $S = T^p|_I$, so no assertion is made across distinct cyclic components. $\square$

20.3 Worked examples

Worked example: Free and bound derivative growth

Write a critical orbit as alternating free blocks and binding blocks. If free blocks contribute at least $e^{\gamma m}$ derivative growth and each binding block loses at most $e^{\beta m}$ with $\beta < \gamma$ in the long-run average, then the total derivative still grows exponentially. Parameter exclusion is designed to retain parameters for which the accumulated binding loss never catches the free expansion.

Worked example: Spectral gap to correlation decay

Suppose a transfer operator decomposes as $L^n = \Pi + N^n$, where $\Pi g = (\int g d\mu)h$ and $\|N^n\| \leq C\theta^n$, $0 < \theta < 1$. Pairing with a test observable immediately gives correlations of order $O(\theta^n)$. The CLT and large-deviation proofs perturb this same operator by e^{tF} .

Boundary case: Positive Lyapunov exponent is not yet a CLT

Positive Lyapunov exponent describes average derivative growth. A CLT additionally needs sufficient statistical decorrelation and control of the return-time structure. In this chapter that extra information is supplied by the Keller-Nowicki Markov extension and spectral gap.

20.4 What this chapter established

The chapter now proves the parameter-selection and typicality spine through the full Phase-Parameter Relation. Jakobson's positive-measure theorem supplies a first stochastic parameter set; the analytic hybrid lamination then transfers the regular-or-stochastic and typical Collet-Eckmann mechanisms between transversals. The Phase-Parameter theorem sharpens that transfer: deep dynamical nest pieces and their parameter windows become asymptotically quasisymmetric, while the parameter windows shrink exponentially along noncentral levels.

The scope distinctions are essential. The Phase-Parameter conclusion is local and asymptotic on prescribed deep Cantor/nest pieces, not a global quasisymmetry of the whole parameter line. Downstream mixing and limit-law theorems still require their own spectral/inducing hypotheses; they do not follow from the Phase-Parameter correspondence alone.

Looking ahead

Chapter 21 returns to the quadratic parameter plane and applies the puzzle, renormalization, and rigidity machinery to local connectivity of the Mandelbrot set in proved regimes.

Mandelbrot Rigidity, Local Connectivity and the Modern MLC Programme

Why this chapter matters

Local connectivity of the Mandelbrot set is one of the central organizing problems of quadratic dynamics. The full MLC conjecture is not assumed here. Instead, Chapter 21 separates several regimes where shrinking puzzle or renormalization geometry is known and proves local connectivity at the corresponding parameters.

The classical route uses Yoccoz puzzles and parapuzzles. Bounded-satellite parameters require neutral/pacman control from Chapter 19. Bounded-type Feigenbaum parameters require new a priori bounds for infinitely renormalizable maps. In each case the conclusion comes from shrinking connected parameter neighborhoods; the Feigenbaum theorem also carries universal dynamical and parameter scaling.

Learning goals

By the end of this chapter you should be able to:

1. explain why puzzle shrinking implies Julia local connectivity;
2. transfer dynamical puzzle moduli to parameter parapuzzles;
3. state the classical Yoccoz MLC regime without assuming global MLC;
4. describe the bounded-satellite neutral-renormalization route;
5. state the bounded-type Feigenbaum MLC theorem;
6. derive scaling universality from renormalization asymptotics while keeping general MLC open.

Conceptual roadmap

a priori bounds $\rightarrow$ puzzle/renormalization shrinking
 $\rightarrow$ parapuzzle shrinking
 $\rightarrow$ MLC at the parameter.

The input producing the bounds depends on the regime: classical Yoccoz puzzles, pacman/neutral renormalization, or bounded-type Feigenbaum renormalization.

Common pitfall

Do not use “MLC” as if it were globally established. Every application in this chapter must carry the exact combinatorial or neutral hypothesis that places the parameter in a proved regime.

21.1 Definitions

Definition 21.1 (MLC). MLC at a parameter c means the Mandelbrot set is locally connected at c ; the global MLC conjecture asserts local connectivity at every c in M .

Definition 21.2 (Feigenbaum parameter). A Feigenbaum parameter is an infinitely renormalizable parameter whose renormalizations have bounded combinatorics and recurrent scaling of Feigenbaum type; the exact bounded-type class is stated for each theorem.

Definition 21.3 (satellite combinatorics). Satellite combinatorics describes renormalizations where the small Julia sets form satellite cycles touching/rotating around a common periodic point, encoded by rational rotation data at each level.

Definition 21.4 (parapuzzle shrinking). Parapuzzle shrinking means the diameters of the nested parameter puzzle pieces containing a parameter tend to zero, which implies local connectivity at that parameter under the standard nested-connectivity setup.

21.2 Main results

Theorem 21.1 (Scaling universality near Feigenbaum parameters). **Statement scope.** For bounded-type Feigenbaum combinatorics in the Dudko–Lyubich theorem, dynamical and parameter scalings satisfy universal asymptotic behavior.

Proof strategy (not part of the proof)

Proof status. derived here.

Strategic outline only. Extract the dynamical and parameter scaling ratios from successive Feigenbaum renormalizations and use the convergence/compactness statement already proved in RES-003.

The mathematical argument begins in the proof environment below.

Proof. The preceding Feigenbaum theorem already contains the required universality statement. Here we isolate it for later use rather than prove a stronger statement from compactness alone.

Dynamical scales.

For a fixed bounded-type Feigenbaum combinatorial class covered by the theorem, the normalized scales of successive dynamical renormalizations have the universal asymptotic behavior proved by Dudko–Lyubich.

Parameter scales.

The corresponding parameter nest satisfies the matching parameter-scaling universality conclusion in the same source theorem.

Interpret “universal” correctly.

Universality is asserted for the corresponding combinatorics and normalization. The theorem does not claim one numerical scaling constant shared by every bounded-type combinatorial class. $\square$

Theorem 21.2 (Classical Yoccoz MLC cases). **Statement scope.** *Let $c \in M$. If $P_c(z) = z^2 + c$ is not infinitely renormalizable, then c has a basis of closed connected neighborhoods in M . In particular, the Mandelbrot set is locally connected at c .*

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Exact Yoccoz–Hubbard scope; internal PL–Yoccoz limb diameter; explicit extremal-metric dynamical-to-parameter modulus transfer; Chapter 9 divergent puzzle annuli and finite tower; neutral/parabolic/tuning case assembly.

The mathematical argument begins in the proof environment below.

Proof. Write $P_c(z) = z^2 + c$. We prove the exact classical parameter statement: if $c \in M$ is not infinitely renormalizable, then c has a basis of closed connected neighborhoods in M . The argument has two genuinely different parts. For all-repelling maps, dynamical annuli must be transferred to parameter annuli with a quantitative modulus estimate. Indifferent parameters are treated by the multiplier/limb geometry; the multiplier-one root requires the finite tuning picture.

Step 1: shrinking limbs and indifferent parameters.

Let U be a hyperbolic component of period k , let c_U be its center, and let $L_{U,p/q}$ be the p/q -limb attached to U . Continue the attracting k -cycle from U through its wake and write

$$\tau(c) = \log(P_c^k)'(z(c))$$

for the branch selected by the marked ray portrait. The Pommerenke–Levin–Yoccoz annulus estimate reconstructed in Chapter 17 says that on the p/q -limb the relevant logarithmic multiplier lies in a disk tangent to $2\pi ip/q$ whose radius is $O(k/q)$. The inverse multiplier section is holomorphic near the corresponding compact arc of the unit circle; at a primitive root one first passes to the standard local double cover. Its derivative is therefore bounded on that compact set. Consequently there is a constant C_U such that, according to whether the root is satellite or primitive,

$$\text{diam } L_{U,p/q} \leq C_U \frac{k}{q} \quad \text{or} \quad \text{diam } L_{U,p/q} \leq C_U \left(\frac{k}{q}\right)^{1/2}. \quad (21.1)$$

Only the fact that the right-hand side tends to zero with q will be used. The analytic continuation and the inverse section here use the hyperbolic multiplier coordinate of Chapter 4; the tangent-disk estimate is the explicit flat-torus argument already owned in Chapter 17.

Suppose first that P_c has an indifferent cycle of multiplier $e^{2\pi it} \neq 1$. If t is irrational, choose rational endpoints $p_j/q_j < t < r_j/s_j$ tending to t . Inside the wake of the unique adjacent hyperbolic component, take the closed multiplier-sector between these two angles together with all limbs attached at rational angles in that sector. Rational parameter-ray landing makes this set a closed connected neighborhood of c in M , while (21.1) makes its diameter tend to zero. If $t = p/q$ is rational and nonzero, c is the common boundary point of the two adjacent hyperbolic components. Take small multiplier sectors at $e^{2\pi ip/q}$ and at 1, adjoin all attached limbs except the distinguished common limb, and use (21.1) again. These unions form a shrinking basis of closed connected neighborhoods of c .

Step 2: the quantitative dynamical-to-parameter annulus transfer.

We now treat an all-repelling parameter. Fix a wake $W_{p/q}$ and a compact subwake avoiding a neighborhood of its root. The rational ray portrait moves holomorphically there by Chapter 4. Let

$$B_i[c'] \supset B_{i+1}[c']$$

be a contributing pair in the critical puzzle, of combinatorial modulus 2^{-m} in the Yoccoz weighting, and let

$$\mathcal{X}_i \supset \mathcal{X}_{i+1}$$

be the corresponding parapuzzle pieces. We claim that a constant $C = C(p/q, W) > 0$, independent of i and m , satisfies

$$\text{mod}(\mathcal{X}_i \setminus \overline{\mathcal{X}_{i+1}}) \geq C2^{-m}. \quad (21.2)$$

Here is the extremal-length proof. Let Φ_M be the parameter Böttcher coordinate from Chapter 4 and put

$$\mu_1 = 2^{m+\nu} |d \log \Phi_M|,$$

where the fixed rank offset ν depends only on the initial wake. The finite-degree critical-value/parameter correspondence gives

$$\text{Area}_{\mu_1}(\mathcal{X}_i \setminus \overline{\mathcal{X}_{i+1}}) \leq C_1 2^m. \quad (21.3)$$

Moreover every crosscut that stays outside M has μ_1 -length bounded below by a constant depending only on the initial wake and equipotential. The only missing curves are those that pass through the small sectors around the moving dividing fixed point $\alpha(c')$.

Choose $0 < r_1 < r$ so that the r -disk around $\alpha(c')$ remains inside the critical depth-zero sector and misses the other depth-one sectors throughout the compact subwake. Compactness away from the wake root gives two uniform positive constants: A , a lower Green-potential bound on the two marked rays outside the r_1 -disk, and $\delta_{p/q}$, the minimum angular separation of the rays landing at $\alpha(c')$ from those landing at $-\alpha(c')$. Since the critical sector does not separate $\alpha(c')$ from infinity, there is an analytic branch

$$\Psi(c', z) = \log(z - \alpha(c'))$$

on the sector and a constant L with $|\text{Im } \Psi| \leq L$ on $r_1 \leq |z - \alpha(c')| \leq r$. Put $z_i(c') = P_c^i(c')$

and $p_i(c') = z_i(c') - \alpha(c')$. In each component meeting a marked boundary parameter, choose measurable sections of the level curves of $\log |p_i|$ that separate that boundary point from $\mathcal{X}_{i+1}$. On those sections set $\mu_2 = |d \log p_i|$ and extend μ_2 by zero elsewhere. By construction the map $c' \mapsto \Psi(c', z_i(c')) = \log p_i(c')$ is injective on each chosen section, so its image lies in the fixed rectangle

$$\{\log r_1 \leq \operatorname{Re} w \leq \log r, |\operatorname{Im} w| \leq L\}.$$

Thus each section contributes bounded μ_2 -area. There are at most 2^m relevant inverse-image sectors, hence

$$\operatorname{Area}_{\mu_1+\mu_2}(\mathcal{X}_i \setminus \overline{\mathcal{X}_{i+1}}) \leq C_2 2^m. \quad (21.4)$$

It remains to bound the length of every curve γ joining the two boundary components. If the outer endpoint has Green potential below A , then γ must cross all of the selected level curves separating that boundary point from $\mathcal{X}_{i+1}$, and μ_2 contributes at least $\log(r/r_1)$. If the endpoint has potential at least A and γ meets M , then the μ_1 -length needed to climb from potential 0 to A is at least A . If γ avoids M , the angular/equipotential separation in the fixed wake gives μ_1 -length at least $\delta_{p/q}$. Therefore all joining curves have $(\mu_1 + \mu_2)$ -length at least

$$\ell_0 = \min\{A, \delta_{p/q}, \log(r/r_1)\} > 0.$$

The extremal-length formula

$$\operatorname{mod} A \geq \frac{(\inf_{\gamma} \operatorname{Length}_{\mu} \gamma)^2}{\operatorname{Area}_{\mu}(A)}$$

combined with (21.4) gives (21.2). This is exactly the place where the former proof had hidden the parameter-space mechanism behind the phrase “phase-parameter transversality.”

Step 3: all-repelling, at-most-finitely-renormalizable parameters.

Chapter 9 already owns the Yoccoz tableau and finite-renormalization geometry. For a nonrenormalizable all-repelling map, its critical nest contains a sequence of pairwise disjoint contributing annuli whose combinatorial moduli have divergent sum; in the nonrecurrent case one may choose an infinite subsequence with a uniform positive dynamical modulus, while in the recurrent case the tableau-fortune argument gives the required divergent weighted sum. Applying (21.2) term by term gives

$$\sum_j \operatorname{mod}(\mathcal{X}_{i_j} \setminus \overline{\mathcal{X}_{i_j+1}}) = \infty. \quad (21.5)$$

Grötzsch’s series law therefore forces the nested parapuzzle pieces to have diameter tending to zero.

If P_c is finitely renormalizable, use the enriched puzzle below the last renormalization. Chapter 9 supplies a priori geometry and shrinking for the last all-repelling nonrenormalizable tail, while Chapter 8 identifies the finite outer tower by straightening/tuning. The same construction of $\mu_1 + \mu_2$ is local to the marked ray sectors and therefore applies to the enriched parapuzzle. Finitely many outer levels change neither diver-

gence of the modulus sum nor the conclusion. Thus in every all-repelling not-infinitely-renormalizable case there are nested parapuzzle pieces $\mathcal{X}_n \ni c$ with $\text{diam } \mathcal{X}_n \rightarrow 0$. Their closures are bounded by finitely many rational parameter rays and an equipotential; Chapter 4 landing/separation implies $N_n = M \cap \overline{\mathcal{X}_n}$ is closed and connected. The sets N_n form a neighborhood basis of c in the relative topology of M .

Step 4: multiplier one, attracting parameters, and exhaustion of cases.

It remains to handle a parabolic parameter with multiplier 1. Such a parameter is the root of a primitive hyperbolic component. If the center lies in a small Mandelbrot copy M' , Chapter 8 tuning identifies M' with the connectedness locus of the corresponding quadratic-like family. Let $\mathcal{X}_n$ be the enriched parapuzzle pieces defining that copy. On the boundary of the hyperbolic component take the two points of internal angles $\pm 1/n$, join them inside the component, and add their landing parameter rays; let V_n be the closed side containing the root. Then

$$N_n = M \cap \overline{\mathcal{X}_n} \cap V_n \quad (21.6)$$

is connected. Any point in $\bigcap_n N_n$ lies in M' , whereas (21.1) forces the portions of M' cut out by V_n —a shrinking piece of the component together with limbs of denominator tending to infinity—to have diameter tending to zero. Hence $\bigcap_n N_n = \{c\}$ and the N_n form the required basis. Repeating the same argument through the finitely many outer tuning levels handles a finitely renormalizable center; for the main-cardioid cusp the outer tower is absent and the limb estimate alone gives (21.6).

Finally, if P_c has an attracting cycle then $c \in \text{int } M$, so small closed disks contained in M already give a connected neighborhood basis. A quadratic parameter is therefore covered by exactly one of the attracting, indifferent, or all-repelling cases above. Under the hypothesis that P_c is not infinitely renormalizable, each case gives a basis of closed connected neighborhoods in M . Hence M is locally connected at c . $\square$

Theorem 21.3 (Bounded-type satellite MLC theorem). **Statement scope.** *For the bounded-satellite infinitely renormalizable quadratic parameters constructed by Dudko–Lyubich, the Mandelbrot set is locally connected at the parameter; the corresponding Julia set is locally connected and has positive area. Their theorem also gives parameter-space self-similarity near Siegel parameters of periodic type.*

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Use Chapter 17 bounded-type boundary geometry, Chapter 19 pacman/maximal-prepacman geometry and the Black-Hole measure engine, and Chapter 20 hybrid lamination; then construct the external tree, seed family, continuation, homoclinic coding and shrinking, followed by the satellite trapping ladder, definite landing, hyperbolic return-scale separation, vanishing escape and Black-Hole assembly.

The mathematical argument begins in the proof environment below.

Proof. The theorem separates into fixed-base external geometry, unstable-family transport and shrinking, and a measure-theoretic area argument. The fixed-base ex-

ternal geometry is supplied by the preceding pacman-renormalization lemmas. The remaining positive-area step combines a satellite-specific trapping ladder with the Chapter 19 Black-Hole machinery.

This was proved in Chapter 19. Theorem 19.6 supplies the pseudo-Siegel external-width and sector-renormalization bounds, while Theorem 19.4 supplies the compatible hyperbolic pacman chart and its one-dimensional unstable direction.

This part of the source Section 5 needs no transcendental black box. By the maximal prepacman and pullback disk lemma the maximal prepacman is exhausted by proper finite-degree branched coverings and the limiting map is sigma-proper, so it has no finite asymptotic values. Let V and W be the two components of $\mathbb{C} \setminus \ell$. On a component U of $g^{-1}(V)$ or $g^{-1}(W)$ there are no critical points; hence $g : U \rightarrow V$ or W is a covering. The target is simply connected, so the covering is univalent and U is simply connected. Exhaust the target by topological disks Y_j with $Y_j \cap \ell$ an arc and lift them to nested proper disks X_j . For every j , the finite graph

$$(g|_{X_j})^{-1}(Y_j \cap \ell)$$

is a tree: a cycle would bound a union of complementary components and force a proper preimage of one of the unbounded half-planes. Passing to the exhaustion gives a connected acyclic locally finite graph $g^{-1}(\ell)$. Finally, if two different lifts U_1, U_2 of (say) V met at two boundary points in $\text{Dom } g$, the closure of $U_1 \cup U_2$ together with the intervening tree arcs would surround a lift of W , contradicting the one-access property. Their common boundary is therefore empty or a single branch point, necessarily critical. This reconstructs the source chess-board rule from the properness established above.

The bounded-type boundary geometry is already owned by Chapters 17 and 19. The lift Z is the universal-cover version of the marked Siegel disk, and the two commuting branches restrict on Z to lifts of the same bounded-type rotation. the Chapter 17 boundary-conjugacy, surgery, and quasicircle lemmas give the quasisymmetric boundary model, while the periodic scaling in Theorem 19.4 makes the lifted boundary globally a quasia-rc. In particular boundary subarcs with endpoints in a fixed compact fundamental interval are compact and contain only finitely many critical points below any prescribed generation.

Fix a return time P . The sigma-proper chess-board pullback tree lemma says that $F_*^{-P}(\partial Z)$ is a tree. A component of its complement mapping to Z therefore meets the preceding bubble tree at exactly one branch point: two distinct attachment points would create a cycle. This point is critical because two complementary lifts meet there, and the restriction away from it is degree one. Following the unique tree-geodesic from ∂Z to such a component records, in order, the critical return times

$$P_1, P_1 + P_2, \dots, P_1 + \dots + P_n = P,$$

which gives a unique finite label $s = (P_1, \dots, P_n)$. Conversely the corresponding successive degree-one pullbacks exist by the maximal prepacman and pullback disk lemma, so every label produces one bubble. If two closed labelled bubbles meet, the later one meets the earlier one only at its root; hence the incidence graph of closed bubbles is

a tree.

A lake of generation P is a component of $F_*^{-P}(\mathcal{O})$, where $\mathcal{O} = \mathbb{C} \setminus Z$. Its coast is a component of the pullback tree and F_*^P maps that coast homeomorphically to ∂Z . Orient the coast by the quasisymmetric boundary coordinate. The two sides meet successive bubbles and define a left and a right bubble chain. If the lake itself meets ∂Z , only finitely many lower-generation branch points lie between that contact arc and its critical root, so after applying the largest lower return one side contains no further branch point and is a finite chain. For a general lake, apply the first return which makes it meet ∂Z ; univalence of the intermediate pullback transfers the same conclusion. This also shows that every proper lake is one of the two side lakes of a uniquely determined bubble label.

It remains to rule out an unbounded labelled limb. Use the marked-sector exhaustion from the renormalization-tiling pseudo-conjugacy lemma and the maximal-prepacman pullback disk of the maximal prepacman and pullback disk lemma. On one fundamental range of sectors, compactness gives a return bound R with the following property: every boundary point whose closest return crosses from one marked sector to the next admits a univalent pullback of the next exterior sector, and the pullback stays in the preceding sector. Periodic renormalization rescales the same statement at every deeper level, with return times $R, R/t, R/t^2, \dots$ in the source power-triple coordinate. Their total is finite. Hence a sufficiently deep representative of every limb is carried conformally into one fixed bounded fundamental sector. Sigma-properness then makes the corresponding inverse component bounded; periodic scaling and dynamical equivalence transfer boundedness to every labelled limb. The accumulating set of each finite bubble chain is therefore compact. A proper lake has compact coast plus one such compact accumulating end, so its closure is compact as well. This reconstructs the source bubble/lake and bounded-limb layer without retaining Lemma 5.10 as an import.

We first prove the landing statement. It is enough to consider a primary critical point $c_P \in \partial Z$, since every finite label is obtained from this case by degree-one pullback. Periodicity gives a dilation A_* with $A_*(c_P) = c_{tP}$. The lake classification in the bubble/lake classification and compact limbs lemma supplies a proper lake $D_P \subsetneq \mathcal{O}$ on which

$$G_P := F_*^{(t-1)P} \circ A_* : D_P \longrightarrow \mathcal{O}$$

is conformal and carries the accumulating sets of the central, left and right bubble chains onto themselves. Each accumulating set is compact by the bubble/lake classification and compact limbs lemma. Since D_P is a proper subdomain of the hyperbolic domain $\mathcal{O}$, the inclusion $D_P \hookrightarrow \mathcal{O}$ is strictly contracting in hyperbolic metric. On the compact invariant union there is therefore a factor $\kappa < 1$ for the inclusion, so G_P expands hyperbolic distances by at least $\kappa^{-1} > 1$. If K is any one of the three compact invariant accumulating sets, then

$$\text{diam}_{\mathcal{O}} K = \text{diam}_{\mathcal{O}} G_P(K) \geq \kappa^{-1} \text{diam}_{\mathcal{O}} K,$$

and hence K is a singleton. The side-chain singletons equal the central one: otherwise the compact tree arc between two of them would have to cross a bounded side lake whose coast separates the corresponding chains, contradicting invariance under G_P .

Denote the common landing point by α_P . Degree-one pullback defines α_s for every label s and preserves the three-chain incidence. Two different labels cannot have the same alpha point, because after cancelling their maximal common prefix they would give two different degree-one preimages of the same bubble meeting away from a critical root, contrary to the sigma-proper chess-board pullback tree lemma.

The scaling copies of one proper lake exhaust $\mathcal{O}$: choose a proper lake whose coast contains a boundary interval about the marked point 0; inverse dilates of that interval exhaust the lifted boundary and the corresponding scaled lakes exhaust the ocean. It follows that every finite-time escaping component meets all sufficiently large scaled alpha points and hence that $\text{Esc}_P(F_*)$ is connected. Components newly appearing between two escaping times are bounded and attach at alpha points.

Embed the bubble tree in the oriented plane. For alpha points α_v, α_w , define the tree order by saying that α_v lies beyond α_w when it belongs to one of the two side lakes at α_w ; define the ambient order by the left/right order of their bubble skeletons. Local finiteness gives a lowest-generation skeleton separating any two finite skeletons. Removing that skeleton separates the plane into exactly the side components given by the bubble/lake classification and compact limbs lemma. Therefore two alpha points are separated in the tree order if and only if they are separated in the ambient order. This proves the order comparison used in the source Proposition 5.20 by planar tree topology alone.

For $\alpha_v \succ \alpha_w$, delete from the connected escaping tree all side decorations branching away between them and call the remaining compact continuum $[\alpha_v, \alpha_w]$. The order comparison makes these continua concatenate at intermediate alpha points and makes scaling equivariant. To see that they are arcs, subdivide a fundamental chain interval at all alpha points up to generation r . Every subchain is carried by a composition of univalent closest-return branches and a power A_*^m to one of finitely many fundamental subchains. The compact marked-sector family in Theorem 19.4 gives uniform Koebe distortion for the univalent factors, while $m \rightarrow \infty$ uniformly with r . Thus the mesh of the subdivision tends to zero geometrically. The alpha order is a complete linear order on the continuum, the alpha points are dense, and the two closed order rays at every point are closed; the elementary ordered-continuum criterion therefore identifies the continuum with an interval. The same distortion estimate shows that escaping time varies continuously and strictly along it.

The dominant alpha points on ∂Z form a bi-infinite ordered chain. Periodic scaling shifts this chain, so the consecutive fundamental intervals shrink to the marked point 0 in one direction and escape to infinity in the other. Their union is the zero external ray. A general segment $[\alpha_s, \infty)$ is obtained by finitely many degree-one lifts of this chain and is again a simple arc with dense alpha points. Finally, two complete rays can split only at a branch alpha point. The order comparison gives a largest common predecessor, and after that point the two rays lie in different side components; hence their intersection is exactly the unique common tail $[\alpha_s, \infty)$. This reconstructs the source order, external-chain and common-tail layer.

At a critical root c_s the local degree is two. Pull back the zero ray R^0 along the two univalent sides of the critical branch. The alpha points, separation orders and external chains lemma gives two simple rays $R_{s,-}, R_{s,+}$ with the same tail from α_s to

infinity and no further intersection. The Jordan theorem applied after cancelling that common tail shows that

$$\{c_s\} \cup R'_{s,-} \cup R'_{s,+}$$

bounds one closed disk; choose the disk containing Z_s and call it W_s . If $F_*^Q(c_v) = c_s$, univalence away from the roots maps the two bounding rays of W_v to those of W_s , hence $F_*^Q : W_v \rightarrow W_s$ is a homeomorphism. Applying the full generation removes the critical root and maps the wake interior conformally to the slit plane $\mathbb{C} \setminus R^0$.

We prove the small-primary-wake estimate without importing the polynomial local-connectivity argument used for calibration in the primary source. On a fixed lifted boundary interval J , all primary wakes are successive inverse lifts of finitely many fundamental wakes under the two closest-return branches. The boundary conjugacy of the quasisymmetric boundary conjugacy lemma is quasisymmetric, so the bounded-type rotation partition on ∂Z has mesh tending to zero in the Euclidean metric. The boundary is a quasicircle lemma gives a quasidisk collar on the actual bounded-type Siegel boundary, and Theorem 19.6 gives the compatible all-scale external-width/sector collars. Therefore each sufficiently deep boundary atom $I_n \subset J$ has an exterior crosscut $\gamma_n \subset \mathcal{O}$ joining its endpoints with

$$\text{diam}(I_n \cup \gamma_n) \leq C \text{diam } I_n \rightarrow 0,$$

with C independent of the atom. Let X_n be the root box cut off by $I_n \cup \gamma_n$; then $\text{diam } X_n \rightarrow 0$.

Remove a slightly smaller root box from one fundamental wake. Its remainder is compactly contained in $\mathcal{O}$. Each inverse closest-return branch is an inverse to a conformal map from one or two proper lakes onto $\mathcal{O}$, so Schwarz-Pick contracts the hyperbolic metric by a factor strictly below one on that compact core. Iterated lifts of the compact remainder therefore either have hyperbolic diameter tending to zero while remaining in a compact part of $\mathcal{O}$, or enter the shrinking root boxes X_n ; in both cases their Euclidean diameter tends to zero. Adding back X_n shows that the entire primary wakes tend to zero in diameter along the inverse-return sequence. Periodic scaling transports the estimate to every boundary level. Compactness of each level then implies that, for fixed $\varepsilon > 0$, only finitely many primary wakes on that level have diameter at least ε . This is the source Lemma 5.29 conclusion derived from the book's quasisymmetric boundary and sector-collar machinery.

At an alpha point α_P , exactly three bubble chains are incident. By the alpha points, separation orders and external chains lemma their three external rays lie in three different primary wakes, so exactly three primary wakes meet there. Follow one boundary ray of W_P toward the root. Whenever it meets the next alpha point, the common-tail theorem identifies the unique neighboring wake sharing the intervening arc. Repeating gives a chain of neighboring wakes whose alpha points converge to c_P : if they accumulated earlier, density of alpha points on the ray would produce another branching point between the alleged limit and c_P ; if infinitely many low-generation roots intervened on a compact boundary arc, this would contradict local finiteness of the pullback tree. Thus the two sides of every primary wake are completely tiled by neighboring wakes.

Suppose now that some $y \in \mathcal{O}$ belongs to no primary wake. Choose two fundamental primary wakes whose union with Z surrounds a topological disk abutting the marked point. Periodic dilations of this configuration exhaust $\mathcal{O}$, so the component Y of the complement of all primary-wake interiors containing y must meet ∂Z at some point x ; otherwise one scaled configuration would separate Y compactly from infinity and force it into a wake. Approaching x from the two sides along ∂Z , choose primary roots $c_{P_j}, c_{Q_j} \rightarrow x$ such that $W_{P_j} \cup W_{Q_j} \cup Z$ encloses Y . The wake topology and primary-wake shrinking lemma gives

$$\text{diam } W_{P_j} + \text{diam } W_{Q_j} \rightarrow 0.$$

The two shrinking barriers cannot continue to enclose the fixed connected set $Y \ni y$ while their roots converge to x ; this contradiction rules out a ghost limb. Hence primary wakes cover $\mathcal{O}$.

The conformal wake map from the wake topology and primary-wake shrinking lemma pulls this primary tiling back inside every W_s , giving the displayed recursive decomposition. At a nonbranch point only one child chain can contain the point; at an alpha point there are exactly three incident child directions. Thus at any fixed level at most three wakes with disjoint interiors contain a given escaping point. Their complements are separated from the point by finitely many wake boundaries, and the recursive tiling fills every sufficiently small component around the point; consequently the union of the containing level- $\geq m$ wakes is a neighborhood. This is the required wake-neighborhood basis.

This follows from the bubble/lake, alpha-point, and wake lemmas above. The source Sections 5.2–5.10 remain the parity/provenance route, but no theorem-sized fixed-base topological input is retained. In particular the long bounded-limb step is absorbed by Chapter 19 marked-sector properness, the alpha landing mechanism by strict hyperbolic-domain expansion, the arc/common-tail mechanism by planar-tree order plus Koebe/scaling mesh decay, and the small-wake step by the Chapter 17 quasimetric bounded-type boundary together with Chapter 19 all-scale exterior collars.

Fix P and write $\rho_{\mathcal{O}}$ and $\rho_{\mathcal{O} \setminus Z_P}$ for the hyperbolic metrics. On a compact subset of $W_P \setminus Z_P$ away from the root c_P , the strict inclusion

$$\mathcal{O} \setminus Z_P \hookrightarrow \mathcal{O}$$

has derivative norm bounded by a constant $\kappa_0 < 1$ in the corresponding hyperbolic metrics. Since F_*^P is conformal from the wake complement onto its image, Schwarz-Pick turns this strict contraction into a definite expansion for the return away from c_P .

It remains to make the constant uniform at the root. Periodicity of the pacman renormalization gives a global scaling A_* for the maximal prepacman. Lift A_* through $F_*^P : c_P \mapsto 0$ on a small critical-value disk to obtain a conformal germ ψ fixing c_P . The germ is attracting; let L linearize it to multiplication by a scalar ν with $|\nu| < 1$. The four local configurations

$$L(W_P), \quad L(Z), \quad L(Z_P), \quad L(\mathcal{O})$$

are forward invariant under this multiplication. Taking inverse dilates globalizes them to scaling-invariant model domains. On one fundamental annulus of the dilation the two model hyperbolic metrics are strictly separated; compactness gives a contraction factor $\kappa_1 < 1$. Scaling invariance propagates the same factor through every smaller annulus. Pulling back by L gives a uniform factor

$$\kappa := \max\{\kappa_0, \kappa_1\} < 1$$

on all of $W_P \setminus Z_P$, hence a return-expansion factor $\Lambda = \kappa^{-1} > 1$.

The same normalization gives a diameter bound for secondary subwakes. Away from c_P there are only compactly many configurations. A secondary wake entering a sufficiently small root neighborhood is, after a power of the dilation, represented in the fixed fundamental annulus; the fixed-base bubble-lake-alpha-wake skeleton lemma's small-primary-wake statement excludes a secondary wake whose image approaches 0 while the wake itself remains macroscopically far from c_P . Thus there is $C_P < \infty$ with

$$\text{diam}_O W_{(P,Q)} \leq C_P$$

for every secondary $W_{(P,Q)} \subset W_P$. Dynamic equivalence of primary wakes and the periodic scaling reduce the constants to finitely many fundamental labels, so the expansion and diameter bounds are uniform after normalization.

Now let

$$W_{s(1)} \supset W_{s(2)} \supset \cdots, \quad s(n) = (P_1, \dots, P_n),$$

be an infinite wake chain. For a secondary wake $W_v \subset W_P$, first apply F_*^P , then the minimal further return F_*^Q which sends it to a scaled primary wake, and finally the unique power A_*^{-m} returning that primary wake to W_P . The resulting first scaled return

$$\chi = A_*^{-m} \circ F_*^{P+Q} : W_v \longrightarrow W_P$$

has the same expansion factor $\Lambda > 1$. After at most two initial levels, $\chi^{n-2}(W_{s(n)})$ is a normalized secondary wake of hyperbolic diameter at most C . Therefore

$$\text{diam}_O W_{s(n)} \leq C\Lambda^{-(n-2)} \longrightarrow 0.$$

The wakes are nested nonempty compact topological disks by the fixed-base bubble-lake-alpha-wake skeleton lemma, so their intersection consists of one point. This is the source Lemmas 5.33-5.34 mechanism, now reconstructed from the fixed topological skeleton and Chapter 19 periodic scaling.

Let R be an external ray and Y its accumulation set. The wake-neighborhood basis in the fixed-base bubble-lake-alpha-wake skeleton lemma implies that Y cannot cross a wake boundary transversely. If Y meets a bubble boundary, the alpha/critical roots supplied by the fixed-base bubble-lake-alpha-wake skeleton lemma are dense along that boundary and separate any two distinct candidates, so connectedness of Y forces it to be a singleton. Otherwise Y lies in an infinite nested wake chain, whose intersection is a singleton by the first-scaled-return expansion and nested-wake shrinking lemma. Thus every ray lands. A periodic ray lands at a periodic point; discreteness of the return-time semigroup on a fixed periodic orbit gives its minimal period.

For $x \in \text{Esc}(F_*)$, if x lies in the interior of a ray segment, the fixed-base bubble-lake-alpha-wake skeleton lemma provides arbitrarily deep alpha points separating x from any different escaping point. If x is an endpoint, the wake basis and the first-scaled-return expansion and nested-wake shrinking lemma give a nested sequence shrinking exactly to x . Hence every combinatorial equivalence class is a singleton. The common-tail theorem from the fixed-base bubble-lake-alpha-wake skeleton lemma then gives a unique arc between any two points of every finite escaping set, so that set is uniquely geodesic; local finiteness in each proper lake makes the hull of a discrete subset a tree.

If the escaping set contained a planar disk D , two different points of a smaller disk $D' \Subset D$ could not be separated by removing a single alpha point: a punctured disk is connected. They would therefore belong to one combinatorial class, contradicting the preceding paragraph. Thus $\text{int Esc}(F_*) = \emptyset$. Likewise, a Fatou component not eventually mapped to Z is disjoint from every bubble boundary and therefore lies in one member of every sufficiently deep wake neighborhood. The first-scaled-return expansion and nested-wake shrinking lemma would force it into a singleton, impossible for an open Fatou component.

Finally suppose an invariant line field is supported on the escaping set. Represent it by an invariant Beltrami coefficient μ , extend μ by zero across Z and its iterated preimages, and integrate $t\mu$ with the normalization fixing the marked critical point, critical value and Siegel marking. Invariance cancels the Beltrami coefficients in the conjugacy equation, so

$$F_t = h_t \circ F_* \circ h_t^{-1}$$

is a holomorphic maximal prepacman. Transport the deformation down the Chapter 19 anti-renormalization tower. The uniform quasiconformal bound and the maximal prepacman and pullback disk lemma give a normal family of normalized anti-renormalizations; every limit is infinitely anti-renormalizable, hence the resulting path lies in the local unstable manifold W^u . On the other hand $\mu = 0$ on the full lifted Siegel disk, so the normalized deformation preserves its multiplier and marked conformal dynamics. By the Chapter 19 fixed-multiplier-motion and same-rotation-convergence lemmas and the stable-set identity in Theorem 19.4, the same path lies in the fixed-multiplier stable slice. The local stable and unstable manifolds meet only at F_* , so $F_t \equiv F_*$.

Thus h_t belongs to the normalized conformal centralizer of F_* . It is the identity on Z by the marked normalization, hence by equivariance it fixes every labelled bubble, alpha point and ray segment. The fixed-base bubble-lake-alpha-wake skeleton lemma's wake basis is dense in the escaping structure, so h_t is the identity there as well. Differentiating at $t = 0$ gives $\mu = 0$ almost everywhere. This proves the no-line-field assertion without importing the source Hairiness Theorem as a separate black box.

The sigma-proper chess-board pullback tree lemma supplies the chess-board tree from sigma-properness, the bubble/lake, alpha-point, and wake lemmas above reconstruct the bubble-lake-alpha-wake skeleton assembled as the fixed-base bubble-lake-alpha-wake skeleton lemma, the first-scaled-return expansion and nested-wake shrinking lemma gives first-scaled-return expansion and nested-wake shrinking, and the landing, geodesic and hairiness rigidity lemma gives landing, geodesicity and hairi-

ness rigidity. Hence the entire fixed-base Section 5 package is internal relative to the certified Chapters 17 and 19 infrastructure.

We reconstruct the source Section 6 mechanism from already-owned tools. First, the maximal prepacman and pullback disk lemma gives a protected critical-value disk and, after sufficiently deep antirenormalization, an exhaustion

$$W_-^{(k)} \cup W_+^{(k)} \longrightarrow D^{(k)}, \quad k < 0,$$

by proper finite-degree branched coverings whose union is the maximal prepacman. The analytic dependence of the unstable family and the collision-free critical orbit from the holomorphic motion and zero-dimensional fixed-multiplier fibre lemma allow the boundaries of the protected disks to be shrunk once so that, for every fixed k , the degree and critical itinerary of these finite coverings are constant on a parameter neighborhood. Thus the finite branched coverings are Hurwitz-equivalent there. This is exactly the finite-degree stability input used in source Lemma 6.1, but here it is inherited from the certified Chapter 19 maximal-extension construction.

Fix a regular value x outside the discrete critical-value set. On each finite covering every solution of

$$F_\lambda^S(z) = x$$

which is not critical continues holomorphically by the implicit-function theorem; constancy of degree and collision-free critical itineraries prevent loss or merger of branches on the chosen neighborhood. Exhausting in k therefore gives a compatible holomorphic motion of all finite preimage sets up to a prescribed escaping time T .

We next identify the boundary of the finite-time escaping set with the accumulation set of those preimages. Let U be a component of $\text{Dom}(F^P)$. Around x choose nested disks $W \subseteq \Omega$ containing at most one critical value. Every component W' of $(F^P)^{-1}(W)$ has uniformly bounded degree, because only finitely many critical points of the protected exhaustion can occur over W . Hence the annulus between the corresponding pullbacks of W and Ω has modulus bounded below in terms of that degree. After uniformizing U by the disk, properness implies that preimages of x approach every prime-end boundary point; the lower annulus modulus forces the corresponding pullback components to shrink. Thus every point of ∂U , and hence every point of $\partial \text{Esc}_T(F)$, is a limit of the moving preimages.

Apply Theorem 6.1 to the motion of the finite preimages. The lambda lemma extends it continuously and quasiconformally to their closure, hence to $\partial \text{Esc}_T(F)$. By the fixed-base external tree and shrinking wakes lemma the base escaping set has empty interior and supports no invariant line field. Equivariance transports any nonzero Beltrami coefficient on the moved escaping set back to an invariant line field on the base set; therefore the coefficient vanishes there. The induced motion of Esc_T is conformal in the source sense. The same argument proves uniqueness: two equivariant motions agree on every sufficiently deep finite preimage set, hence on its closure.

For $P > T$, extend the motion dynamically. On

$$\text{Esc}_P(F) \setminus \text{Esc}_{P-T}(F)$$

the iterate F^T is an unbranched covering onto the already moved $\text{Esc}_T(F)$. Lift the motion through this covering and combine it with the motion already defined on $\text{Esc}_{P-T}(F)$. The lift has the same quasiconformal dilatation. Iterating this construction over dyadic ranges $T < P \leq 2T$, $2T < P \leq 4T$, and so on gives a unique equivariant motion for every finite escaping time for which the critical value has not yet escaped. Openness of this condition follows because membership of the critical value in the complement of the moved closed set is open.

Finally use the renormalization covariance of the maximal prepacman. Every compact parameter arc used later enters, after sufficiently many renormalizations, a simply connected local unstable chart where the preceding motion is unique. Pulling that motion back by the affine/analytic renormalization identifications makes it independent of the chosen depth. Since the fixed-base external tree and shrinking wakes lemma makes the base tree uniquely geodesic with singleton combinatorial classes, the transported alpha points, ray segments, cyclic order and wake labels are canonical. Periodic ray cycles landing at regular repelling cycles continue with their landing points by the implicit-function theorem, so their landing pattern persists. This reconstructs the source Lemma 6.8 and the canonical-identification part of Section 6.3 without retaining Section 6 as a theorem-sized import.

The fixed-base assertions are exactly the fixed-base external tree and shrinking wakes lemma and the family-continuation assertions are the equivariant holomorphic continuation of the escaping tree lemma. Hence the external-ray tree continuation package lemma now depends only on the fixed-base external tree and shrinking wakes lemma; the former source-sized Section 6 motion interface has been discharged.

Here is the reduction. At a parabolic or attracting map choose a forward-invariant topological disk B in the attracting basin containing the postcritical set, and put $X = \mathbb{C} \setminus B$. Every relevant inverse branch maps a proper subdomain of X to X , hence Schwarz–Pick gives strict expansion of the hyperbolic metric. A rational ray is split into periodic pullback segments. The external-ray tree continuation package lemma supplies a periodic blocking ray cycle near the Siegel triangulation and preserves it along the parameter path. If the pulled-back segments meet a fixed compact part of X , hyperbolic expansion forces their diameters to zero. Otherwise they eventually remain in one of the finitely many ends: the parabolic/attracting end or a bounded complementary component cut out by the blocking cycle. The former converges to the distinguished boundary fixed point in the wall topology and the latter contracts in the hyperbolic metric. Thus every rational ray lands; this is the source Theorem 7.6 argument with the transcendental existence/motion of rays isolated in the external-ray tree continuation package lemma.

For the primary parabolic prepacman, the full attracting petal is unicritical. A global Fatou coordinate conjugates it conformally to the full parabolic basin of $z^2+1/4$. Pulling back the dyadic accesses gives a cyclic family of rays landing at the distinguished point α ; adjacent rays bound puzzle pieces and the critical pullback has degree two. Landing persists under the secondary perturbation because the landing cycle is repelling away from the parabolic collision. At the secondary parabolic prepacman, hyperbolic expansion of the nested degree-two pullbacks produces one periodic ray in every repelling petal of the secondary cycle; the unique pair separating the critical Fatou component from the other petals is the characteristic pair. This yields the secondary degree-two

puzzle restriction. When that secondary landing point is regular repelling, linearization gives collars on the two boundary rays; extending those collars through finitely many iterates thickens the restriction to a genuine quadratic-like map. Holomorphic motion of the repelling point and its finite ray segments makes this a holomorphic family.

By the ray landing and the primary/secondary puzzle restrictions lemma the secondary return is a holomorphic quadratic-like family near the chosen parabolic parameter. Apply the internal Chapter 8 straightening theorem and its characteristic-family regularity. The normalized straightening coordinate sends the secondary parabolic map to the quadratic polynomial whose distinguished fixed point has the same rational multiplier. Choose a nearby rational satellite limb. The internal Pommerenke-Levin-Yoccoz limb estimate used in Theorem 21.2 places the corresponding quadratic satellite copy in the straightening chart. Its inverse image is therefore a full small copy $\mathcal{M}_0$ on W^u ; this is the source Theorem 7.9 mechanism.

The thickened domains may be chosen with boundary in fixed linearizing collars of the regular repelling landing cycle. Over the compact copy $\mathcal{M}_0$ those collars vary continuously and remain disjoint, so compactness gives a positive minimum modulus ε_0 . On a hyperbolic component inside the copy, the attracting multiplier is a nondegenerate parameter coordinate by the Chapter 4 multiplier theorem and is constant along a hybrid leaf. Hence the one-dimensional W^u slice is transverse to the hybrid foliation there; the later lamination pullback is made in such a chart and then continued by the lambda lemma.

The characteristic rays from the ray landing and the primary/secondary puzzle restrictions lemma separate the unique critical secondary little filled set from the other members of its cycle. Let W' be the component on the α -side. The inverse branch of the primary return carrying W' strictly into itself is univalent, and α is its only nonescaping boundary point. Denjoy-Wolff therefore forces successive pullbacks Z_j to converge to α . This proves connectedness and forward invariance of the displayed flower exactly as in the source Lemma 8.1.

The quadratic-like collar in the ternary copy and uniform unbranched seed family lemma is added only near the regular repelling landing point. The postcritical set lies in the cycle of secondary little filled sets and those sets are disjoint from that collar. After shrinking the collar, (21.19) follows. Renormalization changes of variables are conformal, so the same modulus and unbranched statement hold on every transported copy $\mathcal{M}_n$.

It remains to project the flower from the maximal-prepacman plane. The external-ray tree continuation package lemma gives the labelled wakes and their holomorphic continuation. Fixed-generation wakes converge under antirenormalization to the corresponding base wakes, while high-generation wakes are uniformly small because nested wakes shrink. Thus every petal outside the Siegel core becomes uniformly small, and the fundamental-domain boundary can be moved along two labelled rays so that complete petals, rather than pieces of petals, lie inside the domain. Choose the standard fundamental-domain quotient used in the Chapter 19 maximal-prepacman/antirenormalization model, gluing the two boundary arcs by the corresponding commuting return. Because the adjusted boundary follows labelled rays and

cuts no petal, the flower descends to the pacman plane inside a Siegel triangulation; its upper part lies in the wall. Pulling that triangulation backward gives niceness at all deeper levels. This is the geometric content of the source Valuable Flower Theorem after the external-tree input has been isolated.

This is now the formal assembly of the ray-tree, ternary-copy, and valuable-flower lemmas above. Consequently the marked satellite seed recognition lemma is no longer a source-sized import: only the global transcendental ray-tree/holomorphic-motion package the external-ray tree continuation package lemma remains external inside the seed branch.

Put $\mathcal{M}_n = \mathcal{R}^n(\mathcal{M}_0)$ on the unstable manifold and let N_n be a small parameter neighborhood on which the transported quadratic-like restriction Q_f is defined.

By Theorem 20.2, hybrid classes of quadratic-like maps form local analytic codimension-one leaves. If a local leaf is written as $\{\phi = 0\}$, then its pullback by the analytic map $f \mapsto Q_f$ is

$$\{f : \phi(Q_f) = 0\}.$$

The transversality clause in the marked satellite seed recognition lemma makes $d(\phi \circ Q)$ nonzero in the one-dimensional unstable direction, so the analytic implicit-function theorem makes this pullback a codimension-one leaf. Distinct hybrid classes remain disjoint, and the lambda-lemma compactness already proved in Theorem 20.2 supplies the local lamination structure. Pulling the leaves back by $\mathcal{R}^{n-m}$ preserves analyticity and transversality. Since Theorem 19.4 identifies $W^s(f_*)$ as the codimension-one same-multiplier slice and gives exponential stable contraction, the pulled-back leaves converge to $W^s(f_*)$ and paste with it to form $\mathcal{F}$.

For the Siegel-side extension, use the compact analytic Siegel-renormalization chart from the Chapter 19 pacman construction. Its image meets $W^s(f_*)$ transversely because the multiplier is a nondegenerate coordinate on the one-dimensional unstable manifold. The same implicit-function argument therefore pulls every sufficiently deep leaf of $\mathcal{F}$ across that chart. This reconstructs the mechanism of the source's Lemmas 9.3–9.4 from already-owned lamination and hyperbolicity tools.

We spell out the hyperbolicity instead of importing a homoclinic-horseshoe theorem. Choose analytic coordinates (s, u) near f_* adapted to the splitting from Theorem 19.4, with s tangent to W^s and u tangent to W^u . After shrinking the chart there are constants $0 < \rho < 1 < \lambda$ and $C_0 \geq 1$ such that one renormalization step contracts the stable cone by at most ρ and expands the unstable cone by at least λ , while the derivatives of the finitely many entrance, gluing and exit charts are bounded by C_0 together with their inverses.

A deep homoclinic return spends N consecutive iterates in this chart, with $N \rightarrow \infty$ as the chosen copy is pulled deeper. Thus the derivative of the first-return map has stable norm at most $C_0^2 \rho^N$ and inverse unstable norm at most $C_0^2 \lambda^{-N}$, up to cross terms that tend uniformly to zero because the pulled-back leaves converge to the stable slice. Taking N large gives invariant stable and unstable cones with contraction factors strictly below one.

For a bi-infinite admissible itinerary, intersect the forward images of a stable graph with the backward images of an unstable graph. The stable graph transform contracts in forward time and the unstable graph transform contracts in backward time, so the

intersection consists of exactly one point. These points form a compact invariant set $\mathcal{H}$ and the cone estimates give uniform hyperbolicity there. The straightening/markings from the marked satellite seed recognition lemma records the same finite branches as the natural extension of the associated quadratic-like renormalization map, hence gives a surjective symbolic coding. Two points with the same code lie on the same forward stable graph and the same backward unstable graph, so the preceding uniqueness forces them to coincide. The coding is therefore injective as well. Since the coding space has singleton connected components, so does $\mathcal{H}$. The constant itinerary yields the unique fixed point g_m . This is the source Lemma 9.5 mechanism with its only new input reduced to the seed the marked satellite seed recognition lemma.

The seed restriction (21.18) is unbranched because the marked flower contains the postcritical set. Before each return to the homoclinic box, an orbit passes through one of the transported seed neighborhoods. Renormalization changes of variables on these stages are conformal, so the modulus and postcritical-exclusion statement descend unchanged to the corresponding deep quadratic-like restrictions. Hence there is one $\varepsilon > 0$ such that every sufficiently deep annulus separating two consecutive little filled sets has modulus at least ε .

We record the elementary shrinking consequence. If

$$K_0 \supset K_1 \supset K_2 \supset \cdots$$

are full compacta in a fixed simply connected domain and each annulus between consecutive levels has modulus at least ε , then conformal radii decrease by a factor at most $q(\varepsilon) < 1$ at each step. Koebe's theorem on the fixed outer chart converts conformal-radius decay into Euclidean diameter decay. Thus $\text{diam } K_j \rightarrow 0$. Applying this to the unbranched critical nest gives the dynamical shrinking claimed in the bounded-satellite dynamical/parameter shrinking lemma.

For parameter space, let P_j be the compact connected piece on the quadratic slice defined by the first j entries of the renormalization itinerary of g_m . The pieces are nested. Their intersection lies in the connected component of $\mathcal{H}$ with that itinerary and is therefore a singleton by the homoclinic horseshoe and singleton coding lemma. A nested sequence of nonempty compact sets in a compact metric space whose intersection is one point must have diameter tending to zero: otherwise a subsequence would contain pairs separated by some fixed $\delta > 0$, and compactness would produce two distinct points in the intersection. Hence $\text{diam } P_j \rightarrow 0$.

This is exactly the lamination, homoclinic-coding, and shrinking lemmas above. Thus the bounded-satellite dynamical/parameter shrinking lemma is no longer itself a theorem-sized source import.

The shrinking connected parameter pieces form a neighborhood basis at the selected parameter. In the dynamical plane, the unbranched nested pieces shrink and their pullbacks away from the critical orbit have uniformly bounded degree and Koebe distortion; consequently every puzzle impression is a point. The usual puzzle criterion for connected polynomial Julia sets gives local connectivity of J_c .

Choose in the periodic neutral model a small safe disk D near the marked critical point but a definite hyperbolic distance from the postcritical set, together with an annular gate A surrounding the regularized Siegel core. Chapter 19 bounded-type

boundary geometry and sector a priori bounds give three uniform facts: an orbit leaving the bounded side of A must cross D within a bounded number of iterates; a fixed subdisk of D returns to the neutral core; and the first-return pullback of a sufficiently deep quadratic-like range occupies a fixed proportion of D .

The homoclinic maps g_m spend a number of consecutive pacman-renormalization iterates tending to infinity in the compact neutral chart. For any prescribed S , choose $S + 1$ well-separated iterates in that block. Pull D to the g_m -plane by the corresponding conformal renormalization changes of variables. The bounded-satellite geometric control lemma and the all-scale exterior collars make the pullbacks pairwise disjoint and preserve one common Koebe bound. The valuable-flower marking and projection lemma says that the valuable flower contains the postcritical set and converges, in these normalized charts, to the neutral core; hence every orbit leaving the flower crosses the pulled-back annular gates and therefore the disks $D_0, \dots, D_S$ in order. The neutral return subdisk and the degree-two pullback $U'_\bullet \in V'_\bullet$ persist under the same compact conformal transport. This proves the whole trapping package using only the pacman compactness and valuable-flower marking established above.

If $J(g)$ already has positive area there is nothing to prove, so for the Black-Hole contradiction assume $J(g) = 0$. Fix a positive-area compact reference set E in the neutral model, separated from the postcritical boundary and contained on the bounded side of the gate A . For deep m , the corresponding set in the g_m -plane has uniformly comparable area. Outside the null Julia set, almost every point of this reference set which leaves the bounded core crosses D_0 . Pulling $U'_\bullet \in V'_\bullet$ back along the first crossing gives quasidisks

$$U_\bullet(z) \in V_\bullet(z)$$

with a uniform collar and

$$\frac{\text{area } U_\bullet(z)}{\text{area } V_\bullet(z)} \geq c_0 > 0.$$

A Vitali subfamily of the $V_\bullet(z)$ has bounded overlap after one fixed enlargement. Summing the density inequality gives a lower bound $c_1 \text{area}(E)$ for points which land in $U_\bullet$. Normalizing by $\text{area}(U)$ yields the claimed η_0 . This is the same landing argument as the definite landing lemma, with the satellite trapping-disk ladder lemma supplying the satellite safe disk.

For the neutral model, choose D so that its closure is compactly contained in the postcritical complement and so that the inclusion

$$U_* \setminus f_*^{-1}(\mathcal{P}(f_*)) \hookrightarrow V_* \setminus \mathcal{P}(f_*)$$

is proper on the orbit segment crossing D . Schwarz–Pick is then strict on the compact closure of D , so the hyperbolic expansion factor is bounded below by some $\lambda_* > 1$. The postcritical sets and the normalized maps converge on the compact neutral chart, while the renormalization coordinate changes are conformal and range in the compact family of the bounded-satellite geometric control lemma. Shrinking the chart once gives one $\lambda \in (1, \lambda_*)$ valid on every transported D_i .

Koebe on the fixed collars gives bounded hyperbolic diameter and bounded distortion of all first-landing pullbacks. Iterating the factor λ makes inverse hyperbolic

diameters decay geometrically; bounded oscillation of the hyperbolic density on the safe collars converts this to Euclidean diameter decay. Thus the safe-return scale-separation statement is exactly the conclusion of the safe-return scale separation lemma in the present satellite geometry.

Assume again that $J(g)$ has zero area. Let $\Sigma \subset D_0$ be the set of points which leave D_0 and never return. An orbit in Σ which escapes the valuable-flower core must cross every D_i . In each D_i , the satellite trapping-disk ladder lemma supplies a forbidden subdisk of definite relative size returning to the neutral core. The uniform expansion and separated return scales lemma places a fixed fraction of their pullbacks on geometrically separated scales. The multiscale porosity lemma the multiscale porosity lemma therefore gives

$$\text{area}(\Sigma) \leq C\sigma^{cS} \text{area}(D_0), \quad 0 < \sigma < 1.$$

For landing branches to D_0 of safe-return depth at most n , the return-domain summation the return-domain summation lemma applies verbatim because the uniform expansion and separated return scales lemma supplies exactly its collar, distortion and scale hypotheses. Hence the union Σ_n of pullbacks of Σ satisfies

$$\text{area}(\Sigma_n) \leq Cn\sigma^{cS} \text{area}(D_0).$$

For points with more than n safe returns, each visit supplies, by the degree-two pull-back in the satellite trapping-disk ladder lemma, a hole of definite relative size consisting of points which eventually land in $U_\bullet$. The uniform expansion and separated return scales lemma again separates a fixed fraction of these holes in scale, so the multiscale porosity lemma yields

$$\text{area}\{z : \text{safe-return depth} > n, z \text{ never lands in } U_\bullet\} \leq C\theta^n \text{area}(D_0)$$

for one $\theta \in (0,1)$. First choose n large, then choose S (hence the antirenormalization depth) large. The exceptional density in D_0 is arbitrarily small. Pulling the estimate back to $V_\bullet \setminus U_\bullet$ along the first-entry branches and using Koebe plus a bounded-multiplicity Besicovitch cover changes it by only a fixed factor. Thus $\xi_m \rightarrow 0$. Notice that the zero-area assumption itself disposes of the residual Julia-set subset, so no additional almost-everywhere landing theorem is imported here.

This is the exponent-two transfer argument already proved in the Chapter 19 transfer, zero-area, and Black-Hole lemmas. Inspecting that proof, primitivity is used only later to construct the Chapter 19 seed family; the Black-Hole comparison itself uses a nice quadratic-like fundamental pair, one uniform modulus/collar bound, exponent-two change of variables, and bounded distortion. The unbranched propagation and shrinking lemma provides the uniform unbranched quadratic-like bounds for the present satellite tower. If the renormalization change of variables is a non-affine conformal map, work in its linearizing coordinate on the fixed collar; the derivative and inverse derivative have bounded oscillation, so they only alter the universal distortion constant. Therefore the same C_{BH} applies throughout the deep satellite family.

Take η_0 from the definite satellite landing lemma and C_{BH} from the satellite Black-Hole transfer lemma. By the vanishing satellite escape proposition choose the an-

tirenormalization depth so large that either the Julia set already has positive area or

$$\xi_m < \eta_0 / C_{\text{BH}}.$$

In the second case $\eta_m \geq \eta_0 > C_{\text{BH}}\xi_m$, and the satellite Black-Hole transfer lemma contradicts the zero-area alternative. Hence $J(g_m)$ has positive area in all sufficiently deep selected cases. The hybrid straightening from the marked quadratic-like restriction to the associated quadratic polynomial is quasiconformal on a neighborhood of the filled set; quasiconformal maps and their inverses have Lusin's N property, so positive planar measure is preserved in the only direction needed here. Thus the selected quadratic Julia set also has positive area. This reconstructs the source Section 9.3 conclusion from the Chapter 19 measure engine and the satellite trapping geometry of the satellite trapping, landing, and scale-separation lemmas above; no additional area theorem is invoked.

This was proved in Chapter 19. Theorem 19.4 gives the periodic pacman point and its one-dimensional unstable direction, while Theorem 19.5 identifies the transverse quadratic-parameter coordinate. In the unstable coordinate,

$$u \mapsto \Lambda u + O(u^2), \quad |\Lambda| > 1,$$

so inverse magnifications converge to multiplication by Λ^{-1} , giving the asserted asymptotic self-similarity.

Consequently the theorem follows from the declared earlier infrastructure. The fixed-base topology, continuation, seed recognition, homoclinic coding and shrinking are established by the preceding lemmas; the satellite trapping through Black-Hole lemmas above together with the Chapter 19 Black-Hole machinery supply the positive-area branch. No further external theorem is used. $\square$

Theorem 21.4 (MLC at bounded-type Feigenbaum points). **Statement scope.** *Every bounded-type Feigenbaum quadratic-like map has beau a priori bounds. For quadratic polynomials this gives local connectivity of the Julia set and, together with complex bounded-combinatorics rigidity, MLC at the corresponding parameter. Combined with bounded-type complex renormalization hyperbolicity, the same bounds yield dynamical and parameter scaling universality for the corresponding combinatorics.*

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Combine the modern beau-bound theorem with complex rigidity and MLC, use the complex natural-extension form of Chapter 15 hyperbolicity to construct the bounded-type horseshoe, prove uniform transverse $C^{1+\alpha}$ holonomy, and derive the resulting dynamical and parameter scaling/self-similarity laws.

The mathematical argument begins in the proof environment below.

Proof. The theorem separates into the beau-bound theorem, the MLC consequence, and the later scaling consequence.

Beau bounds. We reconstruct the Dudko-Lyubich pull-off mechanism in the book. Canonical weighted arc diagrams record the relevant extremal widths. Large width either pulls off through a nonperiodic branch, creates a protected wave handled by the Chapter 16 wave lemma, or localizes in an almost-periodic satellite rectangle. In the last case a spiraling-number calculation on a twice-punctured disk, followed by Grötzsch series estimates and bounded-degree push-forward, gives arbitrarily large periodic amplification as the defect from periodicity tends to zero.

Failure of beau bounds would therefore force repeated exponential growth of these widths across bounded renormalization blocks. Independently, finitely many bounded-type straightening models give an exponential Teichmüller/QC upper bound on the same quantities. Choosing the periodic amplification factor larger than that upper-growth constant gives a contradiction. The inhomogeneous version of the rectangle argument then produces a definite quadratic-like collar, proving beau a priori bounds.

Dynamical shrinking. Uniform annular moduli make the critical renormalization nest shrink. First-landing pullbacks and Koebe distortion spread this shrinking to every Julia point, so the Julia set is locally connected.

Complex rigidity and MLC. We use Chapter 15's rigidity theorem at its source-faithful complex scope. If P_c and $P_{\tilde{c}}$ have the same bounded Feigenbaum combinatorics, the beau-bound theorem applies to both. Chapter 15 then gives a hybrid conjugacy, and uniqueness of the monic centered quadratic straightening gives $\tilde{c} = c$. Thus the full marked combinatorial class of c is a singleton.

Let $\mathcal{P}_n$ be the closed finite-depth parapuzzle classes determined by the first n renormalization blocks. They are nested compact connected sets and

$$\bigcap_n \mathcal{P}_n = \{c\}.$$

A nested compact sequence with singleton intersection has diameter tending to zero. Hence the Mandelbrot traces of $\mathcal{P}_n$ form connected relative neighborhoods shrinking to c , proving MLC at every bounded-type Feigenbaum parameter.

Complex hyperbolicity and scaling. We likewise use Chapter 15's hyperbolicity theorem at the scope actually needed in Lyubich's bounded-combinatorics theory: an arbitrary finite complex renormalization alphabet with a uniform a priori-bound gate, with the one-dimensional unstable line indexed by the natural-extension prehistory.

For a fixed combinatorial bound there are only finitely many renormalization types. Successive tuning realizes every finite symbolic block, and the beau bound just proved gives one uniform modulus on fixed central blocks. Diagonal compactness plus complex rigidity therefore constructs the compact complex horseshoe for every bounded itinerary. Chapter 15 then gives uniform hyperbolicity and a one-complex-dimensional unstable cocycle.

The stable holonomy argument is uniform on this compact horseshoe: exponential stable contraction, transverse nonsingularity, the quantitative lambda lemma and Koebe distortion imply

$$h(z) = h(0) + az + O(|z|^{1+\alpha}), \quad a \neq 0.$$

Hence renormalized dynamical germs converge exponentially to the coded horseshoe orbit, while parameter straightening maps are $C^{1+\alpha}$ -conformal at the Feigenbaum parameter. Their asymptotic scales are the inverse products of the one-dimensional unstable derivative cocycle; for stationary combinatorics this reduces to the usual power λ_u^{-n} .

Thus the bounded-type complex Renormalization Conjecture and the corresponding dynamical/parameter self-similarity conclusions are internal as well. Together with the preceding beau-bound and MLC arguments, this completes the theorem. $\square$

Conjecture 21.5 (General MLC remains open). ***Statement scope.** Global local connectivity of the entire Mandelbrot set is retained as the MLC conjecture and is not used as an established theorem. The proved regimes in Chapter 21 are indexed separately by their combinatorial or neutral hypotheses.*

21.3 Worked examples

Worked example: Modulus accumulation and shrinking

If nested puzzle pieces $P_{n+1} \subseteq P_n$ satisfy $\text{mod}(P_n \setminus \overline{P_{n+1}}) \geq \mu > 0$, then the sum of separating moduli diverges. Grötzsch theory forces the conformal radii, and under bounded-shape control the Euclidean diameters, to tend to zero. This is the geometric core shared by several MLC routes.

Worked example: From puzzle to parapuzzle

When puzzle boundaries move holomorphically with the quadratic parameter and phase-parameter transversality is nondegenerate, a definite dynamical annular modulus yields a comparable parameter annular modulus. Thus a shrinking dynamical nest can produce shrinking connected parameter neighborhoods.

Boundary case: Why the global MLC conjecture remains separate

Theorems for Yoccoz parameters, bounded-satellite constructions, and bounded-type Feigenbaum parameters cover important but restricted combinatorial classes. Their union does not justify a statement about every boundary point of the Mandelbrot set; the chapter keeps the global conjecture as a status item rather than a theorem dependency.

21.4 What this chapter established

The chapter closed its formal theorem spine at the stated source scopes. The proofs deliberately separate elementary deductions from deep imported theorems: every imported theorem is named at the point where it enters, and later conclusions use only the scope actually established.

For study purposes, the most important task is to be able to reconstruct the dependency chain rather than memorize theorem names. In particular, note which hypotheses supply compactness or a priori geometry, which step supplies rigidity, and which step converts dynamical information into parameter or statistical conclusions.

Looking ahead

Chapter 22 begins the final advanced part of the series, after the Chapters 16-21 neutral/typicality/parameter-plane block has been closed.

Part V

Transcendental, Several-Variable, and Frontier Dynamics

Guide to Part V: beyond polynomial compactness

The last Part asks a robustness question: which ideas from one-dimensional polynomial dynamics survive after the most convenient global assumptions are removed?

A transcendental entire map has an essential singularity at infinity. There is no leading polynomial term that dominates every sufficiently large point, and the set of singular values replaces the finite critical-value data of a polynomial. Nevertheless, logarithmic coordinates recover an expanding tract geometry for important classes. Dynamic rays and escaping sets show that external-address combinatorics survives, but in a form adapted to infinitely many tracts and possible asymptotic values.

Hénon maps change the dimension rather than the behavior at infinity. Because a polynomial automorphism of $\mathbb{C}^2$ is invertible, forward and backward escape must be studied simultaneously. The pair of Green functions G^+ and G^- gives rise to positive closed currents, and their intersection produces the natural invariant measure. This is the several-variable analogue of potential theory in Chapter 3, but the analogy has limits: laminations, saddle points, and nonuniform hyperbolicity bring genuinely two-dimensional phenomena.

The final chapter is not a list of fashionable topics. It teaches how to read frontier mathematics responsibly. A result may be published, a current preprint, a theorem in a restricted class, or an open problem. Those statuses are logically different. The book therefore treats source status as part of the mathematical statement whenever the frontier is moving.

The mathematical route through this Part

polynomial escape geometry $\rightarrow$ transcendental tracts and external addresses,
one Green function $\rightarrow$ two Green functions and currents in $\mathbb{C}^2$.

The Part ends by comparing these extensions with the renormalization mechanisms developed earlier and by isolating the questions that remain genuinely open.

What continuity means at the frontier

The earlier Parts develop a relatively closed theorem spine. The frontier is different: publication status can change, hypotheses can be sharpened, and a preprint may split into several later papers. For that reason the provenance appendix should be read as part of the textbook's mathematical apparatus, not merely as bibliography.

Transcendental Entire and Meromorphic Dynamics

Why this chapter matters

Polynomial dynamics has a compact escape mechanism: once an orbit is sufficiently large, the leading term controls it. Transcendental entire maps have an essential singularity at infinity, so no comparable global polynomial estimate is available. Nevertheless, infinity still has a remarkably rigid dynamical geometry. Eremenko's theorem identifies the Julia set as the boundary of the escaping set, and the Eremenko-Lyubich class $\mathcal{B}$ replaces polynomial "outside a large disk" geometry by logarithmic tracts.

This chapter develops that geometry in three stages. First we construct escaping orbits and identify their boundary. Next we pass to logarithmic coordinates and derive the fundamental expansion estimate for class $\mathcal{B}$. Finally, under the additional finite-order tract-geometry hypothesis, we organize sufficiently far-right logarithmic escaping points into ray tails and pull them back to uniformly escaping curves in the original plane. The last implication is intentionally stated with its full hypothesis: class $\mathcal{B}$ alone does not force every escaping point to lie on a ray.

Learning goals

By the end of this chapter you should be able to:

1. define the escaping set and explain why its boundary is the Julia set;
2. distinguish critical values, asymptotic values, and the bounded singular set defining class $\mathcal{B}$;
3. construct logarithmic tracts over the complement of a large disk;
4. derive the Eremenko-Lyubich derivative estimate from Koebe's quarter theorem;
5. derive how finite order supplies bounded slope and bounded wiggling for logarithmic tracts and how these estimates feed the head-start method;
6. state the dynamic-ray theorem at exactly its finite-order or finite-composition scope, including the asymptotic-value pullback alternative;
7. identify what fails when the singular set is unbounded or when the finite-order hypothesis is removed.

Conceptual roadmap

transcendental growth $\rightarrow I(f) \neq \emptyset$, $J(f) = \partial I(f)$,
 $S(f)$ bounded $\rightarrow$ logarithmic tracts $\rightarrow$ Koebe expansion,
 finite order $\rightarrow$ bounded slope/wiggling,
 $\rightarrow$ head-start ordering $\rightarrow$ dynamic rays.

The second line holds throughout class $\mathcal{B}$; the third line is an additional theorem, not a formal consequence of belonging to $\mathcal{B}$.

Common pitfall

Do not confuse “escaping points exist” with “escaping points lie on dynamic rays.” The first assertion holds for every transcendental entire function. The ray theorem used here requires finite order (or a finite composition of finite-order class- $\mathcal{B}$ maps), because its proof needs uniform control of tract slope and wiggling.

22.1 Definitions

Definition 22.1 (escaping set). For an entire function f , its escaping set is

$$I(f) := \{z \in \mathbb{C} : f^n(z) \rightarrow \infty \text{ as } n \rightarrow \infty\}.$$

Definition 22.2 (singular value). For an entire or meromorphic map f , the set $S(f)$ of finite singular values of the inverse is the closure in $\mathbb{C}$ of the finite critical values and finite asymptotic values of f . A point of $S(f)$ is called a finite singular value.

Definition 22.3 (Eremenko-Lyubich class $\mathcal{B}$). The Eremenko-Lyubich class $\mathcal{B}$ consists of the transcendental entire functions f for which $S(f)$ is bounded.

Definition 22.4 (logarithmic tract). Let $f \in \mathcal{B}$, and choose $R > 0$ so large that $S(f) \cup \{f(0)\} \subset D(0, R)$. A component T of $f^{-1}(\{w : |w| > R\})$ is called a logarithmic tract. The restriction $f : T \rightarrow \{w : |w| > R\}$ is a universal covering; after lifting by the exponential map, the tract dynamics is represented by a conformal map to a right half-plane.

Definition 22.5 (external address and address class). For a normalized logarithmic transform $F : \mathcal{T} \rightarrow H$, an *external address* is a sequence $\underline{s} = (T_0, T_1, \dots)$ of tracts. Its address class is

$$J_{\underline{s}}(F) = \{z \in \overline{\mathcal{T}} : F^n(z) \in \overline{T_n} \text{ for all } n \geq 0\},$$

with boundary points included only when the corresponding iterates are defined by continuous extension.

Definition 22.6 (ray tail and dynamic ray). A *ray tail* is an injective curve $\gamma : [t_0, \infty) \rightarrow I(F)$ contained in one external-address class, tending to infinity, such that $F^n|_{\gamma} \rightarrow \infty$ uniformly. A *dynamic ray* is the maximal compatible injective extension of such a tail in its address class; an endpoint, when present, is not part of the open ray unless explicitly adjoined.

Definition 22.7 (bounded slope and bounded wiggling). The tracts of a logarithmic transform have *bounded slope* if constants $\alpha, \beta > 0$ satisfy

$$|\Im z - \Im w| \leq \alpha \max\{\Re z, \Re w, 0\} + \beta$$

for all z, w in a common tract. They have *bounded wiggling* if there are $K > 1$ and $\mu > 0$ such that every hyperbolic geodesic in a tract from z_0 to infinity stays in $\Re z > (\Re z_0)/K - \mu$.

Definition 22.8 (uniform linear head-start). A logarithmic transform satisfies a *uniform linear head-start condition* if there are $K > 1$ and $M > 0$ such that points z, w in a common tract with $\Re w > K \max\{\Re z, 0\} + M$ satisfy $\Re F(w) > K \Re F(z) + M$, and any two distinct points with the same external address eventually become comparable by this inequality under iteration.

22.2 Main results

Theorem 22.1 (Escaping set theorem). **Statement scope.** For every transcendental entire function f , the escaping set

$$I(f) = \{z \in \mathbb{C} : f^n(z) \rightarrow \infty\}$$

is nonempty and satisfies $\partial I(f) = J(f)$.

Proof strategy (not part of the proof)

Proof status. proved here modulo stated analytic input.

Strategic outline only. Wiman-Valiron analytic input; Rouché annulus covering; finite-log-measure admissible radii; compact nested pullbacks; exact invariance; Julia-boundary density argument.

The mathematical argument begins in the proof environment below.

Analytic input used below. The dynamical argument will be proved in full. Its only non-dynamical input is the classical Wiman-Valiron maximum-modulus theorem from one-variable function theory. For a transcendental entire function

$$f(z) = \sum_{m \geq 0} a_m z^m,$$

let $\nu(r)$ be the largest index for which $|a_m| r^m$ is maximal, choose z_r with $|z_r| = r$ and $|f(z_r)| = M(r, f)$, and fix $\gamma \in (1/2, 1)$. Wiman-Valiron theory gives a set $F \subset [1, \infty)$ of finite logarithmic measure such that, as $r \rightarrow \infty$ with $r \notin F$,

$$f(z_r e^\tau) = e^{\nu(r)\tau} f(z_r) (1 + o(1)) \quad (22.1)$$

uniformly whenever the corresponding point lies in

$$V(r) := D\left(z_r, \frac{r}{\nu(r)^\gamma}\right).$$

This is an analytic growth theorem, not a dynamical black box. Everything from (22.1) to the existence and boundary structure of the escaping set is proved below.

Lemma 22.2 (Transcendental growth at large radii). *For a transcendental entire function f ,*

$$\nu(r) \longrightarrow \infty, \quad M(r, f) \geq 2r$$

for all sufficiently large r .

Proof. The central index is nondecreasing in r . If it were eventually bounded, then only finitely many Taylor coefficients could ever be maximal, which is impossible for a transcendental power series; hence $\nu(r) \rightarrow \infty$.

For the second assertion, suppose that there were radii $r_j \rightarrow \infty$ with $M(r_j, f) < 2r_j$. Cauchy's estimate would give, for every $m \geq 2$,

$$|a_m| \leq \frac{M(r_j, f)}{r_j^m} < \frac{2}{r_j^{m-1}} \longrightarrow 0.$$

Thus $a_m = 0$ for every $m \geq 2$, contradicting transcendence. $\square$

Lemma 22.3 (Wiman-Valiron annulus covering). *Fix $K > 1$. For all sufficiently large $r \notin F$,*

$$f(V(r)) \supset \text{ann}\left(\frac{M(r, f)}{K}, KM(r, f)\right). \quad (22.2)$$

Moreover, every compact subannulus on the right is covered by the image of a compact subset of $V(r)$.

Proof. Write $M = M(r, f) = |f(z_r)|$. Let w satisfy $M/K \leq |w| \leq KM$. Choose a logarithm

$$\omega = \log \frac{w}{f(z_r)}$$

with imaginary part in $[-\pi, \pi]$. Hence $|\omega| \leq B_K$ for a constant depending only on K . Set $\tau_w = \omega/\nu(r)$. Fix a small $\delta \in (0, 1)$ and consider the circle $|\tau - \tau_w| = \delta/\nu(r)$. Because $\gamma < 1$ and $\nu(r) \rightarrow \infty$, all points $z_r e^\tau$ on and inside this circle lie in $V(r)$ for large $r \notin F$.

On this circle compare

$$g(\tau) = f(z_r e^\tau) - w \quad \text{and} \quad h(\tau) = f(z_r) e^{\nu(r)\tau} - w.$$

By (22.1), $|g - h| = o(M)$ uniformly. Since $w = f(z_r) e^\omega$,

$$h(\tau) = w(e^{\nu(r)(\tau - \tau_w)} - 1).$$

On $|\tau - \tau_w| = \delta/\nu(r)$, the last factor is bounded away from zero by a constant depending only on δ , while $|w| \geq M/K$. Consequently $|g - h| < |h|$ for all sufficiently large admissible r . Rouché's theorem gives a zero of g inside the circle, and therefore a point of $V(r)$ mapped to w . This proves (22.2).

Now let A be a compact subannulus of the right-hand side. For each $w \in A$, choose $z_w \in V(r)$ with $f(z_w) = w$. By the open mapping theorem there is a closed disk $D_w \Subset V(r)$

whose image contains a neighborhood of w . A finite subcover of A by these image neighborhoods yields a compact set $C \Subset V(r)$ with $f(C) \supset A$. $\square$

Lemma 22.4 (Nested pullback compacta). *Let $C_0, C_1, \dots$ be nonempty compact sets satisfying $f(C_n) \supset C_{n+1}$. Then there exists $z \in C_0$ such that $f^n(z) \in C_n$ for every $n \geq 0$.*

Proof. For $N \geq 0$, let

$$K_N = \{z \in C_0 : f^j(z) \in C_j \text{ for } 0 \leq j \leq N\}.$$

Choose a point of C_N and lift it successively through $C_{N-1}, \dots, C_0$; hence $K_N \neq \emptyset$. Each K_N is compact and $K_{N+1} \subset K_N$. Compactness of C_0 gives $\bigcap_{N \geq 0} K_N \neq \emptyset$. $\square$

Lemma 22.5 (Eremenko escaping orbit). *For every transcendental entire function f , the escaping set $I(f)$ is nonempty. The construction gives an orbit whose modulus grows at least on the maximum-modulus scale selected below.*

Proof. Fix $\gamma \in (1/2, 1)$ and use the exceptional set F from (22.1). Choose R so large that, for all $r \geq R$ with $r \notin F$, Lemma 22.3 holds with $K = 4$, $M(r, f) \geq 2r$, $\nu(r)^\gamma \geq 4$, and

$$\int_{F \cap [R, \infty)} \frac{dt}{t} < 1. \quad (22.3)$$

Choose $r_0 \geq R$ with $r_0 \notin F$ and write $M_{k+1} = M(r_k, f)$.

Assume r_k has been chosen. Since $\int_{[M_{k+1}, 3M_{k+1}]} dt/t = \log 3 > 1$, (22.3) implies that one can choose

$$r_{k+1} \in [M_{k+1}, 3M_{k+1}] \setminus F. \quad (22.4)$$

Because $\nu(r_{k+1})^\gamma \geq 4$, every $z \in V(r_{k+1})$ satisfies

$$\frac{3}{4}r_{k+1} \leq |z| \leq \frac{5}{4}r_{k+1}.$$

Together with (22.4), this gives

$$V(r_{k+1}) \subset \text{ann}\left(\frac{M_{k+1}}{4}, 4M_{k+1}\right). \quad (22.5)$$

By Lemma 22.3, choose compact $C_k \Subset V(r_k)$ such that

$$f(C_k) \supset \overline{\text{ann}}\left(\frac{M_{k+1}}{4}, 4M_{k+1}\right). \quad (22.6)$$

Then (22.5) and (22.6) imply $f(C_k) \supset C_{k+1}$. Lemma 22.4 therefore gives $z \in C_0$ with $f^k(z) \in C_k$ for every k .

Finally,

$$r_{k+1} \geq M_{k+1} = M(r_k, f) \geq 2r_k,$$

so $r_k \rightarrow \infty$. Since $C_k \subset V(r_k)$ and $\nu(r_k)^\gamma \geq 4$,

$$|f^k(z)| \geq \frac{3}{4}r_k \rightarrow \infty.$$

Thus $z \in I(f)$. □

Lemma 22.6 (Exact invariance of the escaping set). *The escaping set satisfies*

$$f^{-1}(I(f)) = I(f), \quad f(I(f)) \subset I(f). \quad (22.7)$$

The forward inclusion need not be equality when an escaping point is an omitted value.

Proof. If $z \in I(f)$, deleting the first iterate shows $f(z) \in I(f)$. Conversely, if $f(z) \in I(f)$, then $f^{n+1}(z) \rightarrow \infty$, so $z \in I(f)$. No surjectivity assertion is involved. □

Lemma 22.7 (Julia boundary of the escaping set). *For every transcendental entire function f ,*

$$\partial I(f) = J(f). \quad (22.8)$$

Proof. We use the standard normal-family Julia package for transcendental entire maps: the exceptional set contains at most one finite point; the full backward orbit of every nonexceptional point is dense in $J(f)$; and repelling periodic points are dense in $J(f)$. These are foundational Julia-set facts, not inputs from the escaping-set theorem.

First let U be a Fatou component meeting $I(f)$, and choose $z_0 \in U \cap I(f)$. Normality implies equicontinuity of the iterates in the spherical metric on every compact subset of U . Given a compact path in U from z_0 to any $z \in U$, subdivide it into finitely many pieces so small that equicontinuity changes every image by less than ε divided by the number of pieces. Since $f^n(z_0) \rightarrow \infty$, the triangle inequality then gives $f^n(z) \rightarrow \infty$. The same argument on compacta gives locally uniform escape. Hence every Fatou component is either disjoint from $I(f)$ or contained in it, and therefore

$$\partial I(f) \subset J(f). \quad (22.9)$$

By Lemma 22.5, $I(f)$ contains an infinite forward orbit. Choose a nonexceptional point $a \in I(f)$ on that orbit. By Lemma 22.6, every point in the full backward orbit of a also lies in $I(f)$. Its density in $J(f)$ shows that every Julia neighborhood meets $I(f)$.

Choose a repelling periodic point p which is nonexceptional. Since its orbit is periodic, $p \notin I(f)$. The complement of $I(f)$ is forward invariant by the contrapositive of (22.7), so every point in the full backward orbit of p is also nonescaping. Density of that backward orbit in $J(f)$ shows that every Julia neighborhood meets $\mathbb{C} \setminus I(f)$. Thus $J(f) \subset \partial I(f)$, which together with (22.9) proves (22.8). □

Proof of Theorem 22.1. Lemma 22.5 proves that $I(f) \neq \emptyset$. Lemma 22.6 records the exact invariance needed for pullback arguments, and Lemma 22.7 proves $\partial I(f) = J(f)$. This is precisely the asserted escaping-set theorem. □

Definition 22.9 (Eremenko-Lyubich class B). **Statement scope.** A transcendental entire function f belongs to $\mathcal{B}$ precisely when $S(f)$ is bounded. If $S(f) \cup \{f(0)\} \subset D(0, R)$, then every component T of

$$f^{-1}(\mathbb{C} \setminus \overline{D(0, R)})$$

is simply connected and $f : T \rightarrow \mathbb{C} \setminus \overline{D(0, R)}$ is a universal covering. Logarithmic coordinates lift these restrictions to conformal maps from tract components to a right half-plane.

Theorem 22.8 (Expansion in logarithmic coordinates). **Statement scope.** Let $f \in \mathcal{B}$ and choose R with the bounded singular set $S(f) \subset D(0, R)$. On every logarithmic tract $F : W \rightarrow H_R$, Eremenko-Lyubich's estimate gives

$$|F'(\zeta)| \geq \frac{\Re F(\zeta) - \log R}{4\pi},$$

so inverse logarithmic branches have Euclidean derivative tending uniformly to zero sufficiently far to the right. Intrinsic tract and half-plane hyperbolic metrics are related by conformal isometry, not by this Euclidean contraction estimate.

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Class-B universal-covering tracts; logarithmic lift; Koebe inverse estimate; exact $1/(4\pi)$ expansion and inverse contraction.

The mathematical argument begins in the proof environment below.

Proof. Choose R so large that the bounded singular set $S(f)$ lies in $D(0, R)$, and put

$$A_R = \{z : |z| > R\}, \quad H_R = \{w : \Re w > \log R\}.$$

Pass to logarithmic tracts.

Each component V of $f^{-1}(A_R)$ is simply connected and $f : V \rightarrow A_R$ is a universal covering. If W is a component of $\exp^{-1}(V)$, covering theory gives a conformal map

$$F : W \longrightarrow H_R, \quad e^{F(\zeta)} = f(e^\zeta).$$

Bound the inradius of a tract.

The exponential map is injective on W , so W cannot contain two points differing by $2\pi i$. Hence no Euclidean disk of radius greater than π lies in W .

Apply Koebe to an inverse branch.

Let $\Phi = (F|_W)^{-1}$. Koebe's quarter theorem on the disk $D(w, \Re w - \log R) \subset H_R$ gives

$$|\Phi'(w)| \leq \frac{4\pi}{\Re w - \log R},$$

and therefore

$$|F'(\zeta)| \geq \frac{\Re F(\zeta) - \log R}{4\pi}.$$

Translate back to the original plane.

Differentiating $e^{F(\zeta)} = f(e^\zeta)$ gives

$$\left| z \frac{f'(z)}{f(z)} \right| \geq \frac{1}{4\pi} \log \frac{|f(z)|}{R} \quad (|f(z)| > R).$$

Thus inverse logarithmic branches have arbitrarily small *Euclidean* derivative sufficiently far to the right. There is no contradiction with conformal invariance: F is an isometry between the intrinsic hyperbolic metrics of its tract and the half-plane. $\square$

Theorem 22.9 (Dynamic rays for finite-order class B). **Statement scope.** *Let f be a finite-order transcendental entire function in $\mathcal{B}$, or more generally a finite composition of such functions. Every $z \in I(f)$ can be joined to infinity by a curve $\gamma \subset I(f)$ on which $f^n \rightarrow \infty$ uniformly. In normalized logarithmic coordinates every sufficiently far-right escaping point lies on a unique ray tail, and maximal compatible extensions of actual ray tails are dynamic rays. Pullback to the original escaping point has the asymptotic-value alternative of Proposition 2.3 in the primary source, so the theorem does not assert that every original escaping point itself lies on a dynamic ray.*

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Normalized logarithmic transforms; finite-order bounded slope/wiggling; exponential and linear separation; uniform head-start; address-order ray tails; corrected composition lemma; asymptotic-value pullback. *The mathematical argument begins in the proof environment below.*

Proof. The proof is easiest in logarithmic coordinates. After increasing the exterior radius and translating, the preceding expansion theorem gives a $2\pi i$ -periodic logarithmic transform

$$F : \mathcal{T} \rightarrow H, \quad H = \{\Re z > 0\},$$

whose inverse branches are uniformly contracting sufficiently far to the right.

Finite order controls tract geometry.

Finite order becomes

$$\log \Re F(z) \leq \rho \Re z + C$$

for $\Re F(z)$ large. A tract contains no pair differing by $2\pi i$, so its hyperbolic density has a uniform lower bound in terms of Euclidean distance to its boundary. Pulling back horizontal geodesics in H and combining this with the finite-order estimate yields two uniform conclusions: tracts have bounded slope, and geodesics from a point to infinity cannot make arbitrarily deep excursions to the left. The latter property is bounded wiggling.

Separation gives a head start.

If z, w lie in one tract, hyperbolic distance and the tract-width bound give an exponential separation estimate of the form

$$|F(z) - F(w)| \geq \exp\left(\frac{|z - w|}{8\pi}\right) \min\{\Re F(z), \Re F(w)\}.$$

Together with bounded slope this implies a linear alternative: sufficiently separated points have one image far to the right of the other. Bounded wiggling rules out reversal of an already large real-part separation. Hence there are $K > 1$ and $M > 0$ such that

$$\Re w > K \max\{\Re z, 0\} + M \implies \Re F(w) > K \Re F(z) + M.$$

Distinct points following one external address eventually become comparable by this relation.

The address class is ordered into arcs.

For a fixed external address, pull back unbounded right-hand continua through the prescribed inverse branches. Uniform inverse contraction keeps the pullbacks within bounded distance of any sufficiently far-right escaping point. A nested-continuum argument gives an unbounded continuum in the address class. The head-start relation is a strict total “speed order” on that class, and its order topology agrees with the ordinary topology on each component. Thus the unbounded component is an arc. The interval from a far-right escaping point to infinity is a unique ray tail, and the head-start inequality forces all iterates to tend to infinity uniformly on the tail.

Finite compositions.

Every finite-order factor has bounded slope and the same type of linear head-start relation. After restricting far enough to the right, their logarithmic transforms compose tract by tract. The separation estimate lets us choose common constants, and each factor’s head-start rule prevents the ordering from reversing. Hence the composite again has bounded slope and a uniform linear head-start condition. This uses the published correction to the composition lemma: bounded slope is required for *every* factor.

Pull back to the original escaping point.

An escaping point eventually reaches a ray tail. Pull that tail backwards along the orbit. A maximal inverse lift either exists for the entire tail or runs to infinity at a finite lifting parameter. In the latter case the image endpoint is an asymptotic value. In both cases the pullback produces a curve through the original point, contained in the escaping set, on which the iterates tend uniformly to infinity.

This last alternative is important: the theorem guarantees a uniformly escaping curve through every escaping point, but it does not say that every original escaping point itself lies on a ray tail. Genuine ray tails have a unique maximal compatible extension in their external-address class; these maximal extensions are the dynamic rays. $\square$

22.3 Worked examples

Worked example: The exponential family as a tract model

Let

$$f_\lambda(z) = \lambda e^z, \quad \lambda \neq 0.$$

There are no critical points because $f'_\lambda(z) = f_\lambda(z) \neq 0$. The value 0 is a finite asymptotic value, since $f_\lambda(x) \rightarrow 0$ as $x \rightarrow -\infty$. Hence the finite singular set is $S(f_\lambda) = \{0\}$, so $f_\lambda \in \mathcal{B}$.

Choose $R > |\lambda|$. Then

$$f_\lambda^{-1}\{|w| > R\} = \left\{ z : \Re z > \log \frac{R}{|\lambda|} \right\}.$$

On this right half-plane, f_λ is a universal covering of $\{|w| > R\}$. Thus the simplest class- $\mathcal{B}$ tract is literally a half-plane. Moreover f_λ has order one, so the finite-order dynamic-ray theorem applies: every escaping point can be joined to infinity by an escaping curve on which the iterates tend to infinity uniformly.

The example also illustrates the cylindrical derivative appearing in the expansion theorem:

$$\left| z \frac{f'_\lambda(z)}{f_\lambda(z)} \right| = |z|.$$

Along the tract, $\Re z \rightarrow +\infty$, so this derivative grows. The general class- $\mathcal{B}$ theorem recovers quantitative growth without relying on the special formula $f'_\lambda = f_\lambda$.

Worked example: Reading the Koebe constant in logarithmic coordinates

Let $F : W \rightarrow H_R = \{\Re w > \log R\}$ be a logarithmic tract map and let $\Phi = F^{-1}$. Because W contains no two points differing by $2\pi i$, every disk contained in W has radius at most π . Apply Koebe's quarter theorem to Φ on the disk

$$D(w, \Re w - \log R) \subset H_R.$$

It yields

$$\frac{1}{4} |\Phi'(w)| (\Re w - \log R) \leq \pi,$$

and hence

$$|F'(\zeta)| \geq \frac{\Re F(\zeta) - \log R}{4\pi}.$$

Thus once $\Re F(\zeta) \geq \log R + 8\pi$, the corresponding inverse branch has derivative at most $1/2$. This calculation isolates exactly where the constant 4π comes from: the Koebe quarter factor and the 2π -periodicity obstruction.

Boundary case: An explicit entire function outside class $\mathcal{B}$

Consider

$$g(z) = z + e^z.$$

Its critical points satisfy $1 + e^z = 0$, so

$$z_k = (2k+1)\pi i, \quad g(z_k) = (2k+1)\pi i - 1.$$

The critical values are therefore unbounded. Hence $g \notin \mathcal{B}$, and there is no single disk containing all finite singular values from which the class- $\mathcal{B}$ logarithmic-tract theorem can be launched. Eremenko's general escaping-set theorem still applies because g is transcendental entire, but the class- $\mathcal{B}$ expansion and the finite-order

ray theorem cannot be invoked from these hypotheses.

Boundary case: Why class $\mathcal{B}$ is not by itself a ray theorem

Bounded singular values give covering tracts and expansion far to the right, but tracts may still have sufficiently complicated geometry to defeat the uniform head-start ordering used to construct rays. The finite-order hypothesis controls slope and wiggling and is the extra geometric input in the theorem proved here. Therefore one must not delete “finite order” from the dynamic-ray statement merely because the map remains in $\mathcal{B}$.

22.4 What this chapter established

Every transcendental entire function has escaping points, and the Julia set is exactly the boundary separating escaping from nonescaping behavior. For class $\mathcal{B}$, bounded singular values allow the exterior dynamics to be lifted to conformal tract maps over a half-plane. Koebe’s theorem then gives a quantitative expansion estimate whose inverse form is a uniform contraction far to the right.

Finite order adds global control of tract geometry. The head-start mechanism converts that geometry and expansion into ordered escaping tails. Finite pullback then gives a uniformly escaping curve through every prescribed escaping point, with an asymptotic-value alternative when the pullback ceases to be a ray tail. Genuine tails have unique maximal extensions, the dynamic rays.

Looking ahead

Chapter 23 leaves one-dimensional transcendental dynamics and turns to polynomial automorphisms of $\mathbb{C}^2$, especially Hénon maps. Escape remains central, but scalar Green functions are replaced by forward/backward Green functions, currents, equilibrium measures, and higher-dimensional hyperbolic geometry.

Hénon Maps and Several-Complex-Variable Dynamics

Why this chapter matters

One-dimensional polynomial dynamics has a single escape direction and a single Green function. A polynomial automorphism of $\mathbb{C}^2$ is invertible, so forward and backward escape become equally important. This produces two filled sets, two Green functions, two Green currents, and their intersection measure. Hénon maps are the canonical nonlinear models in this setting.

The chapter has two complementary goals. The first is structural: explain why every polynomial automorphism with nontrivial degree growth is of Hénon type and build the forward/backward potential theory that replaces the one-dimensional Green function. The second is dynamical: under strong dissipation and a dominated splitting, explain how stable laminations and transverse expansion recover a surprisingly rigid Fatou/Julia picture.

Learning goals

By the end of this chapter you should be able to:

1. distinguish affine/elementary polynomial automorphisms from Hénon-type automorphisms using dynamical degree;
2. compute the inverse and Jacobian of a generalized Hénon map;
3. define G^+ and G^- , derive their scaling laws, and explain why their zero sets are K^+ and K^- ;
4. pass from Green functions to positive closed currents and the equilibrium measure;
5. explain why the equilibrium measure is invariant and why maximal-entropy uniqueness is a deeper theorem than current invariance;
6. define substantial dissipativity and dominated splitting before using them;
7. outline the Lyubich-Peters chain from stable laminations and protected lifts to no wandering domains, no rotation domains, hyperbolicity off parabolic cycles, and $J = J^*$;
8. recognize why the singular limit $a = 0$ is not an invertible Hénon map and therefore has no backward Hénon dynamics.

Conceptual roadmap

$\lambda(F) > 1 \implies$ Friedland–Milnor Hénon normal form,
 Hénon filtration $\implies G^+, G^- \implies T^+, T^- \implies \mu = T^+ \wedge T^-$,
 dissipation + domination $\implies$ stable lamination + protected lifts
 $\implies$ no wandering or rotation domains
 $\implies$ unstable manifolds off parabolics
 $\implies J = J^*$,
 hyperbolicity off parabolic cycles.

Common pitfall

The parameter a in $H(x, y) = (p(x) - ay, x)$ is not a harmless perturbation parameter when it reaches zero. For $a \neq 0$, H is a polynomial automorphism with a polynomial inverse. At $a = 0$, invertibility disappears, so the backward Green function and every theorem requiring a polynomial automorphism leave their stated scope.

23.1 Definitions

Definition 23.1 (Hénon map). A generalized Hénon map is a polynomial automorphism

$$H(x, y) = (p(x) - ay, x), \quad \deg p \geq 2, \quad a \neq 0,$$

up to the equivalent interchange of coordinate conventions.

Definition 23.2 (dynamical degree). For a polynomial self-map F , its dynamical degree is

$$\lambda(F) = \lim_{n \rightarrow \infty} (\deg F^n)^{1/n}$$

when the limit exists. For a generalized Hénon map it equals $\deg p$; for a reduced finite composition it is the product of the factor degrees.

Definition 23.3 (Green current). For an escape-rate function $G^\pm$, the associated Green current is the positive closed $(1, 1)$ -current

$$T^\pm = dd^c G^\pm.$$

Definition 23.4 (equilibrium measure in $\mathbb{C}^2$). When the Bedford–Taylor wedge is defined, the Hénon equilibrium measure is

$$\mu = T^+ \wedge T^-.$$

With the standard normalization it is an invariant probability measure.

Definition 23.5 (filled and Julia sets). Write K^+ and K^- for the points with bounded forward and backward orbits, $J^\pm = \partial K^\pm$, and $J = J^+ \cap J^-$. The saddle Julia set J^* is

the closure of the saddle periodic points.

Definition 23.6 (substantial dissipativity). A degree- d complex Hénon map is substantially dissipative if

$$|\text{Jac } F| < d^{-2}.$$

This is stronger than ordinary dissipativity $|\text{Jac } F| < 1$.

Definition 23.7 (dominated splitting). A dominated splitting on J is a continuous DF -invariant decomposition

$$T_J \mathbb{C}^2 = E^s \oplus E^c$$

for which there are $C > 0$ and $0 < \rho < 1$ such that, for unit vectors $v \in E_x^s$ and $w \in E_x^c$,

$$\frac{\|DF_x^n v\|}{\|DF_x^n w\|} \leq C\rho^n \quad (n \geq 0).$$

Definition 23.8 (periodic Fatou types). An attracting basin is a periodic Fatou component converging to an attracting cycle; a parabolic basin converges to a parabolic cycle. A rotation domain is a periodic Fatou component on which an iterate is analytically conjugate to an irrational rotation action rather than converging to an attracting or parabolic cycle.

23.2 Main results

Theorem 23.1 (Partial hyperbolicity and dominated splittings in Hénon dynamics).

Statement scope. *Let F be a substantially dissipative complex Hénon map with a dominated splitting on its Julia set J . Then the Fatou set has only finitely many components, each attracting or parabolic periodic; there are no wandering Fatou components and no rotation domains. Moreover $J = J^*$, and the dynamics on J is hyperbolic away from parabolic cycles.*

Proof strategy (not part of the proof)

Proof status. proved here.

Strategic outline only. Combine the Lyubich–Peters Mañé engine with the periodic-Fatou classification; use analytic-disk Green-current convergence to obtain density of saddle stable/unstable manifolds, then prove the local transverse-homoclinic horseshoe/closing step and conclude $J = J^*$.

The mathematical argument begins in the proof environment below.

Proof. The proof has four layers: the two-dimensional Mañé engine of Lyubich–Peters, the invariant-Fatou classification, the Bedford–Smillie analytic-disk current argument, and a direct dominated-flow-box homoclinic closing argument.

From domination to a vertical stable geometry.

The dominated splitting and the Jacobian identity imply that the central direction cannot contract exponentially fast at any rate below one: otherwise domination would force a whole bidisk around a Julia point into a normal family. Consequently the stable

line contracts at every rate larger than $|\text{Jac } F|$, hence at a rate strictly below d^{-2} . Graph transform gives a holomorphic strong-stable lamination by vertical-like disks near J^+ .

A finite tubular-covering argument and the holomorphic-motion λ -lemma extend this dynamical lamination to an *artificial vertical lamination* on a thin neighborhood Ω . On a hypothetical wandering grand orbit the artificial and dynamical laminations can be arranged to disagree at no more than one transition. A horizontal line L_0 is chosen transverse to the vertical lamination and avoiding the parabolic periodic points.

Uniformize a hypothetical wandering component and bound its degree.

Inside a wandering component, all derivatives tend to zero on compacta and the true strong-stable foliation admits a global transversal. Stable holonomy analytically continues along every path on that transversal, is uniformly quasiconformal after pushing it into finitely many flow boxes, and acts properly discontinuously. Uniformization therefore straightens the component to a cylinder with vertical stable leaves.

Forward images of horizontal slices have normal limits forming a horizontal lamination. Horizontal and vertical leaves cannot coincide, so compactness and upper semicontinuity of intersection multiplicity give a global finite tangency bound $\deg_{\text{crit}}$. This is the two-dimensional substitute for the finite critical-degree bound in Mañé's one-dimensional argument.

Protected lifts.

Choose a sufficiently thin relatively backward-invariant domain Ω . A protected transverse lift is the part of an iterate of L_0 lying over a planar disk together with a larger protecting disk. Proper-map Schwarz-Pick estimates bound its Kobayashi diameter in terms of the protection ratio and its lamination degree. Compactness gives a uniform bound on how many protected disks are needed to cover one inverse image.

Induction then proves the main estimate: every allowed backward iterate of a sufficiently protected lift has uniformly bounded Kobayashi diameter and degree. There are three cases: lifts deeply inside a wandering component, lifts meeting J^+ or a periodic component, and lifts inside a wandering component whose protecting disk reaches its boundary. The first uses the invariant stable foliation and the finite tangency degree; the second connects the controlled pieces to a true Julia stable leaf; the third joins the center to the nearest boundary point and reduces to the first two cases. The preceding argument reconstructs Lyubich-Peters Proposition 9.3 at the scope used here.

Consequences.

A wandering Fatou component would contain a horizontal disk of arbitrarily large Ω -hyperbolic diameter. A normal subsequence of its forward iterates collapses that disk near a nonparabolic Julia point, where it becomes a protected lift. Pulling back contradicts the uniform diameter bound. Thus wandering domains do not exist; afterward the lift degree improves to one.

Using degree one, horizontal images through stable leaves converge to genuine local unstable manifolds. Their size is uniform away from parabolic cycles. Such unstable disks are incompatible with an attracting rotation domain, and on every compact invariant subset disjoint from the parabolic cycles they give uniform expansion in the center direction. Hence the dynamics there is uniformly hyperbolic.

Periodic Fatou classification.

We also need the Lyubich–Peters classification of recurrent invariant components. For a recurrent invariant component, a recurrent normal subsequence produces a holomorphic idempotent retraction. Rank zero gives an attracting fixed point; rank one gives an invariant disk or annulus carrying an irrational rotation and attracting the component transversely; rank two would force a subsequence of derivatives to converge to the identity, impossible because the Jacobian powers tend to zero.

For a non-recurrent component, subsequential limit maps form a compact semi-group. Rank-one limit images are immersed Riemann surfaces, and a maximal element of the natural divisibility preorder has tangential derivatives bounded above and away from zero. Volume contraction then creates a stable lamination of definite transverse size around that image; normality spreads across the lamination and contradicts that the image lies in the Julia boundary. Hence a maximal limit is a fixed point. If the limit point were not unique, a rank-one limit would lie in its strong-stable manifold. Pulling the backward Green function to the linearizing coordinate gives a subharmonic function of order below one half; Wiman’s theorem forbids the resulting unbounded connected zero set. The whole component therefore converges to one boundary fixed point, and the higher-dimensional Snail Lemma makes its neutral multiplier equal to one. Thus every invariant bounded-orbit component is attracting, rotational, or semi-parabolic. The rotation case has already been excluded by the unstable-manifold argument above.

Saddle density and closing.

For the final Bedford–Smillie step, use the following analytic-disk current convergence. For a small analytic disk in the stable manifold of a saddle point, normalized inverse-image integration currents converge to a positive multiple of the forward Green current. The coefficient is nonzero because inverse iteration expands the stable disk and forces positive backward Green mass. Since the limiting current has support J^+ while every inverse image remains in the same stable manifold, $W^s(p)$ is dense in J^+ ; applying the argument to F^{-1} gives density of $W^u(p)$ in J^- .

In a nonparabolic dominated flow box, late pieces of the stable and unstable manifolds of the same saddle are vertical and horizontal graphs. Their density therefore produces transverse homoclinic intersections arbitrarily close to every nonparabolic Julia point. A direct two-strip cone-field horseshoe near each transverse homoclinic point gives saddle periodic points accumulating on that point. Hence every nonparabolic Julia point belongs to J^* ; closedness and absence of isolated Julia points include the finitely many parabolic cycles as well. Therefore $J = J^*$.

The four layers above prove the theorem at the stated scope. □

Theorem 23.2 (Polynomial automorphism/Hénon normal form). **Statement scope.** *Polynomial automorphisms of $\mathbb{C}^2$ with dynamical degree greater than one decompose, up to polynomial conjugacy and the elementary/affine normalizations in the Friedland–Milnor classification, into finite compositions of generalized Hénon maps.*

Proof strategy (not part of the proof)

Proof status. proved here modulo stated foundation input.

Strategic outline only. Jung-van der Kulk amalgam as a standard algebraic input; cyclic reduction; affine Bruhat double-coset factorization; cyclic-coordinate Hénon regrouping; exact multiplicative degree growth.

The mathematical argument begins in the proof environment below.

Proof setup. Let $\mathcal{A}$ be the affine subgroup of $\text{Aut}_{\text{poly}}(\mathbb{C}^2)$, let $\mathcal{E}$ be the elementary subgroup

$$e(x, y) = (\alpha x + p(y), \beta y + \gamma), \quad \alpha\beta \neq 0,$$

and put $\mathcal{S} = \mathcal{A} \cap \mathcal{E}$. We freeze only the standard algebraic Jung-van der Kulk theorem in its amalgamated-product form

$$\text{Aut}_{\text{poly}}(\mathbb{C}^2) = \mathcal{A} *_S \mathcal{E}. \quad (23.1)$$

Everything from this algebraic normal-form interface to the dynamical Hénon normal form will be proved below.

Lemma 23.3 (Cyclic reduction in the amalgam). *Every polynomial automorphism is polynomially conjugate either to an element of $\mathcal{A}$ or $\mathcal{E}$, or to a reduced alternating word of even length whose first and last factors lie in different factors of the amalgam. In the latter case every positive power is reduced by concatenation.*

Proof. By (23.1), every element outside $\mathcal{S}$ has a reduced alternating word $g_n \cdots g_1$, with each $g_j \in \mathcal{A} \setminus \mathcal{S}$ or $\mathcal{E} \setminus \mathcal{S}$. Choose a conjugate of minimal reduced length. If its length is one, it belongs to one of the two factors. If its length is at least two and the first and last letters belong to the same factor, conjugating by the first letter cyclically permutes the word and places two letters of the same factor next to one another; multiplying those two letters shortens the reduced form. This contradicts minimality. Hence the extreme letters lie in different factors, so the alternating length is even.

For such a word there is no cancellation between its final letter and the first letter of the next copy. Therefore concatenating any number of copies remains reduced. This is the cyclically reduced case. $\square$

Lemma 23.4 (Affine factors reduce to the coordinate swap). *Let*

$$t(x, y) = (y, x).$$

Every affine map in $\mathcal{A} \setminus \mathcal{S}$ belongs to the double coset StS . Consequently every cyclically reduced word is conjugate to

$$g = (t \circ e_m) \circ \cdots \circ (t \circ e_1), \quad (23.2)$$

where $e_i \in \mathcal{E} \setminus \mathcal{S}$, and therefore the polynomial part of each e_i has degree $d_i \geq 2$.

Proof. Translations lie in $\mathcal{S}$, so it is enough to factor the linear part. Relative to the horizontal fibration, the linear parts of elements of $\mathcal{S}$ are the invertible upper-triangular

matrices. Let

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in GL_2(\mathbb{C})$$

be the linear part of an affine map outside $\mathcal{S}$. Then $c \neq 0$. With

$$W = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad U_1 = \begin{pmatrix} 1 & a \\ 0 & c \end{pmatrix}, \quad U_2 = \begin{pmatrix} 1 & d/c \\ 0 & b - ad/c \end{pmatrix},$$

we have $M = U_1 W U_2$. The last diagonal entry of U_2 is $-\det(M)/c \neq 0$, so both U_1, U_2 are invertible upper triangular. Thus $\mathcal{A} \setminus \mathcal{S} = St\mathcal{S}$.

Write a cyclically reduced word with alternating affine and elementary letters. Replace each affine letter by $s'ts$ with $s, s' \in \mathcal{S}$, and absorb the two $\mathcal{S}$ -factors into the neighboring elementary letters. A cyclic permutation, hence a conjugacy, places the word in the form (23.2). Since an elementary map belongs to $\mathcal{S}$ exactly when its polynomial part has degree at most one, reducedness gives $d_i \geq 2$. $\square$

Lemma 23.5 (Cyclic affine coordinates give generalized Hénon factors). *The map in (23.2) is polynomially conjugate to a finite composition*

$$H_m \circ \cdots \circ H_1, \quad H_i(u, v) = (v, q_i(v) - a_i u), \quad (23.3)$$

where $a_i \neq 0$ and $\deg q_i = d_i \geq 2$.

Proof. Write

$$e_i(x, y) = (\alpha_i x + p_i(y), \beta_i y + \gamma_i), \quad \alpha_i \beta_i \neq 0, \quad \deg p_i = d_i.$$

For the successive coordinates under $t \circ e_i$, write

$$(x_{i-1}, y_{i-1}) \mapsto (x_i, y_i),$$

so that

$$x_i = \beta_i y_{i-1} + \gamma_i, \quad y_i = \alpha_i x_{i-1} + p_i(y_{i-1}). \quad (23.4)$$

Use cyclic indices $\beta_{m+1} = \beta_1$ and $\gamma_{m+1} = \gamma_1$, and set

$$X_0 = x_0, \quad X_i = x_i = \beta_i y_{i-1} + \gamma_i \quad (1 \leq i \leq m), \quad X_{m+1} = \beta_1 y_m + \gamma_1.$$

Then (23.4) gives, for $1 \leq i \leq m$,

$$\begin{aligned} X_{i+1} &= \beta_{i+1} \alpha_i X_{i-1} + \beta_{i+1} p_i \left(\frac{X_i - \gamma_i}{\beta_i} \right) + \gamma_{i+1} \\ &= q_i(X_i) - a_i X_{i-1}, \end{aligned} \quad (23.5)$$

where

$$q_i(X) = \beta_{i+1} p_i \left(\frac{X - \gamma_i}{\beta_i} \right) + \gamma_{i+1}, \quad a_i = -\beta_{i+1} \alpha_i \neq 0.$$

Affine changes do not alter polynomial degree, hence $\deg q_i = d_i$.

The initial pair $(X_0, X_1) = (x_0, \beta_1 y_0 + \gamma_1)$ is obtained from (x_0, y_0) by one fixed affine

automorphism C , and the final pair $(X_m, X_{m+1}) = (x_m, \beta_1 y_m + \gamma_1)$ is obtained from the output by the same C . Equation (23.5) therefore says exactly

$$C g C^{-1} = H_m \circ \cdots \circ H_1,$$

with the factors (23.3). This is the generalized Hénon normal form. $\square$

Lemma 23.6 (Exact degree growth of a reduced Hénon word). *For the composition (23.3), put $d = d_1 \cdots d_m$. Then*

$$\deg((H_m \circ \cdots \circ H_1)^n) = d^n \quad (n \geq 1). \quad (23.6)$$

In particular its dynamical degree is $d > 1$.

Proof. Continue the recurrence (23.5) indefinitely, repeating the sequence $q_1, \dots, q_m$ periodically. If $D_j = \deg X_j$, then $D_0 = D_1 = 1$. Since every $d_i \geq 2$, induction gives

$$D_{j+1} = d_i D_j > D_{j-1}$$

when the j -th step uses q_i : the term $q_i(X_j)$ has degree $d_i D_j$, strictly larger than the degree of the linear correction $a_i X_{j-1}$, so cancellation of the leading degree is impossible. After one complete block the maximal coordinate degree is $d_1 \cdots d_m = d$. Repeating the same argument through n blocks gives maximal degree d^n , which proves (23.6). $\square$

Proof of the theorem. An affine map has degree one under every iterate. If

$$E(x, y) = (\alpha x + q(y), \beta y + \gamma),$$

then the second coordinate of E^n remains affine and the first coordinate is a finite sum of affine rescalings of q evaluated on affine functions; therefore $\deg E^n \leq \max(1, \deg q)$ for all n . Conjugation by a fixed polynomial automorphism multiplies degree by at most fixed factors, so every conjugacy class represented in $\mathcal{A}$ or $\mathcal{E}$ has dynamical degree one.

Now let F have $\lambda(F) > 1$. Lemma 23.3 therefore puts a conjugate of F in the cyclically reduced case. Lemma 23.4 converts that word to (23.2), and Lemma 23.5 converts it, by one further polynomial conjugacy, to the finite generalized-Hénon composition (23.3). Lemma 23.6 gives the product formula for its dynamical degree. This is precisely the claimed Hénon-type alternative. $\square$

Theorem 23.7 (Forward/backward Green functions). **Statement scope.** *Let $F : \mathbb{C}^2 \rightarrow \mathbb{C}^2$ be a polynomial automorphism of dynamical degree $d > 1$. The escape-rate functions*

$$G^+(z) = \lim_{n \rightarrow \infty} d^{-n} \log^+ \|F^n(z)\|, \quad G^-(z) = \lim_{n \rightarrow \infty} d^{-n} \log^+ \|F^{-n}(z)\|$$

exist and are continuous plurisubharmonic. They satisfy $G^+ \circ F = dG^+$ and $G^- \circ F^{-1} = dG^-$, equivalently $G^- \circ F = d^{-1}G^-$, and $K^\pm = \{G^\pm = 0\}$ are the forward/backward filled sets.

Proof strategy (not part of the proof)

Proof status. proved here modulo stated analytic input.

Strategic outline only. Hénon filtration; forward/backward escape growth; locally uniform Green convergence; plurisubharmonicity; scaling laws; filled-set zero loci.

The mathematical argument begins in the proof environment below.

Proof. Put

$$u_n^+(z) = d^{-n} \log^+ \|F^n(z)\|, \quad u_n^-(z) = d^{-n} \log^+ \|F^{-n}(z)\|.$$

Use the Hénon filtration.

Outside a sufficiently large bidisk there are forward and backward escape regions V^+ and V^- . A forward-unbounded orbit eventually enters V^+ , and there

$$|\log \|F(z)\| - d \log \|z\|| \leq C.$$

There is an analogous estimate for F^{-1} on V^- .

Obtain geometric Cauchy estimates.

Once $F^N(z) \in V^+$,

$$|u_{n+1}^+(z) - u_n^+(z)| \leq C d^{-(n+1)}$$

for all sufficiently large n . Summing the geometric tail proves pointwise convergence after entry into V^+ . To obtain uniformity on a compact set K , fix the large bidisk used in the Hénon filtration. For points whose first N iterates remain in that bidisk, continuity of the finitely many iterates gives a uniform bound for the corresponding $\log^+ \|F^n(z)\|$; after multiplication by d^{-n} , this initial part becomes uniformly small for large n . Every other point enters V^+ by time N . Compactness of K and continuity of the first N iterates give a finite cover on which the entry time is controlled, and after entry the displayed geometric estimate has a summable tail independent of the point. Combining the bounded initial part with that uniform tail proves $u_n^+ \rightarrow G^+$ locally uniformly. The same argument for F^{-1} gives G^- .

Record regularity.

Each $u_n^\pm$ is continuous and plurisubharmonic. A locally uniform limit of such functions is continuous and plurisubharmonic, so $G^\pm$ have the asserted regularity.

Derive the scaling laws by shifting the index.

For example,

$$G^+(Fz) = \lim_{n \rightarrow \infty} d^{-n} \log^+ \|F^{n+1}z\| = dG^+(z).$$

Likewise

$$G^-(F^{-1}z) = dG^-(z), \quad G^-(Fz) = d^{-1}G^-(z).$$

Identify the zero sets.

If the forward orbit is bounded, then $u_n^+(z) \rightarrow 0$. If it is unbounded, it eventually enters V^+ , where repeated lower growth estimates force positive normalized escape rate. Thus $G^+(z) = 0$ exactly on K^+ . The backward statement is identical. $\square$

Theorem 23.8 (Green currents and equilibrium measure). **Statement scope.** Let $F : \mathbb{C}^2 \rightarrow \mathbb{C}^2$ be a polynomial automorphism of dynamical degree $d > 1$. The currents $T^\pm = dd^c G^\pm$ are positive closed Green currents, and

$$\mu = T^+ \wedge T^-$$

is an F -invariant probability measure with entropy $\log d$. It is the unique measure of maximal entropy.

Proof strategy (not part of the proof)

Proof status. proved here modulo stated analytic input.

Strategic outline only. Green-current construction and Bedford-Taylor wedge; Smillie entropy growth; Pesin unstable conditional measures; Green-current slice Jacobian; BLS Jensen equality; analytic-disk current equidistribution; unique maximal entropy.

The mathematical argument begins in the proof environment below.

Proof. The Green functions are continuous plurisubharmonic potentials, so

$$T^+ = dd^c G^+, \quad T^- = dd^c G^-$$

are positive closed currents. Their functional equations give

$$F^* T^+ = d T^+, \quad F^* T^- = d^{-1} T^-.$$

Because the potentials are continuous, the Bedford-Taylor wedge

$$\mu = T^+ \wedge T^-$$

is defined. With the standard normalization it has mass one, and the reciprocal scaling factors show immediately that $F^* \mu = \mu$.

The remaining issue is entropy and uniqueness. We now prove that part rather than importing it.

Topological entropy.

Put F in cyclically reduced generalized-Hénon normal form. For a large filtration bidisk, a vertical disk whose boundary lies in the forward escape sector acquires boundary degree d^n after n iterates. Projection to the expanding coordinate therefore gives an area lower bound

$$\text{area}(F^n D \cap V) \geq cd^n.$$

Yomdin's local volume estimate converts this into $h_{\text{top}}(F) \geq \log d$; the standard algebraic-degree entropy upper bound gives the reverse inequality. Hence

$$h_{\text{top}}(F) = \log d.$$

The entropy is carried by the compact set of bounded full orbits. Standard smooth entropy theory therefore supplies an ergodic maximal-entropy measure ν .

Identify its unstable conditionals.

Positive entropy for both F and F^{-1} gives one positive and one negative complex Lyapunov exponent. Choose a Pesin unstable generator ξ^u . Its conditional transition probabilities $p_\nu(x)$ satisfy

$$-\int \log p_\nu d\nu = h_\nu(F) = \log d.$$

Slice T^+ on $W^u(x)$ and normalize its mass on $\xi^u(x)$ to a probability η_x . The slice has positive mass: otherwise G^+ would vanish on an unstable disk, contradicting the positive unstable exponent. The current scaling gives a transition probability

$$q(x) = \frac{\rho(Fx)}{d\rho(x)}, \quad -\int \log q d\nu = \log d.$$

On each unstable plaque, Jensen's inequality compares the conditional masses of the atoms of $F^{-1}\xi^u$. Equality of the two entropy integrals forces equality in Jensen, hence the two sets of atom masses coincide. Repeating this for every iterate shows

$$\nu(\cdot \mid \xi^u(x)) = \eta_x$$

for almost every x .

Push one conditional measure to equilibrium.

The analytic-disk current-convergence mechanism established in Theorem 23.1 applies to the Pesin unstable plaque $S = \xi^u(x)$ for F^{-1} ; the positive coefficient is supplied here by the already-proved positivity of the T^+ slice. Thus

$$d^{-n}(F^n)_*[S] \longrightarrow cT^-.$$

Wedge with T^+ to identify c with the T^+ -mass of S . Consequently

$$(F^n)_*\eta_x \longrightarrow T^+ \wedge T^- = \mu.$$

But Birkhoff's theorem, averaged over the same conditional measure, says that the Cesàro averages of these pushforwards converge to ν . Hence $\nu = \mu$.

Thus $h_\mu(F) = \log d$. Finally, any maximal-entropy invariant measure decomposes into ergodic maximal-entropy components, and the argument above forces every such component to be μ . Therefore μ is the unique measure of maximal entropy. $\square$

23.3 Worked examples

Worked example: A quadratic Hénon map and its inverse

Consider

$$H_{a,c}(x, y) = (x^2 + c - ay, x), \quad a \neq 0.$$

The Jacobian matrix is

$$DH_{a,c}(x, y) = \begin{pmatrix} 2x & -a \\ 1 & 0 \end{pmatrix}, \quad \det DH_{a,c} = a.$$

Hence the Jacobian is the nonzero constant a . Solving

$$(X, Y) = (x^2 + c - ay, x)$$

gives $x = Y$ and

$$y = \frac{Y^2 + c - X}{a},$$

so

$$H_{a,c}^{-1}(X, Y) = \left(Y, \frac{Y^2 + c - X}{a} \right).$$

Thus $H_{a,c}$ is a polynomial automorphism of dynamical degree 2.

If $|x|$ is large compared with $|y|$, then the first coordinate behaves like x^2 . This is the simplest model for the forward escape region. If $|y|$ dominates, the inverse formula behaves quadratically in y , giving the backward escape region. The two directions explain why two Green functions are natural in dimension two.

For example, if $|a| < 1/4$, then the degree-two map is substantially dissipative because $|\text{Jac } H_{a,c}| = |a| < 2^{-2}$. This verifies the numerical dissipativity hypothesis in the final theorem, but it does *not* by itself verify a dominated splitting; that remains a separate dynamical hypothesis.

Worked example: Scaling identities force equilibrium-measure invariance

Suppose $G^+ \circ F = dG^+$ and $G^- \circ F = d^{-1}G^-$. Applying dd^c gives

$$F^*T^+ = dT^+, \quad F^*T^- = d^{-1}T^-.$$

Therefore

$$F^*(T^+ \wedge T^-) = (dT^+) \wedge (d^{-1}T^-) = T^+ \wedge T^-.$$

The reciprocal factors are essential. If one accidentally wrote $G^- \circ F = dG^-$, the wedge would scale by d^2 , contradicting invariance. This is a useful normalization check whenever forward and backward Green functions appear together.

Boundary case: The singular limit $a = 0$

At $a = 0$,

$$H_{0,c}(x, y) = (x^2 + c, x).$$

The determinant of the derivative is zero, and the inverse formula from the first example no longer exists. The map collapses information in the y -direction and is not a polynomial automorphism. Consequently the backward Green function G^- , the current T^- , and the equilibrium measure $T^+ \wedge T^-$ from the automorphism the-

orem are not available from the stated hypotheses. The forward first-coordinate dynamics still shadows the polynomial $x \mapsto x^2 + c$, which explains why the singular limit is useful, but it is a different dynamical category.

Boundary case: Dissipative does not mean dominated

The inequality $|\text{Jac } F| < 1$ only says that complex volume contracts. Even the stronger inequality $|\text{Jac } F| < d^{-2}$ does not automatically produce a dominated splitting on J . Domination is a pointwise cocycle statement comparing two invariant directions. Therefore the Lyubich–Peters structure theorem cannot be invoked from a Jacobian estimate alone; both hypotheses must be verified.

23.4 What this chapter established

Positive dynamical degree is the algebraic signal that a plane polynomial automorphism belongs to the Hénon-type side of the Friedland–Milnor classification. Once in that class, forward and backward filtrations produce continuous plurisubharmonic escape-rate functions. Their dd^c -derivatives are the Green currents, and the intersection $T^+ \wedge T^-$ is the canonical invariant equilibrium measure. Bedford–Lyubich–Smillie identify it as the unique measure of maximal entropy $\log d$.

Under substantial dissipativity and a dominated splitting, the geometry becomes much more rigid. Strong-stable laminations, artificial extensions, and protected-lift estimates exclude wandering and rotation Fatou components, build unstable manifolds away from parabolic cycles, and force hyperbolicity there. The same transverse geometry produces saddle points densely enough to identify J with J^* .

Looking ahead

Chapter 24 moves from this established two-dimensional theory to contemporary frontiers: renormalization of unicritical disk diffeomorphisms, critical-quasicircle hyperbolicity, and carefully delimited open problems. The emphasis is not only on what is known, but on the exact hypotheses and source status under which the newest conclusions are valid.

Contemporary Complex-Dynamics

Frontiers and Open Problems

Why this chapter matters

Frontier mathematics requires two kinds of discipline at once. We want to learn the newest renormalization mechanisms, but we must also distinguish a proved theorem from a preprint, a theorem from an open problem, and convergence to a model *space* from convergence to one universal model. This chapter therefore treats source status and theorem scope as part of the mathematics rather than as editorial decoration.

The two formal results illustrate complementary frontier situations. The Crovisier-Lyubich-Pujals-Yang disk-diffeomorphism theorem remains a current preprint. We therefore present it at exact source scope, keep its valuable-chart and stable/center geometry as an explicit frontier-source boundary, and prove only the elementary thinness-to-super-exponential-distance consequence in the book. Lim's critical-quasicircle hyperbolicity theorem is now published; mathematically it is the same theorem already developed in Chapter 16, so here we reuse that theorem instead of duplicating its proof.

Learning goals

By the end of this chapter you should be able to:

1. explain why current source status and exact theorem scope matter at a research frontier;
2. distinguish a regularly unicritical disk diffeomorphism from an arbitrary infinitely renormalizable disk diffeomorphism;
3. define Hénon-like thinness and explain why λ^{cR_n} becomes super-exponential in renormalization depth;
4. state precisely what convergence to the space of one-dimensional unimodal maps does and does not say;
5. identify the stable/center geometric modules in the disk-diffeomorphism theorem;
6. state Lim's periodic-type Corona hyperbolicity theorem at its exact criticality and rotation-number scope;
7. understand why a later published restatement should reuse an earlier exact theorem rather than create an artificial dependency;
8. keep general MLC and the other frontier questions explicitly open where no theorem is known.

Conceptual roadmap

unicritical geometry $\implies$ valuable charts $\implies$ thin Hénon returns,
 $R_n \geq 2^n \implies \text{dist}(\mathcal{R}^n F, \mathcal{H}_0) = O(q^{2^n}),$
 periodic Corona class $\implies$ Chapter 16 hyperbolicity $\implies \dim W^u = 1,$
 frontier claims $\implies$ source audit before scope changes.

Common pitfall

Super-exponential approach to the *space* of one-dimensional unimodal maps does not by itself select a single universal limit. Such a conclusion needs additional compactness, bounded-combinatorics, or a priori-bounds input. Likewise, a theorem proved for periodic-type rotation numbers must not be silently extended to arbitrary irrational rotations.

24.1 Definitions

Definition 24.1 (frontier theorem). A frontier theorem is a recently established theorem in an active research area for which later improvements or publication updates may change the surrounding landscape without changing the statement recorded here.

Definition 24.2 (open problem). An open problem is a precise mathematical statement for which no proof is currently known. It is explicitly labeled as open and is never used as a theorem in a deduction.

Definition 24.3 (source-status gate). A source-status gate is the requirement that a contemporary claim be checked against a current primary or published source, including its hypotheses and theorem scope, before the claim is used as a proved input.

Definition 24.4 (versioned result). A versioned result is a theorem whose recorded statement is not silently enlarged when later strengthenings appear; any genuine strengthening is stated separately with its new hypotheses and conclusions.

Definition 24.5 (regularly unicritical disk diffeomorphism). In this volume, a *regularly unicritical disk diffeomorphism* means a mildly dissipative, infinitely renormalizable C^r disk diffeomorphism in the Crovisier-Lyubich-Pujals-Yang source class, with one nondegenerate quadratic critical orbit and the quantitative forward/backward regularity hypotheses required by their Main Theorem. The phrase is used only in this source-specific sense: it does not mean an arbitrary infinitely renormalizable disk diffeomorphism with one visibly critical orbit.

Definition 24.6 (Hénon-like map and one-dimensional subspace). A normalized map is *Hénon-like* when it has the form

$$G(x, y) = (g(x, y), x)$$

on a bidisk, with $g(\cdot, 0)$ a unimodal map and with controlled dependence on the transverse variable y . The *degenerate one-dimensional subspace* consists of maps $\iota(u)(x, y) = (u(x), x)$, for which g is independent of y .

Definition 24.7 (thin Hénon-like renormalization). A normalized Hénon-like map is δ -thin if, in the chosen C^r normalization, its transverse dependence is bounded by δ ; equivalently for the estimates used here, $\|\partial_y g\|_{C^{r-1}} \leq \delta$. Thinness measures distance from the one-dimensional degenerate subspace rather than distance from any single universal map.

Definition 24.8 (renormalization depth and return period). For nested renormalization domains, the n th return period R_n is the first-return time defining the n th normalized renormalization. If $r_j = R_{j+1}/R_j$ are the relative periods, then $r_j \geq 2$ for a proper renormalization, hence $R_n \geq 2^n$ after normalizing $R_0 = 1$. This elementary growth is what converts exponential smallness in R_n into super-exponential smallness in the depth n .

Definition 24.9 (critical corona and critical quasicircle). For fixed integers $d_0, d_\infty \geq 2$, a (d_0, d_∞) -critical corona is a normalized analytic corona map in Lim's Banach-manifold class whose invariant quasicircle carries the prescribed inner and outer criticalities d_0 and d_∞ . A critical quasicircle is that invariant quasicircle together with its marked critical data. These terms are used only at the source-defined Corona-renormalization scope.

Definition 24.10 (periodic-type rotation and rotational corona). An irrational rotation number θ is of *periodic type* when its continued-fraction combinatorics is periodic under the Gauss map. A rotational corona of rotation number θ is a critical corona whose dynamics on the invariant quasicircle has rotation number θ .

Definition 24.11 (hyperbolic fixed point in a Banach analytic manifold). Let $\mathcal{R}$ be an analytic map on a Banach analytic manifold and f_* a fixed point. The fixed point is *hyperbolic* if the derivative $D\mathcal{R}(f_*)$ admits an invariant stable/unstable splitting with spectrum strictly inside/outside the unit circle. The local stable and unstable manifolds are the corresponding invariant analytic manifolds tangent to those subspaces at f_* . Saying that the local unstable manifold is one-dimensional means that the unstable subspace has complex dimension one.

24.2 Main results

Theorem 24.1 (CLPY unicritical disk-diffeomorphism renormalization (frontier source theorem)). **Statement at the current source scope.** *There are universal constants in the Crovisier-Lyubich-Pujals-Yang source such that, for a mildly dissipative infinitely renormalizable C^r disk diffeomorphism ($r \geq 2$) satisfying their quantitative regular-unicritical hypothesis on the renormalization limit set, every sufficiently deep return admits a controlled C^r valuable chart in which the normalized return is Hénon-like and $O(\lambda_F^{cR_n})$ -thin for some $c > 0$. The same source gives the pointwise description of the local strong-stable and center geometry by regular boxes, critical tunnels, and valuable crescents. Consequently the normalized renormalizations approach the*

space of one-dimensional unimodal maps super-exponentially in the renormalization depth. This is retained as a current-preprint frontier theorem and is not used as a prerequisite for later proofs.

Proof strategy (not part of the proof)

Proof status. frontier theorem from the literature.

Strategic outline only. The full valuable-chart theorem and the stable/center geometry are attributed to the current CLPY preprint at exact source scope. The argument below proves only the elementary implication from the source thinness estimate $O(\lambda_F^{cR_n})$ to super-exponential C^{r-1} distance from the one-dimensional Hénon-like subspace.

The mathematical argument begins in the proof environment below.

Proof. The valuable-chart theorem and the complete stable/center-manifold description are current-preprint results of Crovisier-Lyubich-Pujals-Yang. We keep that source boundary explicit: their quantitative Pesin theory, critical tunnels, valuable crescents, recurrence, ordered cover and Theorem 14.1 are not reproved here as independent theorems.

What is elementary once their thinness estimate is available is the advertised super-exponential approach to the one-dimensional subspace. Write a normalized deep return as

$$G_n(x, y) = (g_n(x, y), x), \quad g_{n,0}(x) = g_n(x, 0).$$

The source theorem gives constants $C, c > 0$ and $0 < \lambda_F < 1$ such that

$$\|\partial_y g_n\|_{C^{r-1}} \leq C \lambda_F^{cR_n}.$$

On the uniformly normalized bidisk, the fundamental theorem of calculus in y and the same derivative bound give

$$\|G_n - \iota(g_{n,0})\|_{C^{r-1}} \leq C' \lambda_F^{cR_n}.$$

Since $\iota(g_{n,0})$ belongs to the degenerate one-dimensional Hénon-like subspace $\mathcal{H}_0^r$,

$$\text{dist}_{C^{r-1}}(G_n, \mathcal{H}_0^r) \leq C' \lambda_F^{cR_n}.$$

Finally every proper relative renormalization period is at least two, so $R_n \geq 2^n$. With $q = \lambda_F^c \in (0, 1)$,

$$C' \lambda_F^{cR_n} = C' q^{R_n} \leq C' q^{2^n},$$

which is super-exponential in the renormalization depth.

This proves the elementary consequence established here from the source theorem. It is convergence to the *space* of one-dimensional unimodal maps, not to one universal map. The full valuable-chart and local stable/center geometry remain explicitly attributed to the cited preprint, and this frontier theorem is not used as a prerequisite later in the book. $\square$

Theorem 24.2 (Lim hyperbolicity theorem for critical quasicircle renormalization).
Statement scope. Fix integers $d_0, d_{\infty} \geq 2$ and a periodic-type irrational rota-

tion number θ . There exists a compact analytic Corona-renormalization operator on a Banach analytic manifold of (d_0, d_∞) -critical coronas with a unique hyperbolic fixed point f_* . Its local stable manifold is exactly the set of nearby rotational coronas with rotation number θ , and its local unstable manifold is one-dimensional.

Proof strategy (not part of the proof)

Proof status. derived here.

Strategic outline only. Versioned theorem reuse: verify 2026 publication/scope parity, then invoke the exact Chapter 16 Corona theorem, inheriting its explicit upstream research interfaces rather than duplicating its proof.

The mathematical argument begins in the proof environment below.

Proof. Fix the criticalities $d_0, d_\infty \geq 2$ and the periodic-type irrational rotation number θ .

Match the theorem with the earlier Corona result.

The critical-quasicircle renormalization chapter already proved, for exactly these data, the existence of a compact analytic Corona-renormalization operator, its unique fixed corona, and hyperbolicity of that fixed point.

Identify the stable class.

That theorem identifies the local stable manifold with the nearby rotational coronas having the same prescribed criticalities and rotation number θ . Thus the stable conclusion here is not a new deduction from the disk-diffeomorphism theorem in this chapter.

Identify the unstable dimension.

The same earlier proof contains the global rigidity and cascade argument that excludes additional unstable directions. Consequently the unstable subspace, and hence the local unstable manifold, has complex dimension one.

Apply the versioning rule.

The present row records the now-published status of the theorem while preserving its mathematical scope. Reusing the earlier theorem is therefore the correct dependency. No claim is made for arbitrary irrational rotation numbers, variable criticality, or a larger ambient function space. $\square$

24.3 Worked examples

Worked example: Super-exponential thinness from return-time growth

Suppose a normalized return satisfies

$$\text{dist}(\mathcal{R}^n F, \mathcal{H}_0) \leq q^{R_n}, \quad 0 < q < 1,$$

and each relative period is at least two. Then $R_n \geq 2^n$, so

$$\text{dist}(\mathcal{R}^n F, \mathcal{H}_0) \leq q^{2^n}.$$

For $q = 1/2$, the first few comparison values are

$$2^{-2}, \quad 2^{-4}, \quad 2^{-8}, \quad 2^{-16}, \quad 2^{-32}.$$

Ordinary exponential convergence 2^{-n} would instead give

$$2^{-1}, \quad 2^{-2}, \quad 2^{-3}, \quad 2^{-4}, \quad 2^{-5}.$$

The double exponent comes from the growth of the return times, not from replacing one exponential estimate by another.

Worked example: A versioned frontier card

Consider two results. The first records the disk-diffeomorphism renormalization theorem at its current preprint scope. Its mathematical content can be used because the statement and argument structure have been checked against the source, but the citation must still identify it as a *preprint*. The second records Lim's Corona theorem after journal publication. The statement itself does not become stronger merely because its publication status changes.

The correct bibliographic treatment is therefore asymmetric: retain the first theorem's preprint status while keeping its exact hypotheses, and record the second theorem at its journal-publication status while reusing the full proof developed earlier. Versioning prevents either status change from silently changing theorem content.

Boundary case: Thinness does not select a universal one-dimensional limit

A sequence can satisfy

$$\text{dist}(G_n, \mathcal{H}_0) \longrightarrow 0$$

while its nearest one-dimensional maps u_n wander inside $\mathcal{H}_0$. For example, take two distinct unimodal maps u and v and let the degenerate part alternate between them while the transverse error tends to zero. Then the sequence approaches the one-dimensional *space* without converging to either u or v . A theorem selecting one universal limit needs additional compactness and rigidity input.

Boundary case: Periodic type is a real hypothesis

Lim's theorem is stated for periodic-type irrational rotation numbers. If θ is merely irrational, the theorem recorded here supplies no hyperbolic fixed point or codimension-one stable manifold. The logically strongest conclusion available from this row is therefore conditional: once periodic type is verified, the Corona theorem applies; without it, this chapter makes no replacement claim.

24.4 Chapter summary and perspective

This chapter illustrates two complementary ways modern complex dynamics reaches the edge of current knowledge. The disk-diffeomorphism theorem is a current preprint whose exact source class and valuable-chart mechanism must be kept explicit; from its thinness estimate we derived the super-exponential approach to the one-dimensional subspace. Lim's Corona theorem is already published and fits directly into the critical-quasicircle theory developed in Chapter 16.

These results do not erase the frontier. General MLC, density of hyperbolicity in its open scopes, invariant-line-field questions, and several rigidity problems remain open. Their role here is mathematical: they mark precisely where the established theory stops and where additional ideas would be required.

Looking ahead

A useful way to read the book as a whole is to trace how increasingly global conclusions are built from local analytic control: normal forms and normal families lead to quasiconformal deformation and puzzles; real and complex bounds create compact renormalization geometry; rigidity identifies limiting models; and, at the frontier, source scope becomes part of the theorem statement itself.

Proof status, sources, and further reading

This appendix records the proof boundary and the principal source for every formal result in the book. It is intentionally more explicit than an ordinary bibliography. The purpose is to let a reader distinguish three different activities: reading the proof given in this book, checking the historical or primary source, and pursuing the larger theory from which an imported engine is drawn.

A source entry does not mean that the proof in the chapter is merely copied or replaced by a citation. For proved here results, it records the literature against which the statement and argument structure were checked. For frontier theorem from the literature results, it records the current source boundary and the exact scope retained for orientation; these results are not used as prerequisites later. Classical and published entries were checked against the cited literature; frontier bibliographic status was rechecked on 8 October 2026.

Scope rule

Every formal theorem in the textbook is stated at a scope checked to be exact or narrower than the cited source. A frontier result remains labelled as a preprint or draft when that is its current status; an open problem is not entered here as a proved theorem.

A.1 Chapter 1: Local Holomorphic Dynamics and Normal Forms

Koenigs linearization theorem. *Proof status:* proved here. *Bibliographic status:* monograph/classical exposition. *Principal source:* Milnor, Dynamics in One Complex Variable, § 6; cross-check lecture-note formulation of Koenigs/Böttcher.

Böttcher local theorem. *Proof status:* proved here. *Bibliographic status:* monograph/classical exposition. *Principal source:* Milnor, Dynamics in One Complex Variable, § 6; cross-check lecture-note formulation of Koenigs/Böttcher.

Poincaré linearization at repelling points. *Proof status:* proved here. *Bibliographic status:* published/stable classical source. *Principal source:* A. Eremenko and M. Sodin, Iterations of rational functions and the distribution of the values of the Poincaré functions.

Multiplier classification of fixed points. *Proof status:* derived here. *Bibliographic status:* monograph/classical exposition. *Principal source:* Milnor, Dynamics in One Complex Variable, local fixed-point theory.

Holomorphic persistence of simple periodic points. *Proof status:* proved here. *Bibliographic status:* monograph/classical exposition. *Principal source:* Milnor, Dynamics in One Complex Variable, local fixed-point theory.

Small-divisor boundary for indifferent germs. *Proof status:* proved here. *Bibliographic status:* monograph/classical exposition. *Principal source:* Milnor, Dynamics in One Complex Variable, local fixed-point theory.

A.2 Chapter 2: Fatou and Julia Theory for Rational Maps

Fatou-Julia dichotomy. *Proof status:* proved here. *Bibliographic status:* published/stable classical source. *Principal source:* S. Bullett, Dynamics of rational maps, § 2.

Density of repelling periodic points in Julia set. *Proof status:* proved here. *Bibliographic status:* monograph/classical exposition. *Principal source:* Milnor, Dynamics in One Complex Variable, classical periodic Fatou classification / density of repelling cycles.

Backward orbit characterization. *Proof status:* proved here. *Bibliographic status:* published/stable classical source. *Principal source:* S. Bullett, Dynamics of rational maps, § 2.

Exceptional set theorem. *Proof status:* proved here. *Bibliographic status:* published/stable classical source. *Principal source:* S. Bullett, Dynamics of rational maps, § 2.

Periodic Fatou component classification interface. *Proof status:* proved here. *Bibliographic status:* monograph/classical exposition. *Principal source:* Milnor, Dynamics in One Complex Variable, classical periodic Fatou classification / density of repelling cycles.

A.3 Chapter 3: Polynomial Dynamics, Green Functions and External Rays

Global Böttcher coordinate for polynomials. *Proof status:* proved here. *Bibliographic status:* monograph/classical exposition. *Principal source:* Milnor, Dynamics in One Complex Variable, § 17.

Polynomial Green function. *Proof status:* proved here. *Bibliographic status:* monograph/classical exposition. *Principal source:* Milnor, Dynamics in One Complex Variable, § 17.

Brolin equidistribution theorem. *Proof status:* derived here. *Bibliographic status:* monograph/classical exposition. *Principal source:* H. Brolin, Invariant sets under iteration of rational functions (Arkiv for Matematik 6, 1965); Milnor; rational convergence supplied by the rational equilibrium-measure theorem proved later in Chapter 3.

Rational equilibrium measure and measure of maximal entropy. *Proof status:* proved here modulo stated foundation input. *Bibliographic status:* published/stable classical source. *Principal source:* M. J. Lyubich, Entropy properties of rational endomorphisms of the Riemann sphere (Ergodic Theory Dynam. Systems 3, 1983); J.-Y. Briand and J. Duval, Deux caractérisations de la mesure d'équilibre... (Publ. Math. IHÉS 93, 2001).

External rays and landing under local connectivity. *Proof status:* proved here. *Bibliographic status:* monograph/classical exposition. *Principal source:* Milnor, Dynamics in One Complex Variable, §§ 16–18.

Connectedness criterion for polynomials. *Proof status:* proved here. *Bibliographic status:* monograph/classical exposition. *Principal source:* Milnor, Dynamics in One Complex Variable, § 17.

A.4 Chapter 4: Parameter Spaces and the Mandelbrot Set: Foundations

Connectedness of the Mandelbrot set. *Proof status:* derived here. *Bibliographic status:* monograph/classical exposition. *Principal source:* Douady–Hubbard Orsay Notes, Chapter 8, Corollary 8.3.

Parameter Böttcher coordinate. *Proof status:* proved here. *Bibliographic status:* monograph/classical exposition. *Principal source:* Douady–Hubbard Orsay Notes, Chapter 8, Theorem 8.1; Milnor escape-rate/Böttcher construction.

Hyperbolic component multiplier map. *Proof status:* proved here. *Bibliographic status:* monograph/classical exposition. *Principal source:* Schleicher, Rational parameter rays of the Mandelbrot set, Corollaries 5.3–5.4; Douady–Hubbard Orsay Notes, Chapters 14 and 19.

Landing of rational parameter rays. *Proof status:* proved here. *Bibliographic status:* monograph/classical exposition. *Principal source:* Schleicher, Rational parameter rays of the Mandelbrot set, §§ 2–4; Milnor, Theorem 18.1 and Corollary 18.4.

A.5 Chapter 5: Quasiconformal Geometry and the Beltrami Equation

Measurable Riemann mapping theorem. *Proof status:* proved here modulo stated analytic input. *Bibliographic status:* monograph/classical exposition. *Principal source:* Ahlfors–Bers (1960), Riemann’s Mapping Theorem for Variable Metrics, §§ 1–4; Ahlfors, Lectures on Quasiconformal Mappings; Astala–Iwaniec–Martin, Chapter 5.

Modulus distortion under quasiconformal maps. *Proof status:* proved here modulo stated analytic input. *Bibliographic status:* monograph/classical exposition. *Principal source:* Ahlfors, Lectures on Quasiconformal Mappings, Chapters II–III; Bishop, Quasiconformal Mappings, geometric-properties chapter.

Quasisymmetric boundary values. *Proof status:* proved here. *Bibliographic status:* monograph/classical exposition. *Principal source:* Beurling–Ahlfors (1956); Ahlfors, Lectures on Quasiconformal Mappings, Chapter IV; Bishop, Quasiconformal Mappings, boundary and quasisymmetry sections.

Normal-family compactness for normalized QC maps. *Proof status:* proved here. *Bibliographic status:* monograph/classical exposition. *Principal source:* Ahlfors, Lectures on Quasiconformal Mappings; Bishop, Quasiconformal Mappings, compactness theorem for normalized K-QC maps.

A.6 Chapter 6: Holomorphic Motions and Teichmüller Theory

Lambda lemma and Ślodkowski extension theorem. *Proof status:* proved here modulo stated analytic input. *Bibliographic status:* published/stable classical source. *Principal source:* Mañé–Sad–Sullivan, On the dynamics of rational maps, Ann. Sci. Éc. Norm. Supér. (4) 16 (1983), no. 2, 193–217, DOI 10.24033/asens.1446; Ślodkowski (1995), Extensions of holomorphic motions. The attached-disc reflection/compactness and continuation mechanism is reconstructed here; only the stated standard analytic tools are frozen.

Ahlfors–Bers holomorphic dependence. *Proof status:* proved here modulo stated analytic input. *Bibliographic status:* monograph/classical exposition. *Principal source:* Ahlfors–Bers (1960), Riemann’s Mapping Theorem for Variable Metrics, especially Theorems 10–11; the Chapter 5 Beltrami-operator/MRMT package.

Holomorphic motion of Julia sets on stable neighborhoods. *Proof status:* proved here. *Bibliographic status:* published/stable classical source. *Principal source:*

Mañé-Sad-Sullivan, Ann. Sci. Éc. Norm. Supér. (4) 16 (1983), no. 2, 193–217, DOI 10.24033/asens.1446, Theorem B proof direction; Slodkowski (1995).

Invariant-Beltrami quasiconformal deformation. *Proof status:* derived here. *Bibliographic status:* monograph/classical exposition. *Principal source:* Ahlfors-Bers (1960); the Chapter 5 measurable Riemann mapping theorem and Beltrami-composition package.

A.7 Chapter 7: Sullivan Theory, No Wandering Domains and J-Stability

Sullivan no-wandering-domains theorem. *Proof status:* proved here modulo stated analytic input. *Bibliographic status:* monograph/classical exposition. *Principal source:* Sullivan, Ann. of Math. (2) 122 (1985), no. 2, 401–418, DOI 10.2307/1971308; Zakeri, Dynamics of Rational Maps, Lecture 21 (Baker-McMullen streamlined proof).

Global Fatou-component classification. *Proof status:* derived here. *Bibliographic status:* monograph/classical exposition. *Principal source:* Sullivan 1985; the Chapter 2 periodic Fatou-component classification (Milnor route already audited).

Mañé-Sad-Sullivan J-stability theorem. *Proof status:* proved here. *Bibliographic status:* monograph/classical exposition. *Principal source:* Mañé-Sad-Sullivan, Ann. Sci. Éc. Norm. Supér. (4) 16 (1983), no. 2, 193–217, DOI 10.24033/asens.1446; Berteloot, CIME lectures (2013), Lemmas 3.1.7–3.1.8 and Theorem 3.1.9; Astorg, HDR, Theorem 1.1.

A.8 Chapter 8: Polynomial-Like Maps, Straightening and Tuning

Straightening theorem. *Proof status:* proved here. *Bibliographic status:* published/stable classical source. *Principal source:* Douady-Hubbard, On the dynamics of polynomial-like mappings, Ann. Sci. Éc. Norm. Supér. (4) 18 (1985), no. 2, 287–343, DOI 10.24033/asens.1491.

Consequences of hybrid equivalence on the filled Julia set. *Proof status:* derived here. *Bibliographic status:* published/stable classical source. *Principal source:* Douady-Hubbard, On the dynamics of polynomial-like mappings, Ann. Sci. Éc. Norm. Supér. (4) 18 (1985), no. 2, 287–343, DOI 10.24033/asens.1491.

Normalized quadratic renormalization. *Proof status:* derived here. *Bibliographic status:* published/stable classical source. *Principal source:* Douady-Hubbard,

On the dynamics of polynomial-like mappings, *Ann. Sci. Éc. Norm. Supér.* (4) 18 (1985), no. 2, 287–343, DOI 10.24033/asens.1491.

Quadratic tuning and small Mandelbrot copies. *Proof status:* proved here. *Bibliographic status:* published/stable classical source. *Principal source:* Douady-Hubbard, On the dynamics of polynomial-like mappings, *Ann. Sci. Éc. Norm. Supér.* (4) 18 (1985), no. 2, 287–343, DOI 10.24033/asens.1491. M. Lyubich, Dynamics of quadratic polynomials I-II / renormalization survey, little Mandelbrot copies section.

A.9 Chapter 9: Yoccoz Puzzles, Principal Nests and Rigidity Machinery

Yoccoz puzzle construction. *Proof status:* proved here. *Bibliographic status:* published/stable classical source. *Principal source:* Hubbard, Local connectivity of Julia sets and bifurcation loci: three theorems of J.-C. Yoccoz, Sections 4-5.

Puzzle Markov and first-return structure. *Proof status:* proved here. *Bibliographic status:* published/stable classical source. *Principal source:* Hubbard/Yoccoz puzzle propositions and tableau rules; standard nice-set first-return formalism.

Yoccoz shrinking theorem in the nonrenormalizable repelling case. *Proof status:* proved here. *Bibliographic status:* published/stable classical source. *Principal source:* Hubbard, Local connectivity of Julia sets and bifurcation loci: three theorems of J.-C. Yoccoz, Theorem II and Sections 6-10.

Finite-renormalization local connectivity and combinatorial rigidity. *Proof status:* proved here. *Bibliographic status:* published/stable classical source. *Principal source:* Kahn-Lyubich, *Ann. of Math.* (2) 170 (2009), no. 1, 413–426, DOI 10.4007/annals.2009.170.413; Kahn-Lyubich, *Ann. of Math.* (2) 169 (2009), no. 2, 561–593, DOI 10.4007/annals.2009.169.561; Avila-Kahn-Lyubich-Shen, *Ann. of Math.* (2) 170 (2009), no. 2, 783–797, DOI 10.4007/annals.2009.170.783.

A.10 Chapter 10: Thurston Theory of Branched Coverings

Thurston characterization theorem. *Proof status:* proved here modulo stated foundation input. *Bibliographic status:* published/stable classical source. *Frozen foundation inputs:* standard finite-type Teichmüller theory, Jenkins-Strebel, the hyperbolic collar theorem, and Grötzsch annulus inequalities. *Principal source for the reconstructed Thurston-specific mechanism:* Douady-Hubbard, A proof of Thurston’s topological characterization of rational functions, *Acta Math.* 171 (1993), no. 2, 263–297, DOI 10.1007/BF02392534, §§ 1–8.

Levy cycle criterion for topological polynomials. *Proof status:* proved here. *Bibliographic status:* published/stable classical source. *Principal source:* Bielefeld-Fisher-Hubbard, The classification of critically preperiodic polynomials as dynamical systems, JAMS 5 (1992), Section 5.

A.11 Chapter 11: Interval and Unimodal Dynamics

Singer theorem. *Proof status:* proved here. *Bibliographic status:* monograph/classical exposition. *Principal source:* D. Singer, Stable orbits and bifurcations of maps of the interval, SIAM J. Appl. Math. 35 (1978), no. 2, 260–267, DOI 10.1137/0135020; de Melo–van Strien, One-Dimensional Dynamics, Theorem II.6.1.

No wandering intervals. *Proof status:* proved here modulo stated foundation input. *Bibliographic status:* standard-monograph complete proof with independent published cross-check. *Principal source:* W. de Melo and S. van Strien, One-Dimensional Dynamics, Springer, 1993, Chapter IV, Theorem A and Sections IV.1–IV.9. Independent cross-check: M. Martens, W. de Melo and S. van Strien, Acta Math. 168 (1992), 273–318. The later van Strien–Vargas real-bounds proof is retained only as a supplementary reconstruction; its 2007 erratum requiring a nice pulled-back target remains enforced.

Signed itinerary order and kneading admissibility. *Proof status:* proved here. *Bibliographic status:* monograph/classical exposition. *Principal source:* J. Milnor and W. Thurston, On iterated maps of the interval, §§ 2–4.

Milnor–Thurston kneading theorem. *Proof status:* proved here. *Bibliographic status:* published/stable classical source. *Principal source:* J. Milnor and W. Thurston, On iterated maps of the interval, in Dynamical Systems (College Park, MD, 1986–87), Lecture Notes in Math. 1342, Springer, 1988, 465–563, especially §§ 5–7.

Entropy monotonicity in the real quadratic family. *Proof status:* proved here. *Bibliographic status:* published/stable classical source. *Principal source:* M. Tsujii, A simple proof for monotonicity of entropy in the quadratic family, Ergodic Theory Dynam. Systems 20 (2000), no. 3, 925–933, DOI 10.1017/S014338570000050X.

A.12 Chapter 12: Real A Priori Bounds

Real a priori bounds. *Proof status:* proved here. *Bibliographic status:* monograph/classical exposition. *Principal source:* T. Clark, E. de Faria and S. van Strien, Asymptotically holomorphic methods for infinitely renormalizable C^r unimodal maps, Theorem 2.1; de Melo–van Strien, One-Dimensional Dynamics, Ch. VI.

Real Koebe principle. *Proof status:* proved here. *Bibliographic status:* monograph/classical exposition. *Principal source:* de Melo–van Strien, One-Dimensional Dynamics, Ch. IV; O. Kozlovski, Ann. of Math. (2) 152 (2000), no. 3, 743–762, DOI 10.2307/2661353, Lemma 2.2.

Cross-ratio inequality and deep C^3 replacement. *Proof status:* proved here. *Bibliographic status:* monograph/classical exposition. *Principal source:* O. Kozlovski, Getting rid of the negative Schwarzian derivative condition, Ann. of Math. (2) 152 (2000), no. 3, 743–762, DOI 10.2307/2661353; de Melo–van Strien, One-Dimensional Dynamics, Ch. IV.

Principal/restrictive-nest bounded geometry. *Proof status:* derived here. *Bibliographic status:* monograph/classical exposition. *Principal source:* de Melo–van Strien, One-Dimensional Dynamics, Ch. VI, Real Bounds and Bounded Geometry; Sullivan real-bounds framework.

Real rigidity precursor. *Proof status:* proved here. *Bibliographic status:* preprint/stable e-print. *Principal source:* M. Jakobson and G. Świątek, Quasisymmetric conjugacies between unimodal maps; W. de Melo and A. Pinto, Rigidity of C^2 infinitely renormalizable unimodal maps.

A.13 Chapter 13: Complex Bounds and Polynomial-Like Renormalization

Complex a priori bounds for real quadratic renormalization. *Proof status:* proved here. *Bibliographic status:* published/stable classical source. *Principal source:* M. Lyubich–M. Yampolsky, Dynamics of quadratic polynomials: complex bounds for real maps, Ann. Inst. Fourier (Grenoble) 47 (1997), no. 4, 1219–1255, DOI 10.5802/aif.1598, Theorem 1.1 and the Complex Bounds Theorem; Kahn–Lyubich covering lemma only as compatible modulus bookkeeping. The high-height case is closed in Chapter 13 by the finite-prefix closest-return decay mechanism.

Compactness of normalized fixed-modulus germs. *Proof status:* proved here. *Bibliographic status:* published/stable classical source. *Principal source:* A. Avila–M. Lyubich, The full renormalization horseshoe for unimodal maps of higher degree, Publ. Math. IHÉS 114 (2011), compact polynomial-like germ spaces.

Pullback and hybrid-rigidity criterion from complex bounds. *Proof status:* proved here. *Bibliographic status:* monograph/classical exposition. *Principal source:* Sullivan–McMullen pullback principle for polynomial-like maps with complex bounds; Ahlfors–Bers compactness developed in Chapters 5–6.

Kahn-Lyubich covering lemma. *Proof status:* proved here modulo stated analytic input. *Bibliographic status:* published/stable classical source. *Principal source:* J. Kahn–M. Lyubich, The Quasi-Additivity Law in Conformal Geometry, Ann. Math. 169 (2009), Covering Lemma. The harmonic-route quasi-additivity proof, Galois quasi-invariance mechanism, generalized separation modulus, and collar reduction are reconstructed here; only stated foundational extremal-length/topology/finite-cover inputs are frozen.

Beau bounds for real-symmetric quadratic-like germs. *Proof status:* proved here. *Bibliographic status:* published/stable classical source, reconstructed at the scope used in this book. *Principal sources:* M. Lyubich–M. Yampolsky, Dynamics of quadratic polynomials: complex bounds for real maps, Ann. Inst. Fourier (Grenoble) 47 (1997), no. 4, 1219–1255, DOI 10.5802/aif.1598, Lemma 2.8, Theorem 4.1 and the Complex Bounds Theorem; A. Avila–M. Lyubich, Publ. Math. IHÉS 114 (2011), Theorem 6.1.

A.14 Chapter 14: Renormalization Operator, Feigenbaum Fixed Point, and Horseshoe

Period-doubling Feigenbaum fixed point. *Proof status:* derived here. *Bibliographic status:* published/stable classical source. *Principal source:* Avila–Lyubich (2011) horseshoe; independent classical calibration: O. Lanford III, A Shorter Proof of the Existence of the Feigenbaum Fixed Point, CMP 96 (1984), 521–538.

Exact dynamical scaling at the period-doubling fixed point. *Proof status:* derived here. *Bibliographic status:* published/stable classical source. *Principal source:* Feigenbaum–Cvitanovic fixed-point equation; normalization cross-check with Lanford, CMP 96 (1984).

Analyticity of fixed combinatorial renormalization branches. *Proof status:* proved here. *Bibliographic status:* monograph/classical exposition. *Principal source:* M. Lyubich, Feigenbaum–Coullet–Tresser universality and Milnor’s Hairiness Conjecture, Ann. of Math. (2) 149 (1999), no. 2, 319–420, DOI 10.2307/120968, §§ 4–5 and Lemma 5.2.

Full quadratic renormalization horseshoe - topological form. *Proof status:* proved here. *Bibliographic status:* published/stable classical source, reconstructed at the scope used in this book. *Principal source:* A. Avila–M. Lyubich, The full renormalization horseshoe for unimodal maps of higher degree, Publ. Math. IHÉS 114 (2011), Main Theorem and Theorem 9.2; Theorems 5.1, 6.2–6.7; Lemmas 9.3–9.4 and Appendix A for the two-sided coding topology.

A.15 Chapter 15: Hyperbolicity, Rigidity, and Universality of Renormalization

Hyperbolicity of the bounded-type renormalization horseshoe. *Proof status:* proved here. *Bibliographic status:* monograph/classical exposition. *Principal source:* M. Lyubich, Feigenbaum–Coullet–Tresser universality and Milnor’s Hairiness Conjecture, *Ann. of Math.* (2) 149 (1999), no. 2, 319–420, DOI 10.2307/120968, Lemma 5.3 and Theorems 5.9, 8.2, 8.3, 8.4.

Exponential convergence along stable classes. *Proof status:* derived here. *Bibliographic status:* published/stable classical source. *Principal source:* M. Lyubich, *Ann. of Math.* (2) 149 (1999), no. 2, 319–420, DOI 10.2307/120968, Hyperbolicity Theorem and Theorem 8.2.

Parameter universality by transversality. *Proof status:* proved here. *Bibliographic status:* monograph/classical exposition. *Principal source:* M. Lyubich, Feigenbaum–Coullet–Tresser universality and Milnor’s Hairiness Conjecture, *Ann. of Math.* (2) 149 (1999), no. 2, 319–420, DOI 10.2307/120968, Universality Theorem.

Rigidity for bounded combinatorics. *Proof status:* proved here. *Bibliographic status:* monograph/classical exposition. *Principal source:* M. Lyubich, Feigenbaum–Coullet–Tresser universality and Milnor’s Hairiness Conjecture, *Ann. of Math.* (2) 149 (1999), no. 2, 319–420, DOI 10.2307/120968, Lemma 5.8 and Theorem 5.9, invoking the quadratic-polynomial Rigidity Theorem cited there.

A.16 Chapter 16: Critical Circle Maps and Renormalization

Real bounds for critical circle maps. *Proof status:* proved here. *Bibliographic status:* published/stable classical source. *Principal source:* Herman–Świątek real a priori bounds for critical circle maps; modern C^3 formulation/exposition: E. de Faria–P. Guarino, Real bounds and Lyapunov exponents, *Discrete Contin. Dyn. Syst. A* 36 (2016), no. 4, 1957–1982, DOI 10.3934/dcds.2016.36.1957.

Rigidity of critical circle maps. *Proof status:* proved here. *Bibliographic status:* published/stable classical source. *Principal source:* E. de Faria–W. de Melo, Rigidity of critical circle mappings I, *J. Eur. Math. Soc.* 1 (1999), no. 4, 339–392, DOI 10.1007/S100970050011; and Rigidity of critical circle mappings II, *J. Amer. Math. Soc.* 13 (2000), no. 2, 343–370, DOI 10.1090/S0894-0347-99-00324-0.

Hyperbolicity of critical-circle renormalization. *Proof status:* proved here. *Bibliographic status:* published/stable classical source. *Principal source:* M. Yampolsky,

Hyperbolicity of renormalization of critical circle maps, Publ. Math. IHÉS 96 (2003), 1–41, DOI 10.1007/s10240-003-0007-1, Theorem 4.4 and the cylinder-renormalization construction.

Universal analytic convergence for critical-circle renormalization. *Proof status:* proved here. *Bibliographic status:* published/stable classical source. *Principal source:* Khmelev-Yampolsky arbitrary-type geometric C^0 convergence together with de Faria-de Melo bounded-type C^r convergence; the book keeps these scopes separate and derives arbitrary-type finite- C^r control only after analytic hyperbolicity.

Critical quasicircle renormalization extension. *Proof status:* proved here. *Bibliographic status:* published 2003–2026; the proof in Chapter 16 reconstructs the Corona global chain, bounded-type rigidity/universality, parameter perturbation and uniqueness, direct collision realization, degeneration compactness, the wave/spreading/trading stages, and the Herman-ring a priori-bound engine from Lim Sections 6.2–8, including localization, controlled cuts, canonical pullback comparison, thick-thin and persistent-fence analysis, strict horizontal loss, exact-denominator completion, and the bounded-degree push-forward. *Principal sources:* W. Shen, On the measurable dynamics of real rational functions, Ergodic Theory Dynam. Systems 23 (2003), no. 3, 957–983, DOI 10.1017/S0143385702001311, Proposition 3.2; W. R. Lim, A priori bounds and degeneration of Herman rings with bounded type rotation number, Invent. Math. 242 (2025), no. 3, 827–893, DOI 10.1007/s00222-025-01369-2, Theorem A and §§ 3–9 (with arXiv v1 Section 7.4 used for the explicit denominator-completion presentation); W. R. Lim, Rigidity of J-rotational rational maps and critical quasicircle maps, Trans. Amer. Math. Soc. 379 (2026), no. 4, 2507–2567, DOI 10.1090/tran/9607; and W. R. Lim, Hyperbolicity of Renormalization of Critical Quasicircle Maps, Comm. Math. Phys. 407 (2026), no. 5, article 88, DOI 10.1007/s00220-026-05580-x.

A.17 Chapter 17: Siegel, Brjuno, Yoccoz and Cremer Dynamics

Brjuno sufficiency. *Proof status:* proved here. *Bibliographic status:* published classical research sources, reconstructed at the scope used in this book. *Principal sources:* A. M. Davie, The critical function for the semistandard map, Nonlinearity 7 (1994), 219–229, DOI 10.1088/0951-7715/7/1/009; T. Carletti-S. Marmi, Linearization of analytic and non-analytic germs of diffeomorphisms of $(\mathbb{C}, 0)$, Bull. Soc. Math. France 128 (2000), 69–85, Theorem 2.1 and Appendix B. The proof in Chapter 17 reconstructs the continued-fraction separation, resonant-block counters, superadditive Brjuno weight, exact divisor increment, analytic scalar majorant, weighted coefficient induction, and convergence argument; neither Davie’s lemma nor Yoccoz sharpness is imported into the sufficiency proof.

Pérez-Marco invariant compacta and hedgehogs. *Proof status:* proved here. *Bibliographic status:* published classical research source, reconstructed at the scope used in this book. *Principal source:* R. Pérez-Marco, *Fixed points and circle maps*, Acta Math. 179 (1997), no. 2, 243–294, DOI 10.1007/BF02392745, Theorem 1, Sections II.2, III.1–III.2 and V.1. The proof follows the elementary rational-rotation route: semilocal Leau–Fatou extension for rational multipliers, density of rational multiplier perturbations, Hausdorff closedness of invariant continua, filling, and nonlinearizable hedgehog identification. The source is retained as historical and comparative reference rather than as an additional theorem-sized prerequisite.

Yoccoz sharpness and quadratic necessity. *Proof status:* proved here. *Bibliographic status:* the historical sharpness theorem is due to Yoccoz; the coefficient-one finite-scale conformal-radius estimate used for this reconstruction is due to Buff–Chéritat. *Principal sources:* X. Buff–A. Chéritat, *On the Size of Quadratic Siegel Disks: Part I*, Theorem 1 and Sections 3–8; J. H. Hubbard, *Three Theorems of J.-C. Yoccoz*, Part I. The proof reconstructs the required finite-scale necessity chain through the multiplier disk, parabolic explosion and collision control, arithmetic good scales, punctured-domain conformal-radius loss, relative Schwarz, the Chapter 6 holomorphic-motion lift, and telescoping.

Closing perspective

The recurring theme of the book is that complex dynamics advances by converting qualitative recurrence into quantitative geometry and then using rigidity to identify the limit. Local normal forms, normal families, Green functions, quasiconformal deformation, puzzles, real and complex bounds, and renormalization are not isolated techniques. They are successive answers to the same question: what structure survives iteration strongly enough to control infinitely many scales?

The frontier remains open in important directions. The point of the final chapters is not to erase that boundary, but to make it mathematically visible. A rigorous textbook should teach not only established theorems, but also how to recognize exactly where established theory ends.